

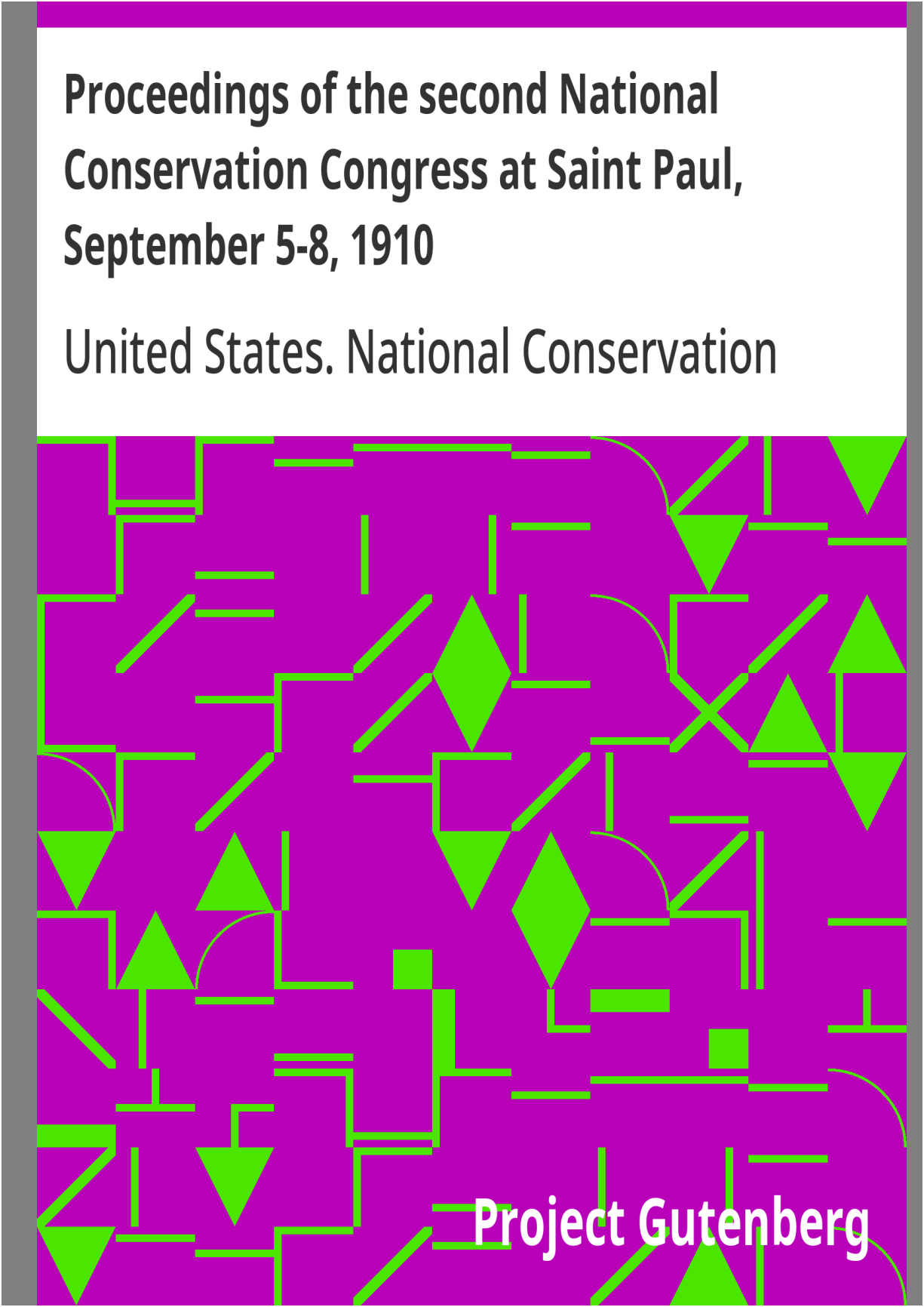
Proceedings of the second National Conservation Congress at Saint Paul, September 5-8, 1910

United States. National Conservation

Project Gutenberg

**Proceedings of the second National
Conservation Congress at Saint Paul,
September 5-8, 1910**

United States. National Conservation



Project Gutenberg

The Project Gutenberg eBook of Proceedings of the second National Conservation Congress at Saint Paul, September 5-8, 1910

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Proceedings of the second National Conservation Congress at Saint Paul, September 5-8, 1910

Creator: United States. National Conservation Congress

Release date: May 5, 2011 [eBook #36031]

Language: English

Other information and formats: www.gutenberg.org/ebooks/36031

Credits: Produced by Bryan Ness, Stephen H. Sentoff and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK PROCEEDINGS
OF THE SECOND NATIONAL CONSERVATION CONGRESS AT
SAINT PAUL, SEPTEMBER 5-8, 1910 ***

ADDRESSES AND PROCEEDINGS

OF THE

Second

National Conservation Congress

HELD AT

SAINT PAUL, MINNESOTA

SEPTEMBER 5-8

1910



BERNARD N. BAKER
BALTIMORE, MD.
President, Second National Conservation Congress

PROCEEDINGS

OF THE

Second

National Conservation Congress

AT

Saint Paul
SEPTEMBER 5-8, 1910

"Let us conserve the foundations of our prosperity"
(Declaration of the Governors, 1908)

WASHINGTON
NATIONAL CONSERVATION CONGRESS
1911

W. F. ROBERTS COMPANY
PRINTERS
WASHINGTON, D. C.



HON. J. B. WHITE

KANSAS CITY, MO.

Chairman, Executive Committee, Second National Conservation
Congress and Third National Conservation Congress

OFFICERS AND COMMITTEES FOR 1909-10

President

B. N. BAKER, Baltimore

Executive Secretary

THOMAS R. SHIPP, Washington, D. C.

Secretary

L. FRANK BROWN, Seattle

Vice-Presidents

JOHN BARRETT, Washington, D. C.

JAMES S. WHIPPLE, Albany

E. J. WICKSON, Berkeley

ALFRED C. ACKERMAN, Athens, Ga.

HENRY A. BARKER, Providence

Executive Committee

J. B. WHITE, Kansas City, Mo., *Chairman*

B. N. BAKER, Baltimore

J. N. TEAL, Portland, Ore.

A. B. FARQUHAR, York, Pa.

L. H. BAILEY, Ithaca

THOMAS BURKE, Seattle

HENRY E. HARDTNER, Urania, La.

W. A. FLEMING JONES, Las Cruces

MRS PHILIP N. MOORE, Saint Louis

MRS J. ELLEN FOSTER, Washington, D. C.

Local Board of Managers for the Saint Paul Congress

HON. A. O. EBERHART, *Chairman*

FRANK B. KELLOGG, *Vice-Chairman*

J. S. BELL, Minneapolis

H. A. TUTTLE, Minneapolis

GEORGE M. GILLETTE, Minneapolis

B. F. NELSON, Minneapolis

L. S. DONALDSON, Minneapolis

JOSEPH H. BEEK, Saint Paul

GEORGE H. PRINCE, Saint Paul

REUBEN WARNER, Saint Paul

PAUL W. DOTY, Saint Paul
THEODORE W. GRIGGS, Saint Paul
W. C. HANDY, *Secretary*

OFFICERS AND COMMITTEES FOR 1910-11

President

HENRY WALLACE, Des Moines

Executive Secretary

THOMAS R. SHIPP, Washington, D. C.

Treasurer

D. AUSTIN LATCHAW, Kansas City, Mo.

Recording Secretary

JAMES C. GIPE, Clarks, La.

Executive Committee

J. B. WHITE, Kansas City, Mo., *Chairman*

B. N. BAKER, Baltimore

L. H. BAILEY, Ithaca

JAMES R. GARFIELD, Cleveland

FRANK C. GOUDY, Denver

W. A. FLEMING JONES, Las Cruces

MRS PHILIP N. MOORE, Saint Louis

WALTER H. PAGE, New York

GEORGE C. PARDEE, Oakland, Cal.

GIFFORD PINCHOT, Washington, D. C.

J. N. TEAL, Portland, Ore.

E. L. WORSHAM, Atlanta

Vice-Presidents

ALABAMA, Hon. Albert P. Bush, Mobile; ALASKA, Hon. James Wickersham, Fairbanks; ARIZONA, B. A. Fowler, Phenix; ARKANSAS, A. H. Purdue, Fayetteville; CALIFORNIA, E. H. Cox, San Francisco; COLORADO, Murdo Mackenzie, Trinidad; COLUMBIA (District of), W J McGee, Washington; CONNECTICUT, Rollin S. Woodruff, Hartford; DELAWARE, Hon. George Gray, Wilmington; FLORIDA, Cromwell Gibbons, Jacksonville; GEORGIA, Hon. Jno. C. Hart, Union Point; HAWAII, Mrs Margaret R. Knudsen, Kanai; IDAHO, James A. MacLean, University of Idaho; ILLINOIS, Julius Rosenwald, Chicago; INDIANA, F. J. Breeze, Lafayette; IOWA, Carl Leopold, Burlington; KANSAS, W. R. Stubbs, Topeka; KENTUCKY, James K. Patterson, Lexington; LOUISIANA, Newton C. Blanchard, Shreveport; MAINE, Bert M. Fernald, Augusta; MARYLAND, William Bullock Clark, Baltimore; MASSACHUSETTS, Frank W. Rane, Boston; MICHIGAN, J. L. Snyder, Lansing; MINNESOTA, Ambrose Tighe, Saint Paul; MISSISSIPPI, A. W. Shands, Sardis; MISSOURI, Hermann Von Schrenk, Saint Louis; MONTANA, E. L. Norris, Helena; NEBRASKA, Dr F. A. Long, Madison; NEVADA, Senator Francis G. Newlands, Reno; NEW HAMPSHIRE, George B. Leighton, Monadnock; NEW JERSEY, Charles Lathrop Pack, Lakewood; NEW MEXICO, W. A. Fleming Jones, Las Cruces; NEW YORK, R. A. Pearson, Albany; NORTH CAROLINA, T. Gilbert Pearson, Greensboro; NORTH DAKOTA, U. G. Larimore, Larimore; OHIO, James R. Garfield, Cleveland; OKLAHOMA, Benj. Martin, Jr., Muskogee; OREGON, J. N. Teal, Portland; PENNSYLVANIA, William S. Harvey, Philadelphia; PHILIPPINE ISLANDS, Maj. George P. Ahern, Manila; PORTO RICO, Hon. Walter K. Landis, San Juan; RHODE ISLAND, Henry A. Barker, Providence; SOUTH CAROLINA, E. J. Watson, Columbia; SOUTH DAKOTA, Ellwood C. Perisho, Vermillion; TENNESSEE, Herman Suter, Nashville; TEXAS, W. Goodrich Jones, Temple; UTAH, Harden Bennion, Salt Lake City; VERMONT, Fletcher D. Proctor, Proctor; VIRGINIA, A. R. Turnbull, Norfolk; WASHINGTON, M. E. Hay, Olympia; WEST VIRGINIA, A. B. Fleming, Fairmont; WISCONSIN, Charles R. Van Hise, Madison; WYOMING, Bryant B. Brooks, Cheyenne; NATIONAL CONSERVATION ASSOCIATION, Gifford Pinchot, Washington.

Standing Committees

FORESTS—H. S. Graves, U. S. Forester, Washington, D. C., *Chairman*; E. M. Griffith, Madison, Wis.; E. T. Allen, Portland, Ore.; J. Lewis Thompson, Houston.

LANDS—Governor W. R. Stubbs, Topeka, *Chairman*; Dwight B. Heard, Phenix; J. L. Snyder, Lansing; Murdo Mackenzie, Trinidad; Charles S. Barrett, Union City, Ga.

WATERS—W J McGee, Washington, D. C., *Chairman*; E. A. Smith, Spokane; Henry A. Barker, Providence; J. N. Teal, Portland, Ore.; Herbert Knox Smith, Washington, D. C.

MINERALS—Charles R. Van Hise, Madison, *Chairman*; Joseph A. Holmes, Washington, D. C.; D. W. Brunton, Denver; John Mitchell, New York; I. C. White, Morgantown, W. Va.

VITAL RESOURCES—Dr William H. Welch, Baltimore, *Chairman*; Professor Irving Fisher, New Haven; Dr H. W. Wiley, Washington, D. C.; Dr J. H. Kellogg, Battle Creek, Mich.; Walter H. Page, New York.



HENRY WALLACE
DES MOINES, IOWA
President, Third National Conservation Congress

CONTENTS

	PAGE
CONSTITUTION	<u>IX</u>
OPENING SESSION	<u>1</u>
Invocation by ARCHBISHOP IRELAND	<u>1</u>
Greeting from CARDINAL GIBBONS	<u>3</u>
Address by GOVERNOR EBERHART	<u>3</u>
Welcome by MAYOR KELLER	<u>13</u>
Address by PRESIDENT TAFT	<u>14</u>
SECOND SESSION	<u>34</u>
Induction of GOVERNOR STUBBS as Chairman	<u>34</u>
Address by SENATOR NELSON	<u>35</u>
Address by GOVERNOR NOEL	<u>48</u>
Address by GOVERNOR NORRIS	<u>52</u>
Address by GOVERNOR DENEEN	<u>59</u>
Address by GOVERNOR HAY	<u>64</u>
Announcement by PROFESSOR CONDRA	<u>71</u>
Address by GOVERNOR BROOKS	<u>72</u>
Remarks by GOVERNOR STUBBS	<u>75</u>
Address by GOVERNOR VESSEY	<u>77</u>
THIRD SESSION	<u>79</u>
Appointment of Credentials Committee	<u>79</u>
Action on Constitution of the National Conservation Congress	<u>79</u>
Remarks by DIRECTOR-GENERAL BARRETT	<u>80</u>
Remarks by GOVERNOR STUBBS	<u>81</u>
Invocation by REVEREND DOCTOR MONTGOMERY	<u>81</u>
Address by EX-PRESIDENT ROOSEVELT	<u>82</u>
FOURTH SESSION	<u>93</u>
Address by MISS BOARDMAN	<u>94</u>
Address by COMMISSIONER HERBERT KNOX SMITH	<u>101</u>
Modification of Credentials Committee	<u>106</u>
Address by HONORABLE JAMES R. GARFIELD	<u>106</u>
Address by EX-GOVERNOR PARDEE	<u>115</u>
Remarks by DELEGATE HARR, of Washington	<u>120</u>
Address by EX-GOVERNOR BLANCHARD	<u>121</u>

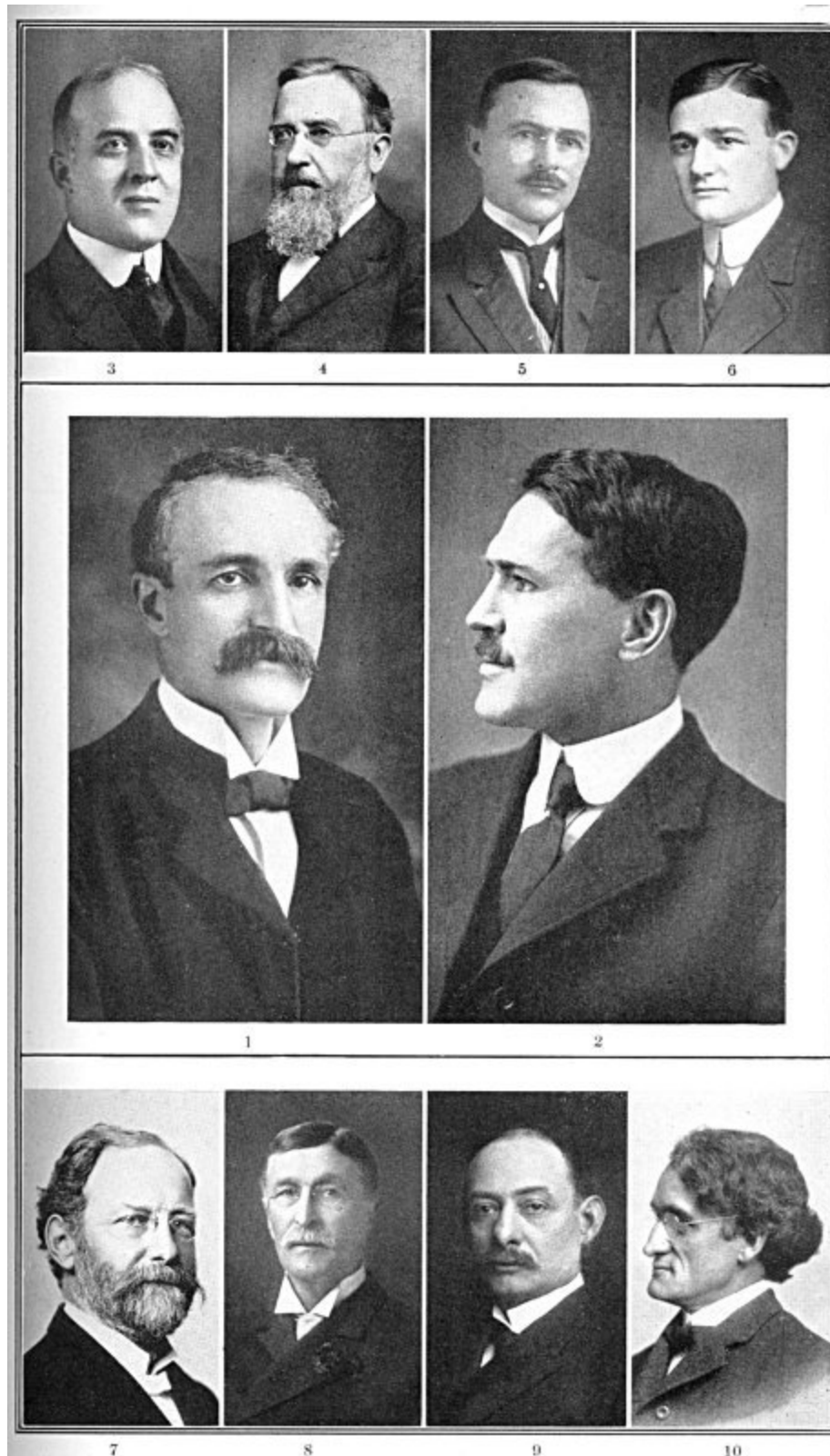
Address by WILLIAM E. SMYTHE	<u>127</u>
Address by WALTER L. FISHER	<u>129</u>
Address by COLONEL JAMES H. DAVIDSON	<u>132</u>
FIFTH SESSION	<u>134</u>
Invocation by BISHOP EDSALL	<u>134</u>
Address by PRESIDENT FINLEY	<u>135</u>
Report of Credentials Committee	<u>145</u>
Address by SENATOR BEVERIDGE	<u>146</u>
Response by GIFFORD PINCHOT	<u>152</u>
Address by PRESIDENT McVEY	<u>152</u>
Discussion by CHAIRMAN WHITE	<u>158</u>
Address by MRS WELCH, of the General Federation of Women's Clubs	<u>160</u>
Address by MRS HOYLE TOMKIES, of the Women's National Rivers and Harbors Congress	<u>163</u>
Address by MRS SNEATH, of the General Federation of Women's Clubs	<u>166</u>
Report by MRS HOWARD, of the Daughters of the American Revolution	<u>167</u>
SIXTH SESSION	<u>168</u>
Induction of SENATOR CLAPP as Chairman	<u>168</u>
Address by PRESIDENT CRAIGHEAD	<u>168</u>
Postponement of Call of States	<u>171</u>
Address by D. AUSTIN LATCHAW	<u>171</u>
Address by JAMES J. HILL	<u>177</u>
Discussion by HENRY WALLACE	<u>188</u>
Address by SECRETARY WILSON	<u>194</u>
Discussion by REPRESENTATIVE STEVENS	<u>201</u>
Address by PROFESSOR BAILEY	<u>203</u>
SEVENTH SESSION	<u>213</u>
Address by PROFESSOR GRAVES	<u>214</u>
Address by ALFRED L. BAKER	<u>222</u>
Address by FRANK H. SHORT	<u>226</u>
Address by DIRECTOR-GENERAL BARRETT	<u>237</u>

Address by HONORABLE ESMOND OVEY	<u>243</u>
Action on time for election and report of Resolutions Committee	<u>246</u>
EIGHTH SESSION	<u>246</u>
Appointment of Nominating Committee	<u>246</u>
Induction of GOVERNOR EBERHART as Chairman	<u>246</u>
Address by DEAN WESBROOK	<u>247</u>
Address by WALLACE D. SIMMONS	<u>257</u>
Address by COMMISSIONER ELMER E. BROWN	<u>264</u>
Address by MRS SCOTT, President of the Daughters of the American Revolution	<u>270</u>
Action in memory of MRS J. ELLEN FOSTER	<u>276</u>
Presentation by MRS HOWARD to GIFFORD PINCHOT	<u>276</u>
Response by MR PINCHOT	<u>277</u>
Address by FRANCIS J. HENEY	<u>278</u>
Address by GIFFORD PINCHOT	<u>292</u>
Expression by GOVERNOR EBERHART	<u>298</u>
Statement by PROFESSOR CONDRA	<u>298</u>
CLOSING SESSION	<u>299</u>
Commencement of Call of States	<u>299</u>
Response by DELEGATE HARVEY, of Pennsylvania	<u>299</u>
Interlude by E. W. ROSS, of Washington	<u>302</u>
Report of Nominating Committee	<u>303</u>
Nomination by CHAIRMAN WHITE	<u>303</u>
Second by GIFFORD PINCHOT	<u>304</u>
Election of and response by HENRY WALLACE as President	<u>305</u>
Election of other Officers	<u>306</u>
Resolution of thanks to retiring PRESIDENT BAKER	<u>308</u>
Response by MR BAKER	<u>308</u>
Report of Resolutions Committee	<u>308</u>
Adoption of Resolutions	<u>312</u>
Interlude by E. W. ROSS, of Washington	<u>312</u>
Remarks by DELEGATE HERR, of Washington	<u>313</u>
Ratification of Vice-Presidents	<u>313</u>

Resolution in memory of PROFESSOR GREEN	<u>313</u>
Resumption of Call of States	<u>314</u>
Response by DELEGATE PURDUE, of Arkansas	<u>314</u>
Response by DELEGATE BANNISTER, of Indiana	<u>314</u>
Response by DELEGATE MILLER, of Iowa	<u>314</u>
Response by DELEGATE YOUNG, of Kansas	<u>314</u>
Response by DELEGATE BAKER, of Maryland	<u>314</u>
Response by DELEGATE THORP, of Minnesota	<u>315</u>
Response by STATE GEOLOGIST LOWE, of Mississippi	<u>315</u>
Response by GENERAL NOBLE, of Missouri	<u>315</u>
Response by CHAIRMAN WHITE	<u>316</u>
Response by PROFESSOR CONDRA, of Nebraska	<u>317</u>
Response by a Delegate from New York	<u>318</u>
Response by DELEGATE NESTOS, of North Dakota	<u>318</u>
Response by DELEGATE KRUEGER, of South Dakota	<u>319</u>
Remarks by DELEGATE JOHNS, of Washington	<u>320</u>
Privileged statement by LAND COMMISSIONER ROSS, of Washington	<u>322</u>
Response by DELEGATE FOWLER, of Arizona	<u>324</u>
Response by DELEGATE HUNT, of District of Columbia	<u>324</u>
Response by DELEGATE BARKER, of Rhode Island	<u>324</u>
Response by PROFESSOR WHITE, of West Virginia	<u>325</u>
Response by DELEGATE WORSHAM, of Georgia	<u>325</u>
Motion for adjournment by DELEGATE MARTIN, of Oklahoma	<u>326</u>
SUPPLEMENTARY PROCEEDINGS	<u>327</u>
Laws that should be Passed, by SENATOR FRANCIS G. NEWLANDS	<u>327</u>
Conservation of the Nation's Resources, by CHAIRMAN J. B. WHITE	<u>328</u>
Practical Aspects of Conservation, by A. B. FARQUHAR	<u>331</u>
Report from Arkansas, by SID B. REDDING	<u>333</u>
Report from Colorado, by FRANK C. GOUDY	<u>334</u>
Report from Florida, by CROMWELL GIBBONS	<u>335</u>
Report from Idaho, by JEROME J. DAY	<u>336</u>
Report from Indiana, by A. E. METZGER	<u>336</u>

Report from Iowa, by A. C. MILLER	<u>337</u>
Report from Louisiana, by HENRY E. HARDTNER	<u>339</u>
Report from Maine, by CYRUS C. BABB	<u>341</u>
Report from Massachusetts, by FRANK WILLIAM RANE and HENRY H. SPRAGUE	<u>343</u>
Report from Missouri, by HERMANN VON SCHRENK	<u>344</u>
Report from Montana, by RUDOLPH VON TOBEL	<u>345</u>
Report from New Mexico, by COLONEL W. A. FLEMING JONES	<u>347</u>
Report from New York, by J. S. WHIPPLE	<u>347</u>
Special report from New York, by HENRY H. PERSONS	<u>352</u>
Report from North Dakota, by PROFESSOR WALDRON	<u>362</u>
Report from Ohio, by PROFESSOR LAZENBY	<u>364</u>
Report from Oklahoma, by BENJ. MARTIN, JR.	<u>365</u>
Report from Oregon, by E. T. ALLEN	<u>367</u>
Report from Rhode Island, by HENRY A. BARKER	<u>368</u>
Report from South Carolina, by E. J. WATSON	<u>369</u>
Report from South Dakota, by DOANE ROBINSON	<u>369</u>
Report from Texas, by WILL L. SARGENT	<u>370</u>
Report from Utah, by O. J. SALISBURY	<u>372</u>
Supplementary report from Utah, by E. T. MERRITT	<u>372</u>
Report from Vermont, by GEORGE AITKIN	<u>373</u>
Report from Washington, by E. G. GRIGGS	<u>375</u>
Report from West Virginia, by HU MAXWELL	<u>376</u>
Report from Wisconsin, by E. M. GRIFFITH	<u>377</u>
Report of the American Academy of Political and Social Science	<u>379</u>
Report of the American Automobile Association	<u>380</u>
Report of the American Civic Association	<u>383</u>
Report of the American Forestry Association	<u>384</u>
Report of the American Humane Association	<u>385</u>
Report of the American Institute of Architects	<u>386</u>
Report of the American Paper and Pulp Association	<u>388</u>
Report of the American Medical Association	<u>389</u>

Report of the American Railway Engineering and Maintenance of Way Association	<u>392</u>
Report of the American Railway Master Mechanics' Association	<u>393</u>
Report of the American Scenic and Historic Preservation Society	<u>394</u>
Report of the Association for the Protection of the Adirondacks	<u>397</u>
Report of the Carriage Builders' National Association	<u>410</u>
Report of the Delaware State Federation of Women's Clubs	<u>411</u>
Report of the Farmers' Union of America	<u>411</u>
Report of the General Federation of Women's Clubs	<u>412</u>
Report of the Lakes-to-Gulf Deep Waterway Association	<u>413</u>
Report of the League of American Sportsmen	<u>415</u>
Report of the National Board of Fire Underwriters	<u>416</u>
Report of the National Board of Trade	<u>419</u>
Report of the National Business League of America	<u>420</u>
Report of the Missouri Valley River Improvement Association	<u>420</u>
Report of the Upper Mississippi River Improvement Association	<u>421</u>
Report of the Washington State Federation of Labor	<u>422</u>
Report of the Western Forestry and Conservation Association	<u>423</u>
Report of the United Mine Workers	<u>424</u>
Timber Conservation, by GEORGE H. EMERSON	<u>424</u>
Forests and Stream-Flow, by WILLIAM S. HARVEY	<u>428</u>
The Conservation of Minerals and Subterranean Waters, by GEORGE F. KUNZ, Ph.D.	<u>429</u>
The Question of Land Titles, by FRANKLIN McCRAY	<u>430</u>
INDEX	<u>431</u>



1. GIFFORD PINCHOT, Vice-President (1910).
 2. JAMES R. GARFIELD, Vice-President (1910).

3. HENRY A. BARKER, Vice-President (1909-10).
 4. A. B. FARQUHAR, Executive Committee (1909).
 5. W. A. FLEMING JONES, Vice-president (1910).
 6. E. L. WORSHAM, Executive Committee (1910).
 7. GEORGE C. PARDEE, Executive Committee (1910).
 8. J. N. TEAL, Executive Committee (1909-10).
 9. WALTER H. PAGE, Executive Committee (1910).
 10. L. H. BAILEY, Executive Committee (1909-10).
-

CONSTITUTION

ARTICLE 1—NAME

This organization shall be known as the National Conservation Congress.

ARTICLE 2—OBJECT

The object of the National Conservation Congress shall be: (1) to provide a forum for discussion of the resources of the United States as the foundation for the prosperity of the people, (2) to furnish definite information concerning the resources and their utilization, and (3) to afford an agency through which the people of the country may frame policies and principles affecting the wise and practical development, conservation, and utilization of the resources, to be put into effect by their representatives in State and Federal Governments.

ARTICLE 3—MEETINGS

Section 1. Regular annual meetings shall be held at such time and place as may be determined by the Executive Committee.

Section 2. Special meetings of the Congress, or its officers, committees, or boards, may be held subject to the call of the President of the Congress or the Chairman of the Executive Committee.

ARTICLE 4—OFFICERS

Section 1. The officers of the Congress shall consist of a President, to be elected by the Congress; a Vice-President from each State, to be chosen by the respective State delegations, and from the National Conservation Association; an Executive Secretary; a Recording Secretary; and a Treasurer.

Section 2. The duties of these officers may at any time be prescribed by formal action of the Congress or Executive Committee. In the absence of such action their duties shall be those implied by their designations and established by custom. In addition, it shall be the duty of the Vice-Presidents to receive, from the State Conservation Commissions and other organizations concerned in Conservation, suggestions and recommendations, and report them to the Executive Committee of the Congress.

Section 3. The officers shall serve for one year, or until their successors are elected and qualify.

ARTICLE 5—COMMITTEES AND BOARDS

Section 1. An Executive Committee of seven, in addition to which the President of the National Conservation Association and all ex-Presidents of the Congress shall be members ex-officio, shall be appointed by the President during each regular annual session to act for the ensuing year; its membership shall be drawn from different States, and not more than one of the appointed members shall be from any one State. The Executive Committee shall act for the Congress and shall be empowered to initiate action and meet emergencies. It shall report to each regular annual session.

Section 2. A Board of Managers shall be created in each city in which the next ensuing session of the Congress is to be held, preferably by leading organizations of citizens. The Board of Managers shall have power to raise and expend funds, to incur obligations on its own responsibility, and to appoint subordinate boards and committees, all with the approval of the Executive Committee of the Congress. It shall report to the Executive Committee at least two days before the opening of the ensuing session, and at such other times as the Congress or the Executive Committee may direct.

Section 3. A Committee on Credentials shall be appointed, consisting of five (5) members, by the President of the Congress not later than on the second day of each session of the Congress. It shall determine all questions raised by delegates as to representation, and shall report to the Congress from time to time as required by the President of the Congress.

Section 4. A Committee on Resolutions shall be created for each annual meeting of the Congress. A Chairman shall be appointed by the President. One member of the Committee shall be selected by each State represented in the Congress. The Committee shall report to the Congress not later than the morning of the last day of each annual meeting.

Section 5. Permanent Committees, consisting of five (5) members each, shall be appointed by the President of the Congress on each of the following five divisions of Conservation: Forests, Waters, Lands, Minerals, and Vital Resources. These committees shall, during the intervals between the annual meetings of the Congress, inquire into these respective subjects and prepare reports to be submitted on the request of the Executive Committee, and render such other assistance to the Congress as the Executive Committee may direct.

Section 6. By direction of the Congress, standing and special committees may be appointed by the President.

Section 7. The President shall be a member, ex-officio, of every committee of the Congress.

ARTICLE 6—ARRANGEMENTS FOR SESSIONS

Section 1. The program for the session of each annual meeting of the Congress, including a list of speakers, shall be arranged by the Executive Committee. The entire program, including allotments of time to speakers and hours for daily sessions and all other arrangements concerning the program, shall be made by the Executive Committee.

Section 2. Unless otherwise ordered, the rules adopted for the guidance of the preceding Congress shall continue in force.

ARTICLE 7—MEMBERSHIP

Section 1. The personnel of the National Conservation Congress shall be as follows:

Officers and Delegates

Officers of the National Conservation Congress.

Fifteen Delegates appointed by the Governor of each State and Territory.

Five Delegates appointed by the Mayor of each city with a population of 25,000, or more.

Two Delegates appointed by the Mayor of each city with a population of less than 25,000.

Two Delegates appointed by each Board of County Commissioners.

Five Delegates appointed by each National Organization concerned in the work of Conservation.

Five Delegates appointed by each State or Interstate Organization concerned in the work of Conservation.

Three Delegates appointed by each Chamber of Commerce, Board of Trade, Commercial Club, or other local organization concerned in the work of Conservation.

Two Delegates appointed by each State or other University or College, and by each Agricultural College or Experiment Station.

HONORARY MEMBERS

The President of the United States.

The Vice-President of the United States.

The Speaker of the House of Representatives.

The Cabinet.

The United States Senate and House of Representatives.

The Supreme Court of the United States.

The Representatives of Foreign Governments.

The Governors of the States and Territories.

The Lieutenant-Governors of the States and Territories.

The Speakers of State Houses of Representatives.

The State Officers.

The Mayors of Cities.

The County Commissioners.

The Presidents of State and other Universities and Colleges.

The Officers and Members of the National Conservation Association.

The Officers and Members of the National Conservation Commission.

The Officers and Members of the State Conservation Commissions and Associations.

ARTICLE 8—DELEGATIONS AND STATE OFFICERS

Section 1. The several Delegates from each State in attendance at any Congress shall assemble at the earliest practicable time and organize by choosing a Chairman and a Secretary. These Delegates, when approved by the Committee on Credentials, shall constitute the Delegation from that State.

ARTICLE 9—VOTING

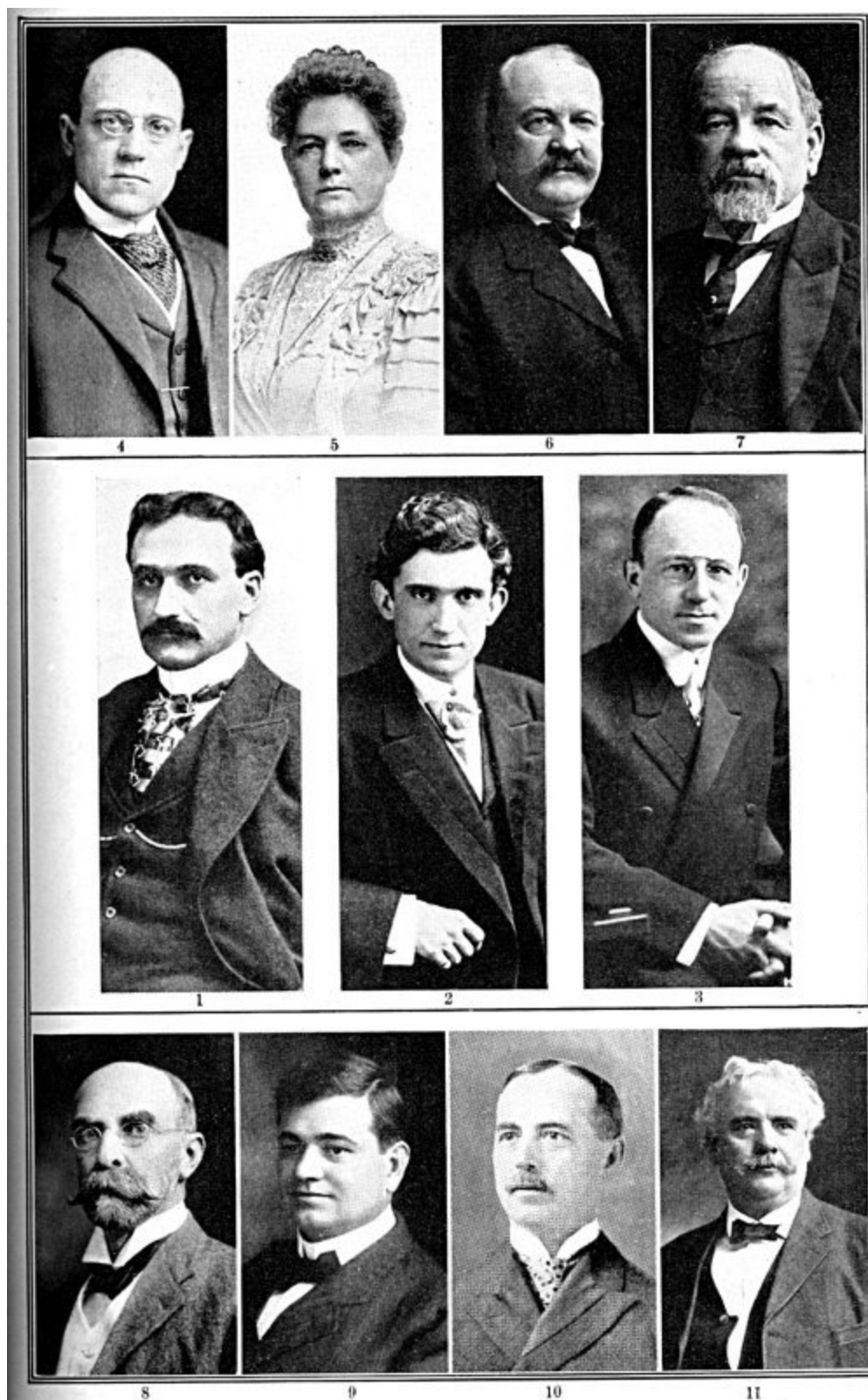
Section 1. Each member of the Congress shall be entitled to one vote on all actions taken viva voce.

Section 2. A division or call of States may be demanded on any action by a State delegation. On division, each Delegate shall be entitled to one vote; provided (1) that no State shall have more than twenty votes; and provided (2) that when a State is represented by less than ten Delegates, said Delegates may cast ten votes for such State.

Section 3. The term "State" as used herein is to be construed to mean either State, Territory, or Insular Possession.

ARTICLE 10—AMENDMENTS

This Constitution may be amended by a two-thirds vote of the Congress during any regular session, provided notice of the proposed amendment has been given from the Chair not less than one day or more than two days preceding; or by unanimous vote without such notice.



1. D. AUSTIN LATCHAW, Treasurer (1910).
2. THOMAS R. SHIPP, Executive Secretary (1909-10).

3. JAMES C. GIPE, Recording Secretary (1910).
 4. JOHN BARRETT, Vice-President (1909).
 5. MRS PHILIP N. MOORE, Executive Committee (1909-10).
 6. FRANK C. GOUDY, Executive Committee (1910).
 7. THOMAS BURKE, Executive Committee (1909).
 8. E. J. WICKSON, Vice-President (1909).
 9. HENRY D. HARDTNER, Vice-President (1909).
 10. JAMES S. WHIPPLE, Vice-President (1909).
 11. W J MCGEE, Vice-President (Editor of Proceedings).
-

SECOND NATIONAL CONSERVATION CONGRESS

OPENING SESSION

The Congress convened in the Auditorium, Saint Paul, Minnesota, on the morning of September 5, 1910, President Baker in the chair, and was called to order on arrival of the President of the United States.

President BAKER—Mr President, your Grace, Ladies and Gentlemen: The honor I have today in opening this great Congress is one that will always be highly treasured, for I feel that what we are trying to do is to make our country great and strong by men who see the Nation's wrongs and are giving their time to this great object. We are meeting today for the purpose of using our very best efforts to assist in protecting the interests of this great country in a way that will best protect every man and woman and child in his or her rights, with justice to all. That our great National resources are in danger of being wasted and not fully preserved for the future, I am satisfied is the thought of all the great minds assembled here today to take part in this Congress.

There is a Great High Power that rules and governs for the best in the world, and I now call upon His Grace, Archbishop Ireland, to open our Congress with an invocation to that Great Power for help, guidance, and direction.

INVOCATION

Almighty and eternal God. We bow before Thee in deep humility. Accept from us, we beseech Thee, from submissive minds and sincere hearts, adoration, praise, gratitude, love, and the promise of abiding recognition of Thy sovereignty and of loyal obedience to Thy laws.

O God, all things are Thine; all things were made by Thee; no thing that was made was made without Thee; "the heavens show forth Thy glory and the firmament declareth Thy power, day to day uttereth speech, night to night showeth knowledge," ever proclaiming that Thou are the Master, that things created are the scintillations of Thy power and wisdom. We are Thine, O God, Thee our Father and our Master; earth and skies are ours

through gift of Thy munificence. "Till the earth," was it said to us, "and subdue it and dominate over the fishes of the sea and the fowls of the air and all living creatures that move upon the earth." Earth is ours, not, O Lord, that we use it at our will and caprice, but that under Thy guidance we bid it turn to our best and truest welfare, to the best and truest welfare of our fellow-men, Thy children all; over all of whom spread Thy love and care.

Grant to us, O Lord, this morning wisdom in our counselings and deliberations, that the intents of Thy providence be our intents, and, Thy will the inspiration of our counselings and our actions.

We thank thee, O God, for the gift to us of America. As to few other lands, Thou hast been prodigal to America of gifts rich and rare. In America skies are serene and health-giving above us; beneath us fields are verdant and fertile; nowhere else are forests more fruitful, hills and mountains richer in imbedded treasure; nowhere else are lakes and rivers endowed with higher grandeur or more ready to proffer to man useful and ennobling service. Of America, through Thy munificence, O God, we are the caretakers. May we be wise and prudent in our duty. We pray that under Thy abiding watchfulness, through our intelligent industry, America grows ever in fairness and in wealth, and be the first and most beautiful of the stopping-places allowed to men in their pilgrimage toward their abiding home in heaven.

Bless, O Lord, America, and bless its people, that they be ever faithful to Thy laws; bless its citizenship, bless its Government, that the spirit of its freedom-giving institutions never die, never lessen in sweetness and in power; that here liberty be ever encircled in order, and order ever wreathed in liberty; that righteousness dominate and permeate prosperity; that whatever the laws we form may be scintillations of Thy own eternal laws—compliance with which is life and felicity, forgetfulness of which is misery and death to men and to nations.

And we pray Thee, O God, send down Thy blessing upon the President of the Republic, upon whose shoulders descends the chief responsibility of upholding the salvation and the dignity of America. We pray that Thou bestow upon him Thy precious blessing. The burthen is heavy, often the

horizon is dark, often the polar star is hidden from which guidance might come; but in Thee, O God, he confideth,—send upon him the wisdom and the strength of Thy Holy Spirit, the wisdom that he may know, the power that he may do, ever Thy will. In Thee, O Lord, in Thy omnipotent hand—prompt to give aid in single-mindedness of purpose and in rectitude of intention—he puts his trust. Be Thou his teacher, be Thou his guide.

Our Father who art in heaven, hallowed be Thy name. Thy kingdom come, Thy will be done on earth as it is in heaven. Give us this day our daily bread, and forgive us our trespasses as we forgive those who trespass against us, and lead us not into temptation but deliver us from all evil. Amen.

President BAKER—Mr President, Ladies and Gentlemen: His Eminence, Cardinal GIBBONS, sends you greeting:

Allow me to say how earnestly I wish the Congress every success in the much-needed work of National Conservation.

It is said that the French and Germans could subsist on what we waste; and I fear that to a stranger visiting our country it must seem that in a hundred years we have wasted more of our natural resources than the nations of Europe have done in all the centuries of their existence. But if we have been reckless in the past, wasting like vandals our rich inheritance, it is also most consoling and full of promise for the future that with the strong aid of our President, of Colonel Roosevelt, and of leading citizens in various parts of the country, we may look for a wiser use of our resources in the near future. And I am the more hopeful of a successful Congress from the fact that there is no political issue involved in the great subject before it which might threaten to divide our counsels and breed discontent, but that the sole motive that actuates the Congress is to conserve and increase our natural resources and thereby contribute to the material prosperity of our beloved country.

It is also decidedly my opinion that we should regard our natural resources as the patrimony of the Nation, a sacred trust committed to our keeping to

be administered for the good of the whole people, and to be transmitted by us, as far as possible unimpaired, to our posterity. By husbanding and using economically the gifts of Nature, we shall have an abundant supply for our own times, and also make suitable provisions for the future. Mother Earth is not only a fruitful mother; she is also a grateful mother, and repays her children for every kindness and tenderness we exercise toward her. And there are also instances on record to show that she is relentless when she chastises.

Did my many duties allow, I should gladly take a more active part in the greatly needed Conservation labors. However, I trust you will feel assured of my entire sympathy and of the hope I confidently entertain of the very great benefits coming to us all as the fruitful result of these devoted laborers. JAMES CARD. GIBBONS.

PRESIDENT BAKER—Ladies and Gentlemen: The opening of the Congress today in Saint Paul is due largely to the kind assistance and friendly welcome of the Governor of Minnesota, His Excellency A. O. Eberhart, who will now extend you a welcome. (Great applause and cheers)

GOVERNOR EBERHART—Mr President, Members of the Congress, Ladies and Gentlemen: When I was invited to appear before this Congress and bid you welcome, it was suggested that I also outline what the people of Minnesota felt when they sought to have this splendid gathering at Saint Paul.

I am sure that no State or city could receive greater honor than to have the President of the United States come fifteen hundred miles to deliver the most important message on Conservation that has ever been presented to the people of this great country. (Applause) Yet I am not going to take more than the twenty minutes allotted to assure you that the only interest this State has in the Conservation movement is that which every true friend of the movement stands for. Last night I cut out the meat of my remarks, this morning the bones, and now there is nothing left but the nerve, and I have scarcely enough "nerve" to deliver it. (Laughter and applause)

The Conservation of natural resources does not consist merely in the preservation of these resources for the benefit of future generations, but rather such present use thereof as will result in the greatest general good and yet maintain that productive power which insures continued future enjoyment. (Applause) While it is true that exhaustible resources like mineral wealth cannot be conserved for both future and present use, except by economic regulations and the prevention of wasteful methods, Conservation deals with their distribution in such a way as to prevent their control by grasping corporations and individuals, who would monopolize them for their own exclusive benefit at the expense of the general public. (Applause)

It follows necessarily that any theory of Conservation which does not provide for the present as well as the future does not cover the entire field and cannot possibly bring the best results. (Applause) From every economic standpoint it is desirable that the present generation should be preferred, since future discoveries and inventions may render present resources of less value and importance to the coming generations.

In its broadest sense the Conservation movement is not limited merely to the consideration of natural resources. Every great convention called to consider the problems involved has widened the scope of the movement so that today it includes the elimination of wasteful methods in almost every field of human activity and the conservation of all human endeavor so as to confer on all mankind the greatest blessings that a bounteous nature and twenty centuries of enlightenment can bestow.

Every consideration of natural resources for the purpose of eliminating wasteful methods, preserving and increasing productive power, as well as regulating operation and control, has for its ultimate object the conservation of human energy, health and life, the securing of equal opportunities for all, and such dissemination of knowledge as will guarantee the continual possession and enjoyment of these blessings. The subjects for consideration by this Congress should, therefore, include not only the restoration and increase of soil fertility, the protection and development of forests, mines and water-powers, the reclamation of arid and swamp lands by irrigation and drainage, the forestation of areas unsuited to farming, the control of rivers by reservoirs so as to prevent flooding, as well as the elimination of

waste in the use of these resources, but also the problems of public comfort, health and life that are so intimately connected with all material and intellectual development. (Applause) Many of these questions will concern home attractions and management, industrial education in the public schools, public highways, State advertising and settlement, pure food, public health, and sanitation.

By far the most important of all natural resources is the soil, and the maintenance and increase of its fertility must, therefore, be given the greatest consideration. (Applause) As long as food is necessary to human life, agriculture must continue to be the most vital industry of man, and the farm will be the most general and indispensable theater of his activity. We must have manufacture, art, schools, churches and government to round out our sphere of civilized existence, but the foundation of them all is the farm. (Applause) From the earth come all the materials for manufactures, the commodities of commerce, and ultimately the support of all human institutions. During the half century just past our country has devoted its energies to the development of manufacturing and commercial industries to such an extent that the scientific methods of agriculture necessary to insure not only the permanency of our institutions but the very existence of human life itself have been comparatively neglected. The pendulum is now swinging back to the farm, and our great Nation is becoming aroused to the fact that its most vital concern is the elimination of soil waste, the promotion of scientific methods of agriculture, and the conservation of that soil fertility which is the foundation of our entire social, political and commercial superstructure. (Applause)

This new birth of agricultural progress comes at a psychological moment. We have developed American manufactures until the \$16,000,000,000 product of our mills and factories exceeds that of Germany, France, and the United Kingdom combined. (Applause) We have built railroads by liberal public and private enterprise until the United States has about one-half of all the railway mileage and tonnage of the world. We have developed banking enterprise and home trade until we have the greatest banking power on earth, and an internal commerce which far exceeds the entire foreign commerce of the globe. We have become the model of the world in our free public schools and our republican form of government. But while we have demonstrated the possession of the greatest agricultural resources on the

globe, and have heretofore supplied the world's markets with an unparalleled volume of farm products, we have wasted a wealth that would maintain our population for centuries. The loss in farm values in nearly all of the older States, as shown by the census records from 1880 to 1900, varies from \$1,000,000 to \$160,000,000 in each State and aggregates the enormous total of more than \$1,000,000,000. Is this not sufficient to arouse the entire Nation and cause such a wave of reform as will put into activity every agency and instrumentality for scientific and progressive methods of agricultural reconstruction?

The unprecedented agricultural growth of the United States, in spite of wasteful methods, has been caused by the extraordinary fertility of its virgin soil, the great inducement offered by States and Nation to promote settlement and cultivation, the rapid growth of favorable transportation facilities, as well as the great demand for agricultural products resulting from the rapid increase of population, wealth and commercial enterprise.

Minnesota affords a splendid illustration of this development process, and I trust that I may be pardoned for using my own State for that purpose, since I am best acquainted with her conditions, development, and resources. Of her 50,000,000 acres of land area, about one-half is actually tilled, constituting the field area of about 200,000 farms whose aggregate area, including lands not tilled, approximates 32,000,000 acres, or 160 acres each. Nearly 4,000,000 acres of her area are covered by 10,000 lakes. This vast farm area possesses a soil unsurpassed by any State or any country in the world. The great glacier of several thousand years ago was generous to Minnesota. Its fine glacial drift almost wholly covers the old rock formations. Coming from many regions and rock sources, it has given to the soil an excellent chemical composition. This, together with the vegetal mold, accumulated for ages, makes the very best of hospitable soils. The incomparable fertility of the Minnesota soil and its ability to withstand fifty years of starvation methods in cultivation is accounted for by the almost uniform mixture of vegetal mold with all kinds of decomposed rock drift, thus making it possible for less than half of the State to produce farm products aggregating the enormous total for 1909 of more than \$427,000,000. (Applause) It accounts also for the fact that, while Minnesota, like all other States, during this period of fifty years has been rather mining the fertility out of her soil than cultivating it, she has

withstood the consequent impoverishment without appreciable shrinkage in farm value. There is perhaps not a single representative in this distinguished assemblage who cannot recall the day when the virgin soil in his locality did not produce from 50 to 100 percent larger crops than it does today, when dense forests covered large tracts now a barren waste, and when the bosom of the earth contained untold millions of mineral wealth now represented on the surface by huge spoil-banks and sunken surfaces. We remember only too well when our fertile fields yielded thirty-five to forty bushels of wheat to the acre, and that the same fields produced only about twelve bushels five years ago. In nearly every community there is found that pathetic omen of decay, the deserted farm—even in this young State.

The economic importance of soil conservation is so great that it can scarcely be estimated. In making my estimates I have taken a very conservative view, and while no absolutely accurate figures can be obtained, the few that I shall give will be found sufficiently reliable to establish the paramount value of soil conservation.

In Minnesota the low tide of soil impoverishment occurred about five years ago. At that time, after several years of apparently unsuccessful effort, the Agricultural College and schools, assisted by the State Farmers' Institutes and the press, succeeded in stemming the tide and arousing considerable interest in new methods of farming along more intelligent and intensive lines. Only within the last year, however, has progress been marked and rapid. When the first State Conservation Congress was called to meet in Saint Paul, March, 1909, nearly every township in the State was represented and all but two counties presented agricultural and industrial exhibits, attracting a total attendance of more than 150,000 people. The wonderful success of that Congress and the enthusiasm it stirred up all over the State gave a great impetus to this new era of agricultural reform in the entire Northwest and insured the complete success of this Congress from a local standpoint. Never before had 6,000 of the most progressive farmers of a State met for the purpose of discussing more intelligent methods of farming, as well as the suppression of wasteful methods in all fields of agricultural and industrial activity.

During the past short period of five years the average cereal yield of this State has been increased more than five bushels per acre; the corn belt has

been extended northward more than 300 miles to the Canadian boundary by the production of hardy and early maturing varieties of corn, yielding the State last year over 60,000,000 bushels, and placing Minnesota among the dozen leading corn States of the Union. It is estimated that plant breeding and seed selection alone last year added about \$15,000,000 to our agricultural products. The cereal production has also affected clover, timothy and other tame grasses, thus largely contributing to the growth of the dairy industry, which has been increased ten-fold in twenty years until it now yields the State \$50,000,000 annually, several counties netting more than \$1,000,000 each. Similar progress has been made in the live stock, fruit, and truck gardening industries, and it is safe to conclude that Minnesota has entered in earnest upon a complete plan of agricultural reconstruction.

But let us consider the opportunities for advancement that are still open, in order that we may determine the economy of soil conservation in terms of dollars and cents. The average yield of Minnesota wheat last season was seventeen bushels per acre. At the agricultural experiment stations the same wheat with improved seed selection and better preparation of soil by crop rotation and tillage yielded twenty-eight bushels per acre, climatic and soil conditions, as well as expense of tillage being otherwise similar, a difference in favor of intelligent farming approximating from five to eight dollars per acre, depending on local conditions. Assuming for the sake of argument that the average difference in the State would not be more than four dollars per acre, it would still increase the agricultural net earning of the State on the basis of the present acreage \$100,000,000 annually. These figures do not take into consideration the further increase of soil productivity by various methods of fertilization other than those resulting from planting crops which enrich the soil with nitrogen, phosphoric acid, potash and calcium, the essential elements of plant growth. Besides, I have not attempted to estimate the value of raising almost maximum yields, where weather conditions are unfavorable, by such drainage, preparation of soil, planting and tillage as will best suit local and climatic conditions. No crop emphasizes the value of seed selection in such unmistakable terms as corn. The average stand of this crop does not exceed 60 percent, which means that the farmer spends 40 percent of his time in the cornfield without result. By selecting the seed in the field at the proper season, testing each

ear before planting, and separating with reference to size, so that as nearly as possible the planter will put three kernels in each hill, the stand can be increased to at least 95 percent. Applying this increase to the 2,000,000 acres of cornfield in Minnesota, it would add approximately 30,000,000 bushels with practically no additional cost of production. That the importance of this matter might be more firmly impressed upon the people of the State, I have issued a seed-corn proclamation designating the time when the seed-corn should be selected and calling the attention of the people to the feed value of the corn product as well as corn fodder, which is of utmost importance in a dry season like the one we are now experiencing. This proclamation has received extensive publicity, and it is safe to say that a large number of Minnesota's 200,000 farmers will heed the note of warning.

Of still more vital importance, if possible, is the maintenance and increase of soil fertility as a source of support for future generations. The soil is the only permanent asset of the farmer, and its net returns in crops constitute his annual dividends. Any impairment of this asset will not only reduce the dividends on which his support depends, but will destroy the productive power of the soil to such extent as to deprive future owners of the most essential means of livelihood. A loss of \$1,000,000,000 in farm values, such as the older States have already suffered, does not mean merely that this vast sum of money has been wasted, but that its annual earning capacity on which thousands should depend for support has been entirely destroyed, and that these thousands have been forced to seek their sustenance from the fields of commerce and manufacture in the large cities. We enact stringent legislation to prevent the impairment of capital in our banking institutions to protect depositors from loss, but the working capital investment of millions in farm property on which all human institutions must necessarily depend for existence has not been safeguarded in any manner whatsoever. Without any organized effort to interfere, we still permit millions of farmers to mine out the fertility of the soil, thus increasing the drudgery of farm life, reducing every source of farm income, converting the producers of the farm into consumers of the city, and thus contributing directly to the great increase in cost of living, the scarcity of farm labor, and the congested conditions that breed disease and crime in our large cities. Apply the situation to the country at large and you will find a situation that is simply

appalling. There are approximately 500,000,000 acres under actual tillage in the United States. Instead of figuring four dollars per acre waste, which probably would be a fair average, we will place the loss at the extremely low estimate of one dollar. This will still make the total loss through wasteful farming methods in the United States reach the enormous total of \$500,000,000 annually. In other words, if the loss were in fact not greater than one dollar per acre, which is unquestionably too low, and that rate could be maintained perpetually without an ultimate depletion of the soil, it would mean that a capital investment of \$12,500,000,000 with an earning capacity of four percent per annum aggregating \$500,000,000 annually, had been completely destroyed.

At the rate of two dollars per acre, which is a low average, we are every year wasting the income from \$25,000,000,000, a sum so great as to be entirely beyond human comprehension. In many of the older States, where farms were sold forty years ago at \$150 per acre, the same farms cannot be sold today for \$25 per acre, sometimes less than the actual cash value of the buildings and other improvements, because the soil has been robbed of its fertility, making it impossible for the owner to earn the most meager living without restoring the vitality of the soil through expensive methods of fertilization.

It is not at all difficult to see how such wasteful methods of farming must affect the entire industrial situation. The younger generation, inspired with the hopes, aspirations, and energy of youth, stirred by the achievements, opportunities, and general prosperity of a truly great Nation, and encouraged by the possibilities of a liberal education, cannot afford to stake its future on the eking out of a mere existence under the shadow of a rapidly increasing farm mortgage or the threatening omen of a deserted homestead. All honor and credit to that farmer's boy who early realizes the handicap placed upon him by the impairment, and oftentimes utter destruction, of the only safe capital investment of the farmer—fertile and productive soil. Should we complain because he goes to the city to seek more inviting and attractive fields of existence after having been robbed of his only means of livelihood on the farm? This is the proper time for us to think it over. In the younger States, where soil mining has been of such short duration as to be incomplete, and the value of the land through settlement, city growth, and increased transportation facilities is constantly growing, the young man,

who has learned intelligent and progressive methods of farming, should have no fear as to the future, for he has the making of a safe investment; but the young lad who, without experience or training, unexpectedly finds himself possessor of a farm where land values have ceased to rise and the soil has been starved until it no longer can yield in abundance, has a white elephant on his hands, and the sooner he can be brought to the realization thereof the better for himself and the entire community.

Where a certain amount of labor should produce thirty bushels of wheat to the acre, it yields but ten, or even less; and when the farmer cultivates his corn, working ten hours per day, four hours thereof is spent in vain, because 40 percent of the field has no corn—not to speak of the poor quality of the corn grown on account of defective preparation of soil, poor tillage, and the lack of necessary nutritive elements within the soil itself. In addition, he has no knowledge as to diversified farming, the value of live stock, dairying, fruit-raising, truck-gardening, and many other means of livelihood which yield large incomes to the possessor of a well-managed farm, nor does he appreciate the enormous waste committed by unnecessary exposure to the elements of farm machinery and buildings.

The young lady faces a similar situation. Every field of employment bids her welcome at wages from \$50 or more per month, and she has already achieved such abundant success in every line of human enterprise, and at the same time enjoyed all the pleasures and delights which bring cheer to the heart of the young, that she cannot afford to even hesitate. Should we complain if she refuses to stay on the farm and take her chances of marrying a \$25 man and a ruined farm plastered all over with mortgages, and be chained in matrimonial bonds of lifelong drudgery to a devastated farm homestead, robbed of everything that contributes to the beautiful and good and true in a woman's life? (Great applause) There is only one answer, and its conclusions are just.

Though I have presented a sad picture, it is not pessimistic. The background is altogether cheerful. Two words express the most simple and effective remedy: intelligent farming. This will not only make farming profitable, but it will surround the home life on the farm with so many attractions as to remove all desire for the deceptive allurements of a city. Intelligent farming does not merely guarantee good dividends on a farm investment, but it

builds good roads to save cost of transportation, consolidates rural schools where intelligent farming, industry and home economics can be taught by precept and example, beautifies the home and its surroundings and fills it with all the attractions that elevate manhood and womanhood, teaches the younger generation the dignity as well as reward of farm labor, and inspires the laborer with the hope of a bright future.

Drainage, farm settlement, good roads, forestry, transportation, industrial education, minerals, cheap heat and power resources, are all important factors in the Conservation movement. Minnesota has successfully drained about 3,000,000 acres in the northern part of the State at an average cost of two dollars per acre, and converted into meadows, grain and clover fields, celery and cranberry gardens, what only a year or two since was a rough wilderness. Every State should have some effective way of making these results known to prospective settlers through exhibits and judicious advertising. No State officer is in a position to bring greater returns to the State than the immigration commissioner, and it is to be regretted that his work is so often crippled by lack of sufficient appropriations.

In marketing produce, distributing material, fertilizer and machinery, the farmers of Minnesota haul annually approximately 20,000,000 wagon loads. Averaging the cost of each load over mostly unimproved roads at \$1.50, the cost of highway transportation in the State aggregates \$30,000,000. Most experts claim that uniformly good roads would reduce this cost one-half, but conceding for the sake of argument that the reduction would be only a third, the net saving to the farmers of the State in one year would be about \$10,000,000. However, this is not the most important result. The building of good roads would build up farm intercommunication and promote the consolidation of rural school districts by making it possible to carry the pupils at all seasons of the year some distance over country roads to the school at a minimum cost.

Several of the north-central border States were the chief shippers of lumber only a few years ago. Now our great forests are largely depleted, and scientific deforestation has become an absolute necessity. One of the most important duties the States as well as the Nation have to perform is the transformation of this vast stumpage area into forests and farms. Practical and scientific reforestation should convert the lands unsuited for farming

into forests, so that every acre would produce revenue and furnish some necessity of life. The dry season of 1910 has particularly emphasized another important duty in this connection, and that is the protection of our forests and settlers from fires. It is a well known fact that enough timber has been destroyed by fire within the last four months to pay for the adequate protection of all our forests for a period of ten years or more, not to mention the great loss of human life, which in itself imposes upon States and Nation the duty of protection. This Congress should be instrumental in stirring public sentiment to such an extent that the various legislatures and the Congress will take immediate steps to stop this needless and expensive waste.

Since mineral wealth is exhaustible, it follows that the interest of the people in this important resource should be guarded against the encroachments of greed with the utmost care. Minnesota furnishes now one-half of all the iron-ore in the United States, and one-fourth of that of the world, exporting this year about 40,000,000 tons. It is estimated that not less than 2,000,000,000 tons of ore has been definitely located, and that the volume of the undeveloped properties is enormous. The State is the owner of very large quantities of ore, and the income from this source alone will increase the State school fund by at least \$100,000,000.

No section of our country could profit more by water transportation than that tributary to this great mineral wealth. The canalization of the Mississippi river system with its 16,000 miles of streams would by cheap transportation bring together the coal fields of the central interior with the iron ore of the North, and produce in the Mississippi valley the greatest iron and steel industries of the world, besides opening up the greatest agricultural and industrial sections to the transportation facilities of the Panama canal.

No commercial nation can long retain supremacy unless it has unlimited supplies of cheap heat and power. In the north-central border States are located peat deposits that should furnish cheap heat and power for untold generations, Minnesota alone possessing more than 1,000,000 acres; and as the source of the three great watersheds of the country, with an elevation of about 1,500 feet over sea and gulf level, there is an abundance of water-power to turn the wheels of manufacture and commerce.

Time will not permit any consideration of the strictly human side of Conservation. We have saved millions of dollars annually by guarding against plant and animal disease, and are just beginning to take note of the untold millions wasted every month through neglect of preventable and curable disease, impure foods, defective sanitation and health inspection in homes and schools, unsuitable playgrounds for children, and the lack of safeguards against railway, mine and factory accidents, all of which come properly within the Conservation scope.

The splendid progress made by Minnesota and other States merely emphasizes the importance of the Conservation movement. Warned by the decay of older nations, we must act before the crisis of exhausted natural resources reaches our Nation and commonwealths. Indeed, warned by signs that are only too plain in our own midst, we must take decisive action without delay. Fortunately, we have passed the pioneer stage of development. Our Nation and commonwealths have all experienced many of the disasters resulting from the skimming of natural resources. Having discovered the vast mines of wealth which surround us everywhere, we must now and forever determine that ignorance, selfishness, and greed shall no longer control our governments and exhaust our resources. (Great applause and cheers)

The problems before us are not merely of tremendous importance, but they are also difficult as to solution. They frequently involve sharply conflicting claims and interests as between the Nation and the various commonwealths. Every State as well as the Nation itself should have a distinct and separate department empowered to deal with all these problems. It matters but little how it should be designated, though it would serve all purposes best to be known as a Conservation Commission. But it is of vital importance that the agency should be given sufficient authority and funds, so as to enlist the strongest and best men in the Conservation service. That such commissions would have sufficient work, and that from an economic standpoint they would constitute good investments, there is and can be no question.

Minnesota, as a distinctly progressive State and a recognized leader in the Conservation movement, heartily welcomes this Congress with its noted guests and speakers. We have the special honor of entertaining and hearing the three truly great men who have contributed so much to the actual

achievements of the Conservation movement, and they are the three most distinguished guests of this Congress, President Taft (applause and the Chautauqua salute), Colonel Roosevelt (applause and cheers), and James J. Hill. Minnesota appreciates this honor and will prove herself worthy thereof. As her Chief Executive, I earnestly hope that the deliberations of this Congress may bring results far beyond our hopes or expectations. I am intensely interested in the Conservation of our resources, and will use all my efforts in securing and enforcing the best possible legislation, believing firmly that the Conservation movement, as here outlined, will promote the general public welfare in a far greater degree than any other, and that it is destined to mark the twentieth century as an era of the greatest industrial achievement for the benefit of all mankind. The people of Minnesota feel keenly their duties and responsibilities with reference to their great heritage of unsurpassed natural resources, and will continue as leaders in the only movement that can insure the perpetuation of our country as the greatest agricultural, industrial and commercial nation in the world. On their behalf, I welcome you to the State. I thank you. (Applause)

President BAKER—It is now my pleasure to call upon his Honor, Mayor Herbert E. Keller, who will welcome you on behalf of the great city of Saint Paul. (Great applause and cheers)

Mayor KELLER—Mr President, Delegates to the Second National Conservation Congress, and Guests: Upon me, as Chief Executive of the city where this body will carry on its labors, the honor of welcoming you devolves. It is a great privilege and pleasure to discharge this duty, and yet my greeting can but inadequately convey to you the appreciation felt by all Saint Paul at being selected as the scene of this great Congress, whose deliberations mark the commencement of a new epoch in the history of our country. (Applause)

The Conservation to and by ourselves as trustees, and the dedication and perpetuation to our children and our children's children as beneficiaries, of the tremendous natural resources of our country is a duty and trust too sacred and too imperative to be disregarded or lightly considered, once the

situation stands revealed in its true light. It is purely and simply a proposition of the greatest good for the greatest number, and the sound judgment of a great people, with the patriotism and unselfish devotion to duty of the founders of our country ever before them, must and shall consider the greatest number to be the countless millions of population to follow after us, and to whom must be handed down a heritage not diminished or impoverished by us, the temporary executors.

We may be likened to children turned loose in some vast Midas treasurehouse and told to go where we would and take what we pleased. A knock at the doors of Congress, a State legislature, or a city council, gives the magical "Open Sesame!" And behold! the lavishing on some private interest or individual of a great National or State property or municipal right or franchise!

The Nation's bounty and generosity has been limitless, for the entire previous history of the whole world provides no precedent for a guide. But, fortunately, thoughtful minds began to work, awakened to what was being done, and the result is the present all-pervasive sentiment and determination to economize, to check improvidence and waste, and to establish a policy whereby future generations, as well as the present, may have equal opportunities to enjoy our natural benefits and advantages; and Conservation is now more than a mere issue: it is an assured, established, sane and universal desire to preserve and perpetuate for ourselves and posterity the treasures of our country.

And so I bid you welcome to the city of Saint Paul. May your labors be fruitful of great good. I know that your stay with us will be enjoyable. Our city limits may be somewhat circumscribed for the immense crowds here this week, but our hospitality and good wishes are as limitless as the ocean. (Applause)

President BAKER—Fellow delegates, I am sure we all extend to his Honor, Mayor Keller, a hearty vote of thanks for what he has done in preparing for this Congress.

And now comes a privilege of which I am very proud—as a southern man all my life—that of presenting to you the President of this great Nation. (Great applause and cheers, the audience rising)

ADDRESS BY THE PRESIDENT OF THE UNITED STATES

Ladies and Gentlemen: Before beginning my formal address, I should like to extend to the President and the Managers of this Congress, to Governor Eberhart, and to the Mayor of the city, my sincere and cordial thanks for the opportunity to come here and address this magnificent audience, and to reach the people of the United States on a subject of the utmost interest to them and to every patriot. (Applause)

Conservation, as an economic and political term, has come to mean the preservation of our natural resources for economical use, so as to secure the greatest good to the greatest number.

In the development of this country, in the hardships of the pioneer, in the energy of the settler, in the anxiety of the investor for quick returns, there was very little time, opportunity, or desire to prevent waste of those resources supplied by nature which could not be quickly transmuted into money; while the investment of capital was so great a desideratum that the people as a community exercised little or no care to prevent the transfer of absolute ownership of many of the valuable natural resources to private individuals, without retaining some kind of control of their use. The impulse of the whole new community was to encourage the coming of population, the increase of settlement, and the opening up of business; and he who demurred in the slightest degree to any step which promised additional development of the idle resources at hand was regarded as a traitor to his neighbors and an obstructor to public progress. But now that the communities have become old, now that the flush of enthusiastic expansion has died away, now that the would-be pioneers have come to realize that all the richest lands in the country have been taken up, we have perceived the necessity for a change of policy in the disposition of our natural resources so as to prevent the continuance of the waste which has characterized our phenomenal growth in the past. Today we desire to restrict and retain under public control the acquisition and use by the capitalists of our natural resources.

The danger to the State and to the people at large from the waste and dissipation of our national wealth is not one which quickly impresses itself on the people of the older communities, because its most obvious instances do not occur in their neighborhood, while in the newer part of the country the sympathy with expansion and development is so strong that the danger is scoffed at or ignored. Among scientific men and thoughtful observers, however, the danger has always been present; but it needed some one to bring home the crying need for a remedy of this evil so as to impress itself on the public mind and lead to the formation of public opinion and action by the representatives of the people. Theodore Roosevelt (great and prolonged applause) took up the task in the last two years of his second administration, and well did he perform it. (Great and prolonged applause)

As President of the United States I have, as it were, inherited this policy, and I rejoice in my heritage (great applause). I prize my high opportunity to do all that an Executive can do to help a great people to realize a great national ambition; for Conservation *is* National. It affects every man of us, every woman, every child. What I can do in the cause I shall do, not as President of a party, but as President of the whole people (enthusiastic applause and cheers). Conservation is not a question of politics, or of factions, or of persons. It is a question that affects the vital welfare of all of us—of our children and our children's children. I urge that no good can come from meetings of this sort unless we ascribe to those who take part in them, and who are apparently striving worthily in the cause, all proper motives (applause), and unless we judiciously consider every measure or method proposed with a view to its effectiveness in achieving our common purpose, and wholly without regard to who proposes it or who will claim credit for its adoption (great applause). The problems are of very great difficulty, and call for the calmest consideration and clearest foresight. Many of the questions presented have phases that are new in this country, and it is possible that in their solution we may have to attempt first one way and then another. What I wish to emphasize, however, is that a satisfactory conclusion can only be reached promptly if we avoid acrimony, imputations of bad faith and political controversy (cries of "Hear, hear," and great applause).

The public domain of the Government of the United States, including all the cessions from those of the thirteen States that made cessions to the United

States, and including Alaska, amounts in all to about 1,800,000,000 acres. Of this there is left as purely Government property outside of Alaska something like 700,000,000 acres. Of this the national forest reserves in the United States proper embrace 144,000,000 acres. The rest is largely mountain or arid country, offering some opportunity for agriculture by dry farming and by reclamation, and containing metals as well as coal, phosphates, oils, and natural gas. Then the Government owns many tracts of land lying along the margins of streams that have water-power, the use of which is necessary in the conversion of the power into electricity and its transmission.

I shall divide my discussion under the heads of (1) agricultural lands; (2) mineral lands—that is, lands containing metalliferous minerals; (3) forest lands; (4) coal lands; (5) oil and gas lands; and (6) phosphate lands. I feel that it will conduce to a better understanding of the problems presented if I take up each class and describe, even at the risk of tedium, *first*, what has been done by the last Administration and the present one in respect to each kind of land; *second*, what laws at present govern its disposition; *third*, what was done by the present Congress in the matter; and *fourth*, the statutory changes proposed in the interest of Conservation.

AGRICULTURAL LANDS

Our land laws for the entry of agricultural lands are as follows:

The original Homestead Law, with the requirements of residence and cultivation for five years, much more strictly enforced now than ever before.

The Enlarged Homestead Act, applying to non-irrigable lands only, requiring five years' residence and continuous cultivation of one-fourth of the area.

The Desert-land Act, which requires on the part of the purchaser the ownership of a water-right and thorough reclamation of the land by irrigation, and the payment of \$1.25 per acre.

The Donation or Carey Act, under which the State selects the land and provides for its reclamation, and the title vests in the settler who resides upon the land and cultivates it and pays the cost of the reclamation.

The National Reclamation Homestead Law, requiring five years' residence and cultivation by the settler on the land irrigated by the Government, and payment by him to the Government of the cost of the reclamation.

There are other acts, but not of sufficient general importance to call for mention unless it is the Stone and Timber Act, under which every individual, once in his lifetime, may acquire 160 acres of land, if it has valuable timber on it or valuable stone, by paying the price of not less than \$2.50 per acre, fixed after examination of the stone or timber by a Government appraiser.

In times past, a great deal of fraud has been perpetrated in the acquisition of lands under this Act, but it is now being much more strictly enforced, and the entries made are so few in number that it seems to serve no useful purpose and ought to be repealed. (Applause)

The present Congress passed a bill of great importance, severing the ownership of coal by the Government in the ground from the surface and permitting homestead entries upon the surface of the land, which, when perfected, gives the settler the right to farm the surface, while the coal beneath the surface is retained in ownership by the Government and may be disposed of by it under other laws.

There is no crying need for radical reform in the methods of disposing of what are really agricultural lands. The present laws have worked well. The Enlarged Homestead Law has encouraged the successful farming of lands in the semi-arid regions. Of course the teachings of the Agricultural Department as to how these sub-arid lands may be treated and the soil preserved for useful culture are of the very essence of Conservation. Then the conservation of agricultural lands is shown in the reclamation of arid lands by irrigation, and I should devote a few words to what the Government has done and is doing in this regard.

By the Reclamation Act a fund has been created of the proceeds of the public lands of the United States with which to construct works for storing great bodies of water at proper altitudes from which, by a suitable system of canals and ditches, the water is to be distributed over the arid and sub-arid lands of the Government to be sold to settlers at a price sufficient to pay for the improvements. Primarily the projects are and must be for the

improvement of public lands. Incidentally, where private land is also within the reach of the water supply, the furnishing at cost of operation of this water to private owners by the Government is held by the federal Court of Appeals not to be a usurpation of power; but certainly this ought not to be done except from surplus water not needed for Government land. About thirty projects have been set on foot, distributed through the public-land States, in accordance with the Statute, by which allotments from the reclamation fund are required to be, as nearly as practicable, in proportion to the proceeds from the sale of the public lands in the respective States.

The total sum already accumulated in the reclamation fund is \$60,273,258.22, and of that all but \$6,491,955.34 has been expended. It became very clear to Congress at its last session, from the statements made by experts, that these thirty projects could not be promptly completed with the balance remaining on hand, or with the funds likely to accrue in the near future. It was found, moreover, that there are many settlers who have been led into taking up lands with the hope and understanding of having water furnished in a short time, who are left in a most distressing situation. I recommended to Congress that authority be given to the Secretary of the Interior to issue bonds in anticipation of the assured earnings by the projects, so that the projects, worthy and feasible, might be promptly completed and the settlers might be relieved from their present inconvenience and hardship (applause). In authorizing the issue of these bonds, Congress limited the application of their proceeds to those projects which a board of army engineers, to be appointed by the President, should examine and determine to be feasible and worthy of completion. The board has been appointed, and soon will make its report.

Suggestions have been made that the United States ought to aid in the drainage of swamp lands belonging to the States or private owners, because, if drained, they would be exceedingly valuable for agriculture and contribute to the general welfare by extending the area of cultivation. I deprecate the agitation in favor of such legislation. It is inviting the general Government into contribution from its treasury toward enterprises that should be conducted either by private capital or at the instance of the State (applause). In these days there is a disposition to look too much to the Federal Government for everything (applause). I am liberal in the construction of the Constitution with reference to Federal power (applause);

but I am firmly convinced that the only safe course for us to pursue is to hold fast to the limitations of the Constitution, and to regard as sacred the powers of the States (great applause and cheers). We have made wonderful progress, and at the same time have preserved with judicious exactness the restrictions of the Constitution. There is an easy way in which the Constitution can be violated by Congress without judicial inhibition, to-wit, by appropriations from the National treasury for unconstitutional purposes. It will be a sorry day for this country if the time ever comes when our fundamental compact shall be habitually disregarded in this manner. (Applause)

MINERAL LANDS

By mineral lands, I mean those lands bearing metals, or what are called metalliferous minerals.

The rules of ownership and disposition of these lands were first fixed by custom in the West, and then were embodied in the law, and they have worked, on the whole, so fairly and well that I do not think it is wise now to attempt to change or better them. The apex theory of tracing title to a lode has led to much litigation and dispute, and ought not to have become the law, but it is so fixed and understood now that the benefit to be gained by a change is altogether outweighed by the inconvenience that would attend the introduction of a new system. So too, the proposition for the Government to lease such mineral lands and deposits and to impose royalties might have been, in the beginning, a good thing, but now that most of the mineral land has been otherwise disposed of—I do not refer here to coal land or gas land or oil land or phosphate land—it would hardly be worth while to assume the embarrassments of a radical change.

FOREST LANDS

Nothing can be more important in the matter of Conservation than the treatment of our forest lands. It was probably the ruthless destruction of forests in the older States that first called attention to the necessity for a halt in the waste of our resources. This was recognized by Congress by an act authorizing the Executive to reserve from entry and set aside public timber lands as National forests. Speaking generally, there has been reserved of the existing forests about 70 percent of all the timber lands of the Government.

Within these forests (including 26,000,000 acres in two forests in Alaska) are 192,000,000 acres, of which 166,000,000 acres are in the United States proper and include within their boundaries something like 22,000,000 acres that belong to the States or to private individuals. We have, then, excluding Alaskan forests, a total of about 144,000,000 acres of forests belonging to the Government, which are being treated in accord with the principles of scientific forestry. The law now prohibits the reservation of any more forest lands in Oregon, Washington, Idaho, Montana, Colorado and Wyoming, except by act of Congress. I am informed by the Department of Agriculture that the Government owns other tracts of timber lands in these States which should be included in the forest reserves. I expect to recommend to Congress that the limitation herein imposed shall be repealed (applause). In the present forest reserves there are lands which are not properly forest land, and which ought to be subject to homestead entry. This has caused some local irritation. We are carefully eliminating such lands from forest reserves or, where their elimination is not practicable, listing them for entry under the forest homestead act.

Congress ought to trust the Executive to use the power of reservation only with respect to land covered by timber or which will be useful in the plan of reforestation (applause). I am in favor of each branch of the Government trusting the good faith of the other (applause). During the present Administration, 6,250,000 acres of land, largely non-timbered, have been excluded from forest reserves, and 3,500,000 acres of land, principally valuable for forest purposes, have been included in forest reserves, making a reduction in forest reserves of non-timbered land amounting to 2,750,000 acres. But had we had the opportunity to include reserves in the forbidden States, the balance would have been otherwise. The Bureau of Forestry since its creation has initiated reforestation on 5,600 acres. A great deal of the forest land is available for grazing. During the past year the grazing lessees numbered 25,400, and they pastured upon the forest reserves 1,400,000 cattle, 84,540 horses, and 7,580,400 sheep, for which the Government received \$986,715—a decrease from the preceding year of \$45,000, due to the fact that no money was collected or received for grazing on the non-timbered land eliminated from the forest reserve. Another source of profit in the forestry is the receipts for timber sold. This year they amounted to \$1,043,000, an increase of \$307,000 over the receipts of last

year. This increase is due to improvement in transportation to market, and to the greater facility with which the timber can be reached.

The Government timber in this country amounts to only one-fourth of all the timber, the rest being in private ownership. Only three percent of that which is in private ownership is looked after properly and treated according to modern rules of forestry (applause). The usual destructive waste and neglect continue in the remainder of the forests owned by private persons and corporations. It is estimated that fire alone destroys \$50,000,000 worth of timber a year. The management of forests not on public land is beyond the jurisdiction of the Federal Government. If anything can be done by law it must be done by the State legislatures. I believe that it is within their constitutional power to require the enforcement of regulations, in the general public interest, as to fire and other causes of waste in the management of forests owned by private individuals and corporations. (Applause)

Exactly how far these regulations can go and remain consistent with the rights of private ownership, it is not necessary to discuss; but I call attention to the fact that a very important part of Conservation must always fall upon the State legislatures, and that they would better be up and doing if they would save the waste and denudation and destruction through private greed or accidental fires that have made barren many square miles of the older States. (Great applause)

I have shown sufficiently the conditions as to Federal forestry to indicate that no further legislation is needed at the moment except an increase in the fire protection to National forests and an act vesting the Executive with full power to make forest reservations in every State where Government land is timber-covered, or where the land is needed for forestry purposes.

OTHER LAND WITHDRAWALS

When President Roosevelt became fully advised of the necessity for the change in our disposition of public lands, especially those containing coal, oil, gas, phosphates, or water-power sites, he began the exercise of the power of withdrawal by Executive order of lands subject by law to homestead and the other methods of entering for agricultural lands. The precedent he set in this matter was followed by the present Administration.

Doubt had been expressed in some quarters as to the power in the Executive to make such withdrawals. The confusion and injustice likely to arise if the courts were to deny the power led me to appeal to Congress to give the President the express power (applause). Congress has complied. The law, as passed, does not expressly validate or confirm previous withdrawals, and therefore, as soon as the new law was passed, I, myself, confirmed all the withdrawals which had theretofore been made by both Administrations by making them over again (great applause). The power of withdrawal is a most useful one, and I do not think it is likely to be abused.

COAL LANDS

The next subject, and one of the most important for our consideration, is the disposition of the coal lands in the United States and in Alaska. First, as to those in the United States.

At the beginning of this Administration there were classified coal lands amounting to 5,476,000 acres, and there were withdrawn from entry for purposes of classification 17,867,000 acres. Since that time there has been withdrawn by my order from entry for classification 77,648,000 acres, making a total withdrawal of 95,515,000 acres (applause). Meantime, of the acres thus withdrawn, 11,371,000 have been classified and found not to contain coal, and have been restored to agricultural entry, and 4,356,000 acres have been classified as coal lands; while 79,788,000 acres remain withdrawn from entry and await classification. In addition, 336,000 acres have been classified as coal lands without prior withdrawal, thus increasing the classified coal lands to 10,168,000 acres.

Under the laws providing for the disposition of coal lands in the United States, the minimum price at which lands are permitted to be sold is \$10 an acre; but the Secretary of the Interior has the power to fix a maximum price and to sell at that price. By the first regulations governing appraisal, approved April 8, 1907, the minimum was \$10, as provided by law, and the maximum was \$100, and the highest price actually placed upon any land sold was \$75. Under the new regulations, adopted April 10, 1909, the maximum price was increased to \$300 except in regions where there are large mines, where no maximum limit is fixed and the price is determined by the estimated tons of coal to the acre. The highest price fixed for any

land under this regulation has been \$608 per acre. The appraised value of the lands classified as coal lands and valued under the new and old regulations is shown to be as follows: 4,303,000 acres valued under the old regulation at \$77,000,000—an average of \$18 an acre—and 5,864,000 acres classified and valued under the new regulation at \$394,000,000, or a total of 10,168,000 acres valued at \$471,000,000. For the year ending March 31, 1909, 227 coal entries were made, embracing an area of 35,000 acres, which sold for \$663,000; for the year ending March 31, 1910, there were 176 entries, embracing an area of 23,000 acres, which sold for \$608,000, and down to August, 1910, there were but 17 entries, with an area of 1,720 acres which sold for \$33,900; making a disposition of coal lands in the last two years of about 60,000 acres for \$1,305,000.

The present Congress, as already said, has separated the surface of coal lands either classified or withdrawn to be classified from the coal beneath, so as to permit at all times homestead entries upon the surface of lands useful for agriculture, and to reserve the ownership in the coal to the Government.

The question which remains to be considered is whether the existing law for the sale of the coal in the ground should continue in force or be repealed and a new method of disposition adopted. Under the present law the absolute title in the coal beneath the surface passes to the grantee of the Government. The price fixed is upon an estimated amount of the tons of coal per acre beneath the surface, and the prices are fixed so that the earnings will only be a reasonable profit upon the amount paid and the investment necessary. But, of course, this is more or less guesswork, and the Government parts with the ownership of the coal in the ground absolutely. Authorities in the Geological Survey estimate that in the United States today there is a supply of about three thousand billion tons of coal, and that of this one-third, or about one thousand billion, are in the public domain. Of course, the other two thousand billion are within private ownership and under no more control as to the use or the prices at which the coal may be sold than any other private property.

If the Government leases the coal lands and acts as any landlord would, and imposes conditions in its leases like those which are now imposed by the owners in fee of coal mines in the various coal regions of the East, then it

would retain over the disposition of the coal deposits a choice as to the assignee of the lease, a power of resuming possession at the end of the term of the lease, or of readjusting terms at fixed periods of the lease, which might easily be framed to enable it to exercise a limited but effective control in the disposition and sale of the coal to the public (applause). It has been urged that the leasing system has never been adopted in this country, and that its adoption would largely interfere with the investment of capital and the proper development and opening up of coal resources. I venture to differ entirely from this view (applause). My investigations show that many owners of mining property of this country do not mine it themselves, and do not invest their money in the plants necessary for the mining, but they lease their properties for a term of years varying from twenty to thirty and forty years, under conditions requiring the erection of a proper plant and the investment of a certain amount of money in the development of the mines, and fixing a rental and a royalty, sometimes an absolute figure and sometimes one proportioned to the market value of the coal. Under this latter method the owner of a mine shares in the prosperity of his lessees when coal is high and the profits good, and also shares to the same extent in their disappointment when the price of coal falls.

I have looked with some care into a report made at the instance of President Roosevelt upon the disposition of coal lands in Australia, Tasmania, and New Zealand. These are peculiarly mining countries, and their experience ought to be most valuable. In all these countries the method for the disposition and opening of coal mines originally owned by the Government is by granting a leasehold, and not by granting an absolute title. The terms of the leases run all the way from twenty to fifty years while the amount of land which may be leased to any individual there is from 320 acres to 2,000 acres. It appears that a full examination was made and the opinions of all the leading experts on the subject were solicited and given, and that with one accord they approved in all respects the leasing system (applause). Its success is abundantly shown.

It is possible that at first considerable latitude will have to be given to the Executive in drafting these forms of lease, but as soon as experiment shall show which is the most workable and practicable, its use should be provided for specifically by statute. The question as to how great an area ought to be included in a lease to one individual or corporation is not free

from difficulty; but in view of the fact that the Government retains control as owner, I think there might be some liberality in the amount leased, and that 2,500 acres would not be too great a maximum.

By the opportunity to register the terms upon which the coal shall be held by the tenant, either at the end of each lease or at periods during the term, the Government may secure the benefit of sharing in the increased price of coal and the additional profit made by the tenant. By imposing conditions in respect to the character of work to be done in the mine, the Government may control the character of the development of the mine and the treatment of employees with reference to safety (applause). By denying the right to transfer the lease except by written permission of Government authorities, it may withhold the needed consent when it is proposed to transfer the leasehold to persons interested in establishing a monopoly of coal production in any State or neighborhood (applause).

As one-third of all the coal supply is held by the Government, it seems wise that it should retain such control over the mining and the sale as the relation of lessor to lessee furnishes. The change from the absolute grant to the leasing system will involve a good deal of trouble in the outset, and the training of experts in the matter of making proper leases; but the change will be a good one and can be made. The change is in the interest of Conservation, and I am glad to approve it. (Great applause)

ALASKA COAL LANDS

The investigations of the Geological Survey show that the coal properties in Alaska cover about 1,200 square miles, and that there are known to be available about fifteen billion tons. This is, however, an underestimate of the coal in Alaska, because further developments will probably increase this amount many times; but we can say with considerable certainty that there are two fields on the Pacific slope which can be reached by railways at a reasonable cost from deep water—in one case of about 50 miles and in the other case of about 150—which will afford certainly six billion tons of coal, more than half of which is of a very high grade of bituminous and of anthracite. It is estimated to be worth, in the ground, one-half cent a ton, which makes its value per acre from \$50 to \$500. The coking-coal lands of Pennsylvania are worth from \$800 to \$2,000 an acre, while other

Appalachian fields are worth from \$10 to \$386 an acre, and the fields in the central States from \$10 to \$2,000 an acre, and in the Rocky mountains from \$10 to \$500 an acre.

The demand for coal on the Pacific Coast is for about 4,500,000 tons a year. It would encounter the competition of cheap fuel oil, of which the equivalent of 12,000,000 tons of coal a year is used there. It is estimated that the coal could be laid down at Seattle or San Francisco, a high-grade bituminous at \$4 a ton, and anthracite at \$5 or \$6 a ton. The price of coal on the Pacific slope varies greatly from time to time in the year and from year to year—from \$4 to \$12 a ton. With a regular coal supply established, the expert of the Geological Survey, Mr Brooks, who has made a report on the subject, does not think there would be an excessive profit in the Alaska coal mining because the price at which the coal could be sold would be considerably lowered by competition from these fields and by the presence of crude fuel oil. The history of the laws affecting the disposition of Alaska coal lands shows them to need amendment badly. Speaking of them, Mr Brooks says:

The first act, passed June 6, 1900, simply extended to Alaska the provisions of the coal lands in the United States. The law was ineffective, for it provided that only subdivided lands could be taken up and there were no land surveys in Alaska.

I do not like to criticise a coordinate branch of the Government. The Executive makes mistakes, and so does Congress, but I do not think it reflects greatly on the intense interest that Congress had in Alaska and her development that they should go to work and pass a law affecting the coal lands in Alaska that didn't operate there at all [applause]. The matter was rectified by the act of April 28, 1904, which permitted unsurveyed lands to be entered and the surveys to be made at the expense of the entrymen. Unfortunately the law provided that only tracts of 160 acres could be taken up, and no recognition was given to the fact that it was impracticable to develop an isolated coal field requiring the expenditure of a large amount of money by such small communities. Many claims were staked, however, and surveys were made for patents. It was recognized by everyone familiar with the conditions that after patent was obtained these claims would be

combined in tracts large enough to assure successful mining operation. No one experienced in mining would, of course, consider it feasible to open a coal field on a basis of a single 160-acre tract. The claims for the most part were handled in groups, for which one agent represented the several different owners. Unfortunately a strict interpretation of the statute raised the question whether even a tacit understanding between claim-owners to combine after patents had been obtained was not illegal. Remedial legislation was sought and enacted in the statute of May 28, 1908. This law permitted the consolidation of claims staked previous to November 12, 1906, in tracts of 2,560 acres. One clause of this law invalidated the title if any individual or corporation at any time in the future owned any interest whatsoever, directly or indirectly, in more than one tract. The purpose of this clause was to prevent the monopolization of coal fields. Its immediate effect was to discourage capital. It was felt by many that this clause might lead to forfeiture of title through the accidents of inheritance, or might even be used by the unscrupulous in blackmail. It would appear that land taken up under this law might at any time be forfeited to the Government through the action of any individual, who, innocently or otherwise, obtained interest in more than one coal company. Such a title was felt to be too insecure to warrant the large investments needed for mining development. The net result of all this is that no titles to coal lands have been passed.

On November 12, 1906, President Roosevelt issued an Executive order withdrawing all coal lands from location and entry in Alaska. On May 16, 1907, he modified the order so as to permit valid locations made prior to the withdrawal on November 12, 1906, to proceed to entry and patent. Prior to that date some 900 claims had been filed, most of them said to be illegal because either made fraudulently by dummy entrymen in the interest of one individual or corporation, or because of agreements made prior to location between the applicants to cooperate in developing the lands. There are thirty-three claims for 160 acres each, known as the "Cunningham claims," which are said to be valid on the ground that they were made by an attorney for thirty-three different and bona fide claimants who, as alleged, paid their money and took proper steps to locate their entries and protect them. The representatives of the Government, on the other hand, in the hearings before

the Land Office have attacked the validity of these Cunningham claims on the ground that prior to their location there was an understanding between the claimants to pool their claims after they had been perfected and unite them in one company.

The trend of decision seems to show that such an agreement would invalidate the claims, although under the subsequent law of May 28, 1908, the consolidation of such claims was permitted, *after* location and entry, in tracts of 2,560 acres. It would be, of course, improper for me to intimate what the result of the issue as to the Cunningham and other Alaska claims is likely to be, but it ought to be distinctly understood that no private claims for Alaska coal lands have as yet been allowed or perfected, and also that whatever the result as to pending claims, the existing coal-land laws of Alaska are most unsatisfactory and should be radically amended (applause). To begin with, the purchase price of the land is a flat rate of \$10 per acre, with no power to increase it beyond that, although, as we have seen, the estimate of the agent of the Geological Survey would carry up the maximum of value to \$500 an acre.

In my judgment it is essential to the proper development of Alaska that these coal lands should be opened, and that the Pacific slope should be given the benefit of the comparatively cheap coal of fine quality which can be furnished at a reasonable price from these fields (great applause); but the public, through the Government, ought certainly to retain a wise control and interest in these coal deposits (applause), and I think it may do so safely if Congress will authorize the granting of leases, as already suggested for Government coal lands in the United States, with provisions forbidding the transfer of the leases except with the consent of the Government, thus preventing their acquisition by a combination or monopoly, and upon limitations as to the area to be included in any one lease to one individual, and at a certain moderate rental, with royalties upon the coal mined proportioned to the market value of the coal laid down either at Seattle or at San Francisco (applause). Of course such leases should contain conditions requiring the erection of proper plants, the proper development by modern mining methods of the properties leased, and the use of every known and practical means and device for saving the life of the miners.

The Government of the United States has much to answer for in not having given proper attention to the Government of Alaska and the development of her resources for the benefit of all the people of the country. I would not force development at the expense of a present or future waste of resources; but the problem as to the disposition of the coal lands for present and future use can be wisely and safely settled in one session if Congress gives it careful attention. (Great applause)

OIL AND GAS LANDS

In the last Administration there were withdrawn from agricultural entry 2,820,000 acres of supposed oil land in California, about 1,500,000 acres in Louisiana (of which only 6,500 acres were known to be vacant, unappropriated land), 75,000 acres in Oregon, and 174,000 acres in Wyoming, making a total of nearly 4,000,000 acres.

In September, 1909, I directed that all public oil lands, whether then withdrawn or not, should be withheld from disposition pending congressional action, for the reason that the existing placer mining law, although made applicable to deposits of this character, is not suitable to such lands, and for the further reason that it seemed desirable to reserve certain fuel-oil deposits for the use of the American Navy. Accordingly the form of all existing withdrawals was changed, and new withdrawals, aggregating 2,750,000 acres, were made, in Arizona, California, Colorado, New Mexico, Utah and Wyoming. Field examinations during the year showed that of the original withdrawals, 2,170,000 acres were not valuable for oil, and they were restored for agricultural entry. Meantime other withdrawals of public oil lands in these States were made, so that on July 1, 1910, the outstanding withdrawals then amounted to 4,550,000 acres.

The needed oil and gas law is essentially a leasing law. In their natural occurrence oil and gas cannot be measured in terms of acres, like coal, and it follows that exclusive title to these products can normally be secured only after they reach the surface. Oil should be disposed of as a commodity in terms of barrels of transportable product rather than in acres of real estate (applause). This is, of course, the reason for the practically universal adoption of the leasing system wherever oil land is in private ownership. The Government thus would not be entering on an experiment, but simply

putting into effect a plan successfully operated in private contracts. Why should not the Government as a land-owner deal directly with the oil producer rather than through the intervention of a middleman to whom the Government gives title to the land? (Applause) The principal underlying feature of such legislation should be the exercise of beneficial control rather than the collection of revenue.

As not only the largest owner of oil lands, but as a prospective large consumer of oil by reason of the increasing use of fuel-oil by the navy, the Federal Government is directly concerned both in encouraging rational development and at the same time insuring the longest possible life to the oil supply. The royalty rates fixed by the Government should neither exceed nor fall below the current rates. But much more important than revenue is the enforcement of regulations to conserve the public interest so that the inconvenience of the lessee shall specifically safeguard oil fields against the penalties from careless drilling and of production in excess of transportation facilities or of market requirement.

One of the difficulties presented, especially in the California fields, is that the Southern Pacific Railroad owns every other section of land in the oil fields, and in those fields the oil seems to be in a common reservoir, or series of reservoirs, communicating through the oil sands, so that the excessive draining of oil at one well, or on the railroad territory generally, would exhaust the oil in the Government land. Hence it is important that if the Government is to have its share of the oil, it should begin the opening and development of wells on its own property. (Laughter and applause)

In view of the joint ownership which the Government and the adjoining land-owners, like the Southern Pacific Railroad, have in the oil reservoirs below the surface, it is a most interesting and intricate question, difficult of solution, but one which ought to address itself at once to the State lawmakers, how far the State legislature might impose appropriate restrictions to secure an equitable enjoyment of the common reservoir, and to prevent waste and excessive drainage by the various owners having access to this reservoir (applause). It has been suggested, and I believe the suggestion to be a sound one, that permits be issued to a prospector for oil, giving him the right to prospect for two years over a certain tract of Government land for the discovery of oil, the right to be evidenced by a license for which he pays

a small sum. When the oil is discovered, then he acquires title to a certain tract, much in the same way as he would acquire title under a mining law. Of course, if the system of leasing is adopted, then he would be given the benefit of a lease upon terms like that above suggested. What has been said in respect to oil applies also to Government gas lands.

Under the proposed oil legislation, especially where the Government oil lands embrace an entire oil field, as in many cases, prospectors, operators, consumers, and the public can be benefitted by the adoption of the leasing system. The prospector can be protected in the very expensive work that necessarily antedates discovery. The operator can be protected against impairment of productiveness of the wells which he has leased by reason of the control of drilling and pumping of other wells too closely adjacent or by the prevention of imperfect methods as employed by careless, ignorant or irresponsible operators in the same field, which result in the admission of water to the oil sand; while, of course, the consumer will profit by whatever benefits the prospector or operator receives in reducing the first cost of the oil.

PHOSPHATE LAND

Phosphorus is one of the three essentials to plant growth, the other elements being nitrogen and potash. Of these three, phosphorus is by all odds the greatest element in nature. It is easily extracted in useful form from the phosphate rock, and the United States contains the greatest known deposits of this rock in the world. They are found in Wyoming, Utah and Florida, as well as in South Carolina, Georgia and Tennessee. The Government phosphate lands are confined to Wyoming, Utah and Florida. Prior to March 4, 1909, there were four million acres withdrawn from agricultural entry on the ground that the land covered phosphate rock. Since that time 2,322,000 acres of the land thus withdrawn was found not to contain phosphate in profitable quantities, while 1,678,000 acres was classified properly as phosphate land. During this Administration there has been withdrawn and classified 437,000 acres, so that today there is classified as phosphate rock land 2,115,000 acres.

The rock is most important in the composition of fertilizers to improve the soil, and as the future is certain to create an enormous demand throughout

this country for fertilization, the value to the public of such deposits as these can hardly be exaggerated. Certainly with respect to these deposits a careful policy of Conservation should be followed. Half of the phosphate of the rock that is mined in private fields in the United States is now exported. As our farming methods grow better the demand for the phosphate will become greater, and it must be arranged so that the supply shall equal the needs of the country. It is uncertain whether the placer or lode law applies to the Government phosphate rock. There is, therefore, a necessity for some definite and well-considered legislation on this subject, and in aid of such legislation all of the Government lands known to contain valuable phosphate rock are now withdrawn from entry.

A law that would provide a leasing system for the phosphate deposits, together with a provision for the separation of the surface and mineral rights as is already provided for in the case of coal, would seem to meet the need of promoting the development of these deposits and their utilization in the agricultural lands of the West. If it is thought desirable to discourage the exportation of phosphate rock and the saving of it for our own lands, this purpose could be accomplished by conditions in the lease granted by the Government to its lessee. Of course, under the Constitution the Government could not tax and could not prohibit the exportation of phosphate, but as proprietor and owner of the lands in which the phosphate is deposited it could impose conditions upon the kind of sales, whether foreign or domestic, which the lessee might make of the phosphate mined. (Applause)

The tonnage represented by the phosphate lands in Government ownership is very great. But the lesson has been learned in the case of such lands as have passed into private ownership in South Carolina, Florida and Tennessee, that the phosphate deposits there are in no sense inexhaustible. Moreover, it is also well understood that in the process of mining phosphate, as it has been pursued, much of the lower grade of phosphate rock which will eventually all be needed has been wasted beyond recovery. Such wasteful methods can easily be prevented, so far as the Government land is concerned, by conditions inserted in the leases.

WATER-POWER SITES

Prior to March 4, 1909, there had been, on the recommendation of the Reclamation Service, withdrawn from agricultural entry, because they were regarded as useful for power sites which ought not to be disposed of as agricultural lands, tracts amounting to about 4,000,000 acres. The withdrawals were hastily made and included a great deal of land that was not useful for power sites. They were intended to include the power sites on twenty-nine rivers in nine States. Since that time 3,475,442 acres have been restored for settlement of the original 4,000,000, because they do not contain power sites; and meantime there have been newly withdrawn 1,245,892 acres on vacant public land and 211,007 acres on entered public land, or a total of 1,456,899 acres. These withdrawals made from time to time cover all the power sites included in the first withdrawals, and many more, on 135 rivers and in 11 States. The disposition of these power sites involves one of the most difficult questions presented in carrying out practical Conservation.

The Forest Service, under a power found in the Statute, has leased a number of these power sites in forest reserves by revocable leases, but no such power exists with respect to power sites that are not located within forest reserves; and the revocable system of leasing is, of course, not a satisfactory one for the purpose of inviting the capital needed to put in proper plants for the transmission of power.

The Statute of 1891, with its amendments, permits the Secretary of the Interior to grant perpetual easements or rights-of-way from water sources over public lands for the primary purpose of irrigation and such electrical current as may be incidentally developed, but no grant can be made under this Statute to concerns whose primary purpose is generating and handling electricity. The Statute of 1901 authorizes the Secretary of the Interior to issue revocable permits over the public lands to electrical power companies, but this Statute is woefully inadequate because it does not authorize the collection of a charge or fix a term of years. Capital is slow to invest in an enterprise founded upon a permit revocable at will.

The subject is one that calls for new legislation. It has been thought that there was danger of combination to obtain possession of all the power sites and to unite them under one control. Whatever the evidence of this, or lack of it, at present we have had enough experience to know that combination

would be profitable, and the control of a great number of power sites would enable the holders or owners to raise the price of power at will within certain sections; and the temptation would promptly attract investors, and the danger of monopoly would not be a remote one.

However this may be, it is the plain duty of the Government to see to it that in the utilization and development of all this immense amount of water-power, conditions shall be imposed that will prevent monopoly, and will prevent extortionate charges which are the accompaniment of monopoly. The difficulty of adjusting the matter is accentuated by the relation of the power sites to the water, the fall and flow of which create the power.

In the States where these sites are, the riparian owner does not control or own the power in the water which flows past his land. That power is under the control and within the grant of the State, and generally the rule is that the first user is entitled to the enjoyment. Now, the possession of the bank or water-power site over which the water is to be conveyed in order to make the power useful, gives to its owner an advantage and a certain kind of control over the use of the water-power, and it is proposed that the Government in dealing with its own lands should use this advantage and lease lands for power sites to those who would develop the power, and impose conditions on the leasehold with reference to the reasonableness of the rates at which the power, when transmuted, is to be furnished to the public, and forbidding the union of the particular power with a combination of others made for the purpose of monopoly by forbidding assignment of the lease save by consent of the Government (applause). Serious difficulties are anticipated by some in such an attempt on the part of the general Government, because of the sovereign control of the State over the water-power in its natural condition, and the mere proprietorship of the Government in the riparian lands.

It is contended that through its mere proprietary right in the site the central Government has no power to attempt to exercise police jurisdiction with reference to how the water-power in a river owned and controlled by the State shall be used, and that it is a violation of the State's rights. I question the validity of this objection. The Government may impose any conditions that it chooses in its lease of its own property, even though it may have the same purpose and in effect accomplish just what the State would

accomplish by the exercise of its sovereignty. That is shown frequently in leases of houses containing a covenant against the use of the house for that which under the law of the State is an unlawful use; and nevertheless, no one has ever contended that that condition, though it be for the stricter enforcement of the State law, is without the power of the lessor as a proprietor of the land which he is leasing.

There are those (and the Director of the Geological Survey, Mr Smith, who has given a great deal of attention to this matter, is one of them) who insist that this matter of transmuting water-power into electricity which can be conveyed all over the country and across State lines, is a matter that ought to be retained by the general Government, and that it should avail itself of the ownership of these power sites for the very purpose of coordinating in one general plan the power generated from these Government-owned sites. On the other hand, it is contended that it would relieve a complicated situation if the control of the water-power site and the control of the water were vested in the same sovereignty and ownership, viz: the State, and then were disposed of for development to private lessees under the restrictions needed to preserve the interests of the public from the extortions and abuses of monopoly. Therefore, bills have been introduced in Congress providing that whenever the State authorities deem a water-power useful they may apply to the Government of the United States for a grant to the State of the adjacent land for a water-power site, and that this grant from the Federal Government to the State shall contain a condition that the State shall never part with the title to the water-power site or the water-power, but shall lease it only for a term of years not exceeding fifty, with provisions in the lease by which the rental and the rates for which the power is furnished to the public shall be readjusted at periods less than the term of the lease, say every ten years.

The argument is urged against this disposition of power sites that legislators and State authorities are more subject to corporate influence and control than would be the central Government. In reply it is claimed that a readjustment of the terms of leasehold every ten years would secure to the public and the State just and equitable terms. Then it is said that the State authorities are better able to understand the local need and what is a fair adjustment in the particular locality than would be the authorities at Washington. It has been argued that after the Federal Government parts

with title to a power site it cannot control the action of the State in fulfilling the conditions of the deed, to which it is answered that in the grant from the Government there may be easily inserted a condition specifying the terms upon which the State may part with the temporary control of the water-power sites, and, indeed, the water-power, and providing for a forfeiture of the title to the water-power sites in case the condition is not performed; and giving to the President, in case of such violation of conditions, the power to declare forfeiture and to direct proceedings to restore to the central Government the ownership of the power sites with all the improvements thereon, and that these conditions may be promptly enforced and the land and plants forfeited to the general Government by suit of the United States against the State, which is permissible under the Constitution (applause). And that by such a provision, *in terrorem*, the edict of States and of the legislatures in respect to these lands might be enforced through the general Government.

I do not express an opinion upon the controversy thus made or a preference as to the two methods of treating water-power sites. I shall submit the matter to Congress with all the arguments, and urge that one or the other of the two plans be promptly adopted.

At the risk of wearying my audience I have attempted to state as succinctly as may be the questions of Conservation as they apply to the public domain of the Government, the conditions to which they apply, and the proposed solution of them.

In the outset I alluded to the fact that Conservation had been made to include a great deal more than what I have discussed here. Of course, as I have referred only to the public domain of the Federal Government, I have left untouched the wide field of Conservation with respect to which a heavy responsibility rests upon the States and individuals as well. But I think it of the utmost importance that after the public attention has been roused to the necessity of a change in our general policy to prevent waste and a selfish appropriation to private and corporate purposes of what should be controlled for the public benefit, those who urge Conservation shall feel the

necessity of making clear how Conservation can be practically carried out (applause), and shall propose specific methods and legal provisions and regulations to remedy actual adverse conditions (applause). I am bound to say that the time has come for a halt in general rhapsodies over Conservation, making the word mean every known good in the world (applause), for, after the public attention has been roused, such appeals are of doubtful utility and do not direct the public to the specific course that the people should take, or have their legislators take, in order to promote the cause of Conservation. The rousing of emotions on a subject like this, which has only dim outlines in the minds of the people affected, after a while ceases to be useful, and the whole movement will, if promoted on these lines, die for want of practical direction and of demonstration to the people that practical reforms are intended. (Applause)

I have referred to the course of the last Administration and of the present one in making withdrawals of Government lands from entry under homestead and other laws, and of Congress in removing all doubt as to the validity of these withdrawals as a great step in the direction of practical Conservation (applause). But this is only one of two necessary steps to effect what should be our purpose. It has produced a status quo and prevented waste and irrevocable disposition of the lands until the method for their proper disposition can be formulated, but it is of the utmost importance that such withdrawals should not be regarded as the final step in the course of Conservation, and that the idea should not be allowed to spread that Conservation is the tying up of the natural resources of the Government for indefinite withholding from use, and the remission to remote generations to decide what ought to be done with these means of promoting present general human comfort and progress (great applause). For, if so, it is certain to arouse the greatest opposition to Conservation as a cause, and if it were a correct expression of the purpose of conservationists it ought to arouse such opposition. (Applause)

Real Conservation involves wise, non-wasteful use in the present generation, with every possible means of preservation for succeeding generations; and though the problem to secure this end may be difficult, the burden is on the present generation promptly to solve it and not to run away from it as cowards, lest in the attempt to meet it we may make some mistakes (applause). As I have said elsewhere, the problem is how to save

and how to utilize, how to conserve and still develop; for no sane person can contend that it is for the common good that Nature's blessings should be stored only for unborn generations. (Applause)

I beg of you, therefore, in your deliberations and in your informal discussions, when men come forward to suggest evils that the promotion of Conservation is to remedy, that you invite them to point out the specific evils and the specific remedies; that you invite them to come down to details in order that their discussions may flow into channels that shall be useful rather than into periods that shall be eloquent and entertaining without shedding real light on the subject (prolonged applause and cheers). The people should be shown exactly what is needed in order that they may make their representatives in Congress and the State legislatures do their intelligent bidding. (Great and prolonged applause)

President BAKER—The Congress is now adjourned to reassemble at 2 o'clock this afternoon.

SECOND SESSION

The Congress was called to order by President Baker at 3 oclock p.m.

President BAKER—It gives me a great deal of pleasure to announce that Governor W. R. Stubbs, of Kansas, has kindly consented to preside at this session. Ladies and Gentlemen, Governor Stubbs. (Applause)

Governor STUBBS—Mr President, Ladies and Gentlemen: I am very grateful for your liberal recognition. And I present to you a man who knows much about the laws pertaining to land in the United States, one better fitted to speak on this subject than any other, Senator Knute Nelson, of Minnesota. I take great pleasure in introducing him. (Applause)

Senator NELSON—Mr Chairman, Ladies and Gentlemen: I could not help thinking this forenoon as I looked at the magnificent audience how every delegate and visitor from abroad must conclude that in one respect Conservation in Minnesota has been a success—Conservation of our prosperous and growing humanity.

I am here to speak briefly of our public-land system, past and present, in the hope that we may derive some lessons from the mistakes of the past and have something to guide us in the future. I shall say little of Conservation in general. My aim will be to draw attention to what I deem of importance for the legislative branch of the Government to do in the future, and I shall do so only in general terms, seeking—on account of my position as Chairman of the Senate Committee on Public Lands—to avoid all matters that will lead to controversy.

As those know who have had experience in public affairs, particularly in legislation, all reforms are matters of compromise. Legislation is largely experimental and those who are most progressive and advanced in seeking reforms for the future often find themselves handicapped by those who would make no change; and the result is oftentimes a compromise in which the reformers get only half a loaf.

The natural resources of our country should be conserved by the individual, by the State, and by the Federal Government. For each there is an appropriate field. The farmer must conserve the resources of his farm; the State the resources of its lands, its forests and its waters; and the Federal Government the resources of its mines, its forests, and its lands with all their appurtenances. When the several forces act in harmony, beneficial results of a far-reaching and permanent value will be attained for the preservation and utilization of our resources. Practical and beneficial Conservation of natural resources on the part of the Federal Government and the State should include and provide for due and efficient utilization of the same for the benefit of the masses of the people. The mere conservation and retention of ownership, the mere securing of a larger price for the resource, may prove burdensome rather than a benefit to the public. The ultimate question is not so much how to hold and conserve as how to properly utilize our resources. The mere holding, or the mere securing of a higher price seems to me to be entirely futile (applause). The aim should not be so much to secure a higher price for the Government as to secure lower price for the consumer and to prevent monopoly (applause). Hence, in the disposal of a resource, care should be taken to prevent combination and monopoly in restraint of trade in respect to the same; and the right, as in the case of railway rates, to regulate the price to the consumer should be retained; in other words, care should be taken and provision should be made that the consumer can obtain the product of the resource at a fair and reasonable rate. To merely conserve and hold at a high price retards development and enables those who have already secured a large share of a resource to monopolize the market and to secure an exorbitant price for the product of the resource. (Applause) The ultimate object of the conservation of a resource should be to utilize it for the best advantage of the consumer. True Conservation means beneficial use—means utilization.

The close of the Revolutionary War found our country with an empty treasury and a large public debt, but possessed of a large quantity of valuable public lands northwest of the Ohio river and elsewhere, ceded by Great Britain, supplemented by a cession from Virginia and some of the

older States, from which were afterward carved great States, though the public domain was at that time regarded chiefly as an asset from which the Government could obtain revenues for its wants and needs.

The first general land law of a public nature for the disposal of our public lands was passed in 1796. This law, after prescribing a system of surveying the public lands, substantially the same as has been since adhered to, provided for the sale of the lands at public auction to the highest bidder, partly for cash and partly on credit.

By the Act of 1800 the minimum price was fixed at \$2 an acre, and land not sold at public auction could be bought at private sale at that price.

The Act of 1820 abolished sales on credit and fixed the minimum price at \$1.25 per acre, at which rate it has since remained. Lands offered at public sale became known as "offered land," and if not sold at public sale could be obtained at private sale or entry at the minimum price.

The result of this system was that, owing to the great scarcity of money in the country at that time among the masses of the people, large blocks of land were purchased by speculators and held by them indefinitely for an excessive profit, and the masses of the people—the settlers, the real home builders—had to purchase the land from these speculators instead of securing it from the Government. The Government got but scant return for its valuable public land. The chief profit was made by the middlemen, those speculators who bought it up in large blocks; they reaped a rich harvest. But in the midst of this system the settlers pressed on to the frontier. They were without money, but they settled on the public lands, squatted there without authority of law; and finally the Government, to help these settlers, to relieve them and give them a little breathing time, in 1841 passed what was known as the general Preemption Law. Under this law the head of a family, a widow, or a single person over twenty-one years of age who was a citizen or had declared his intention to become a citizen of the United States, could secure 160 acres of public land by settling upon, improving and cultivating it, and by paying for and entering the same within from one to three years after settlement, the time of payment in each case depending on whether the land was offered, unoffered, or unsurveyed. This law (the Preemption Act of 1841) was clearly intended to help the pioneers and the settlers, and it

proved of great advantage to them; but owing to the lax procedure that prevailed (under which a man could go on a preemption claim, make a few limited and pro forma improvements, and at the end of six months appear in the land office and prove up and have his final entry made and ultimately get a patent), the Preemption Law itself became a great instrument in the hands of speculators and land grabbers, and in consequence Congress concluded to repeal the law.

The law allowing lands to be secured at private entry was repealed in 1889; the law allowing public sales was repealed in 1891, and the Preemption Law was also repealed the same year. These laws were repealed none too soon, because by that time they had got to be the instruments by which those who were seeking valuable coal lands, timber lands, and other lands would hire a lot of people to go and make preemption claims, and then, as soon as they obtained title, secure the title, whereby thousands and thousands of acres of the most valuable timber and mineral lands, coal lands, and other lands passed into the hands of speculators for little more than a dollar and a quarter an acre, and sometimes even less, for there were various kinds of scrip issued—agricultural college scrip and other scrip to which I will call attention later—put on the market and sold. That scrip would be used instead of money in paying for and entering land; and through it much valuable land passed into the hands of speculators at a cost of even less than one dollar an acre. You who have lived here have all observed that the low price at which the lumbermen secured timber in those early days under the Preemption Law, by cash entry, and under agricultural and other scrip, did not help much to get cheaper lumber. The result was to enable owners of large bodies of pine land to hold them indefinitely for the purpose of securing a higher price for their stumpage.

In 1874 an Act was passed "To encourage the growth of timber on the western prairies." The purpose of this Act was laudable and had it resulted in supplying the prairies with timber the gift of the land would have been amply compensated for. But in its practical operations the law proved a failure. Only a comparatively few of the many claims entered were ever successfully forested, or ever became real timber land. A large share of them were merely taken and held by speculators with no real purpose of complying with the law in respect to the planting, culture, and care of timber. Claimants would secure these claims, enter them under the timber-

culture law, make the first entry, and then hold them just as long as they could, waiting until some land-seeker came around, when they would tell him, "I have a timber claim here, and might relinquish it so you can take it as a homestead; how much will you give me for my relinquishment?" And for a time under this law there was a great speculation in the sale of what we call timber relinquishments. No timber was raised. Speculators had held the land for four or five, maybe six, years as a timber claim. Then when a real settler came along, why, for a consideration of one, two, three, four or five hundred dollars, or whatever the settler was able to pay, the holder would relinquish his timber claim to the Government so that the real land-seeker could secure the same under the Homestead Law, or under the Preemption Law, while that was in force.

In 1862 an Act was passed giving to each State 30,000 acres of land for every senator and representative in Congress for the purpose of establishing and maintaining agricultural and mechanical colleges. In States where there was a sufficient quantity of public lands the State was required to select the land from the public lands in the State, but in States where no such lands could be secured land scrip was issued in place of the land. This resulted in placing an enormous amount of land scrip on the market, which was sold by the State in many instances in bulk to speculators at a greatly reduced price, netting the States from fifty to one hundred cents per acre—perhaps the average did not exceed seventy-five cents per acre. The scrip could be used in entering land or in paying for land under the Preemption and Homestead Laws at the rate of \$1.25 an acre. So far as the States to which the scrip was issued were concerned the grant was a wasteful one. It would have been much wiser and better for the Government to have appropriated \$1.25 per acre in cash to the States instead of giving them the scrip, and reserving the lands which could be entered with the scrip for actual settlers under the Homestead Law. In addition to this college scrip, we have had from time to time various kinds of other scrip, Chippewa half-breed scrip, Sioux half-breed scrip, and Supreme Court scrip, and others that I cannot at this moment recall. Most of this scrip, fortunately, is now about exhausted; very little of it is still afloat and at large. There was also what we called "soldiers' additional" scrip of which there was a great deal; a soldier could take a homestead of 40 or 80 or 120 acres, and if he had no more in his homestead entry, he could take the residue and make up 160 acres anywhere on the

public lands of the United States, without residing on the land; and he could dispose of his interest by power of attorney, by which speculators succeeded in getting that. And the records of our soldiers' homes will show how land speculators have been searching among the veterans for this kind of scrip. Why, I got a letter some years ago from a gentleman in Missouri—I can't recall his name—reminding me of the fact that I had had a homestead; and he told me that I was entitled to forty acres more under my right, in addition to the 120 acres, and that he was willing to buy the land of me. He had hunted up the records, and found a man by my name, but unfortunately the homestead and all the rest transpired and existed in Wisconsin.

In 1878 Congress passed the so-called Timber and Stone Act, originally limited to four western States, but by the Act of 1892 extended to all the public-land States. Under this law land unfit for cultivation and chiefly valuable for timber and stone could be secured in tracts of 160 acres for each entry-man at a price of \$2.50 per acre. Under the law the purchaser is prohibited from buying the land on speculation or in the interest of any one but himself. On its face this law seems fair, harmless, and just, but in its practical operation it proved the means of a good deal of fraudulent land speculation. In the first instance, valuable agricultural land fit for agriculture was entered under the law on the theory that it was only good for the timber or stone on it. In the next place—and there was where the great iniquity occurred—speculators would hire men and women in different parts of the country to go and enter stone and timber claims, furnishing them money through outside friends, and then as soon as they had secured title get a transfer of the land to themselves by paying a bonus of one or two hundred and in some cases up to five hundred and a thousand dollars. Why, I remember how, in a city not a thousand miles away from Saint Paul, one year some twenty-five or thirty school teachers entered stone and timber claims in the State of Oregon! This law finally proved simply a source of speculation to the men who were trying to secure large bodies of timber; and under it a large share of the valuable timber lands now in private holdings were secured. The law should have been repealed immediately; but while the Senate passed a bill repealing it some five or six years ago, the bill failed to pass in the House of Representatives. Since then the Secretary of the Interior has come to our relief to some extent. The

Stone and Timber Act said that this land could be sold at not less than \$2.50 an acre; and up to 1908 the Government had always treated that as the price, and never exacted any more. In 1908 the Interior Department adopted the rule of appraising the lands for the timber and stone on them and selling them at the appraised value, and the result of that policy has been to stop speculation in those kinds of lands. A very limited amount of timber and stone lands have been entered since for now it does not pay big lumber operators or land speculators to hire anybody to go and enter these lands because under this rule they have to pay pretty nearly what the land is worth. But while this administrative order has given us some relief, I am clearly of the opinion that the law should be entirely repealed to the end that we can make suitable provision for the disposal of our stone and timber land under more appropriate legislation and at a fairer rate, both to the purchaser and to the Government. (Applause)

In 1862 Congress passed the Homestead Law, with the general provisions of which most of you are familiar. This law makes a gift of 160 acres to each settler and home-builder who will occupy, improve, and cultivate his claim for a period of five years. Of all our public land laws this has, on the whole, been the most beneficent and productive of the best results. Under its provisions hundreds of thousands of poor and industrious men and women have carved out happy homes for themselves and their children, relieved the pressure of labor in our large cities and great industrial centers, and rapidly laid the foundation for and built up great States in the middle and far West. Judged by results, it is doubtful whether the Government ever received a better return for any of its lands than it has received for these lands given as a free gift, under the Homestead Law, to our farmers and settlers. A happy, prosperous, and industrious rural population will ever prove to be the sheet-anchor of our industrial, social and political well-being, and will ever afford a solid foundation for the integrity and perpetuity of American institutions. The Homestead Law, with all its blessings, had one defect which has, to some extent, marred its usefulness. I refer to the privilege of commutation—the privilege of proving up and paying for the land at \$1.25 per acre prior to the five-year period for final free entry. Originally and for many years after the law was passed, the privilege of commutation could be exercised after the lapse of six months from date of entry. This period was extended to fourteen months some years

ago and this fourteen months' period is still the law. The vice of allowing a homestead entry to be commuted as stated, consists in opening the door to the speculator, who, in the space of fourteen months can secure title to the land on scant and temporary improvements and then move away and hold the land for merely speculative purposes, leaving the surrounding settlers to enhance the value of his land by their continuing and permanent improvements. When they have erected dwellings, barns, school houses, and churches, and have laid out roads and organized school districts, the petty speculator and commutator, who has done nothing to build up the country, stands ready to sell his land at a greatly enhanced price to an actual home-builder and settler. The commutation privilege should not have been included in the law, and should be repealed, in my opinion, as soon as practicable. None but permanent and bona fide settlers should be permitted to secure land under the Homestead Law.

In 1872 Congress enacted a law for the location, purchase, and entry of land containing gold, silver, copper, and other precious metals, commonly called the mining law of the United States, which became a part of the Revised Statutes. Mining claims are of two classes: (1) lode or quartz claims, and (2) placer claims. Both are initiated by discovery, staking out on the ground, and filing notice of location. After these preliminary steps have been taken, claims can be held indefinitely without purchase as long as \$100 worth of work is done each year on each claim; and as a matter of fact, only a small proportion of mining claims, especially placer claims, are ever purchased from the Government. Placer claims are soon worked out and exhausted, while good lode claims are workable and profitable for many years. There is a difference in the size and in the price of lode and placer claims. Placer claims are larger in area and can be purchased at \$2.50 per acre, while lode claims cost \$5 an acre.

In 1873 Congress passed a law for the purchase and entry of coal lands, which also became a part of the Revised Statutes. Under this law every person above the age of twenty-one years, who is a citizen or has declared his intention to become a citizen of the United States, may purchase and enter 160 acres of coal land; and an association of such persons may purchase and enter 320 acres, and an association of not less than four such persons, if they have first expended not less than \$5,000 in working and improving a coal mine on the land, may purchase and enter not to exceed

640 acres in one claim. The price in each case is not less than \$10 per acre where the land is situate more than fifteen miles from a completed railroad and not less than \$20 per acre if the land is within fifteen miles of a completed railroad. Only one entry can be made by the same person or association of persons; and no association or any member of which shall have taken the benefit of the law either as an individual or as a member of any other association, shall enter or hold any other lands under the provisions of the law; and no member of any association which shall have taken the benefit of the law shall enter or hold any other land under the provisions of the law. A preference right of entry for the period of one year is given to any person or association that has opened and improved a coal mine on the public lands. The provisions of the law as to the acquisition and holding of more than one claim are clear and stringent, and have been applied and enforced in the courts in several instances where great corporations have sought, through dummies and otherwise, to acquire vast holdings of coal lands. It is conceded, too, that the minimum price fixed by statute is, in many instances, altogether too low and much below the real value.

In 1907 the Interior Department cured this defect by adopting the policy of classifying and appraising the coal lands and selling them at the appraised value, a value in most instances far in excess of the minimum statutory price. This new policy is continued, and under it something over 100,000 acres of coal lands have been entered. Further legislation is urgently needed in respect to the disposal of our coal lands. If the policy of selling the land is to be continued, not only should the system of appraisal now in vogue be adhered to, but provision should be made to protect the people—the consumers—against the monopolies and combinations in restraint of trade and against unreasonable and exorbitant prices. Many good men, however, who have given the subject great consideration, favor a well-guarded system of leasing instead of sale for coal lands. Their contention is that under a leasing system more careful and less wasteful methods of mining will be pursued and that better protection can be thereby secured for the miner, the consumer, and the Government. I am not prepared to take issue with this contention. A leasing system is clearly preferable where the surface of the land is disposed of for agricultural purposes, for under such a system the conflicting interests of the miner and the farmer can be best

adjusted, regulated, and controlled. Perhaps it would be wise to adopt both sale and leasing systems, leaving it optional with the Government to select the mode of disposal in any given case; for there may be cases where the one method would be preferable to the other.

In 1900 Congress passed an act extending the coal-land laws to Alaska, but the act proved of no value as only surveyed lands could be purchased and entered under the general law, and there was no surveyed land in Alaska, and no provision was made in the act for surveys. By the act of April 28, 1904, the general coal-land laws were extended to Alaska in a more effective and rational manner. Under this act any person or association, qualified to make entry under the coal-land laws of the United States, who opens and improves a coal mine on the unsurveyed public lands in Alaska can locate the land on which such mine is found by staking the same out on the ground, and by filing notice of location in the recording district and in the land office of the district in which the land is situate, within one year after staking out the claim. After these preliminary steps are taken, the locator must cause a survey and plat of the land to be made by a deputy United States surveyor, and thereafter, within three years from date of the location notice he must make application for a patent of the land, prove a compliance with the law, and pay the price of only \$10 per acre for the land. Aside from these provisions all other provisions of the general coal-land law apply to coal lands in Alaska. Under this law between eight and nine hundred coal-land locations have been made, but of these only about thirty-three cases (perhaps a few more) have passed to final entry at the local land office. The time for making entry and securing patents on the other locations has, in most, if not in all instances, lapsed, and they cannot be relocated owing to the fact that by executive orders of November 12, and 27 and December 17, 1906, and by a recent executive order of President Taft, all coal lands in Alaska are withdrawn from location, sale, and entry. This withdrawal was no doubt made in pursuance of recent legislation by Congress and for the purpose of giving Congress an opportunity to enact better coal-land laws for Alaska than those now in force; and such legislation, to my mind, is clearly and urgently needed, and I am in hopes that Congress will take steps at its next session to enact suitable coal-land laws for Alaska in order that the people there may have an opportunity to utilize the coal that is within their own boundaries. (Applause)

By the Act of March 3, 1877, amended in 1891, provision was made for the entry and reclamation by irrigation of desert or arid land in the Pacific coast and mountain States and Territories. Under the original act 640 acres could be entered in one claim, but since the act of 1891 was passed only 320 acres can be entered in one claim under this law. Water for irrigating the land must be secured and the land must be reclaimed and cultivated by means of such water for the period of four years after the preliminary entry, and the price of \$1.25 per acre must be paid before patent can be secured for the land. This law has not proved very effective or beneficial, especially on account of the difficulty, in many instances, of securing the necessary water supply by a single entryman. In many instances the conditions of the law have not been complied with, and as a whole the law may be said to have to some extent failed of its purpose.

In view of the comprehensive character of the general Reclamation Act of 1902, which makes due provision for securing a water supply and provides for limited homesteads under a qualified homestead law, the desert law referred to, could well be repealed. The Federal reclamation system is more certain and effective than reclamation by individuals in isolated cases.

Under the so-called Carey Act of 1894, desert and arid lands are granted to certain States, in limited quantities, for reclamation and cultivation by means of irrigation, this to be done under the auspices and direction of the States to which the grants are made. This law has in some respects proved more effective and of more value than the general desert law, but it cannot be regarded as equal in value and efficiency to the general Reclamation Act of 1902, and therefore it seems to me it is not advisable to make any more grants of this nature to any of the States.

No effective or systematic effort was made to preserve the forests on the public domain until March 3, 1891, when an act was passed giving the President the power to set apart and reserve, from time to time, public lands for forestry purposes. This was supplemented by the act of June 4, 1897, providing for the administration and care of the land so reserved and set apart, which lands are now termed National Forests of the United States. Under this law nearly two hundred million acres of public lands in various States and Territories, including Alaska, have been withdrawn and set apart for forestry purposes and are now embraced, most of them, in our National

Forests and their administration and care has been placed on a sound, workable, and safe basis through the initiation, prudence, and wisdom of our great forester, Mr Pinchot (great and prolonged applause), who has laid the foundation and is the father of our forestry system. This legislation and administrative action came none too soon. Had there been more delay, our timber lands would, long before this, have passed into private ownership and there would have been nothing left for the Government to conserve (applause). No land legislation in recent times has been productive of such beneficent and far-reaching blessings and results as our forestry legislation. While occasionally there has been a little grumbling and friction on the part of settlers and cattlemen as to the administration of the law in some of its details, yet it can be fairly said, when it is borne in mind that it is a new system, that there has been little, if any, valid ground for serious criticism or complaint. The conduct of a few over-zealous forest rangers and a few over-strenuous settlers and cattlemen ought not to militate against the value and usefulness of the forestry system as a whole and in its entirety. (Applause)

Under the act of March 3, 1891, as amended by the act of January 21, 1895, and May 11, 1898, the Secretary of the Interior is authorized, under general regulations to be fixed by him, to grant, without exacting compensation, permits for right-of-way on the public lands for canals and reservoirs which may be used for furnishing water for domestic, public, and other beneficial uses, and for the development of power. Several valuable water-power sites have been secured under these laws as well as under the homestead and timber and stone laws. To put a stop to such practice the Interior Department, in the later part of 1908 and in the early part of 1909, withdrew all power sites from every form of disposal under our land laws and these sites have remained thus withdrawn ever since, except during an interval of a few days or perhaps a few weeks in the spring of 1909; and during that interval no power sites were secured or disposed of. Most of these power sites are of considerable value, and they ought not to be disposed of under any of the existing land laws. Adequate provision should be made by law for the utilization of these water-power sites to the end that the Government may receive fair compensation for the same, and to the end that the public may receive the beneficial use to be derived from the development of any water-power in connection with such sites, at fair and reasonable rates (applause). The problem under our dual system of Government, State and

Federal, is not free from embarrassment, as it is the opinion of men versed in the law that while the general Government may own the power site, with all the rights of a riparian owner, the water in the streams, except for purposes of navigation, belongs to the State, and that the State may allow its citizens to appropriate such water for their beneficial use and thereby render the power site of no value; for without a sufficient supply of water the power site will not be worthy of improvement or development. It seems to me (though perhaps I may err) that the problem of developing and utilizing water-power in such cases can be properly solved only by the cooperation of the State and the Federal Government (applause): the one owning the power site and the other the water in the stream, it strikes me that cooperation is essential and furnishes the only practical solution. And some plan should be devised by which the Federal and State Governments could act in harmony and in unison in such cases. Of course, when the State owns both the water and the power site, the problem is of a less complex character, and is one exclusively for the State to solve except as to the question of navigation. And I may also add in this connection that Congress, at its last session, passed a general law to regulate the construction of dams across navigable waters, by which ample provision is made for protecting the interest of the general Government in all such cases.

Most of our remaining public lands, suitable for agricultural purposes, are within the arid or semi-arid parts of the country. These lands can be successfully farmed only by means of irrigation or by so-called dry farming methods. To aid in developing and successfully farming these lands, the Reclamation Act of 1892 was passed setting apart the proceeds of the sales of public lands within the arid and semi-arid States for the construction of dams, reservoirs, canals, and ditches for the impounding and distribution of water. A considerable number of irrigation projects have been entered upon under the act. A few of them have been completed, but the majority of them are still in an incomplete condition; and there being an insufficiency of funds available for their speedy completion, Congress, at the last session, in order to expedite the work on the incomplete projects, provided for a loan of twenty million dollars, to be immediately available, and to be reimbursable out of the future income of the reclamation fund (applause). This will hasten the completion of the projects and will aid the homestead

settlers of whom there are many, to secure a supply of water on their claims at an early day.

For the purpose of promoting the farming of arid or semi-arid lands by dry-farming methods or otherwise, where no water supply for irrigation is or can be found available, Congress, by the Act of February 19, 1909, provided for enlarged homesteads of 320 acres of non-irrigable lands. The theory on which such legislation was based was this, that such lands to be farmed must be summer-fallowed, so that a crop could be raised only every other year, and therefore a larger quantity of land was needed, as only one-half of the cultivated land could be cropped each year.

Investigations by the Geological Survey have shown that considerable areas of public lands suitable for agricultural purposes are underlain with more or less valuable beds of coal. Such lands, on account of their mineral character, are not technically subject to entry under any other than the coal-land laws of the United States. A considerable number of homestead settlers had settled upon such lands and had made the preliminary homestead entries of the same without any previous knowledge of their mineral character. For the relief of such settlers Congress passed the Act of March 3, 1909, which provides that such settlers may enter and receive a patent for the surface of such land, reserving to the general Government the coal underlying the same to be disposed of under the coal-land laws of the United States. This was supplemented by Congress at its last session by the Act of June 22, 1910, which permits the entry of the surface of coal lands under the Homestead Law, the Reclamation Law, the Desert Law, and the so-called Carey Law, reserving to the Government the coal beds underlying such lands, to be disposed of under the general coal-land laws in existence or to be passed in the future, and authorizing the exploration of the same.

One of the most important of our late land laws and which will prove to be the key to future reforms in our land system is the Act of June 25, 1910, passed at the last session of Congress. This act authorizes the President, in his discretion, to withdraw from settlement, location, sale, or entry any of the public lands of the United States and reserve the same for water-power sites, irrigation, classification, or other public purpose. There was some difference of opinion before the enactment of this law as to the power of the President to make such withdrawals in all cases. This act removes all doubt

and controversy on the subject and enables the President to examine, classify, and appraise the lands and to reserve them for necessary and appropriate legislation by Congress. Many of our lands and their appurtenances are of such a character that they ought not to be disposed of under any of our existing land laws. Good laws are needed for the disposal of our timber and stone, our water-power sites, and our coal, oil, asphalt, and phosphate lands. There was considerable opposition to the passage of this act in both Houses of Congress, and at one time it seemed as though it would not pass, and it would not have passed but for the active, continued, and persistent help of President Taft (applause). He labored for its passage, in season and out of season, to my certain knowledge, and but for his help, I can say with all truthfulness that that important law would never have passed (applause). And since its passage the President has availed himself of it by making new withdrawals, and rewithdrawing many lands which had been withdrawn before but in respect to which some question was raised as to the validity of the withdrawal.

I have not called attention to the various grants of land that have been made, first for wagon roads and canals, and afterward for railroads, nor to the large grants of land that have been made to the several States for educational and other purposes, for the reason that such grants are not likely to be repeated in the future. Provision has already been made, with ample land grants for the admission into the Union of our two remaining Territories, Arizona and New Mexico; and it is not probable that any grants of public lands, except for right-of-way, will be made to any railroad in the future, especially the railroad grants, may seem to have been prodigal and too lavish; but to the legislators of those early days, who were anxious for the speedy settlement and development of our great West, they seemed justified and called for. And it is evident that, in consequence of these grants, the country was more speedily settled and the settlers afforded transportation facilities at a much earlier period than otherwise would have been done. The grants made to the States, especially for educational purposes, have from every point of view been fully justified, and have been, and will continue to be, of great help in maintaining ample and liberal educational facilities in the several States.

In conclusion: I have given you this brief summary and outline of our public land laws, past and present, obsolete and subsisting, in order that

from a consideration of the same we may avoid the mistakes of the past, and gather inspiration and instruction for our future guidance. In view of the diminishing supply and rapidly increasing demand it behooves us to husband, with discrimination and care, all our natural resources, beginning as promptly as possible, and this work must be done by legislation, by administration, and by individual effort. (Applause)

Chairman STUBBS—If there are any pessimistic citizens in the United States they should hear the Senior Senator's story of the lavish management of public affairs in the past, and the splendid change made under that great leader—the greatest man on earth today—Theodore Roosevelt (applause and cheers) and Gifford Pinchot (renewed applause).

I now take great pleasure in presenting to you a typical southern gentleman, Governor Noel, of Mississippi (applause).

Governor NOEL—Mr Chairman, Brother Governors, Delegates, Ladies and Gentlemen: Some months ago I received an invitation to attend this Congress, which I promptly accepted; also an invitation to deliver an address, which I immediately declined. Since entering the hall this afternoon I have been informed of my selection for the first address on my State—each Governor speaking for his State in succession—and my State's views on questions pertaining to our natural resources.

Of course the greatest natural resource of every city and county, as of State and Nation, is the productive energies of its people. Their development, through proper training of mind and heart, should be the chief aim of all people and of the Government. In those resources, however, our interests are the same as those of all other parts of the country, and they open too broad a field for me to enter.

When we take up the question of the natural resources pertaining to our domain, Mississippi occupies a widely different attitude from that of some

States in the Northwest whose Governors are here to speak for them. We are an agricultural people. Not a city in Mississippi will much, if at all, exceed 30,000 inhabitants; more of its population and its wealth, proportionately, than of any other State in the Union are engaged in agriculture. We have no mines, no minerals except some clays and stone, no oil, no gas, no coal. We acquired agricultural lands, and our natural resources are from those lands as agricultural soil and standing timber.

Before the question of Conservation was understood, or at least before it had become of any force in State or Nation, both Mississippi and the Federal Government had parted with their lands and with their forests. Much to our regret now, it is a question of the past, and has to be handled by individuals and by corporations, to whom our lands and timber chiefly, almost entirely, belong. We are interested; we try to regulate our resources in some measure, within the powers of the State Government; but our interest is largely confined to our public lands. We have no coal or metals, our streams are sluggish, and there are few water-power sites. We have little beyond the surface values of the timber and the soil. We are interested in coal because it is necessary for our industries; we are interested in oil because we need it; we are interested in all the elements of the soil spoken of this morning, phosphorus and all the rest. We are greatly interested in all these things notwithstanding the land which contains them happens to be in other States. We have not lost interest in them on that account; and, speaking for our State—which has stood for State rights as it understood them, and stands for State rights still—our only way of securing these rights we believe to be through the Federal Government (applause); our only voice must be through Congress and the President, and we do not care to surrender that to which the Government is now properly entitled. If the choice goes to the State we know how it will go, for past experience has taught that lesson well—local interests will control, and the general good will be subordinated to personal pride and local considerations. We have learned much and suffered much in that line. The Government gave to us, as to others, the sixteenth section of land in every township, one-thirty-sixth of the whole State. We put it in the power of a majority of the householders and patrons of schools in each township to vest the school lands by lease, thinking that local interests, being circumscribed and vitally concerned in education, would at least prevent spoliation of this magnificent donation to

the school children; but we were mistaken. In a great many instances a few who were shrewd and sharp and designing used a law by which a lease could be made from one year to ninety-nine years, and until that law was repealed leased the lands for the largest possible term.

We know that the smaller the area the greater the influence of personages, and of local and private considerations. Therefore, as we look on this question of the Conservation of our natural resources, it is a question of rights, and how those rights can best be maintained and perpetuated; the means, whether through State or Federal Government, is but a minor consideration; and believing that our rights can best be preserved and utilized, now and for all time to come, without waste and without destruction, both for the present and in the future, we think it can best be done under Federal supervision (applause). The only rights we have in coal and oil and metals must be exercised through the Federal Government.

We may not fully understand the water-power problem. It has been said to be only a local issue anyway. We do not understand it that way. The river which rolls by this city smiling, smooth, and clear, after it is joined to the Missouri is muddy, deep, and uncertain; not only all of your waters but all of the waters east of the Rocky Mountains roll past our western boundary. While at some seasons the water is low, at others it is over fifty feet higher, and more than one-sixth in value of the land in our State is subject to overflow. Your waters, which through proper forestation and proper handling by dams and other means would give us a more equable flow throughout the year, come down upon us at a time when we do not need them, and in a degree greatly in excess of any possible need at any time, and we have to bear the sins of deforestation and all of the other evils that come from the wholesale spoliation and destruction of your forest lands (applause). We are vitally interested in that question. We believe in forest reservations; we are sorry we cannot furnish the basis for it in our own State, but so far as the Government lands we have can be availed of for that purpose, we would be more than glad to see the Government take hold of the matter and set our people an example of how forests should be handled and preserved for the present and for the future.

When it comes to water-power, to me, at least, and to many of us, the question of conflict between State and Federal Governments, about which

so much has been said—especially with a view of eliminating the Federal Government—we hardly understand that view of it. We trace our title through the Federal Government (applause). As a lawyer of more than thirty years' practice, whenever I have been given a question for investigation pertaining to the title of land, the first thing I have done was to examine the tract-books to see whether the Federal Government had ever parted legally with its title. If it had not, the question was ended; if it had, then we could proceed to deraign to those properly entitled to it. So when the Federal Government owned the lands and was the source of title, we do not understand how, even though the lands may be within the State, its right as a land-owner is less on a river bank than it is in the interior, or when the Federal Government, as the owner of the lands, should not exercise riparian rights which any other owner tracing title through it might exercise.

Now, we would like cooperation of the States, but we would like the Federal Government to retain where it still possesses them those rights of which the people could not be robbed through control of State legislatures or local authorities (applause). You may say, What interest have we, who are not a manufacturing people, in the mines and the water-powers of other States? Why, we are all in a common country. State lines may be changed; they are accidental; they are artificial; but the national boundary is fixed. When we look for coal or iron, or commercial or industrial products which we do not manufacture, we must look, primarily, within the bounds of the United States. It is within the power of the Government and beyond ours practically, through tariff legislation, to exclude the minerals from outside. We have but one open field, we have but one certain route to any natural or manufactured product, and that is within the boundaries of the Union itself; and we do not want, through monopolization of either coal or oil or water-powers, to be hampered in the protection of the country as a whole so that as consumers we shall have to bear the brunt of evils from which the National Government, through the little influence we might have with it, might protect us, and of which our State government, in the past at least, has been very neglectful. Hence we stand for State rights and Federal control in cooperation (applause). But if it is within the power of the Federal Government, through leasing or otherwise, to retain control of its mineral and coal lands and its water-power sites, to put them beyond the possibility of handling by a State and its legislature, to regulate

corporations' rights so as to prevent monopolization, and at the same time to prevent the Nation as a whole from being deprived of any productive agency in our midst, we want the benefit of it. (Applause)

Our patriotism on this score may be of that questionable type described by Artemus Ward, who said that during the Civil War, when the stress was great, he listened to a magnificent speech from an orator on the subject of enlistment, and became so enthused that when the call for volunteers came he, with others, went up to sign the roll; but when he observed that the orator had not signed nor was likely to sign, because his province was simply that of speaking while other's would be fighting, his own ardor was somewhat cooled, and when he reflected that the orator's eloquence had carried his hearers where he would not go himself, it became cooler and cooler. Still, his patriotism did not entirely vanish, for when his time came to sign the roll for enlistment, he signed it with the name of his mother-in-law and offered her as a sacrifice to his country (laughter). Yet we are not exactly in that category, though we may seem to view the situation from a local standpoint. But knowing of our own condition, knowing of the rights which the Federal Government conferred upon the school children of our State—the sixteenth section and other lands of which you heard in Senator Nelson's address today,—and remembering how in a great many instances, through local influences, legislative or otherwise, the intended beneficiaries were largely deprived of the benefactions intended for them we really think, What has gone is gone, except as a lesson to us; and so far as we are concerned, we shall stand for the right of the people as a whole for the enjoyment of its great resources of coal, of oil, of water-power and other natural wealth, and we want to be protected in such a way that no State or local influence shall be able to take it from us forever (applause). That is our position on this question.

In regard to the water-power question. A while ago I spoke of the Mississippi rolling by; we have never been jealous of the Federal Government's dealings with that river, not a bit (laughter). We are not now. So far as we are concerned, we would be delighted if the Federal Government would acquire the riparian rights, with all the *liabilities*, from one end of the State to the other (laughter). The county in which I live, that part of it in the Delta, as well as six or seven other counties, have had to keep up, without Federal aid until this year—and then only incidentally for

the protection of navigation against some caving banks—for five years more than 200 miles of levee, and it has required an acreage tax of from three to five cents, an *ad valorem* tariff of about ten mills, and a cotton tax besides; and while some of this is among the finest agricultural land in the world, it is almost wrecked by the taxes on it. Missouri has fared better. Her levees are not as extensive as ours; her people put them in good condition, and the general government afterward took charge of them in the interest of navigation; and if the Government will relieve us of the whole burden from the waters which you send down upon us from the North and from east of the Rocky Mountains, and will take the riparian rights from end to end and preserve and use them for the benefit of the whole Nation, all the people of our State will greatly rejoice (applause), and not a voice will be raised on the question of State rights as to any use for the people as a whole to which the Government may put those lands.

So, as we come to voice our wishes, our interests, our desires, they are for cooperation of State and Federal Government, but of absolutely no relinquishment on the part of the Federal Government either of its water-power sites, its coal lands, its phosphate lands, or of any of those other natural resources to which the people of the whole country are looking for future development and prosperity (applause). We are *in* the country, we are a part of it; not merely a part of the Government of the States but a part of the Government of the whole Union (applause), and all that concerns the Union, or any part of it or any of its people, affects us to a greater or less degree. And speaking for our share and our part in the national destiny which invitingly presents itself before us, we say that we stand for Conservation of natural resources by all governmental agencies, State and Federal, which will not only develop now but protect in the future for the proper use and progressive benefit of the people of the whole country to whom they now belong and from whom they should never depart. (Applause)

Chairman STUBBS—Ladies and Gentlemen: I am very glad indeed to introduce to you as the speaker to follow our distinguished friend from Mississippi, the only other democratic Governor in the Congress, Governor Norris, of Montana. (Applause)

You will see whether the views of the southern democrat and the northern democrat are the same after the two get through speaking. (Laughter)

Governor NORRIS—Mr Chairman, and Ladies and Gentlemen: A feeling has prevailed in the West, or did a few days ago, to the effect that no enlarged opportunities were going to be given to express ideas here which were contrary to those held by the program committee of this Congress (laughter). However, I am pleased to note that such is not to be the case, and whether the conference of the Northwestern Governors at Salt Lake City,

recently held, has had anything to do with it or not I don't know. Anyhow, we are thankful for small favors. If it had been the intention and had been carried out, it would have been a mistake, for the reason that the Conservation movement is national in scope, and is a part of no section and no State alone. The Conservation movement—in other words, the public conscience—received its awakening some two years ago, and Theodore Roosevelt did the awakening (applause); and I am pleased to note that the sentiment created by President Roosevelt has ripened into practical action by President Taft (renewed applause). I resent the insinuation that Montana and the Northwest, and in fact the entire West, is opposed to Conservation; in fact, I insist that the Northwest is the leader of the Conservation movement (applause), and that the first practical act in Conservation was taken by a western State, Montana (applause). I am proud of the fact that the first Conservation commission, either State or National, was appointed by me, in the State of Montana (applause). I am further proud of the fact that the first Conservation law, comprehensive in extent, was, under my recommendation, passed by the Legislature of Montana; and in that respect we have led the National Government in the Conservation movement (applause). Therefore, just for a moment, and not desiring to be personal, permit me to state what we have done. And in every respect we have kept step with the National Government and in the majority of cases we have led the National Government, and you can come to us for a lesson as to how to properly conserve the natural resources of the country (applause).

The Legislature which assembled in Montana in 1900 enacted a law conserving the resources possessed by us in our public lands, so generously given us by the Government on our admission. That measure provided for the disposition of the land to actual cultivators of the soil, in 160-acre tracts where irrigated, in 320-acre tracts where it is suitable for dry farming, and in 640-acre tracts where it was only suitable for the raising of hay or for grazing purposes—that is, in the high altitudes, in the mountains. So in that respect we have gone hand in hand with the Government in the passage of the 320-acre homestead act, applicable to entries where irrigation could not be had. In that same law, passed in 1909, some eighteen months ago, Montana forever reserved from sale, and in every patent on every acre of its lands that might thereafter be issued retained the coal rights, and provided for the leasing of those rights from time to time and for periods not

exceeding five years (applause). So today, when President Taft says he hopes Congress will do the same with the Government coal lands, we say, Mr President, we are with you and hope Congress will do this (applause), and if you wish an illustration proving that the title to coal lands can be retained and the coal rights leased from time to time, providing for the right to mine the coal at not less than 12½ cents per ton, come to Montana and we will show you half a dozen coal leases with such provision which have been in force for the last sixteen months (applause). Have we lagged behind the National Government? Oh, no! In fact, we have led the National Government in the matter of Conservation. (Applause)

And as to the metalliferous ores of the mines—the same laws are applicable to State lands that are applicable to Government lands. As to the forests: in the making of those laws, I corresponded, and our commission corresponded, and we made those laws with the consent of, and they were afterward approved by, Mr Gifford Pinchot (applause). There is but one provision which we made then differing from those of the Government. We provided in that law, passed eighteen months ago, that lands more suitable for agriculture than for reforestation should be used for agricultural purposes and not for reforestation purposes. President Taft described this morning how the Government had in the last few months been doing the same thing, so it seems that, after awhile, the Government will catch up to Montana in that respect (laughter and applause).

Now, then, on the water-power question: That same commission is now operating, and it is going to prepare suggestions for submission to the next Montana Legislature with reference to adequate provisions for conserving the waters of the State of Montana, and I have no doubt that the recommendations of the commission will, at the next session, be adopted. We would have done that two years ago except we cannot do all these things at once; our session only lasted sixty days, while Congress is in session all the time (laughter and applause). If we had even six months instead of two years for it, we would have had those water resources conserved long ago (laughter). Is Montana entitled to take a place in the kindergarten class in the school of Conservation? And are we who have conserved our resources to be distrusted as Governor Noel says you must distrust the Legislature and the people of the State of Mississippi? (Applause) I thank my God that I can trust the people of Montana to protect

their own! (Applause) And let me tell you one thing: the whole can never be greater than the sum total of its parts, and the Federal Government can never adequately preserve its resources until you get at least a majority of the people in a majority of the States to so agree, because it takes a majority for the Federal Congress or the Federal Government to act (applause). You start at the wrong end. You have got to start with the people of the State and build up.

Now, are we capable of passing legislation to preserve our water resources? I think we are; and let me tell you some of our plans. In the first place, the water and the land, during the territorial days of each State, belonged to the Federal Government. When the State was admitted, the lands were reserved by the federal Government, but the waters flowing in the streams of the State passed into the control of the State. You heard Senator Nelson, an able lawyer, refer this afternoon to the fact that that was the law. Now, they tell us that you cannot trust the States, you must trust the Federal Government; and yet I listened for nearly an hour to one of the ablest presentations I ever heard of how the Federal Government for a hundred years wasted its resources with all the prodigality of a drunken sailor (applause). Trust the Federal Government! Why, the Federal Government has been the greatest sinner in that respect. I am glad the Federal Government has awakened and is going to preserve its resources, but Montana, at least, woke up a little before (applause). In this matter of the water-power: The most valuable use that water can be put to, or, in other words, the most valuable function that water can perform, is not the development of electrical power; in the semi-arid States it is the applying of that water to irrigation and the reclamation of the arid lands of the West (applause). So bear that in mind.

In the State of Montana—and what is true in that State is true largely in every other State in the West—not one-third of the arable lands that can be irrigated have as yet been reclaimed; less than 2,000,000 acres have been reclaimed in Montana, while there are 6,000,000, in fact there are 10,000,000 acres that can be reclaimed. In other words, there are from six to ten million acres yet to be reclaimed by use of the water that flows in the streams of the State, and that is largely Government land. So that when you talk about conserving the water for water-power purposes, we say conserve it for reclamation purposes (applause); for the reclamation of Government land, too (applause), that may make homes for settlers who will come in

and take it under the Homestead Act. There is the reason why we say that the Federal Government must not by its superior power step in and insist upon using the waters of the streams of the West for power purposes, unless when it so does it makes provision that the rights for irrigation purposes shall forever remain inviolate; otherwise, what does it amount to, the building of a dam across the stream? When the Government conveys the right to build a dam across a stream, it means that the amount of water flowing over that dam will determine the amount of power that may be developed; hence, when that dam is built the Government, if it conveys anything of value, must convey the right to the use of that water, and the right to the use of that water flowing over that dam must accrue as of that date, and forever thereafter the franchise-holder will have the right to demand as a concession from the Federal Government that the same amount of water, all the natural flow of that stream, must go over that dam forever. You thereby absolutely prevent the diversion of any water on that stream above that point for irrigation purposes. The use of water for irrigation purposes does decrease the amount flowing in the stream. That is the reason we object to the Federal Government coming in and taking charge of our water-power and giving it out—we do not care so much about the little income that may be received: that is the reason we are insisting upon the rights of the State.

Now, remember this: In the first instance, there is no contention but what the regulation of water for irrigating purposes is absolutely vested in the State, and that the Federal Government cannot acquire that right; hence a number of irrigators have already appropriated a part of the flow of the stream. The Federal Government grants the right of franchise for the building of a dam. Suppose we assume, for the sake of argument, that it can grant the right to the remaining flow of a stream; it not only thereby forever thereafter prohibits the use of that stream above that point for further reclamation purposes, but the rights of every irrigator, either before or after appropriation is made, comes in conflict, or may come in conflict, with the Federal franchise-holder? In other words, you transfer from the State courts and from the State forum the right of every irrigator to use the waters of a stream to the seat of power of the Federal Government at Washington. In other words, you practically stop irrigation in the arid West when you insist upon having that power (applause). Is that Conservation? True

Conservation demands that every acre of land shall be used for its highest purpose and be made to serve its highest productive function (applause), whether in a forest reserve or out of it. Therefore, in order to serve its highest productive function in the West, water must be applied to the land.

Now, take the 6,000,000 acres of land that may be reclaimed in Montana. If you do not insist upon the Federal Government taking charge of the water-power and preventing its further reclamation, it means 6,000,000 acres of land reclaimed. It is fair to say that each year those reclaimed lands will produce a total of \$25—yea, and if I did not want to be ultra-conservative, I would say \$50—per acre; and at \$25 per acre, you have an annual income from those 6,000,000 acres of land of \$150,000,000. Isn't that worth thinking about? Isn't that a resource worth conserving? Why, the 6,000,000 horse-power that might be developed in Montana is not worth one tithe of that. You say, Give to the Federal Government the right to the water-powers of the State and forever prevent the further reclamation of our land? Why, you are asking of us the most priceless gift that we have to convey—far more priceless than our mines yielding \$50,000,000 yearly, possibly the richest in the world—because you ask us to surrender not \$50,000,000 a year but the opportunity to make \$150,000,000 a year. Has the Federal Government this right? We insist, as a matter of law, that the Federal Government has no authority to grant any right to the use of water on any power site that it may have. If the power site is situated along a stream, the title to the power site rests in the Federal Government and it can grant the right to erect a dam on that site, but the water that flows down the stream by that power site belongs to the State, and unless the State gives you the right to appropriate and take water you will develop no power by a dam-site! (Applause)

Now, is the State ready to surrender any rights that it may have in the waters of the stream to the Federal Government? The State of Montana is not ready to so do, for the reasons I have given. The State of Montana will insist upon every right it has. Let the Federal Government have that which of right or in law belongs to it, but let the State keep that which of right or in law belongs to it (applause). So sure am I that the State has the right to use of its water that I think the next Legislature of Montana will pass a law to regulate the use of water, making its use for power forever subordinate to its use for irrigation purposes, and then say to the Federal Government, You

own your power site, but you do not own the water; we own the water, but we do not own the power site. Your site is worth nothing to you because it is valuable only for power in connection with the use of water. We cannot develop power on that site, but we can go a little farther down the stream and divert that water for the irrigation of land, and it is valuable to us. Now, that is what we mean by the rights of the State in and to the waters of the State. You cannot trust the State? Why not? If you cannot trust the people of Montana to conserve its resources, if you cannot trust the State of Wyoming to conserve its resources, can we trust the State of Maine, or the State of Florida to conserve them for us? What reasons have we to assume that the people of the State of Massachusetts or the State of Louisiana are more patriotic in that respect than are our own people?

The creation of the forest reserves was the greatest act ever performed in recent years. We would not have that act repealed. We have a double purpose in supporting the forest conservation policies. You think of it as valuable for the timber that it will grow. That timber is worth just as much, and will shelter just as many people, in Montana as it will in the Mississippi valley, but we desire it for a further purpose. The forests of these mountains are Nature's reservoirs, builded there by an Omnipotent Creator, and can better conserve the waters that fall in the form of rain and snow than these artificial reservoirs that men may build (applause). We want those waters. The water that comes from our mountains and is conserved under those forests is the very life-blood of the State of Montana. Would you take the water away and stop the reclamation of the arid West? I know you would not; yet you would do so did you not at the same time that you were saving the timber make a provision that the rights to water for power purposes should forever be subject to the rights for irrigation purposes.

Bear this in mind, also. The doctrine of riparian rights does not prevail in the arid West; therefore the owning of the soil on each side of the stream does not convey the right to have the water flow down that stream undiminished in quantity or quality. In other words, the first appropriator is the first in right. I think there has been a misunderstanding as to the position of the West in this respect, as to why we are insisting upon the rights of the State. We insist upon the right of the State to control the waters of the State, not the water-power particularly. There is a decided difference between the waters and the water-power. The waters will irrigate land, the water-power

will develop electricity. Such is the position the West takes. Will you not help us in that, and so help develop the land and make it productive? Do you know it is your own salvation to do so? Ye people of the populous East, where is the produce to come from to feed the ever-increasing millions, unless it be from the reclamation of the arid lands of the West? The time will soon be here, and it is not over four years removed, when we will cease to be a wheat-exporting nation, and in only a few years it must come that the children will cry for bread, and the land must be made to produce it. Therefore we must husband our resources and conserve our water for use for the purpose which will permit the growing of something that will feed human beings; and pine trees do not do it (applause). You of the Mississippi valley who for years have wept great crocodile tears that your lands have been cleared, suppose those lands had not been cleared, whence would come the produce to feed the millions of today? So bear these things in mind that when you come to conclusions you will take all these questions into consideration. And I want to say to you that in the future, as in the past, Montana will not lag in the Conservation movement, but will continue to lead the Federal Government (applause).

A DELEGATE—Mr Chairman, are the propositions advanced by the Governors to be discussed? I see no reference in the program to such discussion, and ask for information.

Chairman STUBBS—The understanding of the Chair is that this afternoon was turned over to the Governors. The intention is to give them an opportunity to relieve their minds this afternoon (applause) and get the way clear for the greatest man you will hear talk in thirty years—Theodore Roosevelt (applause). We are clearing out the brush and getting ready for the real thing that you will have tomorrow. (Laughter and applause)

You can readily see that they have too much water in the South and not quite enough water in the Northwest, judging by the views of the last two speakers.

I now have the pleasure of introducing one of the greatest Governors in the United States, and of one of the greatest States in the Union, Governor Deneen, of Illinois (applause).

Governor DENEEN—Fellow Delegates, and Ladies and Gentlemen: The Governors here have been somewhat confused regarding this program. I was invited by my good friend Governor Eberhart, of this State, to prepare a speech. I have it concealed about my person like a deadly weapon, and I have been wondering whether I dare read it; for if I do, those who follow me will, I fear, have no audience to address, while if I do not follow the text already given to the printer there will be the traditional print-shop "devil" to pay; but I have concluded to talk rather than read, and I hope that my good friends the reporters will publish what I should have said rather than what I shall say. I will follow the example of a very distinguished statesman in our State, who on a great occasion handed his speech to the reporters and said, "Now, having given my speech to the reporters, I shall proceed to ramble;" and so he did. (Laughter)

It is a pleasure to follow the two distinguished gentlemen who have preceded me, the Governor of Mississippi and the Governor of Montana. It is a pleasure to note how the conditions have reversed the attitude of their States regarding State rights (laughter and applause). I am interested in both States. A year or more ago I purchased a farm in Montana where the three rivers join to form the Missouri river, and I discovered after the spring freshets that I now have a farm scattered all the way from Montana to Mississippi (laughter). I am interested in all the States because of that, because I now own property in all. But I cannot quite agree with my distinguished predecessor about the Legislature—we, too, have a legislature (laughter), and whatever value it may have had at one time it is not considered at par at present. (Laughter and applause)

We have a water-power proposition, too, strange to say, even in the flat, level, horizontal State of Illinois. Some time ago when the Government was considering the matter of the Lakes-to-Gulf Waterway, our State supplemented the investigation of the Government in considering the by-

products of that great channel which was to be built (and I hope will be built), and we proceeded on the theory announced by the President this morning; instead of going from agitation to legislation, we considered it better to go on this theory: investigation, then agitation, and later legislation. So our State appointed a very distinguished commission to investigate some of the by-products that would accrue to Illinois by reason of the Lakes-to-Gulf deep waterway.

We soon found we had several questions. First, the matter of reclamation. We have the problem they have in Mississippi, of too much water for too much time out of the year; an even 5,000 square miles of our State is under water too much of the time—an area larger than the State of Connecticut or the island of Porto Rico. We worked out a plan by which, as an incident to the great waterway, we expect to reclaim land which has been estimated to be of the value of \$150,000,000 to the State.

Then we found that in part of that waterway (in 62½ miles of it from Lockport to Utica) there is a fall of 106 feet, and that water-power can be created to the amount of about 130,000 horsepower, worth about \$2,500,000 or \$2,750,000 a year to begin with, and our engineers estimated that by availing ourselves of that power we would be able to contribute to the Government the entire expense of the waterway between Lockport and Utica, and could afford to expend \$20,000,000 in doing so by reason of the by-product that would come to us; and that we would be able, if the Legislature did as it should do, and the Governor did as he should do, and the commission to be appointed would do as it should do—to repay that vast expense in fourteen years as a minimum period, and that in fact we could loan our credit and have the water-power pay for the bonds as they matured. The question was submitted to the people, and after an exhaustive discussion they approved the plan by the largest majority ever registered on any issue in Illinois or in any State in the Union, a majority of nearly 500,000 (I believe it was 497,345 to be exact). Then we presented it to our Legislature. Now, this is the point. When we presented it to our Legislature, what do you think has happened? Why, nothing happened. (Sensation) We have talked, and talked, and talked, but we haven't acted. We have had several sessions, regular and irregular (laughter), on this subject, general and special, but we have failed to act. After the failure of the regular session to act, on December 14 last I called an extra session to determine the State's

part in this water-power and waterway subject. It adjourned on March 2 following (I want you to keep these dates in mind because they are significant); the Legislature was in a deadlock—I am not blaming the republicans for this, although Illinois is a republican State, and I am not blaming the democrats; the fact is that a band of republicans and a band of democrats joined to repudiate the pledges of both parties, and they did it, effectually did it. They adjourned on March 2; on April 29 following (this year) a little corporation with a huge name was formed in our State—the Illinois Valley Gas, Light & Electric Power Company, I believe is the name—you are nearly compelled to take a vacation to pronounce the name all at once—with a capital stock of only \$1000; a huge name for small capital. Then, on May 12 following—thirteen days later—the organizers of the corporation met, and decided they had made a mistake in capitalizing at \$1000; so they made the capital accord with the dignity and length of the name and increased it from \$1000 to \$6,250,000. Since that time they have acquired fifty-year franchises in the following cities: Joliet, Morris, Seneca, Ottawa, Wilmington, Streator, Dwight, Odell, Gardner, Pontiac, Plainview, Yorkville, Coal City, and Bridgewood. Now that has been doing a good deal of work in a warm, humid atmosphere, such as we have in the summer time in Illinois (laughter). They have not only done that, but they have also acquired the other corporations that have had to do with the developing of water-power in Illinois; and not only that, but they have reached out and acquired certain riparian rights necessary to develop fully the power at Marseilles. Now, what will happen? Our sanitary district of Chicago has already expended \$53,000,000 on this channel, and will expend \$20,000,000 more in its full development, and our State will spend \$20,000,000 on its part. In other words, Illinois will contribute \$100,000,000 to this Lakes-to-Gulf Deep Waterway, and a corporation which has not expended one dollar to create this power comes along and puts a toll-gate across it and collects the toll. Bear in mind that none of this power is created by the surface or drainage water of the State; all of it is created by diverting the waters of Lake Michigan to the Illinois and the Mississippi. What would be thought, for instance, if our State should expend \$100,000,000 in building a road from Chicago to Saint Louis and then some one who had not expended a dollar would throw a toll-gate across it and collect a toll of every person and vehicle that passed, and then when he tried to buy our own road back, charge us \$100,000,000 for it?

That would be going some, even in these days of "frenzied finance," wouldn't it? Yet that is exactly what they are doing with the water-power situation in our State. For several reasons (fancied or otherwise; it doesn't take much of a reason to occasion debate) there is a strong effort being made to prevent the State from acting, and our State is in the situation (and Chicago will be in the same situation soon) where we will be compelled, in order to acquire the riparian rights, to condemn them at their market value, and you can see, from the array of towns I read you, that the market value is steadily increasing (I collected their names about two weeks ago, and had not time this morning to wire inquiring whether it was up to date, but give you the list as an indication).

The point I want to make is that our State is a good deal like other States: we are neither abnormally good nor abnormally bad—just an average. Sometimes we are attending to things in such a way that we would prefer to have no metropolitan newspapers to circulate and mislead us; at other times we do things in a grand style in that great State, and we are then very glad that we have such means of disseminating knowledge about what is being done.

In regard to the Conservation movement: I sympathize very strongly with my good friends here from the West. It has been a delightful pleasure to meet them on a number of occasions, on the waterway trip down the Mississippi from Saint Louis to Memphis, then at New Orleans, and again at Washington, where we were all together at the Conservation Conference in Washington called by Theodore Roosevelt. I believe that the Government should not interfere to prevent the full development of the States. A long time ago it was said that he was a benefactor who made two blades of grass grow where only one had grown before, and the man who can put two acres in cultivation where only one was cultivated before is certainly a friend of mankind. So I think we want all the acres put in cultivation by irrigation or dry farming. But the general Government owns certain things: it owns coal lands, oil lands, gas lands, phosphate lands, and forest lands. We heard the President say this morning that the Government owns about a third of the forests that we must have in the north in order to allow the Mississippi to have enough water. The Government owns about a third of the coal, and if I recall correctly, about a third of the phosphate lands, which will become more and more necessary as we develop our agricultural resources. Now the

Federal Government should not permit itself to be put in a position where these great natural resources could be wasted (great applause); it ought to be in a position to develop the States by irrigation, and in all possible ways, but it should not permit itself to be put in the position where a Legislature of a State would take from it power to control some of the very necessities of advanced civilization (applause). They can have a crop of corn every year, they can turn on and off water-power every year, and the rains will come again; if by lack of attention the forests are burned or removed, they can be grown again; but the great Creator provided there should be just one crop of coal for all time, and provided, so far as we know now, that there would be just a certain amount of phosphate lands, and they are for all time and all men. These crops are not growing in Montana just now, they are not growing in other States; and because they were meant for us all, this great continental Republic ought to be able to conserve them so they shall not be abused. We all have the right to use them now, and the Government, in my judgment, should see that there is no possibility of abuse.

It seems very likely that, so far as water is concerned, the State and the Nation will have to cooperate and work together (applause). The State may own the water in Montana because the streams are not navigable, and I assume this is so in Wyoming and Idaho and the other mountain States. The Government at present owns much of the land. The Federal Government may not say to the State, "You cannot use the water because you cannot get in my backyard," and the State may not say, "Water is valueless without the use of the land that is situated adjoining;" so they will have to work together, and they should work together. That is the way it ought to be, and that is the way it will be; and I believe that we here in the West, and in the East and in the South, who have had our States developed by a vast expenditure of these natural resources and vast waste, will have patience and consideration for the views of these men who are somewhat fearful lest we do not permit them to develop their own resources. I believe the Nation will permit them not only to develop the resources, but will encourage them in that development (applause).

Now, just a word about Illinois: I have told you so many bad things about our State that it is not proper to cease speaking without saying some good things. I was delighted with the statements made by Governor Norris about Montana. It is a proud record. It has set a good example to the Government.

Our State has done something, too (laughter). Our State, a long time ago, before we heard of this Conservation movement, had at least six or eight commissions out doing this very work. We have an agricultural experiment station that has explored every foot of our land, I may say, in a phenomenal way; the fact is we are laying off our State in ten-acre plats, and the University of Illinois is surveying each ten acres and making a record indicating the kind of soil, later to give advice as to the development of each ten acres; and the gentleman under whose supervision that is done is a Delegate to this Congress and likely to address you. He is a specialist on soil. And we have had a geological commission that has taken stock of all of our minerals, and although we are a prairie State we are the third in the Union in our mineral output. We are not only locating and taking stock of our coal but showing how to mine it, how to send it by freight, how to store it, and how to burn it—for nine-tenths of its energy is wasted before you get it to the place where you should apply it. We have made a survey of our rivers, studying the fishery question; Illinois river is the second in its output of food products in the United States, being only exceeded by Columbia river in the remote West; it has more than doubled in the last eight years. We have a commission on floriculture and horticulture; and we have an internal improvement commission that is studying every stream in our State and giving the information to our counties and districts for the purpose of forming drainage districts so that the land may be drained and more of it cultivated. In every department—water, soil, minerals—our State has made a most careful investigation, so that we feel we have a complete stock of our resources; we believe, too, in their development, and we are developing them. All the departments of our State work are going along as they should, and our resources are being well conserved.

I have dwelt on a disagreeable feature only because I believe that the example of Illinois should be beneficial elsewhere. We are having trouble in attending to our public utilities, as other States will. Illinois will have expended a hundred million dollars in the making of a water-course that creates water-power, and you are all familiar with the disgraceful story as to how the State has tried to cope with that water-power monopoly through its Legislature and conserve to us what we created ourselves. It is likely that we shall be compelled to see certain corporations or private individuals sowing where they didn't reap, and levying a toll upon a vast expenditure of

money made by our commonwealth; and other States may profit by our experience. (Applause)

Chairman STUBBS—I am very glad indeed to have the opportunity of introducing Governor Hay, of the great State of Washington (applause).

Governor HAY—Mr Chairman, Ladies and Gentlemen: I desire to take this opportunity to thank the good citizens of Saint Paul for seeing to it that the Western States were given representation at this Congress (applause). It was not, and never was, the intention of the managers of this Conservation Congress to allow those who differed with them in opinion to be heard at this meeting, as I know by long correspondence myself with the management. In reading the numerous papers published here in the East relative to the "wild and woolly western men" and their ideas on Conservation, I said to my wife, before leaving home, "It looks to me that I am going down to Saint Paul to get the most glorious spanking a white man ever got." My wife said, "Go down and take it" (laughter). But since arriving here, I am pleased to say that I have found innumerable people who look upon this Conservation question exactly the same way as do the majority of the people of the Pacific Coast.

All that is needed to solve the problem of conserving our natural resources is common sense and the application of the square deal (applause). It is because of a departure from these two essential elements in the consideration of Conservation, that an unsound, unjust, and impracticable policy has been advanced in this country. Common sense has given place to humbug and fairness to intolerance. Instead of calm, dispassionate, logical discussion of the subject, we hear and read on every hand exaggerated statements, misrepresentation, false accusation, dire prophecy, and passionate appeals to prejudice, avarice, and lawlessness. This has given rise to a wholly perverted notion of true Conservation, and has brought about a condition hurtful to the West, and one that, if persisted in, is bound

to prove injurious to the Nation. The only sane and sensible kind of Conservation is that which permits the fullest and freest development of our natural resources under provisions that will perpetuate those resources that can be renewed, and that will obtain the greatest economic good from those that cannot be replaced. But to many of us of the Pacific Coast and Rocky Mountain States, Conservation, as practiced, means to tie up and not to utilize. It signifies to us the letting of our waters run unfettered to the sea for fear some one might develop their power and turn their energy to the benefit of mankind in this generation. To us it means the locking up of our vast forests that they may go to decay or become the prey of the fire king. It means that, to please some bureaucrat, the people of our section are held up to allow the timber trust to secure a profit of a few extra millions each year. It means that our vast coal areas must go undeveloped, and that we be compelled to spend our money with foreign mine owners for fuel, importing the coal at no small expense for the item of transportation alone. It means that the State of Washington is robbed of the use of 500,000 acres of land that the Federal Government granted to it for educational purposes at the time it was admitted to the Union. Conservation as practiced in the past developed into a vast profit-making scheme for certain southern land grant railroads, which under it were given scrip in place of worthless desert land included in forest reservations, treeless since time began and bound to remain treeless to the end of time. And we have seen this scrip brought north and placed upon our timber lands that will cruise from 5,000,000 to 50,000,000 feet per section, and are worth from \$20 to \$100 per acre. This brand of Conservation means to us that 27 $\frac{1}{4}$ % of the total area of the State of Washington paid a paltry \$16,000 into the public coffers in 1909. It means we are called upon to expend large sums each year for policing these Federal reserves, which contribute practically nothing to the cost of State government, while at each session our State Legislature is compelled to appropriate large sums to build roads through Federal reserves. Last year we appropriated \$205,000 for this purpose. To us, Conservation means that settlers within forest reserves who have taken up homesteads in good faith are harassed, browbeaten, and often forced to abandon their claims and lose the fruits of the labor of years. As an illustration of this, permit me to read a letter I received recently from a fellow citizen of mine who, by the way, is a prominent logger, and while a very wealthy man and a large timber owner, is one of that kind of men who came up from the bottom; he started in at

day's wages in the State of Washington a little over thirty years ago. This is what he says:

Speaking for myself and from a selfish standpoint, the present Conservation by our National Government suits me fine, but in the interests of the poor settlers who make our country, a change should be made. Four-fifths of these settlers come out here from Eastern States and endeavor to take up homesteads, but they are so harassed and driven from their homesteads through technicalities and forest rangers under orders that are absolutely foreign to the best interests of our country and the settler, that instead of making good citizens the Conservation laws have made anarchists, and if the thing is kept up, everything that will burn I expect to see burned within the next ten years. You cannot drive a man from his home, with a wife and from one to six children, penniless and hungry and the children in rags, while the land that would support them lies idle and wild just to gratify the theory of some man who may be honest but who is ignorant of the conditions of the frontier. I will name a case of a man I met in Aberdeen, who told me that he tramped forty miles three times to make proof on his claim. He had lived with his family on his homestead for seven years and endeavored to make proof, coming out with witnesses and spending money he needed for his family, only to be told the last time he came out that his hearing was indefinitely postponed. This man came out a good, loyal, American citizen; went back a fire-eater. I know another case on the head of Nooksack river where a man endeavored to take up a homestead on meadow land, and after he made application it was set aside for forest rangers' quarters No. 1. He then tried to take a second homestead and it was set aside for Forest Ranger No. 2; he then endeavored to take a third, and that was set aside for Forest Ranger No. 3. The land is fertile beyond description, but there is nothing living on it, and it is supporting no one.

On the head waters of Skagit river there are tracts of land that will support from three to four hundred homesteads. This is purely meadow land with brush and worthless scrub timber, like all our western Washington meadows. Any five acres of this land will sustain a family in comfort. This land is held in the forest reserve, absolutely worthless

so far as sustaining people is concerned, or paying taxes to the State. If our State is to give up one-third of its taxable property and carry on its government with two-thirds, she has very little interest, if any, in that portion of the State reserved by Conservation, and naturally will not aid in the preservation of the same as she would were the revenue from these resources to become the revenue of the State. Up on Quinault river, ten years ago, there was a flourishing settlement with every prospect for opening up the country. Since this Conservation law has been in force, many of these settlers have left their homesteads, others have been driven off and gone to British Columbia. The United States Government does not build a road into the settlement, and the people are too poor to build out. Take it up in the Northern Peninsula (the greater portion covered by forest reserve), the land would sustain hundreds of thousands of comfortable and independent homes; but today it is a howling wilderness, and the meadow land is as wild as it was a hundred years ago. The people are too poor to build roads in and across the forest reserve, and the Government does not.

I sincerely hope and trust that the people of the East who are not acquainted with the conditions in the State of Washington will permit this State to control and conduct her own Conservation, both water, timber, coal and oil, if necessary, to the best interests of the State and Nation. We have a State that has upwards of ten million horse-power in our waterfalls going to waste every minute. With proper State laws this could be utilized, and so protected that monopoly could not control it. We have millions of tons of cheap anthracite and bituminous coal on our coast. Still, the people of Alaska are buying British Columbia coal and shipping it up to themselves two thousand miles, while the coal is sticking out of the mountain-sides of Alaska and cannot be touched. We are shipping hundreds of thousands of tons of Maryland coal to our navy on the Pacific Coast, in foreign ships, while we, of the State of Washington, are prohibited from shipping our cheap lumber to our own people on the Atlantic Coast, and are compelled, if we ship at all, to ship it by rail to New York and the thickly settled portions of the East at a freight rate that is prohibitive. The only people receiving the benefit of our lower grades of lumber and cheap prices are the Chinese and Japanese. If we were permitted to ship our lumber in foreign

vessels from Washington to New York or other ports on the Atlantic Coast, we could give them lumber that they all need and that we would be glad to sell at a very reasonable figure. It is the fool laws that are oppressing the people, both of the East and the West, and many of them have been made in the interest of monopoly and many through ignorance.

The West is not here to fight Conservation, for, properly directed, it is one of the greatest movements inaugurated in this country since the abolishment of slavery. Our former President instituted many reform movements that, properly directed, mean happiness and prosperity for our people; and of all the movements started by him, in my opinion none means more to the financial welfare of ourselves and our children than Conservation, as vouched for by President Roosevelt (applause). The complaint we have is not against the principle of Conservation, but against the prostitution of that great movement to the impractical ends of certain men out of sympathy with our institutions. They would disregard the rights of the people of the Western States to regulate affairs within their borders; they would retard development of the younger States; they would compel the citizens of the Western States to contribute annually large sums of money to the timber, coal and power companies operating in those sections. While these bureaucrats claim to be working in the interest of the people, they could not better serve the Special Interests if they were employed by them. In the past they laid unusual burdens upon the Western States, and have ruthlessly crushed and brushed aside the honest homesteader who did not have funds to fight or carry his case to the highest court. They are attempting to bottle up and make useless the natural resources of our Western States, and have our local affairs administered through an irresponsible bureau located 3,000 miles away. All the people of the West ask is a chance with the older communities and an honest shuffle—a square deal above the table—and a show to develop our resources and build up prosperous communities made up of innumerable happy homes. I believe the people of the West are as good citizens, and are just as true and loyal to the interests of the Nation as are the citizens of any other locality. As States we do not like to be looked upon as provinces or colonial possessions to be exploited for the benefit of the other sections of this Nation. I have faith enough in the fairness of the citizens of the other sections of this Nation to believe that they do not covet

or desire to rob us of what rightfully belongs to us. We believe the profit arising from the development or exploitation of the natural resources of each State should be applied to the benefit of and to the cost of government of that State.

Let me get this fact set in your minds: 95½% of the national reserves are located within the eleven Pacific Coast and Rocky Mountain States, and 27¼% of the total area of the State I have the honor to represent is taken up by forest reserves, an area in which could be placed the States of Maryland, Rhode Island, Delaware, Connecticut, and the District of Columbia, with room enough to spare to accommodate another Rhode Island. The extreme Conservationist argues that the people of the Western States are not competent or qualified to manage the natural resources within their borders and that a guardian in the shape of a Federal bureau should be appointed to handle them for us. This is a gratuitous insult to the intelligence and integrity of the people of the West. Almost the worst kind of government that can be placed upon a people is a bureaucracy. Let me call your attention to the fact that practically all of the land, mineral, coal, timber, and power-site steals perpetrated upon the people were made when these titles were vested in the Federal Government.

Now, let us deal a little with common-sense Conservation: The people of the State of Washington started a practical system of Conservation long before Conservation became a national issue. The Governor of Montana has said that Montana was the first State in the Union to practice Conservation. Evidently the Governor of Montana is not up on the laws of the State of Washington or he wouldn't have made that statement (laughter). One of the great natural assets of our State is our fisheries. Because of over-fishing it became evident to our people some years ago that, unless proper steps were taken, our fishing industry would be ruined. Laws were passed regulating the taking of fish, and numerous hatcheries were established throughout the State. We are now putting more salmon fry into salt water than is the Federal Government, and today the State of Washington stands first in the Union in the value of the products of its fisheries, all because our people a few years ago started a practical system of Conservation. The expense of enforcing our laws regulating fisheries and the cost of maintaining and operating hatcheries is assessed against that industry. We cannot bring ourselves to consent to turn over the management of this industry to the

Federal Government. In fact, so opposed are the fisher-folk of Puget Sound to Federal control of the fishing industry, which is threatened because of the proposed treaty with Great Britain, that they are fighting the ratification of the treaty by the United States Senate.

Let us now take up the question of the national forest reserves as administered in the western States. I doubt that there is a thinking man who does not love the trees, the deep woods and vast forests of our land; but a tree, like everything else that grows, has its youth, its maturity, its old age and death. A tree not used at maturity decays, falls, and becomes a fire-trap and is a serious menace to standing timber. I believe that when a tree reaches its maturity it should be used and not allowed to go to decay (applause). Failure to make use of our natural resources which are going to waste is the antithesis of Conservation. I believe that all non-forested lands adapted for agricultural purposes should be opened to settlement and homesteaders allowed to file upon them. Within the national forest reserves are vast areas with not a stick of timber on them, and on which timber can never be made to grow profitably. These tracts should be thrown open to settlement. It is people we want in the West, not game preserves (applause); it is happy, prosperous communities, not idle wastes. I would not advise the acceptance of homestead filings upon timbered areas until after the timber is removed and it is found the land is suitable for agriculture. If it is valuable only for timber raising, then the land should be turned over to the State for reforestation. It is the duty of the State to all the States to start a system of reforestation. At the last session of our Legislature, an appropriation was made to start a survey and have maps made showing the areas of our State better adapted for timber-growing than for any other purpose. This work is now well under way. A commission composed of twelve of our leading citizens, interested in forestry, have been appointed to draft a forestry bill to be submitted to the coming Legislature, when, without doubt, the State will start in upon a plan of reforestation; something which every State of the Union should take up. It is the duty of the States to attend to the growing of forests within their borders, and not the duty of the Federal Government. I am not in favor of abolishing the Federal forestry department. This department should stand in the same relation to the State forests as the Department of Agriculture stands to the farming interests of the Nation (applause). We would hardly expect Secretary Wilson to go

around the country, preparing the ground, planting and harvesting our crops, and collecting the revenue therefrom, and we do not expect the Federal Government to go inside of the State and start a system of reforestation where it is absolutely the duty of the State itself to undertake that work (applause).

The greatest infringement upon the rights of the State to handle their own internal affairs is the attempt on the part of the Federal Government to gain control by indirection of our water-power for the purpose of supervising and deriving the revenue from any possible development of the powers. This, by the way, is a policy particularly waged by the National Conservation Association, an organization which is making of this Conservation question a cult, which has practically set up a dogma, and whose members are now quarreling over their claims to orthodoxy. So far about all it has done has been to play into the hands of the power monopoly, which the first apostles of Conservation claim to fear so greatly.

Of all the lame arguments I have heard, the one that the people of the country have not the brains or authority to regulate the charges of any public service corporation, is the worst. We have two means of reaching them: by regulating the rates, and by taxation. No State in the Union was probably ever more troubled than was the State of Washington a few years ago with a railway lobby. In the year 1905 the Legislature of the State of Washington passed a railway commission law, and placed the regulation and control of railroads under this commission. Three years this commission studied the conditions in the State. It was one of the first States in the Union to make a physical valuation to determine the cost of these plants. In 1909 the railway commission of Washington placed an order into effect that saved to the farmers of the State, in the hauling of wheat and other grains alone, \$750,000. At the same time they placed an order reducing the general distance tariffs of the railroads, which cost the railroads of the State \$75,000, and the railroads have never appealed from its decision and those rates are in effect today. In 1909 the railway commission traveled over every mile of road in our State, visited every station, held hearings, and as a result of that trip they made 250 orders ordering new stations, enlargement of waiting-rooms and train facilities; all those things that the people complained about they remedied, and of the 250 orders put into effect—which cost the railroads hundreds of thousands of

dollars—they never have appealed from but 16, and 234 have gone into effect; so the argument that the States cannot control affairs within their own borders, it seems to me, is very fallacious (applause). If we are not competent to handle affairs within our own borders, if we are not competent to regulate corporations, then let us surrender our Constitution and go back to territorial days and let the Federal Government administer our affairs for us. (Applause)

Now, with reference to the water-power bill: The bill before Congress introduced by Senator Smoot, of Utah, and a similar bill introduced by Senator Jones, of Washington, are perfectly satisfactory to the people of the Coast, so far as I know. Governor Norris has explained to you that the beds and banks of all streams, up to the limit of medium high tide and medium high water, belong to the States; they do not belong to the Federal Government. That property is just as much ours as is the jack-knife in our pockets. Senator Smoot's bill provides that all the interest the Federal Government has in this is that it owns the sites. We own the water, we own the power. There is no question about that. The Supreme Court has passed upon it time and time again. The Government owns the sites. The Smoot bill provides that the sites in the Federal reserves shall be turned over to the State government, but that in no instance shall the State pass the fee-simple title to the land, and no lease shall be longer than fifty years. This is perfectly satisfactory, and the people of the State of Washington have no objections to that form of relinquishment to the State.

The high-handed manner in which a Federal bureau attempted to hold up the development of the western States was the result of a false conception of the principles upon which the Government is founded, and a dangerous assumption that honor and efficiency existed nowhere but in one self-appointed guide, philosopher, and so-called friend of the people. I believe it is the intention of those now in authority to administer the natural resources of the West according to law and with some respect for the welfare of the State in which the resources are located. But outside of governmental and administrative circles, an element composed of faddists, dreamers, and enthusiasts is striving to bend popular sentiment to certain impractical and unfair policies of applying Conservation, and it is against this element that the West has taken arms. We want Conservation that benefits all the people, not a Conservation that plays into the hands of a few. Conservation that

does not make use of resources rapidly going to waste is Conservation gone daffy. I have noticed that there are some States down here shouting loud for Federal control of our natural resources. I want to say that those Governors who are here shouting the loudest for Federal control are from the States that have the least amount of natural resources. It is the desire of these people that the revenue received from these natural resources shall be surrendered to the Federal treasury. That is what the western States certainly object to. Some people and papers here are charging that "the interests," whatever you may call them, are favoring State control of the natural resources. I want to say to you that "the interests" are always against local control in any case, and always prefer that monopoly of all kinds shall be placed in the Federal Government and as far away from the people as it is possible to get it.

The address made here by President Taft this morning is in line with the western idea of Conservation as I understand it, and I believe those of us from the West who look at this question as I do endorse the same safe statement that has been made by our great President (applause). Let western men, using up-to-date western methods and familiar with western conditions, deal with and manage western matters. I thank you. (Applause)

Chairman STUBBS—Professor Condra will make an announcement before I introduce the next speaker.

Professor CONDRA—Mr Chairman, and Ladies and Gentlemen: You know that we have State Conservation Commissions and associations representing various States. We have recently perfected an organization of these with a view to cooperation among States and with the Federal departments. The Federal representatives forming our national committees have thought it better not to issue any suggestions to the State delegations, preferring to leave this duty to the committee of the interstate organization, of which I have the honor to be Chairman, as the more democratic method.

We propose that the chairman of each State Conservation Commission or Association call his State delegation together at some stated time and place (in the absence of the chairman the secretary or some other commissioner may act) to organize the delegation and select representatives to serve on the resolutions committee and any other committees, to the end that we may have fair discussion and full representation of all our States.

Chairman STUBBS—I now take pleasure in introducing Governor Brooks, of Wyoming. (Applause)

Governor BROOKS—Mr Chairman, Ladies and Gentlemen: It has been my good fortune to visit nearly every State in this great Union, and to spend considerable time in nearly all the larger cities, though, strange to say, this is the first time I have ever visited this particular spot; and yesterday, while enjoying a beautiful ride through the Twin Cities and around the great parks and other resorts, I felt that my education had been sadly neglected (applause). This is certainly one of the garden spots of the Union, and I think the people here showed the proper spirit when their Governor in his address this morning stated that a State convention on Conservation had been held, at which the attendance numbered some 7,000 people, to consider the proper conservation of the soil and to bring about increased production of the farms. I know that the State of Minnesota is on the right track—that is the important thing, after all. (Applause)

A few days ago the western Governors held a meeting at Salt Lake City, and spent two days discussing this question of Conservation. After full and complete discussion they adopted, unanimously, a brief set of resolutions, which I think express their views in this important matter. Colorado, Utah, California, Washington, Oregon, Idaho, Montana, and Wyoming were represented; and since the resolutions, which have been published in all the western papers, have met with unqualified public endorsement, and as it will only take me about a minute, I am going to read them, as embodying

the views of the western Governors—and, I might add, of 95 percent of the citizens of the great western States:

Resolved, that the Governors of the Rocky Mountain and Pacific Coast States affirm as a platform of principles to be urged upon the National Conservation Congress to be held at Saint Paul, September 5-9, 1910

First, that in legislatively solving the problem of Conservation the National Congress adhere to the doctrine of Abraham Lincoln that the public lands are an impermanent national possession, held in trust for the maturing States.

Right on that point, I wish to refer to the splendid paper read here at the opening of this afternoon's session by that brilliant, honest, and patriotic statesman, Senator Nelson (applause), outlining the public land laws. I call your attention to the fact that at the beginning of this great Nation of ours the Federal Government acquired, by cession from the States, by treaties with the Indians, and by purchase and conquest, all this vast public-land territory, the early idea being that this public domain was to be sold for the payment of the Revolutionary War debt and for the running expenses of the Government; though that early idea was quickly transformed and changed, owing to the insistent demand of the settlers, and the pre-emption laws (with which you are all familiar) followed as the second step. They were a sort of settlement and revenue measure combined; but still the insistent demand of the settlers would not stop, and gradually we reached that stage where the homestead law was passed, and signed by Abraham Lincoln in 1862, giving the settlers 160 acres of land as the result of settlement and cultivation, doing away entirely with the old revenue idea; and under that one law this great State of Minnesota, and every other State in this central country, has developed to a degree unparalleled in the history of human progress (applause). Now, all the West asks is an even break; all the West asks is an equal opportunity. How can we educate our children, how can we maintain good government and good law, how can we do all those necessary and essential things to maintain a high state of civilization and progress, if over one-half of the State is to be held permanently as a Federal resource, giving no taxation or revenue whatever to the support of our State governments? (Applause) It is utterly impossible. We of the West are just as bitterly opposed to monopoly, just as bitterly opposed to any misuse of the

natural resources of this country as any of you gentlemen here assembled (applause); but we do believe that the States themselves can in a great measure work out the safest and best conservation. I might get started here and go on talking, and I do not want to do it; I want to read the other resolutions:

Second, that State government, no less beneficently than National Government, is capable of devising and administering laws for the conservation of public property; and that the National and State governments should legislatively coordinate to the end that within a reasonable period of time the State governments be conceded full and complete administration of such Conservation laws as may be found adaptable to the varying conditions of the several States.

The idea being that conditions vary so tremendously—just as you have heard from the Governor of Mississippi and the Governor of Illinois, the latter of whom told you about a monopoly stepping in and stopping the State development of the water-power along one of their streams. Such a condition is absolutely impossible in the West, because that old law of riparian rights does not apply; there is no law in the West whereby we are compelled to allow the water in the streams to flow by your property undiminished in quantity and undefiled in quality. In the West the law of appropriation applies, the law of use. Under the Constitution of Wyoming, granting twenty years ago, we were given all the water of the State, everywhere and every place; we cannot part title with it, we hold it, and we will always hold it. Talk about monopoly! How absolutely impossible, under the laws of Wyoming! We have used this water wisely and well. I picked out of a paper this afternoon a certificate of appropriation for power granted in 1900, ten years ago: "Whereas, F. V. Andrews has presented to the Board of Control of the State of Wyoming proof of the appropriation of water from Sand creek, tributary to the Redwater territory, for enlargement of Beulah flouring mill ditch, under permit 517 (enlargement for power and milling purposes), now, know ye, that the Board of Control under the provisions of Division 1, Title 9, Chapters 10 and 14 of the Revised Statutes of Wyoming, 1899, has, by an order duly made and entered on the 28th day of December, 1909, in order record No. 4, page 287, determined and established the priority and amount of such appropriation as follows: name

of the proprietor, F. V. Andrews, postoffice, Beulah, Wyoming; amount of appropriation, 145 cubic feet of water, date of appropriation, April 6, 1900. Said ditch so located, the right to use water herein defined, shall not at any time exceed the volume of 145 cubic feet per second, and the right shall at all times be subject to any future regulation and restriction that may be placed on the same by the Legislature of the State of Wyoming." (Applause) It is absolutely impossible to get a monopoly of water-power in the State of Wyoming, and such an instance as referred to by the Governor of Illinois would be impossible. The State of Wyoming could simply refuse to allow that company to use one drop of water; they have the power to do it, it is so provided for in the Constitution, just as the State of Wyoming, if it chose, could absolutely refuse to permit the general Government itself to use one drop of water for power purposes. We have never had any power monopoly in the State of Wyoming, and we do not intend to have.

Third, that experience of the Conservation States demonstrates that dispositions of public property made under existing National Conservation laws and regulations have tended to intrench monopolies and interests menacing the common welfare; and that modifications of such laws and regulations should be promoted by the Conservation Congress.

Our great President this morning stated a great truth, and it came right to the hearts of the western people. You can't understand it here, perhaps, but we realize the importance of Conservation; but we have been talked to death on it. *What we want is action!* We want the people to get busy; we do not want all these things bottled up in cold storage; we want them used for the generation of today. That is the important thing. As it is now in Wyoming, every big coal company in the State is adding an increased price to its coal to the consumer, who is already burdened beyond the point of endurance, simply because there is no further development in these coal lands as they stand today under the withdrawals; every ranchman in the State of Wyoming is paying ten dollars a thousand more for his lumber than he had to a few years ago—ten years ago, five years ago—owing to the fact that development has ceased. The only monopolies that we are troubled with out there are those that are unable to appraise their capital at present simply

because competition cannot come up and meet them on the markets under present conditions.

Fourth, that the elimination from the forest reserves of all homestead and untimbered grazing lands is immediately expedient.

Fifth, the use and control of all water-power inheres of right in the States, within restrictions insuring perpetual freedom from monopoly.

Sixth, that the privilege of American citizens to seek and develop mineral wealth wherever it may be found should be fully amplified and secured by laws.

Seventh, that the idea of deriving Federal revenue from the physical resources of the States is repugnant to that adjustment of constitutional powers which guarantee the perpetuity of the Union. (Applause)

And with only one thought more I leave you: If the western States, never having had the opportunity so far to develop their great natural resources as you people of the East have, as Minnesota and the Atlantic States have, are now to be changed entirely from the time-honored policy that has made these States great and powerful; if now we are to be taxed, as we have been, \$150,000 a year for the forest-reserve grazing privileges, when that same money is used in the great Empire State for forest protection free of cost, then we of the West have a hard row to hoe. We simply ask the same fair treatment as accorded every central and eastern State of the Union. It is not right to tax the West for anything which you would not apply in one of the great eastern States. We want our resources protected, we want them safeguarded for our children and our children's children, but we want the opportunity to make our young States grow and be prosperous, so that we of the West will have those things of which we can be as proud as you people of Minnesota are when you take a gentleman to your magnificent State Capitol, to your great Agricultural College, and to your other great schools—we want the same for our children and our children's children, without Federal interference. (Applause)

Chairman STUBBS—I want to say a word here about a suggestion made by the Montana Governor. I would like to ask Governor Norris if it is not a fact that the Federal Government has led in irrigation in Montana?

Governor NORRIS—Has led?

Chairman STUBBS—Yes sir. Haven't they done a great deal of work to develop your irrigation projects?

Governor NORRIS—For the last three or four years, yes.

Chairman STUBBS—Well, it is within the last three or four years that this Conservation idea has been spreading out, taking root, and going out from Washington; they didn't get started until Theodore Roosevelt got hold of it (applause). As to the Federal Government undertaking to dominate the West and discriminate against the West, I don't believe that it is in the heart or mind of Gifford Pinchot or Theodore Roosevelt or anybody else to do that (applause); but Gifford Pinchot has stood like a rock and fought like a tiger to keep the thieves out of the Alaska coal fields (applause), and you ought to build a monument to his memory for keeping the Cunningham claims off the statute books and from legalizing by Congress, for it would have been an everlasting disgrace to the American Nation to have millions and billions of tons of coal stolen there. What did President Taft say this morning? He said, "We believe in leasing those lands out there in Montana and in Wyoming and all over this country." He does not believe in selling those things; he doesn't believe in turning them over to the State, either. He said as much here this morning (applause). He says, "Lease them for the benefit of the people they belong to."

I tell you this Conservation idea, when it is put on the right sort of basis, is the biggest thing that we have struck in a financial way in a long while; and I tell you right now (I do not know how it happens, but it is a matter of fact) I do know that the great syndicates and the great corporations that want to gobble up all these coal lands and control these power sites, every bloody one of them, want State control. (Applause, and cries of "Right, Right!") And the reason they want State control is because the meshes are too small in the national net; the Federal Government has given them genuine supervision and genuine control of national resources, and I thank God for it, too (applause). I want it to keep coming right along. I would not stand for

one minute to see the West discriminated against; I do not believe in taxing Montana or Wyoming for anything that you would not tax New York or Pennsylvania for; neither does Theodore Roosevelt, for he grew up out in that country and he is one of them and his whole heart is with them; he wouldn't see one iota of discrimination, and nobody else would; but I say to you that it is the great electric power organizations and combinations—it centers down to four or five or six fellows—that are trying to monopolize all the power sites in the United States! That's what's the matter now; and those fellows think if they could get the whole thing in the hands of State legislatures they could dicker and trade with them (applause and cheers). They know they cannot do it at Washington. That is all there is to this whole problem; and I say to you today that the American people ought to build a monument to Theodore Roosevelt and Gifford Pinchot for the work they have done in this line (great applause), to say nothing about the other great work that has been done. I would like to see those Alaska coal thieves sent to jail (laughter and applause), and for my part I do not take any stock in the Ballinger idea of running things up there, either (tremendous applause). If I were President of the United States, I'd kick Ballinger out of that Cabinet in five minutes, that's what I'd do. (Great and enthusiastic applause) We might as well tell the truth about it, too. I say to you that this work has started, and it has started along broad, decent, National lines; the States have plenty to do right now if they will attend to business; they have seventy-five percent of the forests now in private hands with only about twenty-five percent under Federal control, and two-thirds of all the great coal interests of this country in private hands with only one-third vested in the Federal Government; I'd like to see the Federal Government look out for these power sites, and when the contract is made, let it be made in such a way as they can control it. Taft made some good suggestions this morning, and I want to give him credit for it (laughter and applause).

I did not mean to make a speech; I meant to introduce Governor Vessey. (Laughter and applause, and cries of "Go on, go on") We have great men here that are ready to talk, and I must close in a few minutes. Governor Vessey, of South Dakota. (Applause)

Governor VESSEY—Mr Chairman, Ladies and Gentlemen: You can readily see by the color of that man's hair (indicating Governor Stubbs) that he wears the Kansas emblem on his head (laughter and applause) and is not afraid to say something.

Now, in regard to Conservation, I am a good deal like John was the afternoon he was out riding with Mary. For some reason or other he wanted to know whether Mary thought enough of him to marry him, and yet he wasn't quite ready to make her his wife. But he put the question anyway, and she immediately accepted him. They rode along for some distance in silence. Finally she asked, "John, why don't you say something?" He replied, "There's been too much said already!" (Great laughter and applause) And there have been lots of good things said today.

South Dakota is in a peculiar position. It is not in the southern part of the United States, neither is it in the extreme northwestern part; it doesn't even join Kansas (laughter), though it has *some* of the same kind of spirit (applause). The eastern part of South Dakota is a strip of country two hundred miles square, and there is no richer, no more uniform, no better farming land in the United States than that part of South Dakota; the western part of the State goes into the foothills of the Rocky Mountains. In this western part is a great forest reserve; and I want to say I believe that in the State of South Dakota the National Government is doing the best work in preserving the natural forest done anywhere in the United States. Still you find in the western part of our State a great deal of the same spirit that you find in Wyoming, Montana, and Washington. Why? Because of local interests. You see this is largely a local question; and what suits Kansas or Mississippi, somehow or other does not suit Wyoming. It is like the tariff question; and it will probably never be settled until it is settled by an expert commission which will deal with the matter as a whole. (Applause)

I believe largely—very largely, indeed—in State rights. I believe the State should control and own the water-power of streams that are not navigable and that it should be within its province to provide that the waters should first be used for the soil and secondarily be used for furnishing water power to turn the wheels of industry and thereby make the State richer. For we

must admit—just as your great Governor of Minnesota has said—the first duty of the people of the United States is to preserve the soil (applause), because the crop that comes annually from the soil yields the greatest revenue that the United States will ever have; and we must have it, and must have it increased if we expect to support the increasing population of the United States at a reasonable cost so that they can work at reasonable wages and support homes—possibly not of luxury, but of all the comforts that citizens are entitled to.

I appreciate the position that has been taken in the conservation of coal; I appreciate the conservation of timber, of phosphate lands, of oil, and of gas; but I want to say that the same conditions that have been referred to upon this platform with reference to the disposing of power from water-power plants at the lowest minimum cost should apply in the same way to these other natural resources—yet you will notice that in the report of the National Forester it is shown that we have been selling stumpage at market prices. They propose to sell the coal and the gas and the oil, and possibly the phosphate, at market prices. If that is true, it is not real Conservation in the interest of the consumer; because if we only own one-third of the coal and the private individuals who own two-thirds fix the prices, and if the Government follows them in fixing the prices, where does the consumer derive any benefit (applause). The same rule should apply to timber. I can show you, in our own State, where there are parts of the national forests that are ripe and should be cut into lumber, and that lumber should be building homes on our broad prairies. But the price the Government has fixed on the stumpage is too great for mill-men to buy it and manufacture it and sell it, even at the high price of lumber out in that country. Now, who is suffering? The men that are endeavoring to build homes on that prairie. I think we ought to be intelligent on those things. I think we ought to use the timber, and we ought to use the coal, and we ought to use the phosphates, in the upbuilding of this country, and give it to the consumers, if possible, at a price at which they can use it, and not at a price that may be set by the large combinations or trusts that control these products. I thank you. (Applause)

Chairman STUBBS—We were expected to get through here at 5 oclock and it is now ten minutes after 6. I regret that there is not time to allow a dozen or fifteen mighty fine men to continue this discussion. The session is adjourned.

THIRD SESSION

The Congress convened in the Auditorium, Saint Paul, on the morning of September 6, 1910, and was called to order by President Baker.

President BAKER—Ladies and Gentlemen. We have a few minutes before our honored guest Colonel Roosevelt arrives. We shall occupy that time in routine business. At Seattle, where this Congress was formed, the organization was left to an Executive Committee and a Board of Directors. They are now prepared to submit a report; but the first and most important question relates to credentials, on which the Congress at large may properly act.

A DELEGATE—Mr Chairman, I move that the Chair be authorized to appoint a committee of five on credentials.

President BAKER—Gentlemen, you have heard the motion. Is it seconded? (The motion was seconded) If there is no discussion, the motion will be put. All those in favor of the motion will signify their pleasure by saying aye.

A VOICE—What is the question?

President BAKER—The motion is that the Chair be authorized to appoint a committee of five on credentials. All in favor will say aye. Contrary nay. It is a unanimous vote.

The Chair will appoint on that committee Edward Hines, of Chicago, chairman (and will ask him to call his committee together as soon as possible); George K. Smith, of Saint Louis, R. W. Douglas, of Seattle, Charles H. Pack, of Cleveland, Lynn R. Meekins, of Baltimore.

The next important business will be consideration of a Constitution and By-Laws, which Professor Condra will read.

Professor CONDRA—Mr Chairman, Ladies and Gentlemen: I am asked to read the draft of a constitution that you may know that it comes from the State organizations. Your various State committeemen met and adopted the

draft submitted to us by the Executive Committee; therefore the proposed Constitution has the approval of two bodies, one State and one National.

(Professor Condra proceeded with the reading of the Constitution as submitted; after reaching Article VI—)

A DELEGATE—Mr President, as the time is late, and as the Executive Committee have passed upon Constitution and it has been approved by the representatives of the States in the form presented, I move that the further reading be suspended and that the Constitution be adopted. (Applause)

President BAKER—Is the motion seconded? (Several voices seconded the motion) All in favor will say aye; contrary nay. Carried without dissenting voice. (Applause)

Some announcements will now be made by the gentleman from Nebraska.

Professor CONDRA—Ladies and Gentlemen: In order that there may be proper representation of the various delegations in the Committee on Resolutions, it is again urged that all members of each delegation meet and select their representatives. If chairmen of delegations will give us the place and time of meeting we will gladly announce it from this platform. Thus far we have not heard of time and place for meeting of delegations from New Hampshire, North Carolina, Tennessee, Kentucky, Ohio, Minnesota, Kansas, Montana, Wyoming, Utah, or Nevada.

[Several announcements of meetings of delegations were here made.]

President BAKER—We will now listen to an address from Honorable John Barrett, a man known around the world as the Director of the Bureau of American Republics. (Applause)

Mr BARRETT—Ladies and Gentlemen: If I had the fascinating capacity of Governor Stubbs, of Kansas (applause), I might be able to do justice to this

occasion; but I have been sitting in yonder corner, behind three noble Governors each ready to speak, beside the representative of the British government—which today is watching with great interest this gathering—not expecting for a moment that I would be called upon today; and it is only that I may be true to my New England birth and my western training that I rise in response to the suggestions of your Chairman. (Applause) If any reason renders it at all fitting that I should say a word, it is because perhaps I have the honor of representing here today some twenty nations as showing their interest in this great Conservation movement which is sweeping over the wide world (applause). I want to tell you that as this movement grows, under the splendid leadership of the men who are blazing the way, it will become the policy of every American country from Alaska and Canada on the north to Argentina and Chile on the south (applause). We shall hear not only from the United States but from our sister nations of Mexico, Brazil, Argentina, and Chile in this effort to make the world realize that if we are to provide for ourselves and for all men who are to come, we must be minute-men—the minute-men of the present day.

Ladies and Gentlemen, all the world is listening to what was said yesterday, on this platform, and all the world will listen, even more earnestly, to what is said today (applause and cheers); and these two great pronunciamientos on Conservation will be read in every corner of the globe, and you and I will be proud that we have participated in this great movement. (Applause)

[Numerous calls were made for Governor Stubbs.]

Governor STUBBS—Mr Chairman, Ladies and Gentlemen: It gives me great pleasure to be here this morning in anticipation of hearing a great speech from the greatest American and the greatest citizen of the world. (Vociferous applause) I am proud of our country; I am proud of her achievements; I am proud of the great State of Kansas, the greatest State in America (great applause), and I am proud to tell you that we won't meet in a bar-room today (laughter and applause), and that we do not have bar-rooms to meet in down in Kansas (great applause and cheers); and I want to tell you that in Kansas the idea of letting men spend their money for shoes and

clothes and schools and homes has proved a blooming success (laughter and applause and cheers) as compared with the fellow who works by the week and makes ten or twenty or forty dollars and spends it in a saloon Saturday night. (Renewed applause)

You have come here today to consider one of the great problems of the age and you will hear from a master mind, from the great leader of this movement, the policies and the plans and the propositions by which the work will be carried forward. I do not propose to take up your valuable time this morning in any discussion of a question of such splendid proportions that I would not have time to get started nor time in which to stop. (Applause)

Ex-President Roosevelt here entered the hall amid cheers and rousing enthusiasm and mounted the platform.

President BAKER (when silence was restored)—Reverend Doctor J. S. Montgomery, Pastor of Fowler Methodist Episcopal Church, Minneapolis, will now offer an invocation.

INVOCATION

Almighty God, Father of our Lord and Savior Jesus Christ, Thou art the source of all mercy, love, and blessing. Lift upon us all the light of Thy holy countenance.

From the beginning Thou hast never been without a witness in the world, and Thou hast never left us comfortless. Give unto us, O God, the Source of all wisdom, a great measure of Thy wisdom, truth, and blessing. We recognize in Thee the source of every good and perfect thing in all the world. Thou hast opened up this new great world; and on this auspicious occasion, look Thou upon us in mercy. Bless our great land. Grant that every source of material blessing may be conserved to serve all the people;

grant that our citizenship may be blessed and directed from border to border. Remember our country; remember the great Southland, the great Northland; bless the great East and the great West; and may all of our people everywhere have bread enough and to spare, and may we recognize that our supremest duty is not to build up institutions fit for man but to build up man fit for institutions.

Bless Thou the Governors of all the States. Remember our great Government, its legislative, its judicial and its executive branches.

Remember in mercy the President of these United States; and bless Thou our most distinguished guest and most conspicuous citizen in all the world, who is with us this day. Look upon him in mercy, guide him and direct him in wisdom, and grant that no peril may come nigh him.

Bless Thou our flag; may it float on until all nations see the blessings of our great Republic; may it float on until all selfishness dies out of the world's heart; may it float on until all ignorance shall be gone; may it float on until the nations of the earth shall be united in a brotherhood around and about which are wreathed the blessings and the wisdom of Thy holy and undying self.

Be Thou in the deliberations of this great body; grant that wisdom and truth may be uppermost in the minds of all who are here. Accept Thou our gratitude for thy abiding mercy, and at the last, O Lord, gather us all into the haven of eternal rest. Through Jesus Christ, our Lord, we ask it. Amen.

President BAKER—Ladies and Gentlemen: It is now my pleasure to present that citizen of our country who in three continents has evoked the greatest enthusiasm, and who has done for this country no greater service than in forwarding and extending the work of Conservation to protect the natural resources and in carrying out the principles of fair dealing between man and man; our most honored citizen, Colonel Theodore Roosevelt. (Great applause and cheers for many minutes)

ADDRESS BY THEODORE ROOSEVELT

Mr Chairman, and Governor; Governors, and fellow-guests; Men and Women of Minnesota: It is a very great pleasure to me to be here in Minnesota again, and especially to come here to speak on this particular subject of "National Efficiency." (Applause)

Minnesota is one of the States that almost always takes the lead in any great work (applause), and Minnesota has been one of the first to take hold of the Conservation policy in practical fashion; and she has done a great work and set an admirable example to the rest of us (applause)—a work representing a policy well set forth in your Governor's address yesterday—and I am glad that this Congress is held in such a State, where we can listen to such an address made by a Governor who had the right to make it. (Prolonged applause)

Much that I have to say on the general policy of Conservation will be but a repetition of what was so admirably said on this general policy by the President of the United States yesterday (great applause); and in particular all true friends of Conservation should be in heartiest agreement with the policy which the President laid down in connection with the coal, oil, and phosphate lands (applause), and I am glad to be able to say that at its last session Congress finally completed the work of separating the surface title to the land from the mineral beneath it. (Applause)

Now, my friends, America's reputation for efficiency stands deservedly high throughout the world. We are efficient probably to the full limits that are permitted by the methods hitherto used. The average American is an efficient man; he can do his business. It is recognized throughout the world that that is his type. There is great reason to be proud of our achievements, and yet no reason to think that we cannot excel our past (applause). Through a practically unrestrained individualism, we have reached a pitch of literally unexampled material prosperity. The sum of our prosperity in the aggregate leaves little to be desired, although the distribution of that prosperity, from the standpoint of justice and fair dealing, leaves a little more to be desired (laughter and applause). But we have not only allowed the individual a free hand, which was in the main right; we have also allowed great corporations to act as though they were individuals, and to exercise the rights of individuals, in addition to using the vast combined power of high organization and enormous wealth for their own advantage.

This development of corporate action is doubtless in large part responsible for the gigantic development of our natural resources, but it is also true that it is in large part responsible for waste, destruction, and monopoly on an equally gigantic scale. (Applause)

The method of reckless and uncontrolled private use and waste has done for us all the good it can ever do, and it is time to put an end to it before it does the evil that it well may (applause). We have passed the time when heedless waste and destruction and arrogant monopoly are longer permissible (applause). Henceforth we must seek national efficiency by a new and a better way, by the way of the orderly development and use, coupled with the preservation, of our natural resources; by making the most of what we have for the benefit of all of us, instead of leaving the sources of material prosperity open to indiscriminate exploitation (applause). These are some of the reasons why it is wise that we should abandon the old point of view, and why Conservation has become a great moral issue, and become a patriotic duty.

One of the greatest of our Conservation problems is the wise and prompt development and use of the waterways of the Nation (applause). There are classes of bulk freight which always go cheaper and better by water if there is an adequate waterway (applause), and the existence of such a type of waterway in itself helps to regulate railroad rates (applause). The Twin Cities, lying as they do at the headwaters of the Mississippi, are not on the direct line of the proposed Lakes-to-Gulf Deep Waterway, and yet Minnesota, with its vast iron resources and its need of abundant coal, is peculiarly interested in that problem (applause); and the Twin Cities, therefore, have their own real personal concern in the deepening and regulation of the Mississippi to the mouth of the Missouri and on to the Gulf. (Applause)

Friends, I have spoken on how progressive Minnesota is and how progressive these Twin Cities are, but there are other progressive cities in the West, too (applause). I have just come from Kansas City (applause)—it's a pretty live proposition (laughter), and there the merchants themselves have undertaken, by raising over a million dollars, to start the improvement of the waterway lying at their doors so that they shall be able to benefit by it. It is sometimes said that the waterway projects are only backed by people

who are delighted to see the Government spend its money but who are not willing to show their faith in the proposition by spending their own. Kansas City is spending its own (applause). The project for a great trunk waterway, an arm of the sea extending from the Gulf of Mexico to the Great Lakes should be abandoned (applause). Of course, before any project is entered upon, an absolutely competent and disinterested commission should report thereon in full to the Government so that the Government can act in the interest of the whole people and without regard to the pressure of special interests (applause), but subject to the action of such a body the Lakes-to-Gulf Deep Waterway, and the development of the rivers which flow into it, should be pushed to completion vigorously and without delay. (Applause)

In nearly every river city from Saint Paul to the Gulf the waterfront is controlled by the railways. Nearly every artificial waterway in the United States, either directly or indirectly, is under the same control. It goes without saying that (unless the people prevent it in advance) the railways will always attempt to take control of our waterways as fast as they are improved and completed; and I do not mention this to blame them in the least, but to blame us if we permit them to do it. (Great applause and cheers) If Uncle Sam can't take care of himself, then there is no particular reason why any railroad man should act as his guardian. (Great laughter and applause) If he attempted the feat he would merely find himself lonely among other railroads (laughter), and Uncle Sam wouldn't be materially benefitted. Uncle Sam's got to do the job himself if he wants to be protected (applause). We must see to it that adequate terminals are provided in every city and town on every improved waterway, terminals open under reasonable conditions to the use of every citizen, and rigidly protected against being monopolized (applause); and we must compel the railways to cooperate with the waterways continuously, effectively, and under reasonable conditions. Unless we do this, the railway lines will refuse to deliver freight to the boat lines either openly or by imposing prohibitory conditions, and the waterways once improved will do comparatively little for the benefit of the people who pay for them.

Adequate terminals, properly controlled, and open through lines by rail and boat, are two absolutely essential conditions to the usefulness of internal waterway development. I believe, furthermore, that the railways should be prohibited from owning, controlling, or carrying any interest in the boat

lines on our rivers (applause), unless under the strictest regulation and control of the Interstate Commerce Commission, so that the shippers' interests may be fully protected.

And now here another word in supplement: You are the people; now don't sit supine and let the railways gain control of the boat lines and then turn around and say that the men at the head of the railroads are very bad men (laughter and applause). If you leave it open to them to control the boat lines, some of them are sure to do so, and it's to our interest that the best and ablest among them should do so. But don't let any of them do it, excepting under the conditions you lay down (applause). In other words, my friends, when you of your own fault permit the rules of the game to be such that you are absolutely certain to get the worst of it at the hands of some one else, don't blame the other man; *change the rules of the game*. (Laughter and applause and prolonged cheering)

Take the question of drainage, which is almost as important to the eastern States as irrigation is in the western States: Where the drainage of swamp and overflow lands in a given area is wholly within the lines of a particular State, it may be well, at least at present, to leave the handling of it to the State or to private action; but where such a drainage area is included in two or more States, the only wise course is to have the Federal Government act (applause); the land should be deeded from the States back to the Federal Government, and it then should take whatever action is necessary (applause). Much of this work must be done by the Nation, in any case, as an integral part of inland waterway development, and it affords a most promising field for cooperation between the States and the Nation (applause). The people of the United States believe in the complete and well-rounded development of inland waterways for all the useful purposes they can be made to subserve. They believe also in forest protection and forest extension. The fight for our National forests in the West has been won, and if after winning it we now go on and lose it, that is our own affair; but *we are not going to do it!* (Applause) After a campaign in which her women did work which should secure to them the perpetual gratitude of their State, Minnesota won her National forest, *and she will keep it* (applause); but the fight to create the Southern Appalachian and White Mountain forests in the East is not yet over. The bill has passed the House, and will come before the Senate for a vote next February. The people of the

United States, regardless of party or section, should stand solidly behind it and see that their representatives do so likewise (applause). Because our ancestors didn't have sufficient foresight, the Nation is now obliged to spend great sums of money to take responsibilities from the States. We, the people of the East, our State Governors—I have been a Governor of an eastern State myself (applause)—showed that the States in the East couldn't do the work as well as the National Government and we are now getting the National Government to take, at large cost to itself, these lands and do the work the public good requires (applause). When we are now doing that in the East, it seems to me the wildest folly to ask us to start in the West to repeat the same blunders that are now being remedied (applause and cheers). My language shall at least be free from ambiguity.

If any proof were needed that forest protection is a National duty, the recent destruction of forests in the Rocky Mountains by fire would supply it. Even with the aid of the Army added to that of the Forest Service, the loss has been severe. Without either it would have been vastly greater. But the Forest Service does more than protect the National forests against fire. It makes them practically and increasingly useful as well. During the last year for which I have figures the National forests were used by 22,000 cattlemen with their herds, 5,000 sheepmen with their flocks, 5,000 timbermen with their crews, and 45,000 miners. And yet people will tell you they have been shut up from popular use! (Applause) More than 5,000 persons used them for other special industries. Nearly 34,000 settlers had the free use of water. The total resident population of the National forests is about a quarter of a million, which is larger than the population of some of our States. More than 700,000 acres of agricultural land have been patented or listed for patent within the forests, and the reports of the forest officers show that more than 400,000 people a year use the forests for recreation, camping, hunting, fishing and similar purposes. All this is done, of course, without injury to the timber, which has a value of at least a thousand million dollars. Moreover, the National forests protect the water supply of a thousand cities and towns, about 800 irrigation projects, and more than 300 power projects, not counting the use of water for these and other purposes by individual settlers. I think that hereafter we may safely disregard any statements that the National forests are withdrawn from settlement and usefulness (applause).

Conservation has to do not only with natural resources; it has to do with the lives of those who enable the rest of us to make use of those natural resources. The investigations of the Country Life Commission have led the farmers of this country to realize that they have not been getting their fair share of progress and all that it brings. Some of our farming communities in the Mississippi valley and in the middle West have made marvelous progress, and yet even the best of them, like communities of every other kind, are not beyond improvement, and those that are not the best need improvement very much. As yet we know but little of the basic facts of the conditions of rural life compared to what we know about the conditions, for instance, of industrial life. The means for better farming we have studied with care, but to better living on the farm, and to better business on the farm—I mean by that, having the farmer use the middleman where it is to the farmer's advantage and not be used by the middleman chiefly to the middleman's advantage (applause)—scant attention has been paid. One of the most urgent needs of our civilization is that the farmers themselves should undertake to get for themselves a better knowledge along these lines. Horace Plunkett, an Irishman, for many years a Wyoming ranchman, has suggested in his recent book on "The Country Life Problem in America" the creation of a Country Life Institute as a center where the work and knowledge of the whole world concerning country life may be brought together for the use of the Nation. I strongly sympathize with his ideas. Last spring, while visiting the capital of Hungary, Buda-Pesth, I was immensely impressed by the Museum of Country Life, which contained an extraordinary series of studies in agriculture, in stock-raising, in forestry, in mining. It was one of the most interesting places I ever visited, and the exhibits were not merely interesting and instructive, they were of the utmost practical importance; and I felt rather ashamed that I, a citizen of what we suppose to be a very go-ahead country, should be in Hungary and obliged to confess we had nothing at all like that in our own country. I wish we had such a museum in Washington, and some of your farmer congressmen ought to get a detailed report of this Buda-Pesth museum to be printed for distribution as a public document (applause). I would like to see a study made of such museums, so that we may take what is good in them for our own use here in America. (Applause)

As a people we have not yet learned the virtue of thrift. It is a mere truism to say that luxury and extravagance are not good for a Nation. So far as they affect character, the loss they cause may be beyond computation. But in a material sense there is a loss greater than is caused by both extravagance and luxury put together. I mean the needless, useless and excessive loss to our people from premature death and avoidable diseases. It has been calculated that the material loss to the Federal Government in such ways is nearly twice what it costs to run the Federal Government.

One of the most important meetings in our recent history was that of the Governors in the White House in May, 1908, to consider the Conservation question (applause). By the advice of the Governors, the meeting was followed by the appointment of a National Conservation Commission. The meeting of the Governors directed the attention of the country to Conservation as nothing else could have done, while the work of the Commission gave the movement definiteness, and supplied it with a practical program. Now, my friends, so far, I have had nothing but praise to speak of Minnesota; but I cannot continue to speak only words of praise. At the moment when this Commission was ready to begin the campaign for putting its program into effect, an amendment to the Sundry Civil Bill was introduced by a congressman from Minnesota, with the purpose of putting a stop to the work so admirably begun. (Sensation) Congress passed the amendment. Its object was to put an end to the work of a number of commissions which had been appointed by the President, and whose contributions to the public welfare had been simply incalculable. (Voice: "Now, what do you think of Tawney?" and laughter) Among these were the Commission for Reorganization of the Business Methods of the Government, the Public Lands Commission, the Country Life Commission, and the National Conservation Commission itself. When I signed the Sundry Civil Bill containing this amendment, I transmitted with it, as my last official act, a memorandum declaring that the amendment was void because it was an unconstitutional interference with the rights of the Executive and that if I were to remain President I would pay to it no attention whatever (enthusiastic applause and cheers). The National Conservation Commission thereupon became dormant. The suspension of its work came at a most unfortunate time, and there was serious danger that the progress already made would be lost. At this critical moment the

National Conservation Association was organized. It took up work which otherwise would not have been done; if it had not done it we wouldn't have had this meeting here (applause), and it exercised a most useful influence in preventing bad legislation, in securing the introduction of better Conservation measures at the past session of Congress, and in promoting the passage of wise laws. It deserves the confidence and support of every citizen interested in the wise development and preservation of our natural resources (applause) and in preventing them from passing into the hands of uncontrolled monopolies (applause). It joins with the National Conservation Congress in holding this meeting. I am here by the joint invitation of both. (Applause)

When the Government of the United States awoke to the idea of Conservation and saw that it was good, it lost no time in communicating the advantages of the new point of view to its immediate neighbors among the nations. A North American Conservation Conference was held in Washington, and the cooperation of Canada and Mexico in the great problem of developing the resources of the continent for the benefit of the people was asked and promised. The Nations upon our northern and southern boundaries wisely realized that their opportunity to conserve their natural resources was better than ours, because with them destruction and monopolization had not gone so far as they had with us. So it is with the republics of Central and South America. Obviously they are on the verge of a period of great material progress. The development of their natural resources—their forests, their mines, their waters, and their soils—will create enormous wealth. It is to the mutual interests of the United States and our sister American Republics that this development should be wisely done. Our manufacturing industries offer a market for more and more of their natural wealth and raw material, while they will wish our products in exchange. The more we buy from them, the more we shall sell to them. Thank Heaven, we of this hemisphere are now beginning to realize, what in the end the whole world will realize, that normally it is a good thing for a Nation to have its neighbors prosper (great applause). We of the United States are genuinely and heartily pleased to see growth and prosperity in Canada, in Mexico, in South America (applause). I wish we could impress upon certain small Republics to the south of us, whose history has not always been happy, that all we ask of them is to be prosperous and *peaceful*

(laughter and applause). We do not want to interfere, it is particularly the thing that we dislike doing; all we ask of any Nation on this hemisphere is that it shall be prosperous and peaceful, able to do reasonable justice within its own boundaries and to the stranger within its gates; and any Nation that is able to do that can count on our heartiest and most friendly support. (Applause)

It is clear that unless the governments of our southern neighbors take steps in the near future by wise legislation to control the development and use of their natural resources, they will probably fall into the hands of concessionaires and promoters, whose single purpose, without regard to the permanent welfare of the land in which they work, will be to make the most possible money in the shortest possible time. There will be shameful waste, destructive loss, and short-sighted disregard of the future, as we have learned by bitter experience here at home. Unless the governments of all the American Republics, including our own, enact in time such laws as will both protect their natural wealth and promote their legitimate and reasonable development, future generations will owe their misfortunes to us of today. A great patriotic duty calls upon us. We owe it to ourselves and to them to give the other American Republics all the help we can. The cases in which we have failed should be no less instructive than the cases in which we have succeeded. With prompt action and good will the task of saving the resources for the people is full of hope for us all.

But while we of the United States are anxious, as I believe we are able, to be of assistance to others, there are problems of our own which must not be overlooked. One of the most important Conservation questions of the moment relates to the control of water-power monopoly in the public interest (applause). There is apparent to the judicious observer a distinct tendency on the part of our opponents to cloud the issue by raising the question of State as against Federal jurisdiction (applause). We are ready to meet this issue if it is forced upon us (applause), but there is no hope for the plain people in such conflicts of jurisdiction. The essential question is not one of hair splitting legal technicalities (applause). It is not really a question of State against Nation, it is really a question of the special corporate interests against the popular interests of the people. (Tremendous applause and cheers) If it were not for those special corporate interests, you never would have heard the question of State against Nation raised (great

applause and cheers). The real question is simply this, Who can best regulate the special interests for the country's good? (Voices: "Theodore Roosevelt!" and prolonged applause and cheers) Most of the great corporations, and almost all of those that can legitimately be called the great predatory corporations (laughter), have interstate affiliations: therefore they are out of reach of effective State control, and fall of necessity within the Federal jurisdiction (applause). One of the prime objects of those among them that are grasping and greedy is to avoid any effective control either by State or Nation; and they advocate at this time State control chiefly because they believe it to be the least effective (applause). If it grew effective, many of those now defending it would themselves turn around and declare against State control, and plead in the courts that such control was unconstitutional (applause). I had my own experience (applause and laughter); I'll give you an example of it. When I was Governor of New York, there came up a bill to tax the franchises of certain big street railway corporations. As originally introduced, the bill provided that the taxation should be imposed by the several counties and localities in which those corporations did business. Representatives of the corporations came to me and said that this would work a great hardship upon them, that the State authority would be more just, that the local authorities (especially where a railroad ran through two or three towns or counties) would each endeavor to get the whole benefit of the taxation for their own locality, and that, in the name of justice, I ought to agree to have the State and not the localities made the taxing power. I thought their plea just, and recommended and sanctioned the change. The bill was made a law; and those same corporations instantly entered suit against it on the ground that it was unconstitutional (laughter and applause) to take the power of taxation away from the localities and give it to the State (renewed laughter and applause); and they carried the suit up to the Supreme Court of the United States where, during my own term as President, it was decided against them. (Applause)

In the great fight of the people to drive the special interests from the domination of the Government, the Nation is stronger, and its jurisdiction is more effective than that of any State (applause). I want to say another thing, which the representatives of those corporations do not at the moment believe, but which I am sure that in the end they will find out; because of its strength, because of the fact that the Federal Government is better able to

exact justice from them, I also believe it is less apt, in some sudden gust of popular passion, to do injustice to them (applause). Now, I want you to understand my position—I do not think you can misunderstand it. I will do my utmost to secure the rights of every corporation. If a corporation is improperly attacked, I will stand up for it to the best of my ability; I'd stand up for it even though I was sure that the bulk of the people were misguided enough at the moment to take the wrong side and be against it (applause). I should fight to see that the people, through the National Government, did full justice to the corporations; but I don't want the National Government to depend only upon their good will to get justice for the people. (Great applause) Now, most of the great corporations are in large part financed and owned in the Atlantic States, and it's a rather comical fact that many of the chief and most zealous upholders of States' rights in the present controversy are big business men who live in other States (applause). The most effective weapon is Federal laws and the Federal Executive. That is why I so strongly oppose the demand to turn these matters over to the States. It is fundamentally a demand against the interest of the plain people, of the people of small means, against the interest of our children and our children's children; and it is primarily in the interest of the great corporations which wish to escape effective Government control. (Applause)

And I ask you to consider two more things in this connection: Waters run; they don't stay in one State (laughter and applause). That fact seems elementary, but it tends to be forgotten. I have just come from Kansas. Practically all the water in Kansas runs into Kansas from another State, and out of it into other States. You can't have effective control of a watershed unless the same power controls all the watersheds (applause and cries of "Good"), as the water runs not merely out one State into another but out of one country into another. One of the great irrigation projects of Montana has been delayed because the Waters that make the Milk river rise in Montana, flow north into Canada, and then come back into Montana. You can't settle that matter excepting through the National Government (applause); the State can't settle it. So much for what we see here. Now, take the experience of other Nations—of the little Republic of Switzerland. It actually tried what some of our people ask to try; it actually tried the experiment of letting each Canton handle its own waters, and a conflict of jurisdiction arose, and the squabbling and the injustices became such that

about nine years ago the National Government of Switzerland had to assume complete control of all the waters of Switzerland, on the explicit ground that all of the waters belonged to all the citizens of the Swiss nation (great applause). Now, I am not asking that we go ahead recklessly; I am only asking that we do not go backward where other countries have gone ahead. (Applause)

As the President yesterday pointed out, one of the difficulties that we have to meet, in connection with the fight for Conservation, is that our aim is continually misrepresented—that the effect is constantly made to show that we are anxious to retard development. It has been no slight task to bring ninety millions of people to understand what the movement is, and to convince them that it is right. Much remains to be cleared up in the minds of the people, and there are many misunderstandings to be removed. For example, we find it constantly said by men who should know better that temporary withdrawals, such as the withdrawals of the coal lands, will permanently check development. Yet the fact is that these withdrawals have no purpose whatever except to prevent the coal lands from passing into private ownership until Congress passes laws to open them under conditions just alike to the public and to the men who will do the developing (applause). And, now understand me; if there is any doubt whether the conditions are liberal enough to the men who are to do the developing. I always solve the doubt in favor of liberality to those men; I want to give them every chance, I want to give them every opportunity to do well for themselves, but I want to see that in doing well for themselves they also do well for the rest of us. (Applause)

In spite of these difficulties, most of which are doubtless inevitable in any movement of this kind, the cause of Conservation has made marvelous progress. We have a right to congratulate ourselves on it, but there is no reason for believing that the fight is won. In the beginning the special interests, who are our chief opponents now, paid little heed to the movement, because they neither understood it nor saw that if it won they must lose. But with the progress of Conservation in the minds of the people, the fight is getting sharper. The nearer we approach to victory, the bitterer the opposition that we must meet and the greater the need for caution and watchfulness. Open opposition we can overcome, but we must guard ourselves; and you of this Congress must especially guard yourselves

against the men who are really corporate agents but who pose as disinterested outsiders (applause). Now I heartily approve the action of any corporation which comes here openly because it is interested in the deliberations of a meeting such as this, and by its openly accredited agents presents views which it believes the meeting should have in mind (applause); I approve of the corporation that does that, and I would despise any of our people who feared instantly to give the most ample and respectful hearing and real consideration to any such plea thus put forward. (Applause and cries of "Good!") The corporation through its agents not only has a right to be heard, but if it did not volunteer you ought to endeavor to see that its views were presented. My protest is not against the man who comes here openly as the corporation agent, but against the man who comes here openly as something else and really as the corporation agent. (Laughter)

It is our duty and our desire to make this land of ours a better home for the race, but our duty does not stop there. We must also work for a better Nation to live in this better land (applause). The development and conservation of our national character and our free institutions must go hand in hand with the development and conservation of our natural resources, which the Governors' Conference so well called the foundations of our prosperity. Whatever progress we may make as a Nation, whatever wealth we may accumulate, however far we may push mechanical progress and production, we shall never reach a point where our welfare can depend in the last analysis on anything but the fundamental qualities of good citizenship—honesty, courage, and common sense (applause). The homely virtues are the lasting virtues, and the road which leads to them is the road to genuine and lasting success.

What this country needs is what every free country must set before it, as the great goal toward which it works—an equal opportunity for life, liberty and the pursuit of happiness to all of its citizens, great and small, rich and poor, great and humble, alike. (Tumultuous applause and continuous cheers)

FOURTH SESSION

The Congress reassembled in the Auditorium, Saint Paul, after luncheon, September 6, and was called to order by Vice-President Condra.

Professor CONDRA—Delegates, Ladies and Gentlemen: President Baker has asked me, as one of the vice-presidents, to preside pending his arrival.

We are to be congratulated in that we are to hear from many distinguished speakers on many interesting topics this afternoon. We are especially happy in that the first speaker is one who has done much, not only in Washington but throughout the world, for conserving human life through the work of the Red Cross. I have great pleasure in presenting to you Miss Mabel Boardman, of Washington. (Applause)

Miss BOARDMAN—Mr Chairman, Ladies and Gentlemen: Of what value would Conservation be without human life? For the benefit of man's life are given all these energies which are devoted to the Conservation of our natural resources. So at the very foundation of Conservation must lie the preservation of that for which Conservation exists.

It is in this principle of Conservation of human life that the Red Cross has its being. Though first inspired by Florence Nightingale in the Crimea, it was born on the bloody battlefield of Solferino, more than fifty years ago, when Henri Dunan witnessed the terrible waste of human life because of the lack of medical and nursing care. The Red Cross has become one of the great conserving forces of all the world. It acts under the only universal Conservation treaty in existence. One after another all the nations of the world have signed this Treaty of Geneva, first drafted in 1864, revised in

1906, and its provisions extended to naval warfare by the Treaty of The Hague.

The opening words of the Geneva Treaty read: "Officers, soldiers, and other persons officially attached to armies, who are sick and wounded, shall be protected and cared for, without distinction of nationality, by the belligerent in whose hands they are. The belligerent in possession of a field of battle must search for and protect the wounded, and may grant immunity to those inhabitants who have taken into their homes the disabled men. The neutrality of hospitals and ambulances with their personnel, who cannot be made prisoners of war, must be respected, and, for humanity's sake, lists of the dead and wounded must be exchanged for transmission to the families of these men by the authorities of their own country." This wonderful treaty provides its own insignia, and wherever throughout the world the grating doors of the Temple of Janus open wide their terrible portals it flings to the winds of heaven its merciful banner of Conservation of the sick and wounded, the flag of the Red Cross.

The treaty provides, moreover, protection for the volunteer aid societies which have received official authority from their respective governments. These are the three great Red Cross Societies. Recognizing two facts, *first*, that no medical service of any nation can be adequate to the demands of war, and *second*, that at such times the humanity and patriotism of a people become deeply stirred into active life and that this activity should be utilized in such a systematic way as to be of real value in the saving of life for the sake of humanity and for the sake of the country, the members of the original Geneva Conference recommended to the signatory powers the formation of these volunteer aid societies. Thus, the Red Cross had its origin in the purpose of conservation of human life in time of war. How efficiently it has carried out this duty where well organized is shown by a glance at the remarkable statistics of the work done by the Red Cross of Russia and Japan during the late war in the Far East.

I am tempted here to dwell for a moment on one or two facts connected with the Japanese Red Cross. It has today more than 1,522,000 members, and its annual revenue in 1909 amounted to more than \$2,000,000. In spite of the late war which was such a serious drain upon the resources of the country, the Japanese Red Cross never depleted by a single yen its

permanent fund. The report for 1909, just received, gives this permanent fund as more than \$5,000,000, and it has besides in other funds more than \$2,000,000 on hand. By 1913 it plans to have increased its permanent fund to \$7,500,000; and knowing what Japan has already done, we cannot doubt the carrying out of this expectation.

But though since the beginning of history wars have been from time to time the misfortune of mankind, the great forces of nature bring a far more frequent need for such assistance as the Red Cross is able to render. Because of this ever recurring need of organized aid the Red Cross reached out its strong and well-trained arms into this broader field to succor the victims of great disasters.

The charter granted by Congress to the American Red Cross, and which created it the officially authorized Red Cross of our Government, provides that it shall not only "take charge of the volunteer relief in time of war" but that it shall "carry on a system of national and international relief in time of peace, and apply the same in mitigating the sufferings caused by pestilence, famine, fire, floods, and other great calamities, and to devise and carry on measures for preventing same." Under this charter our own American Red Cross is not a private association of certain people, but an officially authorized agency of our Government, responsible to the people, and whose existence Congress may at any time cancel by annulling the charter. Its accounts are audited by the War Department. The chairman and five members of the Central Committee, representing the Departments of State, Treasury, War, Justice, and Navy are appointed by the President of the United States. The State Department is represented because of participation in international relief. The Treasury provides the National Red Cross treasurer, the Department of Justice, the counselor, and the army and navy have their reasons for representation not only because of war association but because, during National disaster relief as at San Francisco, Hattiesburg, and Key West, the Red Cross has the heartiest and most invaluable aid of our army, while in international relief, as in Italy after the earthquake and at Bluefields, Nicaragua, it receives the equally hearty and valuable aid of our navy. Briefly, then, of what does the American Red Cross organization consist? Since its reorganization in 1905, William Howard Taft, now President of the United States, has been yearly elected as its president, and largely to his constant interest, wise counsel, and valuable assistance is its

success due. It has, besides the other usual officers, a national director Mr Ernest P. Bicknell, whose particular duty it is to proceed immediately to the scene of any serious disaster and take charge of or advise in regard to the Red Cross relief work. It has a central committee of eighteen, which elects an executive committee of seven. Under this committee the work of the Red Cross is segregated into three departments for war and for national and international relief, each under a board of fifteen members. The chairman and vice-chairman of each board are members of the central committee.

The war relief board, of which the surgeons representing the army and navy on the central committee are respectively chairman and vice-chairman, has prepared a complete list of every coastwise vessel suitable for a hospital ship, so that such a ship could be chartered at a moment's notice. It has moreover drawn up a complete and detailed list for the equipment of such a ship with estimates of the cost of this equipment and the necessary transformation for hospital purposes. It is studying the questions of civil hospital accommodations for war-time need, of hospital trains, of field hospitals, rest stations, the use of private automobiles for ambulances, and other kindred subjects. A sub-committee, six of whom are members of the board and nine of whom are representative women of the trained nursing profession, and whose chairman is Miss Jane Delano, Superintendent of the Army Nurse Corps, has systematized the Red Cross nursing service, prepared uniform regulations, organized State and local committees, and is fast enrolling the best trained nurses in the country for active service in time of need. These splendid nurses at such times not only undertake the most difficult work under frequently severe hardships, but when on this active duty accept from the Red Cross only half of their usual salary. This Red Cross nursing committee will later take up the plan of providing courses for women in simple home nursing of the sick.

Another sub-committee of the war relief board is the First Aid Committee, the chairman of which, Major Charles Lynch, of the Army Medical Service, is detailed for this particular duty by the Surgeon-General. The work of this committee is the organizing of courses in first-aid instructions throughout the country. On this committee such men as Mr John Hays Hammond represent the mine companies; Mr John Mitchell, the miners; Mr Julius Kruttschnitt, the railroad companies; Mr W. G. Lee, the trainmen; Dr D. A. Mansfield, the sailors' interests; Dr J. A. Holmes, the U. S. Bureau of

Mines. The Y. M. C. A. is also represented on the committee, as it now gives all its first-aid courses in collaboration with the Red Cross. Dr M. J. Shields is employed as the agent to organize these courses among miners. It is expected this autumn that a special car will be donated by the Pullman Company for the purpose of sending with Dr Shields a traveling first-aid equipment and safety-device exhibit. A number of railroads have already most kindly consented to transport this car free of expense to the Red Cross. I may say that in every case of a great calamity, the railroad companies, express companies, telegraph and telephone companies, have placed their services free at the disposition of the Red Cross in a most helpful and generous spirit.

The first-aid courses will soon be extended to trainmen and employees of large industrial concerns, as has been done by the British and German Red Cross. Major Lynch has prepared for the Red Cross a most excellent general text-book on first-aid, also a special book for miners and trainmen, and another, at its request, for the Bell Telephone Company. Furthermore, valuable and inexpensive anatomical charts have been printed for these courses, and small metal boxes hermetically sealed containing first-aid bandages and a leaflet of directions have been made for the society, as well as a larger box for railroad stations, mines, factories, etc. Competitions in first-aid have been held, and prizes and medals awarded. More than sixty thousand posters calling attention to precautions to be taken to prevent personal injury on railroads, and over thirty thousand of a like nature for trolley cars, have been issued by the Red Cross and are distributed on application from various companies.

To spread abroad throughout the country the knowledge of first-aid among our industrial classes, in fact, among all classes of our people, is the aim of this department of Red Cross work. Not only in time of war or disaster will such knowledge prove of great value, but in all of the frequent accidents of daily life will this training be of help. (Applause)

The second board, that of the national relief, has to do with the study, planning and overseeing of relief after national disaster. It is not possible, nor would it be wise, for the Red Cross to maintain a corps of trained workers for active duty after disaster, when such duty comes only from time to time; so to provide itself with an experienced personnel, it has created an

institutional membership consisting of the best charity organization societies of the country. These associations in accepting membership consent to utilize their personnel under direction of this board and of Mr Bicknell, the national director, for active relief duty. For example, Mr Logan of the Atlanta organization, went on Red Cross orders to Key West last September, systematized relief work so as to avoid imposture, unfortunately prevalent at such times, advised with the Mayor and commanding officer of the army post there, arranged that the contributions be mainly expended in rehabilitating the fishermen who had lost their little boats, their only means of earning their livelihood. As each boat was completed, the owner who had been provided with material for his boat and paid a daily wage while building it, was again on his feet, able to support himself, and his name was taken from the list of those being aided.

At the time of the Cherry Mine disaster, Mr Kingsley of the United Charities of Chicago, went immediately to the scene of the disaster, remaining until Mr Bicknell could arrive. Then for several months, at the request of the Red Cross, his assistant and two good women who could speak Italian and Polish to the poor distracted miners' widows, remained at Cherry while Mr Bicknell's plan for permanent relief could be perfected and accepted. By this plan, which is now being carried out, the generous funds contributed by the people of Illinois, by its State Legislature, and by the miners' unions, amounting to about \$300,000, have been consolidated and are being administered by a joint commission so that a pension can be paid to each widow and minor child until the children are of an age to become wage-earners themselves and the fund is exhausted. (Applause)

The national relief board has also had charge of the little Red Cross Christmas stamp—next year to be called a "*Christmas seal*"—placed on the back of letters out of deference to the wishes of the post office department, which has suffered from a multiplicity of stamps issued by others because of the success of the Red Cross stamp. That stalking spectre of pestilence, tuberculosis, had laid its devastating hand on every nation; it invades the palace as well as the hovel, and the youth of the people are its surest prey. With a weapon tinier than the stone in David's sling, the Red Cross sends forth this little seal to do its part. In the last two years it has netted more than \$350,000 with which to war against this grim destroyer. Here again the

Red Cross carries out its principle, the conservation of the *human life*.
(Applause)

The third board is that of international relief with a representative of the state department as its chairman. Two maps hang on the walls of the Red Cross office at Washington, one of the world, the other of the United States with its insular possessions. Starred over these large maps are little red crosses marking the fields of its noble labors for Conservation. Not alone within our own borders lies its merciful service. Far away in Russia, China, and Japan, when famine claimed its thousands of tortured victims, went the Red Cross, aided by the *Christian Herald* of New York, with food for the starving multitudes: when earthquakes in Chili, Jamaica, Italy, Portugal, and Costa Rica brought destruction and desolation, when floods in Mexico, France, and Servia devastated the land, when massacres in Armenia brought suffering, misery, and even death to thousands, when internal war in Nicaragua left regiments of wounded, naked, and starving boy prisoners, our American Red Cross stretched out her helping hand to these, her sister nations in distress (applause). If in Conservation lies thought for men yet unborn, thought must also be given for the men who live today, and the Red Cross recognizes its duty toward the conservation of all human life.
(Applause)

But a moment more on its organization: In over thirty States, boards of representative men, with the Governor in each State as president of the board, have already been appointed, and before the end of the year the boards for all of the other States and for the insular possessions will probably be completed. The duty of such a board is to act as a financial committee for the receipt of contributions of the people of the State in case of war, local, national or international disaster. The Governor being president of the board, may issue an appeal to the people of the State when in his judgment a disaster of sufficient magnitude within the State justifies such an appeal. On the occurrence of disasters without the State, appeals are issued only on advice from the National officers. The Governor or State board may, in case of any disaster within the State of sufficient magnitude, request of headquarters the assistance of the National body. Chapters of the Red Cross may exist in any town, city or county where there are five or more members who pay the annual dues of one dollar. It is the duty of these chapters to respond promptly and vigorously to any request for action on

the part of the Red Cross in time of war or disaster at home or abroad. Appeals issued by the president of the State board or from Washington will state the needs for money or supplies, or both, which the chapter should at once begin collecting. In case of a serious local disaster, the chapter acts as the supply agency for the National director and institutional member, when such member is present. In case no institutional member is at hand, it is expected to take prompt relief measures pending the arrival or instructions of the National director. This, then, in brief, is the organization of the Red Cross for active service: National officers, a central committee, relief boards with their sub-committees; State boards, chapters, and institutional members.

It seems impossible in a non-military country like ours to obtain and retain a large supporting membership with small annual dues, as is done in other countries. When reports of great calamities fill the papers, our people give with wonderful generosity, but the minor disasters, whereby small communities suffer greatly, receive but little notice from our public. If Japan plans to increase its Red Cross permanent fund to \$7,500,000, could not the people of this country raise for our American Red Cross a permanent fund of \$2,000,000? I, for one, believe they will, for New York City alone has already promised nearly quarter of that amount, and this autumn endowment committees of prominent men, appointed by the President of the United States, will make an appeal to our people all over the country to raise this permanent fund for the American Red Cross.

And, last, may I say a word or two for some of the by-products of Conservation in Red Cross service? In the work of the Red Cross first-aid department lies the far-reaching results of conservation of the life of the wage-earner of the family as well as the labor-producer of the country, or in case of his death in disaster, as at Cherry, the administration of the relief funds so that the unfortunate widows can keep their little children at home (applause),—a by-product, the conservation of the family.

The preservation of life in time of war has not only its humane feature but its patriotic reason. In fact, the Japanese Red Cross puts this principle first. The saving of one of the most important assets of any country, that of its young manhood, becomes a by-product of Conservation for the sake of patriotism.

Another by-product is the conservation of communities. Whether some little hamlet or some large city suffers from the overwhelming calamity of fire, flood, storm, earthquake or pestilence, or the still more pitiful disaster of widespread famine settles over a great province or empire, its people are brought down to desolation and despair. Their neighbors suffer as well and there are none at hand to help. Without aid they must die or drift away from their homes like unmoored boats after a storm, to be swamped at sea or wrecked upon the rocks of unknown shores. It is then to these communities as well as to the individual that the Red Cross comes. It calls to the disconsolate "Comfort ye, my people, build again your homes. Sow again your fields; the strong arms of the Red Cross are here to aid you, held up by your brothers of the Nation, yea, by your brothers of the world, if there is need" (applause). On a beautiful silver tablet, presented by an Italian relief committee to the American Red Cross, are engraved in Latin the words of an old Roman historian, "Your bounty has repaired the catastrophe not merely of individual citizens but of entire cities."

And there is one more by-product of Conservation not having so much to do with things material but for the well-being of the world. Is there not need of a conservation of higher things? Above the passion of war, amidst the desolation of terrible disasters, in the dangers of the daily occupations so many of our fellowmen must undergo to earn their livelihood, does not the Red Cross conserve, protect, and extend the great bond of human brotherhood, and, touched by sorrow, make the whole world kin?

Strangely taking its inception on the field of battle, this great international organization of the Red Cross for the conservation of human life was born, has passed from infancy into a strong and noble maturity ever ready to protect and preserve human life, for which the Conservation of all material things has its reason and its purpose. (Applause)

Chairman CONDRA—We shall now have the privilege of hearing the Commissioner of Corporations, called to that responsible duty by President Roosevelt, and continued in his responsibilities by President Taft,

Honorable Herbert Knox Smith, whom I have great pleasure in introducing (applause).

Commissioner SMITH—Mr Chairman, Ladies and Gentlemen: My text is that superb word "power"; and it has no more appropriate place for enunciation than this center of gravity of imperial power, the Mississippi valley.

In our complex civilization there are many things that are necessities of life. Control over any of them represents a power that is essentially governmental. This is plainly true of basic necessities like food, clothing, transportation, heat, and light; it is true also of the natural resources that are back of these. It is no less true of the mechanical power that produces and delivers them. Private control of any one of these, unrestrained either by business competition or by governmental authority, means that irresponsible individuals hold a command over the daily life and welfare of the citizen which the men of our race have never willingly granted to any except their own representatives chosen by them.

For us of our generation, mechanical power is a basic necessary. Our daily existence is borne on its current, and our power demand steadily increases. Our chief present sources of power supply—coal, petroleum, and natural gas—although at present ample, are absolutely fixed in quantity and cannot be replaced. Water-power is the one important source of mechanical power now practically available which is self-renewing. Its importance, therefore, to our present vision, must steadily increase.

Effective restraint, imposed by competition on its control, is becoming more and more improbable. There has been a marked concentration of water-power control in private hands, and this process is advancing rapidly. Public regulation of water-power, the only other alternative, therefore, becomes a necessity.

Electric transmission has worked this change within the last decade. As now commercially practicable, such transmission allows a given water-power to reach a market area of at least 80,000 square miles. It has raised water-

power from purely local work, and made it the vital energy for great communities and distant enterprises. It has brought our water-power resources suddenly within the sweep of great economic forces.

Within these market areas just described, there are strong practical reasons for consolidation of water-powers—what is known as "coupling up." A power plant must be constructed to meet the highest point of its expected demand—the "peak of the load." The nearer the "load" (the power demand) approaches that peak for all the time, the more fully will the entire fixed investment be earning a return. Suppose there are two independent power plants in two neighboring communities where the demand in one community is mainly for power during the day time, and in the other at night. These plants can advantageously combine, throwing the surplus of their joint power by day to one place and by night to the other, thus bringing their normal load in each case up nearer to the peak. Similarly, such coupling up is obviously advantageous in two neighboring watersheds where the excess water-power occurs at different times. In general such combining of varying conditions to produce a closer parallelism of supply and demand is in itself an entirely proper industrial development. We have no reason to oppose it if accomplished by fair methods; we must simply be prepared to regulate such monopolistic power as may result therefrom.

The investigation of developed water-powers now being made by the Bureau of Corporations shows that up to date 18 concerns or closely allied interests control over 1,800,000 horsepower of the water-power developed or in process of construction, and, in addition, over 1,400,000 horsepower of undeveloped water-power. As to undeveloped powers, this information was secured merely as an incident to our main work, and certainly much understates the case. As it stands, however, it makes a total water-power controlled by these 18 groups of over 3,200,000 horsepower. The total water-power in use in the United States in 1908, as estimated by the Census and Geological Survey, was only 5,300,000. And this total includes a very large number of small powers which the Bureau did not include, as it dealt almost wholly with powers of over 1,000 horsepower. The total now commercially capable of development is variously estimated at from 30,000,000 to 60,000,000 horsepower, the smaller figure being the preferable one. The great bulk of both developed and undeveloped water-power lies on the Pacific Coast, in the Northwest and Northeast, and in the

South Atlantic States. Our power demand as measured by the total unduplicated capacity of all prime movers—steam, water, and gas—is now at least 30 million horsepower.

It is obvious that a local monopoly of power covering simply one market area is nevertheless as complete in its effects on the inhabitants of that area as if it covered the entire country. Conditions in separate sections are therefore important. In California, for example, four principal hydro-electric companies dominate the water-power industry. They have a total developed horsepower of 259,000, with probably 500,000 additional undeveloped, and a very strong hold on the most important power markets. And between these four concerns there is also evidence of considerable harmony. This is not a unique case. Conditions somewhat like this exist in the Puget Sound territory, in the southern peninsula of Michigan, in Colorado, in Montana, and in the Carolinas. In each of these sections, one, or at most two concerns are predominant in their control of water-powers, public-service companies, and power markets.

The horsepower figures do not fully represent the extent of actual commercial control. The best powers have of course been developed first. These will always hold a disproportionately dominant position over later developed and less favored powers, because of their lower operating cost and prior hold on the important power markets.

There is also going on a concentration of a wider sort—a process of deep significance, but as yet little recognized. There is a marked progress toward a mutuality of interests among public service companies generally, electric light, power, gas, and street railway concerns. The significant identity of officers and directors in a large number of such companies throughout the United States is very remarkable. This is due in part to specialization by financial houses in given lines of investment; in part to the common employment of certain eminent engineering firms; and in part to relations with certain leading equipment companies. Electric equipment is usually supplied by one of a few great equipment concerns and frequently paid for, at least in part, in the securities of the proposed project. Thus the equipment company acquires interests in widely separated power and light concerns.

Take a single example, the General Electric Company, the most powerful electric equipment concern in the world. Men who are officers or directors of the General Electric Company, or of its three wholly controlled subsidiary companies, are also officers or directors in many other corporations. These other companies, with their subsidiaries, and the General Electric with its subsidiaries, make thus a group interconnected by active personal and financial relationship. This one group includes 28 corporations that operate hydro-electric plants, with at least 795,000 horsepower developed or under construction, and 600,000 undeveloped in 16 different States, a total of 1,395,000 horsepower (equal to more than 25 percent of all the developed water-power in the United States in 1908). This group includes also over 80 public-service corporations, not counting their minor subsidiaries; more than 15 railroads; 6 companies that use their power in the manufacture of cotton goods, with 35,000 hydraulic horsepower developed; and over 50 banks and financial houses, many of them in the first rank of importance. This remarkable financial connection in itself is very significant. Fifty-three General Electric men, in all, constitute this chain of connection. Nor are these men, as a rule, of the figurehead type; their presence on a directorate means something. Of course these facts in no sense always mean identity of control. They certainly do mean a striking degree of non-conflicting interests and personal relationship which makes further concentration easily possible.

This wider concentration is still in a formative stage, developed almost wholly within the last decade. The forces compelling thereto are still operative. It is like a physical solution of chemical elements which is still in suspension but which a single jar may precipitate into crystallization. Water-power, being naturally allied with public-service business, will be included in any movement that affects that business generally. So wide is this interrelationship, and so comparatively few are the constantly recurring names in the directorates, that a few brief conferences, given the necessary impetus, might conceivably at any moment concentrate into definite legal form a sweeping control over the dominant water-powers of the country, as well as their related public service interests.

Here, then, is the present situation of the hydro-electric industry:

(1) It deals with a basic necessity, and its importance inevitably increases as the fixed supply of other sources of power decreases.

(2) Substantial control of mechanical power means the exercise of a function that is essentially governmental in its effect on the public.

(3) Driven by underlying economic and financial forces, concentration of control of water-powers in private hands has proceeded very rapidly. It is doubtful if anything can arrest this process, and a swift advance to a far higher degree of concentration is entirely possible.

(4) Any chance, then, of restraint by competition is rapidly disappearing, certainly over given sections, and public regulation is therefore an imminent necessity.

The extent of such regulation will depend mainly on constitutional limitations. A State, roughly speaking, can at any time exercise a high degree of control over power companies as quasi-public servants. The jurisdiction of the Federal Government covers a wider range geographically, but involves some difficult constitutional questions. Over water-powers on the public lands it has full control. I concede no merit to doubts as to the Government's unlimited jurisdiction there.

As to powers on navigable streams not in the public domain, there is an undetermined constitutional question. It is well settled that no power dam can be maintained on a navigable stream without the consent of the Federal Government. Nearly everyone admits that the Government may impose upon such grants any desired time limitation, and may thus require readjustment of terms at any desired period. But some hold that the Federal Government, in exercising its arbitrary power as grantor, may also impose any further conditions it chooses upon such grant, as, for example, that the grantees shall pay a rental for the power acquired. Others hold that the Federal Government can only impose such conditions as are directly connected with the Federal power over interstate commerce, such as navigation. Even this view would apparently at least permit a rental charge, if applied to navigation improvement. Personally, I am strongly inclined to the former and broader view that any conditions whatsoever may be imposed (applause), both on general principles and on well-established legislative precedents. In numerous bridge and dam acts Congress has used

the broad power and imposed conditions in no way related to interstate commerce. In the California Debris Commission Act, operative since 1893, Congress imposed a straight charge on placer miners for the privilege of emptying their refuse into the streams.

The scope of the Federal jurisdiction is of first importance, because the water-power problem is, in the main, a National one. Much of the power is transmitted across State lines, or is used by interstate carriers. The bulk of the capital that is developing our most important powers comes from interests outside the States where the powers are located, and from the brief survey I have already given of the interrelationships existing between public-service companies it is obvious that State lines and State jurisdiction have no practical relation whatsoever to the sweep of these forces (applause). The hydro-electric industry has been largely nationalized by those who are foremost in it.

The Nation and the State will have to use their full powers to meet the water-power situation. The most effective time to use them is before, not after, private rights accrue. The one certain method is for the State or the Federal Government, to retain its interest, or impose its conditions, at the inception, as a part of the grant. Then public control and private rights go together, as they must if we are to safeguard the public interest in water power. (Applause)

Let there be no unnecessary hampering of hydro-electric development, but let the public be in on the ground floor at the start: for at the start the public must grant the power and for all time the public will be the party chiefly interested in its use. (Applause)

As President Taft very justly said yesterday, when a man talks to you about conservation, you have the right to ask him to specify what steps he desires to take. I am going to specify.

(1) The *status quo* of all water-power still controlled by the Nation or State should be maintained until we know what we have, and can act intelligently thereon.

(2) No water-power grant should be made except for a fixed period, with at least the reserved right to readjust terms at the end thereof. That period,

however, should be long enough to permit adequate financing and complete development.

(3) Complete publicity of accounts and transactions should be required, as well as a record of cost, and the real relation of investment to stock and bond issues.

(4) Power to revoke the grant for breach of conditions should be lodged in a specified public authority. Otherwise there will always be the possibility of protracted litigation to determine the status.

(5) So far as is possible, direct provision should be made against excessive charges and monopolistic abuse.

(6) Public authorities should reserve such constitutional compensation or rental as will establish the principle of underlying public interest.

(7) All public easements of navigation, fisheries, etc., should be safeguarded.

(8) In the case of new grants, all these provisions should be made conditions of the grant.

Finally, the purpose and probable effect on the public of any water-power grant should first be fully ascertained and carefully considered, in order to determine whether public interest justifies beyond a reasonable doubt the surrender by the public of even a part of its power over this great public resource. Where reasonable doubt exists, the surrender should not be made. (Applause)

[During the delivery of the address President Baker arrived and resumed the Chair.]

Honorable JOHN BARRETT—Ladies and Gentlemen: President Baker has requested me to announce that Professor George E. Condra, of Lincoln, Nebraska, has been appointed chairman of the Committee on Credentials in lieu of Mr Edward Hines, of Chicago. (Applause)

Governor Pardee has an announcement to make in regard to the Committee on Resolutions.

Governor PARDEE—Simply that the Committee on Resolutions will meet at the Saint Paul hotel this evening at 8 oclock, in Room 534.

President BAKER—The program in your hands announces that an address entitled "Safeguarding the Property of the People" will be delivered by Honorable Francis T. Heney, of California. He is prevented from being here this afternoon, but will arrive later.

We have now the opportunity of hearing from one whose name has been so closely associated with the work of Conservation that he is regarded as one of its greatest and ablest advocates; I have great pleasure in introducing, to speak on "The Federal Government's Relation to Conservation," Honorable James R. Garfield, of Ohio. (Applause)

Mr GARFIELD—Mr President and Delegates, Ladies and Gentlemen (renewed applause): I appreciate your applause at this time very much, for I fear me at the end of what I have to say it may not be forthcoming.

The subject I have chosen is one that affects very directly what may not merely be talked about but can actually be done by the people of this country in connection with Conservation problems. It was often said a few months ago that Conservation was an enthusiasm—that it was an idea, or perhaps an ideal, and that those who were urging Conservation were not practical men and looking forward to practical work-a-day solutions of their own problems. So I chose to speak on the relation of the Federal Government to Conservation—a very practical subject, one on which we have been working, as well as talking, for a number of years.

There are two good reasons why the Federal Government is directly interested in Conservation. In the first place, it is the largest land-owner in this country; and, in the second place, it has high duties to perform for the interests of all the people of this country. For these reasons, the Federal

Government comes directly in touch with the practical questions of Conservation in dealing with what is left of the natural resources of our public domain. The value of these resources cannot be measured in mere terms of acres. Some 700,000 acres of our public lands remain; but that means nothing unless we know what is contained in or on the land represented by the mere statement in figures. Now we are learning that this great area, both on the mainland and in Alaska, is filled with priceless treasures in the resources needed for the lives of the people of our country; and it is in the handling of these resources—either disposing of them or providing for their use or development—that the Federal Government must deal practically with the problems of Conservation. Only as we know this tremendous area and its priceless treasures do we realize that we must, in the practical handling of these resources, make as few mistakes as possible, and constantly keep in view the interest of all the people as a guide in the solution of any given problem.

Now, we meet with serious difficulties in attempting to decide how best to use the property owned or held by the United States Government as trustee for all the people. We have under our system of government a dual jurisdiction, or rather, two jurisdictions—that of the Nation on the one hand, and that of the State on the other. Yet between these two jurisdictions there is no real conflict; there ought to be no insuperable obstacle to such cooperation between States and Nation as will make possible a wise solution of all questions in which both jurisdictions have duties to perform. We hear much about States' rights, as though the problems of Conservation have brought to life again an old doctrine, as though in some way the Conservationist is endeavoring to take something away from the States. The very opposite is true. There is no effort on the part of the Conservationist to interfere with any duty that the State ought to and can perform. Those duties devolving on the States should be performed by the States; and the people of each commonwealth should see to it that their State representatives not only do what is wise and necessary each year but exercise foresight in dealing with all resources subject to their jurisdiction (applause). That, however, does not mean that the Federal Government is debarred from proper use of the public domain within the areas of the several States; it likewise has great duties devolving on it in so administering its property as to safeguard the interests and the rights of all the citizens of the country.

The State lines are merely accidental in many instances. The States of the old Northwest and the States of the Middle West today were carved out of public territory simply by drawing of lines; they were not political entities in the first instance, but a few people got together and agreed that so many square miles of territory would be made into a State, and whether that State line was drawn here or a hundred miles over there should not determine how we are to deal with the public resources contained within the area.

In the early period of our development there was but little need of giving heed to the questions that are now uppermost in our minds in relation to the public domain. There was land enough and to spare; and the early purpose of the Federal Government was to provide easy methods for getting the public domain (which in those days was considered chiefly useful for agriculture, as it is in the middle West) into farms, and building up commonwealths that are now theatres of agricultural industry. But today the conditions are very different. The remaining agricultural land that can be used without irrigation or drainage is very little in comparison to the needs of our people; and in handling what is left of the public domain it becomes the duty of the Federal Government to see to it that not one acre of land that can be used for agricultural settlement and development is directed to any other purpose—and likewise to see to it that land capable of mineral development or of water development is not stolen from the public domain under the guise of homestead entries. (Great applause)

In order to understand exactly what the Federal Government can do in relation to the use of the public domain, let us keep clearly in mind the powers granted to it under the Constitution, and the laws enacted in accordance with the Constitution by Congress. The Constitution provides that—

The Congress shall have power to dispose of, and make all needful rules and regulations respecting the territory and other property belonging to the United States.

The executive power shall be vested in a President of the United States of America.

* * * he shall take care that the laws be faithfully executed.

Now, in accordance with the provisions of the Constitution, the Congress has enacted the following laws affecting the public domain:

The Secretary of the Interior is charged with the supervision of public business relating to * * * the public lands, including mines.

The Commissioner of the General Land Office shall perform, under the direction of the Secretary of the Interior, all executive duties appertaining to the surveys and sale of the public lands of the United States, or in anywise respecting such public lands.

The Commissioner of the General Land Office, under the direction of the Secretary of the Interior, is authorized to enforce and carry into execution by appropriate regulations, every part of the provisions of this title [the public land laws] not otherwise specifically provided for.

Congress, acting under these general provisions, has from time to time enacted laws affecting portions of the public domain. It has provided the Homestead Act, the Timber and Stone Act, the Mineral Entry Act; provided for the creation of the National Forests; enacted laws relating to the use of the public domain for reservoir sites, for pipe-lines, and for transmission lines; and as the needs of each generation have been made known, Congress, acting for the interests of all the people, has enacted direct legislation for the purpose of providing method for the disposition and use of the public domain.

Meantime, the Executive on his part has performed the duties devolving on him under the Constitution—duties few in number and easily expressed, though of great importance to the public welfare. They are, in brief, to see to it that the laws of the United States are faithfully executed; and he is granted all the executive power that could have been given by the use of the English language. There is no limitation. It is simply "executive power"; whatever that may be was granted to the President of the United States.

One of the great objects for which this Nation was created was to promote the "general welfare." That object was not only stated in the preamble of the Constitution, but was likewise written into the body of the instrument; and the power was specifically granted to Congress to provide for the general welfare of the United States. That was not an idle phrase. The founders of

the Republic recognized that it was impossible for them to foresee all the things that it might be necessary for the Federal Government to do; it was not possible for them to define in specific language all the powers that were to be exercised, nor was it possible for them to indicate to what extent these powers, once granted, might properly and wisely be used; and this welfare clause has made it possible to carry out by both the Legislative and the Executive branches of the Federal Government the beneficent purposes of the founders in ways which they never contemplated or could have contemplated in detail. Fortunately, during the early days of our National existence we had at the head of the Supreme Court a master mind. Marshall was as profound a statesman as he was a great jurist. He recognized with that great far-seeing insight that amounts almost to inspiration, that it would have been to sound the death-knell of the Republic if he, as the chief law interpreter from the judicial seat, should so interpret the Constitution as to tie the hands of the Government and prevent the people from doing the things necessary to make themselves a great and permanent Nation. In one of the earliest decisions involving interpretation of the Constitution (McCullough vs. Maryland. 4 Wheaton 315) Marshall used this language:

Let the end be legitimate, let it be within the scope of the Constitution, and all means which are appropriate, which are plainly adapted to that end, which are not prohibited but consistent with the letter and spirit of the Constitution, are constitutional.

Another sentence in the same opinion sets a standard for judging existing or proposed law; he says—

But where the law is not prohibited, and is really calculated to effect any of the objects entrusted to the Government, to undertake here to inquire into the degree of its necessity would be to pass the line which circumscribes the judicial department and tread on legislative ground. This court disclaims all pretensions to such a power.

Clearly, Marshall saw at that time that if the Supreme Court endeavored to prevent Congress from exercising to the full a power granted under the Constitution, it would at that very moment overstep its legitimate ground and interfere with the functions granted to the legislative body; and in dealing with the powers granted to the Executive, exactly the same rule of

interpretation applies. Now, it is most interesting to notice how from generation to generation Marshall's interpretation has made possible the doing of the things that have been done by our people. In those days it was impossible for men to conceive of the commercial development that has taken place during the hundred years. They could not have realized that within a hundred years we would be a great manufacturing Nation, and that our commercial relations would not be confined to the thirteen colonies but would spread broadcast throughout the entire world.

A striking example of the application of this wise interpretation arose in dealing with the questions of the Philippine government. We there had an entirely novel proposition. The forefathers of the Republic had never contemplated the acquisition by us of territory in the Pacific, or islands elsewhere. Yet when we faced that problem, we found that under Marshall's interpretation, our Constitution was broad enough and big enough, and the powers granted therein were great enough, to permit us to fulfill the Nation's duty to the islands and islanders. President Taft, discussing our work in the Philippines, used this language three years ago:

It is said that there is nothing in the Constitution of the United States that authorizes National altruism of that sort. Well, of course, there is not; but there is nothing in the Constitution of the United States that forbids it. What there is in the Constitution of the United States is a breathing spirit that we are a Nation, with all the responsibilities that any Nation ever had, and therefore when it becomes the Christian duty of a Nation to assist another Nation, the Constitution authorizes it because it is part of National well-being.

That interpretation of the power of both the Executive and the Congress is exactly in line with the power that is exercised by both in dealing with this question of the public domain and the welfare of our people (applause). It would be a childish interpretation of the Constitution to hold that we as a Nation could act for the people in the Philippine Islands as was best necessary for their well-being, and yet within our own confines as a Nation would be prohibited from doing that which is necessary for the well-being and the welfare of our children and their children. (Applause)

The interpretation by Marshall gave vigor to the young Nation. He was not afraid of great responsibilities. He recognized that great responsibilities likewise meant the possibility of great mistakes, but that did not deter him from so interpreting the Constitution as to make possible the doing of the things that have been done. He was not of that class of timid folk who fear to exercise great power lest they may make a mistake. He was not that type, either as statesman or jurist, who because they do not see plainly written in the Constitution specific authority for the doing of every act necessary, therefore hold back and maintain that no such authority exists. This is the type of mind that prevents all progress. The timid man is often side by side with the dishonest man, because the timid man refuses to act from fear while the dishonest man raises the cry, "There is no power," in order to gain for himself that to which he is not entitled, or to escape Governmental jurisdiction or evade governmental regulation of any character. (Applause)

But we are not left simply to academic discussion as to whether the Federal Government has power to deal with the National domain. The Supreme Court has held, over and over again, that the Federal Government, acting through both the Legislative and the Executive branches, has the power to do what is best for the people's interests in handling the public domain. The Court has wisely and properly held that the power granted under the Constitution to dispose of the public domain carries with it every lesser power (applause)—that because Congress has the right to provide for the sale or the gift of land, it can likewise provide for the lease of land under such conditions and regulations as it may prescribe or as it may permit the Executive to prescribe. Therefore, the way is clear for the Federal Government to do whatever may be wise and necessary to protect the interests of the people in the use of the public domain.

Let us take another view of Executive authority. The chief Executive, above all other officers, is recognized and properly held as the great steward, the immediate custodian of the public property and of the people's rights. He is single-headed. He is one upon whom responsibility may be fixed. He is constantly at his desk; he is ever vigilant; he is constantly in touch with the things that interest the people and their rights therein; and as the custodian and guardian of the people's interests, it is to him that we must look for the protection of the public domain. It is not enough that the Executive shall simply carry into effect the specific language of a statute. He must go

farther than that; he must be as aggressive in his vigilance as are those who would take the public property without conforming to the law (applause). The Executive is required to see to it that the laws are enforced. Now, in the enforcement of law he often finds that while the paper record presented to him or to his subordinates by those who seek to acquire the public domain is perfect (there is no difficulty about making a land title good on paper) his duty is only partially fulfilled unless he goes behind the paper record; and when the last Administration took hold of the question of the land frauds, the Executive decided that there was but one way to enforce the law, and that was to see to it that the paper record conformed to the facts in every case presented (great applause). The greatest land frauds that have been perpetrated against the people of the United States were perpetrated because the public officers in years past did not make that direct, careful investigation of the facts and of the condition of the lands which would have enabled them to save for the people hundreds of millions of dollars of valuable property that in the last generation has gotten illegally into the hands of the big interests. (Applause)

The founders of our Republic recognized and understood the vital need of giving ample power to the Executive. It is well to recall what Hamilton wrote when defending the Constitution:

Energy in the Executive is a leading character in the definition of good government. It is essential to the protection of the community against foreign attacks; it is not less essential to the steady administration of the laws; to the protection of property against those irregular and high-handed combinations which sometimes interrupt the ordinary course of justice; to the security of liberty against the enterprises and assaults of ambition, of faction and of anarchy.

There can be no need, however, to multiply arguments or examples on this head. A feeble Executive implies a feeble execution of government. A feeble execution is but another phrase for a bad execution, and a government ill-executed, whatever it may be in theory, must be, in practice, a bad government.

Thus the Executive must be held responsible for much that is done in connection with the administration of our laws. Congress enacts the laws;

they may be faulty; if so, they may be amended. If they are faulty, it is the duty of the Executive to carry them into effect, but to recommend their amendment, alteration, or repeal; but under no circumstances is he fulfilling his duty if he sits supinely by and allows the public domain to be despoiled because the law is not as efficient as he thinks it should be. (Applause)

Much has been said in recent years regarding Executive usurpations. It has been held by those who objected to the new order of things—those who objected to that change in methods by which the public frauds were stopped—that the Executive was usurping powers not granted to him under the Constitution. Now, if it be usurpation to so enforce the law as to prevent dishonesty, fraud, and theft, then there has been usurpation (applause). But I as yet have failed to have presented to me a single instance of actual usurpation. The Executive is as much subject to the courts of the United States as is the ordinary citizen. If the Executive has transcended his power, if he has in his execution of law gone beyond what someone thinks is his power, then the Executive can be haled into court; and over and over again I have said to complainants who came to me when I was in office "All you have to do is to go into the courts of the United States, and if the Executive power that is being exercised is improperly exercised, there in that jurisdiction you can bring us to account." But no one has yet seen fit to bring such an action; and the reason is that there has been no usurpation of executive authority. (Applause)

There is a wide difference between simply being within the law and executing the law. A man may be within the law and yet do absolutely nothing to further the spirit of the law; like an engineer, he is on the track whether he is standing still, going backward, or going forward; but I take it that what we want in executive office is an engineer who stays on the track yet is constantly driving forward the engine (applause). We may rest assured that those who are seeking to acquire the public domain will not be idle if the Executive is standing still. (Applause)

That brings me again to a subject mentioned a moment ago, namely the relation of the Nation to the States; and the Executive here plays an important part. An example will show how the executives of both the Nation and the States should cooperate in working out any given problem: A great water course is a natural entity; the water-shed must be considered

as a unit—otherwise the people within that water-shed will not have equal justice done them in their right to the water. For example, the waters of the Rio Grande rise in Colorado; they cross the line into New Mexico; they then become the dividing line between Mexico and Texas. If we admit for a moment that the power to use and control all the water of the Rio Grande shall be left solely with Colorado because it rises in the great mountains of that State, then we instantly jeopardize the rights of all the people who live south of the Colorado line (applause). If the Chief Executive of the Federal Government had feared to exercise his power to prevent water-power sites and reservoir sites in Colorado from being taken exclusively by Colorado people; if he had been unwilling to exercise the power granted him by the Constitution, then the people below would have had just cause for complaint that the Executive instead of obeying the law was in effect a party to a violation of law in jeopardizing their rights. The only way in which that matter could properly be handled was for the Executive of the Federal Government to withdraw certain lands from sale or entry; and by so doing he made it possible for the people of New Mexico and Texas, and of the Republic of Mexico in conformity with the treaty made by the Federal Government, to have their fair share and just proportion of the use of that water.

The best way to deal with conflicting water rights between States is for the Federal Government to continue to hold every acre of public land capable of use in water development pending agreement with the various States as to how the lands shall be used, to the end that the rights of all the people of each water-shed, rather than the special interests of a few, shall be protected in the use and disposition of that great resource. (Applause)

What I have said in relation to water applies equally to the development of our coal, our phosphates, and our timber. The phosphates recently discovered in the West lie in four States. When the matter was first called to my attention by the report of the Geological Survey and the special report of Dr Van Hise, I was astonished to learn the conditions then existing in our country. Practically all of the mineral phosphates known in the United States were held by one great corporation, and over 40 percent of the products of the Southern mines were being shipped abroad to be used on the fields of Europe; and the same men were already endeavoring to get hold of the phosphate deposits in the West. Therefore I instantly made a

recommendation to the President, and he instantly acted on it and withdrew the phosphate lands (applause). Now that withdrawal was not an interference with the rights of the people of any of those four States, nor was it an act of usurpation, or an improper extension of Executive authority. It simply meant this: that we would hold, prevent the acquisition of those lands under laws not adapted to them, report the matter to Congress, and hold the lands until Congress provided a method for wise disposition of them (applause). And my recommendation was that the phosphate deposits of the country should be disposed of only under lease and with such conditions as would prevent export to foreign lands (applause). We need every ton of our phosphates for our own use. (Applause)

So, if you trace the actions of the Executive and of Congress in dealing with the public domain, you will find that wherever there has been a vigorous execution of law coupled with recommendation of further legislation looking to the welfare of all of our people, there we have made advance along lines that will promote the development of our country in future years; and that wherever there has been laxity in the enforcement of law, wherever we have allowed the interference of big business interests to interrupt the enforcement of law as it should be enforced, land frauds there have crept in and in those conditions we have found the big interests getting control of more than their fair share of the resources of the public domain.

Sometimes we have been accused of being unfair to the big interests. We have been accused of assailing these interests simply because they were big; and we have been charged with raising ghosts to frighten the people, and naming those ghosts water-power trusts, timber trusts, land trusts, or coal trusts, when in reality there was no danger of trust development or of monopolistic holding of these resources. And yet, my friends, if you trace back the history of the acquisition of the public domain you will find that in every instance where there has been a failure to strictly enforce the laws the special interests have slipped in and have gained control of the resources of the public domain. They have never been idle. We ourselves have been indifferent, we have been negligent; and it is not for us now altogether to blame the beneficiaries of our neglect, but we must blame ourselves—and must blame our representatives in office now if by any chance they permit a return to the old conditions. (Applause)

The power of the Executive and of Congress is ample to do all that is necessary to protect the public welfare and the common good. There must be no backsliding in what has already been so splendidly started. We must see to it that our representatives, both in the Senate and in the House, are men who will take a long look into the future—men with imagination. Men with enthusiasm? Yes! Nothing great has ever been accomplished without enthusiasm and without imagination (applause). And we want practical men who will lead us, as I said in the beginning, step by step, to better things. Thus and thus only will the Federal Government exercise to the full the powers granted under the Constitution, and thus and thus only will the people of this country safeguard their property rights, their personal and their political rights as well, and hand down the great heritage that has come to us not only unimpaired but in better condition than we received it. (Great and prolonged applause)

PRESIDENT BAKER—Ladies and Gentlemen: Now that this subject has been so ably opened by Mr Garfield, we are going to call upon another man who has been militant in the work of Conservation—an Ex-Governor who is even more active as an ex than he was as Governor, a sort of characteristic, these days, of prominent men (laughter). I am sure you will have great pleasure in hearing from Ex-Governor George C. Pardee, of California. (Applause)

EX-GOVERNOR PARDEE—Mr President, Ladies and Gentlemen: I hope the Chair will forgive me if I differ from him very radically in one statement that he made, to the effect that *all* of us who have been things (laughter) are now more active than we were when we were things. (Laughter)

I sat here today in this vast Auditorium and saw thousands of men and women and children, gathering to do honor to the man whom we, in common with the rest of the world, consider to be the greatest American now alive (great applause). When I saw those thousands of people filling this great Auditorium, row on row and tier on tier, until the heads of those standing in the topmost row touched the very roof, I thought to myself that

the activities of him who *was* in office are being only continued since he left the office which he filled to our entire satisfaction. (Applause)

I come here this afternoon to discuss the very able paper so well presented to you by him who was once Secretary of the Interior, in the cabinet of the President of the United States (applause); and I hope you will not consider it presumptuous that I should attempt to discuss that very able paper. Mr Garfield was good enough to furnish me with a copy of his address several days ago, and I am free to confess to you that I have given it prayerful consideration and that I can find nothing in it to discuss (applause), because it calls a spade a spade and a thief a thief (applause); and with both of those propositions I have no doubt the ladies and gentlemen here assembled will thoroughly and totally agree. (Applause)

Every now and then we hear of some poor, miserable fool sent to the penitentiary for crimes and frauds against the land laws; but will any one be kind enough to mention to me the name of any principal in such crimes and frauds who, with shaved head and striped suit, is looking through the bars of the penitentiary today? I take it that you will agree with me that the time has come when the rights and duties of the plain American citizen should be again placed within his grasp, and that the rights and duties of the very meanest of us should be regarded as equal to those of the most powerful and the richest and most influential. Our representatives have too often forgotten the fact that they represent the great mass of the people, and that they represent unborn generations of American citizens—that they are plowing legal furrows and building legal fences and making things ready for the coming generations of Americans who will fill this great land of ours.

So when I speak of my own State of California, and say that its people have been robbed and plundered and pillaged; when I say that its government has been debased and corrupted; when I say with shame and with blushes that my native city of San Francisco has been humbled and shamed into the very dust by the corrupting influences of men and public-service corporations who, with us as their benefactors, have turned and stung the breast that warmed them into life; when I say these things I have but to call to your attention conditions which have existed in almost every large city, in almost every State of this Union. (Applause)

Like Mr Garfield, I do not find it in my heart to blame the men who have taken advantage of our laxness; I cannot find it in my heart to blame the two men who own each over a million acres of the best timber land in the State of California for having taken advantage of the laxness in administration of the law in times past—not of the law itself, for the law has been good, and if it had been administered as it should have been administered these two men could not have owned a million acres apiece of the best timbered land in the State of California (applause). But who of us has not heard—in times past more than since the time of Theodore Africanus (laughter)—who of us has not heard those who, perhaps with a selfish interest, have sneered and said, "Well, we're all a little crooked, and why should we take exceptions to the man who is a little more crooked?" when the question of frauds against the land laws was in discussion? I take it that the officials who had those matters in charge should be, as Mr Garfield has so well said, ever vigilant within the law to do those things which the law does not prohibit and not wait for the prods and stings of outraged public opinion that compel them to do the things which they should, in common honesty to the people whom they represent, perform and do for the protection of you and me and your children and my children. (Applause)

I listened yesterday afternoon with mingled feelings to the statements of the gentlemen who four short years ago I would have hailed as brother governors. I heard some most violent utterances concerning the feeling of the people of the Pacific-coast States in regard to State rights. One good brother governor said that 95 percent of the people of the Pacific Coast were in favor of State rights. We had in California on the 16th day of August (less than a month ago) a direct-primary election. At that election there was nominated as the republican candidate for Governor of the State of California Hiram W. Johnson. Out of something over 200,000 votes cast he received over 100,000 votes. His next nearest opponent received 55,000 votes. Mr Johnson's campaign was made on a platform containing three principal planks—Roosevelt, Pinchot, and Conservation. (Great applause) If it be necessary, I can read a telegram from Mr Johnson in which he assures me that he has not yet recanted from his old Rooseveltism, his Pinchotism and his Garfieldism, or his Conservationism (applause); so I think I am safe in saying that instead of 95 percent of the people of at least one Pacific-coast State being in favor of State rights, I am entirely within

the bounds of conservative statement if I say that 80 percent of the people of California have not forgotten the Civil War and remember that the ghost of State rights was laid so many fathoms deep at that time that no ingenious argument of any Governor from the Northwest, the Southeast, or any other portion of this country can revive it and make it walk. (Great applause) If necessary, I could read from this little packet that I have in my hand a portion of a letter from the Grand Master of the Patrons of Husbandry (that is the Grange) of the State of Washington (applause), whose Governor addressed this Congress yesterday afternoon and declared himself and his State as both being entirely in favor of State rights. In that letter the Grand Master of the Patrons of Husbandry of the State of Washington, whose Governor addressed this Congress yesterday afternoon, says that he represents 19,000 of the people of Washington, and that no man has the right to represent them upon the floor of this Congress and say that they are in favor of State rights (great applause). And in this little packet I also have a telegram from the Conservation Association of the State of Washington, signed by its president, which says that its membership in the State of Washington is not in favor of State rights (applause). So, our good southern brethren having forgotten the bloody past (as my Yankee blood has forgotten it), having come again into the Union and declaring themselves loyal sons marching under the American flag and having forgotten the obsolete doctrine of State rights, I think I am safe in saying that the people of the North and Northwest have not changed places with them, but that they believe that the Federal Government should keep and administer the things that belong *to all the people* of the country. (Applause)

We have in California, my fellow-citizens of other States, a great deal of *your* property. We have several millions of acres of National forests that belong to you. They cannot belong exclusively to the people of the State of California until the people of the United States, to whom they belong, give them to us. And I thank God that the National Government, representing the people of other States, has not given those millions of acres of National forests in the State of California to the State of California. For if it had, just as sure as you are sitting here, those acres would have been given over into private ownership, just as thousands and hundreds of thousands of acres of the public lands which were given to the State of California have been squandered with a prodigal hand and given to men who have not obeyed

either the letter or the spirit of the law conveying and granting to them those hundreds and hundreds of thousands and millions of acres of the public lands. (Applause)

Let me instance one case. The Oregon and California Railroad begins at Portland and runs south toward California. The California and Oregon Railroad begins at Sacramento and runs north toward Oregon. They meet somewhere north of the Oregon line. They are both adjuncts of the Southern Pacific. When those roads were contemplated, the Government, by an act of Congress, donated to them 6,000,000 acres of land, much of it covered with as fine timber as grows out of doors—I bar none. In the act of Congress donating that land it was specified that the land should be sold in 160-acre tracts for \$2.50 per acre to all actual settlers who might apply therefor. Was any of it sold to actual settlers? A very few acres of it. Half of the 6,000,000 acres was sold, however, in large tracts to land speculators, to timber corporations, and to people of that kind and class, for \$5.00, \$10.00, \$15.00, \$20.00, \$30.00, \$50.00 an acre. And when the Southern Pacific was brought to bar and asked why it hadn't lived up to the letter and spirit of the law, it said it had. And then we asked, "How do you make that out?" And it said, "Why, only a few actual settlers have applied for the land." And we asked, "Haven't people gone there and attempted to buy that land of you in order that they might settle upon it?" "Oh, yes, but they are not actual settlers." "Why not?" "Because we construe the words 'actual settlers' to mean those persons who had actually settled in that country before the act of Congress was passed." (Laughter and applause.) And when, at the Sacramento session of the National Irrigation Congress, Mr E. H. Harriman was asked why his company was holding 3,000,000 acres of that land grant, he said, "For future generations." And everybody laughed.

Now, let that sink into you. The absolute arrogance, the indecent indecency of that kind of a proposition ought to make the blood boil in the veins of every American citizen who is face to face, or who was seven or eight years ago face to face with the proposition whether or not the American people were to rule themselves or whether they were to continue to be ruled by the "big interests," as Jimmie Garfield puts it. (Applause)

The sun rises every morning, three hundred and sixty-five days in the year, in California; it reddens the cheeks of our girls; it makes our boys strong

and healthy; it brings the gold to the oranges that hang upon our trees. And for all these years we have been thanking God for the rising of the sun in California. "The gentle rain from heaven" has fallen alike upon the just and the unjust out there in California—upon those who deserve to be rained on and those who do not deserve to be rained on (laughter). And all these years we have been thanking God for the gentle rain that falls from heaven. But yesterday as I sat here in this great Auditorium and listened to the Governor of Montana tell what Montana had been doing for this Nation, I began to think (I do not wish to be irreverent in saying it) that we were under no obligations in California to God for the rising of the sun or the falling of the rain; but that we were under great obligations to Montana (laughter and applause) for all the good things that belong to California and Californians. And as my good friend, Governor Norris (to put a name to him) was telling his lurid history of Montana's great doings, I couldn't help but think that as an American citizen some of the things that lie in the State of Montana belong to me, belong to you, belong even to those who live on the hook of Cape Cod or away up in the northeastern corner of Maine or down on the tip of Florida; that those things which belong to the people of the United States even in Montana belong to us all, and that Montana has no exclusive right to them until our representatives in Congress give them to that State, and I am one of those who pray God that it will be a long time before the State of Montana gets from us the exclusive right to, and ownership of, those things that are ours. (Great applause)

Some time ago a good friend of mine, who has never denied when I have charged him with receiving \$20,000 per annum (and by the way, he is a delegate from California to this Congress) as the chief counsel of one of the power trusts of California, said to me, "Oh, how the President is usurping the powers of the Government! Isn't it awful?" But I never could see anything very awful about it when Roosevelt and Garfield and Pinchot and the rest of them were hustling around trying to keep my friend's corporation from stealing from us of California the few things we have left (laughter and applause); nor have I forgotten that, before the time of Roosevelt, Garfield and Pinchot, the corporations represented by my friend did not believe in State rights. But since the time of Roosevelt, Pinchot and Garfield they have begun to sing a different song. That song is State rights. Nor have I forgotten that my friend used to be and still claims to be one of

the most hide-bound republicans that mortal man ever looked upon (laughter and applause). Now he says that the rights of the people of the States are being pillaged and plundered and robbed away from them. I speak again for my State of California when I say that if there is anything in the State of California that the National Government has not nailed down that has not been stolen, I would like to know what it is. (Great laughter and applause)

There are, as you heard Mr Herbert Knox Smith say here on this platform an hour ago, four great power corporations in the State of California. That is so; but there are practically only two power trusts in the State of California. When the Government declared its intention to hold on to the few power sites that are left in the State of California in the National forests, all of a sudden these power trusts wanted all the water-power of California developed in the interests of the people; and they can't say it fast enough or often enough (laughter and applause). But, as you heard Mr Smith say, they have developed and are using only half of the power that they already have in their possession. So when they get gay around where I am, I generally say, "Well, that's all right, but go on and develop all the power you have got *now*; and after you have got that developed, then we'll talk about giving you some more; because I know just as sure as you fellows get an opportunity to lay your hands on any of those power sites in the National forests you'll steal *them* and put them in cold storage, and you'll make my children and their children, so long as there are any children in the State of California, dig up the last dollar that they have to pay you for the necessary electric current to do their business during the next century and the century after that until the end of time in California" (applause). And I, for one, as I say to them, while I am somewhat hardened and calloused by being robbed myself, don't want my children or their children to be robbed into the poor-house and the penitentiary by anybody's power corporation (applause). Therefore I hope and pray that those gentlemen who are so apprehensive that the people of the country will not get, unless they get it through the States, the right to use the things that belong to all the people of the country, will pause until the State of Montana, the State of California, the State of Washington, and all the Pacific-coast States, at least, if not the rest of the Nation, are governed by the *people* of those States and not by the public-service corporations. (Great and prolonged applause)

CHRISTOPHER G. HERR—Mr Chairman: The State of Washington having been mentioned, I wish one minute to speak in behalf of that State.

President BAKER—Is the Gentleman a Delegate from the State of Washington?

Mr HERR—Yes. It has been stated from the platform that the State of Washington believed in State rights. I want to contradict that. As one of the Delegates of the State of Washington, I want to declare my belief that not only the Granges of the State of Washington, as Brother Pardee has stated, but the majority of the citizens of that State, will repudiate any such sentiment coming from anyone in this Congress (great applause and cheers). I want to say that the State of Washington is peopled in part by 25,000 former residents of the State of Minnesota, and that they have full confidence in the National Government—they have full confidence in President Taft, they have full confidence in your Senators Nelson and Clapp, and in Congressman Stevens and the other congressmen of the United States; and I consider it an insult to the Congress and the President of the United States to say that they will not treat the people of the State of Washington as they should be treated. I want to say to you, Ladies and Gentlemen, that the State of Washington will keep step to the music of the Union. (Great applause)

President BAKER—After the next address on the program, any further discussion of the subjects presented will be welcome.

It was gratifying to hear from California through the voice of Ex-Governor Pardee. One of the fortunate features of this Congress is the presence of men of prominence and influence from all sections of the country. Not merely the North and the East and the West are represented, but the sunny South; and we will be pleased to hear from a representative of the great State of Louisiana, who has always been deeply interested in Conservation, and is no less competent to speak on the subject now than when he wielded

the power of Governor of that commonwealth. I have great pleasure in introducing Ex-Governor Newton C. Blanchard. (Applause)

EX-GOVERNOR BLANCHARD—Mr President, Ladies and Gentlemen of the Congress: I am not on the program for a formal address, but I am here to supplement and endorse and support the admirable address delivered you a little while ago by Ex-Secretary Garfield. (Applause)

The times change, and men's opinions seem to change with them. On yesterday, in this Auditorium, I listened to a number of western Governors preaching the doctrine of State rights. For many years prior to the fateful year of 1861, and for four memorable years following it, the question of State rights was forcefully discussed in the forum of the Republic, and afterward practically settled on the battlefield (applause); and we of the South, who went down in that struggle to determine whether these rights of the States were paramount to the authority of the Federal Government, accepted the situation in good faith (great applause)—and we are now marching side by side with the North and the East and the West in that grand procession of progress that makes for the might and power of our great Republic. (Renewed applause)

It seems strange to a southern democrat like myself (applause) that "a voice should come out of the West" (laughter) telling us that this movement for Conservation must be abandoned by the Federal Government and relegated to the tender mercies of the western States (laughter and applause). Gentlemen of the Congress, was the question of State rights, the *real*, genuine doctrine of State rights, behind that demand? No; everyone of you know that it was not. It was a mere pretext; and the history of all nations is full of examples where strong men, having risen to ascendancy and ruling power and wanting to do something not exactly right (some usurpation of power or act of tyranny), first sought a pretext to justify it (applause). Why, then, does this voice come out of the West—a country that in the time preceding and following 1861 was known as "the wild and woolly West," and out of which at that time came not a whisper in advocacy of State rights? Why, now that the "wild and woolly West" has gone and magnificent commonwealths are there, now for the first time comes from the West, in former renegade garb or present robe of splendor, the cry that State rights must dominate the Conservation of the natural resources of the

country? Gentlemen, some years ago a great citizen and soldier of our Republic was the candidate of a political party for the high office of President of the United States at a time when the tariff was the dominant issue, and becoming involved in the intricacies and embarrassing problems of the tariff, he declared, "the tariff is a local issue." Listening to the western Governors last afternoon, I perceived the same idea arising again, only in a different form; for the western Governors would make State rights a local issue.

The natural resources of the United States belong to all the people (applause), not alone to those who happen to live in the States where what is left of the public domain is principally situated today; you and I have just as much concern and interest and proprietorship in the natural resources on and in and springing from the public domain in Wyoming, in Montana, in Idaho, and in other western States, as have the people of those States themselves (applause). Gentlemen, as has been well said already during this Congress, the smaller the community the easier it is for special interests to control it; and that is the reason for this demand that the Conservation of the natural resources in the western States should be turned over to the States themselves. If you want Conservation to amount to anything—if you wish it to go forward in the fullness of development so that what is left of the public domain, of the coal lands, the phosphate lands, the oil and gas lands and the forests belonging to the United States may be preserved and conserved and utilized without present waste and handed down to our children and children's children without exhaustion, then I say the power that should lead in this movement is the mighty power of the Federal Government. (Applause)

When the distinguished and able gentleman who occupies the executive chair in the State of Montana was speaking yesterday, he claimed for his State "the earth and the fullness thereof" in respect to the Conservation of natural resources. He claimed that the movement there had antedated anything done by any other State or by the Federal Government, and to hear his eulogy of what Montana had done in this respect and his absence of expression as to what the Federal Government had done there, one might think, to use the vernacular of the day, that Montana was "the whole cheese" (laughter) in matters of conservation. And yet, when I met the gentleman today and asked him if the Federal Government had not been

doing considerable work in Montana and expending large sums of money to irrigate the arid regions of that State, he admitted that it had. I asked him if the Federal Government had not expended many times more money in doing just that kind of Conservation work in his State than Montana had, and he admitted that it had. I asked him if what the Federal Government had already done in the way of irrigating the arid regions of his State and the projects now under way would not when completed yield to the farmer and the husbandman many hundreds of thousands of acres of valuable land, and he admitted it would and that the aggregate would be more than 600,000 acres (applause). That is what the Federal Government has done and is doing in one western State; and yet that same Governor, and others from the West, advocate that in the matter of Conservation the Federal Government should take a back seat, and permit the States to take the lead in Conservation.

Gentlemen, you heard today from the lips of Theodore Roosevelt a truth that struck me most forcibly, and that was this: It is not so much the question as to who shall take the lead in the matter of Conservation, whether it be the power of the States or the authority of the Federal Government, but which of these powers is best equipped and most able to keep what remains of the public domain and the natural resources from falling into the hands of the special interests and the monopolists (applause). Some of those western Governors, when the imputation was made that if the natural resources were turned over to the States in the manner proposed by them the special interests might handle their legislators, grew virtuously indignant; and yet all of us remember that it has been charged time and time again—and I think no one will have the temerity to deny it—that powerful interests with unlimited money have put forward their own selections for the high office of Senator of the United States and elected them (applause). That has been done repeatedly in the past; and is anyone here bold enough to say that even now there does not sit in the Senate of the United States men from the western States who owe their election to that position through the instrumentality of money? (Applause) No; that is true; and everyone of you knows it is true. If the Legislatures—and I do not mean to imply or to charge that the Legislatures of those particular western States are any more corrupt or more subject to the blandishments of corporations and men of means than the Legislatures

of other States, whether they be North or South or East or West—can be induced through those instrumentalities to elevate men to high position, then I say those Legislatures can be controlled by the same means in other respects; and all of us know that special interests have always out a grabbing hand for what there is in the way of coal lands, in the way of water-power sites, in the way of phosphate lands and oil and gas lands. So I say, gentlemen of the Congress, we had better leave this matter of Conservation in the hands of the Federal Government to lead in this great work wherever the Conservation relates to the natural resources springing from the public domain. I am here to advocate that first; and I am here to say that in other respects, where the State authority finds jurisdiction, there should be cooperation between the States and the Federal Government. (Applause)

We have heard much from these western Governors in their speeches last afternoon relative to the waters in the rivers of their States, and the position was taken that the waters belong to the States. Flowing through the public domain, the land and the water-power sites would belong to the Federal Government, and where that is the case there is good ground for cooperation; but I am far from admitting that those waters belong to the States. There are some decisions of the Supreme Court that so declare, but such decisions were made by the courts under peculiar circumstances and facts differing from the circumstances and facts set before us in the matter of Conservation. Take the great Mississippi; to whom does the Mississippi river belong? Do its waters belong to the States through which those waters flow? Why, don't you know that every drop of water precipitated from the clouds, except that which is taken up by evaporation, every drop of rainfall from the top of the Alleghenies to the summit of the Rocky mountains finds its way through the innumerable channels and smaller streams to the great main trunk that we call the Mississippi river? Don't you know that it is the receptacle for the drainage of half of this great Republic of ours, that much of even the waters that fall in the western part of the great State of New York find their way into the channel of the Mississippi? All of the water thus gathered into the main channel flows by the cities of all the States from Minnesota down to Louisiana, my own State; and all of that water flows through the State of Louisiana to find lodgment at last in the Mexican Gulf. Now, does all the water thus garnered from this immense watershed to flow

through the State of Louisiana belong to the State of Louisiana? If so, we *don't want it!* (Laughter and applause) It fell on these great western States, and too much of it comes down upon us, and we have had a great struggle, extending through many years, to keep that water off our land (laughter). I have known one great flood in Louisiana to cause destruction to the extent of ten millions of dollars. The State of Louisiana alone has expended, by State taxation and levee district taxation, more than thirty millions of dollars since the War in keeping the waters that fell upon your territory off our fertile lands (applause); and not being able to perform the herculean task ourselves, we have appealed, in season and out, to the Federal Government for aid, and a liberal hand has been extended to us. (Applause)

I was for years in Congress from Louisiana and for years a member and chairman of the committee on rivers and harbors of the House of Representatives, and I had to deal with this question. When I went first to Congress the idea prevailed there that the Federal Government had no constitutional authority to appropriate and expend money on Mississippi river except in aid of navigation; it was admitted that could be done under the commerce clause of the Constitution, but Congress denied that it owed any other duty to the river. Myself and others from the lower Mississippi valley, the lands of whose constituents were flooded every now and then by the great river, contended that Congress owed a two-fold duty to the river: to improve its navigation, and to prevent the waters from remaining a terror to those who lived in its lower valley (applause). Congress admitted it owed the first duty, but asked where there was any constitutional authority for the appropriation of public money to redeem private property from the flood and ravages of the river; and it took the representatives and senators from the lower valley States many years—I know I worked at it myself for ten years, in season and out, as a member of Congress—to demonstrate that the Federal Government owed it to the great river to prevent its floods as well as to improve its navigation. In answer to the demand for constitutional authority we cited a principle of law, recognized alike by the civil law system and by the common-law, which long antedated the Constitution of the United States, a principle embodied in a Latin maxim, "*Sic utere tuo ut alienum non laedas*"—so use your own that it shall not become an injury to others (applause). And we asked in that connection, "Who owns the Mississippi river? Does the Federal Government own it? If so, it is its

property as a great feature of our country; and if the proprietorship of the river is in the Federal Government, then should not the Government so regulate and control its own that it will not injure or prove a detriment or damage to those who live in the lower valley?" (Applause) And that argument won.

Prior to 1892, large appropriations were made by Congress for the Mississippi river, all of them with a proviso that none of the money should be expended for the purpose of preventing the floods of the river; and not a dollar was available for the repair and construction of levees. That was the situation in 1882 and on down to 1892, when the argument that the river belonged to the Federal Government and it must so regulate and use it that it should not be a damage and a hurt to us in the lower valley prevailed; and in the river and harbor bill of 1892, at a time when I was chairman of the committee, the Secretary of War was authorized to expend \$10,000,000 on the lower Mississippi from Cairo to the Gulf, and the restrictions and provisos that had hampered the Mississippi River Commission theretofore in the expenditure of money for the two-fold purpose of improving navigation and preventing floods were removed (applause). We wrote these limitations all out; Congress had been educated up to the point where it recognized the second duty it owed to the great river in preventing its floods. The bill passed, and the Mississippi River Commission allotted \$6,000,000 of the \$10,000,000 for levee construction and repairs (applause). We followed this two years later by another bill using the same phraseology and appropriating \$9,000,000 more, and these two great bills, carrying \$19,000,000, with no restrictions on the expenditures for the prevention of floods in the river, have given us along the lower river the greatest and finest levee system ever known in any age or on any river in any country—1350 miles of levees that stay the floods of the Mississippi so that a general flood in the river is a thing of the past; and on every mile of our 1350 miles of levees on the two banks of the river *is the stamp of the Federal Government*. (Applause)

And yet they tell you that these waters do not belong to the Federal Government? They admit that they belong to the Federal Government for purposes of navigation. Congress is committed already to the principle that the waters of the river belong to the Federal Government, because Congress has undertaken to help us to keep those waters off of our lands. But I go

further than that; I agree with my distinguished friend Mr Garfield that the jurisdiction of the Federal Government extends, where the navigable waterways of the United States are concerned, far beyond the point to which they are navigable; it extends to the headwaters of those rivers, and for the very good reason that if the jurisdiction of the Federal Government did not so extend, then where these rivers take their rise some of these western States might undertake to divert from the great Mississippi channel the water needed to supply that river with enough water for navigation purposes. Every river, therefore, must be treated as a unit (applause). That is the view we take of it in the South; and in taking that view we hold to the National idea that water, being one of those natural resources which needs conservation in respect to its greater and wiser use, ought to be controlled by the Federal Government. Water is one of those natural resources that man can do nothing to add to or diminish in quantity; the snows and the rains are the result of great cosmic action—and fortunate it is that such is the case, for past experience in this country shows that if man could diminish the supply he would long since have done so by his neglect and his wastefulness. (Applause)

I have spoken long enough. I wanted to supplement, from the standpoint of the South, the admirable remarks made by the distinguished Governor of Mississippi on last afternoon. We of the South are hand in hand with the Federal Government in this great question of the Conservation of the natural resources; and we look to the Federal Government to lead in that movement (applause). At the same time I repeat that this great movement, so auspiciously inaugurated by Theodore Roosevelt and Gifford Pinchot (applause), needs for its full consummation and for the realizing of the greatest benefits possible the cooperation—with the Federal Government leading—of the Federal Government, the States, and all the people (applause). When we shall have brought these three great agencies into harmonious action looking to proper Conservation, then will our country grow greater even than it is now in all that goes to make up the might and glory of a great nationality of the earth; our country will then continue to present the example of a great continental republic possessed of every variety of climate and production, whose people are as one again, loyally devoted to the perpetuity of the Union, fearing no foreign foe, following the pursuits of peace, serving God according to the dictates of conscience and solving practically the great problems of self-government. (Great and prolonged applause)

[In the course of the foregoing address, President Baker surrendered the Chair to Professor Condra.]

Chairman CONDRA—Ladies and Gentlemen: Before continuing the program, a few announcements will be made.

EX-Governor PARDEE: I again announce that the Committee on Resolutions will meet at the Saint Paul Hotel this evening at 8 oclock in Room 534. Those having resolutions will please write them out, sign them, and hand them in.

Several announcements were made on behalf of State delegations.

Chairman CONDRA: In place of Honorable B. A. Fowler, of Phoenix, Arizona, who was to speak on "Water as a Natural Resource," I call upon a man who has done much for the advance of irrigation, and who organized the first National Irrigation Congress, Mr William E. Smythe, of San Diego, California.

Mr SMYTHE—Mr Chairman, and Ladies and Gentlemen of the Congress: I am called upon at very short notice to speak for our distinguished president of the National Irrigation Congress on water as a natural resource. I need not remind you how valuable this resource is. Some years ago I went to the White House in company with a cabinet officer to confer with the then President of the United States concerning a mooted irrigation question. Secretary Moody presented me to President Roosevelt, saying that I was a democrat interested in the subject of water; whereupon the President turned to me with a smile and said, "What! a democrat interested in *water*?" (Laughter) "Yes, Mr President," I said, "for democrats have sense enough to know that in a country where it seldom rains water is too valuable to drink." (Laughter)

Water is so valuable that we want to guard it carefully as a natural resource. I have but a moment at my disposal, and I am glad to take the advice of the President of the United States who yesterday told us to come out of the clouds, get down to brass tacks, and talk business. He asked us to say what we mean by Conservation, to tell what are the evils that we want to remedy, and explain how we propose to remedy them (applause). In a word, the evil that we want to remedy in the arid States of America is the great evil of permitting men to make merchandise of the melting snow and the singing brook (applause). I stand here to say that no man can possibly be good enough to own the water which another man must use in order to live (applause). It may be that private enterprise can be employed in the form of a construction company to build the reservoir and the means of distribution; but in that case, after our people have paid for the work, and paid for it once and twice and three times, then the Nation should answer our prayer, "Let my people go."

We should have joint ownership of land and water. Today we have a magnificent construction company at work in the seventeen States and Territories of arid America; the name of it is "The United States of America,

Unlimited." (Applause) That construction company turns the work over to the people at actual cost, with ten annual payments, *and without one dollar of interest* (Applause). If the National Government can do that with irrigation, it can do so just as wisely with power; and if it doesn't seem wise for the National Government to do it as a matter of public enterprise, then give us a form of construction company; but in the end, in the day of our children and our children's children and our remote descendants, in the name of God and in the name of humanity, let the people own the water which is essential to their existence. (Applause)

Just one word further. I stand here to endorse what has just been said by one of the few real men whom California ever had the good fortune to put into her Governor's chair (great applause). California is not for State rights; *that* doctrine was trampled to death fifty years ago under the feet of a million armed men. Yesterday it raised its head and stretched out its weird arms seeking to grasp the remnant of the natural resources and turn them over to exploitation by private monopoly. But that will not be permitted. I am here, my friends, to say to you, as Governor Pardee has said, that in this great controversy—the most momentous which has arisen in this country since the close of the Civil War—California and the Pacific slope, and I believe all the splendid States of the Rocky mountain region, stand with that fine young American statesman who during the past few months has thrilled this nation in his fight to save the resources of the people to all the people for the benefit of all the people; that young man who said at Denver the other day that it is more important to help the small man make a living than to help the big man make a profit; that man, who has sounded the highest notes since Lincoln, who has declared that he is in favor of Government by men for human welfare and against Government by money for profit—we stand first, last, and all the time with Gifford Pinchot. (Great applause)

Colonel T. H. DAVIDSON (Delegate-at-large from Minnesota)—Mr Chairman: I noticed scattered through the program of this great Congress the words "General Discussion." We have not limited the time to be occupied by speakers. I now move you, sir, that under the head of "General

Discussion" a delegate shall be entitled to occupy only five minutes, and shall not speak a second time on the same question.

Chairman CONDRA—The rule adopted today fully covers the point, though it has not been put in effect this afternoon. We have two days, perhaps three, for full discussion, and the time will be limited under the rules which will govern tomorrow.

The last speaker on the formal program is one who has been greatly interested in this movement and closely associated with Mr Pinchot. I have pleasure in introducing Mr Walter L. Fisher, a Vice-President of this Congress and of the National Conservation Association and President of the Conservation League of America.

Mr FISHER—Mr Chairman, and Ladies and Gentlemen: I would not take any of your time this afternoon were it not that I, too, have felt the appeal of President Taft for concrete and practical suggestions as to how to solve some of the more difficult of the problems of constructive statesmanship presented in the Conservation movement. The particular point on which I wish to make a suggestion is the relation of the States and the Federal Government to the question of water-power grants.

This question, it seems to me, has been allowed to assume a phase entirely unjustified by the facts. There is, in my judgment, not only no necessary conflict between the interests of the State and the Nation, but there is every incentive for practical cooperation between State and Nation on this matter (applause); and in my opinion the question can never be rightly settled until there is just that cooperation. (Renewed applause)

The Federal Government is the natural agency to which we must look for many of the things which are essential to a solution. There are two phases of the problem, one involving a question of law and the other a question of public policy. As to the strict legal right, it must be apparent that on any stream where the Federal Government owns the riparian property, or on any stream which is navigable in fact or in law, the consent of the Federal Government is absolutely necessary as a pre-requisite to the construction of any water-power works. For myself, I believe that the power conferred by the Constitution upon the Federal Government with relation to interstate commerce absolutely carries the power to make such conditions in any

permit to erect a structure in a navigable stream as the Federal Government may believe it wise policy to insert. The power to make or to withhold the permit, under all the decisions of the courts which have in any way touched that question, implies the power to impose conditions to the permit. There are, I know, those who disagree as to this proposition; but even they will agree on the broader question of public policy which underlies the whole subject. When the Federal Government undertakes the improvement of a navigable stream, it rarely if ever happens that it does not thereby either create water-power or increase potential water-power already existing. It is evident, therefore, that those riparian owners who own existing water-power grants are directly benefitted by the improvement in the navigable water. Whenever the Federal Government protects the headwaters and the water-shed on which the stream depends, it is conferring a direct benefit upon the owners of water-power property along the line; and so with all the other improvements.

You have heard the eloquent Ex-Governor of Louisiana explain what the interest of that State is in the intervention of the Federal Government in the regulation of the Mississippi river. There are few places throughout this country where the owners of water-power grants and those who are interested in all the other uses of flowing water have not appealed to the Federal Government for financial aid or for assistance not financial which that Government alone can effectively render. It must be apparent that in rendering that assistance the Federal Government creates property of value, or enlarges the money value of property already existing. No hardship, then, is done if the owners of this property are required to contribute to the original cost. Not only so, but there can be no justice in the proposition which requires the taxpayers of the United States as a body to pay the cost of the improvement or the protection of any stream when as a matter of fact the people who own the property immediately along the stream will get, in direct money value, a larger benefit than the cost of the improvement.

There are many reasons besides these why the Federal Government *must*, in the very nature of things, be the effective agency to do many of the things which the States can never effectively do, no matter if the whole subject were turned over to them this afternoon. On the other hand, I wish to call attention to the fact, which I believe to be established by experience, that whenever a local community is once aroused to an intelligent appreciation

of its interests and its rights, that local community will better and more effectively regulate local service and local rates than any more remote governmental agency whatever. Herein lies the advantage of local home rule. Now, I am not talking about railroad rates connected with interstate commerce, or about other things which affect more than the local community, but about those things which affect merely particular localities. If a water-power company starts in alongside of a great industrial community and that community is built up so that its industries depend on it, that community itself, once thoroughly aroused and intelligently educated upon the question, will far more effectively regulate those rates in the interests of the public, while at the same time dealing fairly with the corporate or private interests involved, than would the State or the Federal Government. That seems to me a broad, practical proposition which experience has justified.

Now, let us apply the principle to the water-power situation. And my whole purpose in speaking is merely to call the attention of this Congress to a method of treating this question, which will, in my opinion, meet both situations. It is not a novel suggestion; in one of the very last of the water-power grants made by Secretary Garfield, the essential provisions of it were at least hinted at and a preliminary provision made. In my humble opinion, the Federal Government should control the water-power grants on streams that are navigable or where the Government itself controls the riparian property. It should make grants for definite periods of time and should provide for compensation. That compensation as a broad, general rule should be applied to the improvement and protection of the stream and the watershed from which the water-power has been derived, or to other streams and watersheds of like character, for all uses of the water, whether for irrigation on the one hand or for water-power on the other. There should be periodical readjustments of the rate of compensation. In the beginning, and especially in an experimental enterprise, the rate of compensation should be exceedingly low. There should be, as President Taft himself said here in his speech, a readjustment of the rate, say every ten years; and the person or the corporation invited to invest money should be given proper protection in that readjustment. Capitalist and industrial pioneer should be treated not only fairly but liberally, that vigorous development may result. On the other hand, such a grant should contain this provision, or be subject

to this fundamental legal limitation, that the grantee, by acceptance of the grant, acquiesces and will acquiesce in any reasonable regulation of the service and of the rates which may be charged the public that may be provided by the State or by any delegated agency of the State. In that way, the thing in which the local community (the State, its municipalities or minor communities) has the greatest interest will be amply protected and left free to act in its own interest.

Now, what will be the result practically? At the end of the first ten-year period the question of readjusting the compensation will arise. If the local government has not adequately protected private interests, if it has not regulated the rates so that the people are obtaining power upon fair terms and the corporation restrained from making extortionate profits, all the Federal Government will have to do will be simply to increase the compensation. If, on the other hand, the fundamental question is being taken care of and the community in which the water-power is generated and distributed is receiving it at fair terms, the compensation can be left where it is or only slightly increased, depending entirely on the situation.

And this has another side? The Federal Government may possibly at times not be looking after some public interests in particular localities as well as it should, for these same Federal officials who are elected by the method suggested by our friend from Louisiana are the men who are going to control a large part of the regulation of the rates by the Federal Government; so anyone who believes that the delegation of this question to either Federal or State authority is a final solution is equally mistaken in either case. But the method which I suggest will work automatically, because if either State or Nation is alive to the people's interests they will be protected either by the imposition of proper compensation or by the appropriate reduction of the rates. (Applause)

Chairman CONDRA—Fellow-Delegates, Ladies and Gentlemen: In taking note of the remarkable representation from all over the country in this Congress, we should not forget that our President, Mr Bernard N. Baker, is from Baltimore, right on the Atlantic coast and in a southern State; and I

desire to say, with a great deal of satisfaction, that a large part of the success of this Congress is due to his unflagging efforts. (Applause)

We shall close our formal program for the day with a brief address by Colonel James H. Davidson, whom I now have the pleasure of introducing.

Colonel DAVIDSON—Mr Chairman, Ladies and Gentlemen: I shall only detain you a few minutes to make some suggestions which seem to me pertinent.

Many delegates in this Congress seem to have it fixed in their minds that Federal control would settle the questions before us, and other delegates, from the Far West, seem to claim that the States should control absolutely; and to my surprise and great pleasure, I find that the representatives of southern States, like Louisiana and Mississippi, are favoring Federal control. I say to you, Mr Chairman and Delegates to this Congress, that this question is large enough and broad enough to enlist all the statesmanship in the Federal Government and in all the States composing the Union (applause). Reference has been made to that great struggle of nearly fifty years ago, in which I took part for nearly five years from private soldier to brigade commander as a full colonel (being one of but five who advanced in rank from private soldier to a full colonelcy); and I cannot stand up and ask as an American for State rights as against the Federal Government (applause). But it seems to me, Gentlemen, that there is enough for each and all of us to do; and if we, as States, neglect the duties that devolve upon us under the police powers, which all the States have, of regulating internal affairs, including these manufacturing corporations and monopolies, we are weak and are not making full use of the great privileges conferred upon us.

I was interested very much in the discussion by Ex-Governor Pardee; and he pointed out a fact which indicates to my mind that Federal control *alone* is not sufficient. He says that 6,000,000 acres of the most valuable timber lands that ever grew on this continent were conveyed to the Southern Pacific Railway, in a certain sense in trust, to be conveyed to actual settlers at not less than \$2.50 per acre, but that no actual settlers ever went upon that land. It is not charged that the State of California was in any way responsible. There was a case where the Federal Government, and the Federal Government alone, was involved; and yet that valuable property

passed into the hands of that railroad which is the imperial controller of almost everything in California. In the course of the discussion yesterday in reference to the regulation of oil and gas lands it was stated that in California alternate sections had been conveyed to that great organization, and was out of the control of the Federal Government. That is another case where, if California, a sovereign State, had dealt with those things at the proper time and at the inception, it might have been saved some of the great burdens that now rest upon the people of that State.

They speak of four great water-power companies in California, and two water-power trusts. I thoroughly investigated that subject, spending over six months on it three years ago, and I found that water was king in California, yet the water is owned by these four imperial companies. One-half of my life and of my most valuable treasure is my son and his family, now in the San Joaquin valley; and every crevice and cañon, in the mountains, almost, has been pre-empted by these great water-power combinations, and it costs fifty dollars per horsepower per annum for the use of it for pumping or for any other purpose. If the State of California had been alert, and had had proper regulation, it would have seen to it that these monopolies could not take possession of all these cañons and control the water-power against the interests of the people. A board of most distinguished army engineers reported two or three years ago that the cost of generating one electrical horsepower at the falls of Saint Anthony—within ten miles of where I stand—was less than \$6 per annum, and that in the city of Minneapolis to generate one horsepower by steam costs \$42. Is there any reason why these great monopolies that can generate horsepower by water at an expense of from five to six dollars—and I think in California at less—should put it to the people at fifty dollars per horsepower? I hope that one of the results of this Congress will be earnest cooperation between the States and the Federal Government. Let each one be alert.

When the Civil War broke out and President Lincoln called for 75,000 men, the Governors of the different States in the North did not hesitate, nor the Governors in the different States in the South; they immediately began calling for volunteers, making all arrangements to take care of the soldiers, and not an hour was lost. Governor Alexander Ramsey, of Minnesota, tendered a regiment to President Lincoln within an hour after the firing upon Fort Sumter (applause). It was a day for the earnest cooperation of all

the States with the Federal Government. And we are confronting a condition of that kind, commercially and legally, today; and it needs cooperation, without bickering and without lack of confidence, in the most earnest manner, to pass such State laws as are proper and right, and to pass such laws of Congress as will (so far as the General Government has not parted with its rights) control the streams, the lakes, the waters, and the various natural resources in the West. (Applause)

Chairman CONDRA—It is now long after six oclock; and the Congress is adjourned, to reassemble tomorrow morning at 9.30.

FIFTH SESSION

The Congress was called to order in the Auditorium, Saint Paul, on Wednesday, September 7, 1910, at 9.30 a.m.

President BAKER—Ladies and Gentlemen: The State Delegations are requested to hand the Secretary, soon as possible, the names of their nominees for Vice-Presidents of the Congress.

The Committee on Resolutions are anxious to have all resolutions submitted to them at the earliest possible moment in order that they may receive full consideration.

It has been arranged to renew the Call of the States tomorrow afternoon. The first Call of the States was made on Governors' Day (the Second Session), when preference was given to the Governors. Delegations are requested to have a speaker from their State prepared to respond to the call at the Thursday afternoon session.

Now that Delegations are assembled, the Right Reverend Samuel Cook Edsall, Bishop of the Protestant Episcopal Church for Minnesota, will ask the blessing of our Heavenly Father.

INVOCATION

O, Almighty and everlasting God, Who art the giver of every good and perfect gift, we render unto Thee our most humble and hearty thanks for all the blessings which Thou hast vouchsafed unto our country, for our resources of soil, forest, mine, and stream, which Thou hast given into our hands; and we humbly beseech Thee that Thou wilt give unto the President of the United States, the Governors of our States, our legislators in National Congress and in State Legislatures, and unto all those who are in authority, as well as unto all the people whether in public or in private

station, the graces of unselfishness and wisdom; that they may rightly use these bounties to Thy honor and glory and for the good of all mankind; and that Thou wilt so bless and guide the deliberations of this Congress that by all that may be here said and done our minds may be illumined and our hearts stirred to righteousness and obedience to Thy law—through Jesus Christ our Lord. Amen.

President BAKER—Ladies and Gentlemen: We have with us today a truly representative man of our Southland, Mr W. W. Finley, President of the Southern Railway Company, who will address us on "The Interest of the Railways of the South in Conservation." (Applause)

Mr FINLEY—Mr President, Ladies and Gentlemen: The interest of the Railways of the South in Conservation and the interest of the people of the South in Conservation are identical. I will go farther, and state my unqualified conviction that any economic or governmental policy that is, in the last analysis, to the best interest of the people of any community is to the best interest of the railways by which that community is served. Conversely, my conviction is equally strong that any economic or governmental policy that is harmful to the railways is harmful to the communities served by them.

Therefore, Mr President, in all that I say on the topic assigned to me—"The Interest of the Railways of the South in Conservation"—I must be understood as presenting what I believe to be the interest of the southern people.

I am not sure that the expression "Conservation of natural resources" is everywhere understood in its broadest sense. I think that to some minds it conveys only the narrow idea of the withdrawal from present use of some part of those resources. However important that kind of Conservation may be in some localities and under some circumstances, I do not believe there

is much occasion for its application in the part of the United States for which I am expected to speak—the States south of the Ohio and Potomac rivers and east of the Mississippi. I would define the type of "Conservation of natural resources" that should be applied in that section as being the wise use of those resources. In some cases it may involve a measure of present self-denial, as when, in the case of an owner of forest lands, it impels him to cut only the matured timber and leave standing immature trees that have a present market value; but, in that case, it leaves him with an asset which increases in value with each year's growth of the standing timber. In some cases Conservation may mean the use of resources so as to obtain the maximum present profit, as in the case of soils; for I believe that I am supported by the best scientific and practical authority in saying that soils not only preserve, but increase, their productivity when so handled, in the application of fertilizers, the rotation of crops, and the growing of live stock, as to yield the maximum present profit.

The South is interested in the application of Conservation to the wise use to its soils, its minerals, its timber, and its streams. Notwithstanding the wonderful industrial development of the South since 1880, it is still pre-eminently an agricultural section. It is a section, therefore, in which the conservation of the soil is of the highest importance. There is a prevalent belief that the productivity of the soils in those parts of the United States that have been longest under cultivation has been seriously impaired. Statistics do not confirm this belief. Estimates of productions of staple crops per acre have been compiled in the United States only since 1867, and, as there are often wide fluctuations between successive seasons—due to differences in rainfall and temperature—the period covered has not been long enough to afford a basis for definite conclusions. There is also the fact that all available figures are estimates, and consequently are not exact. On their face, however, they do not prove a decline in productivity. This may be illustrated by comparing the production of wheat per acre for ten-year periods since 1867. In the decade from 1867 to 1876 the average for the United States was estimated at 12 bushels; from 1877 to 1886, 12.5 bushels; from 1887 to 1896, 12.7 bushels; from 1897 to 1906, 13.8 bushels, and for the three years since 1906, 14.6 bushels. So far, then, as these figures can be relied upon, they tend to show an increase in productivity, especially as an

analysis by groups of States shows the larger and more uniform increases to have been in some of the older sections of the country.

Similar figures for corn do not show an increase for the United States as a whole, but they show very little decrease. From 1867 to 1876 the average production of corn per acre was estimated at 26.2 bushels; from 1877 to 1886, 25.1 bushels; from 1887 to 1896, 24.1 bushels; from 1897 to 1906, 25.4 bushels, and for the three years since 1906, 25.8 bushels. It is proper to note, in connection with the apparent decline in the fourth decade as compared with the first, that the poorest yield in the entire period was in 1901, when abnormal weather conditions brought the estimated average for the United States down to 16.7 bushels, thus pulling down the average for the entire decade. It is also proper to note that Dr Whitney, Chief of the Bureau of Soils in the United States Department of Agriculture, in discussing these figures, expresses the opinion that, on account of a readjustment of the basis of the Department's estimates in 1881 as a result of the reports of the census of 1880, the figures before that year, both for wheat and corn, were relatively too high.

Estimates of cotton yield per acre have been made by the United States Agricultural Department since 1866. Ten-year averages for the full decades up to 1905 are as follows: 1866 to 1875, 176.4 pounds of lint cotton per acre; 1876 to 1885, 171.4 pounds; 1886 to 1895, 175.9 pounds; 1896 to 1905, 182.6 pounds, and for the four years since 1905, 183.1 pounds. These figures are subject to the same question as to their accuracy that apply to the estimates of wheat and corn production, but, on their face, they do not indicate any impairment of the productivity of the cotton soils of the South. It is noteworthy that the larger and more uniform increases in yield per acre shown by the Department's figures are in the older cotton States.

While statistics of crop yields in the United States do not cover a sufficient period to be of great value in determining the effect of long use on soil productivity, some light is thrown on the subject by comparing yields per acre in the United States with those in other countries where lands have been under cultivation for centuries. Thus, for the ten-year period from 1897 to 1906, inclusive, the average yield of wheat per acre in the United States was 13.8 bushels, in France 19.8 bushels, in Germany 28 bushels, and in the United Kingdom 32.2 bushels. In Germany, statistics are

available from 1883 to 1906, inclusive, showing increases in the average yields of wheat from 18.2 to 30.3 bushels, of rye from 15.4 to 25.1 bushels, and of oats from 27.6 to 55.7 bushels. Similar figures might be cited for other European countries, but perhaps the most conclusive statistics are those collected by Kellerman, a German student of this question, who gives the yield per acre for a large number of German estates, covering long periods of time. I shall cite but one of these—a Schmatzfeld estate with records extending back to 1552. In the period between 1552 and 1557 the annual yields reduced to bushels per acre, were, wheat 12.5, rye 13.2, barley 14.2, and oats 14.8. In the period from 1897 to 1904 these yields were, wheat 45.1, rye 34, barley 50.4, and oats 69.1.

Taking all these figures together, I believe the conclusion is inevitable that, while abuse of soils may impair their productivity, their wise use increases it, and the longer they are properly used the more productive they become. Proper use, such as conserves and increases soil productivity, involves the most approved cultural methods, the application of such fertilizers as may be required for varying soil conditions, the raising of live stock, and, above all, the scientific rotation of crops. There can be little question that the most unwise use to which a soil can be subjected is the raising of the same crop for a long series of years. Some very interesting experiments in continuous cropping and crop rotation, covering a period of sixty-five years, have been carried on at Rothamsted, England. On one plot potatoes were grown for fifteen years. At the end of that period the soil was in such condition that it would not grow potatoes at all. It was then planted in barley, and produced an excellent yield. Another crop followed the barley, and the soil was then in condition to grow potatoes again. On this same experimental farm wheat has been sown for fifty years on the same land without fertilizers, and the yield has gone down from 30 bushels to 12 bushels. On another tract wheat has been grown continuously for fifty years with the use of a complete fertilizer, and an average yield of about 30 bushels has been maintained. On another tract wheat has been grown for fifty years in rotation with other crops and an average yield of 30 bushels has been maintained, showing that, for growing wheat on that particular soil, rotation was equivalent to fertilization. As might be expected, the Rothamsted experiments show the best results where fertilizers are used in connection with rotation, and

justify the conclusion that under continuous use, with proper rotation and an intelligent use of fertilizers, soil productivity can be largely increased.

This is a matter of particular interest to the South, because with our advantages of soils and climate we have an ideal region for soil conservation through crop rotation and intensive farming. There is a quite general impression throughout the North that, except for a few localities in which early fruits and vegetables, tobacco, and sugar cane are grown, the South is a one-crop region devoted exclusively to cotton. This is entirely erroneous. There are many localities in the southeastern States where cotton is not grown at all, and every acre of land in the cotton belt is suited for growing other crops as well. Cotton will continue to be the great staple crop of the South, and with the ever-increasing demand for cotton goods of all kinds, its cultivation will become increasingly profitable, but the southern cotton planter is learning the value of crop rotation; diversified farming and live-stock raising are becoming more general, and the increased supply of cotton demanded by the world will be produced by increasing the average productiveness of each acre as well as by increasing the acreage.

Other things being equal, the conservative use of a raw material, whatever it may be, consists in its manufacture, in the locality of production, through all the stages of preparation for the final consumer. Manufacturing in the South has reached its present growth and is being still further developed on the basis of this kind of conservation of raw material. Industrial development in the South on a large scale may be said to date from about 1880, prior to which time only relatively a small proportion of the raw materials available in that section were advanced through even the first stages of manufacture before being shipped to other localities. It is natural that, at first, only the coarser, and what may be termed the preliminary, processes should have been undertaken. This was the first step in the conservation of raw materials by their manufacture near the source of supply. The South has gone far in that direction, and has already started on the second step, which is the use of the products of primary manufacturing as the raw materials for secondary industries. But a large proportion of southern cotton mill products, lumber, pig-iron, and other commodities, advanced through the first stages of manufacture, are still shipped out of the South to serve as the raw materials of industries in other localities which convert them into articles ready for the final consumer; and southern coal is

shipped to serve as the raw material for power and heat in other parts of the United States and, to some extent, in foreign countries. This is a waste of energy which, under ideal conditions of Conservation would be avoided; and I am glad to be able to say that the present tendency of industrial development in our section is in the direction of its elimination. Substantial progress has already been made in the building up of secondary manufacturing along some lines, and I believe that the most noteworthy progress of southern industrial development in the immediate future will be in this direction, carrying with it an increase in the volume of primary manufacturing through broadening the market for its products.

One of the most valuable of the natural resources of the South is its timber. It is also a resource of which the intelligent conservation will benefit, directly and indirectly, the largest number of people. We have in the southeastern States large and growing industries which use wood alone, or wood in combination with iron, steel, and other materials, as their raw materials. Some of these industries, such as the manufacture of furniture, have enjoyed a phenomenal growth in the past 30 years. There is every reason to expect that this growth will continue and that the variety of wood-working industries will be increased, with the result that they will require an increasing supply of raw materials. As the timber consumption of the United States is now in excess of the annual growth, and as other sections are drawing on our southern forests, it is obvious that if these southern wood-working industries are to survive and are to be handed down to future generations, immediate and effective steps should be taken for the conservation of southern forests. This is the more important for the reason that the same steps taken to insure a perpetual supply of raw material for our wood-workers will tend to stream and soil conservation by increasing stream-flow in periods of drought and by lessening the destructiveness of floods which erode the soil of the upper watersheds and deposit gravel and silt on overflowed lands and in the beds of the navigable parts of the streams.

If we were thinking only of the present time, there would be no occasion for us to concern ourselves with the conservation of our timber supplies. We have ample for the present generation. It is because timber is a crop of slow growth, requiring more than a lifetime to mature most of the species, that timber conservation, if it is to be effective and is to provide for the needs of

those who come after us, must be handled along exceptional lines. It is not the duty of a private owner of forest lands to conserve them unless it is at least as profitable for him to do so as to clear all the timber off of them; but it is the duty of the Government to consider the welfare of future generations as well as of that now living.

The conservation of southern timber supplies is a matter that concerns not only the people of our own section, but those of the entire United States as well. It is a matter of National concern, as, owing to the depletion of their forest resources, the people of other parts of the country must look to the South for an increasing proportion of their timber supplies. It is a recognition of this National interest in the southern forests that has strengthened the support of the proposition for the acquisition by the Federal Government of large tracts of lands in the Appalachian region to be converted into National forests (applause) from which the timber shall be marketed under a system that will result in the perpetuation of the forests. It may be that our Federal Government has no power, under the Constitution, to acquire lands for the purpose of forest conservation; but it is charged with the supervision, improvement, and conservation of our navigable streams (applause), and the evidence as to the effect of forests on stream flow was so conclusive as to lead the House of Representatives, during the last session of Congress, to pass a bill providing the establishment of National forests for the protection of the watersheds of navigable streams. This bill is to be voted on in the Senate on the fifteenth of next February. Whether this plan or some other may be adopted, I think it is of the utmost importance that the campaign of education as to the necessity for the speedy and general adoption of the most approved methods of scientific forestry, which is being so ably carried on by the National Forest Service, should be continued (applause). This is quite important, if the best results are to be attained, because, whatever may be done by the Federal Government, much will remain for the States and for private owners of forests and woodlots to do. If the States and private owners are to do their share, the owners of forest lands, the users of forest products, State legislators, and the people generally should be educated as to the dependence of our future supplies of timber on wise conservation.

The private investor in forest lands buys them with the expectation of making a profit on his investment. He naturally wants to make the largest

possible profit, and to do it as soon as possible. Heretofore, partly as a result of prevailing systems of taxation and the lack of efficient fire protection, self-interest has impelled the investor in timber lands to clean up his holdings to the last dollar's worth of merchantable timber, and to get off the denuded land as quickly as possible, selling it for whatever it might bring. In the early years of our history, when, except in the prairie regions, lands for cultivation could be obtained only by clearing them of timber, this wholesale cutting was more justifiable, and, in some cases now, in locations where the value of the land for agricultural purposes is greater than its value for timber production, it may be the proper method. We have reached the point, however, when, especially with reference to our mountain forests, it may seriously be questioned whether, as a matter of dollars and cents, this method is the most profitable to the forest owner. In view of the present prices of lumber and the practical certainty of advancing prices in the future, I am disposed to believe that we have now reached the point where it will pay the private owner of any considerable body of timber on land having relatively a low agricultural value to adopt conservative methods of forestry (applause). A case in point is that of the University of the South, at Sewanee, Tennessee, which owns 7,000 acres of forest land. In 1899 it was proposed to sell all the marketable timber on this tract, and an offer of \$3,000.00 was obtained. This was rejected, and the University undertook to manage the forest conservatively and market the mature timber from time to time. The result is that, at the end of nine years, instead of having realized only \$3,000.00 from this tract, the University has received from it net profits amounting to over \$18,000.00 above all expenses (applause), including the cost of fire patrol; and instead of having 7,000 acres of cut-over land of relatively little value, it has a continuously productive forest. (Applause)

Whatever may be the decision of our National Legislature as to the proposition for the conversion of our Appalachian woodlands into National forests, I believe it would be a wise and patriotic policy for our State lawmakers to encourage conservative forestry by private owners in every reasonable and proper way. One of the reasons assigned for the failure of private owners to adopt conservative forestry is that in some localities the rate of taxation on timber land is so high as practically to compel every owner to cut the timber as quickly as possible. Another reason assigned is

the general lack of an efficient fire patrol, and the danger that, even if an owner goes to the expense of preventing fire on his own property, his timber may be destroyed by a fire starting on the property of some neighbor who has taken no such precautions. These are matters that come within the province of our State legislators, and I would suggest their consideration of whether it might not be possible to devise a system of taxation that would differentiate between timber lands so managed as to insure the perpetuation of a great National resource and those so managed as to hasten its exhaustion (applause). I would also suggest consideration of the enactment of proper fire laws and the establishment of an efficient patrol, possibly with the expense apportioned among owners of timber lands, as I understand is done in some western localities at a very low annual cost per acre. I would further suggest consideration of the practicability of encouraging the planting of trees on lands of little or no agricultural value. Even under the most encouraging conditions, however, planting of forests by private land owners must, almost necessarily, be on relatively a small scale. As a general rule, therefore, private planting will be limited to the establishment of woodlots on the waste lands of farms; and if reforestation is to be undertaken on a larger scale, it must be done by some Governmental agency. (Applause)

The problem of stream conservation in the southeastern States is very closely connected with both timber conservation and soil conservation. The ends to be sought are a diminution of the volume of water carried by the streams in their flood stages, and an increase in their volume during their low stages. Everything, therefore, which tends to retard the flow of the rainfall into the streams is a conservative agency. Undoubtedly the most effective of these is the natural forest with its soil, composed of porous humus, covered by a blanket of decaying leaves, branches, and fallen trees, and often with a dense mat of underbrush growing among the trees. Such a forest will absorb a large amount of water during a rain-storm, and allow it to seep down gradually into the streams instead of running off in torrents, overflowing the banks of the streams, destroying growing crops and other property, and scouring the soil from the watersheds to be deposited in the lower levels of the streams or at their mouths, shoaling channels or forming bars in harbors. Generally speaking, therefore, every step taken in the conservation of forests is of value in stream conservation; but, if the best

results in the regulation of stream flow are to be attained, other things may be done to advantage. The growth of underbrush having no marketable value is of no benefit to a forest, in fact it may choke out or retard the growth of young trees of valuable species. Such a growth is of great value, however, in retarding water flow, and preventing soil erosion, and, unless cut-over mountain sides are to be reforested, I believe that the growth on them of such species as laurel and rhododendron should be encouraged. (Applause)

Each farmer, especially along the headwaters of the streams, can contribute to a greater or less extent to stream conservation. He can do this by establishing permanent woodlots on those waste lands that are to be found on almost every farm in rolling or mountainous country, and especially on those lands that are liable to erosion. He should, of course, take every precaution to prevent the washing of gullies in his cultivated fields, and where such gullies have already been formed he should so manage as to prevent further erosion. The farmer on the headwaters of a stream cannot be expected to do these things in order to aid in the prevention of flood damages below him. He should be educated to an appreciation of their benefit to himself individually. He will not only be lessening, in some degree, the amount of silt carried down by flood waters, but will be conserving his own soil; and his woodlots will, in a few years, become increasingly valuable as stores of fire-wood and fence-posts, and, eventually, of larger timber. The effect of but a single farmer on an extensive watershed adopting these methods would, of course, be inappreciable, but if thousands of farmers could be led to do so as a matter of self-interest the good results would soon become apparent.

Another method of stream conservation that I believe may be practiced to advantage in some locations in the Appalachian region is the impounding of flood waters in artificial ponds or lakes, to be let out gradually during periods of low water. This is not everywhere practicable, and, I believe, should only be practiced where the benefit will be greater than the damage that will result from overflowing the land included in the reservoir. It would manifestly be unwise to locate such a reservoir at a point where it would submerge a fertile agricultural valley, or where it would render inaccessible a valuable deposit of coal or ore.

One of the great economic advantages of the South is the abundance of its opportunities for the development of hydro-electric power for the operation of its factories, the propulsion of its trolley cars, and the lighting of its cities and towns. If this cheap and efficient power is to be used most advantageously, it is important that the stream-flow by which it is generated should be, as nearly as possible, uniform at all seasons of the year. It is in this connection that reservoirs for impounding flood waters would be of great value. Some of the sites where these reservoirs might be located are so situated that a great and powerful fall of water may be attained. The power plants would often have to be situated at points not suited for the location of industrial establishments, but the power can be carried by wire to factories many miles distant. Where such reservoirs are established the primary purpose will be the generation of power, but they would also serve a highly useful purpose in diminishing the flood level of the streams which they feed.

Your invitation to address this Congress was very gratifying to me, Mr President, not simply because of the high honor which it conferred upon me, but chiefly because the invitation and the suggestion of my topic conveyed a recognition of the interest of the railways of the United States in the Conservation of our natural resources and in all that concerns our national welfare. (Applause) They are interested in soil conservation, because it means prosperity to the farmer and an increase in the volume of farm products to be carried, and also an increase in their tonnage of agricultural machinery and implements and of all kinds of merchandise which a prosperous farmer will buy. They are interested in the conservation of forests and mines, because it means the perpetuation of sources of supply of raw materials which, either in their crude or manufactured state, must be carried to market, and which, in their production and manufacture, bring prosperity to many thousands whose consumption of commodities produced in other localities calls for transportation. They are interested in the conservation of water powers and navigable streams, because cheap power means the development of industrial communities and, while economically efficient waterways mean a loss to the railways of some kinds of traffic, they also mean an increase in general prosperity in which the railways have a share. (Applause)

Conversely, Mr President, the people are interested in the conservation and development of their transportation systems. We have seen that one of the elements of conservation is the manufacture of finished products at or near the sources of supply of raw materials. It is this that enables the people of a community to devote their energies chiefly to those industries for which their locality is best suited and to exchange their surplus production for commodities that can be produced more advantageously in other localities. Transportation makes this specialization of industries possible. Without efficient transportation facilities each community would have to be, to a larger extent, self-supporting, and many of its people would have to engage in the production of commodities which, with our existing facilities for transportation, they can buy more profitably elsewhere. The scale of living would be much more restricted, and many things which are now looked upon as being almost necessities of life would either be unattainable or would be luxuries which only the wealthy could enjoy.

I am glad of the opportunity, Mr President, to speak of the South and for the South before this representative national assembly (applause). Our section is a region of unsurpassed economic strength. Our climate and our soils invite to diversified agriculture, in which there can be produced profitably all the products of the temperate zone and many of those of the tropics. Beneath our soil are stores of coal, iron and other ores, marble and stone for the builder, and clay for the potter and brickmaker. Our forests are sources of great present profit and, under wise conservation, can be perpetuated as sources of wealth for future generations. Our streams flowing from the wooded mountains of the Appalachian region carry the force of millions of horsepower capable of being utilized along their banks or carried in the shape of electrical energy to wherever it can be used to best advantage. The intelligence, energy, and enterprise of our people are attested by the splendid social, agricultural, and industrial structure they have erected on the ruins left by the Civil War. The progress that has been made is but the promise of what will be. The South is a land of present-day opportunity, and its people invite the man seeking an opportunity to work with hand or brain, or the man with money to invest to come to this favored land of busy factories and thriving towns—a land of fertile valleys, forest-clad mountains, and storehouses of mineral wealth. (Applause)

President BAKER—Ladies and Gentlemen: You will no doubt gladly permit interruption of the formal program for a few moments now and then by reports of committees. Professor Condra, Chairman of the Credentials Committee, is now ready to report.

Professor CONDRA—Mr President and Delegates: We have examined the credentials of all Delegates to the Second National Conservation Congress, and find that the duly accredited Delegates entitled to vote in accordance with the Constitution of the Congress number thirteen hundred fifty-one (1351), and that the number of duly accredited Delegates from each State are as follows:

Alabama 1, Arizona 3, Arkansas 4, California 13, Colorado 7, Columbia (District of) 10, Connecticut 5, Delaware 1, Florida 4, Georgia 6, Idaho 10, Illinois 67, Indiana 15, Iowa 78, Kansas 13, Kentucky 4, Louisiana 17, Maine 1, Maryland 8, Massachusetts 3, Michigan 19, Minnesota 631, Mississippi 8, Missouri 25, Montana 20, Nebraska 22, New Hampshire 1, New Jersey 4, New Mexico 1, New York 27, North Carolina 1, North Dakota 77, Ohio 17, Oklahoma 2, Oregon 15, Pennsylvania 16, Rhode Island 1, South Carolina 3, South Dakota 53, Texas 12, Utah 2, Vermont 2, Virginia 3, Washington 26, West Virginia 5, Wisconsin 84, Wyoming 5; total, 1351. Foreign: Canada 2, Mexico 1.

Respectfully submitted to the Congress:

[Signed] G. E. CONDRA, *Chairman*

LYNN R. MEEKINS

GEO. K. SMITH

EDWARD HINES

R. W. DOUGLAS

A DELEGATE—Mr Chairman: I move that the report be adopted and the committee be dismissed.

The motion was put, and was carried without dissenting voice.

President BAKER—Professor Condra will report an action by the Committee on Resolutions.

Professor CONDRA (*reading*)—A motion was made and carried by the Resolutions Committee that resolutions presented to the Congress or to the Committee cannot be received after 5 oclock p.m. Wednesday. All resolutions should be headed with the subject of the resolution and should be signed by the person offering same.

The Resolutions Committee has not yet received the names of the members from Alabama, Delaware, Nevada, North Carolina, South Dakota and Virginia; and the Committee urge that the Delegations from those States act at once. The next meeting of the Committee will be held at 5 p.m. today, Room 534, Saint Paul Hotel.

Mr GEORGE B. LOGAN (*Secretary of the Resolutions Committee*)—Mr Chairman: The Resolutions Committee suggest that resolutions should be grouped under the heads of Land, Water, Forests, Minerals, and Vital Resources; and if those who submit resolutions will simply place the proper heading on each, it will greatly aid the Committee.

President BAKER—Professor Condra will make another announcement.

Professor CONDRA—Ladies and Gentlemen: There is a strong demand for practical consideration of Conservation problems in various States, and for the purpose of discussing these subjects a meeting will be held this evening at 8 oclock in the Saint Paul Hotel. All members of State Conservation Commissions and State Conservation Associations are invited to attend this meeting.

President BAKER—Here is another announcement just handed in: Technical men in attendance are requested to meet in the lobby of the Saint Paul Hotel on the adjournment of the morning session of this Congress. The call includes civil, electrical, mining, mechanical and hydraulic engineers, architects, educators in these sciences, and also geologists and chemists.

Senator Beveridge, of Indiana, will now address us on a subject which ought to be very near the heart of every father and mother—"The Young Man's Idea." I have the pleasure of introducing Senator Beveridge.

[The band here played "The Star-Spangled Banner," while the audience rose and greeted Senator Beveridge with tremendous applause.]

Senator BEVERIDGE—Mr Chairman, Ladies and Gentlemen: The United States IS. (Applause) The American people are a Nation (applause)—not forty-six Nations. (Applause)

In war we fight under one flag (applause) for our common safety; in peace let us strive, under one flag, for our common welfare. (Applause)

Our history is the story of the struggle of the National sentiment of all the people, which special interests for their selfish purposes sought to discourage, against the provincial sentiment of some of the people, which special interests for their selfish purposes sought to encourage. (Applause)

The parent of the provincial idea in American Government was the British crown. The British kings believed that if they could keep the colonists separated by local pride, local prejudice, and local jealousy, the British policy would be easier. They knew that if the colonists were united by common interests, common sentiment, and a common purpose, the British policy would be harder; and that British policy *was* to permit the special interests of the United Kingdom to exploit the people of the divided colonies (applause). And so from King James to King George the British crown sought to keep the people of the Colonies divided—separated by geography for the convenience of the English government; they sought to keep them separated in spirit for the interests of the British manufacturers. Every British law which forced the Revolution was a law to enable the special interests of the United Kingdom to monopolize the markets of the people of the Colonies. Our Revolution was nothing more than the war of the people, for the moment united, against the special interests of the Colonies which had kept them divided.

Now, such is the origin of the provincial idea in America. Washington and his Continentals were the infant National idea in uniform, and manning the shotted guns of liberty (applause). The British and their Hessian and Tory allies were the full-grown provincial idea behind the bayonets of

oppression. Our first attempt at Government was a failure because the British provincial idea still was powerful. The local pride, prejudice, and jealousy of the separate Colonies reasserted itself, after their common danger was past. The result was the Articles of Confederation. Washington said that the Government thus formed was contemptible, and yet it was the provincial idea carried to its logical conclusion; and so it fell. The cruel necessities of the people forced the reassertion of the National idea, and the Constitution of the United States was that idea's immortal child (applause). The Articles of Confederation said, We, the States, form a Government: the Constitution says, We, the *People*, form this Government for our general welfare (applause). And yet into this great "ordinance of our nationality," as Chief Justice Marshall calls our Constitution, there crept defects which the statesmen of that day could not prevent, defects which have caused most of our trouble since, and nearly all of them are due to the provincial idea. For example, few men remember that when the Constitution was adopted, "State rights" was not mentioned in that instrument. Washington had been elected President. The Congress of the United States was in session. The National Government was under way. The Tenth Amendment was adopted to quiet those who were preaching the paradox that the general Government of the people would oppress the people. Noisiest of these was Patrick Henry, then Governor of Virginia, who refused to attend the Constitutional Convention, opposed the ratification of our fundamental law, and was against its adoption. Upon the embers of provincialism he heaped the inflammable brush-wood of excited rhetoric. Being in the Constitution, the State rights provision is as valid as any other amendment. But such is its origin and spirit, and no misinterpretation of the provincial idea of State rights must be permitted to impair the American people's general welfare, waste their resources, plunge the Nation into war, or impede our general progress as a people (applause). Now, as always, the danger has been, and is, not so much that the Nation will interfere with the rights of the States as that the States will interfere with the rights of the Nation. (Applause)

After our present Government was founded, its first conflict with the British provincial idea was in the Whiskey Rebellion of Pennsylvania; the special interests that dealt in rum, under the guise of State sovereignty defied the Nation's laws; but George Washington put down that first State rights rebellion in the name of the Government of all the people (applause). Then

came the special interests' defiance of the laws of the General Government in Andrew Jackson's day, and Andrew Jackson's voice, like the voice of Washington, was the voice of all the people against the voice of the special interests who tried to exploit the people. Next came the special interests that thrived on human slavery, and, in the name of State rights tried to destroy the Government they could not control. But again the National sentiment responded to Abraham Lincoln's call to arms (great applause), and a million bayonets wrote across our Constitution these words of the American people's immortality: THIS IS A NATION! (Applause)

Then came the special interests that robbed and poisoned the people by lotteries, that destroyed the morals of the people by obscene literature. They flourished under State protection. Only the Nation could stop them. Those special interests denied that the Nation had the power to stop them. But the Nation *did* stop them, and the Supreme Court of the Nation upheld the Nation's power (applause). Then came the special interests that sold to the people diseased meats, poisoned foods, and adulterated drugs. Again they flourished under State protection. Again the Nation only could protect the lives of the Nation's people. And again those special interests denied that the Nation had the power, but the Nation *exercised* the power, and today National laws protect the lives and rights of the American people from special interests that were plundering and poisoning and killing them. (Applause)

And it is the same conflict between the National and the provincial idea, for and against the great, necessary, and inevitable reform of the National control of corporate capitalization, on which so largely depend just prices and rates to the people. (Applause)

These are examples of the evils; but nearly every step of progress we have taken has been due to the success of the National idea. For example, President Madison vetoed the first internal improvement bill. He said, in one of the ablest messages ever written—far abler than the diluted State rights doctrine we hear today—that the Constitution gave the Nation no power to build roads, bridge rivers, improve harbors; but the people needed these things in order to win that righteous prosperity which only they can have acting as one people, under one flag—and so Congress passed the internal improvement bill over Madison's veto, and today no one dares

question the Nation's power to make internal improvements; the only question today is how we can best do that work. (Applause)

Again, for a hundred years, the provincial idea kept the quarantine of the Nation's ports exclusively in the hands of the States; but if pestilence entered at a port of one State it attacked the people of other States. The germs of yellow fever did not know State lines when they saw them, any more than a forest fire knows the boundaries between States when it sees them. And so the open grave, the dead on the street, the people's past and future peril, asserted the National idea again for the Nation's safety, and today we have substantially a National control of National quarantine to keep pests and death from our shores, and the States are cooperating.

So you see that the history of the American people has been merely the narrative of the making of the Nation, merely the record of the compounding of a people, merely the chronicle of the knitting together of one great brotherhood. It is an inevitable process, and it is a safe process—except for special interests that seek to exploit all the people. For the American people can be trusted (applause). The combined intelligence and composite conscience of the American people is the mightiest force for wisdom and righteousness in all the world, and no ancient and provincial interpretation of State rights in the name of development must impede our general welfare (applause), no plea for hasty local development must impair our healthy general development (applause), no temporary State politics compelled by the wealthy few must prevent permanent National statesmanship for the general good of all. (Applause)

Affairs that concern exclusively the people living within a State are the business and the problem of that State. Affairs affecting the general welfare of the whole people are the business and problem of the Nation (applause). And even in solving its own problems, every State must remember that its people are an inseparable and indivisible part of the whole American people (applause). Of States as of men it may be written, No State liveth unto itself alone. (Applause)

Just as the idea of provincialism has caused most of our National evils in the past, so it has wrought the waste of our National resources. The provincial idea was that the National resources belonging to all the people

should be handed over for nothing to special interests. This was done under the plea of encouraging individual enterprise and the hastening of local development. And so forests, which once belonged to all the people, have been ruthlessly slaughtered, and upon their ruins have risen the empires of our lumber kings (applause). Priceless deposits of coal and iron and copper and phosphates have been freely surrendered to special interests, and those sources of the people's revenue, which should have flowed into the people's treasury to help pay the expenses of the people's government, have been diverted by the ditch dug by the provincial idea into the treasury of special interests until the multi-millionaire constitutes one of the gravest problems confronting American statesmanship. (Applause)

All this waste and robbery of the people's property must be stopped! (Applause) The hand of waste or theft must not be strengthened by any legal technicality that plays into the hands of special interests and out of the hands of the American people! (Great applause)

Had we kept all the property that belonged to all the people, and compelled special interests who exploited it to pay us a reasonable price for it, that income today would be paying most of our National expenses. Our resources would have been developed and not exhausted, and our whole material evolution would have been rational and sound instead of unbalanced and defective. Had this been our policy from the start, we would have enjoyed all the benefits from our natural resources, and our children today would inherit colossal National wealth and small National burdens instead of the special interests enjoying all the benefits of the people's property and *their* children inheriting colossal fortunes and small private burdens. (Applause)

The Nation must keep and administer for the benefit of all the people the property yet remaining to the people (applause). Every State should help and not hinder the Nation, in doing this great duty (applause). Every State should administer the public property within it, and belonging to it, for the public good. Every municipality should keep and administer the property belonging to it for the public good; and both State and municipality should aid the Nation in keeping and administering for the people the property that belongs to *all* of them.

I want to give you an illustration, very concrete: Many of New York's inconceivably vast fortunes have been expanded by corrupt councils selling watercourses and other property for a mere song to private owners. Had New York kept the property which belonged to the city, instead of squandering it to already multi-millionaires, the city's debt today would not be so vast—and her great private fortunes would not be so vast either (applause). The people's taxes would have been less, and the gigantic unearned incomes of the heirs of great wealth would have been less (applause). And as between the two, the wiser policy have been for the city to keep the property that belonged to all the people of the city instead of selling it sometimes for an infamous price to private owners whose vast wealth, accumulating by the work of the city itself, has raised up in the midst of the American people one of the great questions of the age.

Cooperation of municipality, State, and Nation, in keeping and administering for the general good the property of all the people—this is the policy of common sense and common honesty (applause). Strife and dissension between municipality, State, and Nation, that the reign of pillage may go on and that mighty accumulations of wealth may be upbuilt upon the ruins of the people's resources—*that* is the policy of private avarice and private plunder (applause). Coal, timber, asphalt, phosphates, water-powers—all the property of the people—must be kept and administered for the people by the Government which Lincoln said was "of the people by the people for the people" (applause). Already this greatest of our present-day National policies is well under way. Let any man beware how he retards or hinders it (applause). Already we have saved much of the people's property still belonging to the people. We must save *all* of the people's property still belonging to the people. (Applause and cries of "Good") "Honor to whom honor is due." (Applause) Let us not forget, in this great hour, that the man who, by thought, word, and deed, has wrought for this great reform, until today he stands its National personification (applause), that splendid, courageous, pure, unselfish young American, the President of the National Conservation Association, Gifford Pinchot. (Tumultuous applause and cheers, calls for "Pinchot"; and the audience rose, gave the Chautauqua salute, and continued cheering for many minutes)

For years—and I speak from personal knowledge, because twelve years ago when I entered the Senate I was made the chairman of the then despised

forestry committee—for years Gifford Pinchot has ceaselessly worked and fearlessly fought to keep for the people the property of the people which special interests were trying to steal from the people (applause). And in that Nation-wide battle he has been the field-officer of the man who *first* succeeded in making Conservation a permanent and practical policy of American statesmanship, Theodore Roosevelt. (Great applause. A Voice: Let us vote to give him back his job!)

The soul of our prosperity—even of our very life—is in the idea of our unity as a people. Let municipality, State, and Nation, each act and, within its own province, work to keep what belongs to the people for the people, instead of the municipality, State, and Nation, each within its province, conniving at the waste of the people's property for the upbuilding of the wealth of special interests to the detriment of all the people. The wise, honest and economic administration of the people's welfare means the just advantage which individual enterprise and thrift as of right ought to have. The unwise, uneconomic and dishonest waste of the people's resources for the enrichment of the special few, this in the end, believe me, is the denial of that just advantage which individual thrift, enterprise, and integrity as of right ought to have. (Applause)

The young men of today in working for themselves individually must think and act for what the Constitution calls "the general welfare" of the whole people (applause). After all, only as the Nation is prosperous can any State be really prosperous. After all, only as the Nation is powerful can any State be really safe from foes, foreign and domestic. The young men of the twentieth century in this Republic are not the heirs of the provincial idea which we inherited from the British kings, and which has so hindered our real progress as a people, squandered so much of the people's resources, shed so much of the people's blood. No! The young men of today are the heirs of all the advancement that our struggling millions have made toward their common brotherhood. The young men of today are the heirs of all the victories which heroes and statesmen have won for the general welfare. The young men of today are the heirs of all the unifying influences by which the genius of man has knit this great people into one splendid family. And so the young American of today, when thinking of himself, must think in the terms of the Nation; through his veins must pulse the blood of our general welfare; his every thought and act must be for the common good of all. And

only so can his individual success be well builded; and when it is builded on such foundation, though "the rains descend and the floods come and the winds blow" and beat upon a house thus builded "it shall not fall, for it is founded upon a rock." (Applause)

Why was the American Nation founded? What is the purpose of this Republic? It is to create a greater human happiness than the world has ever known (applause). It is to enable millions of men and women to cooperate in building clean, honorable, prosperous homes. And so let us Americans move forward as brothers and as sisters until we shall give the whole world an example of one great brotherhood in heart and in deed as well as in words. (Great applause)

There were repeated calls for "Pinchot"; and Mr Pinchot, coming forward amidst great cheers and hearty applause, said—

Ladies and Gentlemen, Members of this great meeting: There can be in a man's life but few moments like this, in seeing policies in which he believes and for which he has tried to work so splendidly acclaimed by such a meeting, when at first they were questioned. I haven't anything to say at this time except to thank you most profoundly, and to add that the policies for which this Congress stands are sweeping the country as they are sweeping this body—and that, so far as the United States is concerned, Conservation, I believe, has won out. (Applause) I thank you!

President BAKER—Ladies and Gentlemen: We all know Conservation has, with such a leader, won out. (Applause)

We now take up "A Rational System of Taxing National Resources," by Frank L. McVey, President of the University of North Dakota, whom I have the pleasure of introducing. (Applause)

President McVEY—Mr Chairman and Good Friends: The invitation of the President of the Congress to be present and to deliver an address on the subject of a rational system of taxing natural resources, asked that specific suggestions be made of a practical nature for the improvement of our present laws on this subject. This places upon me a heavy responsibility if the suggestions made are to be accepted in any serious way. The title of the address assigned emphasizes a *rational* system; it implies that the one now in vogue cannot be so designated, and that any system of taxation has a close relation to the Conservation of natural resources. This, if I may put it in so many words, is my thesis.

It is unnecessary for me to go into the need of Conservation, since that has been done in the previous Congress and at various times in the public prints. The question then to which I must devote the time of the program assigned to me is this: How does taxation affect the Conservation of natural resources, and what suggestions of a practical nature can be made for the betterment of the taxation of such resources?

It may be said in the beginning that the difficulties involved in the taxing of natural resources exist to still greater degree in the case of other property. Generally speaking, we have not attained to a rational system of taxation in any field, and we are now attempting to revamp the old system and extend it, by adding to or taking from it. Economic conditions in America have changed from time to time, and these changes have forced upon us a reorganization of our methods, not only of manufacture and of transportation, but also of administration, government, and social organization. Such a condition of affairs is seen today in nearly every State, and attempts are being made to meet it in the specific instance of the fiscal problem by adding to the old system of taxation through the special taxation of corporations, inheritances, royalties, and incomes. The consequence is that so far as natural resources are concerned we have no principle existent in the general scheme of taxation that can be used to meet the new conditions that have arisen in our efforts to conserve our resources. Just as the problems of industrial organization have come upon the States, so now has come the problem of our natural resources. In hazy thinking, and sometimes in indefinite laws, we have attempted to regulate through legislation the great corporations of the present day; and in much the same

manner we shall, by feeling our way, attempt to develop some plan of taxing natural resources.

Sometimes in discussing this question of the taxation of natural resources a great deal of emphasis is placed on the statement that it is the cause of the depletion of timber and mineral lands especially. I think it may be said at the outset that the taxation of natural resources is only one of many factors in the destruction of them. The extent to which this takes place is impossible to say, but the fact remains that the taxation of natural resources may or may not hasten the destruction of forest lands, the exploitation of minerals, and the cultivation of the soil. Where lands bearing timber are owned, interest charges with each year of ownership are piled up, and the same is true of the taxes. Where, on the other hand, lands are held through a royalty contract, the lessee is in a position to carry the lands without special cost to himself except that of the taxes. The consequence is that it is impossible to apply the same principle of taxation to agricultural lands, timber lands, minerals, and water-powers. There must be a differentiation between them, and a differentiation that will clearly meet the various uses to which they are put.

Without question, the general property tax, as it now stands upon the statute books of the different States, does not meet in any true sense of the term the general economic conditions, and the special needs of mining and lumbering in particular. The principle of taxing the product when it is placed upon the market applies particularly to mineral and timber lands, but the same principle in the case of agricultural lands would probably deter their use and fail to meet the needs of revenue as well as working to the discouragement of the agricultural industry. The single-taxers have insisted that the taxation of lands hastens its use, that it forces the owner to develop it; and this is just the thing that is needed in the special instances of agricultural lands and of town lots, but the same principle could not be applied to the other resources of the Nation.

It is possible for the owners of timber lands by following the principles of forestry to modify the product and to keep the land in producing condition indefinitely. Taxation of such land, therefore, should have in view the maintenance of this condition. It must be clearly understood, however, that the fear of fire, interest charges on investment, and the cost of management

will act quite as surely toward the rapid destruction of forests as will taxation. These conditions must also be recognized by the State in the establishment of a fire warden system, and the encouragement of forestation through some plan of bonuses. Where forestation is not practiced, the taxation of timber products under present conditions, whether on stumpage or in transit to the saw-mills, is a serious problem—serious to the local governments because under existing laws logs in transit are taxable where they are owned, and serious to the owners of the timber lands because the fixed charges on their property increase each day without any income from them. As near as can be ascertained, the annual taxes on timber vary from one cent per thousand feet to fifty cents per thousand feet, with an average tax of somewhere in the neighborhood of fifteen cents per thousand feet. Interest charges are probably about twenty-three cents, making a total annual cost of something like thirty-eight cents per thousand feet. In ten years time the tax on each thousand feet of standing timber will amount to \$1.50, which compounded with interest makes a total of \$2.37. When added to the other charges it is probably true that the owner of timber under modern conditions must have at least \$13.02 per thousand feet on his logs delivered at the mill if he is to come out even at the end of ten years with a profit of six percent.

The suggestions which have been made from time to time regarding the taxation of timber have as their fundamental principle the separation of the value of the land from the value of the timber. This plan meets the criticism of the local assessing officers by providing a basis of taxing annually a part of the valuation, and of procuring some income for the local government. If it is understood then that the land may be taxed annually and the timber product when it is cut, we have under this plan a simple scheme of taxation which will unquestionably meet the difficulty that is now urged against the general property assessment of timber lands. Under the old plan of valuing annually the property, it was difficult to secure an appraisement that was satisfactory to anybody; and, what was more, as the years went by the local governments found their assessed values decreasing and the burden of government materially increasing with the decline in amount of standing timber. The annual taxation of the land on which the timber stands meets this difficulty, while the taxation of the product at the time of harvesting

provides a plan that is fair both to the local government and to the owner of timber.

On the other hand, the taxation of mineral properties differs from the taxation of timber lands in that it is not possible for the owner to increase by any plan of Conservation the amount of tonnage that he has in his possession. The Conservation which he might practice is the simple Conservation of saving for a future time. From the point of view of the State the problem is largely one of getting a share of the value of the minerals in the ground. The method that has been generally followed is that of making an appraisement of the mineral lands, which might be very far from or very near the truth. The same principle which is applied in the case of the timber lands, namely, the taxation of the product, should be applied to the taxation of mineral properties. There is no question that the easiest way, and the most satisfactory and acceptable way to all concerned, is a tonnage tax, varying possibly with the character of the ore and the cost of mining, but always depending for the rate and the amount on the ore that has been mined. It will probably be argued, as it has in other instances, that the local governments are compelled to rely largely for their support upon the taxes paid by the owners of mineral properties, and consequently a tonnage tax would deprive them of the regularity of their income. There is much to be considered in this point; but the taxation of the surface on some such basis as that seen in the case of the timber tax would provide a regular income, which would be supplemented by the amount of the tonnage taxes.

The rate of the tonnage tax would not, as in the case of the appraisement of a general property tax, tend to hasten the utilization of the ore. That would be determined entirely by the demand for it in the fields of manufacture. The real essence of the tonnage tax lies in the fact that value found in the ground is distinctly a product of nature, which an *ad valorem* tax cannot recognize, and in consequence the State's right to a share of the value of the earth's products, together with the diminishing value element involved, are overlooked. The protection of the local government, and often of the mineral owner, demands a combination of the tonnage tax and of the local land tax.

When we come to the taxation of water-power we are face to face with a problem that involves even more difficulties than are found in the case of

the timber and mineral lands. The thing here involved is so elusive, so difficult of measurement, and requires such expensive administration, that it is quite conceivable that many years must elapse before an adequate plan for such taxation can be developed. A water-power, however, is perpetual, and in this particular it differs from timber and mineral properties, and is more akin to farm lands. It differs from the latter, however, in this particular, that the work once done in harnessing it is done once for all, and the annual labor expended upon it is not exhausted, as in the case of the farm. Nature, having been harnessed, is able to accomplish the work for which she is called upon.

The first step in any adequate system of taxing water-powers must be their survey. This means listing, locating, and measuring. It means, too, that the Legislature should assume at the beginning all water-powers belonging to the State, and that the acquirement of them must be through lease, as in the case of mineral lands in the State of Minnesota, for example. Several plans have been suggested for the taxation of water-power. One is the measurement of the water flowing over a dam, and another is the taxation of the actual horsepower developed. The latter plan is subject to many criticisms. The development of horsepower depends so largely on the skill of the engineer, on the capital invested, and on the way the water is handled, that it would be far better to measure the capacity of the dam under proper engineering authority and determine a fair rate for the amount of power produced by the water passing over the dam. Of necessity many refinements of this plan would be required; such as the determination of the movement of the stream, the height of the water, the difficulties of harnessing the power; but it is possible, by taking into consideration the general expense of operating a water-power plant, to work out a rate which would be fair to the users as well as to the State. In no instance of Conservation does a greater need of proper taxation appear than in the case of water-power. Nature provides a perpetual force with but little expense after the necessary fundamentals have been arranged, and for the State to receive no compensation of any kind for the utilization of such a great wealth-producer is to bring into existence the greatest possible factor of injustice in the matter of taxation.

It will therefore be seen that a rational taxation of natural resources does not depend on any very great and intricate principle, but that, on the other hand,

the principles involved are comparatively simple. It must be clearly understood as well that the taxation of land for agricultural purposes, for minerals, for timber, or for water-power, must differ in many respects, and that a principle of taxation applied in one case may not work out in the other. But if we keep clearly in mind the purposes for which land can be utilized, and that the fundamental taxation of land as such can be made annually, and that of the product at the time of its harvesting, we have in the three instances of agricultural, mineral, and timber lands a principle that may prove satisfactory when put in the form of legislation. The same idea can be applied to the water-power site; taxation of the land at a nominal assessment and of the water-power on the basis of the amount of water passing over the dam gives us again a principle upon which can be based satisfactory legislation.

It must be remembered, however, that all legislation is compromise in character, and that the recognition of these principles has usually been set aside when it came to the question of legislation. The States have reached a point in the raising of revenue where not only more revenue is needed for the purposes of general social advancement, but where better administration is as essential and necessary as the other. Administration bureaus must be provided in all of the States to furnish the necessary data, if we are to reach some practical basis of conserving our resources through taxation. And tax commissions must be given ample authority, and in addition must have plenty of expert advice and assistance which will give it the necessary endorsement. To my mind, a rational system of taxing natural resources depends largely on administration based upon a few fundamental principles of legislation. It is comparatively not a difficult matter; it is largely a question of willingness to meet the problem; but if the experience of the past has any light to throw upon this subject, it is very clear indeed that legislation will be slow, and that the different interests involved, through fear of some possible advantage likely to be gained over them, will cling to the old system until it is almost too late to produce any results through adequate taxation.

It is my hope that a Congress like this may have some power and some influence in setting aside this attitude, but I fear that an adequate system of taxation will move very slowly when it comes to its formulation in legislation. This is not encouraging, but it is truth; and that after all is what

we are really trying to get at without confusing the issue by arguments favoring present attitudes either of the State or of owners of natural resources. Big views will help solve the problems, little and narrow ones never. (Applause)

President BAKER—Mr J. B. White, Chairman of our Executive Committee, will discuss the question of taxation, especially in relation to woodlands. (Applause)

Chairman WHITE—Mr President, Ladies and Gentlemen: We have listened to a great paper upon this subject of taxation. It is a subject difficult to analyze and very difficult to apply, because each section of the country requires a different form of taxation; each State has different views, and each should apply the remedy according to the local conditions.

I speak as a representative lumberman, and as Chairman of the Conservation Committee of the Lumber Manufacturers of the United States. Now, the lumbermen have asked for nothing in regard to taxation excepting what they have incorporated in a resolution, part of the preamble to which I read:

Whereas, there is a great and growing need for uniform laws among the States in the interest of forest growth, conservation, and protection from forest fires, and for an equitable and helpful system of taxation which will make possible the conservative handling of standing timber.

That is the declaration of the preamble. It asks simply a uniform system of taxation.

I want to say a word for our fathers and grandfathers who have been called the ruthless destroyers of the forests, and I want to say in their behalf that they committed no sin which shall be visited upon their children or their children's children (applause). They cut the forests to make homes for the people; they cut the forests to build our cities and our towns; they sold all they could, they saved all they could, they committed no waste; and it should not be imputed to them that there is a penalty to be paid by their

children or their children's children upon the forests that now stand.
(Applause)

Taxation is regarded everywhere as a part of the cost of a commodity. Every person that buys a foot of lumber, every person that buys a yard of cloth, every person that buys a suit of clothes, or groceries, or anything that is manufactured, is the one who pays the taxes (applause). We are all consumers. We pay each other's taxes, and there is no way of avoiding taxation. It is said that death and taxation are sure. There is no way of avoiding either. The consumer must pay the tax because it is part of the cost.

Now, in regard to the system of taxation; every Nation has its own form. When it is necessary to encourage the growth or manufacture of a product, the States of the world have some way of encouraging it by relief from taxation. Germany has a law putting a duty on American wheat in order that every nook and corner of the waste land of Germany may be made to grow wheat. Now, that is a tax. The people of Germany pay that tax, but it encourages the farmer to grow wheat. And in our own country, when it is necessary to encourage the farmer in the beet-sugar, or any related industry, the Government gives a bounty, and people pay it, and the money is kept at home instead of going abroad for the product. So in timber taxation, it would seem to me that the reasonable way is to tax it as it is cut—let the tax follow the saw. Of course every State will apply the remedy according to local conditions. Louisiana has applied the remedy. She has passed some very good laws, and we are going to hear from the representatives of that State, before this Congress adjourns. We want to consider these things.

There are now so many substitutes for lumber that there will be inducements to let trees stand if they are not overtaxed. A tree must have a hundred years' growth before it can be utilized in the shape of clear lumber in the upper grades. If you tax the tree every year, you are putting one hundred years' taxes upon the timber. We must be reasonable about these things if we would encourage the growing of trees. Any other commodity in the United States pays a tax annually upon the crop, but here, in growing timber, we are paying for a hundred years where we should only pay for one. (Applause)

Some States will not grow trees. Illinois will not grow trees. It would prefer to grow corn. Its land is too rich to grow timber, and the people will grow corn and exchange it for the product of other States which are better adapted to tree-growing and not so well adapted to agriculture. The lands west of the Cascade Range are well adapted to tree-growing on account of the great rainfall, and not so well adapted for other uses. A tree will grow there in forty years to as great a size as it will in eighty years on this side of the Cascade Range. In short, trees will be grown where it pays to grow them, where they are encouraged to be grown, where the people want them grown. We cannot grow trees on sentiment; tree-growing will have to pay; it will have to stand upon a commercial basis. The Government cannot grow trees without its costing something to grow them. Conservation has been wrongly understood.

The great leader of American forestry, Gifford Pinchot, is in favor of development (applause). He said in his speech at Seattle a year ago that there could be greater waste by non-development and by non-use than there had been by the wastefulness of the past. That is true. By non-development and non-use we commit sometimes more waste than we did in the past, for we could not waste when things were not worth anything; a thing that isn't worth saving and whose by-product cannot be utilized is not wasted even if it goes to the burning ground or lies in the woods. (Applause)

President BAKER—Ladies and Gentlemen: You will all be glad to hear from the greatest, grandest, noblest work of God, our good women. I have the pleasure of introducing Mrs George O. Welch, of Fergus Falls, representing the General Federation of Women's Clubs. (Applause).

Mrs WELCH—Mr President, Delegates to the Second National Conservation Congress, Ladies and Gentlemen: In the preparations for this great Congress, there seems to have been no possible item omitted which could in any way contribute to the pleasure or edification of visitors, save in two

particulars; and with these the management had nothing to do. The first is the unavoidable absence of the President of the General Federation of Women's Clubs, Mrs Philip N. Moore, resulting from the accident which befell her in Cincinnati last May, from which she has not fully recovered. The second is due to those two elements which have for years uncounted interfered with man's proposals—time and tide. It is because time must be consumed in crossing the Atlantic and tide reckoned with on the voyage that Mrs Emmons Crocker, of Boston, is not able to be present to speak on "Woman's Influence in National Questions." Her absence is indeed to be regretted, since influence is today women's best asset.

Because of these two regrettable occurrences a great honor and pleasure has fallen upon me. I am proud to be the bearer of greetings to the Second National Conservation Congress from the General Federation of Women's Clubs, an organization 800,000 strong, that may justly claim kinship with this body, since its watch words for years have been Conservation and Service, which are the impulse and purpose of this great Congress.

The inception of the General Federation of Women's Clubs was due to the recognition of the necessity of conserving the energy and strength wastefully expended by scattered clubs remote from each other, which concentrated, might make a tremendous influence for the development of good fellowship and good citizenship. That the General Federation has become of great force I think you will admit, since its President was invited to be one of that first notable Conference called by the President of the United States in 1908 to consider the problems which this Congress is hoping to solve. She was the only woman invited to that Conference of Governors, and it is not vain pride which prompts the mention of the great honor thus conferred upon the General Federation—it is rather an humble sort of pride, since recognition of the work which Women's Clubs are doing carries with it an obligation to greater effort and greater achievement.

The General Federation of Women's Clubs has long been teaching the necessity of Conservation, not only of the natural resources on which the material prosperity of this country depends, but of that vital force which means public health and all that goes with it; of that intellectual force which means education; and of that spiritual force which makes for higher ideals,

wider sympathies, and fuller appreciation of our responsibility for the welfare of our fellow-beings.

In the matter of the Conservation of natural resources, the one which claimed our earliest attention was that of forestry. As far back as 1900 the forestry committee in the General Federation served to bring into mutual recognition and helpfulness the efforts of all the clubs engaged in the work for the protection of forests; and I was proud of the praise given us yesterday by our most distinguished visitor for Minnesota's successful efforts to preserve a large acreage of white pine timber as a National forest reserve. It was a fine and inspiring example to other States engaged in a warfare against the devastating hand of commercialism (applause). And it is another matter of pride that for four years the chairman of the forestry committee of the General Federation was a Minnesota woman, Mrs Lydia Phillips Williams (applause), whose life was devoted to the promulgation of forestry education, and to whose untiring efforts very much of the splendid work done for forestry by Women's Clubs is attributable.

Perhaps the most signal of the triumphs won by the Women's Clubs in the line of forestry was the saving of the big trees of California, after a fight lasting nine years (applause). Those were years of great stress for the women, but we are willing to fight nine years more if need be for the right sort of protection to the forests in the White Mountains and Appalachian ranges (applause). Today we are fighting not alone for the trees that are standing, but for the reforestation of devastated lands and for a stay of the wanton waste of forest products. At our recent biennial convention a whole session was devoted to this phase of the work, showing that our interest is practical as well as sentimental. Since the conserving of forests and the conserving of water supplies are interdependent, the General Federation of Women's Clubs through its committee on waterways is disseminating information, creating interest, and urging legislation for the further protection of these resources.

But the Conservation of natural resources, important as it is, is not the work which represents our heart interest, which appeals to our highest nature; it is not the thing for which we make our greatest effort. It is the problems of life, those affecting the home, society, our children, to which we give our most earnest endeavor. There never was a convention of Women's Clubs

anywhere that did not in some way stress the Conservation of the home, the family, the school, as our greatest need; and it is because we are aware of the grave dangers threatening them, dangers born of our times and fostered by our rapid material growth, that we are endeavoring through organization and concentration of forces to turn the tide into safer channels.

The child has always been the central figure in our deliberations, the one for whom our hardest battles have been fought. The General Federation, through its committees on health, education, and household economy, is carrying on a campaign of education which will give to all children greater opportunity for normal, helpful, happy development. To the child himself, through its department of civics, the Federation is teaching his duty to society and his responsibility to the future. Through its committee on industrial and social conditions it is trying to secure for him safety and efficiency in the great industrial struggle; to protect him against the forces that are pushing him, imperfectly prepared, into the great maelstrom of the workaday world, wasting his young life, minimizing his chances for happiness and usefulness. As long ago as the Los Angeles convention in 1902, Jane Addams, our greatest American woman (applause), pleaded for the protection of the child against the awful economic waste of child labor (applause). She told of little lives by scores and hundreds yearly sacrificed to the god of greed: of conditions in some of the industrial pursuits where for want of a few dollars expended in safety devices, many children were yearly killed outright, or maimed for life. She so touched the hearts of her hearers that a committee on child labor was there created, whose province it was to discover if possible a remedy for these crying evils; at any rate to inform the public of their existence.

Women have worked long and earnestly to ameliorate these conditions, but they must depend on the mutual action of earnest, interested men, such as are sitting in this Congress today, for the enactment and enforcement of the laws necessary to improve a state of things which women have only the power to point out. In the particular case of child labor there can be no accusation of exaggeration or hysteria, since from so unemotional a source as the Federal Government we learn that its recent investigation of child labor shows need of a strenuous and continued effort for the conservation of child life. In the cotton textile industry alone, and along the line of age-limit and illiteracy alone, its statistics show that in a group of States having no

age limit for child laborers, there are over 10 percent of female workers under fourteen years of age, and that in those same States over 50 percent of the children of both sexes so employed are unable to read or write. What worth have forests or mines or any material wealth, gained at the sacrifice of so much vital force?

For the welfare of women and girls, as well as for children, the General Federation is working with all its energy and strength. For moral and social as well as industrial protection it begs cooperation. Against the black plague as well as the white plague it is waging its warfare. For better housing in cities, for improved conditions in rural and remote communities, it is using all its power. What conservation and concentration of effort can do it is trying to accomplish, but it must as yet find its work constantly hampered and hindered by its inability to press to their ultimate accomplishment things which only legislation can effect. A club woman has wisely said that as conditions are today it is the women who suggest and initiate, the men who adopt and complete. This is true; for, after all, women can only point the way.

The Ex-President of the United States told us yesterday that it was a great wrong to allow any body of people to monopolize any good thing. There is, however, an exception to this rule, which I am sure our honored First Citizen would concede to us: Women have long had a monopoly on influence; it has been the one thing accounted their own particular weapon in social warfare (applause). And so I appeal to the men in this audience to yield themselves to that women's weapon when next the General Federation of Women's Clubs or any individual members of the Federation asks them for the enactment of laws which shall tend to the Conservation of the vital forces represented in the mothers of the race and the children who are to be the country's future citizens. The General Federation is, after all, just one more organization trying to make this land a better place to live in, and its people better fitted to live in this better land. (Applause).

President BAKER—The next lady I wish to present represents an association that has done much; Mrs Hoyle Tomkies, of Shreveport, President of the

Women's National Rivers and Harbors Congress.

Mrs HOYLE TOMKIES—Mr President, Ladies and Gentlemen: Greetings to this Second National Conservation Congress from the Women's National Rivers and Harbors Congress, organized June, 1908, and having officers in thirty-eight States and Territorial possessions.

This organization has for its object the development of the meritorious rivers and harbors, the preservation of the forests, and the Conservation of all the natural resources of the Nation. It stands for the establishment by the Federal Government of a definite waterway policy for the improvement of all approved rivers and harbors of the entire country, and also for the adoption of such a policy as will secure not only forest reserves but general forest development. The Congress believes that the development of the waterways of the Nation increases and conserves the people's wealth, *first*, directly, by securing the cheapest mode of transportation; *second*, indirectly, by lowering the cost of transportation by rail; and *third*, by encouraging production. The platform as adopted immediately after organization stated a belief in the need for the Conservation of all the natural resources of the Nation because of the interdependence which necessitated the development of each.

The membership of our Congress is composed of individuals and clubs, representing almost thirty thousand men and women, the latter largely predominating. The work of the Congress, conducted through the Departments of Education and Publicity, is directed by a board of directors representing thirty-nine States and Territories. Voluntarily these women are giving their time, finding in the joy of service for the cause ample recompense.

In the educational campaign, the Congress has culled from the best authorities the strongest arguments and convincing statistics, and has had these printed and circulated in many thousands of copies throughout the length and breadth of the land. In 1908 this Congress secured the

cooperation of the General Federation of Women's Clubs for the promotion of waterway development.

Since organization the Congress has worked incessantly for the passage of Rivers and Harbors bills, and individually for State projects for waterway development. It has worked for the Weeks Bill, and for general National and State development. It urged upon Congress the passage of the bill for the preservation of Niagara Falls in the spring of 1909.

In its educational campaign it has covered the entire question of Conservation, and also urged the non-pollution and the beautification of the streams of our country. It has secured and arranged for large audiences in critical or indifferent centers, for experts to advocate the cause, and it has had speakers at all important public gatherings possible. It has organized Conservation clubs, and secured the addition of Conservation committees in various organizations. It has offered prizes, securing the writing of many thousands of essays by school children upon waterway and forest development. The various State vice-presidents have issued State circular letters, showing how their States were concerned in the cause we represent.

The plan of the Congress to supplement or substitute Arbor Day with Conservation Day met with the hearty approval of the United States Department of Agriculture and the cooperation of many educators, and has been successfully carried out in many States. The resolution of the Congress asking that the principles of Conservation of natural resources be taught in the school and summer normals, has been presented to every State represented in the Congress, Louisiana being the first to immediately pass the resolution unanimously at its State Conference of High School Superintendents, representing forty thousand pupils, and at its State Teachers' Association; Kentucky being a close second, with every encouragement from other States. (Applause)

The same resolution was presented to the National Educational Association in convention at Boston, July 5-9, 1910. Of this resolution, Honorable Elmer Ellsworth Brown, United States Commissioner of Education, to whom we later had the pleasure of listening, wrote in reply to me a pleasant letter in which he enclosed the following copy of his letter to Dr Irwin Shepard, Secretary of the National Educational Association:

DEPARTMENT OF
INTERIOR
BUREAU OF
EDUCATION
WASHINGTON

DOCTOR IRWIN SHEPARD,
Secretary National Educational Association,
Westminster Hotel,
Boston, Mass.

MY DEAR DOCTOR SHEPARD: The preamble and resolution enclosed herewith have been sent to me by the Woman's National Rivers and Harbors Congress, Mrs Hoyle Tomkies, of Shreveport, Louisiana, as President National Educational Association at its Boston meeting. Following our ordinary course in such matters, may I ask you to lay this matter before the committee on resolutions.

You are aware of the conservative position which I take as regards proposals for the incorporation of new studies in our school curriculum, and also as regards the turning aside of our school instructions from the aims of general education to the propaganda of any special cause. The organization presenting this resolution, however, disclaim any intention of introducing a separate new study in the course. The subject which they propose, however, is one so intimately bound up with the geographical conditions and the past history of this country, as well as with our prospect for the future, that it seems to me very desirable that the attention of teachers should be called to it, and that they should be led to see its relation to any proper and adequate treatment of a knowledge of our country. I should think it very desirable, accordingly, that something of this kind be introduced into the platform of the Association of this year, with such adaptation of form and phraseology as the common practice of the Association would suggest.

I am, believe me,

Very truly yours,
[Signed] ELMER ELLSWORTH BROWN,
Commissioner.

As to the action of the National Educational Association regarding the resolution, Dr Shepard wrote to me in part as follows: "I sincerely regret that you were not duly informed earlier of the action, or rather the non-action, of the Committee on Resolutions. I cannot explain their action in this matter. They had a large number of subjects to consider, and the omission of a declaration upon any subject is not to be considered as a judgment against such a declaration, but simply that the Committee did not find it practicable, for reasons satisfactory to them, to include it in the declarations which they offered. Incidentally I may suggest to you the present uncertainty regarding what is meant by Conservation and the wisest policies to be adopted may have led them to defer action in this matter. Let me assure you that we are all deeply interested in Conservation, and believe that it can be profitably brought into the work of the public schools, but many are still uncertain as to the form of such work and the methods by which it can be most profitably introduced into the public school curriculum."

Members of this Congress, there is in this non-action a suggestion potent to us. This indecision, this lack of harmony, should speedily as possible be changed into a definite, harmonious union of Conservation policies (applause). This fall a printed catechism of questions on Conservation adapted to the various grades will become a part of the curriculum of the public schools of Kentucky, and will be tried in various other States.

Delegates have been sent by the Women's National Rivers and Harbors Congress to all important conventions of kindred interests. Since organization it has had representative speakers on the platform of many of the most important conventions. The Congress has furnished lecturers to schools and to various clubs of men and women, and also to the churches, in which latter the subject of "Conservation of Natural Resources from the Moral Standpoint" has proved an appropriate and impressive theme.

In December, 1909, the Congress endorsed the disinterested and patriotic policy of Honorable Gifford Pinchot as Chief Forester of the United States.

(Applause)

This report cannot satisfactorily be closed without mention of the loyal and very enthusiastic support of Conservation being given us by our Hawaiian members, who number several hundred, and who began immediately to put belief into practice. Our State vice president there, Mrs A. F. Knudson, came all the way to Washington to attend our last convention.

These are the general activities of the organization. It would be impossible for me to go into the State activities at this time. Sufficient to say that the message is being given at the fireside, from the platform, in the schools, through the press, all with the idea of perpetuating this Nation—won by the blood of our forefathers—and handing it down in all the glory of its wealth and beauty to future generations. (Applause)

President BAKER—It is a pleasure to present Mrs G. B. Sneath, of Tiffin, Ohio.

Mrs SNEATH—Mr President, Ladies and Gentlemen: After hearing the general purpose for which the women of the General Federation of Women's Clubs have been working, it may seem needless for me to tell what one definite part of this great body is endeavoring to accomplish. I represent Mrs J. D. Wilkinson, Chairman of the Waterways Committee of the General Federation, which is a part of the great Conservation Committee of the Federation, comprising almost 800,000 women in its organization.

Our work is entirely educational. We go into all the schools where we can possibly gain access, and strive to get the matter of preservation of inland waterways taught in the schools as among the great Conservation problems. We have heard from experts all that is being done, all that they are trying to do, all that they are trying to remedy; and we feel that we, as women, have one chief and great duty to perform. You have heard how women strive to

conserve the lives of children, to make them strong mentally, morally and physically. Yet this is not all; the one great problem before the American people today is that of pure food and pure water (applause); and we, as women, must strive in the communities in which we live and the States of which we are a part—and the Nation must come to our aid—to rescue and prevent from contamination the life-giving streams of this country, streams that were given for the benefit of mankind but which man has turned into drainage canals and cesspools. We must have help; we must have it through State Legislatures, we must have it through the Federal Government, else we cannot conserve the lives of those that are dear to us. If a visitor from another land were to say to us, "Your children are being poisoned by their own parents," we would hesitate to believe it; but our children *are* being poisoned—not by criminal intent but by the carelessness of the municipalities in which we live (applause). So I leave with you this one thought: If we accomplish nothing else, if we leave to the men the questions of transportation and navigation and the great problems of irrigation and of water-power, let us work for the purity of our rivers and streams and lakes and inland waterways.^[1] (Applause)

President BAKER—The Proceedings of this Congress are to be published through the kindness of a gentleman in Saint Paul who has guaranteed to have it printed, and all these addresses will go in.

We will now hear from Mrs Jay Cooke Howard, of Duluth.

Mrs HOWARD—Mr President, Ladies and Gentleman: I will keep you only a minute, because you look hungry, and I'm hungry myself. I will simply file my report and tell you briefly what the Daughters of the American Revolution are doing for Conservation.

The D. A. R., being a patriotic society, believe that all their work is in the spirit of true Conservation; but we have a special National Committee, with

a member or members from each State. I represent the chairman, Mrs Belle Merrill Draper, because I am the member for Minnesota. Mrs Draper wrote last fall to all the Governors, asking each what we could do to help the cause of Conservation in his State. When the answers came we went to work, chiefly in three ways: First, in our own meetings, in which we worked up enthusiasm. Second, in the press; the papers in the larger cities have much Conservation matter, but in smaller cities and towns this is not always the case, and you from such places will never know how much about Conservation that you have read—or skipped—was inspired by the D. A. R. Our third branch of work, and the most important one, is with the children. I notice that most of the Governors, whose interesting letters are contained in the report I am filing, preferred to have us turn our attention to the children rather than to the men (laughter). Governor Eberhart's courteous letter mentioned them, and the forests, especially. We have worked through the schools, and also in our own homes. May I tell my own experience? [Voices: "Go on, Go on!"] I felt very proud when my little boy, who had saved eleven cents and did not know what to do with it all, finally said, "Mother, I will give it to the baby; put it in his bank; it will teach him to save." But straws in the family show which way the wind blows in the Nation. Listen to what happened: I provided savings banks, the children conserved their resources, saved their wealth and then somebody came and stole the banks! (Laughter and cries of "Good!")

President BAKER—The Congress stands adjourned until 2 oclock.

SIXTH SESSION

The Congress reconvened in the Auditorium, Saint Paul, at 2 p.m. September 7, President Baker in the chair.

President BAKER—Ladies and Gentlemen: I have the honor of asking Senator Moses E. Clapp, of Minnesota, to preside this afternoon, and to him I now yield the chair. (Applause)

Senator CLAPP—Ladies and Gentlemen: During the course of this Congress much has been said concerning the fact that Conservation applies not only to the material resources of a Nation, but to its productiveness and to its energies; and among those things to which it must under that classification apply is the Conservation of time. Now, I am going to give you a practical illustration of how a loyal adherent can carry out the Conservation of time by omitting a speech, and proceeding at once to the business of the afternoon. (Laughter)

The first entry in the program for this afternoon is an address, "Making Our People Count," by Dr Edwin Boone Craighead, President of Tulane University, whom I take great pleasure in introducing. (Applause)

President CRAIGHEAD—Mr Chairman, Ladies and Gentlemen: In this Republic there is one thing supremely great and sacred, greater than the great Republican party, the party of Lincoln and Grant, greater than the great Democratic party, the party of Jefferson and Jackson, more precious than the Conservation of our natural resources, more sacred than the Supreme Court, or even the Constitution itself—I mean the great American people (applause). To make this people count, not only in the Conservation of our natural resources but also in the enlargement and enrichment of their own lives, is the fundamental, the paramount, problem of this Republic; for ours, it must not be forgotten, is not only a Government of the people and by the people, but also preeminently a Government *for* the people.

The Founders of this Republic were not only scholars and thinkers, but seers and prophets. With profound knowledge of the despotisms that for five thousand years had crushed and enslaved the greatest and sublimest thing on this earth, the individual man, the Fathers of the Republic laid broad and deep its foundations upon an everlasting rock—the inalienable, the ineradicable, the eternal right of man to life, liberty, and the pursuit of happiness. They builded for all time and for all generation of men. (Applause)

The individual man, the individual woman, is by far the greatest and sublimest creation of God that we know of—far greater and grander than any or all the institutions of society. These institutions are the works of the hands of man, they exist for him, and their only reason for being is that they minister to him. Yea, the earth was made for man, and the only reason for the Conservation of its resources is that they may minister unto the needs of the individual man:

Seas roll to waft him, suns to light him rise;
His footstool the earth, his canopy the skies.

In the deliberations of this Congress the words of Ruskin should be uppermost in the minds of all: "There is no real wealth but life;" and by life he meant the perfection of the entire man, body, soul, and spirit. That church is best, that institution is noblest, that civilization is highest, that country is greatest, which furnishes the most abundant life to the largest number of human beings.

The Chinese Empire, which embraces near four hundred million human beings, has existed for five thousand years; yet the countless millions of China, springing up like tropical weeds and sinking back to dreamless dust, have contributed far less to civilization than the twenty thousand Athenians who in the brief Periclean age followed the footsteps of Plato and Socrates. (Applause)

Neither vastness of population or territory, nor richness of natural resources, nor accumulated wealth can alone make a great country. That country is great, no matter how barren its soil, whose children may truthfully repeat the words of the stern old Spartan, who, when one pointing in derision to

the bleak hills of Lacedemonia asked, "What do you grow there?" replied, "We grow men there" (applause). To breed a race of strong men and noble women is the one and only thing that can make a country truly great.

Consider Scotland—a poor and barren country, yet who would dare to call poor the land of Scott and Burns and Carlyle? Who shall estimate the wealth of Scotland's contribution to the world and to America? The sons of her sturdy pioneers who poured down through Virginia and Kentucky and the Carolinas have been worth to this Republic their weight in gold. (Applause)

Take Ireland, that synonym of poverty; and yet how could our great metropolitan cities thrive for a single day without the helping hand of the sons of Erin? Somebody has advised that we buy Ireland, not for her natural resources, not to grow corn and wheat and cotton, but to grow policemen. (Applause)

Coming a little nearer home, take New England with her thousands of abandoned farms, rich only in the variety and ferocity of her climate and the blessed dispensations of our American protection; and yet far from mean have been New England's contributions to the wealth of American democracy. New England, rocky old New England, barren, storm-swept New England, "Land of brown bread and beans," home of the liberty-loving Puritans who, for the sake of the immaterial good, in quest of freedom, crossed the stormy sea, endured the hardships of an untamed wilderness battling with hunger and wild beasts and savages—grand, glorious New England (applause), home of Adams and Webster and Emerson and Hawthorne and Williams and Lowell and Longfellow and Edward Everett and Phillips Brooks (applause)—grand, glorious, immortal New England, by her schools and colleges has almost dominated the intellectual life of this country; and in every part of this vast Republic, yea, in every civilized land under the sun, may be found the sons and daughters of the pilgrims of the Mayflower; scholars, preachers, teachers, missionaries, pioneers who have blazed out the pathway of civilization, established schools and colleges and universities, always and everywhere children of sweetness and light who even on the remotest frontier have kept trimmed and lighted the sacred lamps of learning (applause). Harvard, Yale, Princeton, Bowdoin, Dartmouth, Williams, have contributed more to the dignity of man, given

more to the everlasting glory of the American commonwealth, than all the stock speculators of New York, or all the battleships ever built for the American Navy. (Applause)

Take only one other illustration: Who of you from the waving cornfields of Iowa and Illinois, from the fertile lowlands of the Mississippi, has not wondered, while passing through the Old Dominion and looking out upon her red clay hills, how on earth do these people make a living? Why give me one acre of the best Louisiana soil—and it is nearly all good—and put it down upon the barren rocks of New England, or upon the red hills of Old Virginia, and I would make a fortune selling it for fertilizer (laughter). And yet Virginia has contributed more to the wealth of the American Republic than any other single State of the Union (applause). At the call of what other States did there ever arise a larger band of more gallant men than they who under the leadership of Jackson and Lee withstood for long weary months the combined forces of the Union? And when the War was over, and Virginia found herself in abjectest poverty, she showed to the world that her riches were inexhaustible; for during the next forty years she sent abroad into other States five hundred thousand of her most adventurous sons (applause), and, in so doing, contributed more to the wealth of this Republic than all the gold that was ever dug from the mines of California (applause). I do not wonder that the poorest the humblest son of the Old Dominion, no matter where he finds himself, whether trudging through the snows of Minnesota or loitering perchance beneath the fragrant magnolias of Louisiana—even he, the poorest and humblest, must quicken his steps and lift aloft his head as he remembers, "Mine is the land of George Washington and Thomas Jefferson and James Madison and James Monroe and John Marshall and John Randolph and Patrick Henry and Stonewall Jackson and—towering above them all save Washington only—that matchless military chieftain, great in battle but still greater in defeat as a private citizen, the stainless, the immortal Robert E. Lee." (Applause)

James Russell Lowell said—and said truthfully—that countries are great only in proportion to what they do for the moral and the intellectual energy, the spiritual faith, the hope, the comfort, the happiness of mankind. (Applause)

Chairman CLAPP—Ladies and Gentlemen: It is provided in the program that between the set speeches we will hear briefly from the accredited representatives of the various States, taken in alphabetical order. I now have the pleasure of calling upon the State of Alabama. (Pause) If no one cares to be heard from Alabama, I now call upon Arizona. (Pause) If no one from Arizona, then from the State of *Arkansaw*; and that there may be no mistake on the part of the inhabitants of that State in the termination of the name, I repeat that call in the name of Arkansas. (Laughter)

A DELEGATE—Mr Chairman, I suggest that the call of the States be deferred until 8.30 in the morning, and that it then be taken up as a definite matter of business.

Chairman CLAPP—Will the gentleman make a motion to that effect?

[The motion was made, seconded, put and carried without dissenting voice.]

President BAKER—Mr Chairman: I will be very glad to be here at 8.30. We want everyone to be heard, and I would come here at 6 oclock if desired, though I think 8.30 is early enough. I will be here promptly to open the Congress and hear from the States until the regular speakers begin. Then on Thursday afternoon we have set aside a special time to hear from all the States and all the different organizations represented here.

Chairman CLAPP—Ladies and Gentlemen: During last summer it became my province to distribute nuggets of moral philosophy and political truth to the people of Kansas, Nebraska, and Iowa; and while laboring in that moral vineyard I discovered that there was a newspaper in the Southwest that had an immense influence throughout all that section. We have a representative of that paper with us this afternoon, who will now address us on "The Press and the People"; Mr D. Austin Latchaw, of the Kansas City Star. (Applause)

Mr LATCHAW—Mr Chairman, Ladies and Gentlemen: As a representative of the newspaper profession, before I say anything else, I wish, on behalf of my associates and myself, to thank the city of Saint Paul and its Committee

on Arrangements for the very excellent facilities provided and the thoughtful courtesies extended to the men assigned to cover this Congress.

The subject assigned me is incidental rather than germane to the work of this Congress. It is a big subject, and even if I felt that I could do justice to it I would doubt the appropriateness of using this occasion for the discourse. You are here to consider practical Conservation; to discuss ways and means to develop and, so far as possible, to foster the natural resources of this country, and above all to check and prevent the wasting of them. And it is one striking commentary on the relations of the press to the people that you do not need to give a moment's concern about the publication of your deliberations and conclusions. (Applause)

Yet it does seem fitting that at some stage of these proceedings a little time should be given to the consideration of that far-reaching agency without which the results of this Congress would not reach the public at large; for what you do today will be made known to tens of millions of readers tomorrow. If it were not so, the value of such public-spirited meetings as this would be immeasurably discounted.

However, as a member of the newspaper profession I cannot but feel that my subject would be more appropriately discussed by someone outside of that profession. It might be handled more frankly. It might be made more instructive to both the press and the people. Most assuredly I have not come here to throw stones at my professional brethren, and as for handing them bouquets, that gentle function might be performed with a somewhat better grace by someone outside the family. Still, I shall not be quite so reserved as was an old farmer back in Pennsylvania, whose farm adjoined that of my father when I was a boy, and who always got the worst of it in a horse trade because he was too modest to brag about his end of the proposition.

First of all the newspapers of this country could not have the splendid field they possess, the great opportunities they enjoy and the inspiring attention they command, if they did not appeal to the best read, the most intelligent, and the most responsive people on earth. In no other country is such a large percentage of the public a newspaper-reading public. Nowhere else does the average man know so much about current affairs of all kinds as in this country of ours.

On the other hand, I believe this popular intelligence is reciprocal—that the response the newspapers find for their endeavors is largely due to their efficiency in disseminating the news, in analyzing public questions, and in reiterating the truth. The man who is an habitual reader of a good newspaper owes much to that paper, just as the paper also owes much to him.

It is true that newspapers differ in policies and methods and doctrines, and there are times when the public may be confused rather than enlightened by the different presentations of the same subject, especially if the subject be one of technical complexities, such, for example, as that of the protective tariff. But in the daily run of events and the discussion of them, and in the long run of complex problems, the lines between right and wrong are not difficult to follow. And I am glad to say that from the newspaper point of view, these lines seem to be more clearly discerned than ever before, not alone by the press, but by the people. There has been a National awakening in this country, and the newspapers have had their share of it (applause). There is a broader and franker handling of the subjects of the day. The number of wholly independent papers is constantly increasing, and the number of independent party papers is increasing still more rapidly. The uncompromising party organ will soon be a thing of the past (applause). This greater independence of the press is largely responsible for the increasing independence of the electorate. The time has come when no man's loyalty to his party can be questioned when he honestly disapproves of some legislative measure or official representative of that party.

The chief function of the press is, of course, to present the news, and the news, collectively speaking, is non-partisan. A paper's advertising is non-partisan. If it is the right sort of paper, its circulation is largely non-partisan. And with equal freedom in its editorial policy, a newspaper, especially the big resourceful paper with an efficient and somewhat specialized staff, may make of itself a sort of popular university for its readers, furnishing them with authoritative information, whether obvious in the news or elucidated in the editorials, on the current life of the world.

I am not one of those who believe that a newspaper should confine itself to the mere presentation of the news. That is a great and powerful function, but the paper with a vast audience, with a reputation for honesty and

authority, can make of itself a constructive agency of tremendous power (applause). Also, it can make itself a destructive agency, when the public welfare demands that something should be destroyed (applause).

Of course, we are a busy people, and newspapers must be prepared with reference to our limited leisure. A few papers are conducted on the theory that the public has no time to read anything but the headlines. I am not here to "knock" this class of newspaper. If they do not show a regrettable preference for the sensational or the scandalous, they serve a good purpose in the scheme of publicity. They have greatly enlarged the newspaper audience. Do not forget that. And it is the experience of those who have published this class of papers that sooner or later their readers require more conservatism. As a result there has been a tendency for some time among these papers toward a more dignified style of publication.

But, as I have said, we are busy people. We have need for intelligent digests, authoritative discussions of the subjects of the day as well as news developments of those subjects. An evidence of this need is the fact that, in some of our municipal, State, and National contests in which great issues are at stake, it is necessary, in spite of our boasted and undoubted intelligence, to reiterate salient facts day after day in order to drive them home and make them enter into the conviction of the masses (applause). Sometimes this reiteration becomes tiresome to those of quick perception or ample leisure; but it is a necessary practice on the part of a newspaper that regards itself as an instructive and constructive agency as well as news furnisher. And when a paper thus regards itself it would seem that the ideal and final policy would be one of untrammelled freedom—freedom to support the man or the measure best calculated to serve the public welfare, or to oppose the man or the measure believed to be inimical to popular well-being. A paper thus established, not as an infallible judge but as an intelligent investigator, a patriotic champion, and an enterprising and faithful agency for progress in the community that supports it, can become a tremendous factor for good—a factor that will be taken into account by all friends of the people, and must be taken into account by all enemies of the people. (Applause)

I will not presume to encroach upon the direct business of this Congress except so far as the newspaper hears a relation to it. Every newspaper

publisher has a personal as well as his public share of the general interest in Conservation. The problem of procuring wood pulp at prices that will permit the continuation of the publication of newspapers at the present low rates will soon be serious unless a check is put upon the rapid decrease in the forest area. Wood pulp is made almost entirely from the spruce tree. For years the manufacturers of pulp stripped the forests with little thought of the morrow. The visible supply of pulp timber is becoming limited. Unless tree-growing comes to the rescue, it will not be long before print paper will have to be made from some other material, if a satisfactory substitute can be found, or the pulp will have to be bought from other countries.

I do not know whether you understand how much good timber is handled by newspaper readers. Let me give you some figures: The readers of the paper I represent handle sixty tons of it a day, taking into account the weekly edition. This is, in round numbers, 20,000 tons per year. We are already importing 20 percent of the pulp used in our paper mill. Think of it! In this great, big, new country, once almost covered with mighty forests, we find it advantageous today to import a common forest product from old Germany, where the highest standards of forest preservation and use are to be found. And this pulp, with a protective duty paid, is laid down in Kansas City for less than we have to pay for the domestic product of the same kind and quality. To make the paper for this one mill, the output of which is used exclusively by one paper, a daily average of more than one acre of spruce forest is used.

It is a matter for congratulation that the press of the country has assumed a most friendly attitude toward the Conservation movement (applause). Newspapers still disagree about many things. They have their little differences on the tariff, on the currency system, on corporation regulation, on certain men and particular measures, and they do not agree as to why "Jim" Jeffries didn't come back (laughter); but I have yet to find in a single issue of any paper flat opposition to the Conservation of natural resources (applause). Gentlemen of the Conservation Congress, you have here a movement of National and irresistible sweep, a theme that will endure through successive generations—for if it does not endure the Nation ultimately must perish. The people have grasped this subject spontaneously, and they are ready to study it zealously. Few yet comprehend its scope, fewer still its diversified details; but collectively the people intuitively

understand its vital significance. The country has at last awakened to its gross neglect and waste and prodigality. It has suddenly been reminded of its obligation to future generations along material lines. There is something even more appealing in this than the promptings of altruism: there is the moving sense of parental obligation, of sacred trusteeship. You are to be congratulated—you who are the fathers and prime movers of this great cause—that you have the united press of the country behind you.

And not only is the press with you, but it is ready to do far more than it has been able to do thus far. This movement needs publicity—much publicity. It is new. It must be made familiar. The people must be informed in detail as to the location, the character, and the extent of their resources, and as to the means employed or proposed for the developing and fostering of those resources. The only effective means for the dissemination of this information is the press.

Every year the Government spends millions of dollars on Government reports. These reports are necessary as matters of record and reference, but they are worthless for general reading. Many of the millions expended on these reports could be saved by limiting the number of copies to those that will be used and by leaving the mails unencumbered with the surplus (applause). If a part of the money thus saved were expended in the intelligent preparation of news matter pertaining to the various Government departments, giving to the people the interesting facts as they develop instead of depending on voluminous and unpopular reports for the education of the people in these matters, the work of the Government would be facilitated by popular enlightenment where it is now hampered by popular ignorance. It seems to me there is an opportunity here for the Conservation of our National revenues and our natural resources at the same time.

What is needed is an intelligent publicity bureau or agent in each department and the more important subdivisions, capable of preparing, in news form, as the facts develop, the interesting and instructive features of the department's daily work. This does not mean that all the papers will use all this matter, but some of it would be used by all to whom it is offered, and all of it would be used by some papers. On the whole there would be much wider publicity than could be procured in any other way.

I am not suggesting an untried experiment. Some of the bureaus at Washington have publicity departments. Those of the Agricultural Department and the Geological Survey have been measurably effective, and manufacturers and importers have found large use for the popularized consular reports. But with a single exception there has been no near approach to the possibilities of cheap and helpful publicity in any department at Washington. The exception I have in mind is the Forest Service (applause). Do you know why the country knows so much more about forest conditions and the employed and proposed measures for their improvement than it knows about irrigation, reclamation, the use of the rivers, the potentialities of water-power, or the conservation of coal or oil or minerals? It is because the Forest Service, under the direction of Mr Gifford Pinchot, established a news service of such a character that the press of the country used its output freely and without the cost of one cent to the Government other than the cost of putting the matter in form acceptable to the press. (Applause)

For some reason it was proposed, a couple of years ago, to prohibit, by Congressional enactment, the continuance of this publicity. But the effort resulted only in a complete vindication of the service. It was shown that only legitimate news had been given out, and that this news had appeared in an average of 9,000,000 copies of newspapers per month. These figures were based on clippings procured through the clipping bureaus, and did not include many publications that must have escaped the clippers.

Now, if it had been undertaken to place this same matter before the same number of readers through the medium of the formal and technical reports of the department, the cost would have been more than 100 times as great—and nobody would have read them.

As an illustration that newspapers want more Conservation news than they are getting through regular channels: A number of publishers recently formed a special Conservation service, which they maintained in Washington, whose business it is to follow exclusively the developments of this movement. But this service cannot be made what it should be made if the Government does not cooperate in this policy of needed publicity.

Considering the waste that is incurred in the publishing of Government documents that have no popular educational value, it seems well nigh preposterous that there should not be ample provision, out of a saving that could be made by cutting off this waste, for the publication of matter that the people want and the newspapers stand ready to print free of cost. It would be no more absurd for this Congress to go into executive session, bar these gentlemen of the press from its deliberations, and assume that the official report of your proceedings, which will be printed in the due course of time, would furnish sufficient publicity for the work of this convention. As it is, you have a circulation of tens of millions daily for your output. (Applause)

Chairman CLAPP—Ladies and Gentlemen: We often find a man who excels along some one line of work. The well-rounded man is the one who studies along every line; the truly great man *is* the well-rounded man, the man who studies the forces which make for the conditions in which he lives. We have such a man in this city, of whom we are all justly proud; a man who long ago, in the forge of hope and courage, welded his own fate with the possibilities of the then undeveloped Northwest, and who has lived to see the prophecies born of a study of conditions mature and develop in a splendid empire. It affords me great pleasure to present to you one who will speak on the subject of "Soils and Crops, Food and Clothing"—Mr James J. Hill, of Saint Paul. (Great and prolonged applause)

Mr HILL—Mr Chairman, Ladies and Gentlemen: I do not intend to take much of your time this afternoon, but I hope to bring before you some thoughts that may suggest the practical side of the subject we have to consider at this Congress. In order to make myself clearly understood and to be exact in my statements I will ask your indulgence in allowing me to read what I have to say:

Every movement that affects permanently a nation's life passes through three stages. First it is the abstract idea, understood by few. Next it is the subject of agitation and earnest general discussion. Third, after it has won its way to a sure place in the national life, comes the era of practical adaptation. Mistakes and extravagances due to the enthusiasm of friends or the malice of enemies are corrected, details are fitted to actual needs, the divine idea is harnessed to the common needs of man. In this stage, which the Conservation movement has now reached, the most difficult and important work must be done.

In our own history and in that of other nations we have seen this process many times repeated. Public education was an abstract idea in the time of Plato, a controversy of the Renaissance, and is still only partly realized. Back of all written records lived the man who first saw a vision of government universal, equal, free and just. But the world has not yet achieved the final adaptation of this mighty conception to man as we find him. Democracy is still in the fighting stage.

Only a few years have passed since it first dawned upon a people who had reveled in plenty for a century that the richest patrimony is not proof against constant and careless waste; that a nation of spenders must take thought for its morrow or come to poverty. The first actual Conservation work of this Government was done in forestry, following the example of European countries. It soon became evident that our mineral resources should receive equal though less urgent care. The supreme importance of conserving the most important resource of all, the wealth of the soil itself, was realized. In an address delivered four years ago this month before the Agricultural Society of this State, I first stated fully the problem that we have to meet and the method of its solution. With their great capacity for assimilating a new and valid thought, the people of this country were soon interested. Belief in a comprehensive system of Conservation of all resources has now taken possession of the public mind. What remains to be done is that most difficult of all the tasks of statesmanship—the application of an accepted principle and making it conform in all its general outlines to the common good.

To pack the fact into a single statement, the need of the hour and the end to which this Congress should devote itself is to conserve Conservation. It has

come into that peril which no great truth escapes—the danger that lurks in the house of its friends. It has been used to forward that serious error of policy, the extension of the powers and activities of the National Government at the expense of those of the States. The time is ripe and this occasion is most fitting for distinguishing between real and fanciful Conservation, and for establishing a sound relation of means to ends. (Applause)

We should first exclude certain activities that come only indirectly under the term, "Conservation." The Reclamation Service is one. Its work is not preservation, but utilization. The arid lands of this country have been where they now are, the streams have flowed past them uselessly ever since Adam and Eve were in the Garden of Eden. Irrigation was practiced in prehistoric time. What we have to do is to bring modern methods to the aid of one of the oldest agricultural arts. It is mentioned here because its progress illustrates the dangers that beset Conservation projects proper. They are dangers inseparable from National control and conduct of affairs. The machine is too big and too distant; its operation is slow, cumbrous, and costly. So slow is it that settlers are waiting in distress for water promised long ago. So faulty has been the adjustment of time and money that Congress has had to authorize the issue of \$20,000,000 of National obligations to complete projects still hanging in the air. So expensive is it that estimates have been exceeded again and again. The settler has had either to pay more than the cost figure he relied on or seek cheaper land in Canada. It costs the Government from 50 percent more to twice as much as it would private enterprise to put water on the land (applause). Under the Lower Yellowstone project the charge is \$42.50 per acre, and one dollar per acre annually for maintenance. The Sunnyside project carries a charge of \$52 per acre, and 95 cents maintenance. Under the North Platte project the charge is \$45 per acre, plus a maintenance charge not announced. These projects, in widely separated localities, entail a land charge prohibitive to the frontier settlers to provide homes for those for whom this work was believed to have been undertaken. The pioneer settler who can pay, even in ten annual installments, from \$3,500 to \$4,000 for eighty acres of land, in addition to the yearly fee per acre, must have some other resources to aid him. The work of irrigation would have been more cheaply done if turned over to private enterprise or committed to the several States within which

lie the lands to be reclaimed (applause). This is not a criticism upon any individual. It is merely one more proof of the excessive cost of Government work. (Applause)

Toward the conservation of our mineral resources little can be done by Federal action. The output is determined not by the mine owner, but by the consumer. The withdrawal of vast areas of supposed coal lands tends to increase price by restricting the area of possible supply. Nor can such deposits be utilized eventually except under some such system as is now employed. It is foolish to talk of leasing coal lands in small quantities in order to prevent monopoly. Mining must be carried on upon a large enough scale to be commercially possible. The lessee of a small area could not afford to install the necessary machinery and provide means of transportation without charging for the product a prohibitory price. The land should not be leased by the acre, but by the quantity of coal contained in the land (applause). A vein four feet thick contains about 4,000 tons to the acre; in many fields there are three, four, five, and six veins containing from fifteen to thirty feet of coal, or from fifteen to thirty thousand tons to the acre. What we want is intelligent understanding of the situation (applause). Under too restrictive conditions the coal would remain in the ground indefinitely. The people of the West see little practical difference between a resource withheld entirely from use and a resource dissipated or exhausted. They understand by Conservation the most economical development and best care of resources. It is the only definition consistent with the natural growth of communities in the history of the civilized world.

The prairie States are more interested than any other in the question of cheap fuel. We do not depend on Alaska for our future supply. There is abundant coal on the Pacific Coast nearer to our seaports and commercial centers. Vancouver Island is underlain with it; today, while the railroad companies with which I am connected bought coal lands on Puget Sound, which they still own, we are prepared to burn oil from California instead of coal. I speak of that as a practical reason why we should, before we leap, look to see what the actual conditions are. Then, to say nothing of Nova Scotia on the Eastern coast, there is coal in Spitzbergen, within the Arctic Circle, actually nearer our Eastern markets than the coal of Alaska. While we lament the exhaustion of our coal supply, we maintain a tariff that

compels us to draw upon it continuously. It would be well to cast out this beam before we worry too much over the Conservation mote. (Applause)

The iron deposits of Minnesota, the most wonderful in the world, are today not only furnishing industry in the Nation with its raw material, but are piling up a school fund at home that is the envy of other States and adding more and more every year to the contents of the State treasury. Minnesota is considering the reduction of her general tax levy by one-half. Would it be better if these lands were today held idle and unproductive by the Federal Government, or worked only on leases whose proceeds went into the Federal treasury and enabled Congress to squander a few more millions in annual appropriations? (Applause)

Against some forestry theories the West enters an even stronger plea. What the United States needs is neither reckless destruction nor an embargo upon our splendid Western commonwealths by locking up a considerable portion of their available area. There were, by the last report of the Forest Service, over 194,500,000 acres withdrawn from use in our forest reserves on June 30, 1909. Of this, nearly 58 percent, over 112,000,000 acres, or 175,000 square miles, lies in six Western States. That is an area six-sevenths the size of Germany or France. It is 80 percent of the size of the unappropriated and unreserved land in those six States. How are the cities, towns, and villages in those States to grow if so large a portion of the land is closed to the husbandman? I received today an official statement of the entire amount of public land withdrawn from settlement, and it is astounding. In area it is greater than the thirteen original States; it is nearly as great as New England, New York, New Jersey, Pennsylvania, Delaware, Maryland, Virginia, West Virginia, Kentucky, Tennessee, Ohio, Indiana, and Illinois (applause). And at the same time, we are driving this year not less than 100,000 American farmers to the Canadian Northwest to seek homes there (applause). Now, I say to you that the area of this total withdrawal for various purposes of the public domain is greater than the cultivatable area of the entire Canadian Northwest.

The forest reserves and the lands conveyed by Congressional grants to private interests in Oregon amount to some 50,000 square miles. More than half the area of this great State has been withdrawn by action of the Government in one way or another from cultivation and the enjoyment and

profit of the people of the State. Over one-third of Idaho and 27 percent of Washington are forest reserves. Colorado is almost as badly off; and not more than 30 percent of its forest reserves is covered with merchantable timber, while about 40 percent has no timber at all. On the Olympic peninsula are lands reported to be withdrawn to conserve our water supply where the annual rainfall amounts to something like seven to ten feet (laughter). According to the official report, the cost of administering the Forest Service in 1909 was a little short of three million dollars, and the receipts were \$1,800,000. The deficit on current account alone was over \$1,100,000. The total disbursements were over \$4,400,000, and the actual deficit \$2,600,000. Now, we should be liberal in our grants for the care of our public forests. We should also closely scrutinize the manner of their care. The present season has seen an enormous destruction in the value of the timber in the forest reserves. Our company, for over two months, has had from 800 to 1,000 men at work doing nothing else but trying to put out the fires in the forest reserves. (Applause)

The Forest Service has over 2,000 employees. In 1909 they planted 611 acres, and sowed 1,126 acres more. The West believes in forest preservation. But it believes practically and not theoretically. It realizes that a good thing may cost too much, and is not ignorant of the extravagant financial tendency of every Federal department and bureau. It wants all good agricultural land open to the settler, wherever it may be situated. It wants timber resources conservatively utilized, and not wasted or destroyed.

In connection with forestry interests there is just now much question of the conservation of water-power sites. The demand is that Federal lands forming such sites should be withdrawn and leased for the profit and at the pleasure of the Federal Government. Against this the whole West rightly protests. The water-power differs from the coal deposit in that it is not destroyed by use. It will do its undiminished work as long as the rains fall and the snows melt. Not the resource but the use of it is a proper subject for Conservation and regulation. To withdraw these sources of potential wealth from present utilization is to take just so much from the industrial capital of the States in which they are situated.

The attempted Federal control of water-powers is illegal, because the use of the waters within a State is the property of the State and cannot be taken

from it (applause), and that the State may and actually does, in the case of Idaho for example, perfectly safeguard its water-powers from monopoly and make them useful without extortion has been shown conclusively by Senator Borah in a speech in the United States Senate in which this whole subject is admirably covered. Back in our history beyond the memory of most men now living there was the same controversy over the public domain. Ought it to be administered by the Government and disposed of for its profit, or opened to the people and shared with the States? Let experience determine which was the better guardian. The worst scandals of State land misappropriation, and there were many, are insignificant when compared with the record of the Nation. The total cash receipts of the Federal Government from the disposal of public and Indian lands from 1785 to 1909 were \$423,451,673. The money is gone. It has been expended, wisely or unwisely, with other treasury receipts. It would be interesting to know how much the above sum exceeded the cost of administration. To go back 125 years and dig up the cost of the administration of public lands would be more of a task than I have time for, but I took the last report of the General Government, and in the disbursements of the Interior Department I found that the cost of administering the public lands was in 1907 \$17,421,000, in 1908 \$15,190,000, in 1909 \$14,441,000. Now if we take the entire proceeds of all the public lands sold, including the Indian lands, it averages \$3,400,000 a year for the 125 years during which it has been sold; and we find here that the cost of administering the greatly reduced estate is from three to five times as much as the total receipts would average (applause). But certain limited areas of lands were conveyed to the States for educational purposes. The permanent common school funds, State and local, conserved by the States, amount to \$246,943,349. The estimated value of productive school lands today is \$138,851,634, and of unproductive \$86,347,482. Add to these the land grant funds of colleges of agriculture and the mechanic arts, and the total is merely half a billion dollars. To what magnitude these great funds, now jealously guarded for educational purposes by the States, may grow in time we cannot even guess. Some may eventually provide amply for all educational needs of their States forever. This is one telling proof of the superior fidelity of the commonwealth as custodian of any trust for future generations.

There remains an opportunity and a need of Conservation transcending in value all others combined. The soil is the ultimate employer of all industry and the greatest source of all wealth (applause). It is the universal banker. Upon the maintenance, unimpaired in quantity and quality, of the tillable area of the country its whole future is conditioned. Four years ago, and on many occasions since, I presented the facts and statistics that make land conservation incomparably the paramount issue with all who have at heart the prosperity of our people and the permanence of our institutions. It is unnecessary to repeat in detail what has now become matter of common knowledge and is accessible to all. For the last ten years the average wheat yield in the United States was 14.1 bushels, while in Germany it was 28.7 and in the United Kingdom 32.6. This is a measure of our general agriculture. The cattle other than milch cows on farms in the United States are over 4,000,000 fewer than they were three years ago. The number of hogs declined 7,000,000 in the last three years, and is less than it was twenty years ago. The increase in total value of food products is due to a great extent to higher prices. This failure to conserve soil fertility and maintain the agricultural interest is expressed in recent changes in our foreign trade. These are more than mere balance sheets; since, as you know, variations in international trade balances may produce wide-reaching effects upon all industry.

While our total foreign trade last year was only a little less than the high record made in 1907, the distribution of it was vastly different. For the last fiscal year our imports were nearly \$240,000,000 in excess of those for the same period in 1909, and \$303,000,000 above those of 1908. Our exports were more by \$82,000,000 only than in 1909, and were nearly \$116,000,000 less than in 1908. In 1908 the excess of exports over imports was \$666,000,000; by 1910 it had fallen to \$187,000,000. We are buying more lavishly and selling less because there is less that we can spare—yet, my friends, that \$187,000,000 of balance of trade due to this country is not enough to pay the extravagant traveling expenses of our "globe trotters" who are annually passing from one end of Europe to the other. (Applause)

A glance at the following table of our exports for the last five years in three great schedules dependent directly on the soil tells the whole story:

	Breadstuffs	Meat and Dairy Products	Cattle, Sheep and Hogs
1906	\$186,468,901	\$210,990,065	\$43,516,258
1907	184,120,702	202,392,508	35,617,074
1908	215,260,588	192,802,708	30,235,621
1909	159,929,221	166,521,949	18,556,736
1910	133,191,330	130,632,783	12,456,109

With the exception of the increase in breadstuffs in 1907-8, caused by our desperate need to send something abroad that would bring in money to stay a panic, the decline is constant and enormous. A continuance of similar conditions—and no change is in sight—must mean partial food famine and hardship prices in the home market; an annual indebtedness abroad which, having no foodstuffs to spare, we must pay in cash; and financial depression and industrial misfortune because we have drawn too unwisely upon the soil. This impending misfortune, only the conservation of a neglected soil and all the interests connected with it can avert.

The saving feature of the situation is the interest already awakened in agricultural improvements; an interest which it should be the first object of this Congress to deepen and preserve. Much has been done, but it is only a beginning. The experiment station; the demonstration farm; agricultural instruction in public schools; emphasis upon right cultivation, seed selection, and fertilization through the keeping of live stock, all these are slowly increasing the agricultural product and raising the index of soil values. The work being done by the Agricultural Department under the care of our old Iowa friend, Secretary Wilson—who is a farmer from choice (applause)—is scientifically selecting the good from the bad and the wise from unwise methods, and the information is within the reach of every farmer of this country who will only put out his hand and ask for it. (Applause)

But the work moves more slowly than our needs. The possibilities are great. One might make the comparison with current agriculture elsewhere almost at random, since European Russia is the only first-class country more backward than our own. Take the smallest and what might be supposed the least promising illustration: Denmark's area is about twice that of Massachusetts. It is occupied by more than two and a half million people.

This Jutland was originally land of inferior fertility. What has been done with it? Denmark is now called "the model farm of Europe." Her exports of horses, cattle, bacon and lard, butter and eggs, amounted in 1908 to nearly \$89,000,000. Mr Frederic C. Howe in a recent article says: "The total export trade is approximately \$380 for every farm, of which 133,000 of the 250,000 are of less than 13½ acres in extent, the average of all the farms being but 43 acres for the entire country. The export business alone amounts to nine dollars per acre, in addition to the domestic consumption, as well as the support of the farmer himself." One-half the population are depositors in the savings banks, with an average deposit of \$154. How have these things been accomplished?

First negatively, it has not been done by any artificial means or legislative hocus-pocus (applause). No bounty and no subsidy has any share in the national prosperity. The ruler of the country is the small farmer. He cultivates his acres as we cultivate a garden. He raises everything that belongs to the land. He fertilizes it by using every ounce of material from his live stock, and by purchasing more fertilizers when necessary. There are 42 high schools and 29 agricultural colleges in this little country with a population less than that of Massachusetts in 1900. Whatever else they teach, agriculture is taught first, last, and all the time, to young and old alike. The Dane is a farmer, and is proud of it. England and Ireland and Germany are studying his methods today. No people could imitate them with more profit than our own. (Applause)

Recent good years have brought the average wheat yield per acre in the United States up to over fourteen bushels. Twice that would be considered poor in Great Britain and an average crop in Germany. Therefore twenty-five bushels per acre is a reasonable possibility for us. Suppose we raise it. The present wheat acreage of the United States is about 46,500,000 acres on the average. If it gave 25 bushels per acre, the crop would amount to 1,162,500,000 bushels. At our present rate of production and consumption we may cease to be a wheat exporting Nation within the next ten or fifteen years, perhaps earlier. With the larger yield we could supply all our own wants and have a surplus of 400,000,000 bushels for export. This is no fancy picture, but a statement of plain fact. Is there any other field where Conservation could produce results so immense and so important? Is there

any other where it bears so directly upon our economic future, the stability of our Government, the well-being of our people?

Any survey of practical Conservation would be imperfect if it omitted the almost desperate necessity at this time of conserving capital and credit. This subject deserves full and separate treatment. No more is possible here than to summarize some of the facts and conclusions presented by me to the Conservation Conference that assembled in this city a few months ago. Conservation of cash and credit is important to the farmer as it saves or wastes results of his work, and his work furnishes the greater part of the Nation's wealth. Our States, including cities and minor civil subdivisions, have run in debt about three-quarters of a billion dollars in the last twelve years. Public expenditure is increasing everywhere. Public economy is a virtue either lost or despised. From 1890 to 1902 the aggregate expenditures of all the States increased 103 percent. Boston's tax levy, says Brooks Adams in a late article including this among the serious problems of modern civilization, was \$3.20 per capita in 1822, while now it is nearly \$30. The per capita cost of maintaining the Federal Government was \$2.14 in 1880, \$4.75 in 1890, \$6.39 in 1900, and \$7.56 in 1908. The total appropriations voted by Congress for the four years from 1892 to 1896 were \$1,871,509,578; for the four years from 1904 to 1908 they were \$3,842,203,577. An increase of \$2,000,000,000 in expense for two four-year periods with only eight years between them should give any people pause. Spendthrift man and spendthrift Nation must face at last the same law carrying the same penalty.

If anyone believes that this growth of expenditure is a consequence of the general material growth of the country, let him study the following brief table of comparative statistics. It establishes the indictment of national extravagance:

	Increases			
Wealth	1870 to 1890	116.0%	1890 to 1904	65.0%
Foreign Trade	1870 to 1890	99.0%	1890 to 1908	85.4%
Value Manufactured Product	1870 to 1890	121.0%	1890 to 1905	58.0%

Net Ordinary Exp. U. S. Govt	1870 to 1890	1.4%	1890 to 1908	121.4%
Expenditures of 30 States			1890 to 1909	201.6%

This debauch of capital and credit has sent a poison circulating through the veins of the Nation. Everywhere the individual imitates the profligacy of his Government. Industry and saving are at a discount. Any luxury, any extravagance is warranted if funds for it can be raised by wasting capital or creating debt. There is just so much less money for productive employment: for payrolls and the extension of commerce and industries, and the creation of those new facilities for want of which the commerce of the country is and always must be limited (applause). Hence come also high prices, curtailment of business, distrust, and eventual distress. Hence come waste and idleness, and the increased cost of production that makes both business and employment slow and insecure. Any Conservation movement worthy of the name must place high upon its program the saving of capital and credit from the rapacious hands of socialist as well as monopolist (applause). Extravagance is undermining the industry of this country as surely as the barbarians broke down and looted that mighty empire with whose civilization and progress Ferrero repeatedly insists that ours has so much in common.

We must stand for Conservation everywhere; in the tedious as well as in the interesting application; where it cuts into our pleasure and habits, and jostles our comfortable, easy-going ways of thought, just as firmly as where it is hand in glove with self-interest. This is, above all things, an economic question. It is neither personal nor political. In such petty and partial interests it has found its worst obstructions and encountered its most serious reverses.

The tariff in some respects is a great enemy of Conservation (applause). Whatever we may think of it as a general industrial policy, everyone can see that, by excluding the raw products of other countries, it throws the entire burden of their consumption upon our own resources, and thus exhausts them unnecessarily (applause). This appears clearly when we consider such commodities as we might obtain from Canada, a country that gained nearly 400,000 immigrants from the United States in the nine years up to April,

1909, and has probably taken another hundred thousand since; a country where it is absurd to talk about any actual advantage in the wage scale as compared with our own. The tariff on forest products cuts down our own forests, a tariff on coal depletes our mines, a tariff on any raw material forbids the conservation of similar natural resources here. (Applause).

This Congress announced from the first its purpose to deal with the subject of Conservation in a practical spirit. The present condition of the movement, now in the third stage of its development, demands it. We have to apply the Conservation principle, as we have eventually to apply every other, to our domestic economics; to work it out in the experience and practice of everyday life. How this may be done can be stated in the form of a few conclusions that raise the word Conservation from the name of a more or less vague, diffuse, and disputable theory to that of a practical guide to legislation and administration. (Applause)

Conservation is wholly an economic, not in any sense a political principle (applause). The Nation has suffered and still suffers so much from transferring other economic questions to politics that the mistake should not be repeated (applause). Whoever attempts to make Conservation the bone of a personal controversy or the beast of burden to carry any faction into power or popularity is its worst enemy. (Great applause)

"Conservative" is the adjective corresponding to the noun "Conservation." Any other attitude toward this movement, either radical or reactionary, is treason to its name and to its spirit. It should mean no more and no less than dealing with our resources in a spirit of intelligence, honesty, care for both the present and the future, and ordinary business common sense. (Applause)

Conservation does not mean forbidding access to resources that could be made available for present use. It means the freest and largest development of them consistent with the public interest and without waste. A bag of gold buried in the earth is useless for any purpose. So is an acre untilled, a mine unopened, a forest that bars the way to homes and human happiness.

The determination in each case as to what extent a given resource should be utilized and how far reserved for the future is an intensely practical, individual, and above all a local question. It should be carefully considered in all its aspects by both Nation and State, and should finally rest within

lines determined by proper legislation, as far as may be under the control of local authority. (Applause) Experience proves that resources are not only best administered but best protected from marauders by the home people who are most deeply interested and who are just as honest, just as patriotic and infinitely better informed on local conditions than the National Government can possibly be. (Applause) It is clear that every one of the many problems all over the country can be better understood where they are questions of the lives and happiness of those directly interested.

Behind this, as behind every great economic issue, stand moral issues. Shall we, on the one side, deny to ourselves and our children access to the same store of natural wealth by which we have won our own prosperity, or, on the other, leave it unprotected as in the past against the spoiler and the thief? Shall we abandon everything to centralized authority, going the way of every lost and ruined government in the history of the world, or meet our personal duty by personal labor through the organs of local self-government, not yet wholly atrophied by disuse? Shall we permit our single dependence for the future, the land, to be defertilized below the point of profitable cultivation and gradually abandoned, or devote our whole energy to the creation of an agriculture which will furnish wealth renewed even more rapidly than it can be exhausted? Shall we permit the continued increase of public expenditure and public debt until capital and credit have suffered in the same conflict that overthrew prosperous and happy nations in the past, or insist upon a return to honest and practicable economy? This is the battle of the ages, the old, familiar issue. Is there in the country that intelligence, that self-denial, that moral courage, and that patriotic devotion which alone can bring us safely through? (Applause)

I ask these questions not because there is any doubt of the answer in the minds of the American people, but that it may be made plain what a complex fabric the fates are weaving from the apparently commonplace happenings of our peaceful years, and how each generation and each epoch must render an account for the work of its own days. The unprecedented dignity of this assemblage, its nationally representative character, the presence here of those upon whom great occasions wait, the interest felt by millions who look to it for information and guidance, prove how deep beneath the surface lie the sources of its existence and its influence. Out of the Conservation movement in its practical application to our common life

may come wealth greater than could be won by the overthrow of kingdoms and the annexation of provinces; National prestige and individual well-being; the gift of broader mental horizons; and, best and most necessary of all, the quality of a National citizenship which has learned to rule its own spirit and to rise by the control of its own desires. (Great applause)

Chairman CLAPP—Ladies and Gentlemen: One among the recognized agencies for the spread of information in relation to our agricultural development is a paper published in Iowa by Mr Henry Wallace, who is known to us all. A discussion will now be led by Mr Wallace, and I take great pleasure in presenting him to this assemblage. (Applause)

Mr WALLACE—Mr Chairman, and Ladies and Gentlemen of the Congress: I have been asked to discuss the subject opened up by my old friend—and your friend—Mr James J. Hill.

With very much that he has said, I most heartily agree. He speaks on these and other subjects "as one having authority, and not as the scribes." While listening to him I have been trying to get in my own mind a clear conception of certain fundamental questions that have been discussed at this Congress, and around which the discussion turns. I have been trying to put them in form, pointing out where he and I can agree and where we differ.

I have come to the conclusion that a man has what he had, if he hasn't sold or contracted to sell it, or allowed somebody to steal it; that the United States has the resources that are now in the name of the United States and not under contract to be delivered, and not sold—or stolen—either in compliance with the letter of the law or in violation of both letter and spirit. In other words, there are certain assets or resources that we have and hold; and we all agree that the owner is entitled to the management and use of his assets (applause), and therefore that the people of the United States, as a people, are entitled to the use of whatever resources we may have

remaining (applause). They are not for the benefit of any one man or any combination of men (applause), neither of any State (applause) or combination of States (applause), but for the whole people; therefore we can sell our coal lands or keep them. We will be wise if we keep them (applause). We can sell our forests, or say how they shall lie used, or we can let somebody steal them. We can hold on to our phosphate (and there is very little of these United States that won't be buying phosphates in fifty years) or we can let somebody control and ship it to Europe, to enable the Belgians and the Germans to grow 32 bushels of wheat to the acre while we grow 13 (applause)—and by means of *our* phosphates. Using the language of the President the other day to outline the management of these resources (and he has done it better than any other man I ever knew), we can lease the lands, we can control them, we can prescribe how they shall be used. This much we all agree upon. And we will further agree that the Congress of the United States, our Representatives, must decide how it shall be done.

We can do one of three things: We can deed these lands and these resources to the States, to be used as they think best. We can abdicate our sovereignty—perhaps modifying that to some extent, we can outline what the States shall do and what they shall not do, but that will involve abdicating our sovereignty and will lead to perpetual quarrels between the States (applause), such as now existing, for example, between Colorado and Kansas as to the use of water. Or, as Canada does, as Germany does, as Australia does, as Tasmania does, we can hold to those resources and lease them for money for the benefit of the whole people. (Applause)

Now, my good friend Mr Hill seems to have grave doubts as to the capacity of the United States to handle its business with anything like the same skill with which he handles his (laughter and applause). He tells us that this Reclamation Service is costly—thirty, forty, or fifty dollars an acre, to be paid in ten years without interest—for what? To be able to make it rain just when we want to, and stop it when we want to; that is what irrigation is (applause). And Mr Hill would give five dollars an acre for twenty years if for all time and eternity he, his descendants and his assigns, could make it rain when he wanted to and make it stop when he wanted to (applause). Next to the owner of a quarter-section of land in Iowa I think that the man who owns fifty acres of irrigated land at fifty dollars an acre is a prince of the blood royal (applause and cry of "Good!"). It is the cheapest land in the

United States, in the center of the highest civilization, the best education and the best schools. Mr Hill tells us also that the United States (I guess it was Solomon he had in his mind: he was the brother of a great waster) has received \$400,000,000 or so for its Indian lands—he didn't know how much it cost to acquire them (millions, however)—and that he doesn't know what has become of the money. Well, I found since yesterday where some of it went—to this dam over here between Minneapolis and Saint Paul (great laughter and applause). He tells us that States are more economical than Nations. Now, isn't it a matter of fact that both State and Nation have been playing the part of the prodigal son, wasting our substance in riotous living—and that now we smell the husks?

Gentlemen, the agricultural colleges have wasted a good deal of money. The State of Iowa had a great grant of land for improvement, and I give you my word you could run the whole thing through a barrel if you had enough headway. We have been absolutely throwing away our resources—just like some of our wealthy gentlemen down in New York throw their daughters in the face of titled Nobodies asking them to take them "with the compliments of the author" (laughter). If this country continues to be governed, as it has been governed for the last twenty years, by great combinations of capital that get together in Congress or out of Congress to determine how much tariff they will levy and what else they may do in the way of getting hold of the public domain, it doesn't make a speck of difference whether our resources are governed by the Government or by the States; they will all be stolen anyhow (laughter and cheers, and cries of "Hit him again!")—just as they have been in the past. (Renewed applause)

A VOICE: Conservation ought to have been started a hundred years ago.

Mr WALLACE: You're right. But if the people of the United States have made up their minds that they are going to be in the future a Government "of the people by the people and for the people"; if we mean this in blood earnest (applause) and are willing to sacrifice our party affiliations (cries of "Good, good, good!"); if we are willing to pay money to attend conventions, without going on passes (cries of "You bet!" and cheers); if we are willing to make the sacrifices which always belong to a free government (applause)—then predatory wealth will no longer sit in the seats of Congress, and we

shall have a democracy, a Government of the People instead of a Government of Plutocracy. (Applause and cheers)

Gentlemen, it is just a question whether we have the stuff in us to really be a great self-governing people, a Nation that stands four-square to every wind that blows, that regards a law of the Almighty as supreme law and right and the only manhood worth having as that which comes in obedience to those great laws that govern men in all nations of the world (applause and cheers); it is a question whether we will pay the price for the liberties that our fathers gave us. (Applause)

Now, with about everything that my good friend Mr Hill has said on the conservation of soil fertility I most heartily agree. I get an idea about once a year (laughter), and am able to put it in a way that seems fairly good to me: and for some time past I have been brooding over the thought that the great problem before the American people—a problem involving all other problems that vex us, tariffs, Conservation, trusts, everything—that the great problem we have before us is *how to keep enough skilled labor on the land to enable the farmer to sell his products to the city at a price the people can afford to pay*. Now, just let that soak into you (applause). The problem is to keep enough skilled labor on the farm to enable the farmer to grow the food for this and other nations at a price that the people in the cities can afford to pay. It is the biggest problem before us. It involves all other problems, when you come to trace it down to its roots. The farmer is handicapped by the fact that he no longer tills virgin soil, as his father and his grandfather did, and by the fact that he no longer has timber at his door. We have wasted our magnificent forests of oak and walnut, and given away an empire (for example, in Wisconsin) of the best pine lands that some fellows would put a road through, to get the lumber out under pretense of resisting a Canadian invasion (laughter and applause). Today we are buying fertilizers for all New England, New York, Pennsylvania, Ohio, southern Indiana, all the South, and even for Missouri; it is only a question of time when we shall have to buy them for all our land. Notwithstanding all of the millions of acres that have been put into cultivation every year, our crop production lags behind our population. In the last ten or fifteen years, our production of wheat per acre gradually but slowly decreased until within the last three or four years, when with my friend Secretary Wilson's help we began to do a little better.

The farmer is handicapped by the fact that he is tilling a partially infertile soil; he is handicapped worse in this way: he cannot possibly get, for love or money, the really skilled labor required to maintain the fertility of the soil while he is growing crops (applause). Why, you know how difficult it is in the country to get a hired hand, and you know that a hired girl in the home is a thing out of the question. There isn't a man here ugly enough, if he is a widower, but what could get two second wives where he could get one hired girl (laughter and applause). Now, we cannot use the labor of the city. Let a man go to town and become a lawyer or a doctor for ten or fifteen years, and then return to the country, and what is he good for? He has to serve an apprenticeship for four or five years before he is worth his board. We cannot use the labor of southern Europe except in the wheat fields or in the orchards; farm labor now is *skilled* labor; and we haven't got it. One reason we haven't got it is because my friend Mr Hill has been giving excursion rates up to Canada (laughter and applause)—for the benefit of his railroad, he says—and for the benefit of speculators who can paint a desert to look like the Garden of Eden, and make farmers believe that it is like the land of Egypt "as thou goest unto Zoar." If we could keep on the farm the boys and girls that grow up there we could give the people of the cities food at a price they could afford to pay; but there is the great problem. I will not solve it now, because I would have to discuss the tariff (laughter) and every other blooming thing that allures men to town—including high wages and easy times.

Today the townsman is in trouble. The fact is that he cannot get the farmer's products at anything like the price the farmer ought to have (Voice: "Now you're talking"). The farmer never gets more than two-thirds (Voice: "If he gets that"); frequently he gets one-third. Out in Fresno, California, we found they made a first-class rate at four cents on what I was paying sixteen cents for; the railroad got four cents, the wholesaler four, the retailer four, and the farmer four—and I pay sixteen. And there is another trouble (I am one of the unfortunates so I look at both sides of the question): the farmer in town pays 16 percent, so the merchants tell me, for the privilege of ordering goods by telephone instead of going to the market and getting them; and that is another reason he has to pay so much. But there is still another matter with the city man; it is not so much the high cost of living as the cost of high living and prosperous times (I borrowed that from Mr Hill); for the

man in town now isn't satisfied to live as his father did, or his grandfather, or as he himself did ten or twenty years ago (applause). Why, he wants strawberries from Texas in February, and he wants green peas from Florida, and he wants fresh eggs at the time when hens don't lay, and he wants spring chicken in the coldest weather—and he gets it, but it comes out of cold storage (laughter). That is one reason why the townsman cannot get farmer's products at the price he can afford to pay.

Let us look a little further—but I must not detain you (Cries of "Go on, go on, go on"). This problem has been growing on us for years; ever since the iron rail and steam and electricity enabled us to build cities far remote from the lake or the river or the ocean, ever since we learned to get gold out of quarries instead of out of river sand, ever since human power was multiplied by machinery, ever since railroads netted the country with their systems: there has been a tendency to the development of great cities and a constant decrease in the number of men that work on the farm. We don't *think* now as we used to, because improved machinery (in most cases invented by farmers) has enabled the farm boy of fifteen years of age to do the work of eight or ten men—and at the same time has enabled him to rob the land more effectively than ever before. And this problem would have been met long ago if it had not been right here in this Mississippi valley there is the finest slice of land that the Lord ever made, to be given away by our benevolent Uncle Sam partly to the farmers and partly to the railroads—a country that needed neither spade nor axe to fit it for the plow; for the last twenty years we have been breaking it, mining it, robbing it, and selling its fertility to enable men in the great cities to live cheaply in the Old World and in this country (applause). The people of Kansas invited my good friend Secretary Wilson and me down there to talk about agriculture, and in going from our hotel to the place of meeting we actually fell over bags of bran that were put out there to send to Denmark to make butter and cheese to come back and be eaten in Kansas (laughter). This is the way we have actually been selling, piecemeal, our fertility. Why, you men remember when corn was sold at 15 and even 10 cents a bushel, and oats at 10¹/₂—I myself have sold wheat at 38—lower than the cost of production. The people in cities all over the world have an idea that it was foreordained from all eternity that they should have cheap foods, but they are now waking up to the fact that we have been postponing the day of judgment by

selling foodstuffs for about what the fertilizers would cost, if we had to buy them, to provide bread and meat for the hungry nations. We have sold the buffalo grass on the prairies to the people of Europe, in the shape of beef, dirt cheap; we have built up great cities and States; and the people have all the while thought that cheapness was normal, whereas we are now just getting to the normal basis. For twenty years I could buy bread made from American wheat, in the country on the farm, for three cents a pound, and now I pay five cents in town—and don't get as good bread at that.

The real problem is, *how we are going to furnish bread to the people at a price that they can afford to pay?* I have no hand-me-down solution for that; it is the biggest problem that I know of, and I can venture only some suggestions. First, we can add a little to our production through irrigation. That is a slow process, and limited at best. We can add some more by drainage. We can add a good deal to the yield per acre by better methods of farming. But we are limited, as I have said, largely by the lack of skilled labor. The merchant, the city man, if he is to live on his income, must improve his system of distribution; he must in some way or other, get rid of the go-betweens. Some things will have to be done by railroads and some by Congress, and a number of things will have to be done that they will all say *can't* be done—I'm tired of that story, that you *can't* do anything. Our railroad friends have told us that we can't pass interstate commerce laws, it's unconstitutional; that we *can't* stop the giving of passes and rebates, that it's unconstitutional. Now, we have done all those things. *The people of the United States can do anything that is right!* (applause), though they can't permanently succeed in doing wrong (applause); and these things we have been told we *can't* do we *have done*, and everybody says it is right. Sometimes I take great comfort in watching some of our great "captains of industry," railroad magnates like Mr Hill. To see them you would imagine they had been reading the Psalms of David and saying, "It was good for me that I was afflicted; before I was afflicted I went astray, but now I love"—the Interstate Commerce Law (laughter). The trouble with them is that they turn round and oppose our railroad laws, and the measures brought up by the voice of the people, and insist that they *can't* be enforced.

If the farmers are to sell their products in sufficient quantities to cities at a price that they can afford to pay, the calm and considerate judgment and the earnest cooperation of every class of our people are needed. We have

problems before us that cannot be settled today or tomorrow; they involve questions of deep statesmanship; and they never can be settled until they are settled right, on a basis that is just. And I have this faith in the American people, that notwithstanding all their mistakes and all their follies and all their extravagances and all their partisan differences, down at the bottom they are an honest people, they are an intelligent people, and they are a people that seem to have an instinct of danger and an instinctive perception of what is fundamentally and inherently right. (Prolonged applause)

Mr HILL—I want to apologize to Brother Wallace because I did not make myself entirely understood when I indicated that \$50 or \$42 or \$45 an acre for Government-irrigated land is too high. He says that I would give \$100—and I would, if I had to; but if that land were left with private enterprises, or if the *people of the State* alongside of this \$42 and \$45 and \$50 land were putting water on their land for \$15, I wouldn't charge the settler \$50 or \$42. (Laughter and applause)

Chairman CLAPP—Ladies and Gentlemen: There is a tradition in Washington that the present very efficient Secretary of Agriculture established the Department of Agriculture, because of his long service in that position. I have to dispel that illusion. Nevertheless his service has made that Department what it is today; and I take great pleasure in presenting to you Secretary Wilson. (Great applause)

Secretary WILSON—Mr Chairman, Ladies and Gentlemen: I have enjoyed the two last speeches more than anything else I have heard since I have been here, although I have never attended a meeting anywhere that I can remember where there were so many big men who do things in the world. The greatest regret I have is that there must be more than a hundred men

here well worth hearing who will not have opportunity to speak on account of lack of time.

Mr Hill and Mr Wallace have talked about things that I have not done. Fourteen years ago I went down to Washington with President McKinley to do something with the Department of Agriculture. I could see right well from tendencies that had originated some time previous a growing and a development that now at this present time have come to a head. I saw the necessity for Conservation of the natural utilities of this country, the necessity for Conservation of soils and forests and water-powers and all those things; and I went to work. I have never gone to Congress to get help or money without getting it at once. If I have failed to do something for agriculture, the fault is mine and not that of Congress, because they have never criticized me, except that I have not asked for enough money.

I have found it necessary to educate men, or to have them educated, along new lines. Search history as far back as you see fit to go, and you will find that there has been no education whatever for the farmer. The classical education, so beautifully spoken about by our friend from Tulane University (President Craighead), is a beautiful education; but there is no agriculture in it. It is a difficult thing to change the education of a people; even our religion is interwoven, like our literature, with the old-fashioned classical education. The country was regarded as valuable and the professions went to the country to get new men because the old wore out in the town, and so the farm has always reinforced the professions; and the practice has gone on until today the American Navy is being reinforced even from the farms of Minnesota and Iowa. The average boy who lives in town knows too much about things he shouldn't know, and the boy on the farm or in the country knows little about the things that wouldn't do him any good if he did know them (laughter). My first problem was to organize a Department of Agriculture by training men to go safely where there were but few blazings through the woods.

Mr Hill and Mr Wallace have both spoken wisely of the soil. That is the source of our wealth. When our good people travel abroad, the farmer pays the bill; when you beautiful ladies purchase diamonds—and sometimes bring them back in your hats—the farmer pays the bill (laughter). Of course, since the Civil War the farmer has been keeping the balance of trade

in our favor—has paid all our foreign debts, has paid the cost of our wars, has paid all the expenses of shipments to foreign ports; but a new day has come. While the farm has been producing considerably more and its area has been increasing, certain things have occurred that have a momentous influence on the present and on the future. We have not been producing so fast as we have been increasing in population; it costs too much to get breakfast and dinner and supper, and we eat three times a day. The serious problem which presents itself to us now is that *it costs too much to live*. I never want to see the day come when the American workingman shall be reduced to the condition of the European who makes his dinner on bread alone and still lives. (Applause)

What are the prospects of getting cheaper food to eat? Do we want to bring men from Central America? They are diseased. Do we want to bring them from Mexico? They are not adapted to our climate. We do not care to bring them in much from Canada, because they have no corn up there, and don't eat that kind of food. I see some rays of hope in our leaden sky. The South has in the past suffered from a pest known as the cattle-tick which prevents the development of domestic animals, and they have not given us as much meat as we have shipped to them; but Congress gave my Department money to try to get rid of this tick, and we have been at work for three years and have cleared the pest from the equivalent of an area of three great States, 140,000 square miles (applause), and it will not be many years until all the South is cleared of the cattle-tick. Then the southern States will begin to contribute materially to our food production, because they have a mild winter, they have intelligent people, they have transportation systems; all they need is a little better system of agriculture. We have also been dealing with an invasion from Guatemala for some time, the boll weevil. The question was whether the poor people in that section could sustain life under the burden of this pest, and they came to my Department to go down and do something; and in checking the pest we are meeting the need for improved agriculture and increased production of foodstuffs.

There are two prominent ways of increasing the producing capacity of a people: First, there is Conservation demonstration (we shall be using this word "Conservation" in our prayers if we don't look out). (Laughter) Last year we had 12,500 boys in four southern States, all under sixteen years of age, each of whom grew an acre of corn—the South never grew as much

corn in its history as it did last year—and some of those boys grew over 150 bushels to the acre (applause). They sold it at different prices. They were promised, as an encouragement, free tickets to Washington to see the President and the Capitol, and that the Secretary would give them diplomas. Well, I thought little about this until I marched the boys—looking very serious—each exactly like a man who is getting an LL.D. from a university. The first view of those boys was amusing, but the next one to me was very pathetic. A diploma, you know, is given to a man or a woman who does good work in a college course. Didn't the boy who grew 150 bushels of corn to the acre *do* something? He did; he did the best there was in him; he put his will into the work. I signed the diplomas, and those boys went out as proud as any boys ever went away from a university. This year we have 50,000 boys in the southern States, each under sixteen years of age, each growing an acre of something, each getting lessons and hints in all directions from everybody that can give them, with regard to how to grow crops; we have 400 agents in the South.

Now let me tell you something. You will find in every northern and eastern and western State a minority of good farmers and, I am compelled to confess, a majority of poor farmers. They don't know how to farm; they have yet to learn. Where did bad farming begin, do you think? Why, back in the eastern States where they do everything well—except farming. Now where is there worse farming than there? I believe that the President of Tulane University used to live there; perhaps he can tell us. When I was a boy I went to church on Sunday and to prayer meeting in the middle of the week—I had to (laughter)—but they didn't educate the boys toward the farms; they educated them toward the professions, toward the mechanic arts, toward the factories. And when they were big enough and had an education they left the farm, they left the father and mother there, and by and by when the father and mother couldn't farm any more they rented out the farm—and today the same thing is beginning in Iowa. I can't tell you what is happening in Minnesota; you people who live here must be the judges whether the same robbery of the soil is beginning in Minnesota. A soil robber is a man who grows grain and hay to sell from the farm and puts nothing back; that is what he is, and that is where he originated—back East.

And we began manufacturing in our country at the time we began robbing our soil. The last half-century we have built up our manufactories at an

astonishing rate. Why have we built them up so fast; why have they risen to such tremendous figures? Because our people were fed cheaper and better than the people who worked in factories in any other country. But what is the condition now? Are our people still better fed and more cheaply that work in the factories, that work for the railroads, that work in the mines? No! There is where the trouble comes; that is what has arrested the attention of our people. Every year, maybe oftener (Mr Hill could tell better than I can), the men that work for railroads notify the president that they want more wages because they can't live; and of course he has to raise their wages. While we were feeding Europe, there was no difficulty in getting cheap food here in the United States for our workingmen; but, as Mr Hill told you, and gave you statistics for it—it is pretty hard to follow a man like him, who has all the statistics, and Dr Wallace, who has all the philosophy and wit, but I will do the best I can (laughter)—we are sending less and less food to foreign countries and paying more and more for what our workingmen eat at home. We are not paying off debts any more, though our people are still buying diamonds and pearls—you see the rows we are having in New York when our traveling Americans come back, and want to get their jewels through the custom-house for nothing and hide them and all that; I have no sympathy with it—but we are not discussing the tariff here at all; I never talk politics and won't allow it; I have 12,000 men in my Department and every man knows I'll discharge him in a minute if he talks politics (laughter and applause); we are considering the natural resources of the country and trying to conserve them. (Applause and cries of "Good!")

As the Department grew we organized a bureau for animals, another for plants, one for forests, one for chemistry, and one for soils; and all along the line we have those great bureaus at work. We are the practical fellows who conserve; we are doing it every day. I have just been out among the forests myself four or five weeks, helping to save the Government's property out there. But the great question comes down to the soil. There is no classical college or university that teaches anything about the soil, not one single thing. From the time that Samuel had the school of the prophets at Bethel down to the present day, there never has been anything taught to the people with regard to the soil on which they walk and from which they get their living. I have organized a bureau for it. We are studying the soil all over the country. You might think, to go out on these beautiful prairies, that the soil

is all alike. Well, it isn't; any prairie has probably a hundred different soils, some of them best adapted to grow one plant and some another, some needing one kind of treatment and some another; and the great fundamental question that we must study now is the American soil and its power to produce. (Applause)

With regard to the literature of the farm: There was none when I was a young fellow; there was no college for farmers. I had to get what I did get from observation and from a store of recollection of older men. But now we have an agricultural college in each State. We have an experiment station in each State. We have 3,000 men making research in the Department of Agriculture at Washington, all specialists, the foremost in their lines in the world. When one of those men makes inquiry into something and reports, we put his name to it and print it and send it out to the people without expense. We sent out 20,000,000 pieces last year (applause). And any of you who want anything we have, no matter whether you are farmers or not, you are welcome to it. Some of the best encouragement that we have comes from those who are not farmers at all.

I have told you of the genesis of the soil-robber; is he here in the Mississippi valley? The old-time farmer educated his children, but he educated them to do anything under the sun but farm. When the boy graduated, when he got through with his education, he went anywhere but to the farm. That was until within a few years the custom. The other day I wrote to the dean of the Iowa Agricultural College that several people had applied to me for men to superintend farms, and that a newspaper man wanted a farm expert to go into his office at a good salary, and asked—"How many young men do you graduate this year in a four-year agricultural course?" He replied, and I think he said "We graduated some seventy in a four-year course, but none of them left the State; they are all going back to the farm" (great applause and cry of "Good!"). Those men know something. Now, are you doing that in Minnesota? You have always had a fine agricultural school here connected with your State University, and you have an open door into the four-year academic course in the University; you are doing much for agriculture and education. Yet we are where we are today with regard to scarce food and dear meat because we didn't begin educating the young farmer sooner. But he is going to catch on. There would be a universal introduction of agricultural education into the

common and secondary schools of the country if teachers could be found. That is the great difficulty. Fifty years ago, when Congress endowed agricultural colleges, that was the trouble. They could start the college, they could erect a building, but there was no library, there was no professor who knew anything about agriculture, and the great trouble is a man can only teach what he knows himself. But now, after half a century of effort on the part of the farmers, on the part of friends of the farms, on the part of far-seeing men like James J. Hill (applause), we are getting a creditable agricultural education in this country.

Do not be uneasy about the forests; at the last session, Congress gave me \$400,000 more than they had ever given me before to take care of the forests. Do not be uneasy about the coal, the gas, the oil, and the phosphates; President Taft has withdrawn all those until Congress indicates what shall be done with them. But the soil, Gentlemen, the soil; the big price for meat, the big price for bread; these are things to study. We *can* improve our soil. One of our speakers this afternoon told us that you cannot grow soil. I believe that, once you wash it away. But you can reduce it, beyond the point of profitable production of crops; that you can do, and that is being done. The soil-robber works in Iowa, and I fear he is at work in Minnesota. The old folks have gone to town; and the Lord knows nobody wants them there, because when you want to improve the town with gas and sewer and water and things of that kind, the farmer won't vote for them; he is regarded as a nuisance; everybody wishes he would stay on the farm, and I wish he would. And when the old farmer and his wife go to town, they sell off everything; they rent the farm to a man who has no means to stock it with cattle and sheep, hogs and poultry; he grows grain to sell, he grows hay to sell, and those farms grow worse and worse every year. That is the situation we are in. (Applause)

We are making some progress, some headway. The Government gave to the emigrant from abroad, to everybody who wanted it as long as they lasted, a claim in the rainy belt; but there are no lands left for giving away in the rainy belt. Something can be done in regard to our dry-land farming; something can be done in regard to irrigation. As Mr Hill intimated (in fact, he delivered a great deal of my speech), there is not much being done in the line of irrigation. Take a trip out West and watch the rivers as you cross them, and you will see that we are wasting far more water than we are using

—though in certain neighborhoods in Colorado highly intelligent people are every year building more dams away up in the mountains and saving their winter and spring-flood waters. That is going on and on, and it should go on until all the waters in the mountains are saved for application to the land. Do you remember the history of irrigation in the valley of the Po, in Italy? There are more people to the square mile there than are found in almost any other part of the world. They began at the headwaters of the tributaries and built great dams to hold up the water to an amount suitable for the growing of crops, something like twenty inches or more; and they built on down to the mouth of the Po. Now when there comes a drought like we had this year, they let water out on the fields, and thus get a maximum crop. Without that extra water, at a time of drought their crop would wither and fail. I understand Minnesota has more lakes, more natural reservoirs for holding water than any other State in the Union. Look to it, you Minnesota people; you can, by using that water in a dry year, grow maximum crops.

How do the people of the Old World raise big crops? If you followed Mr Hill's statistics you learned they didn't know as much there once as they do now, for they have raised their crop production from 20 to 30 bushels an acre. He also alluded to the Danes, who by good farming are enabled to sell enormous amounts of farm products. How do they keep that land up? I will tell you what a great many of them are doing. They buy mill-feeds from the United States; they buy bran and shorts, they buy the cottonseed of the South and the flaxseed of Minnesota, and feed their dairy cows. That is a highly intellectual job, isn't it, for an American citizen, to grow food for a Danish cow? But the Dane has his eyes open; he *knows*. He sells \$40,000,000 worth of butter and cheese to England every year, but puts back all the fertility on the farm; and that is what has brought up his little fifteen-acre farm, or his forty-acre farm. He has brought it up by keeping and feeding his cows on our mill-feeds, mind you; and he is prosperous—and we are not so prosperous only because we rob ourselves.

A VOICE—Bran doesn't cost any more in Denmark than in America.

Secretary WILSON—It is American bran, though. And let me tell you something else. The meats you grow up here cost hardly any more in Europe than they cost here, because the retailer over there hasn't got all the

frills that the retail dealer has here, and is satisfied with a smaller profit. (Applause)

Now, Ladies and Gentlemen, I am merely outlining some of the remarks that I prepared and gave to the newspaper people; and I have no doubt you have listened to me as long as you care to (cries of "Go on, go on"). I have enjoyed my visit here. I am on record as saying that these northwestern States, beginning here and extending on west, are the healthiest we have; their waters are good; their climate is fine; they are going to grow vigorous men and handsome women. If we are going to have all their benefits you should conserve your soil, so that your great-grandchildren will have better soil than you have today. Down in Iowa, where I have lived for 46 years, the soil grows bigger crops today than it did fifty years ago; and it is still improving.

You have extended to me the greatest compliment a hospitable people can bestow on a stranger, and that is to give me your attention. I thank you. (Great applause.)

Chairman CLAPP—Ladies and Gentlemen: We will now listen to a discussion by Honorable F. C. Stevens, Member of Congress from this district. (Applause)

Representative STEVENS—Mr Chairman, Ladies and Gentlemen: You are fortunate this afternoon, so far as my discussion is concerned. I was assigned to discuss an address by Senator Dolliver, Chairman of the Senate Committee on Agriculture and Forestry, on the subject of "Cattle, Food, and Leather." We greatly regret the enforced absence of Senator Dolliver, because he is informed on that subject and could have given us a discussion of great benefit. I congratulate myself that I am not obliged to follow him, because I know too little about his subject. So I shall briefly discuss something I do know about.

In the very able address of Mr Hill, and in the very bright discussion of Mr Wallace which followed, there was a general criticism of Congress for undue expenditures of public money. I want to tell this audience that Congress, instead of being extravagant, is often unduly economical of the people's money. The money we spend is what the people want us to spend, and we do not spend nearly as much as they want us to. The estimates that were sent in by the heads of the departments (of which Secretary Wilson is one) aggregated nearly two hundred millions of dollars more than the expenditures which Congress authorized, and the estimates which came from the field officers to the heads of these great departments, for example, like that of Secretary Wilson; from the post-offices scattered throughout the country; from the officers of the War and Navy Departments, scattered all over the world; and from the officers of the State and other departments, were, I will venture to say, nearly two hundred million dollars more still: so that Congress actually did not spend more than two-thirds as much as the people of the United States in their respective localities wanted spent. There is not a single large convention in the United States similar to this—which is one of the most magnificent in the history of this section of the country—that does not call upon Congress for the expenditure of large sums of money, and I will venture to predict that the resolutions, which will be adopted by this Congress will call for a large appropriation from the National treasury. We have in Washington every year a Rivers and Harbors Congress, composed of 4,000 of the brightest, broadest, most patriotic business men of the United States, who go there as delegates, spend their own money to go, and then ask large expenditures from the people's treasury. Scattered all over this country, meeting probably in every State in the Union, are various voluntary assemblages of our People demanding various improvements by the Federal Government, and every one asking for expenditures of the people's money. You never yet have heard of a convention which has met anywhere at anybody's expense asking for a cutting down of expenditures. If there is any one man who is popular in the United States it is the man who calls for the expenditure of the people's money; the men who are the most unpopular, and are condemned and criticised in public life, are those who try to cut down the expenses and be economical with the people's money (applause). I think there ought to be some reform (and I have had some experience); we *are* extravagant; we do spend more money than we ought to, but it is spent honestly, it is spent with

the best of intention, it is spent because the people want us to spend it, and we do not go nearly as far as they ask us to.

Just one suggestion more: It is easy to criticise and ridicule something that a man knows but little about, and I have noticed that in this discussion of Conservation each man is almighty anxious to conserve that which interests *him*; and one of the latest examples of that was afforded by the statement of Mr Wallace in condemnation of the dam between Saint Paul and Minneapolis. Now, in advance I want to state that I am not responsible for that dam; it was there before I entered public life. But there is one thing we are trying to do; we are trying to enforce the principle of practical Conservation, and I wish to call attention to that as a sample of ridicule sometimes seen in the discussion of a subject that really interests the people. The United States thirty years ago started, at the headwaters of the Mississippi, six of the largest storage reservoirs for water in the world, with a capacity of many thousands of millions of gallons of water, designed to improve the navigation of the river and raise it in times of drought eighteen inches here at the levee of Saint Paul. That enormous storage of water in the river should be utilized for the practical benefit of the people of the United States. That is the practical basis for all theories of Conservation. A board of engineers was ordered by Congress to make an investigation of the use of the dam at the Twin Cities, and they have reported that a dam can be built and it has been ordered by Congress and is under construction (it is the one ridiculed). It will be thirty feet high and will yield 15,000 horsepower of electrical energy, worth here \$25 per horsepower-year, making a total value of \$375,000 per annum, at an expenditure in all not to exceed \$2,000,000. It will pay the United States the money that it invests in that dam. It is expected that the United States will sell, for a reasonable price, that electrical energy to the cities of Saint Paul and Minneapolis and the University of Minnesota; these cities can be the best lighted in the world and save a hundred thousand dollars each annually (applause); and, more than that, we will have there the most beautiful lake in the world, extending from the historical falls of Minnehaha below to the great and beautiful University of Minnesota above. That is a practical example of Conservation (applause). Before any of these gentlemen come forward flippantly to ridicule the public works going on in any part of the country, they should realize that there *may be* some things they don't know about. (Applause)

Only one suggestion more (because we all want to hear from Professor Bailey): It is easy to criticise Congress as a whole; it is fashionable to do it; Congress hasn't any friends anywhere; but just remember this: it is a necessary evil; it is the concrete voice of ninety millions of free American citizens; it is the only agency whereby these ninety millions of American people can accomplish their will and desire. We can only run a free Government by the rule of the majority; a majority of one is potent to control this whole great country; 51 percent are in favor of what that majority does, and, 49 percent claim the right to criticise and kick at what that majority does. As this is a free Government they have that right. Now, my friends, we must remember that what displeases us probably pleases 51 percent, and if we had the right to pass the very laws we wanted to on any subject, the chances are that our next-door neighbors, on both sides, would criticise and complain of us, just as we are now doing of other people. The only thing I wish to emphasize is that Congress tries to represent the whole American people, tries to make concrete the voice of the whole American people. It is human, the same as the people are; it makes the same kind of mistakes that the people make; and, after all, the people are responsible for Congress. I thank you. (Applause)

Chairman CLAPP—Ladies and Gentlemen, we will now have an address on "Conservation in Country Life," by Dr Liberty Hyde Bailey, Dean of State Agricultural College, Cornell University, and Chairman of the Country Life Commission. It affords me great pleasure to introduce Professor Bailey. (Applause)

Professor BAILEY—Mr President, Ladies and Gentlemen: Because of the lateness of the hour, and because of the very great treat which you have had this afternoon in the presentation of the fundamental questions of country life, I shall only call your attention to three or four topics which, perhaps, have not been touched by others who have spoken from this platform.

Two great economic and social movements are now before the country—Conservation, and Country Life. The Conservation movement is the expression of the idea that the materials and agencies that are part of the furniture of the planet are to be utilized by each generation carefully, and with real regard to the welfare of those who are to follow us. The Country Life movement is the expression of the idea that the policies, efforts, and material well-being of the open country must be highly sustained, as a fundamental essential of a good civilization; and it recognizes the fact that rural society has made relatively less progress in the past century than has urban society. Both movements are immediately economic, but in ultimate results they are social and moral. They rest on the assumption that the welfare of the individual man and woman is to be conserved and developed, and is the ultimate concern of governments; both, therefore, are phases of a process in social evolution.

Not only the welfare but the existence of the race depends on utilizing the products and forces of the planet wisely, and also on securing greater quantity and variety of new products. These are finally the most fundamental movements that government has yet attempted to attack; for when the resources of the earth shall largely disappear or the arm of the husbandman lose its skill, there is an end of the office of government.

At the bottom, therefore, the Conservation and Country Life movements rest on the same premise; but in their operation, and in the problems that are before them, they are so distinct that they should not be confounded or united. These complementary phases may best work themselves out by separate organization and machinery, although articulating at every point; and this would be true if for no other reason than that a different class of persons, and a different method of procedure, attached to each movement. The Conservation movement finds it necessary, as a starting-point, to attack entrenched property interests, and it therefore finds itself in politics, inasmuch as these interests have become entrenched through legislation. The Country Life movement lacks these personal and political aspects.

These Subjects Have a History

Neither "Conservation" nor "Country Life" is new except in name and as the subject of an organized movement. The end of our original resources

has been foreseen from time out of mind, and prophetic books have been written on the subject. The need of a quickened country life has been recognized from the time that cities began to dominate civilization; and the outlook of the high-minded countryman has been depicted from the days of the classical writings until now. On this side of mineral and similar resources, the geologists and others among us have made definite efforts for conservation; and on the side of soil fertility, the agricultural chemists and the teachers of agriculture have for a hundred years maintained a perpetual campaign of conservation. So long and persistently have those of us in the agricultural and some other institutions heard these questions emphasized, that the startling assertions of the present day as to the failure of our resources and the coordinate importance of rural affairs have not struck me with any force of novelty. But there comes a time when the warnings begin to collect themselves, and to crystallize about definite points; and my purpose in suggesting this history is to emphasize the importance of the two movements now before us by showing that the roots run deep, back into human experience. It is no ephemeral or transitory subject that we are now met to discuss.

All really fundamental movements are the results of long-continued discussion and investigation, but it requires a great generalizer and organizer, and one possessed of prevision, to concrete scattered facts into powerful national movements. The one who recognized the existence of these questions, who saw the significance of the problems, who aided to assemble them, and who projected them into definite lines of public action was Theodore Roosevelt; and he himself has expressed our obligation in this Conservation movement to Gifford Pinchot. (Great applause)

The Conservation movement is now approaching its full; the Country Life movement is a slower and quieter tide, but it will rise with great power. These are the twin economic and social questions that the Roosevelt administration raised for our consideration. (Applause)

They are not party-politics subjects

I have said that these are economic and social problems and policies. I wish to enlarge this view. They are concerned with saving, utilizing, and augmenting, and only secondarily with administration. We must first

ascertain the facts as to our resources, and from this groundwork impress the subject on the people. The subject must be approached by scientific methods. It would be unfortunate if such movement became the exclusive program of a political party, for then the question would become partisan and probably be removed from calm or judicial consideration, and the opposition would equally become the program of a party. Every last citizen should be naturally interested in the careful utilization of our native materials and wealth, and it is due him that the details of the question be left open for unbiased discussion rather than be made the arbitrary program, either one way or another, of a political organization. The Conservation principle is a plain economic and social problem rather than a political issue. (Applause)

The Country Life movement is equally a scientific problem, in the sense that it must be approached in the scientific spirit. It will be inexcusable in this day if we do not go at the subject with only the desire to discover the facts and to arrive at a rational solution by non-political methods. The first recommendation of the Commission on Country Life is that the Government begin taking stock of rural life in order that we may have definite facts on which to begin a reconstructive program.

The soil is the greatest of all resources

The resources that sustain the race are of two kinds—those that lie beyond the power of man to reproduce or increase, and those that may be augmented by propagation and by care. The former are the water, the air, the sunshine, and the mines of minerals, metals, and coal; the latter are the living resources, in crop and live-stock. Intermediate between the two classes stands the soil, on which all living resources depend. Even after all minerals and metals and coal are depleted, the race may sustain itself in comfort and progress so long as the soil is productive, provided, of course, that water and air and sunshine are still left to us. Beyond all the mines of coal and all the precious ores, the soil resource is the heritage that must be most carefully saved; and this, in particular, is the country-life phase of the Conservation movement.

To my mind, the Conservation movement has not sufficiently emphasized this problem. It has laid stress, I know, on the enormous loss by soil

erosion, and has said something of inadequate agricultural practice; but the main question is yet practically untouched by the movement—the plain problem of handling the soil by all the millions who, by skill or blundering or theft, produce crops and animals out of the earth. Peoples have gone down before the lessening power of the land, and in all probability other peoples will yet go down. The course of empire has been toward the unplundered lands.

Thinner than the skin of an apple is the covering of the earth that man tills. Beyond all calculation and all comprehension are the powers and the mysteries of the soft soil layer of the earth. We do not know that any vital forces pulsate from the great interior bulk of the earth. Only on the surface does any nerve of life quicken it into a living sphere. And yet, from this attenuated layer have come numberless generations of giants of forests and of beasts, perhaps greater in their combined bulk than all the soil from which they have come; and back into this soil they go, until the great life-principle catches up their disorganized units and builds them again into beings as complex as themselves.

The general evolution of this soil is toward greater powers; and yet, so nicely balanced are these powers that within his lifetime a man may ruin any part of it that society allows him to hold; and in despair he abandons it and throws it back to nature to reinvigorate and to heal. We are accustomed to marvel at the power of man in gaining dominion over the forces of nature—he bends to his use the expansive powers of steam and the energy of the electric currents, and he ranges through space in the light that he concentrates in his telescope; but while he is doing all this he sets at naught the powers in the soil beneath his feet, wastes them, and deprives himself of vast sources of energy. Man will never gain dominion until he learns from nature how to maintain the augmenting powers of the disintegrating crust of the earth.

We can do little to control or modify the atmosphere or the sunlight; but the epidermis of the earth is ours to do with it much as we will. It is the one great earth resource over which we have dominion. The soil may be made better as well as worse, more as well as less; and to save the producing powers of it is far and away the most important consideration in the Conservation of natural resources.

No man has a right to plunder the soil

The man who owns and tills the soil owes an obligation to his fellowmen for the use that he makes of his land; and his fellowmen owe an equal obligation to him to see that his lot in society is such that he will not be obliged to rob the earth in order to maintain his life. The natural resources of the earth are the heritage and the property of every one and all of us. A man has no moral right to skin the earth, unless he is forced to do it in sheer self-defense and to enable him to live in some epoch of an unequally developed society; and if there are or have been such epochs, then is society itself directly responsible for the waste of the common heritage.

The man who plunders the soil is in very truth a robber, for he takes that which is not his own and he withholds food from the mouths of generations yet to be born. No man really owns his acres; society allows him the use of them for his life-time, but the fee comes back to society in the end. What, then, will society do with those persons who rob society? The pillaging or reckless land-worker must be brought to account and be controlled, even as we control other offenders.

(I know that the soil-depletion idea is now challenged; but I am sure that the Conservation ideal must be applied to soil maintenance even as it is applied to other maintenance. If it transpires that plants hold a different relation to the soil-content than we have supposed, we still know that poor farming makes the land unproductive and that the saving of wastes is a desirable human quality; and we shall probably need to change only our phraseology to make the old statement broadly correct.)

I have no socialistic program to propose. The man who is to till the land must be educated: there is more need, on the side of the public welfare, to educate this man than any other man whatsoever (applause). When he knows, and when his obligations to society are quickened, he will be ready to become a real conservator; and he will act energetically as soon as the economic pressure for land-supplies begins to be acute. When society has done all it can to make every farmer a voluntary conservator of the fatness of the earth, it will probably be obliged to resort to other means to control the wholly incompetent and the recalcitrant; at least, it will compel the soil-robber to remove to other occupation, if economic stress does not itself

compel it. We shall reach the time when we shall not allow a man to till the earth unless he is able to leave it at least as fertile as he found it. (Applause)

It is a pernicious notion that a man may do what he will with his own. The whole tendency of social development is away from this idea. A person may not even have the full control of his own children: society compels him to place them in school, and it protects them from over-work and hardship. A man may not breed diseased cattle. No more should he be allowed wantonly to waste forests or to make lands impotent, even though he "owns" them. (Applause)

Ownership vs. Conservation

This discussion leads me to make an application to the Conservation movement in general. We are so accustomed to think of privileged interests and of corporation control of resources that we are likely to confuse Conservation with company ownership. The essence of Conservation is to utilize our resources with the least waste consistent with good progress, and with an honest care for the children of all generations.

While we not infrequently state the problem to be the reservation of our resources for all the people, and then assume that if all the resources were in private ownership the problem would thereby be solved, yet, in fact, the Conservation question is one thing and the ownership of property quite another. A corporation may be the best as well as the worst conservator of resources; and likewise, private or individual ownership may be the very worst as well as the best conservator. The individual owner, represented by the "independent farmer," may be the prince of monopolists (applause), even though his operations compass a very small scale. The very fact that he is independent, with the further fact that he is intrenched behind the most formidable of all barriers—private property rights—insures his monopoly.

In the interest of pure Conservation, it is necessary to control the single man as well as the organized men. In the end Conservation must deal with the individual man—that is, with a person. It matters not whether this person is a part of a trust, or lives alone a hundred miles beyond the frontier, or is the owner of a prosperous farm—if he wastes the heritage of the race, he is an offender. We are properly devising ways whereby the corporation holds its property or privileges in trust, returning to government (or to society) a fair

rental; that is, we are regulating the corporation and making it responsible to the people. What shall we do with the unattached man, to make him also responsible? Shall we hold the corporate plunderer to strict account, and let the single separate plunderer go scot free? (Applause)

In the last analysis, as measured by the results to society, there is no essential difference between corporate ownership and individual ownership.

The philosophy of saving

The Conservation of natural resources, therefore, resolves itself into the philosophy of saving, while at the same time making the most and best advancement in our own day. We have not developed much consciousness of saving when dealing with things that come free to our hands, as the sunshine, the rain, the forests, the mines, the streams, the earth; and the American has found himself so much in the midst of plenty that saving has seemed to him to be parsimony, or at least beneath his attention. As a question of public action, however, conscientious saving represents a very high development. A high sense of saving ought to come out of the Conservation movement. This will make directly for character-efficiency, since it will develop both responsibility and regard for others.

Civilization, thus far, is built on the process of waste. Materials are brought from forest and sea and mine, certain small parts are used, and the remainder is discarded or destroyed; more labor is wasted than is usefully productive; but what is far worse, the substance of the land is taken in unimaginable quantities and dumped wholesale, through endless sewerage and drainage systems, into the sea. It would seem as if the human race were bent on finding a process by which it can most quickly ravish the earth and make it incapable of maintaining its teeming millions. We are rapidly threading the country with vast conduits by which the fertility of the land can flow away unhindered into the unreachable reservoirs of the ocean. (Applause)

The factories that fabricate agricultural products are likely to be midway stations in the progress of the fertility on its way to the sea. The refuse is dumped into streams; or if it is made into fertilizing materials, it seldom returns to the particular areas whence it came. A manufactory will expend any effort in improving its machinery and practice to enable it to get more

material out of its products, but may do little or nothing to increase the production back on the farms. A sugar-beet or other factory may drain its country until the country can no longer raise the product; whereas, by developing a rational system of husbandry and returning the wastes, as in some European countries, it might maintain the land-balance. Any good milk-products factory should develop sound milk-making on the farms of the region, as any good canning factory should raise the standard of production in the fruits and vegetables that it uses; and this should always be done with the object of preserving and even increasing the land-power. A factory owes an obligation to the open country that supports it.

For these and for other reasons, the city always tends to destroy its province. The city takes everything to itself—materials, money, men—and gives back only what it cannot use or what it discards as useless: it does not constructively build up its contributory country.

City dwelling and country dwelling are the two opposite developments of human affairs. The future state of society depends directly on the finding of some real economic and social balance between the two, some species of cooperation that will build and serve them both. This is the fundamental problem of the social structure. Although city people and country people are rapidly affiliating in acquaintanceship, these poles of society are not yet effectively coming together cooperatively on economic lines. (Applause)

The Conservation of food

The fundamental problem for the human race is to feed itself. It has been a relatively easy matter to provide food and clothing thus far, because the earth yet has a small population, and because there have always been new lands to be brought into requisition. We shall eliminate the plague and the devastations of war, and the population of the earth will tremendously increase. When the new lands have all been opened to cultivation, and when thousands of millions of human beings occupy the earth, the demand for food will constitute a problem that we scarcely apprehend today.

One would think, from current discussions, that the single way to provide the food for the population is to raise more products by moving more people on the land; but this is not at all nub of the question. More products will be raised as rapidly as it pays persons to raise them, and there are now

sufficient people on the land to double its productiveness; and the necessary increase of population will come automatically with increasing profits in the business. Much is said about the necessity of intenser methods of farming, and we all recognize the need; but the chief reason why our people do not raise 300 bushels of potatoes to the acre is that it does not yet pay in most cases to produce the extra yield. The comparative statistics of yields in different countries are useful as appealing to the imagination, but they may be wholly fallacious as guides. What we need is a thorough inquiry into the course of trade from potato-patch to consumer, to see where the profit goes.

We need a greater number of competent farmers, to be sure, whether they hail from the country or the city; the city will still attract those laborers who cannot work alone and who watch the clock, and the city provides the organization or machinery to make them of use; but the real food question and cost-of-living question is the problem of maintaining the producing-power of the earth by means of better farming.

We think we have developed intensive and perfected systems of agriculture; but as a matter of fact, and speaking broadly, a scientifically permanent agriculture on national lines is yet unknown in the world. In certain regions, as in Great Britain, the productivity of the land has been increased over a long series of years, but this has been accomplished to a great extent by the transportation of fertilizing materials from the ends of the earth. The fertility of England, according to authorities, has been drawn largely from the prairies and plains of America, from which it has secured its food supplies, from the guano deposits in islands of the seas, from the bones of animals and men, from the mummies of Egypt (applause). The rotation of crops is not itself a complete means of maintaining fertility.

We begin to understand how it is possible to maintain the producing-power of the surface of the earth, and there are certain regions in which our knowledge has been put effectively into operation; but we have developed no conscious plan or system in a large way for securing this result. It is the ultimate problem of the race to devise a permanent self-sustaining organized agriculture on a scientific basis. The problem is yet unsolved.

We deplore the relative decrease in the exportation of agricultural produce, and seem to think that the more we export the richer we become; but, if our

knowledge is correct, under present systems of farming, the more we send abroad the sooner do we deplete our soils. We properly remove phosphate lands from exploitation and monopoly, but we may remove our phosphates more rapidly by sending our produce in unhindered quantities to Europe. Of course, I am not arguing against exportation and trade, but I wish to point out a fallacy in our common economic speech.

The best husbandry is not in the new regions

The best agriculture, considered in reference to the permanency of its results, develops in old regions, where the skinning process has passed, where the hide has been sold, and where people come back to utilize what is left. The skinning process is proceeding at this minute in the bountiful new lands of the United States; and in parts of the older States, and even also in parts of the newer ones, not only the skin but the tallow has been sold. There are "abandoned" farms from California even unto Maine.

It is persistently said that the old eastern States are worn out, and that the farming in them is wretched. There is reason enough to be ashamed of eastern agriculture, and I hope that our newer regions will not repeat the mistakes of the older States; but the eastern States have most excellent agriculture, more than we are aware. Much of it is very profitable, fully as profitable as any I have seen in the great agricultural West. The acre-efficiency, as indicated by the Twelfth Census, is greatest in the old eastern States. Considered with reference to maintaining high fertility and utilizing wastes, I have not seen better fanning in this country than in many examples east of Buffalo. In the development of our agricultural wealth, the East as well as the West must be reckoned with. We cannot expect to develop widespread self-sustaining systems of farming in the East so long as it must compete with the soil-mining of the West.

We are always seeking growing-room, and we have found it. But now, the western civilization has met the eastern, and the world is circumferenced. We shall develop the tropics and push far toward the poles; but we have now fairly discovered the island that we call the earth (within a year and a half we have reached one end of it and all but reached the other), and we must begin to make the most of it.

Another philosophy of agriculture

Practically all our agriculture has been developed on a rainfall basis. There is ancient irrigation experience, to be sure, but the great agriculture has been growing away from these regions. Agriculture is still moving on, seeking new regions; and it is rapidly invading regions of small rainfall. The greater part of the land surface of the globe must be farmed, if farmed at all, under some system of careful water-saving. Some of it is redeemable by irrigation, and the remainder, representing about one-half the earth's surface, by some system of utilization of deficient rainfall, or by what is inappropriately known as "dry farming." The complementary practices of irrigation and dry-farming will develop a wholly new system of agriculture and a new philosophy of country life.

Even in heavy rainfall countries, there is often such vast waste of water from run-off that the lands suffer severely during droughts. The hilly lands of our best farming regions are greatly reduced in their crop-producing power because people do not prepare against drought as consciously as they provide against winter. It is often said that we shall water eastern lands by irrigation, and I think that we shall; but our first obligation is to save the rainfall water by some system of farm-management or dry-farming.

The irrigation and dry-farming developments have a significance beyond their value in the raising of crops; they are making the people to be conservators of water, and to have a real care for posterity. Agriculture rests on the saving of water. (Applause)

The obligation of the farmer

The farmer is rapidly beginning to realize his obligation to society. It is usual to say that the farmer feeds the world, but the larger fact is that he saves the world. The economic system depends on him. Wall Street watches the crops.

As cities increase proportionately in population, the farmer assumes larger relative importance and becomes more and more a marked man.

Careful and scientific husbandry is rising in this new country. We have come to a realization of the fact that our resources are not unlimited. The mining of fertilizing materials for transportation to a few spots on the earth will some day cease. We must make the farm sustain itself, at the same time

that it provides the supplies for mankind. We all recognize the necessity of the other great occupations to a well developed civilization; but in the nature of the case, the farmer is the final support. On him depends the existence of the race. No method of chemical synthesis can provide us with the materials of food and clothing and shelter, and with all the good luxuries that spring from the bosom of the earth.

I know of no better present conservators than our best farmers. They feel their responsibility. Quite the ideal of Conservation is illustrated by a farmer of my acquaintance who saves every product of his land and has developed a system of self-maintaining live-stock husbandry, who has harnessed his small stream to light his premises and do much of his work, who turns his drainage waters into household use, and who is now troubled that he cannot make some use of the winds that are going to waste on his farm.

The obligation of the Conservation movement

What I have meant to emphasize is the fact that the farmer is the ultimate conservator of the resources of the earth. He is near the cradle of supplies, near the sources of the streams, next the margin of the forests, and on the hills and in the valleys and on the plains just where the resources lie. He is in contact with the original and raw materials, and with the fundamental necessities. Any plan of Conservation that overlooks this fact cannot meet the situation. The Conservation movement must help the farmer to keep and save the race.

The Conservation and Country Life movements will pass through propagandic, economic, and political phases; but they will eventuate into a new alignment of human forces and a redirection of the processes of social development. These results are to be brought about by efforts proceeding along definite lines of action. The Conservation movement is rapidly becoming crystallized into definite proposals. The Country Life movement should be solidified through a definite National organization or commission, that is continuously active. This body should work through all existing rural organizations, placing before them for consideration the specific questions of the day and serving as a clearing-house of discussions that arise in the societies and with the people; and it should make real

investigation into the actual economic and social conditions of the open country, with a view to pointing out the specific practical steps to be taken by National, State, local, and individual enterprise.

The Commission on Country Life made sufficient specific recommendations and suggestions to start a fundamental redirection of effort as applied to rural development. The Report of the Commission will naturally be the diverging-point of future discussions of country-life problems. (Applause)

Chairman CLAPP—Ladies and Gentlemen: The hour grows late, and the Congress will stand adjourned for the day.

SEVENTH SESSION

The Congress was called to order by President Baker in the Auditorium, Saint Paul, at 8.30 a.m. on Thursday, September 8, few Delegates being present, and none responding to an invitation to speak for their States. After waiting some time—

President BAKER—Ladies and Gentlemen: We will now go on with the regular program, leaving the Call of the States for a later time when the Delegations may be more fully represented. In the absence of the Reverend Dr J. A. Krantz, President of the Minnesota Conference of the Swedish Lutheran Church, we will dispense with the public invocation.

Professor Henry S. Graves, Chief Forester of the United States, will now address you on "The Forest and the Nation."

Professor GRAVES—Mr President, Ladies and Gentlemen: The movement for the conservation of our natural resources has reached the second and most critical stage in its progress. The country has expressed in unmistakable terms its approval of the principles of Conservation; there is now before it the problem of the practical application of those principles.

In forestry there is a very general agreement that our woodlands must be protected from fire, that waste must be reduced, and that a future timber supply must in some way be provided. In carrying out these purposes, differences of opinion arise, and it soon develops that with many persons the interest in forestry is confined to the abstract idea and does not extend to its practice. When the requirements of forestry are considered, forest owners usually find that they must make some modification in their methods of cutting, that they must use more care in protection from fire and in saving young growth, and that if they are to secure a new growth of trees after cutting, some investment is necessary. The general public learns that in

order to secure for the Nation the permanent benefits of the forest, National and State expenditures are required.

It is at this point that indifference and even opposition to Conservation arise. Indifference is shown by the public when it fails to make adequate appropriations for public forestry. Direct opposition appears from those who fear that their interests in one way or another may be adversely affected. There is a great deal of misunderstanding in regard to the methods of Conservation, and many have charged that those methods heretofore advocated are impractical. In order to be successfully applied, Conservation must be practical; but at the same time the methods must be such as will actually accomplish its real purposes. To my mind the real significance and value of this Congress is that an opportunity is afforded to make clear the methods of Conservation, and the country will then decide whether it will really be put into practice or become a mere name.

It is not my intention now to dwell at length on the fundamental importance to the country of forest Conservation. To those who know the needs of the people for forest products, the available resources, and the manner in which they are now being used up or destroyed, it must be clear that we are facing a problem which must be met by prompt and vigorous action.

A survey of the forest resources of the world shows clearly that in the long run this Nation must be dependent chiefly on its own supplies. Those who believe that we may destroy our own forests and then draw upon foreign resources of timber are misinformed as to the facts, for those supplies will not be long available. Foreign countries will need for their own use what they can produce, and many of the exporting countries are exhausting their forests just as rapidly as America. The timber supply in this country is being rapidly depleted. We are extravagant in our use of forest products; there is waste in logging and manufacturing, and the loss by fire is a shame to the country. To offset this reduction of merchantable resources the annual production of timber by growth amounts to much less than one-third the average quantity used and destroyed. In other words, we are actually exhausting our forest supplies by use and waste.

There is a sufficient amount of land in the country better suited to forest growth than other purposes to produce all the wood and timber needed by

the Nation, provided the forest is properly handled. This land includes mountain areas where the protection of the vegetation is necessary to conserve water and protect the slopes. The protective benefits of the forest can thus in most cases be secured at the same time as the production of wood and timber. There are, however, certain mountain regions of the West where large trees will not grow, and where the cover of brush and grass must be conserved to protect the slopes and to regulate the run-off of water. In these mountains special reservations must be maintained primarily for protective purposes.

There is but little disagreement in regard to these simple propositions. The difficulty lies in the fact that the people do not appreciate the need of immediate action to put the principles of forestry into practice. The reason why prompt action is not appreciated is that, except locally, the effects of forest destruction have not yet been keenly felt. It is true that the prices of certain grades of lumber have tended to increase. This increase is in part due to the reduction of supplies, but it is due also to the same causes of increased cost of production as have raised the price of other manufactured commodities (applause). The development of railroad transportation and of methods of logging have constantly opened new forest resources and furnished a supply to the public. There are today over 30,000 saw-mills throughout the country cutting timber and competing for the market. Although the prices of lumber may seem high to the consumer it is still true that in some sections the competition among the manufacturers is keeping the prices down to a point where it is hard to market low grades and to utilize in full any but the best trees in the forest. As long as the value of timber is below what it would cost to produce it by growth, the general public will not realize that our supplies are being depleted. It is after the virgin supplies are exhausted—and that will come in a comparatively short time—that the great increase in values will come and the public will suffer. We are urging action now in order that there may be new supplies produced to meet the needs of the Nation at that time. (Applause)

The general public fails also to appreciate the effect of forest destruction on stream-flow and on soil erosion. Some even go so far as to deny the connection between forests and stream-flow. There are many factors which determine the stability of water flow. Climate, character of soil, topography, and vegetative cover, all have an influence on the run-off of water. There

may be a change of conditions of one or more of these influencing factors sufficient to upset the equilibrium established by nature, and alter the manner of run-off of the water in a given watershed (applause). In humid regions, where the old timber is cut off or burned, a cover of young trees or brush often springs up quickly and protects the slopes before the character of the stream channels is changed. A single clearing of the forest may thus have only a small or temporary effect on water flow. The repeated destruction of the cover may, however, result in a permanent change, and finally produce torrent conditions. Thus in the Southern Appalachian province it is not so much the present and past conditions—although those are serious—which demand forest conservation, as what will inevitably be the result of continued destruction of the cover. (Applause)

Where the conditions for forest growth are critical, and the soil and topography are such that the balance of nature is easily disturbed, the effects of forest destruction are much more quickly felt. In certain parts of the West we find already examples of flood and torrent conditions equal to those in France and Asia. For example, in Utah there are watersheds where, on account of the burning of the forests and the over-grazing of slopes, torrent conditions are already definitely established. One of the most extreme and striking instances in the West is found on the watershed of Kanab creek flowing through southern Utah and northern Arizona. As the result of over-grazing, the tributary streams have already become deep washes, and many new and deep gulches have been formed running into the main channel and into the side channels. The water which falls on the surface is quickly carried to some stream or wash which becomes a miniature torrent. The gathering of these together in the main channel makes a flood which is irresistible. The loss from the destruction of dams and bridges, the washing away of arable lands, and the deposit of rocks and gravel on cultivated fields, has been enormous. The restoration of vegetation alone will not cure the evil. It is now an engineering problem to check the torrential flow of water in the various streams and washes.

In spite of the increasing evidences of the effects of forest destruction, the public still fails to appreciate the need of prompt action to prevent the scarcity of timber and to protect the flow of our streams. The time for action is *before* a disaster and not afterward (applause). The small public investments necessary for forest protection are insignificant when

contrasted with the losses and hardships to communities resulting from forest destruction.

The forest problem is peculiarly difficult on account of the length of time required to produce timber of useful dimensions. We are using today trees which for the most part are from 150 to 200 years of age. The time required to produce trees suitable for lumber varies from about 40 years with our most rapid-growing species to over 100 years in many mountain regions. The production of timber requires a long investment. It requires the permanent use of land for forest growth, and a stable policy in handling the forest. At the present time in this country there is great risk from fire, which discourages investment by private capital in the growing of timber. By its very nature, therefore, the problem of forestry presents great difficulties to the average private owner of forest land who has bought the property to market the merchantable timber and not to grow trees.

Forestry nearly always involves an actual investment. Private owners will not as a rule make this investment unless there is clearly in sight an adequate return. On account of the long investment, risk from fire, a burdensome system of taxation of growing timber, and the present uncertainties of market, most private owners today are not practicing a system of forestry which takes into consideration the production of new timber supplies. Many say that if fires are kept out the question of forest production will take care of itself, no matter how the forest is handled, and that all there is to forestry is protection from fire. Let me say, and with all the emphasis I am capable of using, that forest production will not take care of itself. There are cases, and remarkable ones, of natural reproduction of forests even under the worst of abuse. But where there is no systematic provision for reproduction, ordinary lumbering results in the long run in a steady reduction of growth of valuable material; and there are only too many cases of destructive lumbering which leave the land in an unproductive state even when fires do not occur. (Applause)

Forestry is necessary to guarantee to the people the continuous benefits of the forest. The responsibility of working out the problem of National forestry cannot be left with private owners. It is primarily a public question, and the burden of its solution must be largely borne by the public. In the first place those forests owned by the public must be protected and

administered under the methods of practical forestry. These public forests comprise about one-third of the forest area of the country. The remaining two-thirds of our forests are in private ownership, and this includes about four-fifths of the remaining standing merchantable timber. Without doubt the area of the public forests will be considerably increased through the acquirement of areas needed for the protection of public interests, especially in the mountain regions of the East. But the Federal and State forests alone will not be sufficient to produce the supplies of forest products needed by the country. The practice of forestry on private lands, or at least on those areas better suited for forest growth than for other purposes, is a public necessity. I regard the proper handling of these private forests as a public necessity (applause). The private owner cannot escape the responsibility of ownership of an important natural resource; at the same time he cannot be expected to make financial investments in order to provide for a general public benefit. The conditions which prevent him from practicing forestry should be changed. He should be given public aid in protection from fire. There should be a reasonable system of taxing growing timber, and there should be cooperation in meeting the peculiar difficulties of his business which tend to stand in the way of Conservation.

The practice of forestry by private owners may be brought about through assistance and cooperation by the Federal Government and the States. The Government can do a great deal to promote private forestry. It is the policy of the Forest Service to aid in the introduction and practice of forestry on private lands, just as far as its authority permits. This assistance must, however, be largely confined to education, advice, and general cooperation. Through research and experiment, the Government is laying the foundation for the practice of forestry in all parts of the country. The results of the work in forest products will greatly help in the problem of saving waste. The experiments in silviculture are demonstrating the methods of handling woodlands. Direct aid to private owners in the practice of forestry must come chiefly from the States. The proper adjustment of taxes is a State matter. Assistance in fire patrol and fire fighting must come from the States. If on the other hand this aid is given by the States and the Government, and the obstacles now standing in the way of private forestry are removed, private owners should assume their obligations in actually setting to work to practice forestry.

The first necessity is prompt and effective action by the States. As yet most of our States have not assumed their full responsibilities in forestry. In a number of them good forests laws have been enacted; several States are buying lands as public reservations; and in about fifteen States a forest commission or a State forester has been appointed. But the problem of State forestry requires a great deal more than laws on the statute books, or the appointment of a State forester. There must be the machinery to carry out the laws, a thoroughly equipped organization to patrol the State and fight fires, and adequate appropriation of money to make this work really effective (applause). The real test of State forestry will be the development of a forest policy which will be stable, and the providing of the money necessary to carry on the work.

The first duty of the Federal Government in forestry is the proper administration of the forest lands owned by the Nation. A National forest policy has already been initiated. The greater portion of the Federal forest lands have been set aside as National Forests and they have been managed on the principles of practical Conservation. The purpose of establishing these forests has been to guarantee the best possible use of their resources for the people. There is still an impression among some persons that the National Forests are closed reservations, withdrawn from use and development. The keynote of the Federal policy in handling these forests is the use of their resources; but it is the continued use in contrast with that use which exhausts the resources (applause). There are many who assert that the National Forests are retarding development. It is the policy of the Forest Service to encourage the opening up and development of the resources of the forests, but we take the stand that this must be a development which will permanently build up the country. (Applause)

The Federal policy stands squarely for permanent development and maintenance of stable industries, as opposed to mere exploitation which exhausts the resources, and which shortly results in the impoverishment of the region. (Applause)

In administering the National Forests, the first task is to protect them from destruction by fire. In order adequately to protect forests from fire, the first necessity is a system of roads and trails to enable proper patrol and movement of fire fighters, and telephone lines for quick communication.

The second necessity is a well organized force of rangers and guards to patrol the forest and fight fires. Ever since the National Forests were placed under the administration of the Forest Service, the construction of trails and telephone lines has been pushed as rapidly as funds could be secured for that purpose. Although there have already been built 9,218 miles of trails, 1,218 miles of roads, and 4,851 miles of telephone lines, this represents but a beginning of the work when the vast area of inaccessible and undeveloped forests is considered. The Forest Service has a well organized protective service for patrol and fire fighting, though the number of men is still inadequate. Nevertheless it has been possible in ordinary seasons to keep down the fires to a small loss. During the present season there has been in the Northwest an unparalleled drouth and constant high winds that have made fire protection unusually difficult. Innumerable fires were started in the forests from various causes. The woods were dry, and a small spark was sufficient to start a blaze. Where there were roads and trails, the patrol-men were able to reach the fires quickly and either put them out in their incipency or soon mobilize a force of men who brought them under control before they had done much damage. This was well demonstrated by the fact that in the Montana and Idaho districts the majority of railroad fires were put out by the patrol-men employed by the Forest Service and by the railroads in cooperation before they reached dangerous proportions. Many fires were started, also, in the inaccessible portions of the forest where there are no roads and trails. It was often impossible to reach those fires until they had been burning several days, and in many cases had become dangerous conflagrations. The disastrous fires were those occurring under these conditions.

I wish to take this occasion to express my appreciation of the work of those men who lost their lives in these fires, and also of those other men who ever since the opening of this dry season have been fighting these fires, working often day and night, without regard to hours of service—working with a courage, with a singleness of purpose and desire to protect the property of the public, which makes me proud of them. (Applause)

The great lesson of these fires is the absolute necessity for a complete system of roads and trails and of telephone lines in the National Forests. I meet some men who say that forests cannot be protected from fire, and that sooner or later every extensive forest will be burned. The experience in the

Northwest this year only strengthens my conviction that forests can be protected from fire even under the most adverse climatic conditions. But this protection absolutely requires a proper development of the forest in the way of transportation and communication, and an adequate force of men for patrol. The National Forests can be rendered safe from fire but they must be organized for it. This requires extensive construction work at the outset. It requires a large investment in permanent improvement work by the Government. But that necessary expense is insignificant in comparison with the value of the property which will be protected, and the benefits to the communities and industries depending on these forests.

The National Forests are for use, and are administered primarily for the benefit of those States and communities in which they are located. The various resources are opened to use under reasonable restrictions which will guarantee their best continuous service to the greatest possible number of people. The mature timber is cut when there is a demand for its use, but the cutting is conducted under the principles of forestry, so that new growth is established in openings made by lumbering and the continued supply of timber is provided for. (Applause)

The other resources of the National Forests are also being put to use. The grass is utilized under a system of regulated grazing, land more valuable for agriculture than for forest purposes is opened to entry under the forest homestead act, prospecting is allowed without restriction, and legitimate mining is encouraged. It is the aim of the Forest Service to encourage the development of water-powers, and we are endeavoring to work out a practical plan which will facilitate this development by private capital, and at the same time protect the interests of the public (applause). I believe that the use of water-power sites on Federal lands should be under Government control, and I believe that this can be accomplished so as not to prevent the attraction of capital to their development. (Applause)

So far as the National Forests are concerned, Conservation has already carried into the practical stage, for it is being put into actual operation. The National Forests will always stand as a monument to the work of the real

founder and spirit of the Conservation movement, Gifford Pinchot. (Great applause)

There are many opponents of the National Forest policy and of the Forest Service, but I find in most sections of the country that those who are using the National Forests, and who are therefore most vitally interested in them, are cooperating very heartily with the Government in working out the details of their administration. It is through the kind of constructive cooperation which the Forest Service is receiving from lumbermen of the country that the practical management of the National Forests can be made really effective. (Applause)

The burden of my plea today is the need of prompt and vigorous action. Action is required of the general public in giving support for the protection of the National Forests. Action is required by the States in administering the State lands in the interests of the public. Action is required by the States in initiating a system of taxation of growing timber which will not prevent Conservation. Action is required by the States in introducing a system of forest patrol and fighting fires which will permit prompt work in the prevention of the burning of our forests. And action, finally, is needed by private individuals to introduce the practical forestry on their lands just as far as economic conditions will permit.

My suggestion is that the first step is required by the public through action of States and action of the Government. I appreciate that this cannot be accomplished without explaining fully to the people exactly what is required. I appreciate that there is necessary an organized campaign of education which should be carried into every locality of the country. This campaign may and must be practical, and not only the general problem of forestry but also the specific means of solving it must be presented to the people. This educational work may be done in part by the Government; a large amount of it must, however, be carried on through the State officials, through the State forest and conservation commissions, and through National and local associations. (Prolonged applause)

President BAKER—The next subject is "The Stake of the Business Man in Conservation," by Mr Alfred L. Baker, of Chicago.

Mr BAKER—Mr President, Fellow Delegates, Ladies and Gentlemen: Here in this Second Conservation Congress, where are assembled specialists who have given profound study to the different phases of the Conservation of our resources, where are met together scientists in agriculture, forestry, mineralogy and waterways, it is not intended that the remarks of a business man should stumble into the fields of the experts. It is, however, appropriate that he should voice his approving earnestness and vigorous enthusiasm in behalf of the Conservation movement (applause), and voice them to those National benefactors who are holding their shoulders to the wheel of progress. As a delegate to this Congress, representing the business man and with the knowledge of his views, I wish to state with all the emphasis of which I am capable that the business men in this country are heart and soul in favor of Conservation (applause). Owing to the infirmities of human nature a few may faint by the wayside; but the great body and mass can always be depended on to faithfully and loyally support the movement. By so doing they are promoting the proper development of those resources which are not only the foundation of our National prosperity but also the foundation of their own individual success.

The most conspicuous quality in the character of the successful business man is foresight—and he, more than any other member of the community, must realize the necessity of foresight in the management of our National affairs. He himself would never permit the waste or plunder of his own personal resources, and whilst enjoying their daily possession would always take thought for the morrow. The Nation in its control of our resources should reflect the same character and intelligence which the individual shows in the management of his own private affairs. (Applause)

The great body of business men favor the well-known policies of Conservation. They recognize that those resources which are of a public character should be held in trust by the Nation for the benefit of the people (applause) and that those resources of a private nature should be so disposed of that they will be enjoyed by the greatest number for the longest time. (Applause)

They believe in the Government control of water-power (applause) with the cooperation of the States, and in the application of a scientific forestry which will eliminate waste, also in a fire patrol which, *at whatever cost* (applause), will prevent the destruction of our forests and of human life. They believe in better methods of farming and in the improvement of country life so that the bright boy on the farm shall no longer respond to the call of the great city, but find immediately about him equal opportunities for fame and fortune. (Applause) They believe in the continued distribution of information on a large scale that will educate the people and advance their knowledge of Conservation (applause); and finally they believe in the Conservation of public integrity, which is the basal foundation of our National life on which all else depends. (Great applause)

I am not one of those who believe that the Conservation movement should be confined solely to the technical treatment of the forest and soil and the prevention of material waste. The second article in the platform of the first Conservation Congress provides that "the objects of this Congress shall be broad, to act as a clearing house for all allied social forces of our time, to seek to overcome waste in natural, human, or moral forces." I concur in that declaration. (Applause)

We are told that the Constitution of the United States was the unexpected outcome of a conference convened for the sole purpose of investigating our waterways. The charge of irrelevancy might well have been brought to bear upon the discussions which ensued relating to a standing army and the powers of the Federal Government, but in all National movements the importance rests not with their origin but with the extent of their usefulness. (Applause)

However restricted at the outset, Conservation has grown into a larger and more comprehensive movement, and its principles include the conservation of ideals that make for good citizenship (applause). It is in relation to this larger view that I wish to emphasize the importance of the American business man and his influence on our National progress.

In the lifetime of many now living, the land in this great State of Minnesota was divided between two Indian tribes—the Sioux and the Chippewa. These tribes were uncivilized. Intelligence had not arrived at the stage

which produces diversified industry, commerce, and the merchant. The influence of these forces marks the difference between the land of the Sioux and the State of Minnesota today.

The early pioneers who first settled on the Atlantic Coast and then continued their journey across the Continent were all business men, but they were not capitalists. From the eastern States they sought in Europe capital to build up the industries of their locations, and, by the use of this capital and labor rendered the East prosperous; and when these sturdy pioneers opened up the wealth of resources in the West they, in turn, drew upon the East for capital, and by paying for its use and uniting labor with it developed this great country. The descendants of these pioneer business men are the representative business men of today. They are not in an economic sense capitalists. Whilst the capitalist may be a business man, the vast majority of business men are not capitalists. The business man is the one who obtains capital from one source and labor from another source and unites them in an anticipated prosperous undertaking. (Applause)

The material prosperity of the United States is due to our natural resources and the genius of the business man united with the capital of the few and the toil of the millions; but the *creative genius*, the organizing ability, the spirit which animates the partnership, is the contribution of the business man—by his brains, energy, force of character, and toil he has created here in the United States a commercial system of enterprise and a degree of business prosperity unparalleled in history.

If we give the credit of this achievement to the business man, he should also bear the responsibility of the evils which have been engendered (applause). The gravest evils which have developed out of our commercial prosperity are the uncontrolled power of great wealth, the growth of monopolies, and their sinister influence on our political institutions. (Applause)

Industrial efficiency may justify the union of many smaller corporations into one big one, but if it leads to industrial despotism this efficiency is obtained at the sacrifice of industrial freedom (applause). No one nowadays, on the ground of efficiency, believes in a *political* despotism; surely it is equally difficult to believe that any degree of efficiency could justify *industrial* despotism. (Applause)

As early as 1888 so conservative a man as Grover Cleveland expressed himself as follows: "Communism of combined wealth and capital, the outgrowth of overweening cupidity and selfishness, which assiduously undermines the justice and integrity of free institutions is not less dangerous than the communism of oppressed poverty and toil which, exasperated by injustice and discontent, attacks with wild disorder the citadel of misrule." So far as communism of capital is concerned, did not Cleveland's graphic statement adumbrate the conditions as they exist today? Since that time how tremendous has been the growth in the combinations of capital and industry.

But of more importance than the size of the corporations and the combinations of capital is the activity in our political arena of the agents and members of these corporations (applause); they are not there to advocate measures for the welfare of the community, but to obtain for themselves special privileges, to gain some advantage in disregard of the public welfare and merely for private gain. These conditions are precipitating an economic and political crisis, in which the issues are not to be between the two great political parties, but between ranks which are being formed to give battle on these new issues regardless of party lines. (Applause)

To my mind great encouragement lies in the fact that there is rapidly developing a segregation in the ranks of business men. Already many of them, freed from a false sense of class loyalty, or a fear of injury to business, are unwilling to assist by their public support or private esteem that man, however successful or powerful he may be, who by himself or by his agents practices methods which are unfair and opposed to the common good (applause). They no longer respect the citizen who in any way indicates a reluctance to take part in the crusade against bribery and graft, or the one who, by silence, hopes to conceal his public attitude when public sentiment seeks to fasten responsibility where responsibility belongs (applause). This sort of man must come out into the open and declare himself—he must be either with us or against us. (Applause and cries of "Good!")

Even though the advocacy of the control of industrial combinations and the enactment of measures for their regulation temporarily affect business interests, they should not for this reason excite the opposition of the

mercantile world. Those business men who have become convinced of the wisdom of regulation should be willing to follow the example of the intelligent patient who goes through with a necessary operation that in the end he may obtain permanent health and strength. (Applause)

During the last five years there is apparent among business men a larger recognition of their obligations to the community, and there is noticeable among the directors of many of our corporations a stricter sense of trusteeship. An anti-toxin to corruption has entered the very veins of the business world (applause). The phagocytes of health are overcoming the macrophags of decay. This is not a sudden revival, a temporary wave of reform, but a gradual evolution of the moral sense, a permanent advance in the idea of social justice (applause). This moral awakening may show itself politically in an effort toward municipal reform, in legislative and municipal voters' leagues, in a determined resistance to monopoly, or for a larger control and a larger share in the profits of public franchise corporations. But in whatever form it seeks its expression, it is the manifestation of an actively constructive principle which will soon become so effective that the merchant and the man of affairs will overlook the near and personal view which appears on the stock ticker and take the larger view, the view that ultimately provides for the greatest good of the greatest number (applause). This awakened sense of social justice is the new and deeper significance of the Conservation movement. (Applause)

Two years ago the Conference of Governors adopted a declaration of principles which the President said should hang on the wall of every school-house for the education of every citizen who is to become a voter in the next generation.

Since then Conservation has become the watchword of the hour. The widespread use of the word has given to it a meaning undreamed of in the beginning. In the form of an intelligent energy it has applied itself to all the concerns of life from the conservation of the soil and the forest to the conservation of birds, of child life and of health. It enters into our daily life, awakens into an active moral force a renaissance of the old-fashioned virtues—prudence, thrift, and foresight—and gives to them a larger and a National meaning.

Conservation is the intimate and individual message to our day and generation. It marks the advent of a new patriotism wherein love of humanity becomes an integral part of love of country, and where the conservation of our "rocks and rills," our "woods and templed hills," is not a more sacred trust than the conservation of those ideals and principles through which we hope to attain our ultimate National purpose—a Government of enlightened people, enjoying equal opportunities, sharing equal burdens, and rejoicing in the freedom of an Industrial and Political Democracy. (Great applause)

[In the course of the foregoing, President Baker invited Professor Condra to the Chair.]

Professor CONDRA—Ladies and Gentlemen: President Baker desires me to say that his voice has failed. He also authorizes me to announce that the Call of States will be made this afternoon.

I am pleased now to introduce a speaker opposed to the leading objects of this Congress. I ask you to hear kindly any criticism that he may offer. His subject is "The Relation of Capital to the Development of Resources." Mr Frank H. Short, of California.

Mr SHORT—Mr Chairman, Ladies and Gentlemen: I am permitted to speak today for the first time for real money, and apparently in behalf of those who are sometimes denominated "malefactors of great wealth." I observe that one of the Saint Paul papers in announcing this address has referred to me as a lawyer and capitalist. The latter I modestly deny. It is unprofessional for a lawyer to become rich. Good lawyers are scarce and valuable, and judging by the speeches I have heard in this Congress rich men are very common and a great public nuisance. Therefore I hold that it would be a great misfortune for a good lawyer, such as I admit that I am (laughter), to be spoiled by making out of him an ordinary capitalist.

This audience, in listening to my address, will no doubt have in mind the numerous warnings which have been given to them in advance to forestall the evil influences of my humble remarks. I hope none of you will ever have to sustain the painful ordeal of appearing before an audience decorated with hoofs and horns by angels of light wearing crowns and playing harps, who have so kindly bestowed upon me the habiliments of the Evil One. Perhaps, since I have been so excessively featured, I had as well admit the whole horrible truth. First, and perhaps worst of all, I am a Missourian, having committed the indiscretion of being born in the "Show me" State—but not in Kansas. All of my youth was spent in the Middle West in the occupation of a rough rider; and I still enjoy a fight or a footrace as much as though I were a real colonel. Further confessing, I have lived for many years in California and am a lawyer by profession, and have committed the offense of allowing myself to be retained and am now employed by a considerable number of large water companies and electric power companies and other corporations, diligently endeavoring to commit the crime of investing capital under the laws of the western States in the development of the industries and resources of those States.

The difference between a real colonel and a second lieutenant is illustrated by the fact that this admission permits of my being heard under his authority, although industrious efforts by the lieutenant referred to have been devoted to the contrary purpose. I am, however, speaking under the general permission of this Congress, and under no other frank than my unrevoked license as a real though obscure American citizen.

The rights and interests of all American citizens and business institutions under the laws of our country are the same (applause). As a man accumulates property, and his interests and substantial connection with the country and its resources increase, he thereupon becomes just that much more interested in the honesty and integrity of the Government under which he lives, in the perfectly equal and just operation of the law, and above all in the supremacy of the law and similarly in the inauguration, continuation, and perpetuation of good policies.

No doubt we self-governing Americans have all erred, both the poor man and the capitalist; and perhaps it would not be unfair to say that we all ought in humility to bear our equal share of the odium connected with

whatever failures and offenses have been committed during our history, and I am not here to shift any of the burden from one class upon another. Neither am I here to answer denunciations with denunciations. I am handicapped in such debate, for the reason that I acquired my education in the old-fashioned school that was taught to believe that an honest man was one who said little of his own honesty and less of the supposed dishonesty of others.

A convention of this character can be carried on with but little capital, and may travel a good ways on sheer wind; but with all respect to free speech, it takes money to carry on Government and conduct business, and if capital is as timid as it is supposed to be, and if some of our political friends were as dangerous as they sound, all of the money would have been scared out of America before I commenced these remarks on capital. Allow me, however, respectfully to suggest that we of this country are engaged in many vast enterprises; we are responsible to many men and their families for the opportunity to work and to earn a living. We are committed to the completion of many National enterprises of great magnitude. Our crops are none too large, our reserve capital is small and is growing smaller. The general industrial and financial conditions of the country from the point of view of thoughtful men who understand the situation, are not as satisfactory as I wish they were, and those who are gaining fame and ascending to office by wild denunciations of wealth are willing to assume hazards that I do not envy. (Applause)

Honest capital is more secure when governments are made honest and special privileges are denied, when graft is prevented and crimes are punished: and there is never any danger in real reform, but infinite harm can be done by attractive orators of maximum lung power and minimum brains (applause). Honesty is the best policy in large business and in small business, and the most that capital ought to expect or demand, and the most that will be profitable to it in the long run, is to seek and if it can obtain the passage and the enforcement of equal and just laws, the continuation of justice, and the right honestly to accumulate, hold and enjoy property (applause). The relations of capital to Conservation are identical with its relations to all other business. As Conservation tends to increase and continue the natural resources of the country, the fertility of the soil, the perpetuation of the forests, the flow of streams, and all of those conditions

that insure the substantial welfare of the country, the capitalist has an equal interest with all other citizens in Conservation, and the added interest that he can share in a greater degree in the resulting and continuing prosperity than his less fortunate neighbor.

Some excellent things have been done and said in this convention. If "conversational conservation" would cure the evils under which we live we would have no need of doctors for a long time. As against "conversational conservation" I wish now to say a few words about constitutional conservation. From now on I may wander a little from the rich subject that has been assigned to me, but I have been much interested in the suggestion that that branch of the Government that can accomplish the most good for the people should take charge of their business and affairs connected with Government. Unless, however, we have some authoritative source other than the nebulous question of the general welfare to determine where this authority lies, I am apprehensive that most of the resources of Government would be dissipated in fighting over the question of authority.

What I now hold to be true for all time—and you will all agree with me some day—is that that branch of the Government that under our constitutional system is designated as the one having the authority is the only branch of the Government that can benefit capital, conserve or advance the rights of the people, or do justice in any way whatever. Conservation as it was understood in its inception in this country, the preservation of our soils, our forests, and our resources presented a subject of little difficulty, and in connection with which we were all practically in accord and where apparently there would have been no occasion for any serious disagreement. No more new or difficult questions of Government are legitimately involved in Conservation and forestry than are involved in cultivation and farming.

If the device of using the public lands to graft Government onto Conservation had not been invented by some civic genius, we would have had 90 percent of conservation to 10 percent of controversy. But when the landlord seeks to be the governor, especially in America, we get 90 percent plus of controversy and 10 percent minus of conservation. Landlord law and governmental conservation was devised, we are told, to control wealth for the benefit of the plain, small man. Inquire in the vicinity of any forest

reserve, and you will find that there are more plain, small people than there used to be, and they are getting plainer and smaller every day; so apparently the good work will never end.

As briefly as I may, and seriously as I can, I will state the situation that confronts the people of the West, the poor man and the capitalist alike, in connection with the forest reserve. Forest reserves were authorized by Congress for the purpose of protecting forests and conserving the source of supply of streams. Probably one-third of the 200,000,000 acres that have been set apart in forest reserves in the western one-third of the United States are reasonably necessary and suited to these purposes. As to the other two-thirds, they were largely included—and in some instances this is frankly admitted—for the purpose of authority for Government control, to include pasture lands, power-sites, irrigation projects, and the like. If forest reserves had been created to meet the actual necessity which brought them into existence, and if they had been administered with due deference to the rights of the State within which they are situated, to improve and develop its resources without restraint, to construct or authorize to be constructed roads and highways, railroads, telephone and telegraph lines, canals and ditches for the beneficial use of water, and the functions of local self-government had not been assumed to the Federal authorities and denied to the local authorities, I could conceive of no reason why the forestry policy could not have been carried out with great credit and some profit to the Federal Government and greatly to the advantage of the district in which the forests are situated. The pity of it all is that this has not been done. We are told that the sentiment in opposition to transferring from the States to the Federal Government important functions of regulation and control is not unanimous. This is true as to districts not directly affected by the forest reserves; but as to the people within and in the vicinity of the forest reserves, in other words, as to those who have come directly or indirectly in contact with bureaucratic government, the sentiment is about as unanimous as ever existed in America.

That the Forester and those under him honestly desire to benefit the people, especially "the poor, small man," we need not deny; that the actual results have been beneficial, however, we wholly deny. The imperial dominion withdrawn includes territory as large as 20 or 30 average-size eastern States, amounting frequently to one-fifth or one-fourth, and sometimes even

exceeding the latter fraction of the territory within a State, and practically taking over and paralyzing local self-government in certain entire districts of a State. These lands are, and if the policy continues will remain forever, withdrawn from State taxation and revenue, and instead will become a source of expense and burden. First, considering the prime purpose to preserve and protect the forest, what has been the result? The Forester and those under him have my profound sympathy in connection with the recent awful destructive forest fires and the heroic way in which the disaster was met, even though it was not overcome.

For many years experienced and practical men in the West have protested against the policies pursued. Previous to the establishment of the forest reserves the land was pastured by sheep and cattle, admittedly in some instances over-pastured. Frequent fires ran through the country, but in most instances as the country had been closely pastured off and fires had usually recently occurred, these fires did only incidental harm, and in a general way the great forests of the West in many districts—although the result of mere natural processes—as valuable and magnificent as there are in the world, were retained in their primitive and perfect condition. For a good many years now exactly the reverse of this primitive condition has prevailed. Sheep have been excluded and cattle have been limited; falling and decaying timber, the growth of vegetation from year to year, and the accumulation of underbrush and debris have continued; and we have gone on conserving our forests in such a way that we have been accumulating fuel and the elements of destruction, piling up wrath against the day of wrath, until the fires, in spite of precautions, have started, and the destruction that has resulted is inevitable. What is needed now in this particular is a surgeon who has the nerve to amputate the conditions that create fire, and until this is done the danger will go on increasing from year to year and more destruction than benefits will inevitably result. To those who suggest that a sufficient patrol will prevent fires, I respond that they ought to try the experiment of filling a building with powder, putting an ample guard around it, and touching a match to it.

These great reserves have been practically closed to settlement and homesteading. The price of pasturage has been increased, the number of cattle and sheep pastured has been diminished, and the price of meat correspondingly advanced. The price of stumping has been doubled and

trebled, no small mills have been or can be successfully started, and the price of lumber to consumers has been increased. The policy has limited the construction of canals and other appliances for irrigation, and still more effectually limited the construction of like appliances for the diversion of water for the development of electric power. If this water could be diverted for irrigation and electric power under State laws without other restraint, the quantity available in the majority of the western States is so great that the supply would exceed the demand, the price would be lower, the consumption greater, and in every way the people would be benefited. The country would be settled, the people would be more prosperous, the supply of water and electricity would be more abundant and cheaper, and all of the people and all of the industries would be correspondingly more prosperous.

It is gratifying that the line of cleavage and difference between the advocates of bureaucratic control over local industries and the advocates of local self-government have been better defined. Upon the all-important question of the law applicable to this subject, I submit that there is little ground for honest difference. The Supreme Court of the United States has decided practically every phase of the matter over and over again, and the law is settled to the following effect: That the United States Government owns the public lands in each of the States as private proprietor and not as sovereign; that it, the Federal Government, if it seeks to assert any authority in any State, must find its warrant in the Constitution and not in the ownership of the public lands; that the authority of the United States Government to adopt needful rules and regulations in connection with public lands is an authority to protect its proprietary interest and not exercise governmental functions within any State; that every State is upon an equal footing with all of the other States, and for the protection of its own people, its own industries, and the regulation of its own monopolies, each State has all of the powers of any other Government; that the United States Government exercises the same power, and each of the States exercises the same power, "no more and no less," regardless of the existence or non-existence of public land in any State.

The whole pretense made by some that the United States Government can exercise exceptional governmental authority in a State having public lands is a pretense and a pretense only. Under the decision of the Supreme Court of the United States, such a claim has no shadow of foundation, and its

assertion is merely injurious, detrimental to capital, destructive to industry, and can never serve any useful purpose of regulation or otherwise. These principles being fully decided and clearly in mind, it is hard to understand why the issue is raised, and how it is hoped that the policy can be imposed upon the western States or any other States under the Constitution. It has been said with derision that the corporations are appealing to the Constitution. I would to God that neither the corporations nor the American people might ever appeal to anything worse. However much evil may have been taught, no honest man need be apprehensive of injustice if his rights and the rights of his fellow citizens are always measured by a just construction of the Constitution of the United States. (Applause)

We are told, and I think some of our adversaries honestly believe the tale, that all of the remaining resources of the country belong to all of the people. That "all of the resources belong to all of the people" is a slogan that sounds good. Its chief defect is that it is not true, and the next objection is that to assert it now, after pursuing an exactly contrary policy as to four-fifths of the Nation's resources, would be an intolerable injustice. The United States Supreme Court decided a long time ago that the United States Government received and held the public lands as trustee for the benefit of the people and the States within which they were situated, to the end that they might be disposed of to actual settlers at nominal prices in order that the country might be settled, cultivated, populated, and developed; the lands come under the taxing power, and all of the unrestrained functions of State government. These decisions have been reaffirmed, and it has been held that the United States' title and trusteeship as to the public lands is identical in all the States. Therefore it is not true as a matter of understanding or of law that the United States is the unrestrained proprietor of the public lands, but it holds in them a trust; and I submit that no justice can be done or good come from the violation or attempted violation of a trust. Considering the equity of the situation, if the United States is now the owner of the remaining lands and resources for all of the people, it has been such from the beginning of the Government; and having disposed of these resources to the beneficiaries entitled thereto, it is now seriously proposed to seize upon the remaining fraction and hold that fraction for the benefit of all the people, as much as for the benefit of the people and the sections of the

country that have received their proportion as for those who have not received theirs.

The situation might be illustrated by this simple statement: Uncle Sam may be assumed to be the father of four sons; we will name them East, North, South, and West. Uncle Samuel being liberal to a fault and mindful of a trust, has transferred to his three elder sons, East, North, and South, all of their share in his estate. But these elder sons, especially after their industrious younger brother has begun to show the real value of his portion of their father's estate, begin to look with covetous eyes upon the younger brother's inheritance. Finally a deep sense of justice begins to pervade the minds of East, North, and South, and they appear before Uncle Samuel and say, "Father, you have been very profligate in the management of your great estate. You have turned over to us and to our children without needful restriction the whole of the proportion that we can rightfully claim. In the doing of this you have shown great incompetency and have practiced many faults, and behold, you have sinned against Heaven and in the sight of men. We can see no way of atoning for this awful offense except that you shall take and hold that portion of the estate that should descend to our younger brother for the benefit of all of your children. And as a further atonement, having shown in the distribution of your estate to us that you are dishonest and incompetent in the last degree, in consideration thereof we will nominate and appoint you the landlord and guardian, without bonds and forever, of that portion of the estate that, except for this atonement, would have belonged to our younger brother; requiring you, however, to see to it with scrupulous care that we, your elder sons, shall receive from the rents, leases, and profits of this estate our equal shares with our beloved younger brother." Painful as it may seem, these elder brothers seem well nigh unanimous as to this scheme of atonement, and Uncle Samuel seems weak and subject to the influence of the majority. History, however, will record that the Constitution broke the will and the elder brothers were charged with the costs and counsel fees. (Laughter)

If anyone present feels justified in challenging the accuracy or historical correctness of the foregoing statement or its logical application to the situation, he will now please rise and state his case or hereafter forever hold his peace.

The overshadowing political reason why the United States Government must invade the public land States and assert powers of government that it cannot assert in any other States we are told is to control monopolies. As a controller of monopolies not constitutionally subject to be controlled by the Federal Government, and under claims of title to the public lands, the United States Government and its respective bureau chiefs would have St. George, the dragon destroyer, outclassed at the ratio of sixteen to one. It may do as a political issue for a long time, but if the people of the western States had no powers of government or sources of control within themselves, or except through the Federal Government, the public lands, and the heads of bureaus, these people would have little to expect or hope for.

It is gratifying, however, to observe that instead of being helpless and impotent, the western States not only have all of the powers that are vested in any other Government for the protection of their people from monopoly and wrong, but an understanding of their constitutions and laws clearly demonstrates that they are showing themselves far more alert, advanced, and capable in these functions of government than either the Federal Government or the older States in the East. It ought not to be necessary to say to an American audience that it is elementary that the people of a locality can give themselves more honest, efficient, and better government than can be given to them by any remote authority. The reason for this is so simple that the only excuse for attempting to deny it is the ignorance and incapacity of the people concerned to carry on or carry out self-government. The people of the western States alone will suffer if they do not efficiently and intelligently exercise their undoubted authority to supply themselves with good self-government, and efficiently control and direct their own industries and their own monopolies.

About the only argument that is made in favor of Federal control and against local self-government in the West is that the corporations appear to prefer the former. The question is not what the corporations prefer but what the Constitution requires; and, in the next place, the corporations do not deny the authority of the States because they are advised that they cannot and therefore should not attempt to do so, and because they are advised that they must in any event submit to local self-government and that Federal control would be an additional and not a lawful but a wholly unauthorized

usurpation of authority. The American people, of all people in the world, have earned the reputation of being the most obedient to law and the least submissive to usurpation of any people in the world. If some of our wealthy men and some of corporations have offended against honesty and attempted to circumvent, misapply, and misuse the law, these are instances to be regretted, condemned, and punished. The practice should be abandoned, and if not abandoned rigorously prevented; having it, however, religiously in mind that ultimate justice can be done and the law vindicated only by adhering to due process of law.

We are told that Switzerland as a Nation regulates and manages its own power business. Since, however, Switzerland has no more authority or powers of government than California, Colorado, or New York, and since it is probably one-tenth the size of these States and its cantons are about the size of an ordinary western school district, this would not appear to indicate any reason why the western States of the Union could not successfully carry out the same function of government.

Our former President has said to us that he would be as swift to prevent injustice and unwarranted uprising against property as anyone. This I do not doubt, and I am prepared to agree that probably no one living could perform the task more cheerfully or effectively; but in this connection it might not be improper to reflect that the people have been taught, and rightly so, that this is "a government of law and not of men," and we rely upon the equal and continued protection of the law for the protection of our persons and our property, not upon the life or disposition of any man.

We have already referred to the assertion that the remaining resources of the Federal Government belong to all of the people and are to be administered and revenues obtained for their full benefit. We are not, however, deluded with the thought that we are to begin to draw individual dividends. The revenues thus obtained are to go into the Federal treasury (and allow me parenthetically to suggest that the pay-roll will not be far behind the earnings), but if through some oversight a balance should be found in favor of all of the people it will go into the Federal treasury to reduce taxation to the common benefit. Allow me to suggest, and ask all thoughtful people to well consider, that if sufficient revenues were collected and paid into the Federal treasury to prove of great benefit to a hundred millions of people,

the collection and payment of these same revenues will of necessity amount to some slight imposition and burden upon the ten millions of people when they are paid out of their resources and revenues.

While we are considering monopolies it might not be inappropriate to consider that they are of two classes: private monopolies and government monopolies. One of the highest functions of government is to control and regulate private monopolies. It is not always easy, but the undoubted power exists and if properly applied is effective. History records that four-fifths of the exactions and oppressions and human sufferings that have existed in the world have come about when the conduct of business and the sources of supply were confined and vested in the government and constituted a government monopoly. Government monopolies are invariably created for the alleged benefit of the people, and throughout all history have almost invariably operated to the oppression and detriment of the people and ultimately to deprive them of their liberties. In the face of these undeniable records of history, the people of the western States are invited to surrender their control over their industries and their own private monopolies and have substituted therefor a Federal Government monopoly over which they could have no possible control. The western States are asked not only to surrender this control, but along with it to surrender the powers of taxation and revenue over all these great resources. My friends, some of you may congratulate yourselves that these so-called policies are popular, and no doubt to a certain extent they are; we think, however, because they are misunderstood. There need be no misunderstanding between us. You are welcome to your assumption of victory, and to the assumption of defeat for those who adhere to the right of local self-government.

We are correctly told that the ancient doctrine of State rights ended at Appomattox. The doctrine was there ended that the Federal Government did not have all of the power necessary to protect and continue the Nation for the common defense and the general welfare. The undeniable doctrine and right of the American people within the several States to continue an unrestrained local self-government was at that time neither destroyed nor impaired. The right and doctrine of local self-government will endure and continue until, if ever, some common disaster shall terminate and end the National existence as well as the existence of the several States. No question is ever settled until it is settled right. Frankly, today may be yours

but tomorrow is ours. The Constitution of this country is greater and more enduring than any man. Let there be no misunderstanding between us. You should not, but if you would you cannot, deprive the people of this country in any number of States or in any one State of the equal guaranteed constitutional right of local self-government.

In recent months, so numerous have been the complaints and utterances against the courts that it would almost appear that there was a common design to discredit the courts with the American people. For even a longer period there have been recurring attacks upon and denials of the capability and capacity of the representative branch of our government. Even within its obvious jurisdiction the Legislative department has not only been excessively criticized but its very powers denied. The Executive of the country and each of the States, Congress, and each Legislature of each of the States, the Supreme Court and all of the subordinate courts, derive all of their authority from the American people through the Constitution of the United States. He who acts without and in spite of the Constitution acts without authority from the people. Constitutions are adopted to safeguard the rights of all men and to protect minorities from majorities. The question is not, where the Constitution declares the measure of right, what the majority wants, but the question is, what does the Constitution declare; and that is the beginning and the end of the law. The Government under which we have lived is the best vindicated Government in the history of the world. If a democratic people, as we have been told, have destroyed more since the adoption of the Constitution than has been wasted and destroyed in Europe in all of its history, we may admit this and agree that it is wise always to prevent waste; but we can with equal truth assert that if our free people under our free institutions have destroyed more than the people of Europe in their entire history, our people by scientific research and invention have added more to the potential and productive power of the earth and the elements for the benefit and subsistence of mankind than has been added by the people of Europe, Asia, and Africa during the entire recorded history of the world—all since the adoption of the Constitution of the United States.

Whether it be popular or unpopular, it is true that the tendency to belittle the legislative power, to disparage judicial power, and to correspondingly exalt the executive power, is the same evil tendency that has destroyed every free government that has ever existed. It is the same spirit that overthrew the

mild judicial government of Samuel and made Saul of Tarsus king over Israel. It is the same spirit that subverted the free cities and provinces of Greece, and made Alexander, the Macedonian, the sole arbiter of the destinies not only of the people of Greece but of the whole eastern world. It is the same spirit that subverted the Senate and the tribunals of Rome, and made Julius Caesar and his successors the emperors and rulers of the entire known world for succeeding centuries. We may agree that no such events will recur in modern history. But it is the same spirit that brings about such a condition in Mexico that nobody knows or cares when Congress meets or adjourns, because they never pass or suggest the passage of any laws that have not already been approved by the President. They must have a Supreme Court in Mexico, because their Constitution is very similar to our own. For the same reason we assume that they have States, although nobody ever hears of them. Neither do we hear of any one criticizing the decisions of the Supreme Court of that country; nobody has ever suggested that within the last quarter of a century that court has ever decided anything displeasing to the President.

The United States of America today is the world's sole and single exception where the people under a constitution through a long period of years have been guaranteed and have received the equal protection of the law. No guards have been required to stand at our city gates, no bayonets have defended our towns; we have all lived and prospered under the equal protection of equal laws. (Applause)

These institutions are human, they are imperfect and under them errors have been committed, but undeniably under this Government the people have received a larger measure of liberty together with a better distribution of the benefits of industry than was ever received or enjoyed hitherto by any people in the world. We favor that new efficiency that is neither National nor State, that under an equal respect for the Nation and for the State and for each branch of the Government strives for a higher condition of civic virtue, better enforcement and greater respect for the law in all of its branches. I hope and pray that none of us may ever be required to look beyond the years when the Constitution and the law in letter and in spirit are no longer supreme in this country and when we shall have reverted to "that good old simple plan, that each may take whate'er he may and keep whate'er he can." (Applause)

Professor CONDRA—Ladies and Gentlemen: A question has been sent to the Chair: "Will the Congress close this evening?" We do not know; probably the Congress itself will decide. There are several other features in the program, and there will be a report by the Committee on Resolutions. It may be that the Congress can finish all of its work today if you choose to re-convene.

You all know the next speaker, Honorable John Barrett, Director-General of the Pan-American Union. (Applause)

Mr BARRETT—Mr Chairman, Ladies and Gentleman: When the captivating senior Senator from Indiana fascinated us yesterday, and after holding us enthralled by his eloquence ending with that magnificent climax in eulogy of Gifford Pinchot, he left this room remarking to the reporters that he couldn't stay longer because he must go down and look after his State and 3,000,000 people. Now, if some of the rest of us relied on the measure of States and population as a reason for not being here, we would not come at all. For example, I might have said, when invited to take part in the work of this Congress, that I couldn't possibly come because I might neglect that which was best for 21 independent Republics and 160,000,000 people. What I want to say is this—that I would like to multiply twenty times over all the enthusiasm with which Senator Beveridge fired us yesterday, and extend it to many millions of people, in order that the wave started here by him and other speakers might sweep over the whole western hemisphere and remove the slightest question that all these Republics are awake to the practical value of Conservation.

Possibly some of you do not know very much more about the practical work of the Pan-American Union than I knew about the country to which I was first appointed minister some sixteen or seventeen years ago—when I knew as little about foreign affairs as some of us did a few years ago about Conservation. One day the President of the United States, with two United States Senators from North Carolina standing near by—if one of them had

been from North Carolina and the other from South Carolina there wouldn't have been any doubt as to what the conversation was to be (laughter), but as both came from the same State I was in the dark—looked at me and said, "Mr Barrett, I am trying to find some young man who is not afraid of hard work and wants to make a reputation for himself to go off to a distant country, in another part of the world, to settle a case involving several millions of dollars and our treaty rights in the Orient; I am looking for a minister to Siam." Well, I thought that he wanted me to recommend somebody, and was trying to think of somebody in my State that I would like to get rid of and never see again, when he added, "I am thinking of appointing you; what do you know about Siam?" To save my life I couldn't even remember where it was, and I was conscious of the terrible impression I must be making upon the Executive, when with a twinkle in his eye he intimated "I have him this time." Then, a child-memory coming back, I braced myself and said, "Why, Mr President, *I know all about Siam.*" "You do? What do you know about that country?" "Why, Mr President, Siam is the country that produced the Siamese Twins." Whereupon he shook my hand and said he was delighted to get hold of a man of such abundant information. (Laughter)

Now, before proceeding further, let me, as one of the officers of this Congress—although one who has had very little to do with its hard work—join with you in expressing profound appreciation of the splendid hospitality that has been shown the Delegates and all others who have come here to the city of Saint Paul in the State of Minnesota (applause). Moreover, I believe it is only fair and fitting that we should also express our gratitude for the hard work and the devotion to this Congress shown by President Baker and Secretary Shipp and Professor Condra and Chairman White and other men belonging to the Executive Committee. (Applause)

I have been asked, as a resident of the District of Columbia, whether, if this Congress shall go to the East next year, it might not go to the city of Washington, and there arouse the interest and the sympathy of the East. The West is awake; and if it be necessary to secure the cooperation of the eastern sections, and if the Executive Committee hesitates as to where it may go, I can assure them that by the city of Washington, the Capital of the Nation, will be given a welcome akin to that which has been given by the city of Saint Paul.

Ladies and Gentlemen, one feature of this Congress has made a profound impression upon me, of which perhaps too little mention has been made: the cooperation and interest of the women. That was a splendid speech made the other day by Mabel Boardman; other women have spoken well, and others will. I assure you that there is no better omen of the success of this movement than this cooperation by women (applause). And I want to say right here, that whenever I am able to pay a tribute to the courage and the quality of women, I like to do it. It so happened that I was your first minister to Panama, in the days which tried men's souls—where I, as minister, frequently had to preside where three or four splendid boys, graduates from our colleges and high schools, were laid under the wet clay in one grave, all victims of yellow fever. When I went down there with General Davis, then Governor of the Canal Zone, there were some sixteen girls, nurses, picked from all over this country—I think one or two came from Saint Paul or Minneapolis—who had never seen yellow fever before, had never experienced the pestilential conditions faced in Panama when we were "blazing the way" for the present sanitary condition. Well, they came and took up their work; and in a short time the yellow fever spread until men were dying every day in increasing numbers, and both the boys and men came to us and begged that they might return to the United States—in the parlance of the canal work, they had "cold feet," and it was with the greatest difficulty that we were able to hold them there to perform the great task of making the zone sanitary as well as digging the canal that the oceans might be united; but when the yellow fever was conquered, General Davis and I discovered that during all that time of peril and death and threatened desertion, not *one* of those sixteen girls faltered or asked permission to leave her station of duty. (Great applause)

Ladies and Gentlemen, it is a pleasure today to be followed by a representative of the British government who is a credit to his government and to the great man whom he represents here, the Right Honorable James Bryce, British Ambassador (applause). There is nothing more splendid than the thought of the cooperation of this mighty country north of us, Canada, with her 4,000,000 square miles and her ambitious men and women with problems akin to ours; and it is both appropriate and flattering that the British Empire should have responded to the invitation and sent here a

special representative of their Embassy (applause). We are to be congratulated on his attendance.

It seems to me that during the past three or four days I have heard the word "insurgent" used. Am I correct, Mr President?

President BAKER—"Progressive."

Mr BARRETT—I think there have been some references to progressiveness and insurgency. Now, as the head of an international bureau whose constituency is composed of twenty Latin-American Republics, I want to tell you that you don't know anything here about real insurgency (applause). Why, we have men in Central America and South America who could make Murdock and Madison look like picayune persons if they came in competition with them in the matter of insurgency. We have Republics that can give Kansas and Wisconsin and Nebraska and Minnesota cards and spades and all the trumps in the pack, and then beat them out in insurgency. But I want to say this, that in all my experience in those countries as minister and my studies of their history, there has never been an insurgency or revolution, from Mexico south to Argentina which has succeeded without at the same time moving the country forward for its benefit (applause). I do not say this in any political spirit, because I am not in politics; being an international officer, I am neither republican nor democrat, but a citizen of America; yet I do say this, that the spirit of onward movement among men shown thus from time to time is a splendid sign of the progressive type which characterizes the American people, whether they be American of North America or American of South America. (Applause)

Ladies and Gentlemen, it would be a splendid thing today if the voice that has been sounded here on Conservation could be heard by every Pan-American—through that All America comprehending not only our own wonderful land but twenty other Nations, covering an area of 15,000,000 square miles, having a population of 175,000,000 people, and conducting a foreign commerce valued at the magnificent total of \$2,000,000,000 annually. Only a few years ago Latin-America seemed almost like an unknown land; but today these countries from Mexico and Cuba south to Argentina and Chile are making more progress commercially and materially than almost any other section of the world. We hear much of the Orient, of

Japan and of China, whose inhabitants are alien people, alien in philosophy, alien in religion, raising the greatest racial question before the world; but here at the south of us are twenty sister Nations whose peoples have the same ambitions as yours, the same religion, the same philosophy, the same hopes—and yet you and I have been sitting in cozy corners flirting with Japan and China, and neglecting our own sisters in our own family (applause). Last year Argentina—a country half as large as our own splendid land, in a temperate zone, with nearly 7,000,000 splendid white people, having sons whom you would allow your daughters to marry and daughters you would allow your sons to marry—conducted a greater foreign trade than the 50,000,000 Japanese or the 300,000,000 Chinese (applause); and yet we are neglecting them. Now these countries gained independence at the hands of leaders who studied the life of George Washington (applause), and they have continued their existence under the example of such men as Abraham Lincoln. Whether you go upon the high Andes or in the valley of the great Amazon, the names of Washington and Lincoln are known almost as well as those of their own great heroes who helped them to win independence.

Ladies and Gentlemen, it is time that through the cooperation of all these countries we should accomplish protection for them and for ourselves; and we should have in the near future a great Pan-American Conference of Conservation, when all the countries from Canada south will send their representatives to join us in working together to safeguard their prosperity, to safeguard our own, to promote our mutual and several interests until this whole hemisphere from Alaska and the Arctic on the north to Chile and the Straits of Magellan on the south shall present a united force for the benefit not only of ourselves but of those who are to come after us. Is there anything more magnificent than this thought that the twenty-one independent Republics and an independent Nation like Canada should join hands in such a purpose? The details I shall not discuss, but I want it to be a thought that shall sink into your minds.

Now, I wish that I could take all the "hot air" that has arisen in this great auditorium and make a mighty balloon to take you for a trip over our sister countries (applause). I would like to show you Brazil, into which you could place all of the United States and still have room left over for the German empire; I would like to take you up the Amazon, out of which flows five

times the volume of the Mississippi; I would like to take you to Buenos Aires, the capital of Argentina, which has a population of 1,200,000 and is growing faster than any city in the United States with the exception of New York and Chicago—I would like to show you its magnificent boulevards, its splendid public buildings, its schools, its cathedrals and churches; I would like to take you across the Andes over that wonderful tunnel just completed and show you Chile, which if placed at the southern end of California would reach up into the heart of Alaska, in the very infancy of a splendid development; I would like to take you into Bolivia, into which you could put Texas three times and still have room left over; into Peru, which would cover the whole Atlantic Coast from Maine to Georgia; into Colombia, where you could place all of Germany and France; into Mexico, that would cover the whole southwestern section of this country; I would like to take you over all these countries and show you how they are moving forward, prove to you the remarkable fact that during the last fifteen years that part of the world has gone ahead with progress almost equal to ours. Now, if we in this country are going to meet the great problems of manufacturing and the employment of labor and capital in the future, we must aid these countries to conserve their resources to supply our manufacturing plants with raw material. Hundreds of millions of dollars today are keeping occupied by laboring men in this country factories that would have to be closed tomorrow if these countries were unable to supply us with their raw materials—think of that as we remember where we were only twenty-five years ago; and if some God-given influence can empower them to see our mistakes we will find, twenty-five years from now, Brazil and Argentina and Mexico and Canada providing us with those elements which shall make this country forever the greatest power in the world for civilization and for commerce. (Applause)

As I stand here before an audience of the West an inspiration comes for the work we have in Washington that only those can feel whose residence is not entirely in the West. Though born and brought up in New England and later taking my residence on the Pacific Coast, I have been much out of the country representing you abroad; and I rejoice in the ozone of patriotism that I am able to absorb in a State like Minnesota. Time and time again, after trips around the world I have arrived in New York or in Washington hardly feeling that I was in the United States of America; but when I have

crossed the Alleghenies into the Mississippi valley, into sections like this, then I have felt the pulsing of red blood, that impulse and influence which is making our country great; and I am proud today to be able to go back to Washington feeling more capable than ever before for my humble task because of the contact with representative men of the West. (Applause)

There are two personal references that I make before I sit down: When on Tuesday I sat on the platform and saw the personality of the foremost private citizen of the world exerting its influence, the prime thought that came into my mind was, not that he was speaking for the great cause of Conservation, not that he was appealing to the moral sense of our people, but that there stood a splendid, a perfect example of what the young men of this country can do (applause). Is there anything finer than to see a man of his physique, with the glow of health upon his face, the father of a family of which he can be proud, a man with a clear moral life and courageous career, one whose voice has been heard all over the world with respect—is there anything finer than that we should raise up in this country that class of men? And I tell you it would be disgraceful to our country with its 90,000,000 people if we could not produce a man of that kind. It is the personal influence of Theodore Roosevelt, all over this country, not only among our young men, but among our young women, leading to world uplift and to sterling character, that we must have in order to fight the battles that are before us. (Great applause)

And there is this suggestion about his chief lieutenant who has perhaps been the father of this movement: I have known Gifford Pinchot personally, as a dear friend, for many years. It makes my heart well up with joy, it makes my pride as an American citizen more emphatic than ever before, when I think that a man born in affluence of a splendid family, born with every opportunity in the most exclusive circles of New York and Washington, a man who could own his private yacht or spend his time in the gaities of fashionable resorts, a man who could belong to every club and enjoy all its pleasures—that such a man has devoted his life unselfishly to the good of the American people and to the cause of Conservation (great applause). It is a splendid example of true American manhood; and when he speaks here, as he has spoken in other places, the influence that he exerts is not merely for the cause of Conservation but for the highest ideals which you and I have of American manhood. So I rest assured that the cause of Conservation, with

such an advocate as Theodore Roosevelt and such an apostle as Gifford Pinchot, will not be confined within the limits of the United States but will resound through Canada and through Mexico and on south even to the limits of the southern continent; and I foresee that you and I will be proud that we were able to participate in the effort to extend this movement. (Great applause)

A DELEGATE—Mr Chairman: As a member of the Executive Committee of the National Conservation Congress, I ask for the privilege of the floor for the purpose of introducing a resolution.

Professor CONDRA—That will be in order immediately after the response by Honorable Esmond Ovey, Secretary of the British Embassy, which is a part of the presentation now in progress.

I take pleasure in introducing Honorable Esmond Ovey. (Applause)

Mr OVEY—Mr Chairman, Ladies and Gentlemen: When I arrived here on Monday I noticed in the program laid before me a very disquieting item to the effect that a speech would be delivered on the subject of "Conservation as a World-wide Question" by a visiting representative of a foreign nation. I did not think that would mean me, and until yesterday evening was still hoping that some other representative would be found, more adequate than myself, to take the burden from my shoulders. However, no savior has appeared, and I think my best course will be, under the circumstances, to make an entirely clean breast in the matter and tell you that my knowledge on the subject of the technical details of the Conservation of natural resources is very meager. The field of natural resources with which I personally am more occupied is one which is slightly different from that which forms the subject of your deliberations, a field that is perhaps as great and in many ways certainly as important; it is a field which requires neither phosphates nor potash, nor any of these ingredients of which I

unfortunately am so ignorant—it is the field of international relation, and the crop or harvest is the harvest of peace and good will (applause). The duty of the diplomat is to watch this crop ripen. It is a crop which can go on forever ripening and getting greater, but there is, of course, the possibility of some spark dropping; and it is then the duty of the diplomat to attempt, so far as possible to arrest and extinguish that spark before it flames up like these wasteful and terrible conflagrations which occasionally sweep through the forests of this country. In this connection I will point out that in the immediate field of international relation between Great Britain and the United States there has been an exceedingly long period in which there has been no spark dropped (applause); the year after next will, Gentlemen—I may call it to your attention—be the 100th birthday of peace between the two great English-speaking nations of the world. (Applause)

I have the very great pleasure of being here as the representative of my chief, the British Ambassador, Mr James Bryce (applause). The British Board of Agriculture were unfortunately unable to send a delegate to attend this great conference. Mr Bryce himself was the recipient of a very cordial invitation from the President of this Congress, Mr Baker. Mr Baker in his letter stated that should Mr Bryce be unable to accept, he would be glad if a member of his staff could come. Mr Bryce had long pre-arranged and planned a visit to Panama and South America; I can only suppose with his great intelligence Mr Bryce (my own immediate chief) has gone there for the purpose of improving his mind in the contemplation of the achievements of my friend Mr John Barrett (applause). I have been commissioned by Mr Bryce to tell you how very glad he would have been to be able to accept this invitation. Confidentially, I may tell you that, glad as Mr Bryce would have been to be here, I do not believe he would have been so glad as I am to be here myself. (Applause)

Mr Bryce is a man very difficult to represent (applause). His knowledge is encyclopedic. Even if taken by surprise and asked to speak to an audience such as this, containing so many representatives of all the practical, scientific and technical phases of the great problem which is being discussed at this Congress, he would, I am certain, have been able to draw on the great storehouse of his knowledge and give you the benefit of his accurate observation in a technically interesting form. I can, unfortunately, lay claim to no such talents. I will, however, refuse to yield to him in the

enthusiasm—that sort of contagion to which Mr Barrett referred—which I feel here in this great country and in the State of Minnesota on the subject of the noble ideals, the efforts and the aims of these congresses. It seems to me that the idea of careful deliberation and open discussion by persons from all parts of the world in an attempt to arrive at the conclusion and basis on which to build up a policy of Conservation so you can hand down to posterity the great benefits that you enjoy, is a very noble conception.

One of the great characteristic differences between Occidental civilization and that of certain less civilized and advanced Oriental nations is the great quality of foresight, of looking to the future; and this is a quality which you possess in a most extraordinary degree. I do not wish to deny that other people to whom I have referred also possess this quality; I will, if you permit me, give you an instance to prove that it is possessed by them, if in a less perfected form.

There was upon a time a gentleman from some unspecified country in the Far East who had an orchard. To protect this orchard from the prevailing cold northerly winds which destroyed his fruit in the early winter, he built a wall on that side of his property. When he had built his wall he called in a friend to admire it. The friend came and admired it. The wall was solidly built, six feet high, and twelve feet wide. The friend asked him, "Why have you chosen these peculiar dimensions for your wall?" He said, "Ah, I have foresight. I built this way for a reason: my neighbors' walls are frequently blown over by the wind. When mine is blown over, it will be twice as high as it was before." (Laughter) Now, that is not the sort of construction in this magnificent building of Conservation that you are preparing.

Another quality, if I may be permitted to mention it, that I, as a foreigner, have observed, is a great quality which is invariably a concomitant of real progress; it is a certain kind of glorious dissatisfaction with your own achievements, however great they may be (applause). For instance, you have something which is very, very great—your country. You never were satisfied with that, you want to make it very, very good. You have something which is very, very good, the great American people; you want to make them, as far as I can understand, as numerous as possible (laughter and applause). You have your natural resources, which are very great and very good, perhaps the greatest and best on earth, and yet you are not

satisfied. What do you do then? You say, "Let's make them *everlasting*." (Applause) Now, Ladies and Gentlemen, that seems to me a very fine and high ambition on which you have set your minds.

Before concluding, I will venture to tell you about an impression that I received on my way out to Saint Paul, on this my first visit west of Washington. As I looked out of the windows at the flying countryside, upon lake after lake, upon mountain, valley, plain, stream, forest, farm, garden, factory, city, town, I said to myself, "What manner of people then can these well be who have so kindly and courteously asked me to a Congress which is apparently convening for the purpose of conserving the natural resources? What manner of people can these be that by digging, delving, plowing, mining, bridging, tunneling, felling, and building roads and railroads on all these countless millions of acres of rich and fertile land—many of which are protected from approach on the east by apparently uncrossable mountains and unfordable streams and what to lesser intelligence might seem unbridgable rivers—what manner of people may these be who, in spite of these obstacles, in this short period of time, have forced Dame Nature herself to cry out, Gentlemen, please hold steady with me for a moment." (Applause) Such were my thoughts: and it seems to me that the necessity for convening these annual congresses for open discussion of the best means of avoiding unnecessary waste and of giving nature a chance of recuperation affords the highest compliment that it is possible to pay to the enterprise, courage, perseverance, and indomitable pluck of any nation.

Can you, therefore, Ladies and Gentlemen, ask if in view of these facts the Government of Great Britain is interested in your efforts? As Secretary of the British Embassy I myself was instrumental in forwarding to my Government in one year, through the kind intermediation of the State Department, no less than 110 copies of the report of the Governors of 1908 on the Conservation of your National resources, which, if I understand rightly, was one of the first expressions of this great movement—110 departments of that Government interested in this movement. (Applause)

It is my pleasurable duty to inform you that with her own magnificent dominions across the seas, with her great enterprises in forestry, irrigation, agriculture, and mining, in all scientific exploitation of land for the public

good in Canada, in Australia, in India, in Egypt, in South Africa and British East Africa, and in all the other places throughout the world in which Great Britain is now working, the Government which I have the honor to serve is in the heartiest possible sympathy with the great object of your endeavors in conserving for posterity, for people not yet born, the same magnificent heritage which you and we enjoy. (Applause)

Professor CONDRA—All those who wish to say that as Delegates we stand for Pan-American conservation of natural resources, and for good fellowship and world-wide Conservation of all things best for mankind on all lines of industrial development, will please rise.

[The audience rose en masse.]

Professor CONDRA—There was a resolution to be offered at this time.

A DELEGATE—Mr Chairman: I move that the time for the election of officers of the National Conservation Congress for the ensuing year be fixed for the hour of 8 p.m., Thursday, September 8, and that the Committee on Resolutions submit their report immediately following the election of officers.

The motion was seconded by Delegates from Iowa, South Dakota, Utah, Indiana, North Dakota, Wisconsin, and the District of Columbia; and the motion was put and carried without dissenting voice.

Professor CONDRA—A recess will be taken until 2 oclock p.m.

EIGHTH SESSION

The Congress reassembled in the Auditorium, Saint Paul, at 2 oclock p.m., Thursday, September 8, President Baker in the chair.

President BAKER—Fellow Delegates, Ladies and Gentlemen: It has been urged that a nominating committee should be appointed to name officers proposed to be elected by the Congress as President, Secretary, Executive Secretary, and Treasurer. The Vice-Presidents have been chosen by the State Delegations, and their names will be presented this afternoon. So, unless some other course be preferred, the Chair will proceed to form a nominating committee. [After a pause.] The nominating committee will consist of Professor George E. Condra, of Nebraska, as chairman; E. T. Allen, of Oregon; E. L. Worsham, of Georgia; Lynn B. Meekins, of Maryland; and William Holton Dye, of Indiana. Delegates are invited to offer suggestions or nominations to the committee, which will hold a meeting during the afternoon.

I have the honor now of presenting as presiding officer, His Excellency A. O. Eberhart, Governor of Minnesota. (Applause)

Governor EBERHART—Mr President, Ladies and Gentlemen: I am indeed sorry that I am to be engaged elsewhere a portion of this afternoon, so that I cannot take part in the entire program. We have this afternoon an unveiling of a statue in the Capital, and I will necessarily have to take some part in the ceremony; but I shall hasten back just as soon as I can, so that I may hear the speakers who are on the program for this afternoon.

I do not know whether the President of this Congress has made a special effort to secure splendid speakers for this afternoon, but certainly no session of the Congress, either forenoon, afternoon, or evening, has had better, more sincere, and more earnest and efficient workers along the lines of

Conservation interests than those for this afternoon; and for that reason I am indeed sorry that I shall not hear them all.

I want to say to you that the State of Minnesota and the Twin Cities are proud of the Delegates and the guests and the speakers of this convention, realizing that perhaps never in the history of the Conservation movement will there ever be another meeting so important as this, and one that will redound so much to the progressive and effectual work of the movement.

I take great pleasure in introducing to you as the first speaker of this afternoon a man interested in the Conservation movement from the standpoint of public health—Dr F. F. Wesbrook, Dean of the Medical Department of our State University—who will speak on "Life and Health as National Assets." I consider it one of the most important subjects of the Conservation movement. I take great pleasure in introducing Dean Wesbrook. (Applause)

Dean WESBROOK—Mr President, Your Excellency, Ladies and Gentlemen: Short-sighted humanity fails to appreciate nature's gifts until threatened with their loss. This is true of even the greatest of her gifts, life itself. Although belated in our realization of the threatened overdraft on nature's storehouse, a compensatory and irresistible enthusiasm has developed within the last two years which augurs well for the retention by our country of that international leadership so manifestly foreordained by nature's bountiful equipment.

It is significant of our failure to value health, which above all other considerations makes life worth the living, that the first meeting of the Governors in the White House in 1908 failed to provide for the study of health problems. The omission was noted, and in the National Conservation Commission's Report of January 11, 1909, the general schedule gave special consideration to life and health. Only four sections, however, were created in the appointment of the National Conservation Commission. Health was not provided with a special section or with officers. In the North American Conservation Congress, in addition to the Conservation of other

National resources, the protection of game received attention; but among the Commissioners representing the various countries, there was seemingly no one whose training and paramount interest lay in the field of public health. While it is apparent that the initial oversight has been in part repaired it remains to be seen what progress will result from the Second National Conservation Congress, in relation to this, the people's most important natural asset.

The inclusion in the program of a paper entitled "Life and Health as National Assets" must not be taken as evidence that there is any doubt as to the real and assessable value of life and health. Rather are we called upon at this time to realize that they constitute National or public resources furnished by nature and are not to be regarded as strictly personal or private possessions. The individual life has its economic and commercial value to the community and the Nation by virtue of the contribution it may be expected to make to society. This view may perhaps be novel to some. Our ideas concerning the conservation of other natural resources however, have undergone such rapid evolution in the recent past that we may easily orient ourselves to the viewpoint exhibited by the officers of this Congress, that the individual in matters of health, as of other resources, must respect the rights of other individuals and of his municipality, State, and Government. The health aspect of Conservation, which is its most important aspect, cannot and will not be neglected, although it has not been the first to which the attention of the Nation has been directed.

Nor can we dissociate health conservation from the other aspects of the movement, even if we would. The history of man's progress in the knowledge of the natural sciences bears out this statement. Even though we ourselves have broken faith with nature, we are able today to make her fulfil her promises in forestry, agriculture, and other economic matters by the application of our knowledge of those very sciences which may be said to owe their birth to man's search for perpetual life and youth. One can easily imagine that the medieval conservation commission comprised two sections, one on health and the other on minerals. In the former, which undoubtedly was basic and dominated all other considerations, the papers presented dealt with "elixir vitæ" and the "touchstone" whilst in the latter the chief interest was displayed in the "transmutation of metals." At this stage the studies of health and of the control of man's so-called material

assets were carried on hand in hand; and, if we are logical, they always will be.

In any event, man's health depends on the success of his efforts to adapt his environment to his needs, more than it does on the adaptation of himself to his environment. Health interests are fused with social and economic development, but should undoubtedly dominate rather than be dominated by them.

Our lack of interest in matters of health is more apparent than real. It is characteristic of many of us that where our most vital interests are involved, we betray the least public concern. In nothing is this better exemplified than in matters of personal and public health, except it be perhaps in matters of religious belief and practice. Nor should we deem it strange that a similar attitude of mind obtains in matters of health and religion. In medieval times the priest and the physician were one. At the present day, aboriginal tribes combine religion and health, and to too great an extent, perhaps, do our civilized nations fail to discriminate between the two. Particularly is this exhibited in man's cowardly attempt to shift his responsibility for disease and death upon Providence.

One of the greatest causes of lethargy in the conservation of personal and public health is the failure on the part of many to differentiate clearly and sharply between disease and death. The former is really a manifestation of life and vital force, and is capable of modification, prevention, or cure by human agency, since man has shown himself quite able to solve nature's other secrets for the benefit of his comfort or convenience. We conserve health by the application of the same sciences which enable us to conserve our other better recognized but less material natural resources. Disease yields to man's mastery; death remains man's mystery. Even death, however, may be postponed, and Professor Irving Fisher has estimated that over 600,000 deaths occur each year in our country which could be postponed by the systematic application of the scientific knowledge already available. For those who think more easily in terms of dollars and cents, he has estimated this appalling annual National loss at over one billion dollars which can and should be prevented.

We must not be lulled into any sense of well-being by such statistics. There is no royal road to such a goal. Our very success in the eradication of one disease or unsanitary condition may lead to undue optimism in regard to other problems, which later may be found to be dependent on altogether different causes and to require very different methods of prevention or cure. Failure to realize the complexities of modern social activity and economic development, in their relation to health, and, at the same time, to recognize the immense number of variable factors and agencies which are involved in health-protective measures, cannot but lead to disappointment. The individual whose enthusiasm is too easily aroused by the discovery of some hitherto unknown cause of disease, or some new method or theory of cure or prevention, is a source of danger to the commonwealth. The faddist, whether in the matter of such things as food, clothing, fresh air, baths, exercise or other therapeutic agents, as well as the individual who thinks that he has discovered the one cause of all diseases, is to be feared.

Our chief difficulty lies in coordinating the various forces and agencies which are essential to success in the eradication of sickness.

There is no blanket method of preventing all diseases. Quarantine and fumigation are now found to have but a limited application. Vaccination, which is practically an absolute and the only reliable protection against smallpox, cannot be applied to such diseases as malaria, yellow fever, and diphtheria. The use of antitoxin, which prevents annually many thousands of deaths from diphtheria, does not help us in many other diseases. Our knowledge of mosquito-borne disease, which has reorganized life in Cuba, Panama and the Philippines, is not of much practical use in our northern States. As there is no single cause, so there can be no single method either of cure or prevention.

These considerations should not discourage us. They show us, however, the need of further study, and the imperative demand for employing the services of trained physicians, biologists, chemists, engineers, statisticians, sociologists, educationists, and other experts and of coordinating all their efforts. We must steer a middle course, avoiding on the one hand the Scylla upon which those run who become discouraged in the face of what they believe to be the unknowable, and, on the other hand, the Charybdis of that

fateful tendency to minimize the actual complexities of the present day health problem. Fatalist and faddist are equally dangerous.

It is fair to count upon the same progress in the adaptation of physical, chemical, biological, social and other sciences to the diagnosis, cure and prevention of disease as in their application to man's comfort, convenience and economic development. It is clear that the efforts of all the various workers in the different fields must be coordinated; yet the difficulties of coordination are at once apparent. The forces and agencies may be roughly divided into international, National, State, county, municipal and institutional, as well as individual. Each one of these is capable of still further subdivision into two classes, one of which is official or governmental and the other is voluntary. Improvement in public health requires cooperation and coordination of *all* these.

Successful public health administration consists largely in making individuals do what they do not wish to do—or that of which they do not appreciate the necessity—for the good of themselves and others. This brings us naturally to the consideration of another National weakness. We encounter some of the same difficulties in public health work that we meet in the exercise of our other public functions. Rampant individualism is of even greater danger in matters of health conservation than in other affairs of public concern, largely on account of the fact that health is too often regarded as a purely personal rather than a most important public asset. The individualist objects to authority in matters of health control. Consequently he resents dictation as to his personal action, and fails to recognize the need for special training in health administration as in other branches of public service.

Public service of many kinds, and particularly that which relates to the conservation of health in our country, is all too often relegated to voluntary agencies, while in other countries it devolves upon official and governmental agencies. This volitional duty is nobly discharged. The main function of the volunteer should be, however, to afford to the general public object lessons of what is needed and of how progress can be made. In this he rarely fails, although he labors under tremendous difficulty imposed by lack of authority. Funds which are furnished from private sources are frequently insufficient to permit of the employment of experts of the highest

order. Public apathy, on the one hand, and the development of an abnormal interest on the part of voluntary workers on the other, frequently lead to their continuance in service long after they have ceased to be useful, with the result either that the public delays the establishment of an official organization, or, if such an organization be established, there is a conflict between the official and voluntary forces. If municipal health departments, hospital services, police departments, water, school, poor and park boards, and other official servants and representatives of the people were supported by the people and were quick to see and to seize their opportunities, there would be less need of associated charities, of visiting nurses, pure water and milk commissions, tuberculosis camps, play-grounds associations, and other such voluntary organizations. Is it not humiliating that public lethargy made it necessary for Mr Rockefeller to provide funds for the investigation and eradication of hookworm disease?

In Germany, the Government, through its public health service and universities, provides for medical and other research so that Nation has become a leader of the world in scientific health protection and scientific economic development.

Having seen some of the difficulties which stand in the way of satisfactory conservation of the public health, we might perhaps ask ourselves what proof of the possibility of conserving this asset is available. If, at this day and time, the American public is unconvinced of the need and possibility of conserving public health, it is undeserving of the respect of other nations, or even of self-respect. The daily and weekly press, our magazines, and governmental and other publications, have overflowed with information. Our attention has been particularly called to the possibility of preserving the health of men in the field by Japan's experience in the recent war with Russia. Our life insurance companies have been quick to see the practical possibilities of prolonging the lives of their insured and of thus increasing the earnings of their stockholders.

As illustrating our progress, the report on "National Vitality, Its Wastes and Conservation," which was issued by the American Association for the Advancement of Science, is a masterpiece; it was prepared and presented by Professor Irving Fisher, of Yale University. The publications of the various committees of the American Medical Association, and the speech of

Senator Owen in the Congressional Record of March 24, 1910, as well as Federal, State, municipal and other health reports, afford examples of what can be done.

Those who may be skeptical in regard to the ability of our people to compete with older nations in the prevention of disease, should note what has actually been done by Americans under the greatest of difficulties. In Cuba, our Nation overturned the existing order of affairs, and scientific discoveries, made and applied to sanitation by Americans, afforded a lesson to the world. There has been no greater factor in winning the world-wide confidence of other nations than the production of the existing sanitary state of affairs in the Canal Zone by our own citizens. Our work in Cuba, Panama, and the Philippines has served to bring about hygienic conditions in supposedly pestilential regions which are vastly superior to those which obtain at home. What Americans have done for others they have failed to do for themselves, owing largely to the lack of provision of adequate official and governmental agencies and to the failure to coordinate those which exist. Two Americans in Porto Rico showed the possibility of stamping out hookworm disease. The brains were furnished by the United States, and the money by the Island. We have the brains at home, but we refuse to pay the bills.

It is manifest that a full and complete discussion of life and health as National assets is impossible within the limits of a single paper. No attempt need be made to present a complete basis either of comparison or differentiation of health conservation from the other aspects of the National movement. It must be clear to all that in the conservation of lands, minerals, waters, and forests, effort is made to prevent the individual from taking that which belongs to the public. In the conservation of public health, our effort must be directed to preventing the individual from giving to the public something which neither he nor it desires. This is particularly true of infectious diseases. There are many other phases of public health than those which relate to infectious disease, but they cannot be discussed at this time.

I have the honor to be a Delegate to this Congress from both the American Medical Association and the American Public Health Association, which represent factors in the conservation of human life and health concerning

which the public needs more information than it possesses; and with your permission, I shall briefly mention a few important matters:

In the past, individual physicians and local medical associations and societies have brought a scattering fire to bear upon the inactivity and ignorance of the general public in matters which pertain to public health. The public fails to believe in the urgency of health needs, when presented by individuals or groups of physicians, because of its inability to appreciate the motive which leads the physician to urge the establishment of machinery and the special education of officials, as also the provision of funds to carry on work which to the casual observer would mean a diminution of the individual physician's work and income. Physicians who have qualified by postgraduate training in bacteriology, pathology, epidemiology, and in public health, hospital, school and institutional administrative work must be drafted into the direct and official service of the people. This need is increasingly apparent. Others are required who can present evidence of special scientific training in chemistry, engineering, statistical, sociological, charity and other work. At present, great as is the actual need, the demand on the part of the public and the remuneration offered are so small and the possibility of employment so uncertain that universities, technical schools, and other institutions which offer special courses fail to attract students. The public seems to prefer as yet to jeopardize its most valuable asset by employing untrained public health servants who develop efficiency after, instead of before, their appointment. This means a payment in life and health instead of dollars.

The average individual seems willing to pay, and pay well, for a cure when he is sick. Communities pay the cost of epidemics, and will even pay for engineering services in relation to public utilities, such as water supply and sewage disposal; but this is usually done only under the stimulus of some recent or threatened disaster. They, like the individual, want a *cure*, not a *protection*. Clinical experts, life insurance examiners, and consulting and commercial engineers, are all sure of a good livelihood because they can help the individual or community out of difficulties. Sanitarians and municipal engineers are usually left to semi-starvation, because their function is to prevent those same difficulties, without, however, having either available public sentiment or funds to enable them to do it.

Physicians are naturally skeptical of the scientific training and possession of proper ideals on the part of those who have not been especially trained in medicine, and who may have failed to develop the "disease point of view." That they are, however, of a receptive frame of mind can be shown in many ways. The American Medical Association has a number of standing committees, including a Council on Medical Education. This Council, in the endeavor to raise the standard of medical teaching throughout the United States, prepared a standard schedule of minimal requirements, through the agency of ten committees, each of which consisted of ten representative men. One of these ten committees (which had to deal with hygiene, medical jurisprudence, and medical economics) contained in its membership university and college professors of chemistry, physiological chemistry, political economy, pathology, bacteriology and hygiene. There were also executive officers of State and municipal boards of health, and representatives of the Federal Health Service; whilst among the collaborators were engineers and many university professors. Bear in mind that this was a committee of the so-called "medical trust"—the American Medical Association.

Through oversight for which no one is responsible, this Second National Conservation Congress and the American Public Health Association are meeting on exactly the same dates, September 5-9, we in Saint Paul and the Association in Milwaukee—I was just able to get here from Milwaukee. This Association consists of some physicians who are in practice, but more particularly of Federal, State, municipal and institutional administrative officers, as also of laboratory, statistical, engineering, and other technical workers. The membership includes representatives from all of the leading universities and medical and technical colleges. It has three sections, namely, laboratory, vital statistics, and municipal health officer sections. You are familiar with the work of many of its officers and members. Colonel Gorgas, who was responsible for the administrative health work in Cuba, and who has made possible the building of the Panama Canal without undue loss of life, is a member of both associations. The late Dr Walter Reed, who eliminated yellow fever from civilized communities, was vice-president. It is an international association in which Canada, Mexico, and Cuba also participate, and much can be learned by attendance at these annual meetings. One of its chief benefits has been the formulation of

standard methods of scientific procedure, applicable to the suppression of disease in various districts of the several countries.

We in this country are compelled to admit that our neighbors upon the north and south have much in the way of advantage which is denied to our own workers in the United States. In our sister countries, the tenure of office depends on the fitness and training of the incumbent. As a rule the compensation for public service is relatively higher, and the official organizations are better provided with an authority which is commensurate with their responsibility than is the case in our own country. Time will not permit extended discussion of these conditions, but the annual opportunity to compare notes; to tell each other of our successes, as also of our failures; and to help in the formulation of new methods and in an effort toward a higher standard of efficiency, is of untold value. This is, however, a purely voluntary organization maintained for over thirty years at the personal expense of its members in the face of public apathy. This will be realized if I ask, "How many of you knew that we have such an association," and "Did you know that it is now in session"?

There yet remain a few matters of which a general understanding would bring about yet greater cooperation between the doctor and the general public. The medical profession has realized for a number of years that its members must become teachers of personal hygiene to their patients and families, as also to schools and the general public. It is a new viewpoint, and involves the assumption of new responsibilities. The doctor has guarded himself against publicity except through his professional societies and journals and to his students, though ever eager to furnish details of his own discoveries and to recount his failures and his successes to those who could understand and sympathize. This kind of publicity has been regarded, however, by the lay public as a sort of soliloquy carried on in an unknown tongue, and intended for the mystification of that same poor public.

Why there should be any failure of the medical profession, as a whole, to be understood by the general public, it is difficult to see. The general public is composed of individuals, each of whom has a feeling of trust, affection, and possibly of veneration for one or more members of the medical profession. Why then does the public, as an aggregation of individuals, allow itself to become suspicious of the medical profession, an aggregation of physicians?

Why does the public abhor and obstruct the physician in his study of anatomy, dissection, and autopsy on the human body? Why is there so much suspicion of the motives and work, as well as denial of the benefits which accrue to humanity from animal experimentation, when it must be apparent to any right-thinking individual that the extension of a physician's knowledge is possible only by such means? Why must doctors from time to time be themselves forced to urge the necessity of making every hospital a teaching and research institution? A moment's thought would convince anyone that if this be not done, and if medical knowledge be allowed to die out with this generation, there will be no skilled men available for the hospitals and patients of the future. It must also be patent to all that the patients themselves cannot possibly receive such effective care in a hospital in which medical research and teaching are not fostered. Why should the burden of maintaining a high standard of entrance to the profession and of preventing incompetent and untrained persons from assuming the responsibility of physicians rest solely on the medical profession, when the object is the protection of private citizens and public health?

The physicians of the United States are now thoroughly organized. The public should rejoice in this, since it is an attempt to neutralize the narrowing effect of isolation and to foster an exchange of information which physicians offer freely to each other and publish broadcast to the world (applause). County and State associations are affiliated with the American Medical Association, which numbers in its membership over seventy thousand doctors. Just as the individual physician's concern is the care of his patient, so that of the organized medical profession is public health and welfare.

The medical profession is, as a rule, underpaid, but members spend their hard-earned-money and a large portion of their time in efforts to benefit humanity, individually and *en masse*. It is the people's concern to demand a broad education and a thorough scientific training of all students and practitioners of medicine, public and private. It is to their interest to see that every possible facility is afforded for teaching and that a rigid standard of teaching, examination, degree conference, and licensure is maintained. Nothing is more exasperating to the physician of high ideals, whose length and breadth of sacrifice is known to none, than to hear the sneer directed at his profession for its effort to protect the public. The time has come when

the medical profession is in a position to demand that the people exercise discrimination and protect themselves.

One of the first steps toward the betterment of our public health conditions is the coordination of the existing Federal agencies in Washington, of which we are all so proud. When no logical reason can be advanced in explanation of further delay, it is very discouraging to realize that this important matter has been postponed. At the 61st Congress, various bills were introduced, including that of Senator Owen. In support of these bills appeared those who by special training and long experience are recognized at home and abroad as the highest authorities on public health. The whole country is waiting to see what action her representatives will take to protect her most precious asset.

With your permission, I should like to cite some sixteen reasons why the people of the United States should have a department of health at Washington, which were published by the Committee of One Hundred of the American Association for the Advancement of Science:

1—To stop the spread of typhoid fever through drinking sewage-polluted water of interstate streams.

2—To enforce adequate quarantine regulations so as to keep out of the country plague and other similar pestilences.

3—To supervise interstate common carriers, in so far as without such supervision they prove a menace to the health of the traveling public.

4—To have a central organization of such dignity and importance that departments of health of States and cities will seek its cooperation and will pay heed to its advice.

5—To influence health authorities, State and municipal, to enact reform legislation in relation to health matters.

6—To act as a clearing-house of State and local health regulations, and to codify such regulations.

7—To draw up a model scheme of sanitary legislation for the assistance of State and municipal health officers.

8—To gather accurate data on all questions of sanitation throughout the United States.

9—To establish the chief causes of preventable disease and unnecessary ill-health.

10—To study conditions and causes of disease recurring in different parts of the United States.

11—To correlate and assist investigations carried on in many separate and unrelated biological and pathological Federal, State and private laboratories.

12—To consolidate and coordinate the many separate Government bureaus now engaged in independent health work.

13—To effect economies in the administration of these bureaus.

14—To publish and distribute, throughout the country, bulletins in relation to human health.

15—To apply our existing knowledge of hygiene to our living conditions.

16—To reduce the death-rate.

In 1912 there will meet in Washington, on the invitation of the President and Congress of the United States, the International Congress of Hygiene and Demography. This Congress meets triennially in the capitals of the world, and brings together the leaders in health conservation who are officially delegated by the governments of all civilized countries. We have many things to show them of which we can be justly proud. Our Federal, State, municipal and other official health organizations, however, leave much to be desired: and it behooves us, in the few months still at our disposal, to prepare to show the visiting nations our methods and successes. We need many other things, but due recognition and coordination of our Federal health mechanism is the first step which, if we have taken it before the meeting of this International Congress, will best enable us to profit by the experience of the world's experts there assembled.

Nature has been prodigal in her gifts to our Nation. In no respect has she been kinder than in opportunities for health and efficiency. Her very prodigality has rendered us careless and extravagant. It is high time that Americans do as well for themselves in health protection at home as they have done for themselves and others in Cuba, the Canal Zone, Porto Rico, and the Philippines (applause). This demands the creation and maintenance of official organizations to amplify, extend, and ultimately replace the work of our voluntary organizations whose lack of authority prevents their complete success, and whose continuance is an admission of popular inertia and official incompetence. (Applause)

[During the foregoing, Governor Eberhart withdrew, and professor Condra took the chair.]

Professor CONDRA—Ladies and Gentlemen: In the temporary absence of Governor Eberhart I have the pleasure of introducing Mr Wallace D. Simmons, of Saint Louis, who will address you on "Our Resources as the Basis for Business." (Applause)

Mr SIMMONS—Mr Chairman, Ladies and Gentlemen: The President of the United States in opening this Congress called upon the speakers to make definite practical suggestions. The ex-President of the United States the next day emphasized the need of further enlightenment of the people regarding Conservation. It has frequently been my privilege to cooperate with both of them, and I will endeavor to do so now by suggesting a definite plan for spreading enlightenment in a practical manner.

We of this generation have developed a distinctly new type in our American citizenship, one which has no counterpart in the history of any other people, one which has become a most potent and influential factor in our daily affairs: our modern high-class commercial traveler. In any campaign of education, such as I am going to suggest, you can have no more efficient allies than the 600,000 commercial travelers covering this country—not the old-time drummers of questionable methods, but the gentlemen of high character who have won the confidence, the respect and friendship of the merchants and the people generally in every part of this country; and I may add, as a requisite to their success that they are resourceful.

To this development I attribute my having the honor of addressing you today regarding our resources as the basis of our business, because the organization of which I am president employs probably the largest corps of such representatives in the country, and has through them the best system of keeping accurately informed regarding all matters that affect business.

From conclusions based largely upon the observations of the commercial travelers of this country, I will endeavor to outline to you what I believe to be the relationship between our business interests and the question of natural resources; and I believe this phase of the question is most vitally important to the people in whose interest you have gathered here from every State in the Union. The primary reason for that belief—and the one on which all others hinge—is that we are a Nation in trade; a whole people engaged in business. Eighty-odd percent of our people are directly or indirectly dependent for their living on business conditions. The business interest therefore is the greatest interest, collectively, in the country.

Anything which directly affects the living of the majority of our people is not only worthy of our most earnest attention, but should be approached

with due consideration. We should be especially cautious about experimenting with legislation that may interfere with the natural laws of trade. When this is more generally recognized, and the people begin to understand that their individual daily incomes are at stake, they will put a stop to using the business interests of the country as a football for politics.

Not only does there appear to me to be a direct relation between our natural resources and our business, but as I view it our resources are the foundation of our business, or as Mr Hill so aptly put it yesterday, they constitute the capital on which our business is done.

In business we endeavor, by industrious and intelligent use of our capital, to produce as the fruit of our efforts an annual return without impairing the capital—without touching the principle or jeopardizing it in any manner. In private enterprises, the man who assumes the headship of a business organization in which the funds of others are invested as capital, and who then makes a show of prosperity by drawing on that capital to pay what he represents as dividends, is charged with running a "get-rich quick" scheme, and in most States is, by law, held personally liable. I commend to your consideration the consistency of applying that principle where there is involved the capital of all the people—the Nation's resources. (Applause)

If we are a people in trade and mean to continue to be, and if our resources are our capital, can there be any doubt about the wisdom of handling that capital according to the rules of good business? Can there be any doubt where we as a Nation will land if we make annual inroads upon that capital; if we, in the management of the people's business, follow methods which in private affairs bring those responsible before the bar of justice?

We as a Nation take just pride in our business successes; we attribute them to the brains we put into our work, to the thoroughness with which we study what we do and what others have done that we may profit by experience. Is it not well for us thoughtfully to inquire whether the histories of any other nations record the handling of their resources on the "get-rich quick" plan, that we may see what has been the outcome? History is full of such instances; many of them have been pointed out by eminent advocates of this movement. I will therefore not attempt anything but passing reference to some of them. Volumes could be written from evidences found in the Valley

of the Euphrates and of the Tigris, where stood the great Kingdom of Babylonia, the wonder of the ancient world; in the ruins of Palmyra and Palestine; in the Barbary States, once famed as the granary of Rome but now a howling wilderness, because the Mohammedans who conquered it neglected its natural resources; in the ruins of the Cities of the Sahara, whose crumbling courts bring to mind the words of Omar Khayyam—

They say the lion and the leopard keep
The courts where Jamshyd gloried and drank deep.

If we look to history for the other side of the picture—for instances where business prosperity has gone on without interruption as long as natural resources have been conserved and intelligently maintained—we find them so well defined as to lead to but one conclusion. This is illustrated in Germany where they have maintained the fertility of their soil for centuries. It produces more per acre today than it did many generations ago. Their great forest estates have remained intact; they have cut a crop of timber from them regularly every year, producing an annual income, but the capital—the forest estate—is greater and more valuable today than it was before our country was discovered. Fires have not destroyed their forests. They have long since learned the wisdom of applying, "an ounce of prevention," and fortunately have no "pork-barrel" to stand in the way. (Applause)

And we find in our own history many instances where great business enterprises have sprung promptly from efforts to intelligently develop the resources around us. The State of Illinois was passed over by the first settlers as a land of no opportunities. It is today, in productiveness and volume of business, one of the greatest States in the Union. In the States of Utah and Colorado vast areas formerly looked upon as barren and useless wastes, have been, by the intelligent handling of natural resources, made to produce annually wonderful crops of fruit and vegetables, the traffic in which has become a great commercial industry. The development of the Southwest, dependent very largely on one resource—the fertility of its soil—has called into being such lusty young giants as Wichita, Oklahoma City, Dallas, and other cities of that type. In the vicinity of Birmingham, a section which before the War was occupied mainly by cotton plantations—wherein there was nothing that could be properly called business—where generations came and passed to the Great Beyond and never saw the smoke

of a factory or heard the hum of a busy mart of trade, today, with but one generation intervening, we find a live and prosperous modern city, the heart of a great industrial region. The change has come from developing three great natural resources, which up to the close of the War had been allowed to lie idle and unproductive—the forests, the coal and the iron.

Here again we find an example of the business dependence of natural resources one upon the other. The timber from the forests was needed for the mining of the coal, and the coal was needed in the manufactures from the iron ore; and again the forests in the development of means of transportation to the markets of the world.

So there is ample evidence that business activity follows promptly upon the intelligent development of natural resources, and decay with equal certainty follows the neglect or wasteful use of the capital which nature tenders us, and for the intelligent use of which she holds us strictly accountable.

I have frequently been asked by those who know our system of getting reliable information, "How do people over the country feel in regard to Conservation; are they in favor of it in all its aspects, or do they seem to be interested only in certain features?"

As that is a question that has direct bearing on the business of the country, we naturally had made careful inquiry regarding it from Maine to California, and we had learned that the majority of the people do not understand enough about it to hold any real opinion. They have no adequate idea what Conservation means as applied, for instance, by this organization to our natural resources. In spite of exhaustive reports issued by the Government, in spite of scholarly and illuminative articles on the subject, the people generally do not yet understand the real object of Conservation. A busy people in trade do not have time to read Government reports or long speeches on any subject, and of course no one can do justice to even one element of this great subject in a short article. The net result is therefore that there is no general understanding of even the A B C of Conservation such as should be given to the people, such as they would be glad to have, and such as they must have before there is warrant for feeling that the foundation stones of Conservation are so firmly grounded that no transitory

wave of agitation on unimportant details can be successfully used to dislodge them.

The majority have not yet grasped the idea that one of the prime objects of this Conservation movement is to preserve the fertility and productiveness of the soil, on which we all depend for our food supply. They are not aware that already in many parts of this country, where formerly any man who rented farm lands was entirely free to use them with indifference to their future, he is now required by the owners to enter into a written contract which provides just how the land is to be cultivated—how the crops are to be rotated and fertilizers used. The owners of these lands today require their tenants to practice Conservation. (Applause)

The people do not generally understand that when a territory which has been used as a range for cattle is by proclamation withdrawn, as we express it, that does not mean it is no longer to be used for pasturage. Conservation does not aim to suspend use—its object is to perpetuate usefulness in full measure this year, and every year to come. (Applause)

A farmer who owns a pasture—large or small—and rents it for stock grazing, takes due care to cover in his agreement the number of head and the length of time they are to be kept on his land. He makes sure that his pasture is not to be so abused in any one season as to ruin it for the future. He cares for his own land as it is the province of the Forest Service to care for the public land entrusted to their supervision. He practices Conservation because he cannot afford to do otherwise.

It is not widely known that instead of wishing to keep settlers out of the National Forests, inducements are given to get people to settle within their boundaries; homesteaders are free to pasture their domestic stock within the reservation and to cut from the forests the timber they require for building houses, barns and fences. It is not generally understood that making a forest reservation does not mean that no more timber is to be cut there for market; on the contrary, its prime object is to insure continued cutting and selling of it for all time. It is not widely known that the revenue from timber cutting on the public forest lands amounts already to a million dollars a year, and the annual revenue from the pastures puts another million into the public treasury—and that this is only a beginning; or that meanwhile this kind of

revenue-making regulation also affects the regularity of water supply through our rivers and streams—a most vital question as has been shown by many able exponents of Conservation.

When this Nation of business people understands that Conservation is simply another term for business management of the people's capital, the pressure of public opinion will be so strong behind this movement as to brook no interference or delay in the passage and enforcement of the laws needed to begin at once a business administration.

How to spread more widely a correct understanding of such facts is today a most important problem. How shall we reach the people who have not yet been reached, and who in all probability will not be reached by anything published in the usual way?

I have a suggestion to make which I ask you Delegates to take to the Governors who appointed you to attend this Congress; that is, that each Governor summon to his Capitol for consultation, say six of the leading business men of the State, selecting those who in their own business have, by successful use of modern advertising, demonstrated that they have learned from experience how to reach the individual and tell him something they want him to know. Knowing how to do that is just as much a matter of education and experience as are the methods of the Forester or of the politician who is a "past master" at the game. Give the people of your State the benefit of this experience. It can be had for the asking. The business men can be depended on to help whenever called upon. They will be particularly ready in this matter which, in proportion as it is successful, will make for good trade and stable business conditions; and the Conservation of our natural resources stands for more stable business conditions year after year, in that it tends to reduce the chances of losing our new wealth in crops just when it seems to be practically sure.

Ask such a group of successful advertisers to formulate a scheme of reaching the public generally with the kind of information they want and should have about Conservation. Enlist the cooperation of the army of commercial travelers within the State—there are no more loyal American citizens anywhere, none who can do more in such a campaign, none who will more gladly lend a hand when once they are advised along proper lines,

and know how great a factor the Conservation of our natural resources can be in the upbuilding of business and, through it, the general prosperity of our people.

Ask this business council to formulate ways of making known not only the facts about forests and water supply, and the importance of these facts to every individual man, woman and child in the Nation, but why we in the United States average $13\frac{1}{2}$ bushels of wheat per acre, instead of $23\frac{1}{2}$ bushels, as they do in Germany, and $30\frac{9}{10}$ bushels in Great Britain; how this is making homestead lands scarce, and prices high, because we only get half the amount of crops we should get from the land we have under cultivation. When we find our production less to the acre each succeeding year and more mouths to feed, it is time everybody knew why.

Tell them in the simplest and most direct manner possible what is meant by the "pork barrel" in politics—how it is being used to retard the proper development of our natural resources, and why therefore it stands in the way of the Nation's progress. Let them know why we all have reason to thank God that we have in the White House a President who does not let politics silence his tongue on that subject or swerve him from his determination to stop this waste of the Nation's funds. (Applause)

Write up a short story of what Reclamation has done and can do in relieving the situation by opening up to us millions of acres of land which can and will add greatly to our food and meat supply; tell them what has already been accomplished and the progress that is still being made by reclamation work, to the great benefit of the people. Explain in a simple manner that hand in hand with the profitable development of our natural resources must go the development of our great waterways and railroads—that there can be no general prosperity without railroad prosperity; that our railroads and waterways are the connecting links which make our natural resources available, and that the practical value of our natural resources depends largely on the efficiency of our transportation service. (Applause)

Point out to them the lessons which we should get from cases of individual effort along the lines of modern methods in farming; how, for instance, Mr Claude Hollingsworth, near Colfax, Washington, raised this year 45 bushels of wheat to the acre, averaging 62 pounds to the bushel, and of barley $72\frac{1}{3}$

bushels to the acre, when his neighbors, with the same conditions of soil, climate, and rainfall, averaged only half as much; or in South Carolina, where Mr E. McI. Williamson has, by the proper application of fertilizers, modern methods, and little additional expense, increased his production of corn from 15 bushels per acre to an average of nearly 60 bushels, and of cotton from less than half a bale to an average of a bale per acre. Such examples are most convincing, and will do much to arouse interest in the practical value of Conservation.

The conservation of the National health deserves to be emphasized even when we have under consideration this general subject from purely a business standpoint. When we consider that tuberculosis alone costs the people of the State of New York over \$200,000,000 per year, and that it is a preventable disease, and that that \$200,000,000 might be used as capital to give to millions of people profitable and wholesome occupation, the relation of the health movement to the business interests of the country is self-evident.

Of course, this suggestion is based upon entire confidence in the cooperation of the daily press—I have no doubt about that whatever. The newspapers and magazines are not only most potent factors in spreading enlightenment, but they can always be depended on to take enthusiastic hold of any movement that is honestly and disinterestedly for the general good. (Applause)

This whole subject of Conservation is fundamentally a business proposition—a question of managing the people's business with the same care and foresight that we put into private business—a question of using the Nation's capital in a way that will produce a regular, steady and proper income year after year, and at the same time so safeguard the principal that the people of these United States may go on in business indefinitely.

History tells of many peoples who have spent their capital and disappeared from the face of the earth. Let us so organize this Nation's business that it may go on down the centuries as history's exception to the general rule of rise and fall (applause). As we point with pride, honor and gratitude to the signers of our Declaration of Independence and the makers of our Constitution, so may the coming generations of Americans, having in mind

the fates of other peoples, look back with gratitude to us and have occasion to exclaim "See what would also have been our lot had it not been for the foresight and business judgment of our ancestors of the Twentieth Century—worthy successors of the great men who founded this Government of the people by the people and for the people, not only for their own time, but for all time." (Applause)

President BAKER—Ladies and Gentlemen: Nothing is more important to Conservation than education; and I have the honor now to introduce the Commissioner of Education, Dr Elmer Ellsworth Brown, who will address you on "Education and Conservation."

Commissioner BROWN—Mr President, Ladies, and Gentlemen: Every uplift and reform comes back to education. It is uplift carried to the sticking point. It is reform continually going on. In speaking of the educational aspect of Conservation, I am not concerned with anything merely incidental or subordinate, but have to do with a matter as large and vital as any upon which the success of the Conservation movement depends.

It must be admitted on the other hand that education has much to get from the Conservation movement as well as much to give. The schools are learners as well as teachers. To support and further Conservation they will need to learn Conservation facts and doctrines. This Congress and American education are aiming at the same thing in the end—the betterment of American life. What shall it profit to conserve everything else on earth if we fail to conserve the spirit and fiber of our citizens, young and old? That is a view in which Conservationists and educators are fully agreed.

Now, what is our educational establishment, as it stands over against the body of our material resources? It is a group of State school systems, having in the aggregate a certain National character. We cannot insist too

strongly that education is primarily a concern of the States. This group of State school systems represents a combination of public and private agencies, for our State institutions are supplemented by many institutions privately supported and controlled. It represents an extraordinary unity as between elementary education and the higher education, as between the democracy of the lower schools and the science of the universities. It represents, moreover, in all of its grades, an everlasting devotion to intellectual and moral values, as having to do with enlightened citizenship. This is the educational establishment that faces the needs and aspirations with which the Conservation Congress is concerned. There are three or four ways in which I should like to speak of the great work of that establishment as related to your own great work:

1. In the first place, there is the fact that our scholastic education is facing about and turning its attention toward industry and industrial life. This is a new movement in which all States and sections are taking part. It is a change which is attended with the gravest difficulty. No one who is not familiar with the actual administration of schools and colleges can guess how hard a thing it is to introduce a new practice of teaching and make it successful at the hands of many teachers in widely different communities. Yet our educational leaders have addressed themselves to this task with courage and enthusiasm. In 25 States provision is now made for teaching agriculture in public schools. Such provision takes the form of agricultural high schools in Alabama, Massachusetts, Minnesota and Virginia, and in several other widely scattered States. In the best of these schools, there is arising a new interest in all that relates to the soil and the life on the farm. It is no uncommon thing to have class work interrupted by visits from neighboring farmers, who consult the expert teachers regarding drainage and fertilizers and the care of their horses and cows. The boys try out at school the seed corn they are to plant on the home farm, and the girls learn at school to raise poultry and vegetables and make from them appetizing dishes for home consumption. Large provision has been made for consolidated rural schools, and in Minnesota lands are added for instruction in the practice of farming. Oklahoma requires the teaching of agriculture in all public schools, with the cooperation of the normal schools and the agricultural college. This new instruction is spreading in unexpected ways. Columbia University, in the heart of New York City, has begun to offer

courses in agriculture, taking up this work where it left it off early in the nineteenth century. And an agricultural conference has been held at Bryn Mawr College. After that what more is there to be said! (Applause)

But there is still a good deal more. Much might be said about the new trade schools in the cities, and the new instruction in household arts for girls; but I pass these matters by and go back to the farm. What is especially interesting is the freedom with which new modes of teaching have been adopted. Corn contests, potato trains, demonstration farms—our old manuals of teaching knew nothing of these things. Then there is all manner of summer schools, short winter courses, farmers' institutes, and an assortment of other teaching devices. The University of Idaho is employing three field men, a horticulturist, a dairyman, and an irrigation and potato specialist, and is sending regular schools of agriculture about the State on wheels. In Virginia and three or four other States supervisors of rural schools have been appointed. They are making a close study of the resources, industries, and social needs of typical sections of their States, and are lending new life to the effort to make the schools more directly serviceable.

One of the earlier developments of this movement, and one that comes into peculiarly close relations with the Conservation campaign, is the setting apart of a day in each year for planting trees. Nebraska is looked upon as the original center of this movement. A recent report shows the planting of 20,000 trees in a single year in Minnesota, in connection with the Arbor Day celebration in this State (applause). The observance has received a fresh impetus in more than twenty of the States from the publication by the State education offices of attractive manuals offering suggestions regarding the celebration.

The leaders of the new movement in our schools have called for a redirection of rural education. Such a redirection is actually taking place. So much has been begun that it would be easy to believe that the work is done. There are many who suppose that this new education is already in the saddle and is moving triumphantly forward. But that is a mistake. Great changes in education are not brought about so easily. There is a long campaign and a hard campaign before us if the desired ends are to be attained. State superintendents of public instruction, those who are training

teachers in colleges and normal schools, and all who are engaged in this work in supervisory and teaching positions, will need for a long time to come the moral backing and the material support which this influential body can command. That is what they should have without reserve and without stint. (Applause)

The lack of well-prepared teachers of these subjects is one of the most serious difficulties the new movement has encountered. A recent report shows about seventy State normal schools offering regular instruction in agriculture. The Nelson Amendment to the Agricultural Appropriation Act of 1907 provided Federal funds for the training of teachers in the land-grant colleges. At least thirty of these colleges are now offering such instruction. But this work, too, is only begun.

2. And this suggests the second thing that I wish to say. The new movement is making a new demand for men in the business of teaching—strong men, technically trained for their work. If education is to help Conservation, the teaching profession must be enabled to compete with the industries in attracting and holding such men. We are considering both ends of our educational system, the scientific end in the universities and the popular end in the schools. A man who has enough knowledge and skill to train others for an industrial occupation has enough to give him a place in the industry itself. And the industry pays a great deal better than the teaching. It is not necessary that the income of teachers and that of industrial leaders should be equalized. Many men will continue to teach because they prefer to teach. But when the disparity becomes too great, many good teachers, in fairness to themselves and to their families, must give up the struggle and go over into the more lucrative employments. This is what has been going on in recent years. With a rapidly growing population and an increasing body of teachers, we have fewer men engaged in teaching than we had five years ago. We need opportunities in the teaching profession that will attract strong men to face the work before us (applause). I have the highest regard for the work of our women teachers; but both men and women are needed to give us a well balanced public education, and I welcome the alliance of the schools with the Conservation movement, because of the new demand it makes for competent men in the schools.

Let me point out some of the places in our scholastic organization where strong men are needed, for Conservation purposes as well as for educational purposes. It is generally understood that men of the largest caliber are in demand as presidents of technical colleges and universities. It should be equally obvious that such men are needed as State superintendents of public instruction. We have such men, and have had many such in the office of State superintendent—but in many of the States that office cannot attract men as do our college presidencies, because of the short term of service and other limitations with which it is hedged about. We need broad men and strong men as instructors in the technical departments of our higher institutions. Those who deal with our National resources industriously can know but little of the personal strain and sacrifice with which other men have stuck to their task of dealing with these same resources educationally. In our secondary and elementary education there is not only need of specially trained men as teachers, but there is need in particular of specially trained supervisors.

I was in Vermont not many days ago, and there I saw one result of a new law, which provides for the employment of union district superintendents of schools, at a respectable minimum salary. The State superintendent had called together these local superintendents in their annual conference. There were nearly forty of them, where three years before there was not one. Rather young men they were for the most part, though well-seasoned in the responsibilities of teaching. College graduates, alert and ambitious, they gave themselves over to the business which had brought them together, with a heartiness that was vastly encouraging. Other States have made provisions for a similar staff of supervisory officers. New York is one of the latest to take such action. The great States of the West, in which the county is a common unit of school supervision, need in their counties traveling supervisors of special subjects, particularly those relating to the practical business of life on the farm. Such supervisors can become veritable evangelists, bringers of good news concerning the things which make our National resources interesting and full of hope.

3. I have spoken of the new movement toward industrial education in our several States. I have tried to show that this movement is making only gradual headway against great difficulties, but that it can become a strong reinforcement of Conservation and of other public interests *if given a fair*

chance. Now, in the third place it should be said that the Federal Government is concerned with giving it a fair chance. We have no National system of school administration. We do not want such a system. No one seriously proposes to relieve the States of their powers and responsibilities in this matter. But how can the Nation be indifferent to the very stuff out of which it is made? While we have no National system of schools, we have and we are bound to have a National program of education.

It is no new thing that I am proposing. I would simply propose that the program blocked out and entered upon many years ago should be carried out and made as useful as possible. This National program is a simple one. In the earlier days it consisted in the granting of lands for educational purposes. Within the past half-century two additions to this earlier plan have been made. The first of these was the establishment of a central office of information, the Federal Bureau of Education; the second was the annual appropriation of Federal funds for institutions serving a special and urgent National need—the acts for the further support of the land-grant colleges.

Stated now in other words, our whole American scheme of public educational management consists of these four parts: First, the independent school and university systems of the several States, aided by grants of public lands and supplemented by privately managed institutions; second, the free cooperation of the States in educational matters of common interest; third, a Federal education office, aiding the States by its information service and furthering their cooperation; and finally, the distribution of Federal funds, under the supervision of the Bureau of Education.

Let me say a few words concerning that part of this plan with which I have personally the most to do. It is the business of the Federal Bureau to survey the whole field of American education, and make the best things contagious throughout that field. In such a subject as industrial education, it is to study our present needs in the large, and to set before our people the best examples of the successful meeting of such needs in this and in foreign lands. It is to promote unity of effort, by enabling every part of the country to profit at once by whatever has been well done in any other part of the world. As regards such a subject as the Conservation of our National resources, it is to take the broad view which concerns education in all the

States, and to further the common treatment of that subject as related to the geography, the history, and the industries of the American people. Such work as this it is now doing in a preliminary and fragmentary way; but it needs more men—expert and informing men—to make of its educational contagion the really large and transforming thing that these times demand. Give us the men, and we will give the help. When the Nation has made its program, it cannot afford to carry it out on less than a National scale. (Applause)

I have said that our National program already involves a measure of direct Federal aid to education in the States. There is every reason why such aid should be reserved as a last resort. But as a last resort, it has its place in our program. It is doubtful whether the industrial education which the Nation now requires can be adequately carried out without an increase of such Federal participation. But the point to be especially emphasized is this: Any such extension of Federal aid should be based on an accurate knowledge of the needs, and should be made in such ways as will strengthen and not weaken the educational systems of the States. For these reasons, a general investigation of the subject of industrial education in all sections of the country is one of the next things that should be undertaken by the Education Bureau. Such an inquiry has already been recommended from the office of the Secretary of the Interior. It has been urgently requested by the National Society for the Promotion of Industrial Education. Our neighbors of the Dominion of Canada already have a strong commission engaged in a similar inquiry. I earnestly hope that this Congress will call upon the Congress of the United States to institute such an inquiry at the earliest practicable date, and provide for carrying it on in a manner commensurate with the importance of the subject.

When I speak of our National program in education, it is with warmth and conviction. No nation can come to its greatest, industrially and politically, save as it comes to its greatest in education. We have in our American form of governmental relations the basis for the noblest educational structure that any nation has ever erected. In full loyalty to the true relations of State and Nation, we have only to go forward doing generously the things which may rightly be done, in order to have an infinitely varied yet gloriously united educational organization, in which our democracy, our science, and our nationality shall all of them come to their best.

4. Fourthly and finally, what kind of education is it that the new needs call for? I cannot leave the subject without saying a few words on that theme.

Our American schools and colleges have stood in the past for liberal culture. They have taken pride in doing so and they have believed that by so doing they have been serving the ends of democratic citizenship. American education from the beginning has looked the almighty dollar squarely in the face and passed on in serene devotion to spiritual ends. Is all of this to be changed with the new interest in industrial life? Is the technical, in other words, to take place of the liberal? I do not believe it. In fact, no greater calamity could befall our industrial interests. But we are undoubtedly changing our conception of what is liberal and what is technical. We may describe a liberally educated man as one who has learned so thoroughly how the whole world hangs together that he constantly sees his own interests only as related to general and permanent human interests (applause). A technical education, on the other hand, enables a man to do that which most men cannot do, but which has some useful relation to those general human interests. If this is a fair statement, there is no field in which a liberal education is more to be desired than that of our material resources and our industries; for this is the field on which the whole game and drama of human life is to be played, though there is no other in which the temptation to illiberal, narrow, and selfish views is so great. To make the material basis of human society itself a subject of liberal education is one of the greatest things that scholastic enterprise can possibly accomplish. The next step is to join the training for technical pursuits directly to our liberal culture thus broadly conceived, so that every citizen shall add some valuable skill to his more general attainments, and every special skill shall grow directly out of his general knowledge.

This, I believe, will be the great aim of American education everywhere. It is a high patriotic service to further such education. Even in the elementary schools, let our pupils learn that their private interests are to be advanced only in accord with more general interests, and that they are to make their success in life by doing some one thing well for which the world at large has need. We have been, according to our critics, a Nation whose resources were greater and more impressive than our civilization. With such an education as this, we shall be a Nation whose civilization shall overtop all of the natural goods that may ever be discovered or conserved (applause).

Such an education, moreover, could do much to overcome some of the chief obstacles which the Conservation movement now encounters; for it should give us a people who, from engineers and managers to farmers and miners, should not only be masters of their own trades but should pursue them with some positive regard for the public good (applause). Our education is not big enough and virile enough until it can deal with such great National issues as this. I am confident that it will come up to that high measure of power and efficiency, and that already it has begun to carry those larger responsibilities. (Applause)

President BAKER—Ladies and Gentlemen: Can there be higher patriotism than in the efforts of this Congress to protect the rights of all? Conservation is true patriotism; and Mrs Matthew T. Scott, President-General of the National Society of Daughters of the American Revolution, will now address you on this subject. (Applause, the entire audience rising)

Mrs SCOTT—Mr President, Ladies and Gentlemen: In behalf of the National Society of Daughters of the American Revolution, I wish to make my grateful acknowledgments to the Executive Committee (through its President, Honorable Bernard N. Baker) for its courtesy in giving to Mrs Amos G. Draper, the able Chairman of our D. A. R. Conservation Committee who has so splendidly inaugurated and developed this work, and to myself, the privilege and honor of taking part in these splendid exercises. In its last analysis the generic term "Conservation"—in its widest scope, and broadest sense—may be said to be the keynote and touchstone of our great D. A. R. organization. The finest brains and blood and nerve force of the land have been absorbed and found noble expression in various lines of work of the D. A. R. While the Daughters have turned their sympathetic attention to various material branches of Conservation work, we have not neglected the higher intellectual, ethical, and moral Conservation interests; we aim to help preserve the glorious heritage that has fallen to us of self-government, and hand down the birthright undiminished to those who come

after us that the priceless boon of "government of the people by the people and for the people" perish not from the earth. (Applause)

It has been borne in upon me of late that there are two Conservation interests whose importance we have not fully recognized, and they are the conservation of true womanliness, and the conservation of the supremacy of the Anglo-Saxon race on this continent. As to the former, the President of the United States in a recent address at Washington before the annual Congress of the D. A. R., said that woman's place and sphere are on too high a plane to be even discussed. It is surely an inspiration to have the privilege before this splendid assemblage of representing the great patriotic movement, which under the banner of the D. A. R., marches steadily forward, with ever increasing numbers, enthusiasm, prestige, and practical power.

The Daughters of the American Revolution in distinctive and especial ways have lent their organized strength to various good causes, which may all be practically considered as Conservation interests: among other objects, to social uplift, to patriotic education in its widest scope, to placing bounds to the abuse of child labor, to playgrounds, to juvenile courts, to improvement of hygienic conditions in our great cities, to preservation of historic spots and records, to the safe and sane celebration of July Fourth; and to cooperation with the S. A. R. in their noble work for immigrants landing upon our shores and subsequently for these foreigners and their children in the effort to Americanize them and to inoculate them with ideals and principles known in this twentieth century as Americanism.

Much has been done also among the mountain whites of the South. Every mountaineer, child or adult, that in our work we help to educate toward intelligent citizenship—and many of these mountaineers are of Revolutionary ancestry—is a barrier raised against the anarchistic tendencies and the unrest of our great cities; is a guarantee for the supremacy of the Caucasian race in America. Read, if you can secure it, Mr Thomas Nelson Page's plea for the education of the Southern Mountain whites in his magnificent address delivered at Washington before the last Continental Congress! We are also preserving, all over this broad land, landmarks of history—sacred relics of a vanished age—which are object-lessons for our own youth and for the strangers who crowd our shores.

Every monument we rear, every tablet we place, every statue we erect, every old fort or bastion, every Revolutionary relic or Revolutionary soldier's grave we honor, is a tribute to those to whom we owe the imperishable gifts of liberty, of independence, of the right to worship God in our own way. Every fountain or stone recording the trail of the pioneer, the priest, the trader, the soldier, or the devotion of the Revolutionary heroine, is a breath of incense wafted back to the immortals, an inspiration for "tangible immortality" for ourselves, and those who come after us. (Applause)

The Conservation of our natural resources is a subject of intensely practical importance to the D. A. R. Representing as we do the motherhood of the Nation, we feel that it is for us to see that the children of this and future generations are not robbed of their God-given privileges. It is our high privilege and mission to see to it that the future shall be the uncantered fruit of the past. The ideal democracy solemnly dedicated by the Founders, we, as their Daughters, declare shall not be forestalled. As women we cannot be silent and see the high ends at which they aimed made futile by the growth of a grovelling lust for material and commercial aggrandizement. This headlong haste for enormous gain, the total disregard of the future for the present moment, if not stopped will bring us to the condition of the Old World where the fertility and habitability of past ages have been destroyed forever. We feel that it is for us, who are not wholly absorbed in business, to preserve ideals that are higher than business—the outlook for the future, the common interests, and the betterment of all classes. The wasteful scrambling and greedy clutching at our natural treasures has made the present generation rich; but the mothers of the future must be warned by us lest they find that our boasted prosperity has been bought at the price of the suffering, of the poverty, and class war of our descendants. There is no lack of patriotic devotion in the country; but the mere thoughtlessness and inability or unwillingness of the commercial class to drop the interests of the moment long enough to realize how they are compromising the future—this hot haste and heedlessness, it is for us with our larger outlook, to restrain.

Women have already preserved a large National forest in the Pennsylvania mountains; the women of Minnesota have to their credit the Minnesota National forests; it was the women of California who saved the immemorial

groves of the Calaveras big trees. Our own work in behalf of the preservation of the Appalachian watersheds, in behalf of the preservation of historic sites, as well as the efforts being made by various women's organizations to preserve the natural beauty of the Palisades, of Niagara Falls, and of other precious scenic treasures of the Nation, are all steps in the right direction, are all preparation for the larger Conservation interests which the D. A. R. have begun actively to champion. It should be a second nature to women, with the spirit of motherhood and protecting care innate in them, to take an effective stand in the spirit of true patriotism—against the spirit of rank selfishness—the anti-social spirit of the man who declines to take into account any other interests than his own. (Applause)

There is another great world interest that is peculiarly our own as Daughters and descendants of the peace-loving patriots who took up arms a century and a half ago. They were not professional soldiers, but plain citizens hastily rallied together in often-wavering lines of defense of home and country. All the world wondered when at Lexington and Concord, on the village green and at the wooden bridge, the embattled farmers stood across the line of march of the British regular army, and fired "the shot heard round the world." It is the opening decade of the twentieth century of the Christian era; it is time that brute force—the recourse of primitive, barbaric man—cease to be the last arbitrament between great nations calling themselves Christian and civilized, and that the Conservation of peace be established by international arbitration. (Applause)

Again, it is one of the glories of our great organization that we are first, last, and all the time, considering the child. Today in all civilized countries the child is leading the way. I am happy to be able to say that through the instrumentality of our chapters in different parts of the country, interest has been awakened in homeless and dependent children; organizations have been formed for children of foreign birth to teach them respect for the flag, and some things about our form of government. Many chapters provide instructive lectures in their own language for foreigners, who listen eagerly. Many chapters offer prize medals for the best essays on historical subjects—American history especially—and for memorizing our National songs. Nothing is more important than our organized work for the "Children of the American Revolution"—children of American birth and descent—unless it be our work for the "Children of the Republic" in teaching to be American

citizens boys of foreign parentage who come to us with little idea of the difference between liberty and license. For patriotism consists as much in making good citizens as in saving the Nation from bad ones (applause). Every boy of foreign birth or extraction that we can help to transform into a thorough American through this magnificent branch of our work, every lad of foreign birth or extraction that we can help train to become a useful citizen and grow up into honorable manhood as a credit to his adopted land is an added asset to the ethical wealth of the country. Think for a moment what it means to help train these young foreigners in the plastic period of their life in the patriotic principles of their adopted country! A long stride has been taken in their patriotic and civic education, when through the exertions of noble women they have been given some idea of the great principles which are the basis of our form of government.

Another branch of our Conservation work which is especially near my heart, and which I think must be near to the heart of every mother in this broad land, is that in connection with the splendid crusade now being carried on against the evil of child labor. We have attempted, in dealing with this as with every other problem, first to obtain a wide and sure knowledge of the facts, and secondly to avoid everything savoring of the spirit of fanaticism in concentrating our energies on some great constructive policy. The committee on child labor, under the leadership of its noble chairman—the late Mrs J. Ellen Foster, whose life was dedicated to the needs of humanity—has made herculean efforts to bring this matter properly before the attention not only of the D. A. R. but of all the women of our land who are capable of responding to the pathetic appeal of suffering and stunted childhood, that we may wipe away this inexcusable stain on our National honor and this irreparable blight on that product which is more valuable than all the combined harvests of this fertile continent—the splendid American crop of human souls. (Applause)

If in a serener atmosphere than that of the politics of the hour, we as patriotic women can meet and help to solve these and other equally important problems in the eternally feminine way that has always given us power over men—if we would indeed, in the words of the old Athenians, help to transmit our fatherland not only undiminished but better and greater than it was transmitted to us, and if we are indeed unwilling to transmit to posterity mere material possessions unilluminated by divine ideals; if we can

but rise to the height and might of a pure, disinterested, passionless consecration to the principles which time has proved to be the soul of the purpose of the Fathers of the Republic, and on that high level, above the distracting personalities and passing incidents and accidents of the hour, "live and move and have our being" as a National Society, then we shall best establish and preserve the useful influence and leadership in the country to which we loyally aspire. Our interest and work for these great Conservation interests we cannot too often reiterate for our own encouragement and inspiration and for the enlightenment of the public.

As I said before, in the light of recent incidents and experiences, it has been borne in upon me that there are two great Conservation interests we have not yet sufficiently touched. With all the advance in learning, all the discoveries of science, all the enlightenment and uplifting of religion, all the refining of manners, all the acquisitions of men through invention and additions to the facilities for work and comfort of living, all the improvements of institutions providing for the farther and farther spread of well-being among the children of men, still, in the great underlying physical principles of existence, in the "main travelled roads" of humanity from birth to death, there is and can be no essential change. Nevertheless, there are an infinite number of variations and gradations in the product of these eternal operations of nature. Man's battle with nature—for human progress is a constant struggle against natural conditions, a continual re-making of the planet—has been ever accompanied, step by step, by the battle within himself against the contradictions in his inescapable heredity. It is the degree of success in this struggle for the triumph of the spiritual and the intellectual that marks the differences in racial types. Here then are the grand elements of the problem, the condition as well as theory confronting every well-wisher to humanity, every lover of her kind and her country, especially among women. For it is woman who is the divinity of the spring whence flows the stream of humanity—nay, she is the source herself. To her keeping has been entrusted the sacred font. In her hands rests the precious cup, the golden bowl of life. Holier than the Holy Grail itself is this chalice glowing ever, with its own share of the divine fire, its own vital spark from the altar of Almighty power. Never has this office of cup-bearer to creation placed greater responsibility upon woman than in this our own day, and this our own country. Freely we have received, and generously must we

respond; and deeply must we realize what a charge to keep we have—nothing less than the Conservation of the greatest experiment in enlightened self-government the world ever saw. Is that sacred trust to be jeopardized by untried, impracticable, uncalled for innovations upon the institutions of Government sufficing for the Fathers of the Country, and providing for its splendid development thus far? Shall we grasp at a shadow in the stream, like the dog in the fable, and drop the substance to sink away from us beyond recall? Is any *real* interest of the women of the land in danger? Is any *real* interest of women inseparable from the interests of the fathers, husbands, brothers, and sons of the women of the land? Is there any interest of women to be compared in vital importance to themselves, with the conservation of true womanliness?

I plead, as the representative of a great National organization of the women of the land, for the Conservation of true womanliness, for the exalting, for the lifting up in special honor, of the Holy Grail of Womanhood. But not merely the cup whence flows the stream of human life, must we guard and cherish; we must look to the ingredients which are being cast into the cup. We must protect the fountain from pollution. We must not so eagerly invite all the sons of Shem, Ham, and Japhet, wherever they may have first seen the light, and under whatever traditions and influences and ideals foreign and antagonistic to ours they may have been reared, to trample the mud of millions of alien feet into our spring. We must conserve the sources of our race in the Anglo-Saxon line, Mother of Liberty and Self-government in the modern world. I would rather our coming census showed a lesser population and a greater homogeneity. Especially do I dread the clouding of the purity of the cup with color and character acquired under tropical suns, in the jungle, or in paradisiacal islands of the sea alternately basking in heavenlike beauty and serenity and devastated by earthquake and tornado and revolution. (Applause)

I come of the old Virginia stock (applause) which first passed over the Blue Ridge and possessed the great Middle West, just in time to prevent it from becoming Spanish or French or British. Some of the pioneers of Washington's times have stayed on right there, in that eagle's nest of pure Americans where Kentucky, Tennessee, and Virginia meet in the mountains against which Cornwallis' previously invincible raiding column—after devastating the Carolinas—dashed itself to pieces, wiped out by volunteer

mountaineers in that wonderful battle of Kings Mountain which no general planned or even heard of until it was over. Personally, I would be willing to reduce our population-boast by many millions, had the remnant the unadulterated Americanism conserved to this day in these mountaineers' descendants! We may be destined to see our cup of liberty, which we have so generously proffered to the whole world, grow to the proportion of a grand mixing-bowl of races; but if so, will it not at least be wise to see that our own race dominate?

We, the mothers of this generation—ancestresses of future generations—have a right to insist upon the conserving not only of soil, forest, birds, minerals, fishes, waterways, in the interest of our future home-makers, but also upon the conserving of the supremacy of the Caucasian race in our land. This Conservation, second to none in pressing importance, may and should be insured in the best interests of all races concerned; and the sooner attention is turned upon it the better. (Great applause)

[Pending the foregoing, Governor Eberhart resumed the Chair.]

Professor CONDRA—Mr Chairman, Ladies and Gentlemen: At the instance of the President of the Congress, and inspired by the splendid address of Mrs Matthew T. Scott, President-General of the Daughters of the American Revolution and one of the most eminent of American women, I move that the Secretary of the Congress be empowered to prepare a suitable expression of the condolence of the Congress to be sent to the family of the late Mrs J. Ellen Foster, a member of the Executive Committee of the Congress and one of the most militant women of the country in behalf of Conservation.

The motion was seconded by several delegates.

Governor EBERHART—Ladies and Gentlemen: You have all heard the motion. As many as favor its adoption will please rise to their feet. [The entire Congress arose.] The motion is carried unanimously, and the Secretary will be instructed to forward the expression.

While the formal addresses of the women of the Nation to this Conservation Congress are now concluded, there is a little presentation which a lady of our State wishes to make; and in accordance with the instructions of the President of the Congress, I am pleased to introduce Mrs J. C. Howard, of Duluth. (Applause)

Mrs HOWARD—Your Excellency, and Ladies and Gentlemen: Mrs Scott asks me to present this certificate which I hold in my hand, for her and for the D. A. R., to a man whom we all delight to honor.

I used to live in Washington before I grew up and came to Minnesota, where I hope to spend the rest of my life; and there in my time I met many near-heroes and many heroes. I observed that modesty was always a sure sign of the real heroes; and if you had witnessed my efforts with Mr Gifford Pinchot to persuade him to come on the stage and stay there until I could give him this card, you would have no more doubt than before in which category he belongs (laughter and applause). Now, Governor, please don't let him get away while my back is turned (laughter), because I feel he really ought to have this certificate.

Ladies and Gentlemen, this certificate is a tribute by the D. A. R., in the form of a diploma, as you see; it says, in part,

He that planteth a tree is a servant of God. He provideth a kindness for many generations, and faces that he hath not seen shall bless him.

I have intense pride in presenting it to the man who is first in the Conservation war, first in peace, and first in the hearts of all tree-lovers.

[Mrs Howard here presented the certificate to Mr Pinchot amid great and prolonged applause, with cries of "Pinchot!" "Speech!"]

Mr GIFFORD PINCHOT—Mr Chairman, Mrs Scott, and Mrs Howard: There are two reasons, Ladies and Gentlemen, why I am profoundly moved, and delighted to receive this certificate: One of them is—and it is not a bit modified by the fact that you have so kindly, yesterday and today, given me far more credit than I deserve—that I would rather have the good opinion of

the women who are interested in Conservation than that of the men—by far (applause and laughter). The other is, that of all the organizations that have been working for the Conservation movement, for the preservation of the forests and for the extension of the same idea to all our natural resources, there has been none more devoted and more effective than the D. A. R. Besides, of all the women in the D. A. R., no one has been more devoted or more effective than Mrs Howard's mother, Mrs Draper (applause). And in this certificate I have joined together in my mind the kindness of Mrs Scott and the organization which she represents, the good-will of Mrs Draper which I very deeply prize, and that of her daughter, Mrs Howard, who was kind enough to give it to me; and I want to thank them all most heartily. (Great applause)

Governor EBERHART—When our friend Mr Pinchot comes here for the next Conservation meeting, after having seen all the charming ladies who have attended this Congress and worked in its interest, it is to be hoped that there may be still another certificate which he may have in his possession at that time (great applause). I am not saying this for the purpose of announcing any competition on the part of the ladies, but merely because Mr Pinchot himself suggested that he prizes this certificate so highly. But he would, I am sure, prize the other one still more if he got it (laughter).

Some time ago, when it became necessary to send a man of ability, honor, and integrity out West to prosecute land frauds, President Roosevelt looked quite a while before he could find the right one. The instruction under which that man went was that he should prosecute every guilty person, no matter what position in life he held, whether of high or low standing; and the man he sent was eminently successful. After successfully prosecuting those land frauds, he went to San Francisco and continued in the same work with equally great credit and distinction; so that in introducing him to you I am introducing the best-known, the ablest and strongest, apostle of clean citizenship in the United States, a man who stands for a square deal, and who believes in what is best and highest and truest and cleanest and purest in American citizenship. Ladies and Gentlemen, I have the honor and privilege of introducing to you that conserver of clean citizenship, who will

address you on the subject of "Safeguarding the Property of the People," Honorable Francis J. Heney, of California.

[Great and prolonged applause and cheers. Voices: "What's the matter with Heney?" "He's all right!"]

Mr HENEY (after asking an attendant to remove the water pitcher)—Mr Chairman, Ladies, and Gentlemen: As I never take water, I have requested that it be moved over to another table before I commence. (Laughter)

The efficiency of a democracy must ultimately depend on the intelligence of its voters. It was the recognition of that idea which caused the Fathers of this Republic to advocate so strongly the establishment of a public school system in this country. Any effort on the part of any public servant to prevent the voters of this country from having full knowledge of all its public affairs is, therefore, a species of treason, and any failure on the part of any citizen to acquaint himself as fully as possible with our National affairs is a failure to perform one of the duties and obligations which are imposed upon every member of a democracy. (Applause)

Public opinion, it is said, rules the Nation. It might better be said (because it would be more accurate) that public opinion in a democracy *should* rule the Nation; and it might further be said that if we had a real democracy, and a real representative government, public opinion *would* rule the Nation (applause). There are some evidences, however, that public opinion in this country does not have a free chance to operate. I need not mention many instances to convince you. Ninety percent of the people of the United States were opposed to men being permitted to make a profit by poisoning a people; they wanted a pure-food law, and yet it was locked up on the high shelf in Congress for sixteen years until Theodore Roosevelt, with the Big Stick, forced it out (great applause). What public opinion failed to do the Big Stick accomplished. (Renewed applause)

Now, my friends, public opinion should be intelligent; and that requires accurate information. A friend of mine, riding on a street-car in the city of Washington, at a time when the Ballinger-Pinchot investigation was going

on, saw two young men, beyond the voting age, reading the morning newspaper. They had a paper apiece. He was standing close by hanging on to a strap. He heard one of them say to the other, "They are having a great fuss up there in Congress over this Ballinger-Pinchot controversy, aren't they?" "Yes," said the other; "I see that Ballinger has been found three million dollars short in his accounts" (laughter). "Yes, I see that," said the first, "and that they found Pinchot has stolen a *million acres* of public land" (laughter). Whereupon both of them turned to the sporting column to see whether Johnson or Jeffries was predicted to win (laughter). They seemed to have a pretty accurate knowledge, also, of which club was ahead in the baseball game.

Now, my friends, that sort of misinformation is one of the diseases with which we are afflicted in this Republic, and I again call your attention to the responsibility of citizenship; and in that connection I congratulate myself, and I congratulate the Nation, that so many women are beginning to come to places like this, on occasions like this, to learn something about our National affairs (applause), because the future of this country is in the hands of the boys who are now growing up, and, perchance, the girls—who knows what may become of woman suffrage in the next generation? (Applause) Therefore, the more information the mothers have the better opportunity the Nation has of getting intelligent action from the voters.

The subject of my text today is "Safeguarding the Property of the People." Well, my friends, there are just two ways in which the property of the people may be safeguarded: one is by the Legislative arm of the Government, to whom the Constitution of the United States has entrusted the power of disposing of, regulating, and controlling public property; the other is the Executive arm of the Government, to which, under the Constitution, the power is entrusted of enforcing the laws which have been provided by the Legislative body.

Now, it must be apparent to any one that the most efficient Executive must fail in safeguarding the property of the people if the laws provided for that purpose by the Legislative body are loose, inaccurate, or unfitted to conditions. I want to make the charge plainly and unequivocally that, when we come (as we shall in a moment) to inquire into the safeguarding of the property of this Nation, we will find that all the despoiling of the Nation is

directly chargeable upon the Legislative branch of the Government, the Congress of the United States, to whom, under the Constitution, we gave the power of trustees.

In the first place, if unfortunately *our* representatives in the United States Senate—and I use the word "our" figuratively—if the representatives in the United States Senate from each State, respectively, are there in the interest of specially privileged classes instead of in the interest of the average, common man, it will follow that the Executive arm of the Government will be inefficient; and I have discovered that it *is* inefficient in the greater part of the West, where the greater part of the public property of the Nation lies—the Executive arm of the Government *is*, and since the Civil War has been the greater part of the time, utterly inefficient to safeguard the property of the people (applause). But I would be failing in my performance of duty if I failed to tell you why: It is because, while we have entrusted to the President of the United States the appointing of the United States attorneys for the different districts throughout the United States, a rule has grown up in the Senate of the United States which has in effect robbed the Executive of any real power in that respect, and has placed the appointing of such officials in the hands of the United States Senators from the respective States in which those districts lie. (Applause)

What is the result? The result is that if the lumber interests in a particular district are strong, because of having already succeeded in despoiling the people of a large part of their timber interests, they are apt to dominate the election of a United States Senator; and those lumber interests are also liable to dictate, through that United States Senator, the appointment of the United States officials whose duty it will be to enforce the laws of the United States against their benefactors. (Applause)

I would not dare to make such serious charges if I did not speak from absolute experience (applause). When I reached Oregon I found that situation existing in Oregon—indeed, I found on investigation before a grand jury that the then United States attorney was protecting certain men, who belonged to the higher-up class, from indictment, and that he had entered into a corrupt conspiracy with both the United States Senators from that State, by which they had agreed to have him reappointed United States attorney *upon condition* that these men should not be prosecuted (applause).

Moreover, I found that when the first stealing of timber commenced in Oregon and men were arrested for it, a man representing a big and influential timber company had taken to the railroad train about twenty-five men at Portland and carried them up to Salem and had them file openly on contiguous timber claims, each one swearing falsely that he was taking the timber for his own use; and when the matter was exposed immediately and the United States attorney took the matter before a grand jury and indicted the leaders who had instigated those men to go up and make the filings, influential State officials appealed to the United States Senators from Oregon to interfere, and appeals were sent to the Commissioner of the General Land Office and the Secretary of the Interior, so that finally the indictments were dismissed. Shortly thereafter about one hundred men filed on timber claims, under a contract to turn them over as soon as they were acquired, and again the influence of politicians and big business men brought about a failure of justice through an assistant United States attorney, who was the brother of the attorney representing the big interests who had hired these men to make the filings. Case after case of that kind came to my knowledge in Oregon; case after case of that kind has been brought to my attention in four or five other States. All of it can be traced back to the system under which we have been electing our United States Senators. (Applause)

Professor Hadley has well said that the fundamental divisions of power in the Constitution of the United States are between the voters on the one hand and the property owners on the other. That is the fight. That always has been the fight. That always will be the fight in this country. You heard, probably, all of you, that great address by the greatest citizen of the world, made in this hall the other day (applause), in which he outlined those conditions.

Now let us come back, for I want to show you wherein our trouble lies; and I want to show that great genius in railroad building (who is a citizen of your State, and who talked to you yesterday afternoon)—I want to show you and him who is responsible for the "extravagance and waste" of the great natural resources of this country. (Applause)

I have pointed out to you how big business controlled the execution of the laws in practically every place in the West—except, of course, Wisconsin,

Michigan, and Minnesota; in the early days when there was timber here none of these evils existed because these conditions didn't exist; your timber lands were not stolen in Minnesota, Wisconsin, and Michigan; you didn't have United States attorneys suggested by United States Senators who had been selected by owners of large timber tracts or railroads. Some States in the Union have suffered from that, but you never had any such thing come home to *you* (laughter). I congratulate you (renewed laughter). The Nation has had in its possession, owned in common by all of us and our forefathers, 1,800,000,000 acres of land. That is *some* property (laughter); that is more than either you or I possess today (laughter). And that included all of the present Rockefeller oil possessions, it included all of the Northern Pacific's land-grant possessions, it included all of the great anthracite companies' coal possessions, it comprised all of the millions of acres of timber land throughout the United States, including what there was in Minnesota. It belonged to you and me and our fathers and mothers and sisters and brothers. We were pretty rich at that time. We *could* have held on to it and developed it, because I can't believe that if we had offered to pay a patriotic citizen like James J. Hill the sum of \$50,000 a year to build a railroad for us from Lake Superior to Puget Sound and to furnish him the money with which to build it, that he would have refused the job (applause); even had he considered it inadequate compensation for his great ability, his patriotic love of the people of the United States would have led him to do it. (Great applause and cheers) In talking with a banker the other night—one of the Big Four of New York—I asked him if in his opinion Mr Harriman, in the gigantic operations performed by him, was influenced by love of money and the desire to gain filthy lucre, or whether he was influenced by the great gratification of achievement, and he said undoubtedly by the latter; that Mr Harriman would have combined all these railroads for the people of the United States on a salary of \$50 a month, if we didn't want to give him any more, just for the pleasure of doing it. (Laughter and applause) But we have received misinformation, and are receiving it yet, to the effect that there are no patriots in the United States; that no man is willing to develop our coal or our oil or our iron or our water-power or anything else that is left unless we give him everything in sight. (Laughter and applause)

My friends, the way the people of the United States have been treated in regard to this vast property which we owned reminds me of a story I heard about a man down South—a white man. He was going along the river in flood time in the back country, and the river was full of floating logs and refuse and all sorts of timber, and he saw a nigger sitting on the bank—and will you pardon me for using the word "nigger" instead of "colored man," because I have just been making a visit down in Virginia and I suppose I fell into it (laughter); it is not meant as a term of reproach, nor is it used as such there or here—and seeing this negro sitting on the bank, he said to him, "Sam, what are you doing?" "Nothin', Suh." "Whose boat is that?" "That's mine, Suh." "Well, Sam, let me tell you what I'll do; you take your boat and go and haul those logs out of the river there, and I'll give you half of all you get on shore." (Laughter)

It took a little while for that to sink in (laughter). It has taken you forty years to let this railroad proposition sink in. (Laughter)

Right while I am on it, while it is fresh in my mind and in yours: Mr Hill says, "We have been extravagant." Why, my friends, do you know what we gave to Mr Hill? I say we "gave" it; as a matter of fact, we weren't consulted (laughter); we didn't have a referendum on it (laughter and great applause). We gave the greatest land-grant ever given to an individual or a corporation in the history of the world—sixty millions of acres; when I say to Mr Hill, of course I mean the Northern Pacific. We gave outright a strip of land 2000 miles long, 20 miles wide in the States and 40 miles wide in the Territories! Worse than that: instead of giving it in a solid body, we gave every even section, so that in timber lands it carried an immense advantage over anybody else coming in from the outside. Now, it is easy to demonstrate, and I hardly believe Mr Hill would care to deny it—and if he does, I'll get the figures and demonstrate it (applause)—that this land-grant was worth, at a fair figure, ten dollars an acre at the very least. That is six hundred million dollars (applause) of *our* property that we "extravagantly and improvidently wasted," as Mr Hill would call it; and I agree with him. (Laughter and applause)

But what does that mean? Why, the road is 2000 miles long; \$50,000 a mile on an average for the entire road is a very fair figure as the cost of it, making, if I calculate correctly, \$100,000,000, to build it. Let's double that,

and allow \$100,000 a mile for the 2000 miles; that certainly would build and equip the road. That is two hundred million dollars. And we gave six hundred million dollars worth of land, and the railroad was built and now wants *forever* to charge you rates—upon how much of a capitalization? Well, I don't know. But four hundred million dollars profit! Why, that would more than build the Panama Canal—and I wonder that some private corporation didn't do *that* (laughter). It *would*, undoubtedly, if we had been willing to give to it all of the remaining 700,000,000 acres of land that we have left—including Alaska, with the coal mines that Guggenheim wants (laughter and applause). We *have* been "improvident"—or somebody has—with the property of the people.

Now, who was so improvident? Why, Congress; because the Constitution places in the hands of Congress the power to dispose of, regulate, and control the property of the United States; and Congress did it—and *did* us, too (laughter and applause). But not satisfied with that, Congress gave to the Southern Pacific, the Central Pacific, and the Union Pacific 120,000,000 acres more of our inheritance, which we purchased with both blood and money—because the war with Mexico led to a part of the purchase, in which thousands of American citizens were killed, and thousands of American women widowed, and thousands of American children orphaned, while we put fifteen millions of *our* money—our common pot—into the purchase on top of that human blood; and then we "extravagantly and improvidently" gave it away. (Applause)

Not satisfied with that, when we commenced to realize that it was necessary to save the forests of this country—some of the forests which were left—Congress again passed an act, in 1907, called the New Land Act. In 1891 it had passed the law authorizing the President to create National forest reserves. At the same time it had passed a law authorizing the States to select new lands for the school sections which might be included in the National forest reserves. A gentleman in California by the name of Frederick A. Hyde, and another gentleman (who is since dead, and who served a year in jail, just before his death, for defrauding the United States), were actively operating in the State of California in school lands. Now, don't get the idea in your heads from what I have been saying about the way Congress has handled the lands and property of the United States that I am in favor of turning over to the States the power to handle any property in the

hope that it will be better handled, because there, again, my experience teaches me that it will be worse—if possible (laughter and applause). Well, under that law of 1891, Hyde and his companion adopted this system: Where they found that school lands were in reserve (they had a man in the Surveyor-General's office who was looking out for them), they would go down and get bootblacks, and saloon barkeepers, and Tom, Dick, and Harry to sign an application for school lands—under the law of California 320 acres—the law requiring that in making his filing the applicant should swear that he was taking it for his own use and benefit and not for speculative purposes. And at the same time that Mr Bootblack signed the application, he would sign a transfer of his interest, a conveyance of the land, with the date left blank; and a very agreeable notary public would put his seal and acknowledgment upon the affidavit and the assignment, despite the blanks and the absence even of any description of the lands in the application. Then, when Mr Hyde had one or two hundred of these, he would go and take up all those school lands, and have the agent of the State thereupon locate all of these school lands in a body in the finest forest he could find in California—some of the finest that ever grew on earth are there, trees two and three hundred feet high, sixteen to twenty feet in diameter, cutting so many millions of feet to the quarter-section that it would astound even a Minnesota lumberman unless he had been out there and seen it; and those magnificent virgin forests would be separated from public ownership by our "extravagance"—and this, mark you, through Congress passing the 1891 law for the benefit (?) of the schools of the State so loosely drawn that speculators could take advantage of it in this way. So the virgin forests went into private ownership; and Mr Hill will tell you, "What of it? Doesn't that develop the country?"

Why, my friends, they didn't even put the patents on record, because the tax collector of the county would put them on the assessment roll if they did (laughter). And so they grabbed millions of acres, that they had no idea of using in the present; they were holding it for the profit which would come from scarcity of timber through the waste and use which is going on. Why, people living in the very neighborhood of the timber grabbed don't know that it has passed out of Government ownership! And yet those are some of the people who have been living "extravagantly." I believe that some of

them wear shoes that cost the high price of a dollar, and eat bacon that is four-fifths fat. (Laughter and applause)

Let me tell you that extravagance is largely a matter of trying to copy after the Higher-ups. No nation was ever destroyed until it had a large leisure class to set a bad example (applause) in living to the common people; and this Nation has a leisure class which is rapidly growing, and which is more wealthy than any leisure class ever known to the world, civilized or barbarian. Why? My friends, *solely* because Congress has by bad laws permitted all this vast property of the people to get into the hands of the few (applause). There is not a fortune in this country today large enough to be a menace to the liberties of the common people which has not been acquired by despoiling the people through legislation that was either corrupt or the result of such ignorance that it ought to be punished as criminal negligence, or else through unfair discrimination made by common carriers giving one man an advantage over his competitors. (Applause)

Now, I haven't time to finish—I am afraid I have overstepped my time already—(Voices: "Go on, go on," and applause) but I want to "go on" just a little longer (laughter and applause) because I have something on my mind that I want to put on yours. (Laughter)

We didn't lose our great inheritance until after the Civil War. Practically all of the rapes of this Nation by Congress have been committed since the Civil War, and every land law which Congress has placed upon the statute books since 1860 has been vicious—absolutely vicious—in its tendencies, and the Commissioner of the General Land Office and the Secretary of the Interior have constantly, every year, told Congress about it in printed reports and begged and urged Congress to change the laws: *and it has refused to do it!* (Applause)

Of course all members of Congress are not to blame for that; because this fight which Hadley says is going on always, and always will go on, in the division of power fundamentally between the voters and the property owners, has resulted in the property owners having more representatives in Congress than the people ever had. (Applause)

Now, I am not here to abuse anybody. I heard a man tell a homely story last night that went directly to my heart; it's exactly in line with what I think

about most of the men who are responsible for the present condition; I don't say these men are bad, but only that they have a wrong viewpoint—and that was illustrated in the story. This gentleman said that one day his boy brought home a fox-terrier. They had poultry at his home, some brown leghorns and some white chickens. This fox-terrier had been born and raised on a ranch where they had nothing but brown leghorns, and consequently when he went out in the chicken-yard and saw the feed thrown out he rushed out immediately—of course, without being told to do it—and weeded out the white chickens from the brown leghorns and drove them away from the feed and let the brown leghorns have it all (laughter). Now, it wasn't the fault of the dog that the white chickens lost their feed (laughter); we mustn't blame him; that had become second nature, from what we would call, speaking in reference to human beings, *environment* (laughter and applause); and it's a rare dog who can discover for himself that the white chickens ought to have an equal right with the brown leghorns to get *some* of the feed. (Laughter and applause)

When, after the Civil War, business commenced to swing with great strides in this country, owing to the great inventions in machinery, the discovery of the cotton-gin and so many other things that we can't stop to enumerate them, and the growth of the use of electricity in later days, a few men commenced to see business enlarge—and they were not the men who fought in the War, but the men who remained at home and reflected (laughter and applause). Some of them were like the man pictured in one of the illustrated papers where there was a cartoon of Thomas Jefferson signing the Declaration of Independence, with one of the imaginary corporation men of the day—a Tory—rushing in through the door and saying, "Hold on, Thomas, don't sign that document; it'll hurt *business*" (laughter); and these men said, "Let's stop this War, it's hurting business." And there were others who thought the War *made* business, though that was before they had commenced to can beef (laughter). Then after the War, when the men who had made the fight for human liberty and the continuance of equal opportunities in this country came home and went to work, they went ahead satisfied to make a living for their little families in the best way they could, while these *business men* who had remained at home had discovered that *if* a man can get possession of those natural resources which can be turned into energy—the energy which drives

modern machinery, which can do the work of human hands—he can sit back and fold his arms and say to the eighty million people in the United States, "Go ahead; when you want energy to run your machinery, you'll have to come to me and buy it; when your money is gone the eighty millions of you will have to work for me; and when you get to be one hundred and sixty millions, you'll still have to work for me." Now, it requires some imagination to see that, but it is just as fundamentally true as that the earth is spherical—flattened at the poles, as Cook tells us (laughter); and Peary corroborates it. (Laughter)

Let me explain; because I want you to take home something, besides figures, that you will remember. When a man in the old days, when they had no machinery, employed four or five men, he commenced to be a business man; and when he began to put profit in his pocket—even at the rate of only ten cents a day for the labor of each man working for him, if he had five men he was making a clear profit of fifty cents a day, and if he had fifty men the profit was five dollars a day—he got on the road to "big business." If he could have five hundred men and could make fifty cents a day off the labor of each one, he would be making two hundred and fifty dollars a day; and if he could have factories spread out over the United States in which he had an aggregate of ten million men working for him—as in shoe factories when they made shoes entirely by hand—and could make fifty cents a day off each of the ten million men, he would make five million dollars a day. The figures stagger us. Now, with machinery you can take coal, oil, timber, gas, or water-power—those are the energy-creating natural resources—and make machinery run with them; and if you own enough of those energy-creating natural resources to be equivalent to the labor of ten million men, and apply it to the right machinery, you can compete with the man who has ten million slaves to work for him and does not possess this other energy—and you can do better than merely compete, because your water-power doesn't wear out shoes at the toes nor coats at the elbows nor trousers at the knees; so, my friends, the man who owns the water-power is a greater slave-owner—has more energy that can be turned into wealth—than all the planters who owned the colored men of the South.

Now, at the time of the Civil War we didn't understand this great power and the importance of preserving it in the ownership of the people—because it all belonged to us then. There is available—so the report of the National

Conservation Commission says—37,000,000 horsepower in the streams of this country. What does this mean? Why, my friends, the energy expended by an average draft-horse working eight hours a day is equal to only four-fifths of the unit horsepower, as we use it in speaking of water-power, so that it would be equivalent, for an eight-hour day's work, to more than fifty-four million average draft horses. Now, machinery used to be driven by man-power before the draft horse was made to work in place of the man; that was what they did in the old tread-mill before the discovery of steam, which has only been in effective use about a hundred years; and in man-power, what does the forty million horsepower available immediately for use mean? You don't conceive of it, I am sure. A horsepower is equal to the work of at least ten men, and forty million horsepower would be equal to the work of 400,000,000 men! Why, all the people in the United States today are only 90,000,000, including babies. Four-hundred-million-of-men power! And just as sure as the sun will rise, if we permit that to go into perpetual ownership of individuals, the day will come when one corporation will own it all and one man will dictate and dominate that corporation (applause). If you want this country to have material progress at the cost of human liberty, let this source of energy slip out of your hands (applause); but if you want to hold on to any kind of a chance for your children and children's children to have equal opportunities like yours, then follow the policies laid down by Theodore Roosevelt the other day in regard to those energy-producing resources—coal, oil, gas, and water, as well as timber—and this country will be so great that all earlier history will never have told of such progress as the human race will make within these confines. (Applause)

It seems to me that we all ought to be able to realize that no human being in the short space of a lifetime can have earned a hundred million dollars—he cannot have given an equivalent to mankind for \$100,000,000; and when we see the example set by some of these great captains of industry who go over to Monte Carlo and risk a fortune on one bet and one turn of the wheel, and come back to this country and talk about their great benevolence, and then find that the Pittsburg "Survey" found conditions of human life at their workshops so low that it is bound to degrade and pull down the human race—surely it is time to stop and consider. (Tremendous and prolonged applause)

My friends, we must have more democracy in this country (applause). I know this is no place to talk politics, and I am not here for the purpose of talking politics in a partisan sense; but the Conservation of the natural resources for the benefit of the human race—not only the people of the United States—is of such transcendent importance that it rises above all parties and all men (great applause). Why is it that some of these men who have profited by our mistakes and our improvidence in the past are fighting against this Conservation movement? Is it because they fear that we will fail to develop the country rapidly enough? No! Every true Conservationist believes in developing the country rapidly as possible. But we realize the danger, the menace to human liberty, that lies in *parting with the fee title* to all these great energy-producing natural resources; and if we can arouse the people of the United States to a realization and understanding of this question—which, after all, is simple when we get down to it—there will be such a wave of insurgency sweep over this country as will drive the representatives of the special interests out of every public office in the Nation. (Great and prolonged applause and cheers)

Now, in order to illustrate what I have said about what these people—or Congress—have done and failed to do, I must draw your attention to the fact that under the Timber and Stone Act, 13,000,000 acres of the finest timber in the world have been extravagantly and improvidently disposed of and lost to the people through a vicious Act of Congress, and have gone largely into the hands of a few owners; for the repeated reports of the Secretary of the Interior—even the present Secretary, Mr Ballinger—show that ten of the thirteen million acres are in the hands of a few individuals and corporations. Ten million acres! Why, that is equal to two of the smaller eastern States. In 1878, the then Secretary of the Interior, immediately after the Act was passed, said in his report for that year (Report of Secretary of Interior, 1878-1879, pp. XII-XV):

While no legislation applicable to all parts of the country with regard to this subject was had, two bills of a local character were passed, one "Authorizing the citizens of Colorado, Nevada, and the Territories to fell and remove timber on the public domain for mining and domestic purposes," and one "For the sale of timber lands in the States of California and Oregon and in Washington Territory."

In the opinion of the Commissioner of the General Land Office, which is on record in this Department, these two acts are more calculated to *hasten the destruction* of the forests in the States and Territories named than to secure the preservation of them.

Of this act the Commissioner of the General Land Office, in a letter addressed to the Secretary of the Interior, expresses the following opinion:

"It is a fact well known that while almost all the timber-bearing land in those States and all the Territories, except Dakota and Washington, is regarded as mineral, only a small portion is so in reality. The effect of this bill will, in my opinion, be to prevent the survey and sale of any of the timber lands, or the timber upon the lands, in the States and Territories named, thus cutting off large prospective revenues that might and should be derived from the sale of such lands or the timber upon them. It is equivalent to a donation of all the timber lands *to the inhabitants of those States and Territories*, which will be found to be the largest donation of the public domain hitherto made by Congress. This bill authorizes the registers and receivers of the land offices in the several districts in which the lands are situated to make investigations without any specific directions from the Secretary of the Interior or the Commissioner of the General Land Office, to settle and adjust their own accounts, and retain from the moneys coming into their hands arising from sales of lands such amounts as they may expend or cause to be expended. This method will be found exceedingly expensive and result in no good. Experience has shown that the machinery of the land offices is wholly inadequate to prevent depredations."

The "Rules and Regulations" issued in pursuance of the first section of this act are to be found in the report of the Commissioner of the General Land Office, herewith presented. These rules, drawn up with a view to and the intention of preserving the young timber and undergrowth upon the mineral lands of the United States and to the end that the mountain sides may not be left denuded and barren of the timber and undergrowth necessary to prevent the precipitation of the rain-fall and melting snows in floods upon the fertile arable lands in the valleys below, thus destroying the agricultural and pasturage

interests of the mineral and mountainous portions of the country, make it the duty of registers and receivers to see to it that trespassers upon timber lands, not mineral, be duly reported, that upon mineral lands only timber of a certain size be cut, and that young trees and undergrowth be protected, and that timber be cut only for the purposes mentioned in the act. These "Rules and Regulations" will be enforced with all the power left to this department to that end, in order to save what may be saved. But I deem it my duty to call attention to the fact that, as set forth by the Commissioner in the letter above quoted, the machinery of the land offices is utterly inadequate to accomplish the object in view.

After a careful consideration of the above-named Act and its probable effects, I venture the prediction that the permission given the inhabitants of the States and Territories named therein, to take timber from the public lands in any quantity and wherever they can find it, for all purposes except export and sale to railroads, will be taken advantage of, not only by settlers and miners to provide economically for their actual current wants, but by persons who will see in this donation a chance to make money quickly; that it will stimulate a wasteful consumption beyond actual need and lead to wanton destruction; that the machinery left to this Department to prevent or repress such waste and destruction through the enforcement of the rules above mentioned will prove entirely inadequate; that as a final result in a few years the mountain sides of those States and Territories will be stripped bare of the timber now growing upon them, with no possibility of its reproduction, the soil being once washed off from the slopes, and that the irreparable destruction of the forests will bring upon those States all the calamities experienced from the same causes in districts in Europe and Asia similarly situated.

It appears to me, therefore, that the repeal of the above-named act, and the substitution therefor of a law embodying a more provident policy, similar to that of the above-mentioned Senate Bill No. 609, is in the highest degree desirable. If the destruction of the forests in those States be permitted, the agricultural and pasturage interests in the mountainous regions will inevitably be sacrificed, and the valleys in the course of time become unfit for the habitation of men.

The act for the sale of timber lands in the States of California, Oregon, and Nevada, and in Washington Territory, passed by Congress at its last session, is, in a letter addressed to this Department, commented upon by the Commissioner of the General Land Office, in the following language:

"It is a bill of local and not general application to the timber lands of the United States, and adds one more to the already numerous special acts for the disposal of the public domain. The price fixed is too low, as much of the land is worth from five to fifty dollars per acre.

"Under the provisions of the bill the timber lands will, in my opinion, be speedily taken up and pass into the hands of speculators, notwithstanding the provisions to prevent such results. The soil should not be sold with the timber where the land is not fit for cultivation. Only the timber of a certain size should be sold, and the soil and young timber retained with a view to the reproduction of the forests. The bill should have limited the sale of the lands to persons who have farms and homes within the State or Territory, and it ought to have required the purchasers to show affirmatively that they had need of timber for domestic uses."

No less emphatic were later recommendations for repeal or amendment of the Timber and Stone Acts (Report of Secretary of Interior, 1879-80, p. 27):

In my last annual report I discussed the inadequacy of the laws enacted by the last Congress "Authorizing the citizens of Colorado, Nevada, and the Territories to fell and remove timber on the public domain for mining and domestic purposes," and providing "for the sale of timber lands in the States of California and Oregon and in Washington Territory." The opinion I then ventured to express, that the first of these Acts would be taken advantage of not only by settlers and miners to provide economically for their actual current wants, but by persons who see in this donation a chance to make money quickly; that it would stimulate a wasteful consumption beyond all actual need and lead to wanton destruction, and that the machinery left to this Department to prevent or repress such waste and destruction through the enforcement of the rules to be made by the Commissioner of the

General Land Office would be found insufficient for that purpose, *has already* in many places *been verified by experience*; also the predictions made by the Commissioner of the General Land Office with regard to the effect of the second one of the above-named acts. Referring to what was said about these laws in my last annual report, I repeat my *earnest* recommendation that they be *repealed*, and that more adequate legislation be substituted therefor.

It is by no means denied that the people of the above-named States and Territories must have timber for their domestic use as well as the requirements of their local industries. Neither is it insisted upon that the timber so required should be imported from a distance, so that the forests in those States and Territories might remain intact. This would be unreasonable. But it is deemed necessary that a law be enacted providing that the people may lawfully acquire the timber required for their domestic use and their local industries from the public lands under such regulations as will prevent the indiscriminate and irreparable destruction of forests, with its train of disastrous consequences. It is thought that this end will be reached by authorizing the Government *to sell timber from the public lands principally valuable for the timber thereon, without conveying the fee*, and to conduct such sales by Government officers under such instructions from this Department as will be calculated to *prevent the denudation of large tracts*, especially in those mountain regions *where forests once destroyed will not reproduce themselves*. I have no doubt that under such a law, well considered in its provisions, the people of those States and Territories would be enabled to obtain all the timber they need for domestic as well as industrial purposes at reasonable rates, and that at the same time the cutting of timber can be so regulated as to afford sufficient protection to the existence and reproduction of the forests, which is so indispensable to the future prosperity of those regions. I venture to express the opinion that the enactment of such a law has become a pressing necessity, and cannot much longer be delayed without great and irreparable injury to one of the most vital interests of the people. I therefore again commend to the consideration of Congress the bill introduced as Senate Bill No. 609 in the *last* Congress:

"The last clause of the second section will permit any person applying for a tract of timber land and securing a certificate from the Register to sell his right and interest therein *immediately*, and the purchaser, although it may have been obtained by perjury, may be entitled to a patent for the land.

"Section 5 provides that any person prosecuted under Sec. 2461 of the Revised Statutes of the United States, may be relieved of the penalty by the payment of two dollars and fifty cents per acre for the land trespassed upon. This is objectionable, for the reason that the penalty fixed is altogether inadequate, and does not require the payment of costs of prosecution, which are often greater than the penalty to be collected. It should require that the trespasser should pay for the entire subdivision trespassed upon.

"There can be no doubt that if this bill becomes a law it will be taken advantage of, by persons who want to make money quickly, to acquire the timber lands under its provisions at a very low price, and strip the mountain sides of their forest growth as rapidly as possible. How disastrous such a result will be to these States and Territories need not be detailed here."

My friends, every report from 1878 down to the last report this year, tells Congress exactly the same thing, and begs and urges Congress to repeal this Timber and Stone Act. Not only that; every report goes on and tells that large tracts are being stolen and taken fraudulently, and Congress is urged for that reason to repeal it and make a different rule in regard to the sale of the timber, not to hold it but to sell the timber off the land letting buyers take the mature growth, and replanting and reforesting so that the timber will always be there; and Congress failed to act until 1892, fourteen years later. After the above reports went in, with a report of the same kind every year for fourteen years, then, in 1892, with a report before them at the time to the same effect, Congress *extended* the Timber and Stone Act to take in Montana and some other States. *Who got them to do it?* The great amalgamated copper interests are in Montana, and the great smelting interests there wanted *timber*—that belonged to us, and that they could well afford to pay for—and they wanted to get it under this vicious Act, and they *did* get it under this vicious Act; and indictments followed only a short time

ago, but there was failure of proof although everybody knew who was guilty (applause). And, my friends, the Act of Congress in extension of the vicious law, with all these reports before them, cannot be accounted for upon any other theory than that the people of the United States have a minority of representatives in both branches of Congress (applause). Now, after the extension, the adverse reports commenced to come in again; and they have been followed up every year down to the present year, yet that Timber and Stone Act still remains on the statute books unamended and unrepealed! *How can you account for it? I'll tell you how. Why, there is still some timber to be stolen!* (Applause)

Now, I have taken altogether too much of your time. I have not been able to present this matter as satisfactorily to myself as I would have liked on account of the limitation of time—I suppose most of you are glad of that. (Voices: "No, no, no; go on!") I can't go on; it wouldn't be fair to other gentlemen who are here to speak, especially to Mr Gifford Pinchot who is to talk to you immediately after I conclude, and I know you want to hear from him (applause). But I want to say to you that the fight to prevent our natural resources from getting into private ownership is a war that will have a greater influence upon the future of the human race than even the great Civil War in this country had (applause); and I want to say to you, further, that I have enlisted in that war as a private soldier (applause, and a voice: "We'll make you the leader!") for the full term of my natural life. (Great applause)

Governor EBERHART—The next subject for consideration is "The Conservation Program"; and I wish that time would permit me to say some of the fine things I would like to say about the speaker. I will say just one thing: A short time ago I was in the Belasco Theater in the city of Washington and the question of Conservation was up, and this man stood on the rostrum and said to that vast congregation that the time had come when we must forget personalities and men, and work for principles—that it was time for every man interested in the welfare of the Nation to come forward in this Conservation work, forgetting the past, and forget all personal prejudices and jealousies, and work for this one movement; and at

the close of his address he was given such an ovation at the hands of that gathering as he has frequently received here. It is not necessary for me to formally introduce him; you know him as the best friend of our forests—Gifford Pinchot. (Great applause and cheers)

Mr GIFFORD PINCHOT—Governor, Ladies and Gentlemen: I am not tired of receiving your kindness, but I wonder if you are not tired of receiving my thanks! I *do* want to thank you most earnestly for all your kindness; and I have wished all along that one person who has made the fight with me could be here, and that is my Mother. (Great applause)

I shall have to read a good deal of my paper to you tonight, because there are some things I want to say more exactly than I otherwise could; but I will read just as little as possible.

Like nearly every great reform—and Conservation *is* a great reform—the Conservation movement first passed through a period of generalities, general agitation and general approval, when all men were its friends; and it hadn't yet really begun. You have all noticed that when a minister in church makes a general arraignment of wickedness, no particular sinner seems to care very much—it passes over his head, or he applies it to the other fellow; but when he comes down to particular cases, and the special shortcomings, the special desires, the special impulses which control each one of us, begin to be the subject of his oration, then there is a very different situation. Now, it was just so with the Conservation movement. At first everyone approved it, because it touched no one nearly; then it passed into a period of practical application, out of the sweep of the generalities, and at once the men whose particular interests were threatened began to take an active interest in the question, and the opposition began; and with that opened the second period of the Conservation movement.

When this fight began, it was found that the people believed in Conservation all over this Nation, and that fact had to be taken into consideration by the people who were opposing the movement. When there is a general movement of which all men approve, the regular way in which

the attack is made upon it is to join in the approval and then get after the men and the methods by which the general proposition is being carried out. So, now we find that the desire of the opponents of Conservation—and there are not so very many of them in numbers—is not at all that we should abandon the principle of making the best use of our natural resources; they do not urge that we should abandon the ideas of doing the best thing for all of us for the longest time; but the soft-pedal Conservationists *do* demand that Conservation shall be safe and sane. Safety and sanity, in the meaning of the men who use that term most as applied to legislation, means legislation not unfriendly to the continued domination of the great interests as opposed to the welfare of the people (applause); and safe and sane Conservation, as that expression is used by those same men, means Conservation so carefully sterilized that it will do no harm to the special interests and very little good to the people. (Prolonged applause)

I take it, of course, that every friend of Conservation is fully and heartily in sympathy with safety and sanity; that goes without saying, for if there ever was a prudent, safe and sane program, it is that of the Conservation movement, expressing a prudent, safe and sane spirit, and intention as well. But we must never forget that safety and sanity from the point of view of the men who are advocating Conservation—from the point of view of a great gathering like this—means that, first, last, and all the time, the interests of *all* the people shall be set ahead of the interests of any part of the people. (Applause)

Among the things that have been charged against the Conservation movement is this, that Conservation does not know what it wants—that the Conservation movement is an indefinite striving after no one knows exactly what. I want to tell you, on the other hand, that the Conservation program is now, and has for at least two years been a definite concrete attempt to get certain specific things; and that the impression which has been made, or has been sought to be made, that we didn't know what we were after, is wholly misleading. (Applause)

The Conservation program may be found, most of it, in the following reports—the report of the Public Lands Commission of 1905; the report of the Inland Waterways Commission, March, 1908; the great Declaration of Principles adopted by the Governors at the White House, in May, 1908—

one of the great documents of our history; the report of the Commission on Country Life, January, 1909; and the Declaration of the North American Conservation Conference, February, 1909. By the close of the last Administration, the Conservation program had grown into a well-defined platform, and the only important addition of more recent date is a clearer understanding—and we have now a very clear understanding—that monopoly of natural resources is the great enemy of Conservation, and that monopoly always must depend on the control of natural resources and natural advantages of a few as against the interests of the many. (Applause)

None of the men, so far as I know, who are engaged in the Conservation movement, took hold of that side of the fight because they wanted to. I can say, for myself at least, that it was not until I was forced into it by experience that I could not doubt, by being defeated over and over again in trying to get things I knew were right—it was not until the covert opposition of the special interests in Conservation was beaten into me, and beaten into the rest of us, that that end of it was taken up at all. There are troubles enough in this world without any of us hunting a fight; but this fight hunted us (applause), and we are in it yet, as Mr Heney declares.

The principles of Conservation are very few and very simple. That is one of the beauties of this whole movement—that there is nothing mysterious or complicated or hard to understand about it; it is the simplest possible application of common sense. The first of the principles is this: that the natural resources and the natural advantages both belong to all the people and should be developed, protected, and perpetuated directly for the benefit of all the people and not mainly for the profit of a few (applause). The second principle is that the natural resources still owned by the people which are necessities of life, like coal and water-power, should remain in the public ownership and should be disposed of only under lease for limited periods and with fair compensation to the public for the rights granted (applause). I have never sympathized with the ideas we have heard so much of that the people must not try to protect themselves because they are not fit to handle their own affairs, and especially that they cannot handle their affairs in the matter of Conservation. By all means let us have the resources cared for, held in ownership by the people of the States as well as of the Nation, and handled for the benefit of the people first of all. (Applause)

Now, I want to state a few propositions as to each of the four great categories of the natural resources, which seem to me to include not all but a very considerable proportion of the fundamental things that Conservation people seek. It is very likely that some will not agree that these are the fundamental things; but I believe these propositions, taken together, represent fairly the opinion of most of the many millions of men and women who believe in Conservation.

First, as to our waterways: Every stream should be made useful for every purpose in which it can be made to serve the public. We have been in the habit of sacrificing, for example, irrigation to power, or power to the city water supply. Let us study our streams and use them for every purpose to which they can be put. The preparation of a broad plan is needed without delay for the development of our waterways for navigation, domestic supply, irrigation, drainage and power. (Applause)

Second, every water-power site now in State or Federal control should be held in that control (applause), and should be disposed of only under lease for a limited time and with fair compensation to the public.

Third, in the development of our waterways, the cooperation of the States with the Nation is essential to the general welfare. (Applause)

Now, as to our forests: First, all forests necessary for the public welfare should be in the public ownership and remain there (applause). Among these are the National Forests already in existence and the proposed Appalachian and White Mountain National Forests (applause). I am glad to hear you applaud the proposition for the Appalachian and White Mountain forests—we need them (applause). We want also the State forests to be taken care of—the State forests of New York, Pennsylvania, Wisconsin, Minnesota, and other States.

Second, the protection of forests against fire is the duty of State and Nation alike (applause); and that lesson has been driven home this year in a way that I think will make our people understand and remember it for many years to come. I want to pay a tribute in a word, if you will allow me, to the wonderful work done by the boys of the National Forest Service, of the Army, and of the great fire-fighting associations of the West, and by many private citizens, in making what seems to me to have been one of the best,

one of the boldest, one of the most devoted fights for the public welfare of which I know anything in recent years (applause). The way to stop fires in a forest, as in a town, is to get men to them as soon as they begin. The maintenance and extension of forest fire patrol by the Nation and States and by their subdivisions and by associations or private citizens who own timber lands is absolutely necessary. And we must have not only a patrol but a *sufficient* patrol.

Third, the development of existing forests by wise use is the first step in forestry, and reforestation is the second. Practical forestry in our existing forests comes first, tree planting follows; both are absolutely essential if we are to handle this problem right. (Applause)

Fourth: Land bearing forests should be taxed annually on the land value alone, and the timber crop should be taxed only when cut, so that private forestry may be encouraged (applause). Next to fire, there is nothing that so stubbornly stands in the way of practical forestry in this country as bad methods of taxation. (Applause)

Fifth—and I feel very strongly about this: The private ownership of forest lands is in reality a public trust, and the people have both the right and the duty to regulate the use of such private forest lands in the general interest. (Applause)

Then as to the lands: Every acre of land should be put to whatever use will make it most serviceable to all the people (applause). All agricultural land should be put to agricultural use. I have never been one to maintain that forest-bearing land which could be more useful under the plow should be kept for forest uses (applause); I have never been one to maintain, either, that land bearing heavy timber, acquired ostensibly for agricultural uses, should be cut over and afterward abandoned (applause). The fundamental object of our land policy should be the making and maintenance of permanent prosperous homes—that is the whole story (applause). Land monopoly, and excessive holdings of lands in private ownership in great bodies, must not be tolerated (applause). One of the very great difficulties in several parts of our country arises in huge consolidated holdings of land, which make tenants out of men who ought to be freeholders—free men on their own land. (Applause)

Settlement should be encouraged by every legitimate means on all the land that will support homes. That is a fundamental proposition. Thus the tillable land in public ownership, within and without the National Forest, should be disposed of in fee simple to actual settlers, but never to speculators. (Applause)

The first and most needed thing to do for our cultivated lands is to preserve their fertility by preventing erosion, the greatest tax the farmer pays. (Applause)

The non-irrigable and arid public grazing lands should be administered and controlled by the Federal Government in the interest of the small stockman and the homemaker until they can pass directly into the hands of actual settlers (applause). Many millions of acres are now having their forage value destroyed because Uncle Sam exercises no control whatever over a territory vastly larger than any single State—even Texas.

Finally, rights to the surface of the public land should be separated from rights to the forests upon it and the minerals beneath it, and each should be held subject to separate disposal; *and the Timber and Stone Act should be repealed!* (Applause)

As to our minerals: Those which still remain in Government ownership should not be sold—especially coal—but should be leased on terms favorable to development up to the full requirements of our people. I want to make it plain, if anyone should happen not to understand, that the withdrawals which have been made of coal lands and oil lands and phosphate lands are not intended to be permanent; they are intended simply to prevent those lands from passing into private ownership until Congress can pass proper laws for retaining them in the public ownership and having them used there (applause). Until legislation to this effect can be enacted, temporary withdrawals of land containing coal, oil, gas, and phosphate rock, are required in order to prevent speculation and monopoly.

It is the clear duty of the Federal Government, as well as that of the States in their spheres, to provide, through investigation, legislation, and regulation, against loss of life and waste of mineral resources in mining. The recent creation of a National Bureau of Mines makes a real advance in the right direction. And I want here to pay my tribute to the man who has

recently and most wisely been appointed director of that Bureau of Mines, Joseph A. Holmes, one of the best fighters for Conservation that this country has produced. (Applause)

With regard to National efficiency: The maintenance of National and State conservation commissions is necessary to ascertain and make public the facts as to our natural resources. That seems to me to be fundamental. We must have the machinery for continuing this work. Such commissions supply the fundamental basis for cooperation between the Nation and the States for the development and protection of the foundations of our prosperity.

A National Health Service is needed to act in cooperation with similar agencies within the States for the purpose of lengthening life, decreasing suffering, and promoting the vigor and efficiency of our people (applause). I think it is high time we began to take as much care of ourselves as we do of our natural resources. (Applause)

These are not all the things for which Conservation stands, but they are some of the more important. I had meant to speak here of the conflict between State and Federal jurisdictions, which we have seen illustrated in this Congress, but I prefer to speak, not of the conflicts, but of the chances for cooperation (applause). I believe in the Federal control of water-power in navigable and source streams and of water-power sites that are now in the Federal hands. I believe equally that every State has a great duty to its own people in Conservation, and that only by full and free and hearty cooperation between the Nation and the States can we all of us get together to control or develop, as the case may be, those intrastate or interstate agencies which are attempting for private profit to harm all the people (applause). When a question is settled, as I think this Congress has pretty well settled in its own mind certain of the questions relating to the division of the Federal and State work, that is the time to go on and act upon it; and I believe we ought to emphasize here most vigorously the functions of the State as well as the functions of the National Government, always remembering that the Federal Government alone is capable of handling questions which exceed the limits of any one State, and that, as Colonel Roosevelt said here the other day, nearly all of the great corporations have affiliations extending throughout the Nation or at least across State

boundaries. I am as vigorously for the recognition of the State power and the State duty as I am for the recognition of the Federal power and the Federal duty, each in its proper place (applause). But should I at any time see an attempt made to hide behind either one of these powers at the expense of the people, I would not be doing my duty if I didn't stand up and say so.

Just a word in closing: No body like this can get together without firing a man's imagination and heart. I have been at many great meetings, but never at one that seemed to me to contain within itself the possibility and power for good that this one does (applause). I have watched this Conservation movement grow, as we all have; I see it now on the very verge of the most practical kind of results. The clouds have cleared away; we know where we stand; we are ready to go forward, and we know where we are going and how. There has been gathered here a body of men and women whose motive is clearly this, that they propose when they depart to leave this good old earth better for their children than when they found it (applause), and they are carrying that message to the people of the United States more powerfully than it has ever been carried before. If any man or any woman were disposed not to be hopeful about the Conservation movement, I think this Congress would lift them to a new plane; it gives us new hope for the future of our country. I thank you. (Great applause)

GOVERNOR EBERHART—Ladies and Gentlemen: Just a few words before we take a recess until this evening: I wish on this occasion, as it will be perhaps the only one afforded to me, to express my sincere thanks to the officers of this Congress for the splendid manner in which they have done their work. I have never met a more congenial and kindly set of officers than those who are handling this convention (applause), and a great deal of the credit of the success of this convention is due to their personal, persistent, and strenuous efforts. I take it that this is the time at which, as Chief Executive of the State, I should present my acknowledgments. I regret that the President of the Congress, who is always unselfish, has determined that, in order to give the other officers, delegates and guests a chance tonight to be heard, his

own lecture—which we have all been waiting for—shall not be presented at this time.

Among the splendid sentiments which Mr Pinchot has uttered, one of the very best, I think, was that the States and the Nation instead of struggling among themselves as to how authority should be divided, should cooperate (applause) in the Conservation of the resources of the country for the benefit of all the people for all time.

After two or three announcements have been made, we will take a recess until this evening at 8 oclock.

Professor CONDRA—The Committee on Nominations will meet, immediately after this meeting adjourns, in Room 601, Saint Paul Hotel.

Since the report of the Committee on Credentials was received and filed with the Secretary yesterday, there has been an additional registration of 40 or 50 delegates.

It was announced this morning that the Call of the States would be made this afternoon, but it became impossible to do so. President Baker asks me to say that tonight the order of business will be, *first*, the election of officers; *second*, the reception of the resolutions from the Committee on Resolutions; and *third*, special reports from the States—this to continue tomorrow if necessary.

Another suggestion: If any of you have anything to be read from the platform, please put it in such form that it can be read properly and understood clearly. We had an example of misunderstanding this morning, which I regret; and I want to advertise the papers of this city by asking you to read the report in one of them from which you will see the results of that misunderstanding. Do not blame anybody; these things come. Do not blame the ladies of this State for any misunderstanding. I have had too many thousands of womanly women in my classes at the university and elsewhere (and I married one of the most lovely women in the world), and I have too much faith in women to blame them. I blame myself for trying to read a statement which I had not had the time to look at. Let a thing like that not come into this Congress again. Blame no one.

Thereupon Governor Eberhart, for President Baker, declared a recess until 8 o'clock p.m.

CLOSING SESSION

The Congress was called to order by President Baker in the Auditorium, Saint Paul, at 8 oclock p.m., September 8.

President BAKER—Ladies and Gentlemen: The first business in order is action on the report of the Nominating Committee, to be followed by action on the report of the Resolutions Committee. While waiting for these reports we should be glad to hear from some of the States. Washington made a special request to be heard. Is the Gentleman from the State of Washington present?

[There was no response.]

W. S. HARVEY—Mr President: In the absence of the representative of Washington, may the Delegation from the Keystone State, Pennsylvania, be heard at this time?

President BAKER—Colonel Harvey has the floor, and will speak for his State.

Colonel HARVEY—Mr President and Delegates: On behalf of the Commonwealth of Pennsylvania, which it is our honor to represent, we desire to say first of all that no other State in the whole galaxy constituting our Union of States possesses such great natural resources. In some, indeed, the resources may be more varied, but in none are they of such productive and wealth-creating capacity as in Pennsylvania.

Pennsylvania leads all other States in the production of coal, the value of our annual output reaching approximately \$325,000,000 per annum. In the value of its petroleum, natural gas, clay products, and pig iron it has no close second. The annual value of our petroleum production is about \$18,000,000, and of our natural gas about the same, while the value of our pig-iron production reaches about \$235,000,000; of our clays it might be said we have scarcely begun to develop them, yet the value of our clay product is more than \$20,000,000 yearly. We are among the leading States

in the production of cement, roofing-slate, lime, and building stone. Among our other mineral products are graphite, glass sand, mineral waters, metallic paints, mortar colors, and ochre. It will doubtless surprise many to learn that in the year 1907 the total value of all of the mineral products of all of the States west of the Mississippi was more than \$100,000,000 less than the value of the mineral products of Pennsylvania for the same year; and that the value of our mineral products in the same year was equal to almost one-third of the entire value of all of the mineral products of the United States, including Alaska. This also includes gold and silver.

We have thus far spoken only of our mineral resources, but when we add to this our magnificent resources in agriculture (one of our counties leading all others in the United States in the value of its agricultural products), of our timber and our water-power, and more important still, a population second only to that of the Empire State and nearly equal to Canada, it is apparent that we should be vitally interested in the subject of Conservation; and we beg now to be permitted to mention what has been and now is being done along this line:

Our State has for many years had a Forestry Department with a Commissioner and a Forest Reservation Commission, who have purchased for the creation of State forest reserves and paid for up to September 1, 1910, 918,529 acres of land at a cost of \$2,061,872.45 or an average of \$2.25 per acre, and have under contract for purchase about 50,000 acres more. The State also has established nurseries for seedlings, and has turned out thus far 2,500,000; next spring the increased capacity of these nurseries will turn out about 6,000,000 seedlings, and we hope and expect to be in a position within a few years to turn out 20,000,000 each year. These seedlings are being used for reforestation on the State reservations and other lands that have been cut over or denuded, and in time will produce forests from which the State will derive a large revenue. The State has also established a Forest Academy, for which appropriations amounting to \$96,000 have been made; 39 students have been graduated, all of whom, with the exception of two, are now in the employ of the State; 30 students are maintained in the academy; and the course is three years, 10 students being admitted each year. The State has also made provision for protection against, and the extinguishing of, forest fires, and the sum of \$245,000 has been appropriated for this purpose. The State has appropriated for

maintenance and administration of forest reserves since they were first created the sum of \$877,142. In addition to the foresters employed, 41 in number, the State employs 116 rangers and a large labor force.

One of the most important Conservation movements entered into by our State has been the conserving of the health of its citizens by protecting from pollution, through a Water Commission and the State Board of Health, the waterways of the entire Commonwealth. Human life and its preservation from disease and impairment of usefulness and its loss of producing power is the most fundamental of all subjects of Conservation. Pennsylvania has also set an example that we sincerely trust may be followed by every other State wherein forest reserves can be created, by establishing camps for tuberculosis patients, where those who are unable to provide the necessary expense to be cared for in private institutions and in climatically suitable locations can be cared for by the State. Since 1907 Pennsylvania has appropriated to the State Department of Health for the construction of suitable buildings and camps for the treatment of tuberculosis on the reservations of the State, \$3,000,000. The sanitarium established at Mount Alto has treated 3,301 patients, and 115 dispensaries established throughout the State have treated 32,247 patients. The present enrollment at Mount Alto is nearly 800, and of dispensary patients 9,000. This work is under the supervision of the distinguished and capable gentleman at the head of our Health Department, Dr Samuel G. Dixon. The movement for the establishment of tuberculosis camps was inaugurated by Dr J. T. Rothrock about twenty years ago, and his name with that of others who have been influential in this work for the cause of humanity and the conservation of health and happiness will continue to be honored in our State.

Pennsylvania also makes much larger appropriations than any other State in the Union for its general hospitals, furnishing free of cost the best surgical and medical skill to those who are unable to pay for the same, thus saving many lives as well as adding to the bread-winning capacity of every community.

Our Department of Mines is doing a good work in trying to make more secure the lives of the miners and their occupation less hazardous. Our system of factory inspection is doing much to protect the lives of our workers in mills and factories, and the topographic and geologic survey

commission of our State is also carrying on a most important work in the conservation and development of our natural resources.

Pennsylvania has a Forestry Association that has been in continued active existence for 23 years. Its membership extends to every county in the State, and it has taken the initiative and been the organizer and promoter of the measures that caused the creation by the State of forest reserves and a Forestry Commission; and its members have been largely instrumental, through the earnest, persistent, public-spirited devotion to measures and methods, in educating the people not only of Pennsylvania but of other States to appreciate the value and merits of conserving all our natural resources; and what Pennsylvania has done has helped in no small degree to develop conditions that have made possible the present nation-wide movement for Conservation. (Applause)

The State of Pennsylvania has in the above brief statement shown the practical interest it has had for years and will continue to have in the subject of Conservation; and we earnestly assure this Congress of the hearty support and cooperation of the Keystone State in this great cause.

Respectfully submitted, on behalf of the State of Pennsylvania, by Wm. S. Harvey, G. W. McNees, and Joseph C. Righter. (Applause)

President BAKER—Ladies and Gentlemen: We wish to give everybody a chance to speak, and I am willing to stay here all night and all day tomorrow. We shall have some very important business in a few minutes. It might be well under the Call of the States, for speakers to be limited to five minutes (applause). Is that your pleasure? All in favor of a five minute rule will please say "Aye."

[Many voices: "Aye."]

President BAKER—Are any opposed? (After a pause) It is carried unanimously.

A DELEGATE—Mr President: I move you that the States be called in alphabetic order. It will save confusion, prevent Delegates from rising in all

parts of the house, and expedite business.

The motion was seconded, put, and carried without dissent.

Mr E. W. Ross (of Olympia, Washington)—Mr Chairman: Nobody in this part of the house knows what is going on. What is the question before the house?

President BAKER—The question before the house just now was on the motion that the States be called in alphabetic order, which was carried; and the Call of the States is now in order.

Mr Ross—We have expected, since 9:30 oclock this morning, to have the States called in alphabetic order. What is the use in talking to Delegates now about calling the States in alphabetic order at 9 oclock on next to the last day of this Congress? This is the first time since I have attended this Congress that I have heard the Delegates vote on something which pertained to their own proceedings. (Confusion on the floor) Who brought this anyway? Are we to sit here day after day like a flock of cattle and—

President BAKER—The Gentleman is out of order.

Mr Ross—I have traveled two thousand miles, and I had something to say on a proposition germane to what was going on at the time, and I was informed that there would be a time later and a motion was put here and voted on that at 8:30 this morning the States would be called—

President BAKER—The officers were here at 8:30, but there were no Delegates.

Mr Ross—It is now 9 oclock and you talk about—

President BAKER—We were ready at 8:30 this morning.

Mr Ross—I was here and the representative of the State of Washington—

President BAKER—Washington was twice called.

Mr Ross—And he has been—

President BAKER—You are out of order.

Mr Ross—Has been sitting on the rostrum there since 8 oclock this morning, and he hasn't been heard yet!

President BAKER (rapping on the table)—The Gentleman is out of order. Is the Chair sustained?

Many Voices: Yes.

A DELEGATE—Mr President: I make the point of order that the Committee on Nominations was to report immediately after 8 oclock this evening. I therefore call for the previous question and ask that the election of officers proceed.

President BAKER—The Committee will be ready to report in a few minutes.

Mr Ross—Mr Chairman—

President BAKER—You are out of order.

Mr Ross—The gag rule is trying to be enforced, and I appeal to this Congress. That is what we have had from the beginning to the end. Put on your gag rule, and we will go home and never forget it—(Calls from the floor: "Order, order!")

President BAKER—Will the house be in order?

Mr Ross—Put the screws down, the harder you do it the greater the recoil and the rebound, and the boomerang will hit you in the end—

President BAKER—The Chairman of the Committee on Nominations will now report: Professor Condra.

Mr Ross—And I want to say now that when Theodore Roosevelt occupied the platform, myself and 200 delegates walked to the front door and we knocked and we knocked and we knocked—

Many Voices: "You are out of order!"

Mr Ross—and I am tired of the way things have been going on; the representative of the State of Washington has been sitting on that rostrum since 8:30 this morning waiting for the States to be called and the States were—

A DELEGATE—Mr President: I call for the report.

Professor CONDRA—Mr President, and Ladies and Gentlemen: In this committee work we have tried to do our best for the interests of Conservation throughout the whole country for next year and the ensuing years. No member of this committee has been unduly influenced or has any axe to grind whatever (applause)—

[Mr Ross interrupts, and momentary confusion ensues]

Professor CONDRA—As to the Delegates that tried to gain admission to our room this evening, that is a closed chapter and our report is without bias and we hope it will receive your approval (applause). We thought of nominating for the Presidency of the Congress, among others, two persons now on this platform. We consulted them, and they both said it would be better to place in nomination another. One of the two men whom we first thought of nominating is Captain White, the other is Gifford Pinchot. The Committee will ask the former to nominate the President, and the latter to second the nomination. (Applause)

Captain WHITE—Mr President, Ladies and Gentlemen, and Delegates to this Congress: It is a pleasure that comes to man but seldom in life when he can do a great benefit to a people; even if it involves a sacrifice, it is often a pleasure to do it. I did not wish to have my name mentioned, as it has been (nor did I know that it would be), as a possible nominee for the Presidency of this honorable body, nor did I know that my friend Mr Pinchot's name would be mentioned; but in thinking it over, after we were consulted, we both felt like influencing the Nominating Committee to do what was best for the country, this organization, and for all the State associations. The great back-bone of this country is the farming element. It is the farmers who make the country, and to them we must look for prosperity, and when they are prosperous and contented the country is prosperous and the people are happy. So, to that department of Conservation we have looked for a man to act as President of this organization—one who would be satisfactory to the farmers. We found the right man. We are going to put in nomination to this Congress a man in whom there is no guile, who is not only well known in this country but who has international fame; a man who has published for many years one of the largest, if not the largest, of farm journals in the

country; one who was appointed by President Roosevelt as a member of the Country Life Commission, who has lived close to the farmer, who has done perhaps more than any man in his community, making greater sacrifices according to his ability; who has made speeches on many platforms, and during a long life has worked earnestly for the benefit of humanity. I take pleasure in nominating for the Presidency that prince of men, Mr Henry Wallace, of Des Moines. (Great applause)

Mr GIFFORD PINCHOT—Ladies and Gentlemen: I pray your indulgence for a moment while I try to say a little of what I think about "Uncle Henry" Wallace. I call him "Uncle Henry" for the best of all reasons—that when a man has reached his age in a life of usefulness, he becomes, in a sense, the forebear of all the rest of us, and our affectionate esteem naturally expresses itself in calling him "Uncle"; and I say "Uncle Henry" Wallace because I love him. (Applause) I want to add, too, an expression of my highest respect for his character, for his achievement, and, above all, for his breadth of view, which covers intelligently and fully every interest for which this Conservation Congress stands. Mr Wallace lives in the center of the country; his main attention has been given to our central industry. His advice and assistance have been poured forth freely for that class of citizens among us all who have the most to do with the fundamental occupation of conserving the earth and making it forever fruitful; and I deem it to be a most fitting nomination that the Committee has laid before you in suggesting his name.

Before I sit down I want, with your permission, to say a word, also about Captain White. Captain White and Mr Wallace stand together in my mind as two of the finest types of ripened American citizens (applause). I am proud to say that I believe I enjoy the friendship of both. I have been associated with Captain White for many years in Conservation work. He was one of the first of the lumbermen—the very first of the lumbermen, I believe—to take an earnest and effective and active interest in Conservation. It was to his lands that the first class from one of the great forest schools went to study lumbering and forestry on the ground; and at every point his helpful, wise, and effective assistance has been given to the movement for which this great Congress stands. I know that Mr Wallace will not mind my interjecting remarks about another man in seconding his nomination, however irregular it may appear. I wanted to say (and this is the only chance

I have) what I think of Captain White; and I want to add that I shall make only one suggestion to Mr Wallace, if he is elected, and he will accept it or not as he pleases; but I shall certainly advise him to keep Captain White as Chairman of the Executive Committee. (Applause)

Mr President, I take the greatest pleasure in seconding the nomination of Mr Wallace. (Applause)

A DELEGATE—Mr President: I move that the rules be suspended, and that Mr Wallace be elected by acclamation.

The motion was seconded, put, and declared unanimously carried.

Mr BAKER—It gives me very great honor, Mr President Wallace, to present to you the gavel. No man will do more, to the extent of his ability, in supporting your administration and carrying it forward to success. (Applause)

President WALLACE—Mr Baker, and Ladies and Gentlemen: Believe me, this is the greatest surprise of my life. No one had said a word to me about it until a few moments before I came into this room. I believe that if I had had time to think of it I would have declined, but in an unguarded moment, I said if the unanimous choice of this Congress I would do my best to serve you. I know I am undertaking a very great work; I know I shall need all the help of your wisest counsels. I shall probably make mistakes. The man who makes no mistakes is the man who does nothing (applause). I have made mistakes in other undertakings. It is a rule of my life not to mourn over the irreparable past, but to make the best out of the available future (applause); to do one day's work well, and be ready if possible to embrace the opportunities that may come tomorrow.

Now, I feel conscious of my inability to act as President of your organization. I have studiously avoided such offices in the past; I have studiously avoided taking office of any kind or class; but this having been forced upon me, and the offer coming utterly without my knowledge—without a whisper of it, in fact—it gives me an opportunity of service which I will do my best to meet. I shall have to ask you to excuse me from serving tonight, for I am leaving on a train in a very short time. I shall ask you to wait, if I have the Executive Committee to appoint (as I am told I have),

until I have time to study this Conservation movement from the organization's side. I shall make the best selections I can; I will do the very best that lies in me, and that is all that any man can do. (Applause)

I want to say to you that if there have been any factions in this organization, I know nothing of them (applause). I have no part in them. I believe in the Conservation of the resources of the country. I believe that if this is to be done wisely we must imagine ourselves in the position of the men who have differences of opinion here. I realize that the Western people have peculiar difficulties; I realize that their position must be studied from their standpoint (applause)—that whatever help may be given them for the solution of their problems must be given; and if I am to be President of this organization, I will be President of a National organization (applause), and I will know no State (renewed applause), no faction, no party (renewed applause); and, so far as I am concerned, there will be no politics (great applause) in this association.

I thank you for this unexpected and unsolicited honor, and I accept it as an opportunity to serve the American people in this generation and perform a service which will be beneficial to generations yet unborn (applause); for I believe that the mission of this Nation is not to build great cities, not to be a world-power, not to amass wealth untold, but to develop character (applause) and manhood that can stand facing all the storms that blow, that can solve the problems as they come—a manhood that owes its highest obedience not to laws made by mortal man but to the laws made for human guidance by Almighty God. (Applause)

Professor CONDRA—Mr President, and Ladies and Gentlemen: Your Committee nominate for Executive Secretary Thomas R. Shipp (applause), for Recording Secretary James C. Gipe, and for Treasurer D. Austin Latchaw. I move the election of these nominees.

President WALLACE—It is moved that Thomas R. Shipp be elected Executive Secretary. Is that motion seconded?

The motion was seconded from all parts of the house.

President WALLACE—It is moved and seconded that Thomas R. Shipp be chosen Executive Secretary. Are there any remarks?

(Calls of "Question, question!")

Mr ROSS—Mr Chairman: I would like to have a little information on that subject. I would like to inquire whether Mr Shipp occupies any position of trust or profit in the way of emolument under the United States or any State government?

(Calls of "Question!" "Regular order!" "Order!")

President WALLACE—The Chair is unable to give the Gentleman any information on that subject. The question is called for. All in favor, signify by saying "Aye." (Hundreds of voices: "Aye.") Contrary "No." (Pause.) The motion is carried.

VOICES—"Shipp, Shipp!"

Mr ROSS—Mr Chairman—

President WALLACE—Has the Gentleman a motion to make?

Mr ROSS—I was recognized by the Chair and the previous question has not yet been voted upon.

President WALLACE—Has the Gentleman any motion to make the order of business?

Mr ROSS—I rise to a point of order. I have the floor. The Chairman recognized me and the previous question has not yet been voted. I ask for a matter of information.

President WALLACE—The Chair has no information to give except that the Gentleman is out of order.

Mr ROSS—I ask if Mr Shipp occupies a position or employment in any capacity for the United States Government or any State or Territory.

President WALLACE—I don't know. The motion was duly put and was carried.

VOICE—"He is out of order."

President WALLACE—He is. The next nominee is James C. Gipe for Recording Secretary.

Mr ROSS—Does the Chair rule that I am out of order?

President WALLACE—I have, several times.

Mr ROSS—Thank you, sir. That is the cap sheaf.

President WALLACE—Is there a second to the nomination of Mr Gipe for Recording Secretary?

The nomination was seconded.

President WALLACE—Are there remarks on that question?

Many DELEGATES—"Question."

The motion was put and carried, and Mr Gipe was declared unanimously elected Recording Secretary.

President WALLACE—The Committee also recommend the election of D. A. Latchaw for Treasurer. Is there a second to the motion for his election? (The motion was seconded.) Any remarks on the motion?

VOICES—"Question."

The motion was put and carried, and Mr Latchaw was declared elected Treasurer.

Delegate HUNT—Mr President: The District of Columbia moves a vote of thanks to the Nominating Committee who have done their work so well and so pleasingly to this Congress.

The motion received a second, and was put and unanimously carried.

Mr ROSS—Will the Chairman please announce what the motion is? We didn't hear a word of it here.

The DELEGATE—That a vote of thanks be tendered to the Nominating Committee for the work which they have done so well and satisfactorily to this Congress.

Mr ROSS—Mr Chairman, I second that motion. (Laughter)

Colonel FLEMING JONES (of New Mexico)—Mr President: I understand that Governor Pardee is about to submit the report of the Committee on Resolutions, and I have a resolution here which I should like to see embodied in the report.

Mr ROSS—The Gentleman from New Mexico is out of order.

President WALLACE—The Chair thinks the resolution out of order.

Delegate HARDTNER (of Louisiana)—Mr President: I move you that the rules be suspended for the purpose of permitting Colonel Fleming Jones to submit his resolution.

Delegate DYE (of Indiana)—I second the motion.

The motion being duly put and carried, Colonel W. A. Fleming Jones submitted the following:

Resolved, That this Congress express its grateful appreciation of the highly intelligent, unselfish, and successful services of its first President, Mr Bernard N. Baker, of Maryland. Through his untiring effort and his purpose to bring into consultation all the interests of Conservation, the Congress has resulted in a meeting that will be historic in the records of American progress and achievement.

Being formally put, the resolution was adopted unanimously and enthusiastically.

Mr BAKER—Mr President: I wish to express my appreciation, and to have it show in the Proceedings. I have not taken one moment to present anything in which I was directly interested. I thank you very much. (Applause)

President WALLACE—We will now hear from the Committee on Resolutions.

Governor PARDEE—Mr President, Ladies and Gentlemen: As Chairman of the Committee on Resolutions I have been ordered and directed by a majority of the Committee—some 26 or 27 out of about 30 present at the

last session of the Committee—to present the following report, as the report of the majority in the proportions I have mentioned:

RESOLUTIONS OF THE SECOND NATIONAL CONSERVATION CONGRESS

The Second National Conservation Congress, made up of Delegates from all sections and nearly every State and Territory of the United States, met at the call of a great moral issue (applause), now in session assembled in the city of Saint Paul and State of Minnesota, does hereby adopt and solemnly declare the following platform of opinion and conclusion concerning the inherent rights of the People of the United States:

Heartily accepting the spirit and intent of the Constitution and adhering to the principles laid down by Washington and Lincoln, we declare our conviction that we live under a Government of the People, by the People, for the People; and we repudiate any and all special or local interests or platforms or policies in conflict with the inherent rights and sovereign will of our People. (Great applause)

Recognizing the natural resources of the country as the prime basis of property and opportunity, we hold the rights of the People in these resources to be natural and inherent, and justly inalienable and indefeasible (applause); and we insist that the resources should and shall be developed, used, and conserved in ways consistent both with current welfare and with the perpetuity of our People. (Applause)

Recognizing the waters of the country as a great National resource, we approve and endorse the opinion that all the waters belong to all the People (applause), and hold that they should be administered in the interest of all the people. (Great applause)

Realizing that all parts of each drainage basin are related and interdependent, we hold that each stream should be regarded and treated as a unit from its source to its mouth; and since the waters are essentially mobile and transitory and are generally interstate, we hold that in all cases of divided or doubtful jurisdiction the waters should be administered by cooperation between State and Federal agencies. (Prolonged applause)

Recognizing the interdependence of the various uses of the waters of the country, we hold that the primary uses are for domestic supply and for

agriculture through irrigation or otherwise, and that the uses for navigation and for power, in which water is not consumed, are secondary; and we commend the modern view that each use of the waters should be made with reference to all other uses for the public welfare in accordance with the principle of the greatest good to the greatest number for the longest time. (Great applause)

Viewing purity of water supply as essential to the public health and general welfare, we urge upon all municipal, State, and Federal authorities, and on individuals and corporations, requisite action toward purifying and preventing contamination of the waters. (Applause)

Approving the successful efforts of the United States to provide homes on arid lands through irrigation, we indorse and commend the Reclamation Service (applause) and urge its continuance and the extension of the same policy to the drainage of swamp and overflow lands, to be carried forward so far as appropriate through cooperation between State and Federal agencies. (Great applause)

Viewing adequate and economical transportation facilities as among the means of Conservation, and realizing that the growth of the country has exceeded the development of transportation facilities, we approve the prompt adoption of a comprehensive plan for developing navigation throughout the rivers and lakes of the United States, proceeding in the order of their magnitude and commercial importance. (Loud applause)

Recognizing the vast economic benefit to the People of water-power derived largely from interstate and source streams no less than from navigable rivers, we favor Federal control of water-power development (applause); we deny the right of State or Federal governments to continue alienating or conveying water by granting franchises for the use thereof in perpetuity (applause); and we demand that the use of water rights be permitted only for limited periods, with just compensation in the interests of the People. (Prolonged applause)

We demand the maintenance of a Federal commission empowered to deal with all uses of the waters and to coordinate these uses for the public welfare in cooperation with similar commissions or other agencies maintained by the States. (Loud applause)

Approving the withdrawal of public lands pending classification, and the separation of surface rights from mineral, forest, and water rights, including water-power sites (applause), we recommend legislation for the classification and leasing for grazing purposes of all unreserved public lands suitable chiefly for this purpose, subject to the rights of homesteaders and settlers, or the acquisition thereof under the land laws of the United States; and we hold that arid and non-irrigable public grazing lands should be administered by the Government in the interest of small stock-men and homeseekers until they have passed into the possession of actual settlers. (Applause)

We hold that the deposits of important minerals underlying public lands, particularly mineral fuels, iron ores, and phosphate deposits, should be leased for limited periods, not exceeding fifty years, but subject to renewal, the royalty to be adjusted at more frequent intervals; such leases to be in amounts and subject to such regulations as to prevent monopoly and unnecessary waste. (Applause)

We hold that phosphate deposits underlying the public lands should be safeguarded for the American People by appropriate legislation; and we recommend the early opening of the Alaskan and other coal fields belonging to the People of the United States for commercial purposes on a system of leasing, National ownership to be retained. (Applause)

We urge immediate investigation by the Federal Government of the damage done by the smelting of copper ores, and the feasibility of so improving methods as to utilize the injurious by-products in connection with phosphatic fertilizers.

We favor cooperative action on the part of States and the Federal Government looking to the preservation and better utilization of the soils by approved scientific methods. (Applause)

We approve the continuance of the control of the National Forests by the Federal Government (applause), and approve the policy of restoring to settlement such public lands as are more valuable for agriculture. We earnestly recommend that the States and Federal Government acquire for reforestation lands not more valuable for other purposes, and that all existing forests publicly and privately owned be carefully protected by State

and Federal governments. We recognize the invaluable services of the Forest Service to the People (applause), and earnestly recommend that it be more generously supported by the Federal Government, and that State, Federal, and private fire patrol be more generously provided for the preservation of forests and human life; and we appreciate and approve of the continuance of the services of the United States Army in fire control in emergencies.

We favor the repeal of the Timber and Stone Law. (Applause)

We endorse the proposition for the preservation by the Federal Government of the Southern Appalachian and White Mountain forests.

We recommend that the Federal Government conserve migratory birds and wild game animals.

We recommend that both public and private schools instruct the youth of the land in the fundamental doctrines of Conservation.

We realize that the fullest enjoyment of our natural resources depends on the life and development of the people physically, intellectually, and morally; and in order to promote this, we recommend that the training and protection of the people, and whatever pertains to the public health and general efficiency, be encouraged by methods and legislation suitable to this end. Child labor should be prevented and child life protected and developed.

Realizing the waste of life in transportation and mining operations, we recommend legislation increasing the use of proper safeguards for the conservation of life; and we also recommend that in order to make better provision for securing the health of the Nation a department of public health be established by the National Government.

We recommend the adequate maintenance of a National Conservation Commission to investigate the natural resources of the country and cooperate with the work of the State conservation commissions; and we urge the legal establishment and maintenance of conservation commissions or corresponding agencies on the part of all States of the Union.

Nothing in these resolutions is to be construed as questioning the rights of the States or the People of the United States guaranteed under the Federal Constitution. ^[2] (Prolonged applause)

Governor PARDEE—Mr President: Again reminding you that this is a majority report, and that the Committee were told that a minority report would be presented (and I am looking directly at the member of the Committee who gave this intimation), I move you that the report just read be adopted as the voice of this Congress.

Mr G. M. HUNT—Mr President, the District of Columbia seconds the motion of the Chairman of the Committee.

Mr FRANK H. SHORT (of California)—Mr President: In view of the remark that there was to be a minority report presented to this Congress, I think, perhaps, I should say, on behalf of those who have been referred to as the minority (who may be "insurgents" some day) that in view of the provision in the resolutions that nothing shall be construed as contrary to the Constitution of the United States, we do not offer any amendment. We think that no person in this country is entitled to anything that does not belong to him under the Constitution and the law, and we don't think he should ever be offered anything else; and we suppose, if a conflict should arise, that the Constitution will prevail.

A DELEGATE—Pennsylvania rises to second the motion to adopt the resolutions as read.

A DELEGATE—Mr President: As a Delegate from the State of Illinois, I rise to second the motion.

President WALLACE—All in favor of the adoption of these resolutions as read will say "Aye." (A chorus of "ayes.") Contrary, "Nay." (There were no negative votes.) The resolutions are declared adopted.

Mr ROSS—Mr Chairman—

President WALLACE—We will hear you.

Mr Ross—That is all I want, that you should hear me. In view of the fact that this report is presented and heard by the Delegates at this late hour for the first time, and in view of the fact that the report of the Committee on Resolutions and the action of the Congress thereon is all-important and the final result of such a Congress, and in view of the fact that we are to meet tomorrow morning at 10 oclock or half-past, or whatever time it is, I move you, Mr Chairman, that action upon this report be deferred until the convening of this Congress tomorrow morning, in order that the Delegates may be able to read the report, section by section, as it may appear in the newspapers tomorrow morning—

President WALLACE—The Gentleman is out of order.

Mr Ross—So we can act intelligently.

President WALLACE—The Gentleman is out of order. The resolutions have already been adopted.

Mr Ross—Mr Chairman, the motion is made. Will somebody second my motion? I will see if I am out of order.

Mr JOHNS (of Washington)—Mr President: I move to lay the motion on the table. I am from the State of Washington and glory in it, but I do not glory in some of the men that the Governor appointed.

The motion to table was seconded, put, and carried with one dissenting voice.

President WALLACE—What is the further business before the Congress? Are there any other committees to report?

Professor CONDRA—Mr President, and Ladies and Gentlemen: We have neglected one matter of business, and with your permission (at the request of those who have noticed it) I will read the names of the Vice-Presidents selected by State Delegations.

Mr Ross—Mr Chairman: That is what we expected at 8:30 oclock this morning. It is now twenty minutes past 9. I move you that leave to print be extended to the various Vice-Presidents and those designated by States, and

if they can succeed in getting their written speeches printed in the record possibly we will have a chance to read them.

Delegate HERR (of Washington)—Mr President, I have a request to make: that when you print the address read by Lieutenant-Governor (and Acting Governor) Hay, of Washington, you also print, right with it, the dispatch from C. B. Kegley, representing 20,000 Grangers, and also the dispatch from Charles R. Case, representing the organized labor of Washington, both approving the Federal policies of Conservation. I make that as a request coming, as I believe, from the majority of the patriotic citizens of the State of Washington. (Applause)

Professor CONDRA—Mr President, just a moment. This list of Vice-Presidents selected by State Delegations is not the Call of States; we will come to that in a few minutes. The reports by State Delegations might be turned over to the Secretary who will ask that it be printed in the papers in the morning. A few States have not yet sent in the names of their Vice-Presidents, so that the list is not quite complete.

A DELEGATE—Read the list.

President WALLACE—Let us hear the list read.

A DELEGATE—Mr Chairman, I rise to a point of order. The report of the Nominating Committee is the special order at this hour. Having begun on nominations, they should continue until they are closed. The States should be called upon to name their Vice-Presidents. (Applause)

Professor CONDRA—Mr President, if agreeable I will read the names which we have. Reads list. [\[3\]](#)

President WALLACE—Shall we vote on these Vice-Presidents selected by the Delegates from the different States? All who favor the selections will please say "Aye" (Pause). Any opposed will say "No" (Pause). The selections are approved unanimously.

Mr Pinchot wishes to offer a resolution that you will all agree to.

Mr PINCHOT—*Whereas*, Professor Samuel B. Green, Dean of the School of Forestry in the University of Minnesota, and for twenty-two years a teacher

in the State Agricultural College, has recently been called to his reward; and

Whereas, Professor Green for years ranked as one of the most prominent and progressive instructors in Forestry, and has been a great force in the cause of developing and conserving our National resources; therefore be it

Resolved, That in the death of Professor Green the State of Minnesota and the Nation have lost a distinguished citizen, and the cause of Forestry one of its most valuable assets.

President WALLACE—Let us take a rising vote. All in favor of the resolution please rise. It is unanimously carried.

We are now ready for the Call of the States.

The States were then called alphabetically, whereupon the following responded:

Mr A. H. PURDUE (of Arkansas)—As regards Arkansas and Conservation, I will say that that subject with us is not a burning question. People are not yet clamoring for Conservation. Nevertheless, the movement set on foot by those who are promoting it is making itself felt among us, and the thoughtful people of the State are giving it their attention.

Mr O. B. BANNISTER (of Indiana)—I will not take five minutes. I first want to appeal to your spirit of fairness, and express the hope that at the next National Conservation Congress you will not ask Delegates to wait until the eleventh hour of the last day of the Convention. (Applause)

Indiana is represented here by fourteen Delegates. We have spent about \$2500 to attend this Congress (applause). We have sat here for four solid days and attended every single session, and heard the history of things from the birth of our Saviour down to 1908, when Theodore Roosevelt called the Governors together for Conservation, but up to this moment we have not been heard or given an opportunity to talk at all (applause). I just want to call your attention to that fact. I have only had three and a half minutes, and I am perfectly willing that the next speaker shall have my allotted minute and a half. (Applause)

President WALLACE—The President will say that if you live until the next Congress, and I preside over it, you will all have a chance to make your speeches as early in the meeting as possible. (Applause)

Mr BANNISTER—Mr President: It is not a question of speeches; it is a question of voting and being considered as the rank and file, if you please, of this organization.

Mr A. C. MILLER (of Iowa)—The report of the Iowa State Delegation is in the hands of the Secretary.^[4]

Mr YOUNG (of Kansas)—Our report has been filed with the Secretary.

President WALLACE—Ladies and Gentlemen: I am obliged to leave for my train, and I will ask Captain White to act as Presiding officer. (The audience rose as Mr Wallace withdrew.)

Mr BERNARD N. BAKER (of Maryland)—I just want to say "Thank you," for Maryland, for the opportunity of being here.

Mr FREEMAN THORP (of Minnesota)—Mr Chairman, a way has been found (not theoretically but practically and demonstrably) whereby the time of the growth of forest trees is reduced to one-half. The Secretary of Agriculture, Mr Wilson, who merely hinted at this in the last paragraph of his speech the other day, will give you the exact information in detail in his bulletins this winter. It is not a theory or wild guess. It has been demonstrated, only 150 miles from where you sit. There are thousands of trees there, some 11 inches in diameter, grown from seed in fifteen years (applause). There are shown at the Minnesota State Fair today products of the soil grown upon the lightest sandy soils that we have in the State, during the greatest drought ever known, and the corn is as large as any in the State, and the clover superior (applause). This simply shows the new force and new instrumentality which will be at our hands in the great work of Conservation. I will not take your time longer because the information will be given you this winter in bulletins from the Department of Agriculture at Washington. (Applause)

Dr E. N. LOWE (of Mississippi)—Mr Chairman and Fellow Delegates: We from Mississippi have come here to be heard on Conservation and to learn

about Conservation. We are interested in the subject. We have made some progress along the lines of Conservation. I wish to say that the most enthusiastic Conservationist in the State of Mississippi is our Chief Executive, Governor Noel. Practically all that has been done in Conservation in Mississippi has been instigated and encouraged by him. We are at the present time engaged in the study of our soil in cooperation with the Bureau of Soils at Washington. The work has been in progress since December last, and will be prosecuted as rapidly as possible. We have four parties in the field at the present time. In addition to our soil resources we have, in the southern half of Mississippi, a large pine forest; the virgin forest has been depleted very rapidly, and we are realizing the necessity of investigation along that line.

General NOBLE (of Missouri)—Mr Chairman: I have been appointed to attend this Congress as a Delegate from the Trans-Mississippi Commercial Congress, and also as a Delegate from Missouri. I have been associated in the Trans-Mississippi Commercial Congress with Mr Larimore, of Larimore, North Dakota. We were instructed by the President of the Trans-Mississippi Congress to present an invitation to this Congress, which was put in the form of a resolution. I want to say that the Trans-Mississippi Commercial Congress has been in existence for more than twenty years. It has been my privilege and honor to attend many of its different sessions, and I speak whereof I know when I say that it has been one of the most influential bodies in the western country for the advancement of western interests, including among others, the great proposition of the Conservation of our natural resources for the people.

Now, I wish to present this resolution: (Reading)

"Whereas, the Trans-Mississippi Commercial Congress has for its purpose the inauguration and advocacy of great National policies, and has for many years been a friend of the beneficent and progressive Conservation of our natural resources, and that Congress is to meet at San Antonio, Texas, on the twenty-second to twenty-fifth days of November next; therefore—

"Resolved, That the present Congress at Saint Paul does hereby earnestly request the National Conservation Association, by its President, to select and send to the approaching session of the Trans-Mississippi Congress a

delegation of such members as he may deem best (say forty in number), to advocate the cooperation of that body in support of the measures here approved."

I move you, Mr President, that that resolution be adopted as a recommendation of this Congress. You can get no better cooperator and successful worker for Conservation in any portion of this country; so help us to carry into effect the great purposes of this Congress, the Conservation of our natural resources. (Applause)

The CHAIRMAN—Gentlemen, you have heard the resolution.

A DELEGATE—I move that the same be adopted.

The CHAIRMAN—It has been moved and seconded that the resolution be adopted. Are there any remarks?

A DELEGATE—I move that it be referred to the Executive Committee for such action as they see fit, to be reported tomorrow morning.

The CHAIRMAN—The Committee on Resolutions has made its report. This resolution is in order, because it was offered by the Gentleman from Missouri as part of his remarks, and it has been moved and seconded that it be adopted. An amendment has been offered, which was also seconded, that it be referred to the Executive Committee. Are you ready to vote on the amendment?

A DELEGATE—It was not put in the form of an amendment; it was put as a distinct motion.

The CHAIRMAN—Two motions cannot be entertained at once.

General NOBLE—Mr Chairman: May I explain that this is merely a recommendation to the Association. It is not a resolution passed by this Congress to send a delegation; it is a recommendation by this Congress to the body called the National Conservation Association, to act on it or not as they please.

The CHAIRMAN—Gentlemen, we will vote on the original question. All those who are in favor of adopting the resolution as read by General Noble will say "Aye." (The resolution was adopted)

Professor CONDRA—Would it not be in order to hear from the Executive Committee relative to the work in Missouri? I would ask that you grant to Chairman White a minute or two to respond for that State. (Applause)

Chairman WHITE—Gentlemen: I will not take much time, as it is getting late. I may say that Missouri does not have to be "shown."

On behalf of the Executive Committee, let me say this: We have had a pretty hard time in trying to satisfy everybody. I wish to apologize for any shortcomings on the part of President Baker and myself. I was Chairman of the Executive Committee, and had a great deal to do in trying to frame a program that would be satisfactory to every Delegate and everybody else who attended the Congress; but it was a hard thing to do. The President of the United States and the ex-President of the United States occupied two distinct days, and drew great crowds in opposition to the State Fair; since then the Fair has been doing business in opposition to this Congress. Then we had the Governors here, and other great speakers; and the program had to be carried out. I wish to say that this organization will remain in session tomorrow if, at the close of this session, you so desire. I am going to stay here; I have authority to act as President pro tem tomorrow, and President Baker will also stay with me, and we will gladly do all the good we can.

Delegate VON TOBEL (of Montana)—Montana has filed a report.

Professor CONDRA—For Nebraska, I will speak briefly:

I have had the very great privilege and honor of being connected with a Nebraska State Commission for eighteen or twenty years. We have a great variety of resources, mostly agricultural. He who says Nebraska is a poor agricultural spot does not know; he who thinks Nebraska is a sand-hill region does not know. In Nebraska there are four great soil regions. Some of them are very fertile; some 40,000 square miles are unusually fertile, the land values ranging from \$100 to \$200 per acre. We have 18,000 square miles of land worth from \$1.00 to \$5.00 per acre. I am not going to take the time to tell you just how good and how bad Nebraska really is; there is enough of it that is especially good.

We have a number of problems that should be taken up in the way of Conservation, and we have undertaken to do it. We have irrigation, dry

farming, forestation of sandhills and the like; also conservation of soil fertility, and the conservation of lands. Our Commission is non-political; and I believe all States taking up Conservation problems should have non-political commissions. We have in Nebraska, working with the Commission, some ten or twelve committees, with 30 to 40 men at work, studying the problems of the State. We believe in cooperation and thorough investigation, and we believe, further, in contributing that which is suited to those who wish our contributions.

We held a State Congress not long ago in which it was the sentiment, and was declared by the President of the Congress, "We want at this time that there may be made no reference to the controversy now waging in the Nation." And no man on that floor spoke one word pertaining to the controversy. It was said further that, "We wish at this time that our work be non-political, that no man will stand here and talk that he may gain favor, or gain notice in the State, for political purposes;" and with but one exception no man undertook so to talk, and that man was stopped immediately (applause). It was also asked that no man take the floor unless he had a message and facts for the others, such facts as would be worth something to those attending and those at home.

Such is the spirit of Nebraska. We are not the only State, we cooperate with others. We have good features and bad; but we want to learn to do practical things worth while to the farmers, worth while to those who are laboring, and worth while to all the people in the State. One of our committees is working on vital resources. We realize that while we grow wheat and corn for man and beast, we are working chiefly for the elevation of man; and in Nebraska one thing we will see to is that the conditions are suitable for crops, for animals, and for man—and we propose to do our part in conserving the public health, and in looking to better living conveniences and better water supplies in the State.

I have spoken three minutes, but I ask, since I happen to represent the Association of Congresses of the various States, that you join with those commissioners who were in the meeting last night in practical work in the States, and in the United States, so that when we reassemble we will have reports from men who are doing practical work. We ask for reliable cooperation to the end that our investigations will serve as a basis for action

of use to the practical people of our country, especially the farmers. I thank you. (Applause)

A DELEGATE (from New York)—In the absence of our chairman, the Delegation from New York would say, in a word, that we are making progress; that we are with this movement first, last and all the time, and that we hope at the next Congress there may be opportunity, as suggested by the gentleman from Indiana, to draw out fuller information regarding resources from the Delegations who have come from all over the country. Many of the Delegations have come here at great expense. Perhaps no one has listened with greater interest to the able speeches that have been made here than have the Delegates from New York, but we felt, in a representative organization like this, much in the position of the man who, in a legislative body, said that whenever they began to make speeches he went to the committee-room and went to work. We believed that with combined action (as the Chairman has announced) at our next meeting we shall have the speeches and at the same time draw out the resources of the people, and so get down to work and make rapid progress right along. (Applause)

Delegate R. A. NESTOS (of North Dakota)—Mr Chairman: North Dakota has the honor of sending the largest number of Delegates to this Congress with the single exception of Minnesota, which shows that it is very much interested in the movement of Conservation. North Dakota has more coal conserved than any other State in the Union. We have thousands of acres of coal, in seams varying in thickness from 5 to 32 feet of solid coal. All of our resources, with the exception of coal, are in private hands. Our great coal fields, during the last Administration, were put in the hands of the Government, and hereafter no settler can get anything more than a surface right to those coal fields. The coal belongs to the Government. Of course we haven't very much use for coal up there, but we are keeping it. Whenever you get chilly, just raise your hand and we will send down all kinds of coal for all of the hundreds and thousands of our people.

Our chief resource is our soil, which, when properly conserved and developed, can produce one-tenth of the food for this entire Nation with the present population (applause). We have a larger area perhaps of fertile soil than any other State. This is all in the hands of private owners. There is simply one way to conserve our natural resources, and that is to educate the

farmer (applause). There is nothing so cheap as education, and nothing so costly as ignorance. If our State will put half a million dollars into the Agricultural College at the next session of the Legislature, and extend its aid among the different educational institutions of the State, this money will come back in a hundredfold. It is in this direction that we must expect to conserve our resources. The interests of this Nation that lie in private hands are enormously greater than those controlled either by the State or by the Federal Government, and it does not seem to me right that we should spend so much time talking about the rather meager resources of the State and Nation and neglecting the manifestly greater resources that are in the hands of private citizens, because, in the last analysis, this matter of Conservation will be carried out on each and every man's farm. You talk about establishing a National Forest in North Dakota, and already the Government has planted a few acres in the Bad Lands; but forests in North Dakota mean the planting of 10 or 20 acres of quick-growing timber on each man's farm (applause). In that way North Dakota and similar States will carry out their part of the movement for Conservation.

Mr GEORGE W. LATTIMORE (of Ohio)—Ohio, with characteristic modesty, has nothing to say. ^[5] (Applause)

Mr BENJ. MARTIN (of Oklahoma)—Mr Chairman: I appeared for Oklahoma and reported this morning to the Chairman, and I ask that the report be printed in the record.

Mr A. W. KRUEGER (of South Dakota)—Mr Chairman: All of our members who are speakers have left, and there is no one here from South Dakota except myself. I am not an orator, so I will not attempt to make a speech; but when I heard from other States I could not help feeling that I come from a State that has the richest resources in the world. Our greatest resources lie in our inexhaustible soil and its fertility. We have people from most of the States in the Union, and when I have asked our citizens from several of the eastern States, and other rich States like Iowa, Minnesota, Ohio, Wisconsin, and Illinois, "Why are you here?" they said, "Because we have bettered our condition through the State of South Dakota." So I have come to the conclusion that we must have about the best State in the Union (laughter). They tell us that we have more money per capita for our schools than any other State in the Union—but I do not want to make a speech, for I can't do

it (laughter), only to say that I have had the pleasure and great honor to talk Conservation in our State, and the longer I am here the more I am convinced that South Dakota is in hearty accord with the doings of this Congress (applause). I have not been instructed to say this, but from what our State is doing, I cannot see how any true and patriotic American citizen who loves his country, home, and fireside, and who wants to leave them to his descendants none the worse because he lived in the world, can help most heartily endorsing the Conservation of our natural resources, such as forests, natural waterways, water-powers, minerals, coal, oil, and phosphates by the Federal Government. (Applause)

Mr GEORGE H. EMERSON (of Washington)—Ladies and Gentlemen: I come as the calm Pacific instead of the cyclone that at times has swept over this audience. I came prepared with a paper that it was proposed to have placed before you, but it is not propitious at this late hour, neither is the temper of the audience such as to receive it, nor is the time that is allowed me sufficient. I ask, therefore, your permission to file the same.

The CHAIRMAN—Permission will be given. Washington was called this morning but the representative was not present, and Ex-President Baker tells me it was also called again this afternoon.

Two DELEGATES—Mr Chairman—

Chairman WHITE—The Gentleman who addressed the Chair first is recognized. This Gentleman from Washington (indicating).

Mr WILLIAM DOUGLAS JOHNS (of Washington)—Mr Chairman: I would ask of the Delegates here three minutes.

The CHAIRMAN—There are just three minutes left, and you can have them.

Mr JOHNS—Mr Chairman: I wish to tell the Delegates here, for the purpose of showing the necessity of Federal control, how the water-power sites of the State of Washington—the greatest of them—have passed from the hands of the State within a few months, under the administration of Land Commissioner Ross, who has made himself so prominent here this evening. Two corporations have filed on the lower waters of the mighty Columbia, a railroad and water corporation with steamboats plying 100 miles above and carrying freight and passengers, and an irrigation corporation below, using

half of the waters of Columbia River, and all the State of Washington got was filing fees; and Governor Hay wants us to give the balance to him in the same way—the other half of those great waters of the mighty Columbia. The lands secured by the railroad corporation within a few months on the shore—lands worth millions of dollars—were sold by Governor Hay and Land Commissioner Ross for \$10,000, and Governor Hay wants us to turn over more to him for the same purpose. The waters of Chelan River in the Cascades James J. Hill secured (125,000 horsepower) by paying filing fees to the State. No wonder, in his speech, he favored State control! (Applause)

A few days before I left Washington a dispatch came from Port Townsend to the Seattle papers—making a glorious spread—saying that the water-power company, capitalized at a million or two, was going to put in a 6,000 horsepower plant to supply Port Townsend and the neighboring country—and then boasted of the country to show what a good thing it was to invest in. They said the company had secured every water-power site on the river, right up to its eternal glaciers, and that they had been twenty years in securing those sites. Were they doing it for development? Never! They were going to take one lower fall and develop it, and sell the power at a high price. They had secured all the other sites along that river—and for what purpose? To prevent competition until the country grew up by paying taxes simply, holding a water-power site that amounted to nothing until the people were prepared to pay an immense revenue to them. So much for their plea of Governor Hay that he wanted the State developed. The Olympia National Forest, reserved by President Cleveland, was opened in response to a similar complaint as that made by Governor Hay, "You are driving settlers to British Columbia." It contains some of the richest timber lands in the State of Washington and on the Pacific coast. What was done with it? Part of it was covered by scrip, a few quarters were taken by war settlers, the balance by speculators. They sold at from \$600 to \$800 per quarter, a few holding on until within the last few years; and the result was that it has passed into the hands of the corporations. Since the Milwaukee built out there, they burned up much of it; and today you can go into great tracts of that land (I have been through it) and you would never know that a human foot had stepped there—it is as wild as it was before Vancouver sailed along the coast on his voyage of discovery. If the National Forests of the State of Washington were turned over by the United States Government

to the State of Washington and its officials, and the tender mercies of Land Commissioner Ross, they probably would go just exactly as the Olympia Forest went—into the hands of speculators, not to be settled up, not to bring wealth and people and glory to the State, but to be held until timber is valuable, to be kept in primeval wilderness. Gentlemen, I thank you. (Great applause)

A DELEGATE—Mr Chairman: I wish to correct the Chair in his remark that no one was here this morning to present the report from Washington. I happen to be chairman of the Delegation. I know all about the meeting behind closed doors in the Saint Paul Hotel; I am sorry I couldn't tell about it here; but I filed my report this morning at 8 o'clock, and explained that Mr Emerson would speak for our State.

Mr ROSS—The State of Washington has been exhausted—

The CHAIRMAN—It has not been exhausted. We will give you a few minutes.

Mr ROSS—Under the heading of personal privilege. I am not going to take your time to rehash any controversies referred to in the eloquence from the State of Washington. I will take sufficient time, however, to tell you one or two things. The Gentleman, so far as the Delegation from the State of Washington is concerned, speaks for himself and for no one else.

Mr JOHNS—Thank God, I do not speak for you! (Applause)

Mr ROSS—The Gentleman who has just spoken sounded the only discordant note in a meeting of 500 citizens of Seattle where, to a man, they endorsed Richard Ballinger! (Hisses from the house)

He is the only man in the city of Seattle—

Mr JOHNS—The only thing done in Seattle was what Mr Ross did.

Mr ROSS (turning toward Mr Johns)—I quit and allowed you your three minutes, although you were not entitled to appear here at all. Now that you have aroused me to some extent, and since they have kindly consented that I may be heard for a few moments. I wish to tell this vast audience that the State of Washington—and I speak solely in my official capacity, and I am not ashamed of any act I have ever performed—I wish to say that I served

the State of Washington as Assistant Attorney-General for four years, from 1901 to 1905, all the time dealing with our State lands. The people of the State of Washington, on my record there, elected me Commissioner of Public Lands two years ago, and during the four years that I was Commissioner of Public Lands, I made the same fight for the State of Washington that I am making now. I made it in the Navigation Congress, in the Forestry Association—and God knows how many things there are going on that a busy man cannot keep track of—and the people of the State of Washington, every man, woman and child, knew E. W. Ross; they knew his record; they knew his fight; and in spite of all opposition from everybody in the State, like the Gentleman who has just spoken, they nominated me by a popular vote two years ago, and I was—(Commotion in the house, many Delegates leaving).

He says we have given away the water-powers. The State of Washington commenced her Conservation policy prior to November 11, 1889, when we were admitted into the Union. We have one provision in our Constitution relative to water-power that I would rather have in the interests of the common people than all the discretionary powers you vested in all the presidents and all the public officers of the National and State governments. We have a provision in our Constitution like this: the use of water for irrigation and the like shall forever be a public utility. You heard Theodore Roosevelt say that it was the intention of the National Government, upon easy terms, to let the water-power out to private corporations so that the people of the States could have cheap electricity and cheap power furnished by these corporate institutions. And let me say to you, you heard the statements made by Governor Hay, of my State, as to the accomplishments of the Railway Commission in regard to the railway companies. In my State, the State Railroad Commission fixes the proper rate, and that ends it (applause). Our objection to the movement is this: We are not fanatics; we have conserved beyond the possibility of any human agency, State and National governments; we have conserved the people's rights, so that when they need protection of the law it is vested in our Constitution, and all the people have to do is to rise up and enforce it (renewed commotion). That is why we do not wish to surrender those powers to the National Government, or to the discretion of any man.

Take the power proposition which has been mentioned by the Gentleman: Neither the State of Washington, the Board of State Land Commissioners (of which Board I have the honor to be chairman), nor the members of the State Tax Commission had anything to do with the taking of a site by the Hanford Irrigation and Power Company—not a thing. Let me tell you what it was. On Columbia River, some 40 to 60 miles above Kennewick, is what is known as Priest Rapids. The War Department of the United States Government is supposed to control Columbia River. It is navigable for all sorts of crafts both below and above Priest Rapids. By virtue of an act of the Legislature of 1905, the State of Washington conferred upon the Reclamation Service of the United States express authority to appropriate, for its own purposes and the purpose of irrigation, all the waters of Columbia River and every other stream in the State. The Reclamation Service, in compliance with that act, filed upon the waters of Columbia River at Priest Rapids, and, in one particular year, filed an express relinquishment and abandonment of that project. They stated, in cold type, that they would never undertake it. And what next? They consented, in writing, that the Hanford Irrigation Company might have and enjoy it. The Hanford Company went to the War Department of the United States, and obtained a permit to build a concrete dam in Columbia River at Priest Rapids to assist irrigation, and the War Department consented; and outside of that the Hanford Irrigation and Power Company has acquired nothing whatever from the State of Washington. But supposing that the Hanford Irrigation and Power Company is using the waters of the State of Washington for irrigation and power purposes—whenever the people of the State of Washington are convinced that the Hanford Irrigation and Power Company is charging an unjust or unreasonable price for power, or for water for irrigation, or for the annual maintenance fee, thank God we have it vested in the Constitution of the State of Washington that the people can fix the price. That is our style of Conservation, and that is why we object to Federal control. I represent the people of the State of Washington (laughter), and I don't care who says to the contrary, and I am proud to oppose the surrendering of absolute control by the people in favor of the discretion of any man.

Mr E. H. FOURT (of Wyoming)—Mr Chairman, it is now very late. I was not able to attend this morning and submit a report or an address. I will present

this report, and move the Congress that it be printed in the record as a report from Wyoming.

The CHAIRMAN—The paper will be received, if there is no objection. (The paper was handed to the Secretary.)

Mr B. A. FOWLER (of Arizona)—In answer to the call for Arizona, I want to say that at present Arizona is a Territory. One year from now, at the next Conservation Congress, we hope that Arizona will be a State (applause), and that at that time we will make a State report of which you will not be ashamed. (Applause)

Mr G. M. HUNT (of the District of Columbia)—Mr Chairman: I simply want to announce the fact that the District of Columbia is on the map (applause). Lots and lots of folks are under the impression that the District of Columbia only exists from the second Tuesday after the first Monday in December until Congress adjourns (laughter); but, on the contrary, the District of Columbia is on the map 365 days in the year. Further than that, we have a Chamber of Commerce that is working 24 hours a day during that 365 days. Still further, this Chamber of Commerce has authorized and directed me to present an invitation to this National Conservation Congress to hold its next annual session in the District of Columbia. Washington, D. C., is the capital of the Nation; it is your home; it is *your* capital; you have helped to make it what it is, and it is time that you should get there and see how we have spent your money. The Far West has been converted to Conservation, and with the setting of tomorrow's sun the Middle West will have been converted; and we feel that we of the East need conversion, and we want you to come to Washington in 1911 and convert us. I thank you. (Applause)

Mr HENRY A. BARKER (of Rhode Island)—I think, at this late hour, it is not right to take very much time of the Congress. I take pleasure in filing the report of the Conservation Commission of the State of Rhode Island.

About three weeks ago the Legislature of Rhode Island established the Conservation Commission on a new and more efficient basis than that which previously existed. Of course I might spend a great deal of time in telling you that Rhode Island, like every other State that we have heard from so far, is by far the grandest and most splendid of all the States of the Nation (laughter and applause); but I think, under the circumstances, I will

confine myself to reading the last paragraph of the report, I will file in order to show you the position Rhode Island occupies in certain matters. "Rhode Island has awakened to vital things, but even if it had only an indirect interest in Conservation it would still feel that it owed its moral influence to the country as a whole, and that it is not a selfish little 2-cent Republic all by its lonesome, but a part of a great Nation that prefers to be governed from Washington"—I mean Washington, D. C. (laughter)—"rather than from Wall Street. It prefers to belong to a Nation whose prosperity and power and glory need the cooperation and loyalty of every one of its citizens." I thank you. (Applause)

The CHAIRMAN: It has just been called to my attention that several Delegates who have spoken for their States have not handed in their names; they will be privileged to hand their names and addresses, with the remarks that they have made, to the Secretary.

Professor L. C. WHITE (of West Virginia)—West Virginia has been overlooked; it is on the map. I will not take much of your time, Gentlemen; only enough to say that West Virginia has so far fought a losing game on the question of Conservation with reference to our oil and gas resources. The great corporations have wasted natural gas in West Virginia to the value of from \$200,000,000 to \$300,000,000, and this is still going on at the rate of a quarter of million of cubic feet daily. Our late Governor Dawson appointed a Commission on Conservation, and it made an able report; but the legislators, who are largely controlled by the corporations, have taken no notice of it whatever. The only thing actually done in the way of Conservation was the establishment of a State game and fire warden, who has some power in the way of stopping the forest fires—thanks to one great Conservationist, Mr Gifford Pinchot (applause), through whose great influence we have made some advance in the preservation of our natural resources. And the State of West Virginia also owes a debt of gratitude to Dr Joseph A. Holmes, whom the President recently appointed Director of the Bureau of Mines; at his instance an expert was sent from the great laboratory at Pittsburg to the mines of West Virginia to investigate the causes of mine explosions—through whom we learned that the dust of the mines would explode,—and that expert sacrificed his life in a West Virginia mine. The former method of mining has now been entirely abolished, and during the fiscal year ending June 30, 1910, out of the sixty or seventy

thousand miners of West Virginia not a single human life has been lost as the result of dust explosion (applause). And now that Dr Holmes is at the head of that great bureau (placed there against the wishes of some of the members of the cabinet of President Taft), we are sure that other discoveries in certain lines will be made in West Virginia for the conservation of human life. (Applause)

Mr E. L. WORSHAM (of Georgia)—Mr Chairman, I want to report that Georgia, too, is on the map. I am not going to take your time in an attempt to make a speech or even a report. There are a number of problems I wanted to discuss, but in view of the fact that I know all of you have had more Conservation than you can digest in one evening, I forbear. I do want to say, however, that the West and the Northwest are not the only sections of the country which are interested in Conservation. Coming from one of the oldest States of the Union (one of the original thirteen) I can say that there is a greater demand for systematic Conservation in our section of the country than there is in any other. We can appreciate the value of Conservation. Nature has been exceedingly kind to this section of the country in the distribution of natural resources. Georgia was originally the chief gold-producing State of the Union. She still has rich mineral resources. She has water-power enough to run all the mills in the Southern States and then have some to spare. I can appreciate thoroughly what the water-power proposition means, because we are up against that same proposition now, wherein the large corporations are trying to gobble up the water-power sites: and that is one of the main problems of Conservation which confronts the people of Georgia today and will be fought before the State Legislature; and I want to assure you, right now, that we are going to depend upon the National Government for aid in propositions of this kind. (Applause)

We are interested in Conservation, but our time is too valuable to be wasted in the discussion of States' rights, because our people fought out that question forty-five years ago (applause). My father spent four long years fighting on that problem, and we consider that it has been solved to the satisfaction of the great majority (applause).

I want to endorse what Mr Pinchot said this morning in behalf of the work which the women have done for Conservation. I don't know how it is in the

North; but with us the women are the moulders of sentiment, and they have been fighting in this movement for a number of years (applause). We are going to hold a Southern Conservation Congress in Atlanta on October 7 and 8 (applause). That movement is backed by 25,000 women in the State of Georgia, by the various women's clubs, by the press, and by all the leading citizens. If there are any factions, they have not made their appearance as yet. We are going to make that Congress a success. We plead for your cooperation, because we need all the help we can get. (Applause)

A number of telegrams were read.

The CHAIRMAN—If there is nothing further before us—

Mr BENJAMIN MARTIN, JR.—Mr Chairman, Ladies and Gentlemen: The hour is growing late, and it is my pleasure to rise for the purpose of offering a motion to adjourn; but before making that motion, as a Delegate from Oklahoma, and speaking for the other Delegates, I wish to thank the good people of Minnesota, and more especially the people of the Twin Cities, for the delightful manner in which they have entertained us. As we return to our homes, we do so with the most pleasant memories of our visit to this great metropolis. One great inspiration to me, and to most of the Delegates, has been the attendance of ladies at the various sessions. Now, without further comment, I move that this Congress adjourn *sine die*.

Chairman WHITE—Without objection, the Congress will stand adjourned: it is adjourned.

SUPPLEMENTARY PROCEEDINGS

LAWS THAT SHOULD BE PASSED

FRANCIS G. NEWLANDS
Senior Senator From Nevada

Regretting my inability to address the Conservation Congress personally on the subject assigned to me, I submit my views briefly by telegraph.

Conservation legislation necessarily involves harmonious action by forty-seven sovereigns, the Nation and the States, each acting within its jurisdiction. As the legislative bodies cannot confer together, it is necessary that there should be some intermediate organization which will bring about team work. There should be a National Commission and State commissions which can act together, as well as separately, in recommending needed legislation. A reactionary Congress disregarded Roosevelt's recommendations on this subject, but the progressive sentiment of the country will not brook further resistance; and the bill for the appointment by the President of a National Conservation Commission composed of publicists and experts in civil, hydraulic, and electric engineering, in arid and swamp land reclamation, in transportation, and in mining and lumbering, reported by the Senate Conservation Committee at the last session, should surely pass. With Roosevelt as chairman, and Garfield, Pinchot, Newell, and the Chief of Engineers of the Army as members of this Commission, we would have the men who in practical administration have become more thoroughly informed regarding the natural resources of this country than any others.

As to the land laws: It is evident that for years large portions of the public domain have been gradually drifting into private and monopolistic ownership under antiquated and misfit land laws utterly unadapted to existing economic conditions, and therefore stimulating fraud in their evasion and perversion. Legislators outside of the public land States have taken little interest in the subject, relying mainly upon the States involved to

suggest legislation. Had the Senators and Representatives from the public land States counseled together continuously, patiently, and tolerantly regarding the land laws, as they did regarding the Reclamation Act, the confusion and scandal and the prosecutions of the past six years would have been lessened, and a wise solution of needed legislation would have been evolved and accepted by the country. At the next session of Congress such a council of western Senators and Representatives should be held, and the present deadlock of conflicting views ended. In shaping laws regarding the public lands the central idea should be a rational development, without monopoly or waste; the establishment of individual homes upon the agricultural lands; the utilization of the forests and the coal, iron, and oil deposits under conditions that will enlist the aid of needed capital without monopolistic exaction or excessive prices; and the improvement of our waterways, regardless of State lines, so as to promote every use to which civilization can put them, and in that connection secure team work on the part of the various services, National and State, engaged upon them, as well as the cooperation of the Nation and the States, each within its appropriate jurisdiction in the work to be done and the expenditures to be made.

Until comprehensive plans are developed, the Nation should not part permanently with the title to any lands suited for the development of water-power, the promotion of navigation, or the establishment of transfer facilities and sites, but should hold the National properties in such shape that they may be utilized in the working out of comprehensive plans involving the union of National and State powers. In forming these plans it should be borne in mind that the Nation holds the public domain, not for National profit, but in trust for the population, present and future, of the public land States which welcome immigration from other States whose surplus population there finds a resting place. The money realized either from sale or rentals should therefore be applied to the schools, roads, reclamation projects, and other public development of the States in which the lands are located.

The ultimate purpose of the laws should be to gradually substitute State sovereignty for National sovereignty in the direction and control of this great public trust; but great care should be taken not to prematurely turn over the trust to States too weak to resist powerful combinations and

monopolies, or until the organization of adequate public regulation and control is effected.

CONSERVATION OF THE NATION'S RESOURCES

J. B. WHITE

*Chairman of the Executive Committee of the National Conservation
Congress*

In the division of the program set apart for discussion there are many ideas and inquiries crowded upon our minds for expression; and while much will be made clearer to us, there will be many questions that will remain to us unanswered.

Perhaps we may first ask ourselves: Why are we here? What came we here to do? What is Conservation? To whom does it apply? Who are Conservationists? And who are enemies of Conservation? Are there any, and why? What special principles must we subscribe to in order to be known as sufficiently orthodox in creed that we will be received as worthy disciples in this cause? And who but ourselves (and each for one another) shall pass upon our credentials as to our honesty of purpose in this great work? To whom are we answerable but to ourselves, the people? And why should a great congress of thousands of American people meet here, as we are doing this week, on this occasion, when we have a legal Congress in Washington representing every district in this broad land, whose members we have elected to make such laws as are necessary for our present and future welfare?

The answer seems to be that this assemblage represents a popular upheaval of public sentiment, animated and encouraged by those who have thought along advanced lines and are pioneers in this cause in the press and on the rostrum; some of whom have been right, and others of whom were almost wholly wrong. We are here to discuss these features, to winnow the chaff from the golden grain in this agitation of thought which we trust will be the beginning of wisdom, to be crystallized as far as practicable into proper National and State laws for the regulation of Conservation of public resources, and that the people may become awakened to that greater saving principle of personal and private Conservation. It is we, the people, instead

of we, the politicians, who are and should be most in evidence at this Congress.

It has been said that knowledge is power. It is perhaps a better truism to say that action, with knowledge, is power. Knowledge without action would avail little; and action without knowledge would be groping in the dark. But with knowledge and action we can accomplish noble results.

All great reforms and improved conditions spring from the wants, needs, and consciences of a dissatisfied people. Sometimes the needed relief comes through an armed and sometimes through a peaceful revolution. Some man looms up above his fellows from the sea of unrest and his greatness is proven by his devotion to the cause, free from the selfish thought of personal aggrandizement; and by his wisdom and tact he creates confidence in his judgment, in his sagacity, in his fitness for leadership. So few there are who are willing to bear the cross from this high sense of duty and offer themselves a mark for calumny and vituperation, and often in many ways to become a sacrifice to a people's cause! And when one is found, it frequently happens that the public are slow in showing their gratitude and appreciation for what his discernment and discretion saved to a nation; the reward of proper recognition is often withheld until long after he is dead, because he lived in advance of his time.

But there are fictitious and exaggerated issues which are created and developed to huge proportions for the dear people by the sleek politician (and his name is Legion) who sets up his scarecrow of impending woe that he may rush valiantly in and save his constituents and the citizens of a nation from dire calamity, and generations unborn from distress and want. It is not my purpose to attempt to lull to sleep in fancied security those who have been influenced by those suspected of being unnecessarily active in fighting windmills. Always there is need of sound, conservative consideration before taking hasty action, and the people are becoming better informed and more critical in their discriminations, and are learning to know the loud-mouthed pretender from the thoughtful, loyal, public-spirited citizen. People now are doing their own thinking. Time was not long ago when the greatest newspapers of largest circulation manufactured public opinion so successfully that they were the great thinking machines for the country. It was so much easier for the people than doing the actual thinking

and logical reasoning for themselves. People read the editorials of their respective journals in order to get ideas to use in their arguments with each other. I think that as an educator the newspapers then, as now, served a most valuable purpose, but it is infinitely of more help to the thinking man, who criticizes and analyzes what he reads before he accepts and assimilates it as his own. The pen has been mightier than the sword, and the "power of the press" has matured and developed conditions that had to be arbitrated by the sword.

The People Deceived

Much harm has been done by wrong thinking in regard to Conservation, and the people have been deceived and prejudiced; and like a strong man awakening from a sleep they have reached out in alarm to search for and punish, in advance of ascertaining what, if anything, was really the matter. Innocent people and innocent industries were maligned and injured. The public are now finding that they have been deceived by the scheming politicians, and by highly colored newspaper comments, and that "Conservation" has been used as a trick word and is not what they thought it was. They had been led to believe that it was something that someone else ought to do, or should be forced to do, and that they were being robbed because Conservation was not practiced; and that if Conservation laws should be passed as recommended by these ignorant agitators they would be greatly benefited; that everything would be cheaper than they had to pay, and that they could get more for what they had to sell. They never stopped to reason that Conservation without use means holding back from development the natural resources of the country and producing stagnation in business, and that if each succeeding generation should follow the same policy there never would be any improvement.

Those whose education never has extended beyond the three Rs can understand the principles of Conservation in reforestation, reclamation, and restoration—reforestation where it will pay to reforest; reclamation where it will pay to reclaim; and restoration where it will pay to restore to the soil the elements needed, and where forestry will not pay better.

The great American leader of Conservation always has maintained, and especially in his speech at the first Conservation Congress a year ago, that

the first principles of Conservation is development of resources for the benefit of the people who live here now; he stated that there might be just as much waste in neglecting the development and use of natural resources as there is in their destruction by wasteful methods. In the second place, Conservation stands for the prevention of waste; and in the third place, Conservation stands for the preservation and perpetuation of our resources through wise economy and thrift. And its principles apply alike to individuals and to nations. If a policy in any department of Conservation requires great outlay of money in order to develop and conserve for this and future generations, then the Government, the whole people, and succeeding generations may be rightfully asked to bear part of the expense, which could be done by the selling of bonds, and by exemption from taxation some products of growth, like the forests, which are now taxed every year, making the owners pay taxes for a hundred years to get for the market only one crop. No other crop is taxed like this. The owners of any one resource should not bear all the burdens for growing it for future generations; and, if it does not pay, the soil will be used for other crops which will pay better.

True Leaders Misunderstood

As a matter of fact, the true leaders in intelligent Conservation have been misunderstood by press and people. The principle has been attacked as if it meant the non-use of our natural resources by the present generation. Even in Alaska the best known teachers of Conservation urge the development of all the resources for the benefit of all the people. They wish to encourage pioneering on both a small and on a large scale. It is not the purpose of Conservation to preserve from exploration and discovery unknown resources if there can be found for new fields men brave and fearless enough to take the risks of life and of capital. For instance, when in the prospecting of a country's possibilities, and in the risks of life and capital incident thereto, there is called into action every bit of physical or mental energy to meet conditions that seem insurmountable, it is not our purpose to hamper or retard, and say that this risk and cost shall be left entirely to future generations. We are willing to encourage the cutting out of the way, and doing all we can in this generation, believing that the next generation will find new duties suited to its advanced condition and change of needful requirements.

Wise Conservation with use means the maximum of efficiency and profit with the minimum of waste and cost. We do not wish the few to have unfair advantage. We desire each and all to have opportunity according to their talents and their physical or financial abilities, that the known and unknown resources of a country shall not be gobbled up by a few without an equal opportunity to others who can furnish the same needed measure of requirements to insure success. We realize that corporations are necessary to develop a country; that aggregations of capital, made up from large and small stockholders of the people, can accomplish more with less waste than can individuals. They can put in expensive and saving devices and can operate at a far smaller percentage of profit.

For example, in gold mining the individual works his placer claim with simple equipment; he will leave 50 percent as waste. Then the dredging companies will follow, and get half as much more. Then come the big hydraulic companies, reaching benches and levels that men with small capital could not attain. These companies require millions of capital, and they save the waste and are satisfied with a net profit of a few cents per ton. Just so with Alaska's coal; Alaska needs the coal, and all we ask is that some fair method shall be adopted which will best subserve the requirements and will encourage development.

Conservation and economy must enter into our very life, and every effort should be made to get the most out of little; to find a use for what now we are in various ways wasting. In European countries the hard struggles of the masses have produced the saving habit—a virtue we have got to learn and practice individually and as a people before Conservation will become a National success. Unless we halt in our mad extravagances, perhaps there will come to us in some degree those sad experiences of suffering that have put the saving principle into the very fiber of the old world peoples; and we will learn our lesson as they long ago learned theirs. We must each share the burdens of Conservation, and we all likewise will share in its blessings.

Conservation is not any one man's opportunity, prerogative, or privilege. It is for the use and benefit of each and all, and can be practiced in any business or occupation as an important aid to success. It is for peasant and prince, rich and poor, and for the Nation as well as for the individuals. We

must discover some effectual means to prevent disastrous forest fires. We must restore the fertility of our soils.

The Question of Ownership

Whether Conservation is best promoted by individual ownership of certain natural resources is a disputed question, depending on the nature of the public utilities, the location, and other conditions. Government ownership does not mean that the Government is going into business competition with private capital. It means reasonable royalty and fairness and protection to the lessee, to enable him to compete under prescribed conditions favorable and just alike to all parties.

In most cases ownership causes a man to see the need of conservation and economy. The idea of protecting natural resources against waste is not so strong with some, if they do not have possession; while with others it is true that possession gives them the desire and opportunity to see the actual dollars, and they make haste, manufacturing more than the market demands so that only the prime and best qualities find a market, which causes a grievous waste. Especially is this true in the lumber business, but it is not so true in the meat business. The packers of meat products have studied the science of saving and conservation, so that the entire carcass is utilized in some useful manner absolutely without waste. And on the farm the man who is financially able to study and practice conservation of the soil prevents its exhaustion, while his poorer neighbor, lacking the funds for the initial expenses, sells the life of the soil with the crop that he markets, and his farm is soon impoverished. We, or some of us, believe that there should be some way of extending State aid at a low rate of interest to the poor farmer to enable him properly to fertilize his soil; and that the chemistry of soils and scientific agriculture should be taught in the common schools. Thus would the entire country be benefited, and National efficiency strengthened.

Practical Application

The science of Conservation, as a philosophy, is wholly independent of who owns the property; but its successful practical application often depends very much on ownership. Combinations of capital have the advantage, and this needed capital gives greater possibilities for Conservation. Compare the

country butcher and his 50-percent waste with the million-dollar packing house which has no waste. It is not the fault of Conservation that there are extremes in combinations, resulting in trusts or monopolies. They are practicing Conservation in the extreme, in saving of raw material by greater utilization, and by the discovery of new uses for by-products. The Standard Oil Company is another example of the very fineness of division and subdivision of by-products, which finds a place in therapeutics and in the arts, and appear in vaseline, paints, dyes, and a hundred other valuable chemical products. This is Conservation. But there are hundreds of ways where Conservation can be practiced to a profit in every occupation of life, to the physical, intellectual, and moral betterment of mankind. As corporations are made up of many individuals to do certain things that are necessary to be done, which it would be impracticable or impossible for any individual to do alone, is it not best to recognize them as artificial individuals, subject to the control as well as to the protection of wise laws, which permit no individual to prosper at the expense, discomfort or injury of another individual?

Conservation, as a living, vital principle stands out beyond and above selfish partisan politics; and no man or combination of men will ever be able to make a political issue of it any more than you can make the gospel of spiritual salvation a political issue. But, like the gospel of spiritual and physical health, it demands the homage and acceptance of all. There will be many men of many minds, crystallizing by their combinations into different sentiments, and advocating different methods. It is so with churches. But their central doctrine of salvation will continue to be the basis of creed. And ours is Conservation, that the country and its people may continue to prosper and progress, and that the principle and practice of love and charity, which make up the Golden Rule, shall not cease to influence the hearts of men. The great question to each one should be: Where and how does Conservation apply to me?

We are here to build the temple, and to bring men up to the standard which we now unfurl to the world.

The sower goeth forth to sow; some sow to discord and strife, and some to peace and harmony; some sow to love and some to hate; some sow to adversity, and some to prosperity; some sow to selfish greed, and some to

philanthropy and public good; some sow to prudence and Conservation, and some sow to extravagance and waste.

"There is that scattereth, and yet increaseth; and there is that withholdeth more than is meet, but it tendeth to poverty."

"Be ye not deceived. For whatsoever ye sow, that shall ye also reap."

PRACTICAL ASPECTS OF CONSERVATION

A. B. FARQUHAR

Executive Committeeman National Conservation Congress

In the forum of argument the Cause of Conservation, as a general principle and in every detailed application, has already won. When taken on its own merits, hardly a voice is now raised in opposition. Yet it has still its enemies, none the less pertinacious or dangerous that their antagonism is based not on public but on private interest—enemies who are carrying on a bitter contest by indirect methods, and clouding the issue by starting side questions. We have all heard of the medical practitioner who covered his general ignorance of pathology by the device of inducing fits, which were his specialty. So, when anybody finds the Conservation policy an obstacle to his pecuniary interest, it is an easy expedient to cover his inability to overthrow or confute that policy on point of principle by a display of his skill in exposing real or fancied weaknesses on irrelevant points.

Instances of this effort to secure an advantage by an adroit befogging of the question will occur to everyone who has followed the discussion of our subject. Some of these have been so often exposed that it would be only wearisome to allude again to them, were it not for the importance of being on guard at all times on all points against the crafty contrivings of the enemy.

Let us first consider the confusion of Conservation with hostility to corporations; of encouragement to enterprise, with license to destruction of natural resources. It is true that we have heard and read some vigorous protests recently against grants of timber and water-rights to certain business corporations, whose profits would be sharply curtailed by preventing or limiting their free use of the country's forest and river wealth.

Perhaps those protests were stronger because a combination was to profit by the grants than they would have been had the beneficiaries been dissociated individuals; what is certain is that they were decidedly stronger because of a belief that the concessions were to be granted without exacting for them their full money value. Yet, even allowing that objections founded on the principles of Conservation may have been mixed in this case with objections on other grounds, it is contrary to common-sense to apply to the former the invalidity or the validity that may be discovered in the latter. Arguments for Conservation are no more or less sound because aggregations of capital in large industrial undertakings are dangerous or useful to the public. As Conservationists, we are not concerned either to palliate the frequent dangers or to depreciate the frequent valuable services to the public from such aggregations. But we are concerned to see that the National resources are not squandered for private gain, if our best efforts can prevent it.

Another example of a confusion of the question, or raising a false question, is when a saving of natural resources is identified with locking them up; as though energy conserved were to be understood as energy neutralized. Quite the contrary: Conservation means so treating our resources as to get the most we can from them. In the case of our forest wealth, as of any other in which there is growth and decay, a judicious degree of consumption of the product is a necessity. After the period of most rapid development has been succeeded by a slower rate of increase, approaching that when losses by accident and by interior degeneration are barely made up by the growing season, there is obviously a gain in removing the tree and leaving its place to be filled by something that will continue to grow. In a properly conserved forest no tree is left standing until attacked by decay, but each is cut when its value is greatest, so that the best management is recognized not so much by the largest amount of standing timber as by the largest yearly gross increase in growth of timber. It is somewhat similar with water-power. "The mill never grind with water that is past," so the power has to be used when it is within our grasp, or be lost. Conservation of water-power is maintenance of the amount of such power that so appears, and disappears. It is for use; Conservation for disuse, in this case, becomes an absurdity. Even as to mineral resources, which cannot be replaced when once withdrawn, Conservation calls only for economical use, not for neglect or insufficient

use. The argument is that we have quite as good a right to the enjoyment of the gifts of Nature as our successors have, but not a right so much better than theirs as to render a wanton waste of our patrimony justifiable, or even pardonable.

In connection with this same identification of forest Conservation with neglect of the forests, we should consider the hasty tendency shown in some quarters to regard the terrible forest fires of the last few weeks as proof that we are safer without a woodland than with it, and that therefore our policy is at fault. But our policy is care, and not neglect. We would avoid such calamities by preventing them. The recent fires appear to have been due very largely to carelessness, from meal-preparation or from smoking in presence of dry combustible material; and in some instances to actual incendiarism, which was, where it occurred, a crime no less serious than wilful murder. These forests might have been safeguarded by an expenditure of one-tenth the loss by fire, and at no danger to life. Fires from locomotive sparks—a far too frequent nuisance—should no longer be classed with unavoidable accidents; for the masses of dried grasses, etc., that make the sparks dangerous could and should be removed from the track vicinity; or the locomotives provided with efficient spark-arresters, or oil used instead of coal. From strictly non-preventable accidents such as lightning, general conflagrations must be exceedingly rare; and thus the cure for the entire evil is within the reach of capable caretakers. Where an efficient forest guard is provided, as experience in European countries and some American States has demonstrated, these disastrous visitations are no longer dreaded. To look on a universal removal of our Nation's tree-wealth as the only remedy, is like the old proposition to drive rats from the granary by burning the barn.

One more illustration will be enough. It is a great advantage to any cause when the opposing advocate can be found in error on any point, no matter how trivial or how far aloof from the essential issue; for then the convenient presumption of "false in one thing, false in all," can be raised, and the weakness of the said cause most effectively disguised. This expedient is at least as good as the famous rule in pleading: "When you have no case, then abuse the plaintiff's attorney." It is readily possible, I dare say, to show that some of our Conservationist champions may have in a few cases underestimated the life yet remaining to our National resources. Estimate in

these matters is necessarily uncertain, few or none of the elements of the calculation being well ascertained. This is particularly true of our mineral supply. The time when our coal, for example, may be expected to become practically exhausted cannot be set within a hundred years, for we cannot gauge the stock in veins out of sight, nor foresee the rate at which the use of coal will be accelerated. But it would seem sufficient to know that, at the present rate of extraction (the amount mined in 1906, as the figures show, excelled all previous records, and yet the increase in 1907 over 1906 was more than the total annual supply 10 years earlier), no finite quantity could indefinitely survive; hence more attention must be paid to avoiding the present lavish waste in mining, as well as applying water-power and wind-power wherever adapted to do the work now done by burning coal. These economies should be introduced now from choice, not left till the disappearance of our fuel-supply drives us to them by force. It is a wretched business to allow our boast that "The country's coal supply is good for many hundreds of years yet" to uphold us in the reckless waste we now make of it.

But I attempted to cover the whole field of Conservation in an address at our first Congress, and will not repeat but, in conclusion, will touch upon that form of Conservation which all will concede to be of supreme importance—the Conservation of the most precious asset of the State: its men, women, and children. If the life and health of citizens is sacrificed, by vicious measures or by simple neglect, no saving of any other of our possessions will at all avail us. The importance of efforts, on a National scale, for the maintenance of public health has been proved by ample experience, and we should see to the continuance, and especially to the proper organization, of such efforts. An important step in that direction is the proposed formation of a Department of Health under the Federal Government, as contemplated in the bill so ably championed by Senator Owen at the last session of Congress. Our race is a prey to epidemics which extend far beyond State boundaries, arising from causes that often require long-continued and expensive investigations for their determination and their counteraction; and it is obvious that any effectual work against them must be under charge of the General Government. The clear and cogent reasons for this view have again and again been given by sanitary experts, and it is needless to repeat them. The matter is of sufficient importance to

call for action from our Congress, and a resolution favoring a Department of Health, at the National Capital ought surely to meet with no opposition. I would propose the following resolutions:

Resolved, That this Congress declares its hearty approval of the opinion made public last week by our honored Chief Executive, President Taft, that his party and his Administration are pledged "to make better provisions for securing the health of the Nation. The most tangible and useful form that this can take would be the establishment of a National Bureau of Health, to include all the health agencies of the Government now distributed in different departments."

Resolved, That we accept, in principle, the "Health Department" bill of Senator Owen now pending, and strongly recommend that that measure, suitably amended where necessary, be enacted into law.

Resolved, That our Secretary be directed to communicate a resolution advocating a Department of Health to the members of the National Senate and House of Representatives, and that our own membership be urged to use all their individual influence to aid the passage of the measure hereby recommended.

REPORT FROM ARKANSAS

SID B. REDDING

Secretary Arkansas Conservation Commission

Progress has been made in the Conservation movement in Arkansas through the recent organization of the Arkansas Conservation Commission. Up to this time we have had no legislation along Conservation lines, and our Conservation Commission is one whose members serve without compensation. The Commission was appointed by our present Governor, and its officers are George W. Donaghey, *Chairman*, and Sid B. Redding, *Secretary*. The Commission has effected a permanent organization, and its membership includes some of the leading business and professional men of our State. The Arkansas Legislature will convene in January, 1911, and at that time Governor Donaghey will perhaps recommend legislation covering a fixed Conservation policy for our State.

REPORT FROM COLORADO

FRANK C. GOUDY

President Colorado Conservation Commission

The Colorado Conservation Commission is composed of thirty-six members appointed by the Governor of the State February 17, 1909, with Mr Frank C. Goudy designated as Chairman. The Commission met on call to organize March 11, 1909, at which time Mr Goudy, the Chairman, was elected President of the Commission for the ensuing year, and the following subjects were fixed upon as embracing the general scope of Conservation in Colorado, viz: Lands, Waters, Minerals, Forestry, and Natural History. Standing Committees were appointed and put in charge of these five divisions of labor.

The organization is composed of men holding all shades of opinion concerning Conservation. Some think the Federal Government should turn over to the State all the public domain within its borders, together with its natural resources of every kind; that the State should own and control the public land and all it contains. Others hold that these transfers from the Federal Government to the State should be made, but that they should be subject to conditions to be named in the grant, providing adequate protection against monopoly and other objectionable control. Still others believe there should be cooperation with the General Government, at least until such laws are enacted as will assist in the work of Conservation and until the State is better prepared, financially, to meet the expense necessarily attending such a work. There are many others who believe in the continued Federal control of the public domain and its resources.

Including the first meeting, five sessions have been held, each occupying two days. In connection with the several sessions already held, considerable labor has been performed. Many papers have been read, numerous addresses have been delivered, and the Standing Committees have made sundry carefully prepared reports. Of the papers read, more than half have been given by persons not members of the Commission, not for lack of readiness on the part of our own people, but to divide this feature of the work with the public at large. It has never been difficult to secure speakers either inside or outside of the Commission. The Commission itself is

composed of a body of more than ordinary intelligence. The Annual Meeting was marked with a banquet to emphasize the passing of the year. The last meeting, April 18-19, 1910, was devoted entirely to the subject of the water-power resources of the State. Numerous letters from men prominent at Washington were received and read, and five papers were given by persons interested in the subject. All phases of the question were presented, and the most mature thought of the present time was elicited.

One of the duties of the Secretary of the Commission is to take notice of any unlawful waste or destruction of natural resources and report the same to the proper authorities. This work has been sufficiently pursued to disclose a field calling for special attention—one that calls for legislative recognition, authority, and assistance.

It may be of interest to the Congress to know something of the resolutions that have been adopted by this Commission. A brief abstract of the elements of a few will suffice to show how the body stands on the subject of Conservation.

1—A hearty endorsement of the general policy of the Government in control and conservation of the resources of the Nation.

2—Hearty cooperation between the State and Nation in Conservation.

3—That all plans of Conservation should safeguard against monopoly.

4—That in disposing of water-power sites, all franchises should be limited to a reasonable period to prevent monopoly and regulate charges.

5—That in taxing forested lands, no account should be taken of the timber until it is cut and sold.

6—That all afforested lands over one acre and not over ten acres on a tract of 160 acres should be exempt from taxation for a period of ten years.

7—That the State, by proper laws and reasonable appropriations, should cooperate with the General Government in the protection of the forests within the State from fire and lawless depredations of every kind.

Among other things, a committee has been appointed to prepare and submit to the next meeting a brief and clear statement, for general circulation, as to

what Conservation is and what it is not; what it stands for, and what it seeks to do. The purpose is to clear away the haze of misunderstanding and misapprehension in the public mind concerning it.

The Commission is about to publish a full report of its proceedings, covering the five sessions already held.

In closing this statement, it may not be out of place to say that nature has been lavish of resources in our State—they are many and abundant, but in a certain measure undeveloped, and, so far, we have had no leisure to take up matters not directly and specifically local to Colorado, except in cases where they are necessarily general.

REPORT FROM FLORIDA

CROMWELL GIBBONS
Florida Conservation Commission

The spirit of Conservation prevaieth everywhere in these modern times, and for the reason that during the past several years vicious attacks have been made upon the National resources throughout the length and breadth of our land, and to such a marvelous extent that our whole people have awakened to the fact that something must be done and at once if we wish to preserve our general resources sufficiently to care for those we expect to come after us, and who are dependent on our country for an honest and successful living. We have been greedy and selfish in the past, and now is the time for us to curb this vicious appetite and think of those who are to come hereafter. Modern times have come to stay, but the spirit of Conservation will grow until we have accomplished the grand results of providing proper protection to our forests, mineral wealth, lands, water-power and waterways, and last but not the least our various climates that God Almighty has given us to conserve the health of our people.

Much is said as to the methods to be adopted and what necessary legislation should take place to obtain actual results of Conservation. The idea of giving absolute control over the forests, the inland waterways, and the public lands confined within the States to the National Government is repugnant to me and I believe to all of the people of my State. It has too

much the tinge of centralization of power in the Federal Government, and we have had enough of this already. The notion that giving the States power and control is in favor of the special interests is ridiculous when we look back and know what has already been done by the Federal Government giving away some of its most valuable resources to the trust-monopoly corporations of the country, and we view in comparison what the States have done where they have controlled many of these resources. I have but to call your attention to what we are doing in Florida and have done the past several years in the way of Conservation. We realized some years ago that our public lands were fast being absorbed by the railways operating in our State, and that the time would soon be at hand when our people would be unable to secure homesteads, and immigration to our great State would be unable to place that energy with the soil of our State and bring about the development of resources we were entitled to through the natural course of developments. We had within our borders a vast empire of land, over 4,000,000 acres of fertile land known as the Everglades, all of which was looked upon by the land grabbers as not worth 15 cents an acre; but greatly to his credit, to Governor Napoleon B. Broward, now our nominee for United States Senator, is due the reclamation of this property and a saving to the State of lands now valued at over \$35,000,000. Against great political odds and vigorous contests, the policies of Governor Broward were endorsed; and after much litigation through the State and Federal courts we have been able to conserve this vast area of land by drainage under State supervision and at the expense of the State. The policy was greatly doubted, but it has proven a grand success in that the State, securing title to these lands, successfully sold half of the same at a price sufficient to build the necessary dredges and pay for the work of the draining of the entire tract of property by carrying the surplus waters from Lake Okeechobee through the trunk canals to the waters of the Gulf and to Atlantic Ocean. The work that is now going on has accomplished over 100 miles of main canals with locks to preserve sufficient water for the purpose of irrigation in dry spells. When the work is completed, which will be inside of three years, the State will have provided over 275 miles of canals with the lateral canals approaching the properties of the various owners, all of which will not only result in giving the necessary drainage and irrigation but will also furnish water transportation to the Gulf and to the inland water route from Key West to Jacksonville (a distance of about 500 miles), as well as deep-sea

connections at the various ports along this route. By this State Conservation we are giving to the people one of the richest bodies of fertile lands in the United States, a territory greater than the States of Rhode Island and Connecticut combined, every inch of which will grow either sugar-cane or truck of all kinds through winter and summer. These lands will produce at least three crops a year, and to the industrious citizen who desires to live in a country that will give renewed youth and a climate unexcelled and a living independent of the world. I know of none that can be found better located to give the results than this vast empire known as the Everglades, so promptly conserved by our State Government and our people.

We are also interested in another line of Conservation, and that is the preservation of our pine forests and the prohibiting of the destruction of our sapling trees which have been attacked by those who are greedy for wealth and have no regard for the future. Much will be accomplished in this direction, as our people are absolutely opposed to the complete destruction of the forests, as it will provide no future for our timber markets; and destruction of our pine forests would undoubtedly affect our climate, which by all means should be conserved as well as the timber for the building of our homes of the future.

It may be as well for me to call your special attention to the fact that, in addition to this great work of Conservation I have related in reference to the saving of public land and our efforts in the direction of saving the forests, our State has for several years aided in a public way in the building of the great inland waterway along the Atlantic coast within the State of Florida. These canals have been made during the past 15 years, until now we have a thorough water route from Saint John's River southward connecting streams and inlets until there has been dug over 300 miles of canal, giving this great waterway and enabling the people along the eastern coast sections a cheap means of transportation to the railway center of the State at Jacksonville.

I think that we have done our part toward Conservation, and all under the jurisdiction and authority of our State. There has been no effort at graft, but all have worked in harmony in the interests of the public welfare, thus demonstrating (so far as Florida is concerned) that she is able to control her own affairs; and all she asks of the National Government is its aid and assistance.

REPORT FROM IDAHO

JEROME J. DAY

We believe that, in the interest of the United States and the State of Idaho, the agricultural land within the forest reserves of Idaho should be opened to settlement and made available for home building; and that the decision as to whether land is good agricultural land or not should be referred to those who are primarily agriculturists, rather than to those who are primarily arboriculturists, and to those who are familiar with farming in Idaho rather than to those who are familiar only with farming in general.

We believe that the protection of the forests of Idaho and the safety of life and property in Idaho require that good roads be built along the lines of streams leading into the heart of the forest reserves; and that the land along these roads, whether valuable for agriculture, timber, or mineral, should be open to entry, with such provisions in relation to habitation and improvement as will secure the presence, between the months of June and September, of a local fire-fighting force, consisting of men who know the country, have a financial interest in the locality, and are skilled in the use of the axe and in methods of fighting a forest fire.

We believe in the separate classification of coal lands, oil lands, phosphate and mineral lands; and we believe in the administration of those lands in such a way as to prevent waste, promote safety in mining, and defeat monopoly. We do not believe in a policy for revenue in relation to these lands. The revenues to be derived should be incidental, and belong of right to the State of Idaho.

We believe in the cooperation of the State and the National Government in the conservation and utilization of the water-power within the State of Idaho.

We will recommend that the Governor of Idaho call a convention or congress to consider questions relating to home Conservation in Idaho, and to recommend policies and legislation and a system of administration for all forms of public wealth that lie within the borders of the State.

REPORT FROM INDIANA

A. E. METZGER
Indiana Conservation Commission

The Indiana Conservation Commission was appointed by former Governor Frank P. Hawley shortly before he left the gubernatorial chair. The Commission as appointed by Governor Thomas R. Marshall, his successor, consists of nine members with Mr Henry Riesenberg as chairman. The Commission, through its chairman, made an effort to get a bill through the Legislature, making an appropriation for the use of the Commission, but it failed to pass. Governor Marshall was repeatedly urged to set aside a small sum out of the contingent fund so that an investigation could be made and published, but this the Governor has repeatedly declined to do, and it is thought he is not very favorably disposed toward the cause of Conservation. Having no means the Commission could do absolutely nothing, and hence may be said to be in a state of "innocuous desuetude."

Mr Riesenberg, the chairman, has, however, lectured on the subject throughout the State, visiting many points, giving his time freely and defraying the expenses out of his own pocket. He has also written innumerable articles for the papers of Indiana, and these, together with his lectures, have served to keep the subject alive; and Indianians are probably as well informed and as fully alive to the subject as people in any other State.

REPORT FROM IOWA

A. C. MILLER
Chairman Iowa State Drainage, Waterways and Conservation Commission

I have been asked to prepare for your consideration and information a history of the Conservation movement in Iowa, reviewing briefly the work done by the State Drainage, Waterways and Conservation Commission.

The sentiment toward the Conservation of our natural resources has been developing gradually for a number of years, keeping pace with the development that has been aroused throughout the country. So far as Iowa is concerned, it was augmented greatly through the efforts of the commercial bodies throughout the State, and especially of those of the city of Des

Moines, when during the year 1907 they inaugurated a movement which had for its object the securing of an appropriation by Congress to be used in surveying three of our principal streams in order that we might determine whether or not they were subject to improvement for navigation, and for the further purpose of ascertaining the value of the water-power which might be developed if the rivers were improved for navigation.

Great interest was manifested by our people, and we were finally successful in securing an appropriation by Congress for a survey of the larger of the three rivers, the Des Moines; and the United States Government has at this time a corps of some 30 engineers at work. They expect to finish their work early in 1911.

The Thirty-third General Assembly of Iowa convened in December, 1908, and remained in session until the following spring. The sentiment had been aroused to such an extent at this time that there seemed to be a general demand for the creation of some kind of a Commission to take these matters up and work them out intelligently for the good of our people and report with recommendations to the next General Assembly. It seemed hard, however, for all to unite on a general plan. Portions of our State demanded a Commission for dealing with the question of drainage only, leaving it to other Commissions to handle the question of water transportations, forestry, and water-power. A compromise was finally made, and this Commission was created. It is composed of seven members appointed by the Governor: A. C. Miller, *Chairman*, L. W. Anderson, E. A. Burgess, A. F. Frudden, T. W. Keerl, Thomas H. McBride, and W. H. Stevenson.

Duties Defined

First—To investigate the present condition of public drainage in Iowa and the benefits which can be derived from the best drainage engineering practice, the most economical administration of drainage projects, and a more economical best method of procedure to bring about the development of the water-power of those benefits may be secured.

Second—To investigate the present condition of all overflow of flood-plain lands of Iowa, showing losses due to floods in the destruction of farm crops, the losses due to the destruction of property in the cities and towns and built-up districts, the losses due to the withdrawal from crop cultivation of

such flooded lands, and recommending the proper methods of preventing such flood conditions.

Third—To investigate and survey at least one representative Iowa river to ascertain the available dam sites and the potential water-power and report the best method of procedure to bring about the development of the water-powers of the State, at the same time retaining the ultimate control of the water supply as a property of the State.

Fourth—To cooperate with the United States survey provided by act of Congress and investigate the possibilities of navigation upon the rivers or upon adjoining lands by canal, and to secure the aid of the Government experts when practicable in the several matters investigated by this Commission.

Fifth—To investigate the questions of forests and their preservation and culture in the State, especially with reference to the influence of forests on the flood conditions of the rivers and the erosion and waste of the soils.

Sixth—It is the clear intent and purpose of the act providing for the Commission that the close interrelation of the several phases of river development shall be shown, and the necessity for a broad, comprehensive treatment of our rivers shall be studied and reported upon.

Seventh—The general question of the relation of the State to the preservation of the fertility of the Iowa soils.

Eighth—The general question of the wise and conservative development and use of the mineral resources of the State, especially with reference to the mining of coal.

Ninth—The general question of the nature and condition of such lakes in Iowa as now belong to the State, and the relation of lakes and streams to the preservation of such varieties of fish, birds, and native animals as are desirable which now belong to the State.

Members Serve Without Pay

We have but \$5,000 to carry on the expenses of our two years' work. Out of this must be paid our secretary and office expenses, and the cost of printing

our reports and whatever field work is done by the engineers. It would seem that the great State of Iowa, producing annually more than \$600,000,000 of wealth could well afford to have been more liberal in appropriating for this work. The task assigned surely is no small one.

Work of the Commission

We have begun to compile our report for the printers, and expect to present a report that will be valuable for future reference and we hope of much interest to our people.

We have investigated the drainage conditions over the State quite extensively, and undertaken to ascertain the number of acres of land not available for agriculture through lack of drainage. We will have recommendations to make pertaining to this question, but have not yet worked out any plan for financing cost at reduced rates. This will be considered later, but the Iowa farmer has money and is not so much interested in having the interest rates for carrying his debt reduced, as he is in getting the first cost reduced.

Water-Power

We have investigated and surveyed a number of water-power sites and are more and more impressed with the importance of the State looking after them and seeing that the control does not slip away. No estimate has yet been made as to the value of the undeveloped water-power of the State, but I will venture that it will be shown to be several millions of dollars. If not looked after, it will be but a short time until it will be under the control of individuals or private corporations. Almost invariably wherever our engineers have gone, they either find the engineer for some crowd of individuals—for some corporation—on the job, or find that he has preceded them. I will cite one example in our State: On Cedar River, at Moscow, individuals are planning the construction of a great dam which will store an immense body of water. They have 7,500 acres of land already acquired, adding greatly to the power developed by the natural flow of the river itself. When this dam is completed, it will turn from the channel into a canal practically the entire discharge of the river at low stage, carrying it around the country to the city of Muscatine, with an average fall of about ninety feet, developing 25,000 horse-power, and finally discharging the water into

the Mississippi, never returning to the original channel from which it was taken. A syndicate plans to finance this proposition on a basis of \$3,000,000, and if unmolested the probability is that it will be carried out successfully. The Commission, however, is powerless, being a temporary creation with its duties defined. We, therefore, can only call attention in our report and urge upon our Legislature that it take some action toward protecting our people in their rights in these matters.

We are working in perfect harmony with the United States Government engineers who are on the Des Moines River work at this time. We expect much from them in the way of information that we can use in our report. Iowa is interested in some 900 miles of navigable streams, either touched by her borders or within her territory, and we hope some day to again reap the benefit of being able to load and unload freight at docks within the corporate limits of our beautiful capital city, as we did for many years in its early history; and not only Des Moines but all the cities bordering on the great Mississippi or the equally great Missouri.

Iowa occupies a proud position among the States today, rich in fertile soil, rich in minerals, coals, and shales, blessed with a happy and contented people; if given the benefit of improved waterways like the Mississippi, the Ohio, and the Missouri, making them great highways; and if at the same time permitted to improve our tributary rivers and the water-powers of this great Central West, it will make a mighty empire of itself—and Iowa, magnificent State that she is, will be in the center of it all.

We are not yet so far advanced with our work that I am able to tell you what the Commission will or will not recommend to the next Legislature. I am firm in the belief that a permanent Commission should be created, with a liberal appropriation for carrying forward the work. Possibly the field now covered by our Commission should be divided. It would seem that the drainage interests of the State would be of sufficient importance to justify the employment of a State engineer, and possibly the question of drainage would receive the entire attention of some State board. There is a great work that can be done by our Commission in the future if the State should see fit to make it permanent and appropriate the money to carry forward the work. The beautifying of our meandered lakes is something that is attracting the attention of our people, and would prove a popular move if started.

They are also becoming much interested in the treatment and handling of soils, and much good would ultimately result to our farmers if this phase of Conservation was handled intelligently and carefully. Professor Stevenson, of the Iowa State Agricultural College, a member of this Commission, is recognized as an expert in this line of Conservation work, and I believe that his part of the report when published will be instructive and interesting. I can only hope that enough interest will be aroused throughout our State to influence the next Legislature to put the Commission on a permanent basis, furnishing the means to carry on the great work.

REPORT FROM LOUISIANA

HENRY E. HARDTNER

Chairman Louisiana Conservation Commission

Louisiana was the first State to create a commission for the Conservation of Natural Resources by legislative enactment, and enjoys the proud distinction of being the first to enact sane and comprehensive laws tending to conserve, protect, and perpetuate the natural resources of the State. In 1908 the Legislature created a Conservation Commission, whose duty it was to report to the Legislature in 1910 as to the conditions of the various resources and to recommend necessary laws for their use and preservation.

The Commission went to work with a will, holding meetings all over the State for the purpose of arousing the people and educating them in the great work. The lumbermen were our friends from the beginning; so were the owners of timber lands, and operators in the production of oil, gas, sulphur, and salt; the farmers dependent on the streams for irrigation purposes soon saw the benefit to be derived from a policy that would protect and perpetuate our natural resources, and also gave us their hearty cooperation. This great work accomplished, the people as a whole soon realized that 80 percent of the proceeds of the forests and rivers was expended for labor and supplies, and joined heartily in the movement; and thus we were prepared to ask the Legislature that certain laws be enacted.

Honorable Harry Gamble, our efficient Secretary (and a member of the Commission) prepared the various acts, and with such care that they will stand the test of any court. It was my pleasure as a member of the

Legislature from the newly created parish of Lasalle to introduce and handle a number of the Conservation measures. Governor Sanders, one of the greatest men in the United States, who recently resigned a United States senatorship to which he had been unanimously elected because the people needed him at the helm of the State Government, gave his hearty support to every measure bearing on Conservation. But with all the serious obstacles removed, and the advantage of a friendly administration, our bills could not be made effective without a constitutional amendment; and so we faced a real crisis.

In order to raise a sufficient fund to protect our forests from fires and for reforestation purposes, and to prevent the gas and oil fields from being recklessly exploited and wasted, it was necessary to levy a license-tax on timber and minerals severed from the soil. Our resources being in the hands of individuals and corporations, it was just and proper that they contribute to the cost of the work for preserving their properties, and the people through the State would enact and carry such laws into effect as would benefit all. To pass a constitutional amendment is not any easy matter; and thus the real work began. The Constitution of the State, which provided for a license-tax on nearly every profession or business, had left out lumber and minerals, probably because it has only been in recent years that there was any development along such lines. That part of the amendment referring to natural resources was as follows: "Those engaged in severing natural resources, as timber or minerals, from the soil or water, whether they thereafter convert them by manufacturing or not, may also be rendered liable to a license-tax, but in this case the amount to be collected may either be graduated or fixed according to the quantity or value of the product at the place where it is severed."

When the amendment came up for final passage I spoke in part as follows: "The whole Conservation program as recommended by the Conservation Commission, of which I had the honor to be chairman, is dependent on this amendment of Article 229 of the Constitution. In carrying out the idea of Conservation, as in carrying out any other governmental policy, it is necessary to raise money. In order to introduce a forestry system and to protect your forests from fires, it is necessary to have money to employ persons informed along these lines whose special duty it will be to look after that kind of business. Now in order to do this, it seems no more than

fair that the persons who are profiting by the depletion of our natural resources should contribute to the payment of these bills; but before that can be done, it is necessary to change the Constitution.

"Article 229, as originally made, exempts manufacturers. Notwithstanding this fact, in 1902 a general license act was passed in which the Legislature, in their wisdom, saw fit to levy a license-tax on the manufacture of lumber. When it was attempted to collect this tax, it was carried to the Supreme Court which held that a license-tax levied on the manufacture of lumber could not be collected for the reason that manufacturers were exempt under Article 229, and the sawing of lumber was a manufacturing business. The court did not say that the attempt of the Legislature to levy a license-tax on the manufacture of lumber was inequitable or unjust, but merely that it was unconstitutional according to Article 229. This bill, from and including lines 12 to 24, attempts to change the Constitution so that the tax may be levied on the severing of trees from the soil. It is to be noticed that there is no attempt to levy a license-tax on the manufacture of lumber, but it is proposed to change the Constitution so that the license-tax may be levied on the cutting down of trees in forests.

"As stated before, the Conservation Commission, after having investigated this question for two years and examined the laws not only of the United States but of foreign countries, has reached the conclusion that those persons who are engaged in the exhaustion of the natural resources of the State, in justice to the State which permits them to do business under this law, in justice to the people, and in justice to future generations of the State, should bear a slight additional tax in order to restore and protect those resources.

"This, Gentlemen, is the reason why you are asked to change Article 229 of the Constitution. You are already acquainted with the facts connected with the natural resource depletion of this State, and I will not now discuss that question. I am simply explaining to you, to the best of my ability, the necessity of changing the Constitution as proposed in this bill in order that we may have the proper source to raise a revenue in order to carry out Conservation policies."

We succeeded in passing the amendment, and then passed the License-tax or Revenue Act which provides the following taxes: $\frac{3}{4}$ cent per 1,000 feet log scale on fine and hardwoods severed from the soil; 1 cent per 100 stave bolts; $\frac{3}{10}$ cent for each telegraph and telephone pole; 1 cent each for piles; $\frac{1}{8}$ cent per cup per year for extracting turpentine from growing trees; for production of oil, $\frac{2}{5}$ cent per barrel; for natural gas, $\frac{1}{5}$ cent per 10,000 cubic feet; for mining sulphur, 2 cents per ton; for mining salt, $\frac{1}{5}$ cent per ton. The license-tax on timber will yield about \$20,000 annually, and the same amount will accrue from mines and mining. The Conservation Commission will use these funds for the protection and perpetuation of the State's natural resources.

The Forestry bill, which we consider a good one, was then passed. There are no restrictions as to size-limit in cutting timber. Ample provisions are made for a complete fire patrol system and methods for preventing loss by fires. In Louisiana and all southern States, denuded lands will reforest naturally if fires are prevented, and a good crop can be grown in from 25 to 40 years. For any one who will engage in the business of growing timber, especial inducements are held out. The assessment on the land is fixed at \$1.00 per acre for 30 or 40 years, and the growing timber is not taxed during that period. The Deputy Forester must be a man practically and theoretically educated in silviculture, and under the State Forester has supervision of forestry work. Consent is also given to the United States to acquire by gift or purchase not exceeding 100,000 acres for a National forest reserve; the State may also acquire by gift or purchase lands for forest reserves.

Act 254 provides for the establishment of a department of mining and minerals, including oil and gas production, authorizing the prohibition of unsafe and wasteful mining and the appointment of a supervisor of minerals on recommendation of the Conservation Commission.

Act 265 to "establish a Board of Commissioners for the protection of Birds, Game, and Fish," empowers them to employ wardens, officers, and assistants, and to provide means to carry the Act into effect; gives them complete control and management of all the waters of the State, such as the Gulf of Mexico (within the jurisdiction of the State), all lakes, bays, sounds, rivers, streams, passes, bayous, creeks, lagoons, and ponds by granting

management and control of all fish, shell-fish, oysters, diamond-back terrapin, turtles, shrimp, crabs, and alligators; and provides for oyster, game, and fish reserves by granting them control of birds, game, and fur-bearing animals, etc. Birds, game, and fish are among the greatest natural resources of the State, yielding an enormous food supply and a large revenue.

Act 57 declares that waters found in the bayous, lagoons, lakes, bays, and rivers to be the property of the State. The idea is that the State will not permit any one to create a monopoly of this resource, which belongs to the people.

Act 280 provided for the creation of a Commission for the Conservation of Natural Resources.

Act 333 provided for conservation of natural gas and oil by preventing waste.

A number of other Conservation measures were enacted into laws, 29 in all, but I cannot touch upon them at this time.

We are proud of our success in inaugurating safe and sane policies for Conservation; we are proud of our Governor, J. Y. Sanders, who urged the passage of the various bills; we are proud of our lumbermen, timber owners, gas and oil operators, and miners who recognized the need for Conservation and the justness of our bills, and assisted in their passage. And above all we are proud of our people as a whole, who are so wide-awake on the question of Conservation of natural resources.

REPORT FROM MAINE

CYRUS C. BABB

District Engineer Maine State Water-Storage Commission

The two principal resources of the State of Maine are its forests and its water-powers. Of its total area of 30,000 square miles, 21,000 square miles, or 70 percent are in forest lands. Over 1500 lakes and ponds are located in the State, covering 2200 square miles of water surface, and not including the innumerable little ponds of an acre or two in area that are located in all

directions. There are in the State one lake to each 20 square miles of territory, and one square mile of lake surface to each 14.3 square miles of territorial area.

Although the State ranks 35 in area, and 30 in population, it ranks third in the Union in water-power development, having, according to the U. S. Census, a total of over 343,000 horsepower in use. It is surpassed only by New York and California in total horsepower.

The State has always conserved its water-power. The Supreme Judicial Court of the State has held as follows:

It is a rule of law peculiar to this State and Massachusetts under the Colonial Ordinance of 1641-7 that all great ponds—that is ponds containing more than 10 acres—are owned by the State.

While private property cannot be taken for public use without compensation, the waters of great ponds and lakes are not private property.

Under the ordinance, the State owns the ponds as public property held in trust for public uses. It has not only the *jus privatum*, the ownership of the soil, but also the *jus publicum*, and the right to control and regulate the public uses to which the ponds shall be applied.

The authority of the State to control waters of great ponds and determine the uses to which they may be applied is a governmental power, and the governmental powers of the State are never lost by mere non-use.

Early Investigation

Maine has always been in the forefront in the investigation and conservation of its resources. Thirty years before the National Government authorized its first geological investigations, and over forty years before the Federal Geological Survey was established, the State of Maine had made such a survey. By Act of the State Legislature, March 28, 1836, a geological survey of the State was authorized under the direction of Dr Charles T. Jackson, State Geologist. The investigation was continued for three years.

The results of this geological survey, considering the difficulties of transportation at that time and the non-existence of accurate maps, are interesting.

A detail survey and report on the natural history and geology of the State was made in 1861 and 1862 by Ezekiel Holmes, Naturalist, and C. H. Hitchcock, Geologist. Reports were made on the zoology and botany of the State, but the most interesting and detailed reports treated of the geological resources.

A hydrographic survey of the State was authorized by the Legislature as early as 1867. The resulting report of Mr Walter Wells is considered as authority even to the present day.

Present Organizations

At the present time there are two organizations in this State working along geological, topographic, and hydrographic lines. They are known as the Maine State Survey Commission, and the Maine State Water-Storage Commission. The first organization was authorized by Act of the State Legislature March 16, 1899. Its powers were subsequently amended and enlarged by an Act approved March 23, 1905. It is authorized to cooperate with the U. S. Geological Survey, and its work includes the topographic and geological surveys of the State.

The creation of the State Water-Storage Commission was authorized by Act of the Legislature April 2, 1909. His Excellency, Governor Fernald, at the Conference of Governors in May, 1908, was so impressed with the importance of the objects and recommendations there brought forth that, at the next meeting of the State Legislature, he advocated and finally approved the Act creating said Commission. This Commission is directed to collect information relating to the water-powers of the State, the flow of rivers and their drainage area, the location, nature, and size of the lakes and ponds in the State, and their respective value and capacity as storage reservoirs, with a view to conserving and increasing the capacity of the water-powers of the State. The Act further provides that every person, firm, or corporation before commencing the erection of a dam for the purpose of developing any water-power in the State, or the creation or improvement of a storage

reservoir, shall file with the Commission certain prescribed engineering plans.

The first report of the Commission to the Legislature is asked to show, in so far as time will allow, a comprehensive and practical plan for the creation of such water-storage reservoirs as will tend to develop and conserve the water-powers of the State, and to report the necessary steps that should be taken by the State to further conserve and increase them. The Commission is further requested to ascertain what lands can be purchased by the State and the cost thereof, with information as to their value as forest reserves or for conserving the water-powers of the State, or for reforestation; and further to investigate the question of denuded, burnt-over, or barren lands in the State, and their extent and value, with a view to their purchase by the State for reforestation.

By an agreement dated December 1, 1909, between the Director of the U. S. Geological Survey, the Chairman of the State Survey Commission and the Chairman of the State Water-Storage Commission, the work of the three organizations in the State is brought under one direction. This agreement provides for a cooperative survey of the natural resources of the State; that said survey shall include the continuation of topographic mapping, a determination of the amount and availability of water resources, their present development and the best methods of their future utilization; also the further determination of geologic resources. The executive officer, under the terms of this agreement, is a duly appointed employee of the U. S. Geological Survey, with the title of District Engineer.

State Highway Department

This department was authorized by legislative Act of 1907. The appropriation for the work is based on a tax of $\frac{1}{3}$ mill on the State valuation. Provision is made in the law whereby the State will aid financially, on a sliding scale, the various towns if they raise money for highway construction purposes. On the average it may be said that for every dollar appropriated by a town, the State will pay an additional dollar. The law further provides for a limitation of the amount that the towns may raise for this purpose, based on the valuation of said town. The sliding scale of appropriation by the State is as follows: to towns having a valuation of

\$200,000 or less, the State will pay two dollars for each dollar appropriated by said towns; to each town having a valuation of over \$200,000 and less than \$1,000,000, one dollar for each dollar appropriated by said town; to towns having a valuation of over \$1,000,000 and less than \$1,200,000, ninety-two cents; to towns having a valuation of over \$1,200,000 and not exceeding \$1,400,000, eighty-five cents; to towns having a valuation of over \$1,400,000 and not exceeding \$1,600,000, eighty cents; to towns having a valuation of \$1,600,000 and over, seventy-five cents for each dollar appropriated by the town; and to unincorporated townships, one dollar for each dollar appropriated.

State Forestry Department

This department was created by legislative Act of 1891 through the appointment of the State Land Agent as Forest Commissioner. This Commissioner is directed to institute an inquiry and to report as to the extent to which the forests of the State are being destroyed by fires and by wasteful cuttings, and the effect of such action on the watersheds of the lakes and rivers and on the water-powers of the State. His principal duties, however, are the supervision and control of measures for the prevention and extinguishment of forest fires in all plantations and unorganized townships in the State. An efficient fire-fighting organization is now in operation in the State under this department, and during recent years valuable tracts of timber have been saved that would otherwise have been destroyed.

Other Organizations

There are other departments and organizations that are doing very valuable work in the preservation of the natural resources of the State of Maine. Many pages could be written on their results but at present a number of them will only be mentioned by name. Included in this list are the Departments of Inland Fisheries and Game, Department of Agriculture, Bureau of Industrial and Labor Statistics, State Board of Health, and Department of Harbor and Tidal Waters.

REPORT FROM MASSACHUSETTS

FRANK WILLIAM RANE
State Forester

HENRY H. SPRAGUE
Chairman Metropolitan Water and Sewerage Board

While we do not have an authorized Conservation Commission in Massachusetts, we nevertheless have many wide-awake and active State officials and commissions in charge of work which in the total answers the same purpose to the Commonwealth.

Massachusetts is noted for her excellent roads, and she is constantly enlarging the mileage. The Fish and Game Commission is perfecting our laws and encouraging modern protection and management of both fish and game. The propagation and dissemination of each is a large part of their work.

General agriculture is undoubtedly improving and various rural industries such as apple raising, cranberry growing, asparagus culture, and various specialties are receiving renewed attention. The State Agricultural College is growing in influence and value to the State.

The increasing population of the State has made it necessary to set apart and protect many of the ponds and streams throughout the Commonwealth for the purpose of water supply. During the past fifteen years the Commonwealth has expended more than \$41,000,000 for the acquisition and construction of Metropolitan works in order to provide the city of Boston and surrounding municipalities with water. One of the storage reservoirs constructed for the "Metropolitan District" is the largest reservoir in the world built up to the present time for the purpose of providing domestic water supply. Large sums have been spent not only for the direct protection of the reservoirs from pollution, but also in acquiring and improving large marginal areas of woodland, and in the planting with trees of many hundreds of acres of cleared lands which have been acquired. Cities and towns outside of the Metropolitan District have made and are making like provisions for obtaining and preserving their water supplies.

Under recent legislation the gradual metering of all water services in the Metropolitan District is required, and more vigorous inspection has been introduced; so that in the past year or two a material reduction in the total

consumption has been effected notwithstanding the increasing number of water takers.

In the building of the great Wachusett reservoir for the Metropolitan Water-works provision has been made for the utilization of the power which may be generated by the fall of the water over the dam to the level of the aqueduct through which the water is conveyed into the Metropolitan District. Machinery for a power plant is about to be installed in the power house already erected, by which it is estimated that from 2500 to 3000 horsepower may be generated and disposed of, not only at a profit to the District, but also to the advantage of the local industries.

While the State has permitted the taking, for the benefit of the municipalities, of the necessary sources of water supply by the exercise of the power of eminent domain, it has adopted the policy of compelling the husbanding of the waters by the prevention of unnecessary and wasteful consumption, and of utilizing the power generated by water works for the benefit alike of the works of the mechanical industries of the Commonwealth.

For conserving forest, park, and shade trees, Massachusetts has undertaken the great task of suppressing the ravages of the gypsy and browntail moths. This work has now extended over a period of years, and eminent entomologists concede that nothing equal to this undertaking has ever before been attempted. As many as 2700 men at one time have been employed by the State in this work. Massachusetts has spent millions of dollars in the work, and it is not only a protection to our own people but equally prevents the dissemination of these pests to other States. Parasites have been collected and introduced from foreign countries, and everything possible undertaken to assist in the work. Our improved high-power spraying machines with new and improved devices for destroying these insects will undoubtedly prove of great value in future spraying undertakings throughout the Nation.

The forestry work meets with continued whole-hearted support at the hands of our people. The work of reforestation is becoming more popular each year, and great good is bound to result therefrom. Our forest fire laws are proving to be workable and hence practical. The poorer towns are receiving

State aid in the purchase of fire-fighting equipment, and the wealthier towns are equipping themselves. The past year, as heretofore, the Legislature has been inclined to assist the State Forester in his various endeavors.

REPORT FROM MISSOURI

HERMANN VON SCHRENK

Chairman Missouri State Forest Commission

The Forest Commission of the State of Missouri was appointed a year ago for the purpose of making recommendations to the Governor concerning a future forestry policy for the State.

The Commission, after a thorough study of the conditions prevailing in the State, prepared a report to the Governor, the principal feature of which was the recommendation that a State Forest Board be established with a State Forester. In submitting its report to the Governor, the Commission suggested a bill, modeled after what appeared to be the best laws already in force in other States. The Commission called particular attention to the necessity for establishing fire guards and doing educational work among the people of the State. The report and the bill were sent to the Legislature by the Governor with a strong recommendation that the bill be passed. Owing to the enormous amount of other business on hand and the lateness in the year, the Legislature did not have time to fully consider the bill, and it will come up again at the next session.

The Commission has investigated the forest resources of the State in a general way, and feels that there is a large field for the work of perpetuating forests, especially in some parts of the State where the land is more or less unfit for agricultural purposes. The Commission has furthermore planned the organization of a State Conservation Association, this to be organized sometime this fall along lines similar to those of Associations already existing in many States.

While the Conservation work of this State is as yet in its infancy, the general interest awakened is very large, and the Commission anticipates large practical results during the coming year.

REPORT FROM MONTANA

RUDOLPH VON TOBEL

Chairman Montana State Conservation Commission

Probably none of the Governors of States who attended the Conference of Governors called by President Roosevelt in May, 1908, returned to their constituents more thoroughly imbued with the principles of Conservation, or more fully determined to put those principles into practice in this State, than Governor Norris, of Montana. Almost immediately, acting on the suggestion of Governor Folk at the Conference, he appointed a Forestry Commission, consisting of Judge Lew A. Callaway, of Virginia City, Ex-Governor Robert B. Smith, of Kalispell, and Ex-Senator Paris Gibson, of Great Falls.

It soon became apparent to Governor Norris, in view of the most unsatisfactory condition of the land laws of the State, that there was work along the lines of Conservation of a broader scope than was comprehended in the plans laid down for the Forestry Commission, and he appointed what was known as the State Lands Commission, which was expected to draft a bill covering all State lands, except timber lands, to present to the Legislature. This Commission consisted of Honorable David Hilger, of Lewistown, Ex-Governor B. F. White, of Dillon, and Honorable Charles S. Hartman, of Bozeman. Subsequently, Mr E. M. Brandagee, of Helena, was appointed to fill the vacancy on the Forestry Commission caused by the death of Ex-Governor Smith, and Mr Rudolf von Tobel, of Lewistown, was appointed on the Land Commission to fill the vacancy caused by the resignation of Mr Hartman.

After several meetings had been held by each of these Commissions, it was found impracticable to separate the work of the two without duplicating much of it and causing some conflict; so the two were consolidated, and thereafter worked together in the preparation of a bill covering the entire land holdings of the State to present to the Legislature. Such a bill was prepared, submitted, and passed by the Legislature, and approved by the Governor, March 19, 1909, and is now the law of the State.

This Act places all State lands under the control of the State Board of Land Commissioners, consisting of the Governor, Secretary of State, Attorney General, and Superintendent of Public Instruction. It provides for the appointment of a Register of the State Land Office, a State Land Agent, a State Forester, and other minor officials. The duties of the Register are to attend to the sale of lands, and he is the chief of the office. The State Land Agent's duties are, generally, to examine all lands in the field; and the State Forester has general charge of the timber lands of the State.

The Act further provides that no timber land shall ever be sold, except only such as, after being cleared, would be more valuable as agricultural land, than it would be for the growing of timber; and that only the merchantable timber in the forests of the State shall be sold from time to time. It also provides for the reforestation of the lands as occasion may require. The State Forester is made the general Fire Warden of the State, and the Deputy Forester, all peace officers, and the Game Wardens, are made Deputy Fire Wardens, charged with the duty of protecting the forests of the State, all being liable to forfeiture of office for neglect.

The Act provides for prohibiting the sale of lands known to be coal lands, and provides that mines may be opened in the coal lands of the State and worked on the royalty basis, the minimum royalty being fixed at ten cents per ton; it provides that every patent issued for State lands shall reserve to the State the coal, oil, gas, and other minerals contained therein, with the right to enter upon the land and extract the same: thus reserving to the State all coal and other minerals in State lands, whether the same are known at the present time to exist or not. It also provides for the location of water-rights by the State for irrigation of State lands and provides for the location of mining claims on State lands in practically the same manner as it provided for the location of such claims under the Federal Statutes.

This, in brief, is an outline of the work accomplished by the Commission.

Owing to the facts that the timber lands of the State are not in one compact body and that large tracts of timber land lying adjacent to the State forests are owned by private parties and corporations, the experiences of the past summer in fighting forest fires, has demonstrated that all owners are not equally interested in preventing the destruction of the timber upon their

lands; at any rate that they are not equally willing to pay the expense of preserving it. It was found that while some few corporations, owning large tracts of timber land, furnished their quota of men and money to protect their interests, by far the larger number either declined or neglected to furnish either, throwing upon the State the burden of protecting the timber of private owners in order to protect State property; and it is the intention of the Commission to recommend and urge upon the Legislature the passage of an Act requiring private owners of timber land to protect their forests, and in case of their failure or neglect to do so, authorizing the State to do so and to charge the expense thereof to the land.

Inasmuch as the State has a large quantity of timber land within the National forests which is unsurveyed, and which if surveyed would be school sections, but which the Secretary of the Interior has decided belongs to the National Government until surveyed, the State derives no benefit whatever from the land and will not derive any until the same has been officially surveyed. The Commission proposes to recommend the passage of an Act ceding to the Federal Government all of the lands within the National Forests which would be school section, upon Congress granting to the State a like area of equally good timber land, in one or more compact bodies so located that the State can obtain some benefit therefrom. This method of handling the matter, I understand, was favorably considered by Mr Pinchot while in office, and also by President Taft.

The Commission also has in mind the preparation of a bill looking to the conservation of the waters of the State. While Montana has many valuable water-powers, most of which are still undeveloped, the principal use of water in the State is, and always must be, for the irrigation of the land; nevertheless, much of the water of the State is available for power purposes which could not be made available for irrigation. Under a long line of decisions of the Supreme Court of the United States, beginning with the case of *Martin vs. Waddell* (16 Peters, 367) decided by Chief Justice Taney in 1842, down to the case of *Kansas vs. Colorado* (206, U. S.), the beds of all navigable streams below high-water mark, together with the waters flowing over them, belong absolutely to the State, subject only to the right of Congress to regulate commerce, and are subject to State control. On the other hand, the land bordering upon such streams all belonged to the general Government originally, and in many places available for power sites

the lands bordering on the streams still belong to the General Government. In order to develop these power sites the work must be undertaken by both State and Nation, or by their joint consent; and it is hoped that some legislation may be secured in the State and in Congress regulating this joint control. Much has been said and written in regard to the compensation due the Government, either State or Nation, from the owners of developed power sites such as we have in Montana; but the Montana Commission is more interested in the power to regulate rates than in the power to exact compensation for the use of the waters, for the reason that all compensation paid to the Government must eventually come from the consumer, and in any event would be comparatively small, while the regulation of rates to the consumer is the only power necessary to complete control and the prevention of monopoly—although it is believed that some compensation should be exacted. Such legislation would eventually conserve the undeveloped water-powers of the State, but other questions arise as to those sites which have already been developed.

There are four dams across the Missouri river in Montana, either completed or in process of construction, each of which utilizes, or is intended to utilize, the entire flow of the river. All of these powers were developed under special Acts of Congress passed after Montana became a State; but in no case was the consent of the State obtained, or even sought. The Commission has not yet decided whether it will attempt to bring these developed powers under State control or not, and of course has not devised any method of doing so (in case it should be deemed advisable to attempt it), although individual members of the Commission—including the writer—have expressed themselves as decidedly of the opinion that the owners of these developed powers, not having obtained any consent from the State for the construction of their dams or for the use of the water, may be brought under State control. The Montana Commission looks upon this water conservation as its main work for the immediate future.

On the whole, the Commission feels that it has already accomplished considerable in the way of practical Conservation, but that there is much more to be done, some of which it hopes to be able to accomplish at the coming session of the Legislature during the first of the coming year.

REPORT FROM NEW MEXICO

COLONEL W. A. FLEMING JONES

I come from a Territory that for sixty years has been knocking at the doors of Congress, seeking admission to the sisterhood of States. The treaty of Guadalupe Hidalgo provided that our Territory should be admitted to Statehood "at the proper time" (which was to be judged by the Congress of the United States), and to the enjoyment of all the rights of citizens of the United States according to the principles of the Constitution. The implied requisites for admission are population, taxable wealth, and the desire of Statehood. All of these we have in abundance, including a population that exceeds by far that of any of the States at the time of their admission, with the single exception of Oklahoma, and something that is by no means generally known is the fact that our Territory has fewer foreign-born citizens per thousand than any State in the Union. However, the present Congress has enacted legislation under which we may be admitted, and our Constitutional Convention is now in session, framing a fundamental law that I am sure will meet with the approval of Congress and the President. But for the fact that the best brains of our Commonwealth are engaged in the work of framing this Constitution, a much larger representation would have been present here.

New Mexico is proud of what she has done in the cause of Conservation. The Act of the Thirty-eighth Legislative Assembly creating our Conservation Commission is broad in its scope and is a model for those States which have not enacted any such legislation.

I hope to attend the Third National Conservation Congress, not from a Territory whose people are wards of the Government and not considered capable of the management of their own affairs, but as the representative of the Great State of New Mexico, the forty-seventh star in our flag.

REPORT FROM NEW YORK

J. S. WHIPPLE

Chairman State Forest, Fish and Game Commission

New York may well be called the Empire State because of its great population, its railways, canals, navigable rivers, agricultural development,

and diversified industries. It also has within its boundaries vast forests that give it an important place among the States of the Union in regard to woodland products, fish, and game.

No statement regarding the Conservation question in New York would be complete without first referring to a few of its assets and their stupendous value. Those to which I refer will readily indicate the importance of New York as a field for the protection, development, and use of natural resources.

The State has an area of 50,203 square miles, or 32,129,920 acres. Of this great territory 27 percent is occupied by forests, a proportion nearly the same as that of the forest area of Germany. There is standing in New York about 41,500,000,000 board feet of timber; the output of our forests last year was 1,064,000,000 board feet. There are 2,308 saw-mills. The value of our forest product in 1907 at the mill was \$24,000,000. In the manufacture of wood pulp New York leads all other States. Last year 245,000,000 board feet of domestic logs were used for pulp, and that was only about 20 percent of the total amount used. New York also leads in the number of paper-mills. It has approximately 170 establishments for the manufacture of paper. The paper and wood-pulp industry is represented by a capital investment of about \$57,000,000.

New York's vast wilderness contains much large game. Over 6,000 deer and 100 bear are killed each hunting season. The annual commercial value of fur and game animals and game birds approximates \$750,000. We rank third as a fish-producing State; the products of all species, including shell-fish, amounts to about \$40,000,000 annually, the annual shell-fish product being valued at about \$12,000,000.

The Adirondack Park contains 3,313,564 acres, the Catskill Park 576,120 acres, and 1,641,526 acres of land are owned by the State, of which one-third is virgin forest or that which is now equally good. Twelve large rivers wholly within the State have their source in the Adirondacks. The course of each is marked at frequent intervals by falls or rapids, and they, with others outside of the Adirondacks (excluding the Niagara and Saint Lawrence), have a natural horsepower already developed of 630,000; they are capable of furnishing at least 1,500,000 horsepower. This estimate would indicate

that there is still 880,000 horsepower running into the sea wasted. It has been estimated that New York State would derive a revenue of over \$15,000,000 annually from its fully developed water-power if controlled and sold by the State. Besides the Adirondack rivers there are the Delaware, Susquehanna, Chemung, Alleghany, Esopus, Genesee, and many other rivers of great value.

New York has over 500 miles of canals, or about 25 percent of the total canal mileage of the United States, over which there are transported annually some 3,500,000 tons of freight. Mineral production is considerable. The mining of iron ore is a well developed industry. One of the largest known iron ore deposits in the world is located in the Adirondack wilderness. Gas, oil, garnet, graphite and many other mineral products are marketed annually to an amount over \$5,000,000.

Only three other States yield a greater total value of agricultural products. New York ranks first in average value of production per acre. One-ninth of the hay and forage of this country are raised in New York, and the animal industries are of enormous value. Our hay-producing acres are worth \$93,000,000. New York has 226,720 farms with an aggregate area of 9,522,000 acres, valued at \$1,070,000,000, furnishing employment for 373,650 persons. The annual product of these farms is worth \$345,000,000. New York has 30 acres of tree nurseries capable of producing 12,000,000 trees annually, and will double that acreage during the next year. We have taken the lead in the establishment of tree nurseries, in planting, and general work of tree propagation.

Work Accomplished

These are some of the factors which make Conservation of natural resources in New York State very important. The work is being carried on by various State Departments rather than by any single commission. Governor Charles E. Hughes, and the Departments under him, gave great impetus to the work during his term as Governor. Besides \$101,000,000 authorized for canal improvement and \$55,000,000 for good roads, over \$2,000,000 is expended each year by the State in Conservation work as represented by the activities of the Forest, Fish, and Game Commission, the Agricultural Department, and the State Water Supply Commission.

All sections of the State have been awakened, and active steps are being taken in every direction. New York was first to achieve an onward movement in the preservation of its natural resources when in 1885 it led the way in the establishment of State Forest Preserves, and inaugurated the policy of protecting her forests for the health and recreation of the people and the protection of water sources. The same leadership has been continued in control of water by statute creating the State Water Supply Commission in 1905 and vesting it with jurisdiction over the water supply of the State.

Water

Water is now recognized as one of the most valuable economic resources of the earth, and the importance of measures for public control to secure full benefit of hydraulic resources to the people is being realized very rapidly as the great educational propaganda now carried on in New York progresses. The powers of the Water Supply Commission extend to the progressive development of water-powers of the State for the public use under State ownership and control. It also has the power of improving, straightening, and dredging the channel of any water course of which the irregular flow is shown to be detrimental to public health and safety. Four great reservoir projects have been located and surveyed; many other propositions have been tentatively examined, so that all water storage possibilities of the State are approximately known.

I want to say just a word about the granting of franchises, especially in respect to water-power rights in perpetuity. We have become so accustomed to the idea of a non-controllable ownership of our natural resources that even our agents in the Legislature have seemed at times not to fully appreciate the importance of State control and the rights of the people at large. No agent of the people has any moral right, nor have the people themselves, to bind by water rights in perpetuity future generations who will have their own problems to solve and their own lives to live. It is therefore of first importance to understand our relationship as trustees toward these public resources. Are they ours to do with as we please, to use or waste as we see fit, or are they ours to use to the best advantage and with the least waste; and is it our duty to pass them on unimpaired, improved if possible, for those who are to follow us? It is self-evident that this world

was not made for us alone. After us countless millions will come and go. Could it have been intended that during our temporary occupancy we should have such a complete control of God's gifts to Man that we, by our own act or legislative will, could determine for all time how these blessings might be used or enjoyed? We may give them away, we may deprive the people of their rights in them; but when on the one hand a road leads to safety and on the other a way to danger, there should be no hesitation about which we should take. New York has improved on its old policies, which can best be illustrated by an extract from an address by Governor Hughes:

"Water-power privileges have been granted in the past without any provision for a payment to the State in return for what the State gives. These grants have frequently been made without proper reservations or conditions and without anything constituting a suitable consideration. They have amounted simply to donations of public rights for private benefit. It does not fetter individual enterprise to insist upon protection of the common interest and due payment for what is obtained from the public. Last year on the grant of a franchise to a development company which was to develop power from Saint Lawrence river it was insisted that provision should be made for compensation for the privilege upon a sliding scale according to the power developed. And thus it was established that hereafter in the State of New York public privileges, on terms of justice to the investors and the public alike, must be paid for."

Proposed Legislation

Last year a measure prepared for the purpose of relieving the tax burden on reforested land was presented to the Legislature, but it failed of passage. This effort will be renewed until the much desired result is obtained. Timber should be treated as a crop and taxed when cut. Timber owners and tree planters should be encouraged to conserve and plant by making the carrying charges less, that better management may be had and more planting done.

The leasing of camp-sites on State land, the building of good roads through the Forest Preserve, and the removal of dead and down timber were all submitted in the shape of constitutional amendments, but the Legislature also failed to sanction these propositions. The public mind is not yet ready for complete and comprehensive Conservation in New York, to have which

requires a change in our Constitution. The need is urgent, but, I regret to say, not fully appreciated.

Agriculture

The Agricultural Department is performing a splendid work in soil Conservation. It assists in the preservation and protection of trees and in planting work, as well as the fostering of farm crops and the husbandry of meat products. The College of Agriculture is devoted to the cultivation of intelligent and scientific methods in all branches of crop production. Fertilization of the soil, destruction of injurious agents, and new methods of intensive farming, are all taken up in the various branches of the Department. In the State College of Agriculture there were enrolled last winter nearly 1000 students. We have two experiment stations with over fifty scientific men on their staffs. We have three lower-grade agricultural schools, and the State is conducting farmers' institutes, which have held more than a thousand sessions in the past season.

Forests

All the foregoing endeavors are closely related to the continued life of our forests, and in many respects are dependent on them. A producing soil we must always have, or life of all kinds will become extinct. Without a fairly regular supply of water a producing soil is impossible; producing farm land is impossible. Hence if our water sources do not perform their natural functions, we cannot get along very well. The absence of forests in a mountainous State like New York will prevent a regular flowing water supply, necessary to the demands of good soil productivity; therefore, forests very largely hold the key to the whole Conservation situation as it bears on the life, health, and general welfare of the people of New York State. The question of timber supply, water-power, health resorts, and atmospherical conditions, as affected by the forests, are matters of secondary consideration in view of the indirect but vital influence forests have on our soil production. Neither soil nor water can be totally destroyed. They may become impaired and unavailable on account of irregularity in rainfall, but to some degree they will always perform their natural services for mankind. The forests, however, might suffer total obliteration as they have in many sections of the Orient and Occident. Wherever this calamity

has occurred, we find soil and water have reached their minimum of usefulness. While we could not exist without water or soil, that does not mean that they are the most important subjects for Conservation in my State. The question of having to exist without them is entirely eliminated; they will always be there in some degree of efficiency or inefficiency. They will always be with us in their efficient state if we exercise reasonable care in the use of our forests. On the other hand, it is within the scope of possibility that our forests might be destroyed to all practicable purposes, and history points out that soil and water supply would then be of slight utility in a mountainous country. The forest is the controlling resource, like the governor of an engine without which the engine would destroy itself. Hence forests in New York State by their influence upon soil and water flow occupy the position of first importance among our natural resources to be conserved.

The waste of our forests has been appalling, both by lumbering and conflagration. The great "burns" found through all our mountains furnish striking evidence of gross carelessness and indifference to the value of this great resource. It is time that these acts of colossal folly were stopped. Supreme selfishness on the one hand and deadly indifference on the other are at the root of it all. Some people do not understand the great danger of total forest destruction threatening certain of our watersheds. It takes 50 to 100 years to grow a mature tree. The average soil may increase about one inch in a century. It requires soil to grow trees, and fire, the great enemy of the forest, destroys not only the trees but the soil as well. On two or three occasions in the past seven years the Adirondack Park has come dangerously near being wiped out by fire. Rain alone has saved it. In 1903 and again in 1908 several large fires burning at the same time threatened to unite and destroy the entire park. No human agency can combat successfully a great forest conflagration when once it is under way. In 1908, 177,000 acres of land was burned over in New York State; the loss approximated \$644,000. In 1903, 500,500 acres were burned, and the loss was more than \$1,000,000. Loss of soil and reproduction was not considered in the estimated loss and never is.

It is logically evident from the history of forest fires that prevention is the right objective in seeking to remedy this great evil. Methods of protection after fire starts will fail when certain commonly occurring weather

conditions prevail. In New York we have devised an effective forest fire-fighting organization, based on the principles of prevention. The Adirondack and Catskill sections have been divided into four districts, three in the Adirondacks and one in the Catskills. A superintendent was appointed to take charge of each district. Under him there were assigned regular patrolmen and special patrolmen, and to a certain extent the superintendent cooperates with supervisors of towns. The aggregate number of men engaged in this work this year is 356. In addition to this the supervisors in every town in the State of New York are responsible personally for damages caused by forest fires in their respective towns, if they are negligent in putting them out.

I met the Boards of Supervisors of the various forest preserve counties and discussed with them ways and means of fighting fire, explaining the law and showing their responsibility. This action was followed by good results. The superintendents were in turn assembled at Albany, and properly instructed as to their duties and the relationships to be carried on between their subordinates and themselves. Twenty observation stations were erected on high points, and equipped with strong field glasses, range finders, maps, and telephones. The whole territory has been covered with telephone lines. These stations have proved an incalculable benefit in the apprehension of fires when they are in an incipient state. We have also added to the fire-fighting apparatus portable fire extinguishers. These are very useful in checking a fire at the beginning. Old trails and tote roads are kept clear of obstructions to make the woods more accessible. The whole system is chiefly valuable in that it is based on the fundamental principles of early discovery, immediate alarm, and prompt action. Over 250 fires were discovered and extinguished last year so quickly that they attracted no public notice, and the damage done was unappreciable.

Another step was taken by the Forest, Fish, and Game Commission when the question of oil-burning locomotives running through the Forest Preserve was called to the attention of the Public Service Commission. After an exhaustive investigation, oil as fuel was substituted for coal by order of the Public Service Board. This order required that the railroads should install oil-burning engines for use between 8 a.m. and 8 p.m. from April 15 to November 1 each year, all engines to be inspected by representatives of the Commission. Coal-burning locomotives still run through the Forest

Preserve at night which, on account of the heavy dew, it is thought in most seasons does not materially increase the fire risk; but it is doubtful whether in an extremely dry season coal-burning locomotives would not set fires at night as readily as they do during the day time. The partially restricted use of coal as fuel was the best change obtainable at the time the order was promulgated.

The third factor contributing to reduce fire danger was the provision of the new law requiring the lopping of tops of all coniferous trees felled in the forest preserve. The value of this provision is realized when it is understood that the tops of trees felled a decade ago, when not lopped, are still ready to burn, while the debris of lopped trees disappears entirely as a fire menace in the same period of time because they lie flat on the ground, absorb moisture and rapidly decay.

Scenic assets have a tangible value. Figures have been adduced showing that \$200,000 was paid in fares to Niagara Falls to the New York Central Railroad in three months. The visitors to the Adirondacks leave nearly \$8,000,000 behind them each season. These figures seem to suggest the culture of the esthetic, as that side of the problem is very remunerative. There ought to be as much attention paid to the acquirement and preservation of places of natural beauty, public usefulness, and historic interest, for the full enjoyment and use of all the people, as there is for the preservation of natural resources that have only a commercial value. To this end the people of the State of New York and New Jersey have established an interstate park, and by statutory enactment preserved for all time the picturesque and historical palisades of the Hudson, and many acres of woodland. To this end Mrs Harriman gave 10,000 acres of wild wooded land and \$1,000,000 to the State last winter, to which the State of New York added by bond issue \$2,000,000 for the enlargement of the interstate park. By statute also about 53 square miles of the historic Highlands of the Hudson south of West Point have been saved and set aside for park and forestry purposes. Watkins Glen, a beautiful part of Schuyler County near Seneca Lake, has been purchased by the State, and its scenic beauty preserved. A reservation has been established in the Thousand Islands of Saint Lawrence river and one at Niagara Falls preserving these beautiful places to the people for all time. Without such places pleasant to the eye and conducive to health, a numerous portion of the race thus deprived of

opportunity for exercise, for recreation, and the quiet enjoyment of nature's great gifts of beauty that have existed for the full and untrammelled benefit of former generations, we must become a nation of human derelicts rather than a nation of healthy-bodied men and women. We must have these resources to keep up the physical standard of men and women, and more so in the future than in the present because conditions of living are changing rapidly in America. In 1800 only 3 percent of the people dwelt in the cities or large towns; in 1900 more than 33 percent lived amid urban conditions.

President Roosevelt never said a more striking thing than when he gave as the definition of civilization something to this effect: "The prime difference between a civilized and an uncivilized people is that civilized man looks beyond his own immediate needs, and even beyond those of his lifetime, and provides for generations yet unborn."

In considering the principles of Conservation, development comes first, using and improving the natural resources of our country for the benefit of the people. The second principle is the prevention of waste. Conservation comprehends the substitution as far as possible of materials for those that are exhaustible. Conservation reaches out into a wide field, and, as often said, it means the "greatest good to the greatest number for the longest time." Conservation advocates the use of foresight, prudence, thrift, and intelligence in dealing with public matters. It means the application of common sense to our public affairs. Conservation guarantees progress, efficiency, supremacy, perpetuity, the life of the Nation. There is no interest of the public to which the principles of Conservation do not apply.

SPECIAL REPORT FROM NEW YORK—WATER RESOURCES OF THE STATE

HENRY H. PERSONS

President State Water Supply Commission

The people of the State of New York have a deep natural interest in the important economic problems now brought so forcibly to the attention of the American people through the Conservation movement. That interest is properly manifested at this time because, in all probability, no other State in the Union is invested with conditions so favorable and opportunities so

promising for the early accomplishment of material progress in the practical conservation of one of its most valuable natural resources. In New York State the surface water supply as a natural resource is second in value only to the land itself, which indeed owes its value largely to the existence of an abundant natural water supply. It must be conceded that the value of water for potable and domestic purposes cannot be estimated in dollars and cents, constituting as it does a necessity of life for which no substitute exists. Its money value is represented by whatever it costs to obtain the supply, be that much or little.

Aside from any such consideration as this, water is practically the only natural resource within the State of New York for the development of power, that great and fundamental requisite to the prosperity and comfort of a civilized community. The State does not have enough coal of its own to operate its existing iron mines, to say nothing of mining the whole of the valuable deposit, estimated at 300,000,000 tons. This condition is compensated for in a large measure if not altogether by the fact that, in addition to the existence of an abundance of water, the profiles of the streams and the general topography of a large portion of the State are naturally favorable for the establishment of hydraulic power developments and the construction of storage reservoirs for the regulation of the flow of the streams.

The State has taken a notable step forward by assuming certain regulative powers over the disposition of these resources, and by the institution of a systematic inventory of them to determine the extent not only of the supply but of existing developments and present uses, and the possibilities for additional uses and new developments. It has also made extensive studies to determine the possibilities for water storage, the necessary complement to extensive power developments within the State.

Development of Water Conservation as a State Policy

A brief statement of the most important historical facts leading up to and determining the present status of water conservation within the State seems pertinent, and will doubtless be of assistance in furnishing a clear prospectus of the controlling conditions and the complicated problems

involved in the formulation of a comprehensive and practicable plan for the regulation of these waters.

In 1902 a special Act of the Legislature created the Water Storage Commission. That Commission was directed to make surveys and investigations to determine the causes of the overflow of the various rivers and water courses of the State, and to determine what, if anything, could be done to prevent such overflow. The serious nature and wide extent of the floods occurring at more or less frequent intervals in a large number of streams throughout the State had long been a source of anxiety to the residents of the flooded districts owing to the injuries and dangers occasioned by the sudden overflow.

The failure to take proper measures of a corrective nature earlier was not due in any sense to a lack of interest, intelligence, or energy on the part of the citizens of the State. The interest was usually localized, owing to the fact that ordinarily the entire State does not suffer from floods at the same time, so that while small communities had made some attempts to secure relief there had been no State-wide movement or concerted action in that direction. Several obstacles usually rendered individual and local remedies comparatively difficult and ineffective. The complexity of the hydrographic problems usually involved in a study of flood conditions, together with the expense incident to a technical investigation to determine the causes and means of relief, constitute one of these obstacles. Small municipalities cannot usually see their way clear to employ a hydraulic engineer to investigate such problems, and conclusions arrived at, or remedies applied without such a study are likely to result in an unsatisfactory manner. Furthermore, the proper remedies, when ascertained, usually require for their execution the acquisition of land and water rights which individuals or minor municipalities have no power to condemn. Another obstacle arises from the fact that the distribution of the burden of expense for any particular improvement can scarcely be made equitably, or the payment of the amount enforced by any means other than the power of assessment.

These were the conditions which led up to the demand for a State investigation and the creation of the State Water Storage Commission. That Commission, after about a year's investigation and research with a remarkably small appropriation at their disposal, submitted to the

Legislature an extremely valuable and comprehensive report on the flood conditions of the principal streams of the State. The report pointed out that storage reservoirs constituted the only practicable solution of the problem in the majority of instances, and recommended the construction of several such reservoirs at points where conditions were known to be favorable. Having submitted its report, the Water Storage Commission automatically ceased to exist.

The next step in the development of the water-storage movement was the creation of the River Improvement Commission by act of the Legislature in 1904. The creation of that Commission was the only practical outcome of the valuable report on the causes and remedies of floods in New York rivers made by the Water Storage Commission in 1903. The River Improvement Commission was invested with power to make preliminary investigations, plans, and surveys for the regulation of the course of any stream, of which the restricted or unrestricted or irregular flow should be shown by petition of local residents to be a menace to the public health and safety of the community. If the improvement appeared to be of sufficient importance and the Legislature approved, the Commission was then authorized to carry out the project and to assess the cost of the same according to the benefits received by the various individuals and the properties benefited. To provide for carrying on the work pending the collection of such assessments, authority was given the Commission by the act to issue certificates of indebtedness, or to sell bonds, to be retired on the collection of the cost from the beneficiaries. That Commission was composed principally of State officers as ex-officio members, and while its work was excellent its progress was unavoidably slow.

While the River Improvement Commission was still in existence, the State Water Supply Commission was created in 1905; the primary object of its creation being to insure an equitable apportionment of the sources for public water supplies among the various municipalities and civil divisions of the State. The Legislature apparently had a very clear conception of the need for such a State agency and hence created the Water Supply Commission with those specific powers. It soon became apparent that this Commission was in better position than the River Improvement Commission to study flood conditions, involved as they were with the general subject of water supply; so that by Act of the Legislature in 1906

the River Improvement Commission was discontinued as a separate board, and all its powers and duties were transferred to the State Water Supply Commission.

The jurisdiction of the Water Supply Commission was thus considerably broadened to include the study of water storage on a large scale. Its powers and duties were subsequently extended to an investigation of water-powers within the State, and the preparation of a plan for their general development. The Commission is therefore engaged in three distinct but closely related lines of work: (1) the apportionment of municipal water supplies; (2) the improvement of rivers in the interest of public health and safety; and (3) the formulation of a plan for the general development of the water-power resources of the State.

Municipal Water Supplies

In practically working out a comprehensive plan for water conservation, the State has rightly begun with the matter of public water supplies. Previous to the establishment of the Water Supply Commission, the laws of the State permitted any city, village, or other municipal corporation to acquire or condemn lands for sources of water supply practically at will, and without regard to whether its plans were just and equitable to other municipalities and their inhabitants that might be affected thereby. Thus, a large city armed with the power of eminent domain might take territory from a smaller community regardless of the present or prospective needs of the latter for the water sources thus appropriated. In fact, the people of the community invaded did not always have the foresight to realize that they would sooner or later require those sources for themselves. It can readily be seen that such a course might involve a serious menace to the future growth of the smaller community. Fear of such procedure led to the passage of special prohibitory laws for many localities, particularly those adjoining New York City, against what was feared might be the ruthless exercise of the great power of the larger community. The effect of such legislation, involving as it did so much hostility between the different localities of the State, proved that the then current practice afforded but a partial, inadequate, and unfair method of administering the distribution of sources of water supply.

Provision for a pure and adequate supply of water for domestic purposes for all its inhabitants is one of the first duties of the sovereign State. Through its important effect upon public health alone, the general use of pure water is a matter of the gravest importance to every man, woman, and child regardless of local divisions of government or grouping of citizens. It was with a realization of these principles that the Legislature of 1905 wisely determined to delegate the power of control over the selection of sources of public water supply to a permanent commission which, by the aid of constant and special consideration of this subject, should become expert in controlling such selection so as to insure equity, among all the inhabitants and civil divisions of the State, and the resulting unimpeded prosperity, growth and comfort of each and every community. The law, therefore, provides that no municipality, or person, or water-works corporation engaged in supplying the inhabitants of any municipal corporation with water shall have power to acquire lands for any new or additional sources of water supply until its plans have been submitted to, and approved by, the Water Supply Commission.

In passing upon plans thus submitted to it, the Commission is empowered to determine: (1) whether the proposed plans are justified by the public necessities of the community; (2) whether the plans are just and equitable to other communities, special consideration being given to future as well as present needs for water supplies; and (3) whether the plans make fair and equitable provision for the determination and payment of any and all damages, both direct and indirect, which will result from their execution.

Under the operation of this law, which appears to have set a precedent among the States of the Union in the general State administration of water-supply resources, there has resulted a smoothly adjusted progress in the development of public water supplies, without further need of appeal to the Legislature for the drastic prohibitory special legislation formerly so much sought after.

It is thus well established in the public law of New York State that the control of sources of water supply is a State function, and that all persons or municipalities must apply to the central State Government and receive permission to take what may be determined to be a just share from the State's total supply of this indispensable resource. It must, therefore, be

evident that the State should aim toward an ideal of administration of its water resources which would secure fully and impartially the rights of each and every one of its inhabitants and all of their local groupings to a just and equitable share of the public waters. This problem becomes especially complicated under our modern conditions of civilization which in promoting the growth of enormous cities, call for engineering works of the greatest scope and magnitude for the purpose of providing the requisite quantity of pure and wholesome water.

One of the most recent and familiar illustrations of this fact is the present vast undertaking of New York City, which at a cost of about \$161,000,000, is going 90 miles to the Catskill Mountains to secure a water supply which its engineers estimate will be sufficient for its needs for only a comparatively few years. In this great project, as well as in the case of many others not so great, there is involved a large element of hardship and damage to the locality invaded, in the necessary taking of private property for the larger public water supply by constructing immense storage reservoirs which permanently occupy the lands thus acquired, and furnish no considerable means of support and prosperity to the region—as is the case when land is acquired for railroad purposes.

This project of New York City constituted the first important case to come before the Water Supply Commission for its official approval. After extended and careful consideration of all the manifold interests involved in this remarkable project, and after a protracted series of hearings, the suggestions of the Commission with regard to the protection of the rights of all the other municipalities and people affected were incorporated into law, and the project received the sanction of the Commission. Under the authority thus given New York City has entered upon its work of constructing the most pretentious municipal water-supply system in the United States.

Subsequent to the New York City petition, many other applications from villages and cities, large and small, have been passed upon. By the accumulation of special knowledge resulting from comparing the problems of different localities, the Commission has been able to bring to the aid of the smaller communities of the State a fund of experience and counsel which in not a few instances has proved of great benefit and assistance. The

Commission aims to make its practice simple, expeditious, and inexpensive; and the technical points involved in each application are carefully passed upon by a competent engineer.

A complete census of all existing water supply plants and systems has been made and is revised from time to time, and the progress of each applicant whose plans are approved is carefully followed. Construction work involving expenditures of \$230,000,000 has been passed upon by the Commission and undertaken by the municipalities of the State. This has entailed the official consideration by the Commission of 85 separate applications, in connection with each of which public hearings are conducted.

Numerous complaints have been filed with the Commission alleging unsatisfactory domestic or fire service both on the part of municipalities and water companies. The source of dissatisfaction seems to be the lack of foresight on the part of the municipal or water company officials, as a result of which they have obtained an inadequate supply or insufficient pressure. There are many instances of this condition in the State. There are also many consumers who object to excessive rates which they claim are imposed upon them by water companies. On the other hand, some of the companies themselves have attempted to secure legislation to provide that the State shall be the arbitrator in the adjustment of water rates. These conditions seem to point to the conclusion that in the comparatively near future the State will have to assume control over these matters. A certain degree of this sort of control is exercised in an indirect way at present in the case of applications which are before the Commission for consideration, but no jurisdiction lies with the Commission unless the acquisition of lands for a new or additional source of supply is involved.

River Improvement for Health and Safety

A number of river-improvement petitions presented to the River Improvement Commission and still pending at the time that Commission's powers were transferred to the Water Supply Commission involved the construction of storage reservoirs in the Adirondack forests. The River Improvement Commission had considered the constitutional questions involved in the utilization of State forest lands for storage reservoir

purposes, and had reached the conclusion that the force of a clause in the Constitution prohibiting the removal of timber was paramount to all exercise of the police authority of the State to protect the public health and safety; and it had declined further to consider any petitions involving the utilization of State forest lands for reservoir purposes. The Water Supply Commission on the other hand has held that the statutes relating to river improvements in the interest of the public health and safety are not sufficiently comprehensive to afford a proper basis on which to advance systematic water conservation involving water-powers. The existing river improvement law has the health and safety element as its basis, whereas the carrying out of a comprehensive conservation policy would be of greatest financial value to the existing and new power developments, owing to the regulating effect of storage reservoirs on the flow of the streams. For this reason the Water Supply Commission has not urged the execution of river improvement projects involving water storage, under existing statutes, and has recommended to the Legislature that the advancement of such projects should await the determination of a definite State policy and the formulation of a thoroughly comprehensive plan by means of which the storage reservoirs shall constitute a source of income to the State, even after the bonds are retired. Several projected improvements therefore await the enactment of a more suitable statute.

Meantime, however, an important project calling for rather different treatment had arisen in the proposed improvement of the Canaseraga creek, the most important tributary of Genesee river. This project originated with the River Improvement Commission, and the Water Supply Commission inherited and actively carried on the consideration of the problems involved. For the last 22 miles of its course this creek flows through a broad, fertile valley. Owing to the steep declivities of the upper water-shed and the resulting suddenness and severity of floods in the valley, a large portion of these flat lands were submerged two or three times a year, and the channel had gradually become filled with silt which raised the prism to such a height that the stream itself and its banks were actually higher in places than the adjacent land. In times of flood the stream overflowed and the water would stand for several days at a time over the low areas, in a large measure destroying such crops as were in a growing condition and effectually deterring the farmers from cultivating the lands thoroughly and

systematically. The project of improvement which, after due course of public hearings and consideration by the Water Supply Commission received the official approval of the Legislature, contemplates the straightening, widening, and deepening of the channel of the stream, so as to afford a much more capacious flood prism and to shorten the length of the stream through the flooded district by about six miles. At the same time lateral ditches are proposed to be constructed to carry off the overflowing waters from the lower adjacent lands in order to protect them permanently from any serious or protracted inundation.

This project did not involve the use of any State forest lands, nor did it affect any water-power developments. The fact was readily established that the proposed improvement was of great importance to the public health and safety of the community, and also of great importance, from a financial point of view, to the prosperity and general welfare of the community on account of the benefits that would accrue to agriculturists from the protection to be afforded by the proposed improvements against flood damages. The machinery involved in the working out of the project was put in operation and from time to time various obstacles were encountered which had to be surmounted by amending the law. Gradually the statute has been so moulded that it is now thought to be in practical working order, and the proposed Canaseraga creek improvement is actually provided for and financed; the bonds having been sold at a good premium. The actual work of the construction of the proposed improvement will probably be begun in the near future.

The practicability of the method having thus been established the Water Supply Commission believes that the State now has a method by which floods may be mitigated if there are no water-powers or State forest lands involved. On the other hand, the solution of the problem where these complications do exist, is much more difficult. In the cases of the Genesee, Hudson, and Raquette rivers, petitions for the improvement of which have been filed under the public health and safety statute, very little real relief can be afforded by straightening or enlarging the channels of the streams. Water storage appears to be the only practicable solution, and the water-powers which would be improved could afford to bear a larger share of the cost of improvement than those who would benefit from flood control.

Water-Power and Water Storage

The most recent extension of the jurisdiction of the Commission, under which it is investigating the water resources of the State, contemplates three principal lines of operation. These are: (1) To collect information relating to the water-powers of the State; (2) to make plans for such specific developments as the Commission deems available; and (3) to make such other investigations and studies as will enable it to devise a comprehensive and practicable plan for the general development of the water-powers of the State for the public use and benefit and the increase of the public revenue under State ownership and control. In accordance with this statute, the Commission has proceeded to investigate in great detail the conditions governing rainfall and run-off of streams within the State, and has maintained a number of observation and gaging stations in cooperation with the United States Weather Bureau and the United States Geological Survey. A detailed investigation was also made by competent engineering employees to determine the number, capacity, equipment, and other material information relating to practically every water-power in the State. A general investigation of topographic conditions has also been made and practically all promising storage opportunities have been located and their approximate possibilities determined. A number of great reservoir projects have been surveyed and mapped in great detail. In many instances borings have been made to determine the character of foundations for dams, and complete detail plans of the dams and other structures have been prepared. The financial phases of a number of these great projects have been gone into in detail, and an exhaustive study of the constitutional and other legal aspects of the problems involved has been made by the Commission, and the required comprehensive plan has been prepared.

In spite of the great natural advantages which New York State possesses in its interior streams with their enormous possibilities for power, developed and undeveloped, the fullest utilization of these possibilities can never be realized under existing conditions. Every river in the State exhibits such irregularity of flow that the water-power which may be economically developed from the present minimum flow is far below the average which can be attained by means of scientific regulation. The difference between maximum and minimum flow of most of our streams when stated in figures is startling to the layman. The Hudson, which is more or less typical of the

streams of the State, has a maximum recorded daily discharge of 100 times its least daily flow. The Genesee, which is much more flashy, has a maximum daily discharge about 400 times the minimum daily flow. On the other hand the Oswego, which is naturally more or less regulated by storage in the "Finger Lakes," has a maximum discharge about 20 times the minimum. The yearly discharge of some of the rivers in a wet year is nearly double the yearly flow of a dry year. On a great many streams as much as three-fourths of the volume of yearly flow usually runs off in the spring and early summer months. These remarkable fluctuations of stream flow are principally attributed to the uneven distribution of precipitation through the year, which unfavorable conditions are undoubtedly aggravated by the varying conditions affecting evaporation, which is generally greatest in the months of least precipitation. Over a large portion of the State, the greater part of the annual precipitation occurs in the winter and spring months. Considerable water is temporarily stored in the snow banks, and is usually reduced to the equivalent of rain simultaneously with the customary heavy rainfall of the early spring months. It is quite common for millions of cubic feet of water to run over the falls and dams in the streams during these spring freshet periods which, if it could be stored until the drier summer and fall months, would be of wonderful utility in not only maintaining a higher rate of flow in those dry months, but also doing away largely with the damage and inconvenience incident to the sudden run-off of flood waters in their natural condition. These conditions point to the necessity for large water storage reservoirs as the only practical means of accomplishing any considerable degree of regulation.

The investigations of the Water Supply Commission have shown that there is an installation of water-wheels having a capacity of about 830,000 horsepower within New York State, of which amount about 200,000 horsepower is at Niagara Falls. The average daily output of the plants is about 620,000 horsepower, including 145,000 at Niagara Falls. There are in all more than 1,800 hydraulic power plants within the State, many of which are equipped with steam auxiliary power plants. The total capacity of these auxiliary plants is about 124,000 horsepower. The investigations have indicated a total development of about 1,500,000 horsepower to be economically feasible within the State. This would be uninterrupted continuous power, exclusive of Niagara river and the portion of Saint

Lawrence river not under the jurisdiction of New York State. A considerable part of this amount is represented by that which would be added to the existing developments by the regulation of the flow of the streams. A number of individual opportunities exist for considerable new developments, some of the more important of which are a 30,000 horsepower on Genesee river at Portage Falls, a 30,000 horsepower on Sacandaga river at Conklingville, a 32,000 horsepower on Raquette river at Colton Falls, and many others ranging from 1,000 to 20,000 horsepower.

The investigations of the Commission have shown that the construction of large storage reservoirs for impounding flood waters may be beneficial in many ways. Probably not all of the possible advantages would result from the construction of any particular reservoir. The extent and variety of benefits may be summarized somewhat as follows:

- (1) The equalization of stream-flow by storing the water during wet seasons and using the same to increase the volume of the stream through dry seasons;
- (2) A consequent large increase in the power value of the stream, due to augmenting the low-water flow, and thus doubling or trebling the dependable flow for power purposes;
- (3) A consequent decrease in the height of freshets, thereby reducing the great pecuniary damages caused by the periodic recurrence of floods;
- (4) By increasing the low-water flow of polluted rivers a dilution would result which would improve the sanitary conditions on the stream;
- (5) Navigation would be benefited by a higher stage of water on the lower reaches of the rivers;
- (6) The extension of transportation facilities, often to an important and desirable extent, by navigation on the proposed reservoirs;
- (7) The low lands of the river valleys could be made somewhat more tenable, and their agricultural products increased by reducing the contingency of floods;

- (8) The perpetual submergence of extensive tracts of swamp lands, which are now unsightly and a menace to health, would be possible;
- (9) The creation of extensive lakes with beautiful shores offering desirable locations for permanent homes and great attractions to summer visitors seeking recreation and health; and
- (10) Inestimable indirect benefit to the State due to the stimulation of industrial enterprises, the increase in number and prosperity of the people, and the creation of taxable wealth by the progressive development of water-powers.

Among the more promising opportunities for the inauguration of a State policy in storage reservoir construction is that offered by Genesee river. The Commission's investigations have shown that it is practicable to build a reservoir with a dam near Portage, which would be about fifteen miles long and over a mile wide, with a total capacity of about 18,000,000,000 cubic feet at a cost of about \$4,500,000. The regulation of the stream by this reservoir would not only practically do away with disastrous floods in the Genesee valley, but would add power worth at least \$200,000 a year to the existing developments at Rochester, and develop at least 30,000 horsepower in connection with the dam; the value of water-power at Mount Morris would also be greatly enhanced, and the nuisance created by the present polluted condition of the river below Rochester would be abated. Other opportunities are offered on Sacandaga river and other tributaries to the Hudson, on Raquette river, and on Black river, where a system of several reservoirs is proposed. Many smaller projects are also under consideration. It is estimated that \$20,000,000 would be sufficient to build the reservoirs whose construction is justified under present conditions.

Problems Involved With Water Storage

There are in general two acceptable methods of reducing or preventing floods. The storage of the water which constitutes the hood wave, or a considerable portion thereof, is doubtless preferable if there is a site for a reservoir of sufficient capacity and the construction is not too expensive.

The other method consists of widening, deepening, and straightening the channel. In recent years, the public has been rather generally educated to believe that storage reservoirs constitute the universal and easily applied remedy. There are many rivers in New York State on which this method may be used effectively, but on many others the absence of basins of sufficient capacity or the excessive cost preclude the possibility of complete flood control in this manner.

The problem of absolute flood control is, however, more complex than the foregoing simple statement would imply. One complication arises from the fact that the damage from floods in New York State is often increased by the formation of ice gorges. The formation of these gorges cannot be prevented by an ordinary system of storage reservoirs, although the temporary holding back of the ice in a reservoir would in a few cases undoubtedly be of some assistance. It seems that the most effective method of dealing with this condition consists of keeping the ice broken up on the reaches of the stream where gorges are most likely to form, and thus provide a clear passage for ice brought down by floods. This method would probably work hardships or inconvenience to the ice harvesters on some of the rivers; but the protection afforded to property would doubtless more than offset the disadvantages. The State has entered upon a policy of protecting property in this manner along the Hudson below Albany.

Another condition by which floods are greatly aggravated is the obstruction of the channel by insufficient bridge openings and other structures. The cause may be ignorance as to volume of flood run-off, or in the struggle to realize a large ultimate income from a small present investment the possibility of occasional damages may be carelessly disregarded. This encroachment on the channels of streams should be a matter for official regulation, and deserves more public attention than has yet been given it in this country.

Perhaps the complication which involves the most difficult problems of construction and operation of flood-control works is that of combining adequate flood protection with equalizing of stream-flow for the development of power and other purposes. To materially ameliorate flood conditions on large rivers usually requires the provision of an enormous amount of storage; logically, the larger the proportion and the greater the

capacity, up to a certain limit, the better the control. On many streams it is doubtless feasible to build systems of reservoirs which would entirely do away with destructive floods, provided the reservoirs be intelligently operated solely for flood control. It must be frankly admitted, however, that the ideal use of storage for flood control is not entirely consistent with the best use of the same storage for equalizing the flow throughout the year. For the purpose of ideal flood control, the reservoirs should be emptied of accumulated flood waters immediately after the flood has subsided and as rapidly as possible without swelling the stream to dangerous proportions, in order to have the storage available for another flood. On the other hand, for the purpose of equalizing the flow as completely as possible throughout the year, the reservoirs should only be drawn upon when necessary to supplement the natural flow in the stream in order to maintain the desired average flow. Theoretically, if the extremes both of the rate and volume of flow of the stream can be determined (which usually requires very long records of discharge), and if sufficient storage be provided for the absolute equalizing of the flow, the solution of both problems would go absolutely hand in hand, and flood control by storage would be synonymous with ideal equalization of stream flow. The most practical solution, where conditions will permit, seems to be to provide an excess of reservoir capacity so that the portion of the reservoir above a certain elevation may be reserved entirely for flood control while the portion below that elevation may be used for equalizing the flow of the stream. This plan has been proposed by the Commission in the case of the projected Portage reservoir on Genesee river.

Undoubtedly the greatest economic problem involved in a study of flood control is that of the adjustment of the relative rights of the residents of the upper and lower sections of the river valley. From the point of view of each the matter calls for different modes of treatment. The up-river resident believes the solution of the problem will be found in facilitating the passage of the flood by his district. This may result in discharging a great volume of water on the communities down-stream at a time when it would swell the crest of the flood in that section. The down-stream resident naturally has to contend with a much larger volume of water, so that to restrict it to a channel of moderate dimensions is out of the question, and he therefore prefers an arrangement whereby the surface waters from the upper stream

may be at least temporarily stored in the basins containing the lands of his up-stream neighbor. The Water Supply Commission has held that the proper disposition is the improvement which will work the greatest good to the greatest number, provided there is a distinct economic advantage to the community in the river valley as a whole.

The matter of municipal water supply is likely to be involved in some of the great storage projects, also the water supply for the canals of the State. Several streams on which water storage is practicable are at present or will be in the future used as sources for canal water supply. The plans contemplated by the Commission would result in insuring the sufficiency of these supplies, but the uninterrupted maintenance and protection of a constant water supply during the navigation season is undoubtedly essential to the proper operation of the canal system. In these times of extensive municipal water supply systems, it seems reasonable to assume that there may be instances in the practical working out of a comprehensive plan of water conservation where the project of water supply for a municipality or group of municipalities may be combined with a water-storage project to good advantage. At any rate the careful and prolonged study which has been made of municipal and domestic supplies by the Water Supply Commission has given it a full appreciation of their prime importance, and the Commission believes that in any water-shed the question of municipal water supplies should be given first consideration.

It has been believed by many that the State, in the exercise of its police power, could construct storage reservoirs which involved the use of some of the State's forest lands in spite of the constitutional provision that "The lands of the State, now owned or hereafter acquired, constituting the Forest Preserve as now fixed by law, shall be forever kept as wild forest lands," and that "They shall not be leased, sold, or exchanged, or be taken by any corporation, public or private, or the timber thereon be sold, removed or destroyed." The River Improvement Commission considered the constitutional question thus involved, and reached the conclusion that the force of this prohibitory clause in the constitution was paramount to all exercise of the police authority of the State to protect the public health and safety, and it declined further to consider any petitions involving the utilization of State forest lands for the construction of storage reservoirs. The Water Supply Commission has held practically the same view of this

question and has accordingly recommended to the Legislature that the Constitution of the State be so amended as to permit the flooding of State forest lands for the purpose of constructing storage reservoirs which are to be forever owned, maintained, and controlled by the State for the public use and benefit and for the purpose of providing a public revenue.

The conservation of the water resources of the State on a broad and comprehensive basis, which shall give practical consideration to the most favorable natural opportunities and produce the most beneficial results necessarily involves the flooding of relatively small areas of State forest lands in the Adirondacks. The surveys indicate that 55,000 acres of State land would be required for a complete system of water storage, including many reservoirs likely to be built only in the distant future, if ever. Even this total of 55,000 acres is only 3.9 percent of the State's holdings within the boundaries of the Adirondack Park; of this amount about four-fifths is low swampy land or is under water, and only one-fifth, or eleven thousand acres, is of any considerable value for forest purposes. This question of the amendment of the Constitution is under consideration by the State Legislature.

The drainage of swamp lands is another problem which tends to complicate rather than simplify the water-storage situation. There are within the State extensive areas of swamps whose owners would like to have them drained and reclaimed for agricultural purposes. Some projects of this character have already been carried out, but the questionable constitutionality of most drainage laws has interposed to retard any very widespread reclamation movement of this character. Here again the desires of the up-stream and down-stream residents do not harmonize. The down-stream riparian owner, especially if he operates a water-power, objects to the drainage of those marsh lands on the ground that they constitute a natural storage reservoir which operates to steady the flow of the stream. His solution of the problem would be to build dams across the outlets from these great swampy tracts and thus increase their capacity for storage. In some instances it appears to be entirely feasible to do so, while at the same time it seems equally practicable to secure the necessary storage by raising the surface of some existing lakes and subjecting them to some fluctuation. The question enters as to whether it is not better to flood a comparatively small additional area

around the shores of existing lakes in order to secure the required storage and then drain and reclaim swamp lands for agricultural purposes.

By special act of the Legislature in 1909 the Water Supply Commission was given jurisdiction over certain local improvements to streams which contemplated the betterment of both the sanitary and scenic conditions. Certain lakes in the State are bordered with large areas of unattractive swamp and stump land which the local residents would prefer to have permanently submerged. It is claimed that the scope of improvement would include not only benefits to the conditions affecting the health of the community, but that such improvements would in a number of instances result in rendering the region more attractive, especially to summer visitors seeking recreation and health. It is also pointed out that in some instances the interior navigation on some of the lakes would be materially improved, and that the community would materially benefit from the improvement in this manner. The State has already carried out some improvements of this nature, and it seems quite probable that there are possibilities of a number of similar improvements. The problem does not appear to enter into the larger storage reservoir projects, and has not been given very extended consideration by the Water Supply Commission.

New York State and her citizens are justly proud of her scenic falls. Of these the American Falls of Niagara are doubtless the most widely known. There are, however, other falls on streams within the State which constitute local attractions of great interest in their respective communities. The Salmon Falls on Salmon river in Oswego County, the series of falls in Letchworth Park on the Genesee, and High Falls on the Ausable are prominent examples. The Water Supply Commission entertains a deep appreciation of the esthetic value of these beautiful masterpieces of the hand of nature, and believes intrinsically in their preservation. This attitude of the Commission is exemplified in the plans for the proposed Portage Falls power development, which provide for a flow greatly in excess of the minimum flow over the falls for a period of twelve daylight hours in each day. On the other hand, the Commission sees also the wonderful amount of quiet comfort which would be afforded to modern civilization by electric light and the many other applications of power which can be generated by the waters running over some of the falls of the State. The major part of the surplus water is wasted in the spring months of the year, and does not

contribute in any appreciable measure to the scenic beauty of the falls; on the other hand, the natural flow of the streams frequently is reduced to such a low rate that the falls lose something of their attractiveness. It will doubtless prove practicable in connection with power developments at some of the naturally attractive falls in the State to insure a larger minimum flow in the dry weather as well as to conserve the great amount of power at present running to waste over the falls in the wet season.

In humid climates irrigation is admittedly more or less of an experiment. Its financial feasibility seems to depend on its being considered a matter of insurance against the failure of crops in seasons of low rainfall. There have been a number of scattered experiments carried on at different places in the State, but the plants used, especially in the older experiments, were comparatively complicated and expensive. The equipment for one particular set of experiments cost about \$500 per acre. More recent experiments have been conducted in sections of the State where the precipitation is light during the growing months, and in fact throughout the year, and with a less expensive and a more generally practical equipment. In a few instances, which have been brought to public attention, the experimenters have been able to raise excellent orchards and garden products by means of a comparatively inexpensive irrigation plant, whereas other portions of the gardens and orchards of the same farms did not produce results nearly as satisfactory. One successful experimenter claims that he has made 20 percent interest on his investment by the installation of a small irrigation plant. The precipitation records show that there are portions of New York State where the rainfall during the crop-growing months does not amount to more than one-fourth or one-fifth of the water which is applied to the same crops where irrigation is conducted on a broad scale. The subject has not been entered into in great detail by the Water Supply Commission owing to the fact that its statutory jurisdiction does not seem to justify such a study, but it appears that the possibility of such use of at least a portion of the water supply of the State should be borne in mind and its development carefully watched in connection with the formulation of a general plan for the conservation of the water resources of the State by means of storage reservoirs.

Need for Comprehensive Plan and Definite Policy

The importance of a fixed policy establishing State leadership and control in the matters of water Conservation cannot be overestimated. Without it, there is no place for consecutive and correlated action, either executive or legislative. In the past the State has had no policy of power development, either under public ownership or by encouragement and regulation of private or corporate development. Unlike many other States, New York has never, under general laws, granted the right of eminent domain to individuals or corporations for the purpose of flooding lands to create storage ponds and develop water-power. Moreover, it must be conceded that in view of the doubtful constitutionality of the "mill acts" of other States, and particularly in view of the strength of the modern sentiment demanding universal sharing in the benefits of natural resources, this State is not likely in the future indiscriminately to grant its power of eminent domain for this purpose. Unless the State shall define its policy and enter upon the work of carrying it out, this feature of its natural resources must largely remain in its present undeveloped condition, or be subject to the same haphazard and uncontrolled methods of utilization that have governed in the past. If we are to permit private interests to build storage reservoirs for power purposes on any broad and satisfactory plan, it can only be done by amending the Constitution. As adequate reservoirs cannot be generally constructed for power purposes by private enterprise without constitutional amendment, and possibly not then, the better way to accomplish this object is for the State itself to announce its policy and undertake its performance in the interest of all classes and citizens.

Development by the State ensures the fullest possible utilization of the power possibilities of each stream, whereas development by uncontrolled private enterprise often involves waste of resources. Private capital, seeking the greatest possible immediate return on the investment, naturally confines its attention to the most concentrated portion of a given fall. The less precipitous portions of the fall above and below, involving a large unit outlay in development, are consequently apt to be neglected, and in too many cases permanently wasted, because no other enterprise is likely to undertake their development afterward, even if the rights of the company already on the spot would permit this to be done. On the other hand, the State, with its greater power and scope, and with financial resources enabling it to defer the return on its investment, could undertake the

construction of the more extensive works necessary to develop the full extent of the fall in the supposed case. Without amplifying the point, it should be clear that the State is the only authority with sufficient power to ensure the complete development of each and every stream so that every foot-pound of energy represented by its falling waters may be given up when necessary to the service of man.

The prime inclusive reason for the exercise of State authority over the control of stream-flow for power development is that under modern social and economic conditions this step is necessary to ensure the equal participation of all citizens in this form of natural wealth, which is peculiarly the heritage of the whole people. Some of the more particular supplemental reasons for State control have been mentioned in the foregoing. It appears that from all points of view the State is the proper authority to undertake and carry out the conservation of its own water resources.

The State Water Supply Commission is engaged in studying the subject of conserving the falling waters in the rivers and streams of the State. In a country where all of the streams both great and small fill their banks in the springtime after heavy rains, and then decrease in volume all through the dry months so that they become in most instances worthless as power streams and of but little value in many other ways, it is clear that storage reservoirs of large capacity, the size depending, of course, on the water-shed in each case, must be built, if wasted water and worthless streams are to be turned into valuable assets. The building of storage reservoirs requires available areas to flood, favorable sites for dams, and scientific knowledge to supervise the construction of such dams and reservoirs. There must be, also, some general head to locate and plan such reservoirs on a broad and comprehensive scale, so as to store the largest possible amount of water in each given case; otherwise opportunities for economic development will be lost and money wasted. The plan should be so feasible and comprehensive as to include every profitable storage possibility, be it either great or small. The plan must permit of doing the work by reservoir units, and at such places as make promise of early and satisfactory return. With such a plan all who are interested in using to the best advantages that which is our own, and saving and conserving for the future that which justly belongs to our children, can work in harmony. Such a plan will enlist the people of every

locality in the possibilities of water storage in their own developments, and at the same time not interfere in the least with the developments of a similar character in other parts of the State.

A plan that will enlist such an interest and make possible such a systematic development of a great and wasted natural resource, the Water Supply Commission has been trying to devise. It makes no claim to perfection, but it does claim that it has devised a workable plan for saving and conserving this wasted energy for both public and private use and so as to provide a public revenue. The plan includes the building of storage reservoirs by the State which shall be owned and controlled by it. The scheme is to use the stored water to equalize the flow of each stream upon which it is built, and charge the users of the stored water for the additional power such stored water gives to mill owners further down the stream. This does not contemplate charging a mill owner anything for the power he now has, but only for the additional power he gets by reason of the equalized flow of the streams due to using the stored water when he needs it most.

The Water Supply Commission as a part of its last annual report to the Governor and Legislature submitted a bill providing for a systematic development of the water-power resources of the State under State control. This bill contemplated the return of a net revenue to the State and accordingly provided for the assessment of benefits upon individuals and properties benefited by reason of the construction and operation of storage reservoirs. Many of the provisions of this bill were new in principle, and it was to be expected that a measure of such far-reaching effect would meet with some opposition. Although the bill provided for contracts to be entered into with respect to payments for benefits to be conferred, and the power of assessment was only to be resorted to in order to forestall an unwilling beneficiary from blocking the progress of a great public enterprise, such a provision met with disapproval in the Legislature and the bill was not advanced. The Commission believes that as the Legislature becomes more familiar with the problems involved, it will approve of this policy. For these reasons, the bill with amendments in other respects will again be submitted to the Legislature in connection with the next annual report.

REPORT FROM NORTH DAKOTA

C. B. WALDRON
State Agricultural College of North Dakota

While Conservation means the same to all people, namely, the perpetuation of those resources and conditions that make a prosperous existence possible, yet each Commonwealth must develop its own best means for bringing this about.

While it is wise for the Federal and State governments to take what steps they may to prevent the wasteful destruction of certain natural resources like our minerals and forests, yet if all this be done and with the thoroughness that the most ardent of us could demand, still the great problem of Conservation taken as a whole would scarcely be touched. The utmost that the Government can do directly, though of considerable magnitude in itself, is relatively of small importance. Even meetings like the present one have a significance and value only as they inaugurate and vitalize Conservation movements more important and extensive than any Government can ever hope to bring about by direct means.

This principle applies to the greatest degree in instances in which control of the natural resources has already passed to the individual owners. It applies with even added force when such ownership lies in agricultural lands. The reason for this lies in the fact that of all natural resources the soil is by far the most important, and, further, that conservative principles and practices apply with greater directness and profit there than in any other field. The conservation of this season's plant food and soil moisture means next season's crop. Through plant and animal breeding the more prolific and profitable strains are conserved, and through battle with plant and soil diseases and with pests of all kinds we conserve the purity of our soil and the crops that we grow. Such active and constant exercise of Conservation as this may be, in a field that directly affects our entire population in the most vital and direct manner possible, is a matter for our most earnest consideration.

What is being done to train the great body of mankind to whom this important task of Conservation is entrusted; and are the present measures adequate?

Aside from legislation pertaining to weeds, plant diseases, and insect pests, there is little that can be done directly to enforce Conservation measures. The friction encountered in enforcing even this body of laws indicates the difficulties that arise when public restrictions come into conflict with private enterprises. True, it is a crime to waste the fertility of the soil on which the very existence of the race depends; but until all our traditions change, the only punishment that will be visited upon the offender is not from the legally constituted State but from nature herself. He whose will is to rob and skin the land may not be reached by legal process, but he must be taught that the penalties which an outraged nature exacts are as inexorable as the Blind Goddess ever pronounced.

While there always will be fools that can learn only in the school of experience, yet the great majority are glad to find an easier and cheaper way.

Back of the Conservation of the farm must lie the education of the farmer; and greater than all the other problems of Conservation is this one. We are barely entering upon this field, for the reason that the fund of knowledge upon which this education is to be based has been but recently acquired. Our knowledge of the soil in its relation to plant growth, the control of plant diseases, and the laws of plant improvement, have all come to us in recent years. Still, much as there is yet to determine, there is already a vast fund of knowledge of untold worth; but means are not yet provided for making it useful and effective.

Speaking for North Dakota, such natural resources as she possesses, aside from her soils, are being well protected and conserved through public measures already in force. Her vast fields of lignite coal underlain with valuable clays have been withdrawn from homestead entry, and hereafter only surface rights in these lands will be granted.

Such forests as the State originally had have long since passed into private hands, and the land has mostly been cleared for farming. In North Dakota, forestry, like agriculture, will be operated by the individual land owners for their direct if not immediate benefit. It may be found advisable to plant public forests in parts of the Bad Lands and other rough areas, but by far the greater part of tree planting will be done upon small areas on the individual

farms. The State already encourages such planting by a bounty paid in the remission of taxes. This is not enough. The land owner in most cases does not know what trees will prove the most profitable, nor how they may best be grown. Here again the one necessity is education. Object lessons in tree planting should be established in each community, and all pupils in the public schools should be shown how to grow a grove of trees. Such a system would produce immeasurably greater results in the way of timber production than would come from the public forests, important as these doubtless are.

But agricultural education will conserve something more than the fertility of the soil and the vitality and purity of our crops. It means also the conservation of a prosperous, virile, self-dependent, and intelligent people. It means a prosperous people, for no cost of education of the right kind was ever known to impoverish a people, and no expenditure rightly made could ever equal the gain. Conservation can never be expected of the ignorant. Conservation is but the larger and more altruistic expression of the term known as thrift; and ignorance and poverty know it not. The means for extending and improving agricultural education will develop and expand in the same measure that we apply ourselves to the problem.

Agricultural colleges have not rendered the assistance that they should in extending agricultural education, because their field has been too restricted. Excellent as their instruction may be, it reaches only a very small percentage of our people directly. Their scope and activities must be enlarged till their influence is felt in every community. They should not be shut out from participating in the work of general education as they now are in many instances. In a measure we repudiate the findings of science, and discount the progress we have made, in not providing a wider application for our researches. There is at present no adequate means for the dissemination of the vast body of knowledge that alone will save to us our own great underlying industry of agriculture.

The world has oftentimes tried the experiment of building a State upon other foundations than that of a conservative agriculture and an intelligent and prosperous agricultural class, and always with the same fatal outcome. The grandeur of cities, the glory and might of great armies, the highest culture in the arts, and the noblest of religions and philosophies, will not

suffice to save the nation that knows not nature and defies her laws. That State but hastens the day of its own destruction that fails to train its citizens in the right use and management of their land holdings. No jealous interest of whatever worth in itself should be given consideration at the expense of that which maintains all of our interests.

North Dakota has been favored by nature with a soil so productive that, properly tilled and conserved, it will feed one-tenth of the present population of the entire Nation. It is an asset such as few nations ever possessed, and it should be so safeguarded that its great contribution to the Nation's existence may steadily increase. The one way to do this is to teach the land owners that Conservation in agriculture means not only patriotism and good citizenship but prosperity as well, that useful education at any price is always cheap and ignorance costly, and that no values can be more stable and certain than those lying in productive farm lands.

The patriotic sentiment that leads men to sacrifice time and money that our natural resources may be conserved is most commendable. Of still more service is he who aids in developing a system of education that shall teach men to conserve the natural resources entrusted to their own hands. The task is a great one, but not beyond the range of possibility; and upon its successful accomplishment rests the welfare of the whole Nation.

REPORT FROM OHIO

WILLIAM R. LAZENBY

Ohio State University

Chairman Executive Committee of the Society for Horticultural Science

The welfare of our country, as well as that of the States composing it, depends on a wise Conservation of its rich and varied natural resources. Many of these resources have been so bountiful, and apparently so inexhaustible, that we have drawn upon them without a thought of their limitations or the dire effects of their exhaustion.

Speaking especially for Ohio, I trust it will be understood that by "Conservation" I mean an honest effort to make that State a good one to live in for all of us now there, and for all who may come after us.

In addition to the three problems named below, other Conservation questions will doubtless require attention; but for these, every instinct of justice and humanity insists that we accord them instant and earnest consideration.

1—*The Forestry Problem*

I place this first, because the influence of the forests is so far-reaching, and we have no clear-cut, well-defined policy in Ohio designed to preserve, improve, and extend our forests.

Ohio has an area of 41,000 square miles, and has been tremendously rich in hardwood timber. We have cut down this timber most improvidently, with no effort to restore the supply, and so far as the State is concerned are now on the verge of a timber famine. In 1900, according to the Twelfth United States Census, Ohio ranked seventh as a lumber-producing State, being exceeded by Michigan, Wisconsin, Pennsylvania, New York, Minnesota, and Maine. Since then she has dropped to the nineteenth rank, and bids fair in the near future, unless prompt and vigorous action is taken, to have so little timber left as not to be rated at all. The effects of this wholesale removal of our forests may be briefly summarized as follows:

(1) We are compelling those who come after us to pay an almost prohibitive price for lumber, and are likely to see an end of some of the most important wood-consuming industries of the State. As a source of wood supply our forests touch the interests of all. We are a universally wood-consuming as well as food-consuming people.

(2) The recent floods in the river-valleys of Ohio, which have caused losses of life and of property valued at millions, have followed and will continue to follow the denudation of our hills by excessive tree-cutting, followed by fire.

(3) In many places the erosion or wash caused by the rapid run-off of the rain and melting snow is reducing the deforested hills to barren wastes, and is covering much of the fertile soil of the valleys with sterile sand and gravel.

The forest problem is the great Conservation problem in Ohio. It affects the State, because it concerns every citizen of the State, and it can only be

solved by action of the State and the Nation.

2—*The Waterway Problem*

In my opinion this question comes next in importance. By waterways I mean not only navigable streams and canals, but power sites on non-navigable as well as navigable streams. If the forests are properly managed, water will be an unfailing source of power. No few men, nor any special interest, should control these sources of power, for this means a control of all industry that depends on power. Our waterways may not be so enormously valuable as those of some other States, and this is all the more reason why they should be conserved for the public good.

We shall be needlessly mortgaging the future by allowing any special class or interest to use our waterways and water-power sites without making some direct payment for these valuable privileges. This is important not only for State revenue, but as a recognition of the principle that what belongs to the people should not be absolutely surrendered to private interests. There is great value in our undeveloped water-power. An engineer's inventory of all the waters of the State, with their possibilities of power, would cause Ohio to sit up and take notice.

If forests and waterways were properly conserved, we would hear less from railroads and power companies of the enormous bill of expense from floods at one time, and loss from low water at another.

3—*The Mineral Problem*

Ohio is rich in coal, oil, gas, stone, clay, sand, and other mineral resources. These should be carefully catalogued, so that the people could know more about the material assets of the State.

Mineral lands should be sold only to those who are prepared to develop them, and under conditions that will prevent the improvident waste of reckless exploitation. For the present it is probable that the actual development or working of the mineral properties of the State can best be done by private interests acting under some public control, but the State has no moral right to permit such valuable privileges to pass from its control for nothing in return. It is only by some form of National and State

Conservation that we can secure an abundant and continuous supply of such primal necessities as wood, water-power, and coal.

The control of animal diseases and of insect and fungus pests that are spread by interstate transportation, and the preservation of migratory birds, which are our best allies in fighting injurious insects, are vital subjects for the consideration of a National Conservation Congress. The control and destruction of enemies and the protection and multiplication of friends by the concentrated and cooperative action of the States are subjects that clearly come within the scope and interest of National Conservation.

Conservation can only be effective by good laws faithfully executed. By proper legislation we can encourage the reforestation of our denuded hillsides and stimulate the planting and care of valuable timber trees through relieving such land from undue taxation. Timber should be taxed like other property, when cut; but to tax land and its timber crop every year is manifestly unjust.

In order to rightly conserve our forests we should furnish good opportunities for young men to become well trained in forestry. For this our schools of forestry must be well equipped. I am pleased to state that Ohio has made a splendid beginning in this direction; and there is no reason, if properly supported, why this centrally located State should not have one of the best forestry schools in the country.

What is needed to properly investigate the conditions and formulate a Conservation policy for the State is a good Conservation Commission. In addition to this, we need more thought, more study, more science, on the part of the public, concerning the natural resources of the State, with less blind devotion to the old ways and means of doing things, which if ever judicious, have long ceased to be so.

REPORT FROM OKLAHOMA

BENJ. MARTIN

I have the honor to represent as a Delegate to this Congress the Muskogee Commercial Club of Muskogee, one of the leading organizations of Oklahoma, under the influence of which the city of Muskogee grew from a town of 4,000 inhabitants in 1900 to its present population of 30,000.

A distinguished citizen of a neighboring State, on a recent visit to our city, constituted himself a Grand Jury and indicted each citizen of larceny. He charges that Oklahoma for years had been stealing from the other States of the Union some of their best brain and brawn, until now we have approximately two millions of the choicest sons and daughters of the American Republic. To this indictment we now offer ourselves for arraignment before this Congress, and plead guilty, and we are ready to receive our sentence without a plea that justice be tempered with mercy. As to other charges of wrongdoing on the part of some of Oklahoma's distinguished sons, which have been much heralded in the press, I most emphatically enter a plea of "Not guilty," either in law or morals; and time will completely vindicate them.

The resources of Oklahoma are vast, far beyond the conception or knowledge of those who have resided within her borders for many years. Conservation is of particular importance to us, for yet our resources are practically in their virgin state. We heartily join hands with you of our sister States in this great movement, in my opinion due to the work and wisdom of Gifford Pinchot more than any other American citizen. However, his ideas and earnestness were very fully and heartily appreciated by that foremost American, Theodore Roosevelt, to whom for his great work in inaugurating and fostering Federal Conservation we give honor.

Chief among our resources are the vast variety of agricultural products which grow in great abundance. In the same field may be seen growing enormous yields of corn, cotton, oats, wheat, and alfalfa. No other State can excel Oklahoma in the production of these products. We join the great corn-belt of Illinois and Iowa in singing the song of Whittier—

Heap high the farmer's wintry hoard,
Heap high the golden corn;
No richer gift has autumn poured
From out her lavish horn.

Let other lands exulting glean
The apple from the pine,
The orange from its glossy green,
The cluster from the vine.

We better love the hardy gift
Our rugged vales bestow,
To cheer us when the storm shall drift
Our harvest fields with snow.

The following extract is from the First Biennial Report of the Oklahoma State Board of Agriculture:

"Oklahoma is the greatest country on earth, not only because we can grow everything here that can be grown anywhere else in the United States, but because many crops we can grow here are decidedly more profitable than are crops of like character in many other sections of the country."

We join our sister States of Pennsylvania, West Virginia, Tennessee, Alabama, and others in the endeavor to conserve their vast deposits of coal, not solely from patriotic motives, but also because of our extensive coal, oil, and gas fields, only a small part of which have yet been developed. The supply of timber in the eastern and southeastern portions of our State is worthy of the consideration and protection of the Conservation movement. Particularly rich is our State in its streams of water and its water-power. The principal rivers are the Arkansas, the Grand, the Verdigris, the Canadian, the Cimarron, the Washita, and the Red, the latter forming the boundary between Oklahoma and Texas. These streams within themselves contain great resources, yet in the virgin state, awaiting but to be developed and utilized by American genius.

I know of no more appropriate way of closing my statement than in the words of Colonel John A. Joyce—

The rolling hills and mountains,
Without their forest dress
Will soon bring to the Nation
Great hunger and distress;
And if we do not listen
To the scientific strain,
The soil of grand Columbia
Will be washed away by rain.

Brave nature in her glory
Works for animated things,
And tells the old, old story
Of feeding serfs and kings;
But man, obtuse and greedy,
Will not listen in his pain
To the poor, and weak, and needy,
Who must live by sun and rain.

We must save the soil and water,
Or a desert there will be
For wife, and son, and daughter,
In this land of Liberty.
And the Congress of the Nation,
Must now listen to the brain
Of our scientific sages
Who would husband soil and rain.

REPORT FROM OREGON

E. T. ALLEN

Assistant Secretary Oregon Conservation Commission

Oregon's chief Conservation advances of late have been the passage of progressive water laws, by the effort of the State Conservation Commission, and the progress of private timber owners in the prevention of forest fires. The most urgent task now on hand is to secure more liberal State aid in forest protection.

Immediately following the Conference of Governors at the White House in 1908, Governor Chamberlain appointed for Oregon a Conservation Commission of 15 members. This semi-official Commission was reduced to 7 members, and given statutory standing and a small appropriation, by Act of Legislature filed February 23, 1909. Its work is "To ascertain and make known the natural resources of the State of Oregon, and to cooperate with the National Conservation Commission to the end that the natural resources of the State may be conserved and put to the highest use."

No legislative session has been held since the statutory Commission was appointed. In its earlier form, however, it recommended and secured the passage, by the same Legislature which gave it official standing, of a workable law for the development of Carey act projects, and one for complete State control of waters within the State. Both have proved excellent, no defects of importance having developed.

The Oregon water law, in particular, is generally regarded as an example of good State action. It is based on the police power of the State to preserve the public peace and safety of its water users. Under this law, rights to the use of water for power development are limited to a period of 40 years. A simple and expeditious method is provided for determining early water rights, protecting existing rights, and acquiring new rights. Prior rights are determined by a Board of Control consisting of the State Engineer and the division superintendents of the two water divisions into which the State is divided. Established rights are protected by a water master in each district of a division, acting under the direction of the division superintendent. He may make arrests and compel the installment of suitable devices for controlling the use of water. New rights are granted by certificate of the Board of Control, after proof, under a system based on priority of application and beneficial use. Water for irrigation is made appurtenant to the land irrigated. Oregon also has a law providing for a State tax, on a horsepower basis, upon water-power projects.

Oregon has a non-partisan State Board of Forestry, consisting of representatives of the industries and agencies chiefly concerned in forest management and protection; also an excellent forest code, so far as punitive and regulative provisions are concerned. It lacks appropriation or machinery to make this code effective. To secure such provision by the next

Legislature is the chief present work of the Commission. The Commission works under the plan of attacking one point at a time, instead of dissipating efforts among all the improvements needed. Water and water-power were felt to be the most urgent, forestry is considered next, and when the forest laws are made satisfactory, other branches of Conservation will receive concentrated effort.

There is also an Oregon Conservation Association which, under the same plan, is now chiefly devoted to carrying out the work of the State Board of Forestry for which no appropriation exists. Its secretary is secretary of the State Board, and the funds of the Association help to pay postage and clerical help derived by the State.

Under an alliance called the Oregon Forest Fire Association, affiliated in turn with the Western Forestry and Conservation Association embracing five States from Montana to California, a large number of the private forest owners of Oregon cooperate to secure better protection from forest fires. These owners spend from \$50,000 a year upward for patrol and fire-fighting, their employees having authority from the State as fire wardens.

Among the Conservation problems to be taken up next in Oregon are the protection of fisheries, good roads, improvement in technical methods in irrigation and dry-land farming, topographic surveys, and inventories of State resources.

REPORT FROM RHODE ISLAND

HENRY A. BARKER

Chairman Rhode Island Conservation Commission

This Conservation Congress has been so very generous with its invitations that it happens that about every organization in which I am interested has been asked to send Delegates. As a result, quite a good lot of them have been so kind as to bestow this honor upon me—most of them prudently waiting until they found out that I was coming anyhow. For that reason my desk in Providence is adorned with a nice little pile of beautifully engraved cards, each telling me that this City of Saint Paul takes pleasure in extending its hospitality, etc. Along with each of them came other cards to

warn me that if I wanted hotel accommodations I had better speak quick. So I spoke with reasonable speed—and eminently satisfactory results; but I am glad I did not have to find accommodations for all of the Delegates that I seem to be.

I want to say, also, that if it gives the cordial City of Saint Paul pleasure to extend this charming invitation, the pleasure is entirely mutual; I am delighted to accept the hospitality.

I am glad that I need not report at this time for anything except the State of Rhode Island, and I am sure you will be. You may ask, "What has Rhode Island to conserve?" In reply I want to tell you that no State in the Union in proportion to its population has so much that needs conserving. Some of our friends from the Far West tell us heartbreaking things about how the Government has reserved or restricted so much of the western area that there isn't enough left to make farms and villages on. I think I heard day before yesterday that in the State where I attended the First Conservation Congress last year there were Government reservations as big as Massachusetts and Rhode Island combined—though I should say these wouldn't necessarily look so very big when painted on the map of Washington, or seriously hamper the operations of its people. And we have this sad condition contrasted with that of the happy East where the Government owns no reservations at all; but back in the East we do not realize that this is a good fortune. Never having had any land in our part of New England owned either by the State or by the Nation, we have been somewhat frantically endeavoring to have them secure some for the good of our people, even though it now has to be bought. Everybody knows how earnestly we wish that the Government might have done for us at the beginning of our settlement just what the Government is able to do, and is doing, for the West today. There isn't any talk of "State rights" in the East. It is a question of the States' necessities. The Eastern States are all working to their utmost to get the Government to undertake certain enterprises like the Appalachian White Mountain reservations, that are of an interstate character; but each State expects to cooperate for as much of the remaining work as it can.

You will be glad to know that Little Rhody is trying to do its share. It always does its share. It always matches the Government, at least dollar for

dollar, on any public improvement work. Just now it is spending a million dollars on the harbor of Providence to match another million that the Government appropriated last year. That is the kind of "State rights" the Government gives it. But not much compared with what the railroads are putting in.

The formal establishment of a Conservation Commission was almost the very last act of the Rhode Island Legislature at its special session, only about two weeks ago. We didn't expect, of course, to be quite so much up to date, or so early in any new field, as our brethren in Montana for example, though we have had a Conservation Commission, rather informally appointed by the Governor, ever since that notable gathering of the Governors at Washington, and work that such a commission would naturally do has been going on, under other names, longer than I can remember.

The aim of the new Commission is to secure the maximum of efficiency and the minimum of politics. I do not know what the political affiliations of its members are, or if they have any, and I do not believe the Legislature knows. It is made up of ex officio members, to bring into efficient cooperation several well-established departments that have long dealt with some phase or other of Conservation. The head of the Bureau of Industrial Statistics, which is conducting a State survey of natural resources, including soil analysis; the Secretary of the State Board of Agriculture; the Director of the Experiment Station of the State College; the State Forester; and the Secretary of the Metropolitan Park Commission—these departments will now contribute their efforts to a common purpose. The State Forestry Department, with advice from the National Forest Service, has been getting some very up to date forest laws passed, and the Park Commission has made a visible beginning to secure for public use and preservation some necessary recreation places for the over-crowding population of the Providence "Metropolitan District," which has about four-fifths of the population within about four-fifths of the area of the Twin Cities combined.

The State College, assisted by the U. S. Bureau of Soils, has been showing such farmers as care to take notice that southern New England is a very different sort of place agriculturally than it has been the habit to suppose, and that at least three ears of corn may be made to grow, where, previously,

one went to the dogs—or the hogs. The very fact that there are more ever-hungry mouths to feed and more manufactures to the square inch in southern New England than there are anywhere else makes this necessary. We must care for every drop of water that falls on our hillsides. The cities need it; the manufacturers need it (and can use it first); the great bleacheries—that furnish about all the textiles that all of you use and wear—need all they can have; and the people need the lakesides and the river banks for recreation as in the past.

At present our markets get most of their "fine Rhode Island turkeys" from Vermont and their "new-laid eggs" from beyond the Mississippi. A large part of the Rhode Island greenings and Massachusetts Baldwin apples come from Oregon and Washington, though not because they refuse to grow in their native habitat. But much of the soil must have put back into it those elements which previous unscientific generations robbed it of. And here is an amusing paradox: With a population growing in density faster than in any other State of the Union, and with more markets just around the corner, there are, nevertheless, more acres of forest-covered lands and more acres of unutilized lands in Rhode Island than there were 50 years ago—and more in proportion than in almost any other State in the Union.

Well, that's where Rhode Island comes in, in this Conservation movement; and it has come in none too soon. If it had only had a wise and paternal Government to help it administer and develop its natural resources a century ago, the cost of living would be less today for every one of its inhabitants.

Rhode Island has awakened to vital things, but even if it had only an indirect interest in Conservation it would still feel that it owed its moral influence to the country as a whole, and that it is not a separate selfish little two-cent republic all by its lonesome, but a part of a great Nation that prefers to be governed from Washington rather than from Wall Street: a Nation whose prosperity and power and glory need the cooperation and loyalty of every one of its citizens.

REPORT FROM SOUTH CAROLINA

E. J. WATSON
Commissioner of Agriculture
Chairman State Conservation Commission

South Carolina Commission's full report delayed, so report briefly by wire. Active work has been done. A preliminary forest survey has been made, and a complete measure for conservation of forests and protection against forest fires has been introduced in the General Assembly and will be pushed during the coming session. Active steps have been taken toward drainage and reclamation of coastal lands, and a measure to provide for a complete system under the direction of the State Commission is now being prepared for introduction in the Legislature in January next. Conservation of human resources has been greatly advanced in the past two years, following the enactment of complete factory inspection laws. No State is giving more attention to conservation of all her resources at this time than is South Carolina. I am heartily in sympathy with everything making for Conservation, and greatly regret I cannot be with you at the Congress.

REPORT FROM SOUTH DAKOTA

DOANE ROBINSON
Secretary Conservation Commission of South Dakota

The South Dakota Conservation Commission, consisting of Senator Robert J. Gamble (Chairman), Eben W. Martin, Samuel H. Lea, O. C. Dokken, and Doane Robinson (Secretary), was appointed by Governor Coe I. Crawford in August, 1908, and has been continued by Governor Vessey.

The Commission made a preliminary report on the resources of the State in December, 1908. It has been unprovided with funds, but the newspapers of the State and of the Northwest have been open to its use, and from the beginning the policy was adopted of furnishing a weekly letter, educational in its nature, pertaining to the State's resources and their Conservation. These articles have received very wide publicity, both within and without the State.

The Commission acted as Executive Committee of the South Dakota Conservation and Development Congress called by Governor Vessey and

held at Pierre June 29-July 1, 1910. This was an exceptionally successful Congress, in which nearly two thousand citizens participated. Every county was represented, and the interest was very marked. The program consisted of addresses and papers educational in character, many speakers of national reputation participating. An annual Congress is contemplated.

REPORT FROM TEXAS

WILL L. SARGENT

Secretary Conservation Association of Texas

The interests of Conservation in Texas are promoted largely by a voluntary organization of citizens, the Conservation Association of Texas. The Association held a Congress at Fort Worth in April last, at which much enthusiasm was manifested, and plans and policies were adopted, largely in the form of resolutions. The substance of these resolutions forms the body of this report.

We lay especial stress on the dirt roads of our State. Considering our great farming interests and their numerous and increasing yearly output, and the impassable condition of roads during certain seasons, we urge upon our county and State authorities the immediate betterment of our Texas roads by drainage, split-log drag, top-gravel dressing, or other up-to-date methods.

As the services of a large number of experts are necessary for the intelligent guidance and direction of all plans of Conservation in all lines, and as intelligent workers are necessary for the effective carrying out of such plans, we urge upon our legislative authorities, as the necessary foundation for all Conservation the better financial support of our great public school system, the introduction of agricultural and industrial studies into these schools, and the better equipment and maintenance of our higher educational institutions, and that more substantial financial support be accorded to the Agricultural and Mechanical College, and the Department of Agriculture, and that adequate appropriation be made for those institutions and for farmers' institutes to the end that the supply of experts and leaders may be made more nearly adequate to the needs of our rapidly growing State.

We know from past experiences that the overflow of our rivers and streams have resulted in washing away not only a great deal of rich and fertile soil, thereby injuring the lands of our farmers, but that these floods have destroyed crops running into millions of dollars in value and brought destruction and ruin to hundreds of our most worthy citizens. We earnestly recommend that the Legislature shall pass such laws as will constitutionally and in practical and adequate way prevent or curtail such losses in the future, the details of which can be worked out at the proper time and in an appropriate way by the legislative body itself.

We deplore the wasteful methods of lumbering practiced in Texas and look with dismay at the early day (say fifteen years) when all our best timber will be cut and unobtainable except at great cost, when the cut-over land, littered with dead branches and decayed treetops, will be annually burned over, the humus destroyed and the soil become unfit for cultivation and washed into the streams. We also apprehend with dismay the direful effects resultant upon our Texas climate when the timber is gone and the forest area has become a grassy, burned-over waste. We urgently recommend to the people of Texas that they call upon the Legislature for the establishment of a forestry department, under charge of a trained forester, and under control of the State Agricultural Department; and it shall be the duty of said forester also to lecture in both the University and the Agricultural and Mechanical College, and take charge of all forestry work in the State, and his work shall be in connection with the Forest Service of the United States Government, for the saving of the forest remnant in our State and the replanting of the cut-over area on lands not suitable for agricultural purposes.

We believe in a strict conservation and preservation of the public domain of Texas in a way that will best encourage homesteaders, and that all laws made for the protection of the State and the people against fraudulent entries or the illegal acquisition of the public domain on the part of private citizens or corporations should be strictly enforced, and we recommend to the next Legislature the passage of a law making it a felony against all persons knowingly and fraudulently entering into conspiracy to acquire any portion of the public domain in violation of the laws of Texas made for the benefit of actual settlers.

Recognizing the importance of fish as a food supply for our people, we indorse such laws as have already been enacted for the purifying of our rivers and lakes and such further legislation along that line as conditions demand, and recommend that hatcheries for the propagation and protection of fish be established and maintained by the State.

We indorse the work of the Texas Audubon Society in behalf of the wild birds of Texas, and urge that the next Legislature shall enact laws for the better protection of the birds, to the end that their extermination be prevented, so that they may be allowed to increase in numbers, delighting the world with their beauty and song, and also serving the economic purpose for which they were created, namely, the protection of crops by the extermination of insect enemies.

We congratulate the farmers of Texas for adopting modern methods in tilling the soil and in a diversification of crops. The great and beneficial results that have come to them through this system have clearly demonstrated its practicality.

The Legislature is asked to pass a law covering the features now partially covered by several independent laws and providing for a State Department of Engineering, which department shall be authorized to make surveys, maps, and estimates looking to the reclamation of overflow and wet lands anywhere within the State, and further being authorized to examine and approve all the plans and estimates of such improvements before said improvements can be accomplished, by this means being empowered to mutually protect all interests involved, whether these interests are at present active or in the future probable.

In order to carry out most economically the Conservation of the wealth latent in the soil and water supply of Texas, we recommend the enactment of legislation which will provide means and instrumentalities for a soil and water survey of the State as a basis for the earliest possible development of such wealth for the common good.

We recognize in the reclamation of our arid lands one of the greatest factors in the future development of the State, because of the million acres of fertile lands that can and should be reclaimed by irrigation. Recognizing all vested rights, we encourage the conservation, storage, and equitable distribution of

natural and flood waters of streams, artesian wells, springs, rainfall, and other sources of water supply. We favor a uniform system of irrigation laws that will give security for the investment of capital in the development of irrigation projects, and at the same time fully protect and safeguard the users of water and define the rights as well as the obligations of the enterprises delivering the water to them. We favor the State never parting with title to her water-power and the control of her streams to corporations or private individuals; we favor legislation that will secure the aid of the State in its conservation and reclamation work, such as the construction of reservoirs to be used for power, for irrigation, as well as for domestic and other purposes. The State is requested to enact a law creating an irrigation commission, acting under the direction of the Commissioner of Agriculture, whose duties shall be fully defined by statute.

We heartily endorse the purposes and objects of the National Conservation Association, and urge all the friends of Conservation in Texas to cooperate by becoming members of the National Conservation Association.

Recognizing that the prosperity and the happiness of our people depend on the utmost protection of their health and the protection of their domestic animals from disease, we recommend that the Legislature appropriate sufficient funds for the maintenance of the State Board of Health and the State Sanitary Board.

Recognizing the great value of the experiment stations and demonstration farms located in the various agricultural sections of our State, we indorse the work of the stations already established, and recommend that a law be passed authorizing the County Commissioners of each county to provide, at their discretion, for such stations and demonstration farms, in order that the most approved methods of agriculture may be exemplified and new facts may be determined.

We believe it would be advisable for the Congress of the United States to pass a law repealing all laws authorizing the sale of any of the public domain in the United States and its Territories, including the Philippine Islands and other possessions, and in the future only sell the surface for agriculture and stock raising purposes, and forever retain title in the people of the United States of the timber and of all minerals and all coal, oil, gases,

phosphates, water and water-powers, to be worked under control of laws passed by Congress by paying a reasonable royalty to the people for the same.

REPORT FROM UTAH

O. J. SALISBURY

Vice-President Utah State Conservation Commission

The Utah State Conservation Commission was authorized by an Act of the State Legislature approved March 22, 1909. The Act prescribed the powers and duties of the Commission, and appropriated a certain sum annually to be expended for the purposes thereof. Pursuant to the said Act the Governor of the State duly appointed a Commission, consisting of seven members, who organized and began active operations about the first day of October, 1909.

Such legislation was called for and enacted on account of the pressing necessity of devising ways and means of preserving and protecting the abundant, varied, and valuable natural resources of our young and growing State; and it was a source of gratification to this Commission to find that such resources had suffered comparatively little waste in the years past, and that the duties required of the Commission were to ascertain the character and extent of the State's resources, and to work along lines of Conservation and protection rather than those of restoration.

The Commission prepared and issued a preliminary report on the resources of the State late in the year 1909, and 2000 copies were distributed to our State legislators, to Government departments, Conservation associations, public libraries, etc. Owing to the short time in which the Commission had to collect data and prepare the report, it was somewhat limited in its scope and general in its character.

The Commission has now in course of preparation a complete map of the State, showing the National Forests, ownership of public lands (whether Federal or State), character of the soils with analyses thereof, with other information to enable it to make an intelligent and accurate report to the Governor and State Legislature at the coming session in 1911, suggesting

and recommending such legislation as will best conserve and protect the State's natural resources to the benefit and advantage of our citizens of present and future generations.

The amount of the annual appropriation for the purposes of the Commission is \$3,000.00. There was expended during the year 1909 the sum of \$211.55, and during the year 1910 the sum of \$2,767.62.

It is the intention and purpose of the Commission to continue along the lines upon which it has started, to ascertain the extent and character and point out the location of the agricultural, mineral, power, and other natural resources of the State, and to place before the public such information concerning these resources as will enable the home-seeker, the investor, the manufacturer and all those seeking industrial pursuits adapted to our State, to secure for themselves some of the advantages which the development of such resources offers.

SUPPLEMENTARY REPORT FROM UTAH

E. T. MERRITT

Delegate from Utah

The State of Utah has not yet undertaken any great work in the matter of Conservation of public resources, although a Commission has been created with the Governor as chairman. An office is maintained and the gentlemen of the Commission are giving earnest thought and study to the issues involved, feeling that they want to be sure they are right before they go ahead. However, the General Government has been very liberal in the attention it has given us, and we find our phosphate lands, the public coal lands, lands adjoining streams suitable for power sites, and practically every acre of our forest lands have been withdrawn from entry. And yet we feel that we have no quarrel with the Government in these matters. We believe that just as soon as equitable and reasonable methods have been devised for the sale or lease of the first three named they will be placed in such a position to be of practical use and benefit to the people, as they should be; in other words, we do not believe they will be bottled up or pickled or preserved for future generations, but under wise and equitable laws and administration will be converted to the use of the people.

The forest reserves are properly cared for in Utah, and their use and administration is equitable and fair. Mr Pinchot told us when he began his administration that while no doubt mistakes would be made and some inconvenience suffered by the people, yet he wanted it understood that the forests belonged to the people, and that the purpose of the Government was not to exploit them for revenue or for glory or for the fun there was in it, but rather to take care of them for the use and benefit of the people, especially for the people who had conquered and developed the adjoining country; to conserve the water supply, and to perpetuate and care for all the resources and homes of the people. He further told us that whenever we could suggest betterment of the Service in the interest of the people, such suggestions would be gladly welcomed. Such promises have been faithfully carried out, and we believe the Government has been a kind parent to the State of Utah. We see no reason for a quarrel as to the rights of the State and those of the Government. We think there is plenty for both to do, and at least to us there is profit and benefit for us to go hand in hand in cooperation with the Federal Government in the development of our State.

We believe that only by the General Government can the problem of water-power sites, particularly on large or interstate streams, be handled. The history of Utah shows that some years ago the adjudication of water-rights was in the courts of the several Judicial Districts of the State, and that in the course of their procedure it was a common thing for all the water of the stream to be decreed to the several owners residing within that Judicial District, absolutely without regard to the rights of other citizens using water from the same stream, although residing in some other Judicial District. We changed our laws, placing the acquirement and adjudication of water-rights in the State Engineer. We found this a big improvement, but we still find ourselves in the matter of interstate streams entirely at the mercy of the fellow above us. Of course the fellow below can take care of himself. The lesson is obvious. We maintain that only the General Government can properly and rightly hand out justice and equity in the matter of power sites and water-rights as affecting interstate streams.

We have found cooperation with the General Government immensely valuable to us in the matter of experiments in the drainage of water-logged or alkali lands, measurement and recording of the flow of our streams, the

eradication of disease among our livestock, and in fact in every department where cooperation has been tried.

We are suffering today in Utah, as in many other parts of the country, from mistakes and carelessness of the general Government in the handling of the public resources, but this is also true of ourselves in our own administration; and we are very glad to see an awakening on this subject. The people of Utah, in common with all of the people of the whole country, are deeply interested in the subject of Conservation in all its phases, and believe that the great mistakes of the past, both National and in our own State, will not be repeated.

REPORT FROM VERMONT

GEORGE AITKIN

Vermont Conservation Commission

The Commission on the Conservation of the Natural Resources of Vermont has no statutory existence, but was originally appointed by Governor Fletcher D. Proctor in support of the general Conservation movement instituted by the Conference of Governors at Washington in May of 1908. The Commission has been continued by parole of Governor George H. Prouty.

It has recognized and been in absolute sympathy with the principles fundamental to Conservation work, namely, that conservative use and, where practicable, the intelligent maintenance and restoration of natural resources are indispensable to the continued prosperity of State and Nation and of inter-nations; that State boundaries or National boundaries do not confine and limit natural resources; that it has become the sacred duty of State and Nation to take measures for the preservation on the people's account of all the means of their life, welfare, and comfort, including soils, water, minerals, and forests; these to be safeguarded as public utilities to be used and treated in the interests of future as well as of existing generations, and to be stripped of every vestige of monopoly and trust.

Apart from the conservation of these necessary and material things, we have been interested in the advancement also of what is nearly as, if not

more, important, the conservation of health, the retention and improvement of our self-governing opportunities, the equalizing and qualification of educational opportunity, and of every phase of civic, moral and social advance. Vermont is mainly interested directly in the conservation and right use of public health, of its soil, of its forests and woodlots, of its water supplies, of its quarries of granite, marble, and slate, of its game and fish, and in its steadfast attention to educational opportunity and the administration of justice. For the greater part it possesses a very widespread individual ownership and control in all its natural resources and their development and use. It has for decades prior to the so-called Conservation movement supervised and fostered all these economies through legislation; so that it may be said that the State has gradually but definitely applied the principles of Conservation to its affairs and its resources for many years prior to the existing discussion of the subject. This is true in connection with quarrying, agriculture, forestry, and water supplies, though it should be added that Conservation subjects have been much more prominently considered in recent years with increasing advantage to the farmers of the State and also with an increase in manufactures.

Our method of legislation and the machinery of our self-government represent an evolution and are the result of much and intimate public discussion, and they are working out good economic results. Perhaps this may best be indicated by a reference to the legislation passed in 1908. There was enacted a law which abolished the Board of Agriculture, and substituted in its place a Board of Agriculture and Forestry, consisting of the Governor, the director of the State Agricultural Experiment Station, and two citizens known to be interested in the advancement of agriculture and forestry. The disbursement of the appropriation under this Act was left discretionary between agriculture and forestry, and the results in the brief elapsing period since its passage have been very gratifying. In addition to this there were acts sustaining the work of the State Agricultural College, providing for increased support of agricultural fairs, for the acquisition of forest reserves, for the appointment and maintenance of a State Forester, for the more definite supervision of all agricultural interests, and for a more direct inspection of cattle and of dairies. The appropriations of 1908 included increased provision for the conservation of agricultural, forest, and dairy interests, for the care of game, for education and public health, and for

the investigation of the water resources of the State. Special attention was given to amendments of the law which aim to safeguard forests from fire and game from extinction, and to prevent the loss or misuse of water for domestic, power, and transportation purposes. This, however, was not an accident of recent agitation, but more particularly an evolution; and it operates, so far as Vermont is concerned, in a true appreciation, use, and care-taking of its local resources.

There has been special consideration given of late to public health, and laws were enacted governing the inspection of animals, supervising control of contagious and infectious diseases, suppressing adulterations of foods and drugs, advancing the working plans of the State laboratory of hygiene, more closely regulating the practice of medicine and surgery, forcing more specific duties on health officers everywhere, defining the practice of optometry—in short, all the means by which a State government may advance the well-being of its citizens through the application of what has been made known in science touching all these questions.

The State also advanced the well-being of its people by conserving their natural resources, material or acquired, through the creation of a Public Utility Commission, whose work has since demonstrated the need and value of its existence by its influence in behalf of the public of their use and service. We hold here that one of the most effective Conservation measures is that which gives the people the best service at the lowest cost of all the applications of natural resources, as interpreted by science, which nature bestows in the way of power, water, light, and drainage. We wish to state positively, however, that these problems cannot be treated as accidents of public experience, but as subjects of legislation and public treatment which define themselves in their true relationship to property rights and individual rights and to public necessity by the process of evolution.

This is illustrated by the way in which forestry conservation was instituted in Vermont many years back, when a few men of foresight took an interest in the subject, formed a society, and kept bringing attention to the subject until it was made a part of the law and in equal standing with agriculture in this State, and is now apparently an assured State subject of continued standing as much as other subjects of legislation, like education, public health, the preservation of game, and the administration of justice.

The expansion of the granite and marble industries of Vermont has been so great as to give it rank among the foremost producing States of the Union, and in the art and quality of its material and work it is foremost in all respects.

In self-government, as affects all the things which make living conditions naturally satisfactory and profitable, there has been marked increase in the conservation of all the living opportunities afforded by the State; but it is again emphasized that this has been in due course of growth and not the incidental recognition of a possibility. Our people have been conservative, rational, and human in the development of their chance, their natural resources, and their duty in regard to these, and have not required either through neglect or by any lapse of their rights the service of the National Government in this regard, least of all through any material modification of the relationship defining State rights and State duties. There is a greater disposition here to accept direction as concerns the husbandry of our resources from science than from politics, and to insist that the care and supervision of such matters will best conserve our interests and our happiness if left to the judgment, regulation, and control of our own folks.

There has been in the past few years a marked increase of income per acre from cultivated land in Vermont, and a relatively greater income per acre than in the leading agricultural States, due, no doubt, to more intense farming, and there has also been an increase in the output of dairy products, while quarrying and stone-cutting manufactures have multiplied and taken a strong grasp on market opportunity. At the same time the great glory and strong defense of our State, its forests and its woodlots, have been conserved, and planting and scientific cutting have more and more become the rule. The reports from the stone industries indicate a growing demand for the manufactures of the State in granite, marble, and slate. The reports from agriculture indicate an increasing tillage and a larger financial return, an advance in the price of land, and vastly improved living conditions of the farm. The report from all the State commissions charged with the supervision of public health and the real life interests of the people supply increasing evidence of improved water supplies, of municipal lighting and power ownership, of increased transportation facilities, of reduction on accident hazards, and of steady advances in the art of and provision for public instruction.

In forestry, which is one of the greatest natural resources of Vermont apart from its vast contribution to the beauty of the State as a great natural park and game preserve, there has been the most marked advance. The office of State Forester was established in April of 1909, since which date its occupant, Mr A. F. Hawes, has made sixty-three addresses upon the subject in various parts of the State before numerous associations, agricultural societies, and forestry conventions. The State nursery under his direction has become one of the largest in the United States, today containing over 3,000,000 trees, and there have been sold within the past year—a remarkable exhibit for a State of our size—750,000 trees, distributed through every county in our State. Private timber holdings have been examined, detailed advices for handling many forests have been furnished, and in many instances trees have been marked for cutting by State advice on private lands. Besides this, there have been established two State forests of 800 acres which will be treated as subsidiary reservation nurseries to the one established at Burlington.

Attention has been and is being given to all details relating to the promotion of agriculture, forestry, dairying, minerals, and water powers, so that it is possible to advise you that Vermont is wholly alive to all natural, moral, educational, industrial, civic, and political propositions as they stand related to the Conservation of everything that will best promote the well-being and happiness of its people.

REPORT FROM WASHINGTON

E. G. GRIGGS

Chairman Washington Delegation

On behalf of the Washington delegation, of which I have the honor of being Chairman, I desire to congratulate this Congress and every delegate on the opportunity afforded us in hearing that grand interpretation of Conservation so ably presented by President Taft. It will live as an epic, and should be translated throughout the land.

Since that opening day I have been thrilled and electrified by this theme of Conservation, which is but another name for Patriotism, the husbanding of the Nation's resources.

The country is stirred by that same feeling which I sometimes think aroused our Fathers before the Civil War. Let us profit by the great forward steps they made in the determination of State and Federal rights. To us it has fallen to solve these patriotic, philanthropic, and commercial questions of the day.

I deplore the interjection of demagoguery and personal political advancement. I believe there is a sane, safe and sound Conservation that we can all practice. Above all things, let us eschew politics and throw a little more of that unselfish, self-sacrificing effort into this great fight for the Nation that characterizes our friend and collaborator, Gifford Pinchot.

We should leave this Congress united in this one idea at least, that we will stop the Nation's waste and encourage its development, so far as it lies within our power.

Eighteen years ago I left the State of Minnesota and this delightful city which was my home, to do my share in the development of the Pacific Slope—"I love its rocks and rills, its woods and templed hills." Wild horses could not drag me back to Minnesota, where fifty years ago my father pioneered, and is yet interested—not that I love Minnesota less, but only that I love Washington more. You have grown and developed great cities. Do not forget to let us do likewise.

We no longer say, with Greeley, "Go West;" we say, Come West. Under the classic shades of our noble forests and within easy access of the snow-capped peak of Mount Tacoma—that mother of water-powers and protector of forests—we are solving *our* pioneer problems, and we are not lagging behind in the race.

Our citizenship is of the highest type and from all of your States, for it is composed of that progressive element that first made your own cities famous—and did not back out of big problems. We are no longer savages devastating the frontier and Uncle Sam's patrimony. He is no longer "rich enough to give us all a farm;" but we are citizens alive to the big problems of the day—and we are the virgin State in which Conservation and common sense can be practiced before it is too late. I predict for the State of Washington—with wise Federal and State legislation—a shining example of what horse-sense and Conservation will bring about.

If we sell our common lumber at the mills on Puget Sound for \$8 to \$10 a thousand, which is two to three dollars less than we got 15 years ago, and have to pay \$600 to \$700 for a team of horses in Minnesota today that 15 years ago we could buy for \$200 to \$300, is it any wonder that we lumbermen of the West are interested in Conservation?

Rich beyond measure in timber, coal, fish, mines, and agricultural lands, the great State of Washington is with you and your commissions that must finally work out and crystallize wise and patriotic legislation. Let us Nationally inventory our stocks and resources, unify and codify our laws affecting taxation and irrigation, liability and responsibility—develop our interstate commerce, and promote the general welfare.

REPORT FROM WEST VIRGINIA

HU MAXWELL

Chairman State Conservation Commission

Near the close of 1908 Honorable W. M. O. Dawson, then Governor of West Virginia, appointed a commission of three members, Neil Robinson, James H. Stewart, and Hu Maxwell, to prepare a report for the guidance of the Legislature in framing laws for the Conservation of the State's resources. The report was ready for the Legislature when it convened in January, 1909. It recommended a number of changes in existing laws, and the enactment of several new ones. Its principal recommendations were as follows:

1—A forest law providing for the prevention and suppression of fire, and for the care of woodlands and watercourses. A draft of the proposed law was included in the report.

2—A law to lessen the waste of natural gas, by requiring the plugging of wells when not in use, and saving the gas from others instead of permitting it to blow into the air. It was urged that effort be made to check the leak from gas mains.

3—For the purpose of checking the tremendous loss of by-products in coke making a law was recommended, to take effect five years from its passage,

prohibiting the erection of any other than by-product ovens, but placing no restrictions on any ovens then in use, so long as they might last.

4—The State was urged to cooperate with the Federal Government in all reasonable ways for the improvement of navigable rivers in the State, and in the protection of mountain forests and the building of storage reservoirs to check the rush of floods and improve low-water conditions.

5—The establishment of an engineering school was recommended for the special purpose of educating men to develop and conserve the State's resources. It was pointed out that much of the practical work of Conservation does not depend so much on the enactment of laws as on the training of men to do the work. In this connection it was shown that vast quantities of low grade coal, which is now unmarketable, is thrown away or left in the mines, though it would be sufficient, if manufactured into producer gas, to furnish power to drive much of the machinery in the State and in surrounding regions. If the State's water-power were fully developed it would be sufficient to turn every wheel in the State, but this development cannot be brought about by laws alone; it must depend largely on trained men.

6—Better game and fish laws were recommended to take the place of the old laws which had failed to produce the desired results.

7—It was urged that prompt investigation be made of the question of municipal water supply in the State with the view to the prevention of pollution of the running streams.

8—It appearing probable that certain valleys in West Virginia would respond in a satisfactory way to irrigation, it was recommended that experiments be carried out to test the matter.

9—The State's natural scenery is such that it might be made a valuable asset, in connection with the protection of forests and streams, and the Commission recommended that the fact be borne in mind in laying out new roads, so that full advantage be taken of all scenic possibilities.

10—An immigration agency was recommended for the purpose of bringing into the State desirable immigrants who will cultivate the farms which suffer from neglect in many parts of the State.

11—Changes in road laws were urged which would make possible the building of permanent, durable, desirable highways in place of the gullies and precipitous paths which in many parts have been tolerated as roads from the earliest settlement of the region down to the present.

12—The purchase of land by the State in each of the congressional districts was recommended for farms to serve as models and object lessons for the surrounding farmers; their management to be in the hands of trained agriculturists.

The Legislature which convened in January, 1909, considered one or two of the recommendations of the Commission. A forest and game law was enacted, though it was not the measure which the Commission recommended. The law, however, is a good one so far as it goes, and if its provisions shall be carried out, much good may be expected.

No steps were taken by the Legislature to lessen the waste of natural gas or to save the by-products in coke making. A new highway law was enacted, and a State commission was appointed to study the road problem.

REPORT FROM WISCONSIN

E. M. GRIFFITH
State Forester

Governor James O. Davison appointed the Wisconsin State Conservation Commission July 24, 1908. The seven members appointed were men whose positions gave them a considerable knowledge as to the natural resources of the State, and the Governor gave the Commission full authority to call upon any State department for detailed information.

During the summer of 1908 the Commission held several meetings in the Capitol, and reports were prepared on the three most important and pressing Conservation problems in Wisconsin, viz: water-powers, forests, and soils. A full report covering these three subjects was then made to the Governor,

and this the Governor transmitted to the Legislature in February, 1909. The Commission made the following recommendations:

WATER-POWERS. 1—That franchises for water-powers be granted under a general statute.

2—That the issuing of such franchises be placed in the hands of the railroad rate commission, or similar board, under conditions to be provided by a general statute.

3—That such franchises be in the nature of leases for a long term of years. Such leases should be renewable on equitable terms. Rentals should be low, and should be applied to the extension of the State forest reserve.

4—That a reasonable Conservation charge be levied on all developed water-powers on rivers of which the headwaters are protected by forest reserve lands, the income from such charge to be applied to the extension of the State forest reserve.

5—That the survey of the water-powers of the State be completed in cooperation with the United States Geological Survey.

FORESTS. 1—The State Conservation Commission regard it of the utmost importance that the State forest reserve, located about the headwaters of the more important streams of the State, be greatly extended. At the present time the opportunities to make such extensions are much more favorable than they will be in the future, and therefore the Commission recommend that immediate action be taken to secure such extensions.

2—The State Conservation Commission recommend to the Governor that, in view of the large increase in area of the forest reserves since the last session of the Legislature and the probability that in the future such holdings will be materially added to, the annual appropriation of the State board of forestry for administrative purposes should be largely increased.

3—The State Conservation Commission also approved the following principles as adopted at the Lake States Forestry Conference, held at Madison, December 10, 1908:

"Resolved, That forest fires being one of the greatest enemies of the State, and thus akin to riot and invasion, the Executive power of the State should be employed to the utmost limit in emergencies in their suppression and control for the protection of the lives and property of the people.

"Resolved, That we advocate the patrol system as the only satisfactory method of preventing forest fires, and the commanding factor in fighting them.

"Resolved, That we recommend the retention of the fire warden system with the county, rather than the town, as the unit, as being essential in securing interest and responsibility among the people most affected.

"Resolved, That in all districts covered by State fire patrol a reasonable portion of the expense for such patrol should be placed upon the unoccupied, unimproved, or wild lands, whether forest or cut-over land, preferably in the form of an acreage tax.

"Resolved, That the expense of the local fire warden service, and the help called out for the suppression of fires, should be borne wholly or in part by the county or town, but the payment should first be made by the State to insure promptness.

"Resolved, That all officials, including public prosecutors, charged with the enforcement of fire-protective measures, should be subject to severe penalty or removal from office for non-performance of duty.

"Resolved, That the successful prosecution and a commensurate punishment in case of conviction often cannot be secured in the locality where the offense has been committed, and in order that the law shall be enforced, in the interest of justice, and under authority of the attorney general, a change of venue should be permitted.

"Resolved, That it is the sense of this meeting that lands containing forests should be taxed in the usual manner so far as the land is concerned, said land to be assessed as if it contained no timber; but the forest products should be assessed and taxed only when they are cut and removed, and then in an appropriate manner; that the harvest timber tax should be based on a stumpage value determined by the value of the forest product at the place where it is assessed, less the cost of placing it there."

SOILS. The State Conservation Commission recommend to the Governor that a soil survey of the State be undertaken and carried on at such a rate as will give a general view of the soils of the State in about five years. The Commission call especial attention to the immediate need of such a survey in the central and northern parts of the State, the soils of which are now coming rapidly into agricultural use; and also to its necessity on lands which may be included in a forest reserve and which should be devoted to forestry or agriculture according to the nature of their soil.

Let us see what were the results of these recommendations. A number of bills were introduced in the Legislature of 1909, seeking franchises to dam navigable streams and to create reservoirs and reservoir systems; but acting upon the recommendations of the Conservation Commission, all such bills were referred to a special committee of the Legislature on "Water-powers, Forestry, and Drainage" which has carefully investigated the development of the water-powers of the State and will report either to a special session of the Legislature or to the regular session in 1911. Undoubtedly the issuing of such franchises will be placed in the hands of a competent board or commission. All forestry bills introduced in 1909 were referred to the same special committee of the Legislature. Two members of this committee have made their report, and include the following recommendations in regard to the forestry work of the State:

1—An act to provide a State tax of two-tenths of one mill for each dollar of the assessed valuation of the taxable property in the State, to be collected annually for a period of twenty years, the tax when levied and collected to constitute "a forestry investment fund" to be used for the purchase, improvement, and protection of the forest reserve lands.

2—An act to provide for the piling and burning of white Norway and jack pine slash.

3—An act to provide for the employment of an efficient fire patrol by the State board of forestry.

In accordance with the recommendations of the Conservation Commission, the Legislature in 1909 passed an Act providing for a soil survey of the State, and this work is being done by the Geological Survey and College of Agriculture, for the purpose of ascertaining the character and fertility of the developed and undeveloped soils of the State, the extent and practicability of drainage of the swamp and wet lands of the State, and the means for properly conserving and increasing the fertility of the soil of the State.

It will be seen from the above that the work of the State Conservation Commission has already shown important results, and it is believed that the Legislature and people of Wisconsin have now begun to realize clearly the urgent need and also the means which should be taken to conserve the great natural resources.

REPORT OF THE AMERICAN ACADEMY OF POLITICAL AND SOCIAL SCIENCE

The Conservation of natural resources is a subject in which an American academy of political and social science must necessarily have a keen interest. The primary purpose of the American Academy being to assist in the right solution of the political and economic problems confronting the people of the United States, it has actively cooperated with those individuals and organizations that have done most to give impetus to the Conservation movement.

At the White House Conference called by President Roosevelt in May, 1908, the American Academy was one of the National organizations represented. The following November, the Academy devoted one of its regular scientific sessions to Conservation, the chief address of the session being delivered by Mr Gifford Pinchot, the Chairman of the National Conservation Commission. The Academy was also represented at the Conference which met in Washington in December, 1908, upon the invitation of the National Conservation Commission.

The most valuable aid the American Academy has given the Conservation movement was rendered by the publication, in May, 1909, of a comprehensive volume containing eighteen papers especially prepared by

men prominent in the Conservation movement. The scope and character of this volume are indicated by the following list of papers and contributors:

Forestry on Private Lands—Honorable Gifford Pinchot, U. S. Forester, and Chairman National Conservation Commission.

Public Regulation of Private Forests—Professor Henry Solon Graves, Director Forest School, Yale University.

Can the States Regulate Private Forests?—F. C. Zacharie, Esq., of the Louisiana Bar, New Orleans.

Water as a Resource—W J McGee, LL.D., U. S. Inland Waterways Commission; Member National Conservation Commission.

Water Power in the United States—M. O. Leighton, Chief Hydrographer, U. S. Geological Survey.

The Scope of State and Federal Legislation Concerning the Use of Waters—Charles Edward Wright, Assistant Attorney to the Secretary of the Interior.

The Necessity for State or Federal Regulation of Water-power Development—Charles Whiting Baker, C. E., Editor-in-Chief Engineering News, New York.

Federal Control of Water Power in Switzerland—Treadwell Cleveland, Jr., U. S. Forest Service.

Classification of Public Lands—George W. Woodruff, Assistant Attorney-General for the Department of the Interior.

A Summary of our Most Important Land Laws—Honorable Knute Nelson, U. S. Senator from Minnesota; Chairman of the Senate Committee on Public Lands, and Chairman of Committee on Lands, National Conservation Commission.

Indian Lands: Their Administration with Reference to Present and Future Use—Honorable Francis E. Leupp, Commissioner of Indian Affairs.

The Conservation and Preservation of Soil Fertility—Cyril G. Hopkins, Chief in Agronomy and Chemistry, University of Illinois Agricultural Experiment Station, Urbana.

Farm Tenure in the United States—Henry Gannett, Geographer U. S. Geological Survey.

What may be Accomplished by Reclamation—Honorable Frederick H. Newell, Director U. S. Reclamation Service.

The Legal Problems of Reclamation of Lands by Means of Irrigation—Morris Bien, Supervising Engineer, U. S. Reclamation Service.

Our Mineral Resources—Honorable George Otis Smith, Director U. S. Geological Survey.

The Production and Waste of Mineral Resources and their Bearing on Conservation—J. A. Holmes, Chief, Technologic Branch U. S. Geological Survey; Member National Conservation Commission.

Preservation of the Phosphates and the Conservation of the Soil—Charles Richard Van Hise, President of the University of Wisconsin.

There were 5500 copies of this volume published, and its wide distribution at a most opportune time caused it to have an exceptionally effective influence. By the end of 1909 the edition was practically exhausted, and a new edition became necessary. The Canadian members of the American Academy, it is interesting to note, were particularly pleased to receive this publication.

It is the belief of those most active in the work of the American Academy that the question of the Conservation of American resources outranks all other economic questions now before the people of the United States. It is especially important that National and local organizations should cooperate as fully as possible in educating the public as to the present condition of our resources, the manner in which they are being used, and the measures that should be taken to make these resources of permanent as well as of present value to the American people.

Respectfully submitted,
[Signed] EMORY R. JOHNSON,
Chairman
FREDERICK C. STEVENS
WM. B. DEAN
W. A. FLEMING JONES
WM. L. WEST
CHARLES W. AMES
Committee

REPORT OF THE AMERICAN AUTOMOBILE ASSOCIATION

When the American Automobile Association was originally honored with an invitation to the National Conservation Congress it promptly accepted with two objects in view; *first*, to influence, if possible, the advocacy of a good highway construction and maintenance policy throughout the United States—National, State, and local—in its program in order to broaden and help the movement itself, and *second*, to enlist the friends of Conservation in advancing highway construction; in other words, to make the theory of Conservation cover not only the care and perpetuation of natural resources, but all broad economic activities, throughout the length and breadth of the country, concerning the care and betterment of property, whether natural or artificial. The resident in the East must feel that only by bringing within the scope of the Conservation movement these somewhat narrower and more artificial economic measures can any wide and deeply interested following be secured in the more thickly settled eastern States, as most questions of bulk ownership and management of natural property in this section have long since been settled in law and in fact. If you adopt this theory and definition of Conservation, and thereupon, among other efforts, give your help to advance the matter of good roads, then the advocates of good roads all over the country will have gained an ally, and you will have secured new friends.

The American Automobile Association is devoting the major part of its time, means, and enthusiasm to advancing and coordinating the activity of good highway construction and maintenance, and to the preparation and

enactment of good National, State, and local legislation regulating traffic on these highways all over the country. The Association is organized in the large majority of all our States, with a large local following in every center, and with an effective central management cooperating with the most important like bodies abroad and with such associations at home as the U. S. Office of Good Roads; National Grange, Patrons of Husbandry; Farmers' Educational and Cooperative Union; and League of American Wheelmen. It consists of State organizations in most of the States, comprising approximately 250 local clubs and over 30,000 members. It is an active force engaged in useful educational and constructive work to better our National life by improving in an intelligent and public spirited manner a very important branch of transportation. It is and has been for some years the leading spirit in this work, as witness the organization of the National Good Roads Convention with the above-mentioned cooperating associations to be held in Saint Louis toward the end of this month.

Transportation, broadly considered, has been the greatest ruling economic force in every civilization created by man. Its absence or limitation ever makes for barbarism or the decadence of the people so confined. It is the pioneer and prime moving force in the creation of progress and enlightenment. Each stage of the world's history that has witnessed some pronounced advance in transportation methods has been swiftly followed by a more than proportionate advance in progress, in wealth, and in happiness of the people affected. Witness the march of wealth and education following the practical operation of the steam railway in the later half of the last century, and the further advance following the practical perfection of electrical transportation during the last quarter of the same century. Steam has provided transportation for the great bulk of world life; electricity opened the way for relatively lighter and cheaper transport, thus opening sections otherwise not accessible for economic reasons. The motor-car and the public highway have crowned these achievements by providing a means for speedy, cheap, safe, and agreeable transport to any corner of the country, the qualities just described constituting the essence of what is best in transportation.

The public highways in the country, however, which premise the reasonable use of motor transportation, have not advanced either in quality or quantity with the means of transport itself during the past fifteen years. The very

existence of steam transport when this country was young and sparsely settled and poor and badly developed, and even of electrical transport at a later day, had in themselves limited the development of a reasonable highway system, when comparison is made with other older countries of like wealth, population, and civilization. In earlier days military necessity did not compel this Government to build National highways for the movement of troops—the railroads did that. Economy of transport did not compel the several States to build highways—the railway, the steamboat, the electric tram cared for that. It was not until the advent of the practical modern motor-car that the almost savage condition of this country with respect to highways became apparent. Since then, say within the past ten years, the force moving all over the country toward reasonable highway development, maintenance, and regulation (which had its great inspiration in the army of motor-car tourists acquiring a knowledge of the geography and the beauties of this country by a new and independent method of travel, and which has more recently turned into a flood of growing purpose and organization for better highways because of the conviction of the farmer and the business man of the United States of their economic value in reducing the cost of ton-mile detail haulage to the lines of bulk transportation), as well as toward the moral uplift of the entire farming and country life, due to releasing the country resident from the unhealthy isolation of former times—this force must now be recognized and satisfied, and this Conservation Congress is a logical forum for exploiting and advancing these aspirations.

A recent phase of this great new interest and industry has been the abuse heaped upon it by certain special interests that have been touched by the change the motor-car has wrought over the country. The Reverend Sam Small once remarked that if you threw a brick in the dark and heard a dog howl you knew that you had hit him. The misrepresentation and denunciation and apparent lack of understanding of the true meaning of this new interest seems to come near those financial and bulk transportation interests—with their affected fear of largely mythological mortgages—from which the motor-car user in the aggregate has detached some profit either in transport or in investment. It needs no fine intelligence in these times to understand the weight and purpose of this opposition which has assumed an almost proscriptive right to the collection and handling of the loose money

of the unorganized individual all over the country. What is this doctrine that the banker has become the censor of the individual's needs and actions with his own money? Have the farmer and the business man of this country recently become so poor or reckless or so much in debt as to apologize to their fiscal agents for the purchase of a motor-car with their own money or lose credit? Does this not logically lead to an equal apology and loss of credit for owning a decent home instead of a miserable one, or wearing good clothing, or eating good food, or getting a good education, or buying a carpet, a piano, or any of the other things which in the sum constitute the high environment of American life? The tens of thousands of users of motor-cars that are today deriving health and pleasure and, in a far greater number of cases than generally known, profit from the purchase and use of motor-cars, are deflecting interest and capital from channels which have long enjoyed them to their great benefit. That is the origin of the detraction of the motor-car industry and the individuals who created it and who are enjoying it today.

Fair and intelligent consideration is not generally given to the fact that speedier transportation wherever possible is inevitable in human history; that, when a farmer or a doctor or a real estate agent, or a business man of any sort, finds that, at the same cost, he can do, with the same personal effort per day, four times more work in a motor-car than with a pair of horses, provided decent roads exist—when this fundamental economic fact reaches the masses, then good roads teeming with motor-cars and trucks and reasonable universal legislation will be demanded and gotten. When added to this, the same investment provides the means of winging off where fancy leads on a healthful and charming tour or visit, who shall deny that the individual is wise to avail himself of this new facility?

Finally, sufficient weight is not given to the fact that every ton of freight in this broad country must be carried from its primal source, not once but several times, to a railroad or steamboat or tram, before it reaches the goal of the final user. The perfected motor-wagon and truck made in quantity at reasonable cost, provided the good highway exists everywhere, is the inevitable source of such reasonable transport: and, from the standpoint of utility, or effectiveness, or congestion of street areas, or speed—from any standpoint whatsoever—it is as distinct an advance over animal traction as was the electric tram thirty years ago over animal traction in that field of

enterprise. The millions of dollars going into this industry spread out through the people, irrigating the total prosperity of the country through its appropriate channels, just as money spent on everything else the individual buys throughout the country, adds its appropriate quota to our National prosperity, and should be quite as immune from attack and misrepresentation.

Good highways and highway legislation are today a generally recognized National necessity. If this country were now through concerted action, Nationally, in States, in counties, and in cities, to spend enough money to put its streets and highways in a comparable condition with those of England or France, and to replace the great percentage of animal traction and motor-cars as now made, to carry the bulk of detail tonnage on these highways, it could not in any other manner or with any better advantage to the coming generation, as concerns its wealth, happiness, and profit, invest this enormous sum or, in any other manner, not only add to the value of country property but influence so positively and so speedily an increase in the happiness and general content of country life in the United States.

In conclusion, it is respectfully urged that the project of good highways and reasonable uniform State and National legislation governing their use should be incorporated in detail in the program of this National Conservation Congress and every kindred association throughout the length and breadth of the land.

Respectfully submitted,
[Signed] POWELL EVANS
*Chairman, A. A. A. Conservation
Committee*

REPORT OF THE AMERICAN CIVIC ASSOCIATION

I have already had the honor of presenting some statement of Rhode Island's interest in the Conservation movement, and of the ways in which she proposes to demonstrate it. But I also bear messages from the American Civic Association and other organizations. Perhaps one might think, on first consideration, that there was nothing very closely related, or perhaps related at all, in the purposes of the Conservation Commission of the State of

Rhode Island and those of the American Civic Association, the Providence Board of Trade, the Metropolitan Park Commission of Providence Plantations, the Atlantic Deeper Waterways Association, and the Rhode Island Chapter of the American Institute of Architects; yet I bring you greetings from all of these. I want to tell you that they are all working with all the enthusiasm there is in them for some phase or other of the mighty movement for Conservation.

Some people have said—half contemptuously perhaps (I am afraid so)—that Conservation is made to cover about every kind of a movement there is on this great footstool, but perhaps the statement is about true so far as these movements are concerned with the preservation and development of any of the great assets of nature or artificial achievements of man that are necessary or useful to the well-being of our own or future generations. Whether we are considering the forests upon the mountain sides that control the floods and affect the farms and the water-powers and the navigable streams below, or are thinking how to plan and lay out and construct our towns and cities so that they shall most worthily and efficiently fulfill their two great purposes as places (1) to live happily in and (2) to work most successfully in, we find their principles overlapping and leading from one end of the line clear to the other. You cannot separate them, and it is not worth while to try.

The interests of the American Civic Association, of course, are not restricted to any State or section. Its activities are Nation-wide. "For a Better and More Beautiful America" is its motto, and it believes that a more beautiful America is bound to be a better and more prosperous America. It believes also that the Conservation of beauty means the Conservation of patriotism; and its distinguished president has paraphrased a well-known utterance of Ex-Mayor McClellan to the effect that "The country healthy, the country wealthy, and the country wise, may excite satisfaction, complaisance, and pride: but it is the country beautiful that compels and retains the love of its citizens." It is the love of country that lights and keeps glowing the holy fire of patriotism, and this love is excited primarily by the beauty of the country and the environments of the citizens.

The American Institute of Architects believes that when a thing is most usefully done it is most beautifully done. It believes that Conservation deals

with two great departments closely related in human endeavor, and that you cannot divorce the necessity of city planning from the development of the resources of nations. A properly planned structure, whether it be of a single building or of a whole city, with all its homes and shops and streets, means the Conservation of the people's efficiency through all the generations that shall ever come to dwell therein. Similarly, the park movement, as we see it scientifically promoted, is almost wholly a measure of Conservation. It is not, as the previous generation believed, primarily to tack on ornate luxuries to the urban fabric, but to preserve the necessary recreation places that would otherwise be obliterated, but without which the race of city-bred dwellers cannot survive. It is to safeguard human efficiency and happiness.

The Atlantic Deeper Waterways Association, whose president, Honorable J. Hampton Moore, has bidden me extend his greetings, calls for things that mean much Conservation of effort. Its project would remove much of the material burden of unnecessary cost. There is Conservation of vast energy and the saving of huge National burdens in the present eastern ambition for the fuller improvement of harbors and development of connecting inland waterways. Let me tell you how the improvement of the harbors related to the handling of at least 80 percent of the \$1,500,000,000 worth of all our imports, for this is the proportion that comes into the eastern harbors of the Nation. It relates to the transportation of products of the eastern States worth over \$14,000,000,000 a year—of 85 percent of all the cotton that the Nation raises, and 58 percent of all our manufactures; to the 765,000,000 tons of merchandise that has to be transported through these States in which more than 50 percent of all our people dwell, and then transferred in various ways for the equal benefit of the other 50 percent. No item in the cost of our existence is of more importance than that of transportation.

Well, of course, the Board of Trade is interested in all these things, though it looks upon them primarily as they bear upon the up-building of a city. It believes that it is working to assist the logical development of a city of glorious possibilities where certain services to the Nation may best be performed. If there were not sound economic reasons for the up-building of a great city at any given place, it would be foolish and wicked to attempt by artificial means to talk it into being, or try to force it by the hothouse method of overheated air. But if you have the necessary natural assets and

opportunities that but await intelligent handling, why here comes the need of Conservation as a vital obligation.

[Signed] HENRY A. BARKER *Delegate*

REPORT OF THE AMERICAN FORESTRY ASSOCIATION

No organization can more appropriately than the American Forestry Association make its statement and its appeal to this Congress; for it is the first of our Conservation organizations. It has a past of nearly thirty years to which it can point with pride of real achievement; an active and efficient, though not a noisy, present; and a future of ever enlarging opportunity.

In a very real sense we may say that the work of this Association, through years of much misunderstood effort, under the able guidance of the great leaders of the American forestry movement, made this Congress possible; for it was through the study of forestry and its relation to the country that the whole problem of our National resources came to be understood. The man who has given the Conservation of natural resources its impetus, with the help of his distinguished chief, then President of the United States, was the recognized leader, the apostle and evangelist, of the forestry movement; and today no portion of our natural resources holds a more important place than the forests. They are inseparably linked with soils and waters, both of which depend on them in great measure; and as a product of the soil, nothing exceeds the forests in value and in necessity to human welfare. Forests, like agricultural crops, belong to the renewable class of products, and their maintenance involves much more complicated and permanent problems than the non-renewable products like metals, coal, oil, and gas. Therefore we conceive the field of our Association to be vital and lasting, and so broad, many-sided, and far-reaching as to amply justify the existence of an organization dedicated to the advancement of scientific forestry for the best utilization of our forest lands for all time.

Our appeal is to the citizen who desires to promote the economic and moral welfare of the Nation, for moral welfare comes only through good economics and such management of natural resources as makes for prosperity; to the lumbermen and to all manufacturers who use forest products, for to them this is a subject that touches the permanence of their

industries; to the educator who looks beyond mere culture and believes that our education must more and more fit men and women to cope with the complex problems of modern life. In this last connection we shall soon announce plans, recently set on foot, for giving practical and definite assistance to those teachers who wish to bring the fundamental principles of forestry into their work, but who do not know how. We shall try to show them how in a systematic and practical way.

Our work is independent of that of the Government, but is conducted in close touch with it. As an independent body of citizens we can do and say what Government officials cannot do and say. Our program embodies: (1) An equitable system of taxation which shall not unduly burden the growing crop; (2) adequate protection against fire, which will reduce this greatest of forest perils to a minimum; (3) the practice of scientific management upon all existing forests; (4) the planting of all unoccupied lands which can be utilized more profitably for forestry than for any other purpose; and, (5) the whole to be brought about through harmonious adjustment of functions between the three classes of owners—National, State, and private. We do not believe that either one of these agencies is to be relied on alone. Each has its place. I say this because our position in this regard is often misconceived. I may add (to correct another misapprehension) that we do not believe in putting under forest land more valuable for agriculture. Forestry and agriculture are not rivals. They go hand in hand.

One specific object to which we have given much effort for several years is the establishment of National Forests on the great interstate water-sheds of the Northern and Southern Appalachians. The conditions, which are acute for the thickly populated East, can only be handled by the united action of the National and State governments and private owners. The central cores of the White Mountains and the Southern Appalachians clearly require National care and management. With this and cooperation of the States and private owners with the National Government, we can save a rare country of beauty, health, and productiveness from being made a depopulated waste. We begin to see the light. In the House of the last two Congresses we have passed a bill, after fighting to a finish the reactionary element which has controlled that body and throttled legislation framed in the public interest. In the Senate we have a strong working majority which can only be beaten, as in the Sixtieth and Sixty-first Congresses, by filibustering in the last

hours of the session. If we are not cheated of our reward next winter we shall mark a new step in the progress of American forestry by making the National Forest system really National.

The Association now has about 6600 members; it maintains an office in Washington, where a close watch is kept upon National legislation, and through its correspondents, upon State legislation. It provides lectures, issues bulletins on important subjects, conducts a correspondence bureau, and publishes a monthly magazine, *American Forestry*, which is contributed to by the best authorities in the country, and is the only popular magazine of its class of National scope. We enjoy the cordial cooperation of the U. S. Forest Service and of the various State forest bureaus.

We look forward confidently to a future in which the practice of scientific forestry will become general throughout the United States, when our forest lands will be clearly defined and permanently maintained in productive growth, when waste lands will cease to play so large a part in our National statistics, when the production of the forests will cease to be so much less than the consumption of forest products, and when the National wealth will be contributed to largely each year from this source. But even with this hopeful outlook we cannot see that our work will ever be done, and we welcome the assistance which this Conservation Congress can give us.

[Signed] EDWIN A. START
Executive Secretary

REPORT OF THE AMERICAN HUMANE ASSOCIATION

The Committee on Conservation of National Animal Resources (the same being a sub-committee of the National Conservation Commission of the Federal Government) have the honor to report as follows:

The animal resources of the United States constitute a large proportion of its natural productive energy. This country has hundreds of millions of dollars invested in horses, mules, cattle, hogs, sheep, and chickens. These constitute natural resources which are producing a larger percentage of wealth and a larger proportionate return for capital invested than almost any one other resource. Furthermore, the actual means of sustaining life is more

dependent on these resources than on all others combined, for aside from the food value of the cattle, hogs, sheep, and chickens, and also aside from the other products which are received from them, agricultural operations would be rendered largely inoperative if the assistance of the larger animals were withdrawn. In this way the products of the soil upon which man is so largely dependent for sustenance would be materially affected, and without the assistance of these animals the supply would diminish to the extent of actual starvation for vast numbers of the world's populace. Even if mechanical contrivances should replace the labor of beasts, the cost would be enormously increased; and the natural fertilizing products being removed, the productive value of the soil would also be progressively decreased.

From whatever point we look at this important question, the value of our animal resources is so great and so fundamental that the Nation may well give its best energies and most discriminating intelligence to their protection and conservation. It has been estimated that through the humane treatment and care of horses the average life of these useful creatures can be easily increased from 20 to 25 percent. This likewise means a proportionate increase in the results derived from their labor, which in the aggregate would amount to hundreds of millions of dollars a year. The same is also largely true of the increased value of other domestic animals as the result of humane and considerate treatment, which in all instances would greatly prolong their lives.

The American Humane Association has been greatly interested in promoting the more merciful treatment of range stock, which in the past have been largely left to shift for themselves during the cold, bleak winters of the Northwestern ranges. This has resulted in the death of vast numbers of livestock. A recent report of the Department of Agriculture indicates that over 1,000,000 domestic animals die in the United States each year from hunger and exposure.

Another department in which the humanitarian societies of the United States have been largely interested which bears directly on the conservation of a great natural resource, has been the protection of the fur seals. These interesting and valuable animals, through piratical efforts employed in their destruction, have become partially exterminated, and a great source of

National wealth has been almost annihilated. From vast herds, numbering a great many hundreds of thousands, the seals have been reduced until their rookeries in the islands of the northern Pacific belonging to the United States have been almost depopulated. Friends of the Conservation policy have earnestly protested in Congress against this inhumane and economically unwise course, and during the last session legislation was passed and signed by President Taft, which would insure the ample protection of the seals. Grave fears are expressed at the present time lest this result should be endangered by unwise administrative measures which are threatened. I earnestly hope that the second National Conservation Congress will speak in no uncertain terms in regard to this important question, so that the seals may be restored once more to their original numbers and productive value.

This Committee will not undertake to present all the activities in which we have been interested which bear upon this subject, but content ourselves with showing the great importance of this particular phase of Conservation. We trust that this Committee will continue for another year, and that the results of this Congress will be felt in every portion of the United States.

Respectfully submitted,
[Signed] WILLIAM O. STILLMAN,
Chairman
M. RICHARD MUCKLE
ALFRED WAGSTAFF
JOHN PARTRIDGE
SAMUEL WEIS
JOHN L. SHORTALL
GUY RICHARDSON
Committee

REPORT OF THE AMERICAN INSTITUTE OF ARCHITECTS

The Committee of the American Institute of Architects on the Conservation of Natural Resources has the honor to report as follows:

A wide and increasingly active interest in the subject exists among the officers and members of the Institute. The Committee believes that few, if any, of the great National organizations touch the subject of Conservation at so many points, or are more vitally interested in its wise and efficient progress, or can be more directly helpful in the application of the principles of Conservation in a great series of important industries.

The construction of modern buildings, either for residential or business purposes, involves the use in one form or another of practically the entire list of materials included under the general meaning of the term the "natural resources" of the country, excepting only agricultural land and foodstuffs; and in common with all other thinking citizens, the architects realize that the continued prosperity of the building interests is in the long run dependent on the wise use of these resources. Exact statistics of the great building industry of the country are not obtainable; but a somewhat extended inquiry recently made led to an approximate estimate of the amount of money expended upon buildings in the United States per annum at an average of not less than \$1,000,000,000, practically all of which passes under the hands of the architects in the specifications of materials to be used and in certification as to quality and cost.

Among the materials used are metals, including iron and its various products in rolled steel, sheet metal, pipe, castings, and machinery, with copper, lead, graphite, zinc, nickel, silver, and even gold; lumber in enormous quantities and of all kinds; clay products, such as brick, terra cotta, roofing tiles, drain tiles, floor tiles, and porcelain; stone, including granite, marble, limestone, sandstone, and other quarry products; cement, lime, sand, glass, oils, gums, hemp, bitumen, asphalt, asbestos, barytes, and many other minerals; woven cotton, linen, wool, and other fibres. There are also used coal and water-power, and above all that greatest of all resources of the Nation, the labor of Man, both skilled and unskilled. This but briefly suggests the variety and extent of the interests represented in modern building. Therefore the profession of architecture, represented by the American Institute of Architects, has a most real interest in this great topic, and can and does wield a very potent influence upon the use of the products of mine, quarry, factory, and field.

It has been stated, with a large measure of truth, that if the architects will study the economic use of lumber and specify or permit the use of short lengths (such as 2-foot and 4-foot lengths as against 12-foot and 14-foot lengths) where such are structurally permissible, that a quarter of the lumber cut per annum could be saved without lessening the amount of lumber used in building. If the architects specify concrete to the exclusion of steel, the steel market is affected; if brick or clay products, the cement market is affected; if copper or sheet iron, or lead, or tile, or slate, or pitch, or even thatched straw, for roofing instead of shingles, the number of shingles used is correspondingly reduced. It is obvious that if the architects will substitute clay products or concrete or steel for lumber now used in building, no more effective method of conserving our lumber supply could be devised.

Materials used in buildings are not necessarily lost to the future, however. On the contrary, a certain class of materials, such as steel and other metals, are thus preserved, though temporarily withdrawn from use. Who shall say that other needs and other customs of building of a future time will not be as different from ours as ours are from those of former times? Indeed it is not wholly fantastic to prophesy that the skyscrapers of today may become the iron mines of tomorrow.

The architects are only indirectly employers of labor, but as such they can, more fairly and with less self-interest than any other class, observe the conditions under which labor in the building trades is employed. Your Committee believes that the great annual losses by reason of accidents to men engaged in the building trades are largely preventable; that laws governing the construction of scaffolding, hoisting apparatus, derricks, and other machinery used in quarrying or manufacturing and building, should be passed where they do not already exist, and should be rigorously enforced everywhere; that mechanics and laborers should be taught not to take unnecessary risks but should suffer their fair share of blame if they do, and that they should be encouraged by the public authorities in all reasonable demands for the opportunity to pursue their avocations without unnecessary hazard of life and limb.

The architects believe in the Conservation of buildings once they are erected, and to this end that fire-proof construction should be adopted

wherever possible. In all American cities today fire is a constant menace, and the annual loss from this cause both in life and property is appalling. The strict enforcement of wise building laws will largely prevent this loss; but some concession in taxation to those erecting fire-proof buildings might be found feasible, whereby a premium would be given to those owners of buildings who contribute to the greater safety of life and property by erecting fire-proof structures—or on the other hand an increase of taxation might be made on those erecting buildings which endanger the lives and property of their neighbors and whose flimsy structures make necessary the present large public expenditure for fire-department service in our cities.

This Committee, in common with those who have from the beginning promoted the cause of Conservation, believes in the *use* of our natural resources, not in their *abuse*—in their equitable distribution and development in the hands of the people or in the hands of the Government, not in locking them up in the hands of a few; and that if corporate capital can develop them better than individual capital, then that it should be so done only under restrictions that will safeguard the interests of the people and be subject to Governmental control and limitation, while at the same time giving the capital engaged absolute assurance of protection, security, and reasonable profit. This Committee believes that *use* does not mean *waste* or loss, nor does it mean that reckless spendthrift policy which would squander in a generation, or less, the vast natural resources of this Nation, or permit these resources to be monopolized.

The American Institute of Architects is heartily in sympathy with the principles of the Conservation of our natural resources—and will do its part to advance those principles.

[Signed] CASS GILBERT
Chairman

REPORT OF THE AMERICAN PAPER AND PULP ASSOCIATION

As long ago as 1898 the officers of the American Paper and Pulp Association, realizing the importance of maintaining a perpetual supply of pulpwood, devoted the annual meeting of that year principally to a

discussion of the science and practice of forestry, then almost unknown in the United States. At that meeting addresses were delivered by Doctor Fernow, then Chief of the Government Forestry Bureau, by Mr Gifford Pinchot, his successor, and by Mr Austin Carey, now connected with the Forestry Department of the State of New York. Mr Hugh J. Chisholm, then President of the Association, in his annual message said:

"Those among us who have weighed the matter carefully are well aware that if we as a Nation are to take and permanently hold the foremost place in paper making, we must begin at once to husband our resources. Fortunately, the science of forestry, until recently but little known and heeded less, is ready to point out the way, and we shall learn from three of the best authorities of the country, not only why we should, but how we may, put in practice the principles of forestry. I hope that everyone will go away resolved directly or indirectly to do what he can to secure a rational use of this mainstay of our business."

The attitude of the Association, in the past twelve years, has been to exert its influence in every way possible in the encouragement of forest Conservation. Every year resolutions have been adopted urging timber land owners in the paper industry to practice conservative methods; and at the same time attention has been called to the vital importance of preventing forest fires, and in more recent years the subject of taxation of timber lands has also received attention.

Not only has a universal sentiment in favor of Conservation been created in the industry, but practical results have been accomplished. It is not too much to say that our timber land owners, with possibly here and there an exception, have been for a number of years all conducting their operations so as not to impair the reproductive capacity of their lands. In the first place, they have carefully studied their holdings, in many instances being assisted by the Forest Service at Washington; they have thus become enlightened as to how far cutting timber can go without jeopardizing the future. In the next place, they have voluntarily limited the size, or the diameter of trees, below which no cutting shall be done. They have very generally, although to just what extent cannot be definitely estimated, adopted the method of felling trees with the saw instead of the axe, and have in other ways sought to bring the waste down to a minimum. But

perhaps in no way have they done better service than by encouraging legislation and the enforcement of it for the prevention of fires.

It is roughly estimated that the paper makers own in the United States about 5,000,000 acres, consisting mostly of spruce timber lands. While this is insufficient to afford a natural growth equal to the demands, the deficit is made up by purchases in the United States and by importations from Canada, and the use of other kinds of wood. There is still much more spruce cut for lumber than for pulpwood, but the paper makers are continually adding to their holdings, and there appears to be a readjustment of prices going on which is leading to the substitution of pulpwood production for lumber production.

The example set by paper makers is being followed by other timber land owners, so that we may confidently say that no timber lands of any moment are in any sense being denuded for the production of pulpwood. Less than 2 percent of the consumption of wood in this country is domestic pulpwood, and with a continuation of the conservative methods now in vogue, there need be no fear of diminution of our forests by the paper industry. In fact the perpetuation of the industry in the United States depends largely on the perpetuation of the forests of the United States, so that the paper manufacturers have every incentive to maintain them. The use of hemlock and other kinds of wood for pulp making has greatly increased, thus tending to relieve any drain there might be on the supply of spruce. As most of the paper mills are dependent on water-power, the manufacturers have still further incentive to protect the water-sheds. The Forest Commissioner of Maine has stated—

"Since the advent of the pulp and paper industry in Maine, covering a period of less than twenty years, the system of handling our forest lands has been completely revolutionized. Prior to ten years ago, in cutting logs in the woods, it has been demonstrated by actual tests and measurements that only from 60 to 65 percent of the volume of the lumber trees actually cut was saved and utilized for lumber purposes, while since that period on account of the paper industry it has been demonstrated by later measurements and experiments that from 80 to 85 percent of the volume of lumber trees is actually utilized, and what is of far greater importance is the fact that crooked, seamy and defective trees, as well as all of the undersized trees

formerly cut and destroyed in swamping and in making yards and landings are now utilized. * * * Fully one-half of the whole territory of Maine has never as yet produced one single log for pulp and paper production. I refer to Saint John River drainage, where the same wanton system of lumbering, although possibly in a somewhat lesser degree, is being followed as was followed through the long period from 1860 to 1900. Were this territory fully developed for lumbering by means of proper railroad connections or water facilities, it is safe to assert that conservatively managed, as the paper companies are endeavoring to do today with the best knowledge obtainable, it would supply the entire demand for all the mills now located in Maine indefinitely."

In the State of New York all the paper makers who own lands in the Adirondacks have an Association, including many other lumbermen, which has cooperated with the State authorities in securing legislation which would foster conservative cutting and the prevention of fires.

The International Paper Company, owning nearly a million acres of forest lands in New England, New York State, and elsewhere in the United States, has always conducted its operations with a view to the future supply. In eleven years this company has cut less than two-tenths of a cord per year per acre, which is believed to be less than the natural growth. Two years ago this company started a nursery in Vermont, and each year it has been putting in transplants in increasing quantities in Maine, New Hampshire, Vermont, and New York State, supplementing its own supply by purchases of seedlings and transplants at home and abroad. This replanting is being done on abandoned farms, pasture lands and burns. On their other holdings no replanting is necessary, as there is always sufficient growth left for reproduction. Some other companies have done replanting, but in general conservative cutting and protection from fire render extensive planting unnecessary.

The paper industry has acted on its own initiative, and while self-interest may have actuated it the result is none the less beneficial from the public point of view, and the policy is more apt to be followed permanently than if impractical law, attempting to make Conservation compulsory, were passed.

[Signed] E. W. BACKUS
Delegate

REPORT OF THE AMERICAN MEDICAL ASSOCIATION

The most important interest which this Nation has to guard is human life and health. The conservation of National vitality is fundamental to all plans for the conservation of property and material welfare. As the life is more than meat and the body more than raiment, so is the preservation of health and the avoidance of unnecessary sickness and death of far greater importance than any other interests. Realizing this, the American Medical Association, the National organization of the American medical profession, has been in hearty sympathy with the Conservation movement from its inception. Composed of 52 State and Territorial associations and 1997 local branches with over 70,000 members, this Association has for years advocated the conservation of human life through the abolition of preventable diseases and the betterment of sanitary and hygienic conditions with a view to making the future work of the profession prevention rather than cure. For the accomplishment of these purposes it is today carrying out a number of important lines of work:

1—The American Medical Association has, since its organization in 1847, labored constantly for the elevation of medical schools and of the standard of medical education. Especially during the last five years it has, through its Council on Medical Education, carried on a system of inspection of medical schools with the publication of reports thereon, which has materially raised the standard of medical education and has eliminated a considerable number of low-grade institutions. It is obvious that any increase in efficiency of the medical profession of the present or of the future cannot but result in increased economy of health. The Association is glad to report that medical education in the United States is today upon a higher plane than ever before, and that the public is coming more and more to realize the value of a thorough scientific training for those who undertake the care of the sick.

2—Through its publication, *The Journal of the American Medical Association*, it is constantly laboring to improve the economic condition of the profession, recognizing as a general principle the fact that a poverty-

stricken doctor is a dangerous doctor, both to the profession and to the community. The physician who is not able to procure proper instruments and drugs, or who through poverty cannot keep up with the progress of the profession or secure the necessary books and medical journals for his instruction, may and often does become an actual danger to his patients. Proper efforts on the part of the profession for its own material well-being will result in a better class of physicians and consequently in better medical services to patients.

3—One of the most important activities of the Association in the past five years has been the work of our Chemical Laboratory established for the investigation of pharmaceutical preparations offered to physicians for administration to patients, and for the analysis of so-called patent medicines sold directly to the public. This work has been carried on through the Council on Pharmacy and Chemistry supported by the Association, and has resulted in a much-needed reform in pharmaceutical products. Many preparations which were carelessly, ignorantly, or fraudulently compounded, as well as many others which were sold under false representations, have been investigated and the results published to the medical profession. Although much yet remains to be accomplished, the reform in pharmaceutical preparations has already resulted in an enormous amount of benefit to the people through the enlightenment and education of the profession on this important question. An investigation of "patent medicines" has also been carried on, and many of the preparations offered to the public have been shown, by chemical analysis, to be fraudulent; some are positively harmful, some are harmless but are not as represented; while extravagant, absurd, and impossible claims, false testimonials, and misleading advertisements, are common to many of these preparations. The Association, by its work, has exposed many swindlers and fakirs, and as a result has earned their bitter antagonism.

4—In addition to investigating and exposing frauds in pharmaceutical preparations, the Association has also established a bureau for the collection and preservation of material regarding medical frauds and fakes—including fraudulent "cures" for tuberculosis, cancer, paralysis, locomotor ataxia, and other diseases—which are advertised to the public through false representation, leading not only to an enormous loss to the people through money paid to the swindlers without any beneficial results, but also to great

loss of life and economic loss through illness owing to the victims of these frauds being deprived of proper treatment. The Association is cooperating with other organizations and with the proper authorities for the detection and punishment of these frauds and for the suppression of this most despicable kind of swindlers—those who prey upon the sick and, as a means of extorting a few dollars of blood money, take advantage of the natural desire of the sick or dying to recover health. It has been estimated by the National Association for the Prevention of Tuberculosis that the money loss alone to the people of the United States through fake consumption cures amounts to \$15,000,000 annually. Probably the loss to sufferers from cancer and other incurable diseases is as great. This robbery of the sick and helpless should no longer be tolerated in any civilized country.

5—The Association has maintained a committee for the past four years on the prevention of ophthalmia neonatorum or blindness in infants due to gonorrheal infections, a preventable cause of a large percentage of existing blindness. The United States Census for the blind and deaf taken in 1900 states that 11 percent of the total number of blind lost their sight before the completion of the first year of life, and that in 25 percent the cause of blindness was due to this form of infection. The committee of the Association has been laboring for four years past, and is still at work, endeavoring to educate the public so as to secure proper legislation for the prevention of this form of blindness.

6—Through its State and county branches, as well as through its official publications and its connection with State boards of health and other agencies, the Association has been endeavoring to educate the public on the importance of better hygienic and sanitary conditions and laws, with special reference to pure food and water; proper ventilation of houses, stores, schools, factories, and work-shops; the prevention of avoidable accidents; the development of parks and playgrounds; and the avoidance of the evils of intemperance and excesses. Realizing the importance of this work and the inadequacy of existing methods for bringing practical instruction on sanitary and hygienic questions before the public, the Association at its last annual session established a Council on Health and Public Instruction, the special function of which shall be to place before the people, through the public press, magazines, pamphlets, public meetings, addresses, moving

pictures, and every other available means, the best information obtainable as to the preservation of life and the avoidance of disease. The significance and importance of this action on the part of the organized medical profession of the country can hardly be overestimated. It means that physicians as a class have taken up seriously and systematically the prevention of disease and the education of the public as to how the elimination of avoidable diseases can be secured. With the cooperation of the newspapers and of the people many preventable diseases which have for centuries claimed a fixed toll of human life can be practically eliminated, and hundreds of thousands of lives saved each year.

7—While the Association has labored for the enactment of any laws, either State or National, which were for the benefit of the public health, it stands particularly committed to legislation on three subjects. These are: (a) Adequate State laws insuring purity of the food supply, (b) such State laws as will increase the efficiency of State boards of health and enable them to combat and suppress unnecessary and controllable diseases, and (c) such legislation as will provide an adequate plan for the collection and preservation of vital statistics, in order that proper data for the study and prevention of diseases may be available. It is not to the credit of this country that in half of our States human beings are born and die without any legal recognition of the fact, that not even as much attention is paid to the birth of a human infant as is given to the birth of a race-horse, a pedigreed bull, a blooded dog, or even an Angora kitten. It is not to our credit as a civilized Nation that human beings die and are buried without any legal recognition or record being made of the cause or manner of their death. It is in no sense to our credit that in many communities diphtheria, scarlet fever, and cerebro-spinal meningitis decimate the infant population yet no one knows, nor is it anyone's business to find out, how many deaths result from these epidemics, or how many persons die from various diseases in the course of a year. Proper birth registration lies at the basis of social organization, and has been so recognized for years by European nations, yet it does not exist today in this country. Vital statistics, showing the relative health, morbidity, and mortality of various sections, are of the utmost importance, since healthfulness is recognized as one of the best business assets which a town and county or a State can possess. Yet through lack of proper laws we have today death registration alone in only half of the

Nation, and practically no registration of births whatever. This disgrace on our civilization, which is the wonder and amazement of European nations, should be at once removed by the passage and enforcement of uniform laws in all of the States.

8—The following resolutions were adopted by the House of Delegates of the American Medical Association, June 7, 1910:

"Resolved, That the principles of the Owen Bill, having for its object the creation of a National Department of Health, now pending in the Senate, and similar bills introduced in the House by Representatives Simmons, Creager, and Hanna, be, and are hereby, heartily approved by this Association, and the cordial thanks of the medical profession of the United States, officially represented, are hereby tendered to Senator Robert L. Owen, Irving Fisher, and their co-workers for their able and unselfish efforts to conserve and promote the most important asset of the Nation—the health and lives of its women, its children and its men—properly understood the greatest economic question now confronting our people.

"The members of this Association stand for pure food, pure drugs, better doctors, the promotion of cleaner and healthier homes, and cleaner living for individuals, for the State and for the Nation. We believe this to be held as equally true by the reputable and informed physicians of all schools or systems of practice.

"We welcome the opposition of the venal classes, long and profitably engaged in the manufacture of adulterated foods, habit-producing nostrums, and other impositions on the people, to the extent of hundreds of millions of dollars annually, and express our sympathy for the well-meaning men and women who have been misled and worked into hysterics by the monstrosly wicked misrepresentations of a corrupt and noisy band of conspirators, who are being used as blind instruments to enable them to continue to defraud and debauch the American people.

"Medical science is advancing, especially on its life-saving side, with a rapidity unknown to any other branch of human knowledge. It is known of all men that our members in every community in the United States are unselfishly working day and night, instructing the people how to prevent tuberculosis, typhoid fever, and the other diseases from which physicians

earn their livelihood. Therefore, we welcome and will wear as a badge of honor the slanders of these unholy interests and their hirelings."

The American Medical Association, representing as it does the medical profession of the country, stands pledged and committed to any measure which will improve the public health and preserve the lives of our people. Believing as it does that health and life is our greatest National asset, and that no nation is truly great whatever its material possessions that cannot boast of strong and healthy citizens, we ask the support and approval of the American public and of this Congress in the efforts which are being made for the preservation of human life.

[Signed] J. T. PRIESTLY, Des Moines
F. F. WESBROOK, Minneapolis
A. R. MITCHELL, Lincoln
CHAS. S. SHELDON, Madison
F. R. GREEN, Chicago
Committee

REPORT OF THE AMERICAN RAILWAY ENGINEERING AND MAINTENANCE OF WAY ASSOCIATION

In October, 1908, the National Conservation Commission invited the American Railway Engineering and Maintenance of Way Association, in connection with other technical bodies of this country, to be represented at the Conference in Washington, and to assist the National Conservation Commission with suggestions concerning advisable lines of inquiry, nature of report to be made, and possibilities of accomplishment on the part of the Commission. Acting upon this invitation, the Board of Directors of the Association appointed a Special Committee to cooperate with the Commission. This Committee consisted of eight members of the Association, selected from widely separated sections of the country.

The Association, through its Committee, was represented at the joint Conservation Conference held in Washington beginning December 8, 1908;

and the Committee has been keeping in touch with the Conservation Commission through Mr Pinchot and the Secretary, Mr Thomas R. Shipp. Several meetings of the Committee have been held, and in March, 1909, the Committee was addressed by Dr Joseph A. Holmes, of the Commission.

In March, 1909, the Committee, through its Chairman, requested Mr Pinchot to furnish, through cooperation with the Forest Service, suggestions as to the best methods to be pursued by the railroad companies for the prevention and control of forest fires, with statistics of the loss from such cause, and urged upon the Commission the importance of endeavoring to effect reduction in the tariff on cross-ties and in lumber rates, in order to make it possible for the railroad companies to import ties and save thereby the home supply. The cooperation of the Committee was offered with the forest-products laboratory at the University of Wisconsin, or with any of the National or State organizations.

On May 13, 1909, an elaborate report was transmitted to the Committee by the National Conservation Commission, through Secretary Shipp, containing valuable suggestions as to the possibilities of railroad companies assisting the work of Conservation by thorough methods of prevention and control of forest fires and the cultivation of timber for railroad purposes, by the use of sawed instead of hewed ties, the use of treated timber and the extension of the supply of creosote, and other features, many relating to timber resources. This report was transmitted by the Committee to the American Railway Engineering and Maintenance of Way Association, published by the Association, and distributed throughout the country in one of its bulletins. Dealing directly, as it does, with those features of Conservation that affect the railroad companies and their patrons, and having a circulation among railroad officers covering the United States, as well as large portions of Canada and Mexico, the results should be exceedingly beneficial to the cause of Conservation.

In March of this year the American Railway Engineering and Maintenance of Way Association, recognizing the growing importance of the Conservation movement, established the Special Committee as one of the Standing Committees of the Association, at the same time largely increasing its personnel and bringing into membership a number of prominent railroad officers of this country and Canada.

The work of the Committee has been divided into sub-committees for the purpose of specialization; these, with an outline for investigation are as follows:

No. 1—Tree planting and general reforestation

a—Extent of existing forests considered in connection with increase of growth and consumption

b—Judicious selection of tree varieties for planting, and locality and soil conditions considered; possibility of value from growth on cut-over land

c—Methods of planting and cultivation, with cost of same, considering possibilities from cut-over lands

d—Anticipated results at maturity from trees so produced

e—Methods and costs of caring for and protecting existing forests

No. 2—Coal and fuel-oil resources

a—Extent of existing supplies, considered in connection with consumption

b—Extent of waste in production

c—Economic consumption, giving consideration to practical use of by-products

No. 3—Iron and steel resources

a—Supplies of raw material, considered in connection with consumption

b—Waste in production

c—Best methods of protecting finished products from destructive influences

The Committee will continue on the lines of investigation as shown, and holds itself in readiness to cooperate with the National Conservation Commission and its kindred and subsidiary organizations, as well as other National societies, for the furtherance of the great principles of Conservation of the Nation's resources.

The Committee:

A. S. BALDWIN,

Chief Engineer Illinois Central R. R. Co. (*Chairman*)

MOSES BURPEE,

Chief Engineer Bangor and Aroostook Railroad

W. A. BOSTWICK,

Metallurgical Engineer Carnegie Steel Company

E. F. BUSTEED,

General Superintendent Canadian Pacific Railway

E. B. CUSHING,

Southern Pacific Company

E. O. FAULKNER,

Manager Tie and Timber Department, Atchison, Topeka and Santa Fe System

W. F. H. FINKE,

Tie and Timber Agent Southern Railway

J. W. KENDRICK,

Vice-President Atchison, Topeka and Santa Fe System

A. L. KUEHN,

General Superintendent American Creosoting Company

G. A. MOUNTAIN,

Chief Engineer Canadian Railway Commission

WM. McNAB,

Principal Asst. Engineer Grand Trunk Railway

C. L. RANSOM,

Resident Engineer Chicago and Northwestern Railway

[Signed] A. S. BALDWIN
Chairman

REPORT OF THE AMERICAN RAILWAY MASTER MECHANICS' ASSOCIATION

In behalf of the American Railway Master Mechanics' Association I wish to thank the officers of the National Conservation Congress for the courtesy shown our Association by inviting our President, Mr C. E. Fuller, to attend this Congress. Mr Fuller was unable to be present, and it was therefore my good fortune, as First Vice-President, to take his place.

As you no doubt are aware, the membership of the A. R. M. M. Asso. is composed of the heads of the mechanical departments of practically every railroad in the United States and a large number from Canada, and all of us are heartily in sympathy with the Conservation movement that has had such wonderful growth during the five years it has been before the public. The enormous amount of lumber, coal, etc., that is used by the railways makes it imperative for them to use it as economically as possible, and great efforts are being made, by education, to use a pound or a ton of coal so that the greatest efficiency may be obtained therefrom. During the calendar year ending December 30, 1909, the company I am connected with used 4,193,617 tons of coal in its locomotives and power plants; we have a large force of instructors, including master mechanics, road foremen of engines, and traveling firemen who are continually riding the engines and giving the enginemen the benefit of their experience in the proper method of handling the locomotive so that steam will not be wasted, and that only the proper amount of coal will be shovelled into the firebox to produce the desired results. The use of feed-water heaters, superheaters, and compound locomotives has been hastened by the desire to get as much use out of the heat in the coal as possible; the feed-water heater and superheater promising the best field for economy in locomotive practice. The lignite fields of Wyoming are being opened by using this kind of coal in locomotives that have been specially designed to burn it. Heretofore it was necessary to haul coal from southern Iowa to Wyoming, a distance of about 800 miles, which was a very wasteful operation; a good deal of this will be dispensed with by using lignite coal in the territory near which it is mined. So that a comparatively poor grade of coal can be made better, a washery, with a capacity of about 1800 tons per day has been erected and put in operation, which washes out a large percentage of the slate and other impurities in the

coal; this means that a ton of washed coal has a greater heat value than the same amount of unwashed coal would have.

The question of conserving the life of the ties used has had due consideration, and a treating plant has been in use for nearly six years which is expected to increase, by treatment, the life of ties about 40 percent, besides enabling us to use an inferior kind of timber as ties, that before was considered impractical; the importance of thus prolonging the life of ties will be appreciated when I say that for the calendar year ending December 30, 1909, we used 2,996,957 ties. Other wood was used in the same period as follows: piles, 83,201; posts, 382,556; lumber, 56,172,000 board feet. It therefore makes it very necessary on account of the constantly increasing price of lumber to reduce the amount used and wasted. The use of concrete has enabled us to make things of that material, which a few years ago would have seemed impossible; floors in roundhouses and shops, which rapidly deteriorate (when made of wood), on account of moisture, are now made of concrete, which stands up admirably in that service.

We are enormous consumers of oil, and the same care is exercised in its use as with coal and lumber—in fact, under present conditions, it is absolutely necessary that the greatest economy be instituted in the use of all kinds of material as a matter of self-preservation.

During the time I have spent at your meetings, it has been quite a revelation to notice the intense interest that has been manifested by everybody on the subject of Conservation; and as the representative of the American Railway Master Mechanics' Association I wish to assure you of our heartiest cooperation in the work. Again I thank you for the opportunity of being present.

[Signed] H. T. BENTLEY
First Vice-President

REPORT OF THE AMERICAN SCENIC AND HISTORIC PRESERVATION SOCIETY

The suggestions of the Committee of this Society appointed to cooperate with the National Conservation Association must naturally be determined

by the objects for which the Society exists. It is the aim of the American Scenic and Historic Preservation Society to protect the interesting features of the natural landscape, to save from obliteration all historic places and objects, to erect suitable historical memorials where they are needed, to promote the beautification of cities and villages, and otherwise to develop in the people a regard for the beautiful in nature and for the historic in human institutions, cultivating this general field by means of free lectures, literature, prize competitions, correspondence, and other educational means as well as by using influence to have places and scenery preserved as parks and reservations. The interest of this Society, therefore, lies not so much in the fields of economic production as in the less definite regions of historic appreciation and artistic sensitiveness to surroundings. The report of its Committee on Conservation will naturally not deal with the direct economic questions with which most other cooperating societies and organizations would naturally be concerned.

The Committee desires first to express its appreciation of the work of the National Conservation Association and to pledge itself to cooperate with that Association in the furthering of its work. The Committee holds itself in readiness to cooperate in the enterprises originating from the National Conservation Congress and the National Conservation Association so far as they are within the proper province of the American Scenic and Historic Preservation Society. The Committee feels that the establishing of the National Conservation Commission, and its successor, the National Conservation Association, marks a distinct advance in utilizing for the good of all the people the resources which really belong to all the people, and which should be used for their welfare, rather than exploited for the interest and gain of a few persons or wasted and despoiled by the thoughtlessness of the people themselves.

The Committee holds that all natural resources should be protected and utilized in a scientific and unselfish way, and that the heritage of the earth should be passed over to our descendants with the least possible loss consistent with wise use in the present generation. Its special interest in the question, however, lies in the belief that all this effort should harmonize with the preservation of the beauty of the natural landscape and with the Conservation of all places and scenes of historic interest.

It is too little appreciated that every natural object makes a two-fold appeal to the human mind: its appeal in the terms of its physical or material uses, and its appeal to our sense of beauty and of personal satisfaction. As the people progresses in civilization, the public mind becomes constantly more sensitive to the conditions in which we live, and the appeal to the spiritual satisfaction of life constantly becomes stronger. It is, therefore, of the very first importance that whatever is done by the National Conservation Association shall be executed in the feeling that not only shall the physical needs of life be met, but that the earth will constantly be made a more satisfactory place in which to live, and that the lessons of history must exercise an increasing influence.

It is important that we not only save our forests in order that they may yield timber and conserve our water supplies, but also that they may adorn and dominate the landscape and contribute to the meaning of scenery. It is important that our coal supplies be not only conserved for their use in manufacture and the arts, but also that smoke does not vitiate the atmosphere and render it unhealthful, and discolor the objects in the landscape. It is of the greatest importance that water supplies be conserved by storage reservoirs and other means, but this Conservation should be accomplished in such a way as not to menace health or offend the eye or destroy the beauty of contiguous landscape; the impounding of waters without regard to preserving natural water-falls, streams, and other scenery, is a mark of a commercial and selfish age, and is a procedure that cannot be tolerated in a highly developed society. It is important that regulations be enacted regarding the operation of steam roads through wooded districts not only that the timber may be saved, but also that the natural beauty of the landscape may be protected from fire and other forms of destruction. The fertility of the soil must be saved not only that products may be raised with which to feed and clothe the people, but also that the beauty of thrifty and productive farms may be saved to the landscape. The property-right in natural scenery is an asset to the people, and the best Conservation of natural resources is impossible until this fact is recognized.

On this point we call attention to the following paragraph in the report of the Commission on Country Life: "In estimating our natural resources, we must not forget the value of scenery. This is a distinct asset, and it will be more recognized as time goes on. It will be impossible to develop a

satisfactory country life without conserving all the beauty of landscape and developing the people to the point of appreciating it. In parts of the East a regular system of parking the open country of the entire State is already begun, constructing the roads, preserving the natural features, and developing the latent beauty in such a way that the whole country becomes part of one continuing landscape treatment. This in no way interferes with the agricultural utilization of the land, but rather increases it. The scenery is, in fact, capitalized, so that it adds to the property values and contributes to local patriotism and to the thrift of the commonwealth."

It is especially important, in the opinion of this Committee, that the National Conservation Congress and the National Conservation Association lend their influence to the establishment of reserves in all parts of the country for the preservation of natural features of great scenic interest, for the protection of birds, animals, and native plants, and also for the Conservation of the lessons of history. The Committee earnestly requests that in the program of the activities of the National Association these questions may be given their due consideration.

*What the American Scenic and Historic Preservation Society has
Accomplished*

Having now stated its general position and its outlook on the subject of the Conservation of our natural resources, the Committee cites, by way of illustration, a few of the things that the Society has accomplished.

The American Scenic and Historic Preservation Society is the medium through which Honorable Wm. Pryor Letchworth, of Portage, gave to the State of New York a superb tract of 1000 acres of land embracing the famous Portage Gorge of Genesee River, including the three picturesque Portage Falls. This property, which cost the owner about half a million dollars, will pass into the official custody of the Society, as Trustees for the State of New York, on Mr Letchworth's decease. Letchworth Park, as it has been named by the Legislature, possesses not only remarkable scenic beauty, but also high scientific and educational value. The geological strata here exposed have given the name to that extensive formation of rocks known as the Portage Group, and the vegetal and bird life of this

reservation is remarkably varied and of the greatest interest to students of natural history.

The Society also secured the purchase by the State of New York, and is official custodian of, the famous Watkins Glen at the head of Seneca Lake. This property embraces about 105 acres of land, and includes rock exposures that have received the attention of the United States Geological Survey and prominent geologists for many years. It presents one of the most remarkable examples of stream erosion in the eastern States.

Through the intercession of the Society, the State of New York has purchased and committed to the care of the Society 35 acres of land on the promontory of Stony Point on the Hudson River. Here, in addition to an interesting exposure of primitive rocks and varied flora, are the historical associations of General Anthony Wayne's exploit during the Revolutionary War, which evoked the admiration of the leading military men of America and Europe. In like manner the State has purchased and committed to the Society's care a small reservation on Oneida Lake embracing the remains of Fort Brewerton.

Ten years ago, Governor Roosevelt requested the Society to represent the State of New York in concerted measures with the State of New Jersey for the Conservation of the Palisades of the Hudson. As the result of this initiative, the State of New York appropriated about \$450,000, the State of New Jersey about \$50,000, and the Honorary President of this Society, Mr J. P. Morgan, gave \$125,000, and today the picturesque cliffs on the western side of the lower Hudson for a distance of thirteen miles have been rescued from defacement and are in the care of the Palisades Interstate Park Commission. As a sequence to this work, and a result of the general sentiment developed in favor of scenic and historic preservation, Mrs Edward Harriman recently gave to the State of New York 10,000 acres of land on the western side of the Hudson for a State Park, and she, together with Mr Morgan, Mr John D. Rockefeller, Mrs Sage, and others, have supplemented the gift with over \$2,500,000 of money.

Ten years ago, the Society secured legislation by means of which a reservation of 35 acres at the head of Lake George was made by the State,

for the purpose of preserving scenery and the ground made historic by events in the Colonial and Revolutionary Wars.

The long and difficult campaign for the preservation of Niagara Falls, in which the Society had an honorable part, is familiar to all, and need not be repeated here.

Many other instances could be cited in different parts of the country, some connected directly with the Society's work, and all the result of the general sentiment which has been developed during the past 25 years in favor of conserving natural scenery and creating urban and extra-urban parks for the benefit of mankind. Not the least important of these in their bearing on conditions of life are the city parks. In New York City, for example, the Washington Headquarters Park and Joseph Rodman Drake Park were created at the direct instance of the Society; and the famous Central Park, in the creation of which our late President Andrew H. Green, as Controller of the Park, was an important factor, has been protected against invasion by race tracks and many other artificial encroachments by the vigilance of the Society. Among the gifts of city parks by private individuals stimulated by the sentiment created by the Society's work may be cited a series of parks embracing about 500 acres and costing with their improvements a quarter of a million dollars or more presented in 1907 to the city of Utica by Mr Thomas R. Proctor, a Trustee of the Society. In 1909, another member of the Society, Mr Henry H. Loomis, gave to the city of Geneva (New York) about 26 acres of woodland for a city park. In Jamestown (New York) a park system has been developed largely under the influence of a Trustee of this Society. In Colorado Springs, within two years, there have been two remarkable expressions of this general sentiment which has now become so general that no one Society can claim direct connection with its results. We refer to the series of completed parks, boulevards, and paths, embracing over 1500 acres of superb scenery, given to that city by General W. J. Palmer; and the gift of the famous Garden of the Gods to the same city by the heirs of the late Charles W. Perkins, of Iowa. These two gifts have placed Colorado Springs in possession of what is probably the most remarkable series of city parks of the kind in the United States. The sentiment created by this Society has also expressed itself in the beautifying of many cities by the improvement of open spaces, public greens, and church yards, and by the erection of monuments and drinking fountains.

Of State parks as distinguished from city parks, those which have received the most attention from this Society, outside of the five reservations under its immediate control and the Palisades Interstate Park, have been the State Park at Niagara Falls and the Adirondack State Park. The State Reservation at Niagara Falls, comprising 112 acres of land and 300 acres of land under water, and including the American Fall and half of the Canadian Fall, was created in 1885; and it was partly on account of the lessons taught by that reservation that the President of the Niagara Commission, the late Honorable Andrew H. Green, ten years later founded the American Scenic and Historic Preservation Society. In the long campaign for the protection of Niagara Falls from the inordinate diversion of their waters and the disfigurement of their environment the Society has taken a leading part. The Adirondack Park now comprises over 1,500,000 acres. Here, also, it has been necessary to maintain a constant campaign to protect the forests from destruction by fire, artificial flooding, and the illicit removal of timber.

In the far Southwest the efforts of the Society have been directed chiefly to the extension of the Grand Canyon preserve, and the protection of the Hetchhetchy valley—a part of Yosemite National Park—from what we believe to be an unnecessary project for flooding a part of the National Park for the purpose of supplying water to San Francisco.

In conclusion, we may say of the movement at large for the preservation of remarkable works of nature for the instruction and enjoyment of the people, that it is older than the organized movement for the Conservation of the material resources of the country; and if it cannot be said that one is the outgrowth of the other, it is true that both are necessarily closely inter-related and that each should proceed with full regard for the other's welfare.

The Conservation Committee:

L. H. BAILEY (*Chairman*), Ithaca

CHARLES M. DOW, Jamestown

HENRY E. GREGORY, New York City

EDWARD HAGAMAN HALL, L.H.D., New York City

SAMUEL V. HOFFMAN, New York City

THOMAS P. KINSFORD, Oswego

GEO. FREDERICK KUNZ, PH.D., SC.D., New York City

WILLIAM P. LETCHWORTH, LL.D., Portage
THOMAS R. PROCTOR, Utica
COLONEL HENRY W. SACKETT, New York City
CHARLES DELAMATER VAIL, L.H.D., Hobart College, Geneva

[Signed] L. H. BAILEY,
Chairman

REPORT OF THE ASSOCIATION FOR THE PROTECTION OF THE ADIRONDACKS

The Association for the Protection of the Adirondacks, with headquarters in New York City, was formed ten years ago before the word "Conservation" as now used had acquired its present meaning. In the light of the present use of that word the object of this Association might properly be expressed in the title "Association for the Conservation of the Natural Resources of the Adirondacks."

"The Adirondacks," in a general way, is the term used to describe a region of about 12,500 square miles in northern New York, lying between Lake Champlain on the east, Lake Ontario on the west, Saint Lawrence river on the north, and the Mohawk on the south. In the heart of this region the State has, by statute, delimited an area of about 3,300,000 acres, or 5,156 square miles, under the title of the "Adirondack Park." Within this more restricted area lie the principal mountains and the principal forests of the State. The State owns about one-half of the area of Adirondack Park, and its policy is progressively to acquire the remainder.

The work of the Association for the Protection of the Adirondacks for the past decade has been directed toward the preservation of the natural conditions and the material resources of Adirondack Park for the benefit of all the people of the State. During this period, what is now known as the movement for the Conservation of natural resources has developed, although "Conservation" in fact, if not under that name, was well begun in New York State a quarter-century ago when, in 1885, the Legislature established the Forest Preserve.

In the State of New York, the natural resources, as that term is commonly understood, to the conserving of which public attention is now chiefly directed, are the forests and the waters. While the forests lie chiefly in the Adirondacks, the streams and water-power sites lie chiefly outside of Adirondack Park; but in the protection of the Adirondacks is involved the water question as well as the forest question, for three reasons: *First*, because many streams take their rise in the Adirondacks; *second*, because of the intimate relation between the forest covering of water-sheds and stream-flow; and *third*, because there are a few possible reservoir sites situated on State lands in Adirondack Park which are coveted ardently by private interests strongly represented in the State Legislature.

The natural resources of the Adirondacks, however, are not limited to the forests and streams. In a State embracing a tenth of the population of the United States, including a city embracing a twentieth of the population of the Union—a State and a city in which the vocations of life are pursued under the highest nervous tension—the Adirondacks possess natural resources for the conservation of human vitality (for the recuperation of health and the recreation of personal energy) which are no less important to the welfare and prosperity of our people than the cultivation of a timber supply or the development of hydraulic power. In addition to these considerations, two other elements enter into the question of Conservation in the Adirondacks: One is the preservation, for purposes of science and sportsmanship, of the natural wild species of animal life which have become extinct not only in other parts of New York, but also generally throughout the eastern States; the other is the preservation of the scenic beauty of this great mountain resort, which is seriously threatened in ways hereinafter to be mentioned.

These latter considerations of health, recreation, and esthetic delight are not less entitled to recognition because they cannot be measured in terms of board feet or amperes convertible into dollars and cents. Rest and recuperation are not the exclusive needs of men of large expenditure of brain and nerve force, nor is actual positive pleasure conceded to be the exclusive privilege of men of large means. The principle contained in the ancient command to do upon six days all that thou hast to do and to rest upon the seventh day is receiving a wider application in modern industrial conditions which constantly tend to shortened hours of labor on the six days

and a larger recognition of every man's right to a measure of the possible joys as well as the inevitable labors of living.

Therefore, to conserve the Adirondacks as a health and pleasure resort for the people at large as well as for a source of a timber supply and the fountain head of important water supplies is the object of our Association.

Forest Conservation

Without entering into statistics of the relative area of forested and denuded lands in New York, or the relative rate of forest removal and forest growth which is so disproportionate as to threaten the complete denudation of the State within 20 or 25 years, we may mention something of what has been done in the way of practical forest Conservation in the State, partly by the aid of this Association.

There are six principal ways in which the forests can be conserved:

- 1—By restriction of commercial lumbering
- 2—By prevention of timber stealing
- 3—By control of forest fires
- 4—By building good roads
- 5—By replanting
- 6—By prevention of flooding

1—Commercial Lumbering. There appear to be three ways of reducing the danger of the denudation of private forest lands, namely, (a) to educate the owners as to the unwisdom of indiscriminate and wholesale cutting; (b) to convert private lands into State lands by purchase, and thus bring them under the protection of the Constitution which forbids the cutting of trees on State land; and (c) the passage of laws offering inducements to, or imposing some restrictions on, private owners for the purpose of reducing their cut. Of these three methods, good progress has been made with the first two; the third has been attempted only in a mild way and without effect.

In the past dozen years, the private owners of forests in New York have awakened to a lively sense of the shortsightedness of the policy of cutting everything in sight. Prior to about 1890, roughly speaking, lumbermen as a usual thing took nothing less than two-log trees, leaving all that were under

12 inches in diameter on the stump. But with the improvements in machinery and processes for the manufacture of wood pulp, not only was the range of cutting extended from poplar to spruce, hemlock, pine, and balsam, but the lumbermen also disregarded size limits and cut all the trees of certain species, large and small. This close cutting was disastrous in both its primary and secondary effects; it left no provision for future growth, and it thinned the forests so much in places that further damage was inflicted by wind and ice storms. In the closing years of the last century signs of an awakening to the dangers of this policy appeared. In 1898 the Division of Forestry of the United States Department of Agriculture issued Circular 21 entitled "Practical Assistance to Farmers, Lumbermen, and Others in Handling Forest Lands," conveying an offer to cooperate with owners in the preparation of working plans for forest lands which presented conditions favorable for systematic and conservative management. One of the first private owners to appreciate the wisdom of adopting the more conservative course recommended by the Government was the late Honorable William C. Whitney, owner of a tract of 70,000 acres in Hamilton County. Prior to 1898 he had been cutting down to a diameter of 8 inches three feet from the ground; but in 1898, after securing expert advice, he raised the limit to 10 inches, which was maintained until last spring, when lumbering on that preserve was finished. The result of this judicious policy has been that there is now a fine growth of young trees on the property, which in a few years will come to merchantable size. In 1900 the State of New York appropriated \$2,000 to enable the Forest, Fish, and Game Commission to take advantage of the Government offer to the extent of working out the theory of conservative forest management on a selected tract of land known as "Township 40 of the Totten and Crossfield Purchase," embracing Raquette Lake in Hamilton County. This could be only a theoretical demonstration as applied to State forest land, because (for very excellent reasons) the State adopted a Constitutional Amendment in 1894 which provides that—"The lands of the State now owned or hereafter acquired constituting the Forest Preserve as now fixed by law shall be forever kept as wild forest lands. They shall not be leased, sold or exchanged, or be taken by any corporation, public or private, nor shall the timber thereon be sold, removed or destroyed" (article VII, section 7). But while only a demonstration on paper of a theory and not a demonstration in fact, the result of the practical study on the ground and the consequent publicity of the conclusions was of value,

for it attracted the attention of lumbermen to the diameter limits below which it is wasteful in the long run to cut. Township 40 is a virgin forest, and taking into consideration all the conditions of that particular tract—character and density of growth, rate of reproduction, proximity to outlets, cost of logging roads, camps, and stream improvements—it was calculated that 12 inches was the most advantageous minimum diameter to be used. In the following year a forest working plan for townships 5, 6, and 41 was worked out with a similar result, namely, the recommendation of a 12-inch minimum limit. Since that time conservative lumbering has been adopted on several private properties other than the Whitney preserve. One of the most notable cases is that of the International Paper Company, the owner of very extensive woodlands, which for sometime past has not cut trees less than 10 inches in diameter.

It may be said with confidence, therefore, that the campaign of education in forest matters during the past ten years in this State, and the mathematical demonstration of the wisdom, from the practical business standpoint, of placing limitations on the cut, are bearing fruit. Not only is the system of culling or selection tending to supersede wholesale tree-cutting of all sizes, but there is also reason to believe, from the latest available statistics, that in 1908 there was an actual change in favor of a reduced cut.

In the past decade there has been material progress in forest conservation by the enlargement of the forest land holdings of the State. During this period, the State has purchased about half a million acres of forest land, and its Forest Preserve, on January 1, 1910, embraced 1,641,523 acres, of which 1,530,559 were in the Adirondack mountains and 110,964 were in the Catskill mountains. Much of the land acquired during the past decade has been lumbered land, and has contained little merchantable timber. The purchases have had the advantage, however, of increasing the area of wild land which, so long as the present forestry section of the State Constitution shall stand, will at least have the chance to produce a new forest without risk of destruction. In pursuing the policy of building up its Forest Preserve, the State has shown in times past regrettable and costly procrastination, with the result that it has bought denuded land at twice the price at which it could have bought forested land. In this respect, the State still lags behind what many believe to be the rate at which the State's holdings should be increased. The signs of encouragement under this head are evident not only

in the increased aggregate area of the State Forest Preserve, but also in the improved methods of administration. In times past, the forest administration has been so lax, not to characterize it more strongly, that while with one hand it was spending large sums in purchasing land, with the other it was parting with State property on flimsy pretexts, with the result that in some years, while purchases were being made, the State's holdings were actually decreasing instead of increasing. Weak compromises, by which the State parted with its timber and retained the land, involved transactions in which the State apparently bought a second time land which it already owned; and the purchase of land at exorbitant prices from favored friends, were practices of the past, the abatement or abolition of which is not the least encouraging evidence of the Conservation movement in this State.

In legislation, little has been attempted in the way of offering inducements to lumbermen to restrict their cut, and nothing has been done in the way of compulsion. In 1893 and again in 1894 Honorable Roswell P. Flower, then Governor, in a message to the Legislature recommended the enactment of a law which would provide for some reasonable compensation to such owners of private forests as should consent to cut no trees except under conditions imposed by the State; and a Law was enacted embodying that idea, and it now forms section 43 of the consolidated Forest, Fish and Game Law of 1909. This section provides that the Forest, Fish and Game Commission may "contract that lands within the Adirondack Park not owned by the State shall, in consideration of exemption from taxation for State and county purposes, become public as part of the park in like manner as State lands. Such a contract must provide against the removal of live timber except spruce, tamarack, or poplar, more than twelve inches in diameter three feet from the ground, and may reserve to the owner the right to clear not more than one acre within each hundred acres of land, and may contain such other reservations for occupancy as may be agreed upon. The approval of the commissioners of the land office must appear on any such contract by the certificate of their clerk. Such contract shall be recorded in like manner as conveyances made by commissioners of the land office." This law has proved no inducement to forest owners, and has been ineffective in limiting their cuttings.

Our Association has considered the subject of legislation providing for some discrimination in the taxation of forest lands which, by lowering the

rate of taxation on immature forests, should offer an inducement to forest owners to allow their young timber to stand and grow; but as yet no satisfactory plan has been worked out. There is another phase of this question, however, which is attracting increasing attention in neighboring States, but which as yet has received little consideration in New York, namely, the compulsory restriction of timber cutting by legislation. Two recent judicial decisions on the power of a State to regulate the use of the natural resources of private land bear with much force on this subject. The Senate of Maine requested the Supreme Court of that State to give, for its guidance, an opinion upon the following question:

In order to promote the common welfare of the people of Maine by preventing or diminishing injurious droughts and freshets, and by protecting, preserving, and maintaining the natural water supply of the springs, streams, ponds, and lakes of the land, and by preventing or diminishing injurious erosion of the land and the filling up of the rivers, ponds, and lakes, and as an efficient means necessary to this end, has the Legislature power under the Constitution, by public general law, to regulate or restrict the cutting or destruction of trees growing on wild or uncultivated land, by the owner thereof, without compensation therefor to such owner?

With the exception of one justice, who declined to give an opinion for constitutional reasons, the opinion of the Court was unanimously in the affirmative (Opinion of the Justices, 103 Me. 506).

The other decision referred to was in the case of *Hathorn vs. Natural Carbonic Gas Co.*, involving the use of the mineral waters at Saratoga Springs. The State of New York passed a law entitled "An act for the protection of the natural mineral springs of the State and to prevent waste and impairment of its natural mineral waters." The object of this law was to prevent the practice of artificially accelerating the natural flow of mineral waters for the purpose of extracting the carbonic acid gas for commercial uses. In the case in question, the Court of Appeals, with one dissenting voice, decided that the part of the statute in question was constitutional, and affirmed an order of the lower court restraining the defendant from doing what the law forbade. Judge Haight, the dissenting justice, differed from the majority, though not on the general proposition of the State's right to

regulate the use of the springs; he based his objection on the ground that the statute in question did not attempt to regulate the production of the mineral waters in order that the public might enjoy the medicinal properties contained therein, but absolutely prohibited the pumping of carbonated waters throughout the State for the purpose of extracting carbonic acid gas. On the general question of the police powers and the conclusion that the Legislature may by statute regulate the use of the waters, Judge Haight was in full accord with the majority. "Surely," he said, "the State, under its police powers, may, in the interests of the people, protect such great gifts of nature to mankind."

Decisions like these would seem to be finger-boards pointing in the direction of compulsory Conservation if an enlightened self-interest or public spirit on the part of private forest owners do not accomplish the same purpose.

2—*Timber Stealing*. A very practical form of Conservation in which this Association has had a leading part has been the prevention of the unlawful removal of timber from State land. In 1905 reports reached us to the effect that in the face of the plain prohibition by the Constitution private parties had made deliberate arrangements with contractors to lumber on State land, and that these operations were being carried on with the ample knowledge if not actual collusion of the then Forest, Fish and Game Commissioner and his subordinates. To verify these rumors, the Association sent to the Adirondacks in the dead of winter a representative, who, using snow-shoes when other modes of travel were impossible, penetrated into the depths of the forests, and found the lumber men in active operation on State lands. As the investigation progressed, it developed that between 15,000,000 and 16,000,000 board feet of timber had been removed unlawfully from State land during the preceding year, with the knowledge of the authorities whose duty it was to prevent it; and that it was done under a well-understood system of friendly cooperation by which the timber thieves, technically called "trespassers," were permitted to go through a form of confessing judgment and paying for the timber at a rate so low as to make the transactions profitable for the trespassers. Not only was the mandatory legal penalty not exacted, but the so-called confessions of judgment were allowed to be made before country justices of the peace in amounts greatly exceeding their jurisdiction, and the timber was permitted to be removed

from the State land in direct contravention of the Constitution. From the perfection with which the system was then working it was apparent that the illicit practices were of long standing; but the exposure by this Association resulted in the retirement from office of the Forest, Fish and Game Commissioner and the Chief Game Protector, and the effectual stopping of this form of depredation.

3—*Forest Fires*. Substantial progress has also been made in the direction of Conservation by fire prevention. The history of forest fires in this State may well prove of interest to other States having virgin forests. The most prolific source of forest fires in New York hitherto has been the steam locomotive. Before the introduction of the steam railroad in the Adirondacks, forest fires were infrequent and of small extent. With the construction of every new railroad using coal or wood for fuel, fires became more numerous. The danger from this direction was apparent 30 years ago, but with an indifference for which a costly penalty has been paid, the State failed to find a remedy until within the past two years. In the tenth United States Census, Professor Charles S. Sargent, speaking of the forest fires in the United States during the year 1880, said: "In the State of New York, the total area burned was, in acres, 149,491; and the value of the property destroyed, \$1,210,785. Of the causes to which these fires were attributed, 37 cases were reported as originating from clearing land; 43 cases as originating from sparks from locomotives; 22 cases as originating from hunters." The "First Annual Report of the Forest Commission of the State of New York for the Year 1885" said: "The statistics show that in New York State at least, more forest fires are traced to railroads than to any other cause." Three years later (1888) the State Forest Commission was so alarmed at the danger of fires from railroads that it formally declared the extension of railroads into the forests to be a calamity. It declared—

The extension of railways into the Forest Preserve proper cannot but be regarded as a calamity, and it is respectfully submitted that it would be most expedient to put a check upon their further encroachment by proper legislation. * * * A further extension of 'better means of transportation' by railway or steamboat threatens more danger to the forest than it promises benefit to the public. * * * Complaints are loud against all railroads as being instrumental in scattering fire.

These warnings are cited not so much as an argument against the introduction of railroads into forest lands—which can now be done with safety by the use of oil fuel or electric power—as to show how early was the realization of the danger of forest fires from railroads.

About 1892 another railroad, the Mohawk and Malone, was built through the heart of the forests, and the testimony taken from old woodsmen in the fire investigation in 1908 showed that the building of the road was followed by the inevitable train of fire. The annual fire area in the Adirondacks which had previously ranged from a few hundred acres up to 25 square miles, increased to 80 square miles in 1899 and to 940 square miles in 1903. In 1908 an area of 277 square miles was burned over in the Adirondacks alone. The maps of the large fires of 1903 and 1908, showing the burned areas chiefly bordering the lines of railroads, were strong object lessons as to the principal source of the fires, however the railroad companies might attempt to disguise them. In 1908 public sentiment on this subject became aroused as never before. It was felt that whatever may have been the excuse for permitting the advent of coal-burning or wood-burning locomotives in the forests 30 or more years ago, the further toleration of these fire-spreading agents was little short of criminal since electricity and oil fuel had been developed as practical agencies for developing power. In the year last mentioned, therefore, the Forest, Fish and Game Commissioner, backed up by the Association for the Protection of the Adirondacks, applied to the Public Service Commission for an order to compel the railroads running through the Forest Preserve to use oil-burning locomotives during the fire danger season. The railroads, as was to be expected, earnestly protested against the innovation; but the evidence was so convincingly against them that the Public Service Commission ordered the use of oil fuel, and the installation of oil-burners has made an encouraging beginning.

With the removal of this prolific cause of forest fires, the enforcement of salutary laws which had practically been a dead letter, the enactment of certain new laws providing for toppling by lumbermen, etc, and the organization of an improved fire-fighting system by the Forest, Fish and Game Commission, it is believed that New York has taken a long step forward in the direction of conserving her forests from fire.

4—*Forest Conservation by Good Roads.* The natural conditions in the Adirondacks which for so many years made the mountain wilderness impregnable by civilization and to a great extent preserved that region from the denudation which has characterized the more thickly populated parts of the State have also retarded the development of road building. The road system of the Adirondacks is therefore rudimentary. Avenues of communication are comparatively few, and such as exist are not systematically connected and are generally of poor quality. Until recently, this comparative impenetrability of the forests has doubtless tended toward their preservation; but conditions have changed to such a degree in recent years that the construction of good roads in the Adirondacks seems to be desirable both for the greater enjoyment of the Forest Preserve as a health and pleasure resort and for the greater safety of the forests themselves. The increased appreciation of the Forest Preserve as a refuge in summer time, the great progress made in methods of highway travel, and the increased facility which good roads would afford for visiting the woods, are in themselves strong reasons for the extension of the present highway system in that region. When, to the foregoing considerations, are added the very practical value of roads as fire lanes and the advantage which they would afford in reaching forest fires, the argument for their construction becomes very strong.

The attitude of this Association with reference to the bearing of the Constitution on the subject of roads in the Forest Preserve is that if good roads be necessary to keep the forest lands as "wild forest lands," in the words of the Constitution, they should be allowed. An opinion of Attorney General O'Malley, given to the Forest, Fish and Game Commissioner on or about June 22, 1910, however, has been interpreted to mean that no roads can be built on Forest Preserve land under the Constitution. The question was raised by the Superintendent of Roads in Franklin County, who asked permission to use stone from a ledge of rocks on neighboring State land for road purposes, promising not to cut away any timber or otherwise damage the land. In his opinion, the Attorney General said in part, "It was clearly the intent of the framers of the Constitution to preserve the lands constituting the forest preserve in their natural state, and therefore you have no authority to permit county officials to use the stone in the ledge referred to." In order, however, plainly to authorize the construction of highways in

the Adirondacks, our Association caused two alternative propositions to be introduced in the Legislature of 1910. One was in the form of a bill providing that when validated by an amendment to the Constitution it should be lawful to construct upon State lands in the Forest Preserve any of the State highways described in section 120 of Chapter 330 of the Laws of 1908, and any of the county highways designated upon a map already prepared by the State Engineer and Surveyor, as provided by law and approved by the Legislature by Chapter 715 of the Laws of 1907. The bill limited such highways to a width of 4 rods, provided that they should be built and maintained under the supervision of the State Highway Commission, and imposed certain other conditions with reference to keeping the highways clean, removing inflammable material, the exclusion of railroads, the public inspection of maps of routes, etc. This bill, if enacted, was not to become effective until validated by an amendment to section 7 of Article VII of the Constitution specifically referring to it by chapter number and year.

After that bill was introduced it appeared that the same end might be attained, without becoming complicated with other questions relating to section 7 of Article VII, by amending section 12 of the same article referring to Highways. We therefore caused to be introduced a Concurrent Resolution to amend section 12 of Article VII of the Constitution by inserting after the first sentence these words: "Any county having part of the forest preserve therein shall receive its equitable apportionment of highways. Highways within the forest preserve shall be opened or improved in the same manner as other highways in the State, except that they shall not be laid out to a greater width than 100 feet or improved for a greater width than other highways in the State under similar conditions." But this proposition, so highly desired by the State Highway Commissioners, by the Forest, Fish and Game Commission, by the local communities in the Adirondacks, and by the visitors to the Adirondacks, was smothered in the Ways and Means Committee through the opposition of the Chairman, who was also the majority leader in the Assembly, who is financially interested in water-storage, and who was evidently determined that no legislation beneficial to the Adirondacks should be passed until the private interests which he represented had secured what they wanted in the way of

permission to build storage reservoirs on State lands. For this reason, then, forest conservation by road building is at a standstill.

5—Replanting of Denuded Areas. Constructive forest conservation, that is to say, the building up of new forests to take the place of those removed, has made some progress in New York, but not so rapid as could be wished. The fault has not been that of the Forest, Fish and Game Department, but of the Legislature which has not furnished the means for the liberal prosecution of this work. The State has good nurseries and expert help, but lacks means to prosecute this branch of its work in the manner which its importance warrants. Fortunately, private owners are taking up the subject of replanting effectively. The International Paper Company, for instance, has adopted the policy of tree-planting to renew its crops, and has a large nursery at Randolph, Vermont, from which it is distributing young plants to different sections of the country, including the Adirondacks, where it owns and controls lands. Within the limits of the Adirondack Park there are about 120,000 acres of State land which should be replanted, and in the Catskill Park about 30,000 acres. As to the cost of replanting: last year the Forest, Fish and Game Commission sold about 1,000,000 trees to 180 private parties for reforestation, and a careful analysis and average of their reports by the Forest, Fish and Game Commissioner indicate that reforestation cost these parties, including cost of stock, expressage, and labor, \$8.50 an acre. We are informed that the State could reforest to advantage from 2,000 to 2,500 acres a year, and could supply material for planting at least 30,000 acres a year on private land.

The importance of conservation by reforestation becomes apparent when one takes into consideration the relative rates of forest removal and forest reproduction. In the United States at large, we take from our forests each year, not counting the loss by fire, three times their yearly growth. We take 36 cubic feet per acre for each 12 cubic feet grown. We take 230 cubic feet per capita, while Germany uses 37 cubic feet and France 25 cubic feet. In the State of New York we are cutting away our trees five times as fast as they grow, and at the present rate of denudation, the State will be rendered practically barren of forest growth—except in the Forest Preserve—within 20 years, unless there is a decided change in the proportion between tree-cutting and tree-planting.

6—*Tree Destruction by Flooding.* A source of tree destruction of no inconsiderable extent in the Adirondacks in years gone by has been flooding by lumbermen's dams. The seriousness of this phase of the forest question has been greater than the area of destruction might indicate, for the reason that, in addition to the loss of the trees killed, unsightly and unhealthy conditions have been produced which have robbed certain regions of important elements of value. Prior to the adoption of the Constitutional Amendment of 1894, which prohibited the removal or destruction of timber upon the lands of the Forest Preserve, it had been the practice for nearly fifty years to build dams in the Adirondack region either for the purpose of driving logs or in connection with canal feeders. Those were days of prodigality, when the great North Woods stood in almost their pristine condition, and when the lumbermen, in the presence of thousands of square miles of luxuriant forests, thought nothing of killing thousands of trees by drowning. Almost every dam, therefore, that was built in the woods, set back the water upon forest land and killed trees. A dam built at Forestport in 1848 and subsequently enlarged killed so many trees that the State had to appropriate thousands of dollars simply to remove the dead trunks. About 1879 the State built a dam at Old Forge on Moose river, which is the outlet of the famous Fulton Chain of lakes, and subsequently built a dam at the outlet of the Sixth lake of the chain. These dams raised the water in the various lakes from one to six feet, blighting the adjacent timber and producing a scene of desolation the vestiges of which are still evident after a lapse of thirty years. In 1886 and 1887 the State built a dam on Beaver river at Stillwater, raising the water 9 feet. Great areas of timber land for a distance of 20 miles were flooded and the trees killed. The whole basin became filled with a tangle of drift-wood; great swamps were created beyond the flow line, springs were covered up and polluted, and the region rendered so unhealthy that land became unsalable. Lovely lakes and ponds were submerged, and favorite camp-sites obliterated; feeding grounds for game were destroyed; and hunting in that vicinity was ruined. The magnitude of the damage may be judged from the fact that one of the adjacent property owners, Mr Wm. Seward Webb, sued the State for \$184,350 damages. The claim was settled by the State buying from the claimant 75,377 acres, including the damaged area, for \$600,000. In 1865 the building of a dam was authorized on Oswegatchie river at the mouth of Cranberry lake; this dam created a reservoir of 13 square miles flooding

thousands of acres of land, destroying large quantities of timber, and creating unsightly and unsanitary conditions. About 1882 a dam was built on Raquette river below the Tupper lake outlet, with the result that soon the region between Big Tupper and Little Tupper lakes looked as if some terrible blight had fallen upon it. The scene in 1893 is thus described in the Forest Commission's report:

The serious and extensive damage caused by the dam arrests the eye, presenting one of the saddest and most desolate pictures of destruction ever witnessed. No forest fire or devastating cyclone or ruthless axe of the charcoal burner ever wrought such ruin or left such a blasted scene as this. For ten miles the lands along the Raquette river are covered with the white and ghastly skeletons of the noble trees which once made this spot a sylvan paradise. The bare trunks, bleached by the sun and storm, the gnarled roots and gray, scrawny limbs thrust sharply forth, recall to mind one of Dore's pictures in the "Inferno." The traveler gazes on it all with amazement, and then gives vent to the strongest words that a righteous indignation can supply. And this was once one of the most beautiful rivers in all the wilderness.

Illustrations of this sort could be multiplied to show the spirit of indifference to tree destruction in the past, and conditions which are now forbidden to be repeated upon State land. The Constitutional Amendment adopted in 1894, prohibiting the destruction of trees in the Forest Preserve, was aimed at this evil among others, and has been one of the most valuable instruments in this State for forest conservation.

Water Conservation

The subject of water conservation in the State of New York presents five different aspects:

- 1—The development of hydraulic or electric power,
- 2—The improvement of commercial waterways,
- 3—Flood prevention,
- 4—Sanitation,
- 5—Domestic use.

As might be expected in the largest manufacturing State in the Union, there is in New York a very general appreciation of the importance of water storage for the development of power for industrial use; therefore, of the different phases of the water-storage question now pressed upon public notice, that one probably commands the most attention at the present time.

1—Power Development. The Association for the Protection of the Adirondacks is chiefly concerned with this question as it bears on the Adirondacks; but owing to the fact that waters originating in part in the Adirondacks flow in many cases to great distances beyond that region, it is impossible to treat the subject as one of purely localized interest. The question naturally arises, What proportion of importance is there between the question of water storage in the Adirondacks and water storage in the State at large? On the face of things, the proportion seems small. The watersheds of the whole State aggregate 30,476,800 acres, while the State lands within the Adirondack Preserve with which we are chiefly concerned comprise only 1,530,559 acres, or less than 5 percent. A comparison of possible water-power developments shows a similar disproportion. The Fourth Annual Report of the State Water Supply Commission says that "With the complete utilization of all storage possibilities an eventual development amounting to not less than 1,500,000 horsepower, exclusive of Niagara and Saint Lawrence rivers, is possible for the entire State." If, to this estimate be added the existing 200,000 horsepower development at Niagara Falls, 100,000 horsepower as the resource of the lower Niagara, and 400,000 horsepower for the Saint Lawrence, an eventual total of 2,200,000 horsepower for the whole State does not seem to be beyond the range of possibility. From figures derived from the various sources it would appear that about 7½ percent of this development would require encroachment upon State land in the Adirondack Park, which is now forbidden by the Constitution. When it is considered that attention has been concentrated for several years on the resources of the principal Adirondack streams, while the possibilities of the rivers outside of the Adirondacks have not yet been completely explored, there is much reason to believe that were the census of the hydraulic resources of the State complete it would be found that the ratio of the power possibilities of State Forest lands to the power possibilities of the whole State is about the same as the ratio of the respective water-sheds, or about 5 percent. There are two or three reasons,

however, why the question of water storage in the Adirondacks assumes an importance quite out of proportion to this ratio. One is the acknowledged fact that the majority leader of the larger house of the State Legislature is personally interested in water-power developed from Adirondack waters, and desires to have the Constitution amended so that State lands may be flooded for the benefit of his own as well as other private corporations. This powerful member of the Legislature has the sympathetic support of the Speaker of the Assembly, who stands sponsor for a power corporation on Genesee river, on the banks of which the Speaker lives. With the water-power interests thus strongly represented in the Legislature, and with some of them casting covetous eyes on State land from which they are restrained only by the Constitution, it is not surprising, perhaps, that in the public agitation of the water-storage question such statements should be made as that "the most important single obstacle to the carrying out by the State of this conservation policy" is "the necessity of amending the Constitution" so as to permit the flooding of State land.

Now the attitude of this Association—and this may be of interest to other States where the same question may arise—is as follows: At the outset, the Association opposed amending the Constitution for the purpose of permitting the flooding of State lands on two grounds; *first*, on account of the disastrous consequences to the forests which have invariably followed the construction of reservoirs in the past, and *second*, because it involved the principle of using public lands for private purposes without any guarantee of proportionate returns to the people whose domain was thus used. For several years the Association, with the unquestionable support of public opinion, maintained that position for the reason that there appeared to be no safe way of compromise.

During the past year, however, as the result of painstaking study of the problem by the New York Board of Trade and Transportation and our Association, a plan of legislation was evolved which it is believed may safely be adopted, and which, while conserving the public interests in the Adirondacks, will permit a reasonable use of State land for the purpose desired by the water storage people. The first problem encountered in working out this plan was presented by the fact that if the Constitution were amended generally so as to permit the flooding of State land, nobody could foretell to what extent or in what manner the lands might be flooded. It was

therefore decided to prepare a law which should prescribe all the limitations and regulations in advance, and which should contain a provision that it should not become effective until validated by a constitutional amendment. Then, after this law had been enacted, it was proposed to adopt an amendment to the Constitution referring to the law specifically by chapter number and year, and permitting what was provided therein and nothing more. In pursuance of this plan, such a bill was drafted and introduced in the Legislature at its session which closed in May, 1910. It provided that storage reservoirs might be built upon State lands in certain specified watersheds at certain specified points; that the flow-lines should be accurately surveyed and permanently monumented; that the total area of State land flooded should not exceed certain stated amounts—approximately 3 percent of the total area of the Forest Preserve; that all trees, stumps, and other organic material should be removed from within the flow-line; and certain other conditions designed to protect the public interests in the construction, maintenance and use of the reservoirs and the water-power developed therefrom. The law was not to become effective until validated by an amendment to the Constitution, and the constitutional amendment was to consist simply of an addition to the present section 7 of Article VII to the effect that "The provisions of this section may be modified as provided in chapter — of the laws of 1910, but in no other respect whatever." By this plan it was believed that the safeguards would be erected in advance, and in voting for a constitutional amendment our citizens would know exactly what they were voting for. The bill, however, was defeated through the influence of the majority leader of the Assembly, and instead a concurrent resolution to amend the Constitution, proposed by him, passed the first of three requisite stages of adoption. The provisions of this amendment and the utterances of its author clearly reveal the attitude of the water-power interests represented by him, and present an issue of importance to every State in which the question of Conservation under State auspices may arise. This issue, in brief, is whether, after the State has granted the use of land already belonging to the people and has acquired additional land in the exercise of its power of eminent domain; after it has furnished the capital for building storage reservoirs and for managing them when built, the profits shall accrue only to the private individuals or corporations benefited thereby, or whether the State itself shall derive a reasonable revenue from

its lands and reservoirs for the relief of taxation, or for public improvements, to the consequent benefit of all the people?

The Constitutional Amendment proposed by the water-power interests in the last Legislature provides only that the actual cost of the water storage shall be paid by the private beneficiaries, leaving to them all of the profits and advantages; and the author of the amendment publicly declared himself as opposed to the periodical regulation of charges for the use of water thus conserved, or to paying anything more than the bare cost of construction and administration. On the other hand, the proposition of this Association left the question of State revenue open for future legislation without any inflexible constitutional provision one way or the other. There the matter rests at the present moment. The issue remains to be fought out in the future, possibly in the Legislature of 1911, possibly at the polls the following November, and possibly later. At present the signs of the times are not encouraging to the belief that private interests will be given such valuable privileges without some reasonable return to the people from whom they are derived.

2—*Improvement of Waterways.* Water conservation for the improvement of commercial waterways has little connection with the Adirondacks. The principal waterway improvement now in progress in New York State is the enlargement of Erie Canal at a cost of \$101,000,000. Very little of the water for the canal comes from the Adirondacks, and the construction of reservoirs on State forest land is not required to augment the supply.

3—*Flood Prevention.* The three principal streams within the borders of New York—the Genesee, Mohawk, and Hudson—are subject at times to disastrous floods. These are in no small part the result of human folly. In the first place, the indiscriminate denudation of forests of the greater part of the State has removed one of the most valuable natural regulators; and it is the universal complaint that such denudation has resulted in the spasmodic flow of streams which are dry or low at one season and raging torrents at another. In other cases, as for instance at Rochester, on the Genesee, the river has been obstructed by bridge piers unscientifically placed, which obstruct the flow of water and cause great damage. The Hudson, from the confluence of the Mohawk to Albany, is also subject to floods, and as the headwaters of the Hudson rise in the Adirondacks it has been argued by those who desire

to have storage reservoirs for power purposes in Adirondack Park that the Constitution should be amended so as to permit the building of reservoirs in the Adirondacks to control the floods of the Hudson. As a matter of fact, the statistics furnished by competent engineers show that 75 percent of the floods at Troy and Albany are due to waters which do not originate in the Adirondacks, but can be controlled along the Mohawk; and that of the remaining 25 percent over half (say 15 percent) are due to water originating along the Hudson and its tributaries outside of Adirondack Park. So far, then, as flood control is concerned, it has little bearing on Conservation in the Adirondacks.

4—*Sanitation.* Except as a subterfuge, there is practically no connection between the subject of water conservation in the Adirondacks and sanitation. The Hudson is so polluted from Troy southward with sewage that the fish have been almost exterminated, and the industry of fishing on the Hudson which thrived within the memory of living men has almost disappeared. Sanitation of the Hudson from the head of navigation southward cannot be effected by storage reservoirs in the Adirondacks. The only prominence which the question of sanitation ever had in connection with water conservation in the Adirondacks was from five to ten years ago when persons who desired to build storage reservoirs on State lands, for the purpose of driving logs or developing power, used the plea of "public health and safety" in petitions presented to the River Improvement Commission to disguise their real purpose.

5—*Domestic Use.* There are those who think that in time the Adirondacks may be drawn upon for municipal water supplies for cities in the Hudson valley. The extent to which New York City has reached out for her water supply during the past 70 years would seem to lend color to such prophecies. In 1842 New York City introduced a water supply from the Croton Reservoir 40 miles distant; at the present time it is building a great reservoir in the Catskill Mountains 90 miles distant. Many people believe that eventually New York will be forced to go to the Adirondacks 200 miles away for a pure water supply, and that the resources of the Adirondacks should be preserved against that need and should not now be parted with for private use when there is the possibility that in the future they will be required for all the multifarious uses of human existence in the great

metropolis. Water conservation in the Adirondacks for municipal use, therefore, is important chiefly with reference to the future.

Scientific Forestry on State Lands

As persons unfamiliar with the history of the Forest Preserve in New York may wonder why the State does not utilize commercially the timber growing on State lands, it may not be inappropriate to conclude this report with a brief explanation of the reasons for the iron-clad restriction placed by the Constitution on the removal of State timber. The reason for this restriction is two-fold: First, it is not apparent that there are enough trained foresters yet available or that the problem of the conservative handling of State forest lands for commercial purposes is yet sufficiently understood to warrant the State in undertaking scientific forestry; and second, the citizens of the State are not confident that if the removal of timber were permitted, the people at large would derive any benefit from it.

1—Lack of Practical Men. At a public meeting held in the American Museum of Natural History in New York under the auspices of this Association on April 25, 1907, Professor Henry S. Graves, then Director of the School of Forestry at Yale University and now Chief Forester of the United States, speaking on the subject of scientific forestry on the State lands in New York, said: "It would be exceedingly difficult at the present time to secure trained men with adequate experience to carry out a plan of successful forestry." That situation with respect to the dearth of practical foresters still exists and promises to continue until relieved either by the more general teaching of forestry in colleges and schools or by a more general training in the field, or both. Another drawback is the lack of systematic study and knowledge of our Forest Preserve. With the exception of Township 40 and adjacent territory, and possibly a few other tracts, little has been done in the direction of examining the land to determine its value, the amount and character of timber, the growth of trees, and the local conditions which are factors in the profitable management of the forests; nor has anything yet been done toward preparing a comprehensive plan for the whole Preserve.

A concrete illustration of the impracticability of scientific forestry under existing conditions is afforded by the experimental forest in Franklin

County established under an act of 1898. The hopes entertained in regard to this experiment were well set forth in the message of Governor Black to the Legislature on January 5, 1898. The Governor pictured in graphic terms the desirability of enlarging the Forest Preserve as a health resort and a conservator of the northern New York water-sheds, and referred to the rapid inroads made upon the forests by commercial lumbering, and to the protection which the Constitution extended to State lands. He argued that, properly managed, the State forests might be made productive of a substantial revenue; but, he said, "The Constitution should not be amended until the people have learned prudence instead of waste, and have equipped themselves with knowledge and experience adequate to the care of this great domain. Our conditions here are not like those in Germany and France, but in what respects they differ, few can tell." Then, with a view to the acquisition of this necessary knowledge and experience, he recommended the following plan:

There are students here who have made a careful study of the forests, their capacities and their needs. The number of these gentlemen I understand to be increasing, for through the labors of several of our citizens of great generosity and public spirit, the subject has been studied and discussed, and upon the general ignorance relating to this question there is beginning to be some light. The knowledge necessary to the proper treatment of the woods must come largely through experiment. It cannot be had unless the means of acquiring it are provided. I believe the means can be secured best through the purchase by the State of a tract of ground covered with those trees which are to be the subject of experiment. Such a tract the State could set apart and gain from it the knowledge which will enable it by and by to deal with the millions of acres it has already and will in the meantime acquire. The time will come when the State will sell timber to the lumbermen, spruce to the pulp mills, reap a large revenue for itself and still retain the woods, open to the public, protecting the sources of water, growing and yielding under intelligent cultivation. The management of this experiment should not be subject to the vicissitudes of politics. It should be placed in charge of the Regents, or of the Trustees of Cornell University, or of some similar body not subject to political change. The State should pay such reasonable sum as may be needed to administer

the plan. Reports should be made to the Governor and the Legislature annually of progress and results. The income from the tract so acquired should be paid to the State and the land itself should become the absolute property of the State, and a part of the Forest Preserve at the expiration of a period named. I believe such a plan would be soon, if not at once, self-sustaining, for the trees now ready to be cut would produce immediate revenue, and such revenue would be repeated at short intervals. The benefits could be hardly overstated, and in this direction, as in many others, the wisdom of New York entering upon a comparatively new and untried field would be finally approved.

Following Governor Black's recommendation, the Legislature of 1898 enacted a law pursuant to which 30,000 acres of forest land in Townships 23 and 26 in Franklin County were purchased for \$165,000 and conveyed to Cornell University for the purposes of a "New York State College of Forestry;" and in the years 1898 to 1902 sums aggregating \$110,000 more were appropriated for salaries of the Director and instructors in the College of Forestry and for working capital for improving, maintaining, and administering the College forest. With a view to making the forest self-sustaining, the University on May 5, 1900, made a fifteen-year contract with the Brooklyn Cooperage Company by which it agreed to deliver to the company annually one-fifteenth of the wood and timber standing in the College forest. The details of this contract and the litigation which ensued are not essential to the present statement, but the results of the experiment were highly important; instead of yielding the State a revenue, all of the moneys appropriated were used up except about \$9,000 of working capital, while about 3,100 acres of forest land were denuded and only about 440 acres replanted. The results were so obviously disappointing that in 1903 Governor Odell vetoed the appropriation of \$10,000 for that year, and since then no appropriation for the College of Forestry has been made except one of \$5,000 in 1903, exclusively for the purpose of removing the underbrush and for replanting trees. Soon thereafter (June, 1903) Cornell University discontinued the College of Forestry. In his message to the Legislature in 1904, Governor Odell, speaking of the School of Forestry, said: "Its operations had for their object the substitution of valuable growths for so-called worthless timber, but this has resulted in the practical destruction of all trees upon the lands where the experiment was in progress. No

compensating benefits seem possible to the present generation. The preservation of the forests is primarily for the protection of the water supply, and this is not possible through the denudation of the lands. Therefore this school failed of its object, as understood by its founders—a failure which was not due, however, to the work of the University, which followed out the letter and the spirit of the law."

Mr Justice Chester, of the Supreme Court of the State of New York, in his opinion rendered in June, 1910, in the case of the People of the State of New York against the Brooklyn Cooperage Company and Cornell University,^[6] said that there could be no net revenues from the College Forest, as the expenses exceeded the income. He also pointed out how, under the operation of the contract, practically the entire College Forest would be denuded for the benefit of a private industry and not for the promotion of education in forestry. "There is proof in the case," he said, "that 500 acres were sufficient for conducting experiments on the 'clear cutting' system of forestry as distinguished from the 'selection' system."

Notwithstanding the failure of the forest experiment, Governor Odell in 1904 hoped that the Forest School would be continued: "Because," he said in his message, "with the lapse of years, a proper understanding of scientific forestry will become more and more a necessity." What Governor Odell said remains true. But what is needed is not only scientific knowledge but also knowledge of local conditions. A high order of theoretical knowledge was brought to the management of the Cornell tract, but the experiment failed for lack of knowledge of local conditions and business prudence.

2—Lack of Confidence that Benefits will Accrue. The second obstacle to the introduction of scientific forestry upon State lands is the lack of confidence that if the forest products were utilized any benefit would accrue to the people generally. The feeling may be understood in the light of the history of the Forest Preserve. In its beginnings, this was not a deliberately planned institution, but grew up in haphazard fashion, without forethought or system. Once the State owned nearly all the land within the Adirondack wilderness, but prior to 1883 there were no laws which prevented the State from parting with its lands, and large areas were sold to private parties for almost a song—lands which the State has gradually been buying back ever since at constantly increasing prices.^[7] In a message to the Legislature in

1882, Governor Cornell called attention to the shortsightedness of this policy, in these words:

By far the greater quantity of land within the Adirondack wilderness proper belongs to the State. Individual ownership is now confined to a few hundred thousand acres. Heretofore it has been the practice of the State, with questionable policy, to sell its wild lands at nominal prices to private parties, who have gone on, in most cases, and cut off the marketable timber where accessible, and then abandoned to the State the clearings, worthless generally for agricultural purposes, thereby escaping the payment of taxes. Forest fires have followed and raged with destructive fury, denuding the mountains and checking the flow of springs and streams that supply the navigable waters to the north and the Hudson river to the southward. Furthermore, many of the lakes, the natural reservoirs of the mountain courses, have been damaged by dams and overflow, so that the shores of those lying within the working timber limits present the effects of irreparable injury.

In 1883 a law was enacted which prohibited the sale of any State lands in the counties of Clinton, Essex, Franklin, Fulton, Hamilton, Herkimer, Lewis, Saratoga, Saint Lawrence, and Warren, and by subsequent acts the counties of Oneida, Washington, Delaware, Greene, Sullivan, and Ulster were added to the list. Prior to that year the State had recovered about 800,000 acres of land which the owners had permitted to be sold for taxes—patches of land scattered here and there without any system or studied continuity. After the passage of the laws forbidding the sale of State lands the value of the lands began rapidly to appreciate, and private parties, desiring to acquire it endeavored to circumvent the law prohibiting the sale by attacking the State's tax titles. With the aid of pliant State officials, these efforts in many cases were successful, the State either parting entirely with its title or, retaining the title to the soil, parting with the title to the timber. In this manner the State lost about 100,000 acres of land. A report made to the Comptroller in 1895 showed that these cancellations were made with disregard of the law and the rights of the State. As the result of all the tax-sale transactions of the State, it has acquired about one-half of its present forest-preserve holdings in the Adirondacks. The other half was acquired by

purchase. The first actual appropriation of money for the purchase of land for forest purposes was \$10,000 appropriated in 1883 during Grover Cleveland's administration. In 1885 the Forest Preserve was established by law, and since then the building up of the Forest Preserve has proceeded with more intelligence and upon a more definite policy. Up to the present time, the State has spent about \$3,800,000 on the purchase of lands for the Adirondack and Catskill forests.

While the Forest Preserve was thus being evolved, other evils than the illegal cancellation of State titles developed. While the statutes—subject to change at any time at the wish of the Legislature—forbade the sale of State lands, there was nothing, to prevent the sale of the timber on the land. In 1893 Governor Flower, whose friendship for the forests was unquestionable, recommended to the Legislature that "the State could acquire considerable revenue by granting permission to fell trees above a certain diameter on State land." But the policy thus proposed with the best of intentions was a disastrous one, for the reason that with the reckless lumbering methods employed the lumbermen would destroy fifty trees while taking out one.^[8] By 1894, with the juggling in titles to State lands, the destruction of trees in lumbering operations, the killing of trees by flooding, the creation of unsanitary conditions by dams, and the general misuse and mismanagement of the State forests, conditions became intolerable, and the Constitutional Convention of that year adopted the stringent section before quoted (page [399](#)). Every word was carefully weighed, and designed to meet some phase of the situation. The necessity was so obvious that it was adopted without a dissenting vote by the Convention, and subsequently was overwhelmingly ratified by the people.

Since then, persistent efforts have been made by the lumber and water-power interests to impair this safeguard, but without success. We do not believe that the time has yet come to relax this section of the Constitution with respect to timber cutting; for while it is true that during the past few years conditions in the management of the Adirondack Forest Preserve have greatly improved and the public confidence in the possibility of the proper utilization of our forests had begun to take root, it is an unfortunate fact that that confidence has received a severe set-back by the course of legislation in 1910 with reference to the use of Adirondack waters. When the controlling powers in the Legislature are hostile to the idea that the State

shall derive a revenue from its waters, it cannot be said that the auspices are propitious for the State's deriving any revenue from its timber. We do not believe that the people of the State are prepared to part with their forests upon the terms upon which they are asked to build storage reservoirs and furnish water-power to private interests, that is to say, for the bare original cost of the timber.

It therefore appears to be the part of wisdom for the people to defer scientific forestry on State lands while the present attitude of the legislative mind continues, and to preserve their forests intact until the prospect of deriving a revenue from them is better.

[Signed] WARREN HIGLEY,
First Vice-President
EDWARD HAGAMAN HALL,
Secretary

REPORT OF THE CARRIAGE BUILDERS' NATIONAL ASSOCIATION

Soon after the Conference of Governors called by President Roosevelt in the White House, May 3-15, 1908, the Carriage Builders' National Association appointed a Committee on National Conservation, which has submitted two reports adopted by the Association. The last report, recently adopted, covers the items in which the carriage trade is most vitally interested. In addition to data taken from the Report of the National Conservation Commission, it summarizes the work and opinion of our Association on the important subject of Conservation.

A late census report showed in its lumber cut a total of 203,211,000 board feet of hickory as compared with 9,255,000,000 feet for all hardwoods. This would indicate that the hardwood forest at present contains a little over 2 percent of hickory; probably as much as 4 percent for the entire hardwood area. The forest of the eastern half of Kentucky has been estimated recently to contain about 5 percent of hickory. The lumber cut does not show the large quantity of hickory which is cut and shipped in the form of round billets, rived or split spoke stock, etc. This form of material is frequently culled from the forest ahead of the lumberman, and tends to cause the low percentage of hickory in the lumber cut before noted. Including this with the 203,000,000 feet of hickory lumber would raise the total cut to at least 350,000,000 feet per year.

Add to this hickory cut for fuel in localities with no transportation facilities, and the heart, pecky, and other portions wasted, and the total soon amounts to 400,000,000 feet. If hickory forms 3 percent of this forest (much of which is culled already for hickory—the lumber cut alone showing a little over 2 percent) there would be a total stand of 12,000,000,000 feet of hickory. Much of this is mature timber, with an annual growth of less than $1\frac{1}{2}$ percent. Hence there may be figured a growth of less than 180,000,000 feet against a consumption of about 400,000,000 feet. Though this is to some extent speculation, when supported by increasing difficulty in getting hickory timber and with rising prices, it is nevertheless sufficient to indicate

that a thorough study of the growth of hickory is one of the important steps in attempting to plan relief measures.

The report made to President Roosevelt was enthusiastically received, and an organization was formed to bring about a campaign of education among the people of the United States on National Conservation of our resources. In turning over the office of President to William H. Taft, Theodore Roosevelt recommended to him strongly the work of National Conservation, and reports through the press have shown that he is very enthusiastic and is taking a live interest, notwithstanding some of the newspaper reports regarding the controversy between some of the members connected with the Association, which, in our judgment, has been a splendid advertisement for the cause.

We are also pleased to report that the National Hickory Association of the United States (whose membership is composed largely of the members of our Association) have taken a great interest in this work of Conservation, and have taken an active interest with the National Conservation Commission appointed by President Roosevelt in making up their report. They also held an enthusiastic meeting in Cincinnati last April, passing resolutions to work toward the end of having a permanent National Conservation Committee appointed by the Government, and also in the various States.

Your committee recommended that all our members take an active interest and cooperate with the members of the National Hickory Association and the National Conservation Association, and offered the following resolutions which were adopted:

"Resolved, That we heartily endorse the work of the National Hickory Association and assure them of our hearty cooperation.

"Resolved, That we favor the maintenance of Conservation Commissions in every State, to the end that each commonwealth may be aided and guided in making the best use of those abundant resources with which it has been blessed.

"Resolved, That we also especially urge on the Congress of the United States the high desirability of maintaining a National Commission on the

Conservation of the Resources of the Country, empowered to cooperate with State commissions, to the end that every sovereign commonwealth and every section of the country may attain the high degree of prosperity and the sureness of perpetuity naturally arising in the abundant resources and the vigor and intelligence and patriotism of our people.

"*Resolved*, That a joint committee be appointed by our chairman, to consist of six members of our Association, whose duty it shall be to work in harmony with the State and National Commissions and the National Hickory Association."

Respectfully submitted,

[Signed] H. RATTERMANN, Cincinnati,
Ohio

Chairman

J. D. DORT, Flint, Mich.

DANIEL T. WILSON, New York City

E. W. M. BAILEY, Amesbury, Mass.

GEORGE H. BABCOCK, Watertown, N.
Y.

WILLIAM A. SNYDER, Piqua, Ohio

W. P. CHAMPNEY, Cleveland, Ohio

D. M. PARRY, Indianapolis, Ind.

MAURICE CONNOLLY, Dubuque, Iowa

LUCIUS GREGORY, Chase City, Va.

Committee

REPORT OF THE DELAWARE STATE FEDERATION OF WOMEN'S CLUBS

As the one Delegate from the State of Delaware, I feel that I must speak a word for her. Delaware has an enviable list of great names, from Cæsar Rodney, whose memorable ride turned the scale in the vote for liberty, with Thomas F. Bayard and John Clayton, down to the present time, when we have a man like Judge George Gray to be proud of.

The Delaware State Federation of Women's Clubs, which I represent, goes hand in hand with the women of sister States in this great movement. Our Legislature has appointed a State Forester—and the Granges and our Agricultural College at Newark are working to improve our soil and crops, while our women are supplementing their efforts wherever they can. We are cooperating with the Red Cross in the fight against the White Plague, and have succeeded in having a child labor law enacted, and are now working for a juvenile court. We have offered prizes to the public school children for the best essay on waterways; and we are beautifying our waterfronts and securing pure water. We have no great forests, but we raise the best peaches in the world and are rapidly coming to the front in apple culture, and we are going to keep up a ceaseless educational campaign, so that our people will realize the importance of conserving our natural resources.

I consider it a great honor and privilege to represent the women of Delaware at this great Congress, and thank you for your attention.

[Signed] CORNELIA R. HOLLIDAY

REPORT OF THE FARMERS' UNION OF AMERICA

It is a matter of great regret to me that the National Convention of the Farmers' Union occurs almost simultaneously with the gathering of the Second National Conservation Congress.

I regard the question of Conservation as one of the very greatest now before this country. I regard Gifford Pinchot as the father of the Conservation idea in America. I believe that future generations will credit his activity in awakening the American conscience to almost criminal extravagance in exploiting our resources as one of the most practical displays of patriotism in National history.

I trust that the deliberations at Saint Paul will be attended by much progress and profit. Let me beg also that while you concentrate on resources, you do not overlook the conserving of that greatest of our resources—the American Farmer. I regard his uplift of first importance to the present welfare and destiny of America.

I shall hope that such steps as you take during the current session will be of far-reaching influence in directing the vital thought of an aroused people.

[Signed] C. S. BARRETT
President

REPORT OF THE GENERAL FEDERATION OF WOMEN'S CLUBS

It gives me great pleasure to report to this Congress the work undertaken and accomplished by the Waterway Committee of the General Federation of Women's Clubs during the sixteen months of its existence.

Every State federation in the Union was asked to assist in this movement by adding to their standing committees one called Waterways; and ready responses came from many States. The work as outlined for each State falls under three departments, Civic, Educational, and Publicity. In this way the work can be systematized and developed along the lines to meet the needs of each locality.

We have been told that our country stands foremost in waterway richness; with its many splendid rivers and great lakes, as it is well nigh girdled by oceans. Plans are rapidly maturing for the celebration of the short route to the East through Panama in 1915. From the dawn of history to the present time, civilization has followed the water routes; all the great cities are on, or in close proximity to, waterways. The date of the rapid reaching of railroads in every direction throughout our land was the signal for the neglect and non-use of water highways, until in the majority of cases the river fronts have been absorbed for railroad ways. There are now scarcely any good terminal facilities to be found for water transportation. To meet the problems confronting us in regard to our waterways, women resolved that there must be instituted a campaign for education, such an education that the awakening resulting therefrom shall become a force of tremendous energy.

Man must know that in giving development to a stream it must be improved from its source to its mouth, and for its every use. Storage dams should be built at every available point. The fish raised in the reservoirs thereby

created will soon pay for the outlay in construction. It is estimated that by fully conserving the waters and utilizing the water-power developed in connection with storage and other works, three times as much land can be reclaimed in the western half of the United States. Such dams will decrease largely the annual damage from flood waters, with which we are so familiar, as well as regulate a more even stream-flow. A larger and purer water supply will be assured; water for irrigation in the more accessible regions will be afforded. An improved stream provides cheaper power for manufacturing purposes, stimulates various industries, and thereby furnishes larger fields of employment. If the limitation of streams as self-clarifiers were better understood there would be such protection given to them and their water-sheds that there would be no more refuse, laden with typhoid, cholera and inflammatory intestinal germs given to them, especially if the great distances these germs travel and their tenacity of life were better known. The developed stream affords water for transportation when the stream is navigable, which affects both the producer and consumer from the remotest section to the heart of the Nation.

It costs no more to develop the average stream than to build a railroad of the same mileage, but the improved stream carries 125 times as much freight per year as can be carried by rails, and at one-sixth the cost. Some 75 percent of the total freight commodities originating on the traffic lines in the United States consist of heavy raw materials, the staple productions of the farms, the forests, the mines, and the live stock ranges of the interior. These are commodities where economy of transportation is a prime essential to production. The even stream-flow which comes from improvement gives moisture to the agricultural lands along the banks; the trees at the head waters and outlining its meanderings testify to the interdependence of forests and streams. An improved river system as outlined in these suggestions also necessitates drainage of all lowlands, save those suffering from the encroachments of the sea.

At a glance we readily see that the development of waterways affects the Nation at large and man individually in a more vital way than any other of the natural resources. The idea is generally prevalent that the development of our Nation's waterways is pre-eminently man's work, and that there is nothing for the women to do. Yet there is not one phase of waterway development that does not directly or indirectly touch every home of this

Nation. Who is there, then, to say that it is not the duty of every woman as mother and citizen to inform herself thoroughly on so vital a subject that she may be among the most active educators in this great campaign? In almost every great sociological and reform movement, women have been the originators; and today they are the dynamic forces which destroy the evils that are opposing civic righteousness. Shall the homemaker refuse to protect her household from one of the greatest sources of physical infection which follows in the wake of modern indifference to pure water supply? Purity in water means health, impurity means sickness and death.

Every year millions of dollars are spent by Americans in travel in the older countries. We read beautiful descriptions of voyages down the Rhine. Along the Thames the Victorian embankment adds glory to London. The little River Seine with its many canals, making Paris, though inland, one of the greatest ports in France, remains beautiful throughout its length; flowing through the center of Paris, it has been kept decorative, banked with foliage and flowers, skirted by long lines of graceful masonry, with pleasure promenades, bordered on either side with beautiful statuary and sparkling fountains. Does it not fill your heart with a sense of mortification to compare these water fronts of European cities with the water fronts of our American cities? Public beauty excites that love of country which is at the very foundation of true patriotism. Let us resolve within ourselves to reverse these conditions, and bend our energies to improve and make of our waterways the most beautiful in the world.

Reports from the 39 States now in active work along these lines have shown great returns from the efforts put forth. We have 619 federated clubs showing definite results of their undertakings. In one State a splendid reference library on "Waterways" has been established; in another a great warfare was waged for pure drinking-water, the women going to the polls and making a fight for the sand filtration plan. Sixty-three clubs have reported making sanitary and parking water fronts as their especial work with splendid results. Prizes have been offered in many States to school children for the best essay on "Inland Waterways": over 5000 children in one State alone entered this contest. Placing Conservation in the public schools has been accomplished in several States; in every State great work is being done along educational lines, with the hearty cooperation and support of the superintendents and teachers. This subject has been given

place on 150 programs of State, district, and local meetings of various organizations; and many speakers have addressed schools and club assemblies. The press has been most courteous in every State in its cooperation with this Committee; 101 different articles have been published in all the prominent newspapers throughout the States. The Waterway Committee of the General Federation have sent delegates to waterway conventions in a number of States. There is scarcely a club in the Federation that has not given at least one number on its program, if not the entire program, to the Conservation of our natural resources.

Fifty thousand circulars and pamphlets have been sent from the Chairman's office and distributed throughout the States by the different chairmen. The great demand for waterway literature from every quarter convinces us of the growing interest in this subject. Thus we stand as strong allies in this great Conservation movement.

[Signed] MRS J. D. WILKINSON,
Chairman Waterways Committee
(Reported through MRS G. B.
SNEATH)

REPORT OF THE LAKES TO GULF DEEP WATERWAY ASSOCIATION

I bring greetings from three different bodies allied in this work: the Business Men's League of Saint Louis; the Missouri Waterways Commission, of which I have the honor to be Chairman; and the Lakes to the Gulf Deep Waterway Association, of which I have the honor to be President. On behalf of Governor Hadley and the State of Missouri, I wish to extend to this Congress the assurance that Missouri is for the policy of Conservation of natural resources in the way in which it is understood by most of you; that is to say, she is for the economical development of her resources in the highest degree, and at the same time for the preservation of the rights of the people in the control of those resources.

Some time ago, following out the policy advocated by Mr Gifford Pinchot and by President Roosevelt, Governor Hadley appointed the Missouri Waterways Commission to examine and report upon the water resources of

the State. In this department, Missouri is richer than many other States in the Union. Located in the center of the most fertile valley in America, she possesses two great rivers; the Mississippi, forming her entire eastern border, and the Missouri, exactly bisecting the State, connecting her two great principal cities. In addition to these there come down out of the Ozark Mountain region a series of smaller navigable rivers, the Osage, the Gasconade, the Big Piney, the Current, the Black, the White, and many smaller streams flowing into the great rivers and enabling boats to reach almost every part of the interior. In the course of time all of these rivers will be very much improved, and many of them made navigable. The sources of these streams are in the Ozarks, and they are fed by the most beautiful springs which are known to exist in America; one of these springs, named after our Governor, discharges, it is estimated, 50,000,000 gallons a day, even in the driest season—an amount equal to the entire consumption of a city of probably 50,000 inhabitants. There are many more which flow from 5,000,000 to 10,000,000 gallons a day. You cannot go a quarter of a mile along any valley road in the Ozark region without coming upon a spring oozing out of the limestone or sandstone cliffs, and adding its limpid waters to some brook or river. The crest of the Ozarks is 2,000 feet above the sea, more than 1,500 feet above Saint Louis, and all of these streams flow pell-mell down the hills to their navigable portions; so that the State has a very large amount of latent water-power. It is well to remember that the Ozarks remain forested, and that it is in the shelter of these forests that the waters gather to form the abundant springs and streams.

The Missouri Waterways Commission has employed one of the best-known hydraulic engineers in America, Mr M. L. Holman, to make a preliminary survey of these and other resources; and on this he is now engaged. When this has been completed, a report will be made to Governor Hadley embodying a policy for the control and development of this power, and this policy, it is expected, will be recommended to the next State Legislature by the Governor with the view of securing legislation conserving at the same time the water resources and the people's rights in them.

This is not, of course, the full extent of the Waterway Commission's work, for we have also to consider the use of the streams for navigation, a department in which the State is as much interested as the Federal Government, although we are not allowed to tamper with the navigable

rivers themselves. We are also to consider the reclamation of swamp lands, the preservation of soil, and the general use of water, which is today the Nation's greatest asset. In the last Congress an appropriation of \$1,300,000 was made for Missouri river, which means as much to Missouri as a part of its Conservation work as it does to the cities and the Nation for its value to navigation. Both the Missouri and the Mississippi are great devourers of soil. The Missouri will tear out an entire farm and ruin a farmer in an incredibly short space of time when it is changing its bed. The application of revetment to the banks and the contraction system in the effort, certain of success, to obtain a 6-foot permanent channel between Kansas City and Saint Louis, will return to the farmer, it is estimated, more than the entire outlay in additional capital wealth represented by the rich accretions of the Missouri bottoms. The securing of this appropriation and the very large appropriations also for the Mississippi fronting the State and leading from this beautiful city of Saint Paul all the way down to the Gulf of Mexico, has been largely stimulated by the work and activity of the Lakes to the Gulf Deep Water Association; and many of you will remember how much that organization has had to do with the doctrines of Conservation.

This reference to the Lakes to the Gulf Deep Waterway Association may be pardoned, when it is remembered that this Association has always stood for the complete utilization of the waterways for all purposes for which they are available, and that it has thereby become one of the most effective Conservation agencies in the world. It may interest you to know that we of the Lakes to the Gulf Waterway Association played an historic part in the early history of Conservation in this country. In October, 1907, the Association chartered a fleet of steamers and carried President Theodore Roosevelt from Saint Louis to Memphis to show a President of the United States for the first time the necessity of improving the inland waters. One of the steamboats which made that trip was the General McKenzie, and the passengers on the McKenzie were the Inland Waterways Commission appointed by President Roosevelt, upon the suggestion of our Association, to examine the question in hand. One of the members of this Commission was Gifford Pinchot; another was Mr Frederick H. Newell, head of the Reclamation Service; another was Dr W J McGee, Secretary of the Commission; another was Herbert Knox Smith, head of the Bureau of Corporations; and another was Alexander McKenzie, always a friend of the

waterways. On the steamer Alton, escorting the President, were the Governors of 22 States; and still another vessel bore about 75 members of the Federal Congress.

The second night out from Saint Louis was a stormy night; the rain fell in torrents, and the vessels made their way with great difficulty through the intricate channel of Point Pleasant reached from New Madrid southward. On that memorable night Gifford Pinchot and his associates in the Inland Waterways Commission came aboard the steamer Alton, and on the deck of that steamboat, protected from the storm by canvas awnings, held the historic meeting that gave birth to two great movements: Conservation, and the House of Governors. As a result of that meeting, where the policy of Conservation was fully laid out, President Roosevelt announced in his speech at the Lakes to the Gulf convention in Memphis that he would call a meeting of the Governors, and did call this memorable meeting of May 15-18, 1908, at which public sanction was given to the Conservation movement, and the House of Governors became an established organization. We have always felt that the place of the Lakes to the Gulf Deep Waterway Association in bringing about this meeting is one of the proudest achievements that the Association has on its records, and will live in history.

The Lakes to the Gulf Deep Waterway Association has always felt the necessity of allying itself with the Upper Mississippi River Improvement Association, the Ohio River Improvement Association, and the general Conservation movement for the best development of all river channels. The Mississippi today has the largest storage reservoirs in the world, although they are almost equaled now by the storage in the Salt River Irrigation Project in Arizona. But because of the cutting and burning of the forests, and the failure of the Government to complete the reservoirs, the Mississippi this year has been unnavigable above Saint Louis through the whole summer season. Nothing but conservation of the head-waters—and it must be remembered that adequate attention should be given to the forests about the head-waters—can prevent a recurrence of that circumstance in the next drought. The reservoirs which are now established should be supplemented by others on the Wisconsin, the Flambeau, the Chippewa, the Minnesota, and all the other streams flowing into the upper river, and some scheme for conserving the waters of the Ohio, although it will come at great

expense; and the Tennessee also must be dammed and reservoired, both to withhold the floods and to conserve the water for dry-season navigation. Costly as these reservoir systems may be, it will require but little figuring to show that, again in league with the Conservation policy and a light charge by the Government on the water-power in these navigable streams, they will return interest and sinking fund on the cost of the improvements. Here in Saint Paul, and between here and Minneapolis, we have an illustration of the great lack of proper development in the series of falls and rapids—not half of which is properly utilized—on which the Government has spent much money and for which the people receive no return whatever.

But the Lakes to the Gulf channel is a magnificent illustration of Conservation. It requires, as in Illinois, the cutting of 100 miles of canal through rock and riverbed, and the building of dams which will develop 150,000 horsepower; and the use of the money from that power now going to waste will pay the entire cost of this expensive rock channel (this in itself is an ideal example of Conservation). In the Mississippi reach between Grafton and Cairo, which is to be deepened to 14 feet or more by three large dams, will be developed more than 600,000 horsepower, and this in return will also pay for the cost of the work and a surplus besides. Below Cairo the improvement of the river contemplates—and the present appropriations are carrying this out—the revetment of the banks in every bend, which will save to the Nation in soil an amount every year which it is impossible to calculate, but which is worth many millions of dollars; will allow the building of levees close to the waterfront without danger of their caving in, and so reclaim possibly 100 square miles of additional land in the Delta; and will make a permanent and safe drainage system for the great swamps along the river, from which a few years' crops will more than pay for the entire Lakes to the Gulf Deep Waterway.

Swamp drainage, storage to prevent floods, storage to provide water-power and better channels, the establishment of suitable banks and good levees—all of these are a part of the Conservation policy that was launched on that memorable trip on which Theodore Roosevelt inspected the Mississippi.

[Signed] W. K. KAVANAUGH,
President

REPORT OF THE LEAGUE OF AMERICAN SPORTSMEN

The Committee appointed by the League of American Sportsmen to make recommendations to the National Conservation Congress beg leave to report briefly as follows:

Federal Laws

The United States should enact laws so that in addition to those now in force, the following will be possible:

The protection by the United States Government of migratory birds and fishes.

The setting apart and protection of game refuges, parks, and breeding grounds, and scientifically caring for same. Some of these should be established in the forest reserves now existing that are suitable for this purpose, and competent caretakers put in charge. The Wichita Reserve is a good example to follow. Marsh lands and water should not be forgotten, as all bird and forest life must be considered.

Trained Government game-keepers or experts should be provided, that can be furnished upon applications received from State or private game parks—same to be paid by the applicant served.

The States

The States should each and all set apart game refuges and parks and care for them practically. Competent care-takers and trained game-keepers should be put in charge. These game refuges for wild life should be distributed as generally in each State and cover as wide an area as possible; for it must not be forgotten that the song and insectivorous birds are as important to save and find refuges for, as is what is usually denominated "game."

The game laws of the States should be as nearly the same as geographical and local conditions will permit.

The enforcement of the game, bird, and fish laws, together with the care of game preserves, should be divorced from politics. At present in most of the States the selection of a game warden is based not upon training or fitness

for the position, but is the reward of party or personal political fealty. Should by chance the appointee show adaptability and really study the subject of game protection, by the time his education is well under way and he has become valuable to the State, the political wheel turns again and some one else is to be rewarded.

So-called game laws to be enforceable must be practical and have the sympathy of the people. Therefore, the work of education must be continued and amplified by both the State and Federal powers to show, *first*, the value of bird life to the farmer and all the people as insect and weed-seed destroyers; *second*, the value of game and fish as food products; *third*, their value as an incentive to a life out-of-doors and health; *fourth*, the value to the State because of the tourist and sportsmen's travel attracted thereby (statistics on this subject should be gathered by both Federal and State authorities, and given constant and wide publicity); *fifth*, the non-resident hunting and fishing license should be made as nearly alike in the several States as possible, and a reasonable amount of fish or game allowed to be taken home by the terms of said license; *sixth*, resident licenses issued by the State should furnish funds for carrying on the work of game, bird, and fish protection and propagation, and we recommend a careful consideration of this subject by those States that have not already such laws in force; and *seventh*, the so-called spring shooting of water-fowl should be stopped.

All of which is respectfully submitted:

[Signed] WM. B. MERSHON, Saginaw,
Mich.

Chairman

JNO. F. LACEY, Oskaloosa, Iowa

F. SHAROIR, Stamford, Conn.

J. H. McDERMOTT, Morgantown, W.
Va.

J. ADAMS BROWN, New York City

R. D. EVANS, Washington, D. C.

Conservation Committee

REPORT OF THE NATIONAL BOARD OF FIRE UNDERWRITERS

Since the commencement of the Conservation movement, the National Board of Fire Underwriters has been deeply interested in the governmental and associational activities aiming to foster and protect the natural resources with which our country has been so bountifully blessed. Our representation at the Washington Conferences of 1908 indicated our sympathy with the propositions presented, and the continuance of our Conservation Committee is a manifestation that we have been and are ever ready to cooperate in a furtherance of those principles which you as an organization stand pledged to advance. We believe that unless there is an intelligent development and utilization of our natural resources, the comfort, prosperity, and happiness of future generations will be seriously impaired, and we are in hearty accord with all legislation having for its object the preservation from destruction of Nature's gifts and Man's handiwork.

The address which this Committee presented to the Joint Conservation Conference sought to set forth some very important facts concerning the excessive fire waste which persists in the United States and suggested remedial measures, which we still firmly believe, if adopted, would materially diminish the grievous loss of life and the tremendous and unnecessary destruction of created values by fire. We therefore beg to reaffirm those suggestions at this time, as follows:

The present fire waste in this country is an unnecessary National calamity, and to reduce it it is essential—

First—That the public should be brought to understand that property destroyed by fire is gone forever, and is not replaced by the distribution of insurance, which is a tax collected for the purpose.

Second—That the States severally adopt and enforce a building code which shall require a high type of safe construction, essentially following the code of the National Board of Fire Underwriters.

Third—That municipalities adopt ordinances governing the use and keeping of explosives, especially inflammable commodities, and other special hazards, such as electric wiring, the storing of refuse, waste, packing materials, etc. in buildings, yards, or areaways, and see to the enforcement of such ordinances.

Fourth—That the States severally establish and support the office of fire marshal, and confer on the Fire Marshal by law the right to examine under oath and enter premises and to make arrests, making it the duty of such officer to examine into the cause and origin of all fires, and when crime has been committed requiring the facts to be submitted to the grand jury or proper indicting body.

Fifth—That in all cities there be a paid, well disciplined, non-political fire department adequately equipped with modern apparatus.

Sixth—That an adequate water system with proper distribution and pressure be installed and maintained. In the larger cities, a separate high pressure water system for fire extinguishment is an absolute necessity, to diminish the extreme imminence of general conflagrations.

The publication by the U. S. Geological Survey of Bulletin 418, known as "The Fire Tax and Waste of Structural Materials in the United States," is worthy of high commendation, and we believe a wider distribution of this pamphlet and the preparation and dissemination annually of similar information, will materially serve to awaken the public to a realization of the enormous values in utilized resources which are destroyed by fire beyond recall, and cause action to be taken by States, municipalities and individuals to enact such laws and regulations as will make for the exercise of greater care and forethought in the preservation of materials produced from our natural resources. It must be evident that the conservation of our forests and mines will fail of its full results if the utilized products therefrom are to continue to be unnecessarily destroyed by fire to a degree that is a National disgrace.

We share the pride of all our fellow citizens in the remarkable growth and prosperity of this country, in the extensive building operations, and in the increased commercial values; but, if we would conserve those natural resources which have been the principal foundations of our success, we submit that it is equally important to adopt and enforce such measures as will lessen the steadily and rapidly increasing fire waste of our utilized resources.

The National Board of Fire Underwriters has for years devoted its energies and activities principally to the reduction of the fire waste and the safeguarding of life and property. Standard rules and lists of hazardous and protective devices and materials are distributed free of charge, the results of the tests conducted at the Underwriters' Laboratories are made known to anyone evincing an interest, a model Building Code, prepared under the advice of experts in construction and engineering, has been urged for adoption in every municipality of the country, and as a result our advice and cooperation are sought in the revision and adoption of the building laws of our cities. Under the immediate direction of our Committee on Fire Prevention, expert engineers investigate the fire-fighting facilities and structural conditions of our cities, submitting copies of the reports, with suggestions for improvements, to the officials of the city visited and to the press; the expense of the work of this Committee alone, for the last six years, has amounted to \$432,742.

We have persistently endeavored to influence the introduction of improved and safe methods of building construction, to encourage the adoption of better fire protective measures, to secure efficient organization and equipment of fire departments with adequate and improved water systems, and to have adopted rules regulating the storage and handling of explosives and inflammable products; and we contend that successful efforts along these lines will very largely lessen the fire waste of the utilized resources, the destruction of which at the rate of over \$216,000,000 annually (1900-1909, inclusive) is one of the greatest drains upon our natural resources and one which can be corrected, if the Nation, State, city, and citizen will cooperate along the lines indicated above.

The destruction of our utilized resources by fire is increasing at such a rapid rate that the subject of its reduction should be very prominent in the minds of the people. Losses recorded for the past thirty-five years, not including forests, mine or marine fires, total the enormous sum of \$4,906,619,240. Unrecorded losses, if obtainable, would materially increase these figures. These annual fire losses run from \$64,000,000 in 1876 to \$518,000,000 in 1906. In 1907, a normal year, our recorded losses were \$215,084,709, and our estimated fire defense cost \$241,401,191, or a total amount equaling about 50 percent of the value of the new buildings erected that year in the entire country. In 1908, also a normal year, our ash-heap cost \$217,885,850,

and the relations of defense-cost and fire loss to new buildings remained about the same. Our contributions to fire that year were over \$1,250,000 each day of the year, a sum equal to the operating expenses of our Government, including those of our army and navy, for the same year; and in 1909 we gave to fire over \$25,000,000, more than was spent in that year for the same governmental functions.

No one organization can effect the needed reform. Since 1880 the population has increased 73 percent, while the fire loss for the same period increased 134 percent. The National Fire Protection Association and the National Credit Men's Association are spreading the doctrine of reform in the recklessness with which our utilized resources are destroyed by fire. Each organization should be encouraged. Membership is open to all in the former, and in the latter to the business men and merchants of our cities. The work, however, is carried on without State or municipal cooperation and therein lies the chief reason of delayed success.

If the office of State Fire Marshal were created by every commonwealth, and that official and his deputies were given power to enforce good fire-prevention laws, to investigate and if necessary prosecute cases of arson or criminal carelessness in the starting or spreading of fires, to ascertain the cause of every fire, and by the distribution of literature to educate the citizen to the need of care and forethought in the protection of his property, a distinct conserving of the utilized resources in that State would follow.

If our municipalities will enact and enforce improved and safe methods of building construction and cause the removal or reconstruction of existing structures which constitute, because of their construction, a menace to adjoining properties, our cities will be freer from the imminent conflagration which now threatens them. Eliminate defective chimney flues, unprotected external and internal openings, excessive areas, weak walls, and combustible roofs; prohibit the storage of rubbish, and demand the safe use and handling of dangerous inflammable liquids and oils; regulate the use of explosives; and the destruction of our values, created from the natural resources but enriched many-fold by human toil, industry, and skill, will be materially diminished.

If the citizens of a community, as members of their local civic bodies and boards of trade, will create in such organizations a Committee on Fire Prevention, whose duty it shall be to study the subject and awaken among their associates a realization of individual and communal responsibility, and if our boards of education will emulate the action of the State of Ohio in prescribing primary education of the school children as to the chemistry of fire, the causes of fires in our homes and how to guard against them, and how to extinguish incipient fires or hold them in check while awaiting the response of the fire department, a preparation will be made in that community which will check the constantly increasing fire waste.

And so while this Congress discusses and formulates policies for the Conservation of our natural resources, it should, at least, as representing the official, professional, commercial, and industrial life of the Nation, distinctly and emphatically advocate such regulation as will preserve those resources which are the embodiment of the thrift and industry of our people—the utilized resources—from unnecessary and wasteful destruction by fire.

Respectfully submitted,
[Signed] A. W. DAMON, Springfield
Chairman
GEO. W. BABB, New York
C. G. SMITH, New York
W. N. KREMER, New York
R. M. BISSELL, Hartford
R. DALE BENSON, Philadelphia
R. EMORY WARFIELD, New York
Committee

REPORT OF THE NATIONAL BOARD OF TRADE

In response to the invitation of this Congress, the National Board of Trade, which participated in the Conference of Governors at the White House in 1908, is permitted to take part in its deliberations. The National Board of Trade, as its name implies, is National in character, and is composed of a large number of Boards of Trade, Chambers of Commerce, and other organized bodies representing many of the large commercial and industrial

centers of the entire United States. It was organized 42 years ago for the purpose indicated in the following declaration: "The National Board of Trade was formed for the purpose of promoting the efficiency and extending the usefulness of the various Commercial and manufacturing organizations of the United States of America, securing unity and harmony of action with reference to business usages and laws, and especially the proper consideration of and concentration of opinion upon questions affecting the financial, commercial, and industrial interests of the country at large, and to provide a concerted action regarding National legislative measures and Governmental department affairs."

It will be seen from this declaration that the object of the National Board of Trade is to attempt to harmonize public opinion on National questions. About 15 years ago it became impressed with the wanton wastefulness and public neglect of our National forests, and resolutions were adopted inviting public attention to and legislation for the preservation and conservation of the timber resources of the United States. In a very short time it became evident there were other important questions involved in the regulating of forests, primarily the grave necessity of creating forest reserves and protecting them from depreciation by Government control and administration; and the establishment of a Bureau of Forestry was advocated. The National Board of Trade was also a pioneer in advocating the reclamation of arid lands and the drainage of swamp and overflow lands and practical reforestation, and adopted resolutions urging legislation to this end.

The activity of the National Board of Trade in promoting the measures it has advocated consists of the printing and the distribution of many thousands of copies of reports of committees and resolutions, as well as large numbers of its annual report in permanent book form, which of itself constitutes a valuable commercial library of reference; these publications have been sent to Members of Congress and the officials of the National Government, to State officials and members of State Legislatures, and to mayors and other officials of many cities having more than ordinary interest in public-welfare questions. The dissemination of this information has required a great deal of time and the expenditure of no small sum of money, and the National Board of Trade and its constituent members, together with

all others interested in its work, appreciate the patriotism and generosity of its President, who has done so much to carry on its work.

The commercial interests of the entire country are thoroughly alive to the merits of, and are earnestly championing, the cause of Conservation of all our natural resources. Economic use that does not destroy, but protects and fosters reproduction where reproduction is possible, prolongs and perpetuates the industries dependent on natural products for their maintenance; and these compose the larger part of all our manufactures. The National Board of Trade in its 42 years of existence has been the exponent of the principles upon which alone permanent trade and commerce can be maintained and extended—high standards of commercial honor and integrity, and doing unto others as we would that others should do unto us.

There are in this Congress, on the invitation of its officers, Delegates from National organizations which have contributed greatly to various phases of Conservation problems, which are now crystallizing into a National policy. So far as we are informed, it appears from the report of the Committee on Credentials and other committees that have been announced that no representation has been given these Delegates to enable them to participate in the active work of the Congress. We, as Delegates from the National Board of Trade, representing the commercial interests of the entire country, recommend that in case invitations are extended to National organizations to be represented at future congresses that suitable provision be made for their representatives to participate in the practical work.

The National Board of Trade rejoices with this Congress in the advanced thought that the campaign of education has created in the minds of the American people, and it also feels great satisfaction in that it has for many years earnestly advocated and been instrumental in the adoption of the wise, beneficent, and economic measures that are in the interest of not only the present generation but of generations yet unborn.

Respectfully submitted on behalf of the National Board of Trade,

[Signed] A. T. ANDERSON, Cleveland
WILLIAM S. HARVEY, Philadelphia

*(Chairman Committee on Forestry,
Irrigation, and Conservation)*

REPORT OF THE NATIONAL BUSINESS LEAGUE OF AMERICA

I deeply appreciate the privilege, and am not insensible of the honor, of briefly addressing this great Congress of representative men in every field of human endeavor, who are met to plan for the Conservation of our natural resources.

First, I wish to emphasize the fact that the patriotic men who are planning Conservation today are mostly not the men who will execute. The men who are to conserve our lands and waters and minerals, and perpetuate our forests, are now running around in knickerbockers, or being rocked in the cradles of the Nation. They and their children and their children's children, down along the line of centuries, will carry out the vital precepts and principles of this great Conservation movement—this timely warning cry against careless National extravagance, this imperative codicil to the Declaration of Independence.

There are some resources we cannot restore, but may conserve or substitute. As one door closes another opens. Coal, iron, copper, and other products of the mine, when once consumed cannot be reproduced; but for all time the tree may be perpetuated—the friendly, faithful, useful tree that conserves the rain-drop with its treasures of light, heat, power, and life-giving properties for vegetation, and fills the world with inspiring beauty. The restoration and preservation of our forests, then, and an adequate policy of accomplishment, become of the weightiest importance.

In this connection I beg to suggest the American farmer boy. It is proposed to organize the farmer boys and young men of this country into a great National body, to be known as the Tree Planters of America. The plan involves instruction and actual practice in tree-planting and tree-culture, with suitable prizes for excellence and results. It aims to permanently check the wastefulness of go-as-you-please forestry now evident from every car-window in this country. In brief, without entering into details, the suggestion seeks to organize all farmer boys from twelve to twenty years of

age as Tree Planters, in every commonwealth, county, and township of the United States; with the cooperation of the Forest Service at Washington, Governors of States, and the proper official heads of town and county governments.

The plan in general unifies the individual, the State, and the Nation, into one vast organized body for the practical reforestation of the country. The system once made operative will become an inseparable part of the life of the farmer of the future. It is kindred to the splendid educational and philanthropic work of Mr Bernard N. Baker, the ideal and actual President of this Congress; and I hope it may merit your approval as *one* practical means to the end we all are aiming at.

The time for talking has gone by. The time for action has come. Therefore let us begin at the foundation and organize the coming men who are to do the actual work of reforestation. The mind of the American boy is plastic. The impressions he receives remain to the end. Teach him, then, to practice those things that make for permanent universal betterment; for with his brain and brawn he determines the destiny of this great American Republic.

[Signed] AUSTIN A. BURNHAM
General Secretary

REPORT OF THE MISSOURI VALLEY RIVER IMPROVEMENT ASSOCIATION

When the Missouri Valley River Improvement Association was organized in August, 1906, practically no one in the valley thought the Missouri navigable in its then unimproved state, and only a few people believed it worth while to solicit Government aid in trying to make it navigable. The general impression seemed to be that the Missouri had outlived its usefulness. Compare this feeling with the sentiment that exists today! The people of Kansas City and the entire Missouri Valley have become awakened to the great possibilities of this river as a means of cheap transportation. Through the efforts of our Association and the people of the valley, the Congress of the United States in 1907 made an appropriation of \$400,000 for the improvement of the Missouri; in 1909 Congress made

another appropriation of \$555,000, and in June, 1910, still another of \$1,465,000 for improving the river from its mouth to Fort Benton.

So great is the interest in the Missouri river project that the people of Kansas City recently raised a fund of over \$1,000,000 for the purpose of navigating the Missouri with modern and up-to-date boats especially adapted to this river. Experiments are now being made with different kinds of boats to determine which are the most practical. With the opening of navigation in the spring of 1911, we hope to have a modern boat line in operation between Kansas City and Saint Louis. In addition to raising \$1,000,000 for navigating the Missouri, Kansas City at her bond election in the spring of this year, voted \$75,000 bonds for the improvement of her harbor.

The sentiment in favor of improving and navigating the Missouri was brought about to a great extent by some of the business men of Kansas City who in 1906 organized a boat-line company to maintain regular steamboat service between Kansas City and Saint Louis to demonstrate that the river was navigable even in its then unimproved state. This company, not waiting to build boats suited to the river, bought two old boats, and in 1907 and 1908 operated them with great success, carrying freight between Kansas City and Saint Louis at two-thirds of the railroad rates. When the people of Kansas City saw what could be done with the antiquated type of boat, they became interested in navigating the river with first-class steel-hull boats, built especially for the Missouri—which resulted in the organization of the Million Dollar Boat Line.

A movement is now under way to organize a company for the purpose of building a large dam across one of Missouri's streams within 120 miles of Kansas City. It is proposed to put up a plant that will generate 30,000 horsepower; this to be transmitted to Kansas City and sold to the consumers at the low price of one cent per kilowatt-hour. The largest consumers of electric power in Kansas City are now paying $2\frac{1}{2}$ cents and the smaller consumers from 8 to 10 cents per kilowatt-hour. The proposition has the appearance of being feasible, and if it can be carried through it means a great deal to the future growth of the Missouri Valley, as it will furnish cheap power to prospective manufactories.

Respectfully submitted,
[Signed] JEROME TWICHELL
Chairman

REPORT OF THE UPPER MISSISSIPPI RIVER IMPROVEMENT ASSOCIATION

How to conserve the natural resources of every land has become an absorbing theme throughout the civilized world, and I think no one is more alert in reference thereto than the inhabitants of the former Northwest Territory and of the Louisiana Purchase. They are of the salt of the earth; yet notwithstanding their power they have permitted constant encroachments by predatory greed and covetousness, mostly by the corporate monopoly rampant world-wide in this Twentieth Century. It is thus fitting that this magnificent assembly of progressive public-spirited Delegates from nearly every avocation and locality should here gather at the head of navigation of the great flowing stream that drains the most fertile valley on this mundane sphere. Viewing these fertile lands, it would be most natural to expect that the rights of this people declared by the law of Congress enacted in 1787 should be deemed wise, especially this provision:

Article IV. The navigable waters leading into the Mississippi and Saint Lawrence, and the carrying places between the same, shall be common highways and forever free as well to the inhabitants of the said Territory as to the citizens of the United States and those of any other States that may be admitted into the Confederacy, without any tax impost or duty therefor.

The Association that I represent has labored during the past decade to so awaken public sentiment in this valley that a six-foot channel will be provided from here to the Gulf; and I bring the message to you that we have aided much in arousing the people from lethargy to a forceful activity for cheaper transportation by inland waterway improvement, which has been assured to this upper river within the succeeding dozen years by Congressional action at the last session.

The problem to be grappled with now is how best to regain for the public the landings for boats, which we find have been obtained and are largely held by private interests antagonistic to thorough use of the stream. Generally for a mere pittance the landing rights, to the thread of the stream, passed to private ownership needlessly and without any consideration to the original grantor, the Government. Each city and village along the river is now up and doing, as is this city of Saint Paul in providing a municipal wharf at enormous expense; they are now fully apprised of the importance of these holdings, which we ardently hope will be regained for free public use, so that improved machinery for loading and unloading cargoes of modern boats and barges by a single power lift may become effective, as may be seen along the Rhine. When this is done, boats will again ply this great river and its tributaries, carrying the abundant products of every kind that this valley annually produces at a much cheaper rate than by rail.

We, who people this Central Northwest, were pioneers in opposing rapacious transportation rates; it was the Granger movement hereabouts, nearly forty years since, that aroused the law-making powers to the necessity of conferring on State and Federal commissions the power to regulate rates; and further results are yet to be hoped for in the regulation of charges for freight, passenger, express, sleeping-car, and mail service, together with telegraph and telephone charges. This valley between the Alleghany and Rocky Mountains was ordained by nature to supply foodstuffs for a goodly portion of the globe's people; and with the opening of the Panama Canal, along with the development of our inland waterway transportation, the problem of traffic rates must be solved.

While the general Government has been using the people's money to improve rivers and build canals, no sooner does the Government undertake to develop power incidental to some praiseworthy project than it finds that the water-power was absorbed by private interests, which were at all times alert to obtain grants in perpetuity (now worth millions) without any regulation to redound to the people's good—as shown by the reports of our waterway conventions. The best sites are already taken away from the people; shall we bend every energy to save what remains? This should be all changed in future grants of power-rights in flowing water; a census of the Nation's water-power resources should be taken, and all grants hereafter should be determined, with the respective values of the same, for use at

equitable rates. When once the law-makers realize that the people are truly in earnest about Conservation, a halt will be called upon reckless legislation in the interest of exploiters; then sincere citizens may be induced to stand as legislative candidates, without fear of being pilloried by a subsidized press and venal poll-workers at every turn in a canvass.

Our waterway improvement conventions in this valley have spoken plainly, and the rivers and harbors are faring better than ever before—in fact, our efforts along these lines have done wonders to bring to the people, by acts of Congress, what is justly their own. Will the Conservationists array themselves against all law-makers who have proven recreant by their attitude toward clean-cut legislation in aid of Conservation throughout the United States? Smooth words, without conscientious acts in the interests of our lofty aims, should meet with a lasting rebuke! "Fight it out on this line if it takes several summers," should be our slogan.

[Signed] M. J. McENIRY

Chairman *Conservation*
Committee

REPORT OF THE WASHINGTON STATE FEDERATION OF LABOR

The Washington State Federation of Labor will not be represented by any of the Washington State Delegates at the Second National Conservation Congress. We are, however, deeply interested in the question of conservation of natural resources for the people, and as President of this organization, with a membership of over 20,000, I believe I am expressing the sentiment of the workingmen of this State when I say that I am in entire accord with the declaration of views and recommendations of the Governors of States and Territories of the United States, as adopted at the Conference of Governors, called by President Roosevelt, in May, 1908.

Our vast forests, our water supply (for irrigation and power projects), and our fisheries are of inestimable value to the people if properly developed under a control that will make the very best use of them with due regard to their future possibilities and greatness. Forestry, irrigation, and water power are to a great extent dependent on one another in their successful

development, and the magnitude of the undertaking requires the hearty cooperation of State and Nation if it reaches the degree of success that we hope for.

I trust that the Congress will strengthen and perfect plans adequate for the protection of the people's interests and the development of these resources with an eye to their future greatness.

Respectfully submitted,
[Signed] CHAS. R. CASE
President

REPORT OF THE WESTERN FORESTRY AND CONSERVATION ASSOCIATION

Let us concede that Conservation means that we, as a people, should manage all our resources with the intelligence and prudence that an individual should devote to managing his own property. Let us use them profitably, as he would; neither destroying or wasting them unnecessarily, nor giving them heedlessly to anyone who needs them less and will use them less to our advantage. But let us not, during excursions into Constitutional problems, State rights, and other bewildering issues, forget that first of all comes protection from destruction and waste! The great danger now is that our resources will disappear while we are deciding to whom they shall belong.

It is of this kind of Conservation alone, the Conservation that conserves, that I bring you a message from the Pacific Northwest. The Western Forestry and Conservation Association does not decry the necessity for wise action by State and Nation in the safeguarding of water-power, minerals, and lands; but the settlement of such affairs is not our function. I come only to tell you of the work of the most perfectly organized and successful Conservation movement ever undertaken by private individuals in this country—the forest-protective associations of the Pacific slope. We talk little, but we work, spend money, and accomplish.

In our five States from Montana to California stands half the merchantable timber in the United States, the majority in private hands. The control of

this stupendous community resource entails grave responsibilities. To preserve it for the fullest use, to replace it when used, if possible—this is the timber-owner's duty. His ownership is largely a public trust. Nowhere else has he realized this so promptly and acted so adequately as in the Pacific Northwest. I have come to report his stewardship, and to show you that you need not wonder whether he will follow the Conservation banner.

The Western Forestry and Conservation Association has no individual membership. It is the central medium or clearing house for a dozen subsidiary associations of timber-land owners, representing millions of acres, who cooperate in order to apply to the best advantage the most modern and efficient systems of forest protection. Through this means they employ a trained forester to assist them in solving problems of reforestation, forest legislation, education, and like matters demanding expert knowledge or central facilities. Its meetings are attended not only by delegates from these timber-owners' organizations, but also by the leading State and Forest Service officials and representatives of the public Conservation associations. All work in the closest harmony to devise and execute practical and effective policies. There are no dissensions at these meetings; no question as to who is most competent by right of law or geography. Every man there, be he a humble officer of the Forest Service, State Forester, or timber owner, is there because he wants to do his own part, with his own hands or money, in preserving the magnificent forests of the West. He knows what he is talking about, and the rest are mighty glad to hear him.

But we do not stop with meetings, and herein is perhaps our chief difference from a great many advocates of Conservation. You have all read of the recent fires in our northwestern country. They have been greatly exaggerated, the area injured really being very limited. Nevertheless, while we talk here of generalities, bands of weary, half-blind men are still battling to prevent fresh outbreaks; the smoke still curls over the blackened forms of those who met a fearful death to save the lives of others; scores who fought till they could fight no more still lie bandaged and sightless in the extremity of mortal agony. No honor is too great to do these heroes. We of the West owe a sacred debt to them, one and all, and not least to the men of the Forest Service whose training made them as efficient as they were brave. We want more, not fewer, of them. But side by side with the bravest,

equally efficient, equally trained and disciplined, worked the patrolmen of our fire associations. Conservationists employed by private effort. We have had no time to prepare nice statistics, for our fire fighters have something else to do; but I venture to say that our Associations' expenditures for forest protection this year will be over \$300,000. In the Coeur d' Alene fires alone, a single one of our Associations put 850 men in the field.

And yet this is not much to boast of. There should have been no fires to fight. The way to prevent fire is to prevent it, not fight it when almost or quite beyond control. The only solution of the fire question is better enforcement of better laws, better public sentiment, and better patrol. There must be an organized force of trained and vigilant men, ample in numbers during the dry season to reach all fires in their incipency. It is in this that our Associations now lead all other agencies. They handle the fire situation in a much better and more comprehensive manner than even the Government has ever done, because they spend three times as much money per acre for patrol. Thoroughly excellent as are the methods in the National Forests—they are identical with those of the most progressive practical timberman—Congress does not sustain them adequately.

Our own system is by no means perfect yet. Although in the territory covered by our Association in Idaho, Washington, and Oregon we have perhaps 500 organized and equipped patrolmen, each authorized to hire help when needed, there is still much unorganized area, and not all timbermen within our territory contribute as they should. We need more men and more money from our own brethren, and heartier cooperation from public, State, and Government. But we confidently expect to get all this, just as we have in greater measure each year in the past. And when, as already in Washington last year, one Association protects 8,000,000 acres with a loss of but 1,000 acres; when this small loss was caused by less than 6 fires out of 1,200 extinguished; when in this historic year of 1910 we have controlled our countless fires so that actual disasters can be counted on the fingers, and our loss as a whole is insignificant—we feel that no one has done more to prove his willingness and competence to practice Conservation that counts than the northwestern forest owner.

The northwestern timberman approves all measures that will give the greatest number of people the greatest permanent opportunity to profit by

the fullest use and least waste of all our resources. Thus they will be most prosperous and use most lumber. He is doing more than anyone else, Government or State, to protect both old and growing forests from wasteful destruction, so there may be most lumber to use. I take it this is Conservation.

[Signed] E. T. ALLEN
Forester

REPORT OF THE UNITED MINE WORKERS

I regret very much that serious complications in the mining industry of our country, together with an enormous amount of important matters requiring my immediate attention, makes it impossible for me to keep my engagement to address the Conservation Congress on the subject, "Are we mining intelligently?" I am intensely interested in the subject of conserving our natural resources, but I am still more interested in protecting the lives and health of our people. We are not mining intelligently, as I will explain by letter.

Success to the work of the Conservation Congress! It means much for the future generations of this greatest industrial country on earth.

[Signed] T. L. LEWIS
President

TIMBER CONSERVATION

GEORGE H. EMERSON
Hoquiam, Washington

To save our Nation's resources is the wish of all; to save our timber is the special wish of all timber owners—no one is so much interested as he who has his private gain or loss joined to his interest in the public good.

The American people are a prodigal folk. They have looked upon their resources as inexhaustible, their lands as unlimited. They have called upon all nations to come, and to all comers they have given lands, mines, timber, water-power. Has this course been right? Up to a point in our development,

yes; of late, no—most emphatically NO! These resources are entrusted to us as a heritage for our children and generations yet to come. "America for Americans" should have been sounded 25 years ago; had it been, there would today be no cry of approaching timber shortage.

What more absurd disposition of our timber land could have been made than the laws under which it has passed to private hands? The Homestead and Preemption acts, framed for prairies, requiring the settler to live on and cultivate the soil, have been extended to our forests, and to comply with their terms, thousands of men have withdrawn from vocations by which they were increasing the wealth of the Nation, and with blankets and provisions strapped on their backs and axe and compass in hand have worked their weary way through the pathless forests to vacant Government lands, on which they filed. Then with axe and fire they spent months destroying the property they proposed to acquire title to—destroying the resources of the Nation instead of increasing its wealth; and in doing so, fires reached beyond their control and destroyed still other timber. The law and the ruling of the Land Office have made this destruction one of the considerations of acquiring title. Settlers must prove they "have cleared and planted and maintained a residence on the land;" that is, they must prove they have cut and burned a certain amount of the Nation's timber, and have wasted or—worse—employed in destruction certain of the Nation's time, and this to acquire title to land upon which they could no more live than in the middle of a desert! Lands whose only value was in timber they were compelled, in part, to destroy; and this where they never intended to settle, other than to comply with the letter of the law, and never expected to return after acquiring title. The months or years wasted in complying with these foolish laws they might better by far have been spent in jail at the public expense. It would have cost the Nation far less, and would have been less dangerous to life than the lonely existence remote from other human beings, where any accident to limb costs a life.

Sometimes there was an actual settler who wanted a farm or a pasture. He considered the timber only in the light of its cost to remove, and with axe, saw, and fire, he proceeded to its destruction. And why not? That which cost nothing looked to be of no value! Timber appeared as free as air and sunshine.

Later the lumberman came, and up to 1885 our Government offered him in Washington hundreds of thousands of acres of the best-timbered land for \$1.25 per acre. Michigan and Wisconsin had been so offered, and mostly sold. The lands of the Northern Pacific could then be had at \$2.50 per acre and paid for in the bonds of the Company, then worth half their face. The lumbermen looked upon the timber as inexhaustible. Only that near water could be harvested by known methods; only the best of the trees could be sawed and sold at a profit; only western markets appeared possible. What wonder fires were set to burn the choppings and make pastures? No people save that which cost nothing, and for which they have no use and cannot sell. When things become of value they are conserved, and when of enough value they are manufactured or grown; and the ratio between cost and selling price regulates the supply of things manufactured or grown.

Up to within a few years there has been plenty of timber land that could be taken under the Homestead, Preemption, or Timber and Stone Acts, or scripted or bought of the railroads. The blame, then, for the waste of our timber has been with the laws that made it valueless. The men we have sent to Washington to make our laws have given this timber to all comers of all nations. They are the men our people should hold responsible for the waste of our resources. These same men now tell us, "We are on the verge of a timber famine," and that the lumbermen are wantonly wasting the Nation's timber. Is it not the old cry of "Stop thief!" sounded by the culprit? By their acts they have made this timber valueless. Had the Government estimated the cost of growing a timber crop and sold its timber at about that price, timber would have been protected, conserved, and replanted, and its use would be as in Europe, about 60 feet per capita per annum, instead of 600 feet as in America.

Since our timber has taken on a value, its destruction by fire has greatly decreased. Timber owners now use precautions, and employ fire patrols. So, too, with harvesting; it is cut cleaner, sawed with thinner saws, manufactured with better appliances, and great saving has been effected in every branch of the industry—all because of greater values. Now, if just tax laws were passed, taxing no crop until harvested, and taxing reforested land as stump land; if rates of interest were lower, and if stringent fire laws and careful patrol were enforced; if stumpage was a little higher or labor a little lower, or the railroads were to make a reduced rate on low-grade products,

the law of supply and demand (or the ratio of cost to selling price) would reforest old choppings. Toward these things we are rapidly advancing, and before our timber is exhausted we shall have reached this point.

If our Government would hold her reserved timber at cost of reproduction, and protect the timber of the Nation by import duty, the question of timber shortage in America would soon be settled. Instead, they threaten reduction of its present value and increase of its waste by the removal of duty on imports. There is no way to conserve any commodity but to give it value, and no way to make people manufacture goods or grow crops except to offer a price that covers cost and a profit.

If the public would buy lumber of strength and durability suited to the purposes required, instead of ordering grades better than needed, they would help the Conservation of our timber far more than by essays and speeches. The most unreasonable of all buyers are our Government officials; with them there seems to be no purpose for which ordinary lumber is suited. So, too, if our State legislators would pass just tax laws, they would make a grand move toward timber Conservation. Instead, counties are spending hundreds of thousands of dollars—which the timber owners must pay—estimating the number of feet of standing timber, so as to be sure they find it all and tax it out of existence. This generation owes posterity laws that will save some of our present timber and leave to them growing timber crops instead of charred and desolate stump lands telling only of their fathers' greed and lack of foresight.

Wonderful tables have been prepared showing the upward tendency in prices of timber lands. Far better prepare a table showing the cost of growing a timber crop, and causes that have deprived it of its legitimate value. Water always rises to its level when the pressure is removed. Timber-value level is costly to produce. The greatest pressure to hold timber values down in the past have been our land laws; first the Federal laws for the sale of timber, second the State laws for taxes—and lack of all laws for protection and planting.

Our Nation is still a prodigal. She taps the fuel supply of future generations and allows the gas to burn and the oil to run to waste. More of the timber of the Nation has been burned for clearing and pasture than has been sawed by

the mills; but when the lumbermen are accused of destroying their property, or not utilizing all that will return cost for their labor, they are accused of lack of good intelligence—and that we resent. New England and New York have a greater area in timber than they had 50 years ago. Nearly every town site has a saw mill that supplies local demand and makes shipments to nearby cities. The few days I spent in New Hampshire last spring, and the auto trips I took through the places I knew in my youth, impressed these facts with force. Rail trips through Connecticut, Rhode Island, Massachusetts, New Hampshire, Vermont, Ohio, Pennsylvania, and Maryland revealed the fact that thousands of acres once under cultivation are now in timber, and that old forest lands are reproducing. Pine groves, cut when I was a boy, are being harvested, and fields where I picked rocks every spring are growing beautiful pine forests; the present owner of the old homestead in New Hampshire has put in a little saw and shingle mill to cut trees that were not sprouted when I left the old farm. The small saw-mills that are supplying the local demand are cutting the largest of the new growth, and the supply of that portion of the States where the timber was once exhausted will hereafter be adequate to local demands. As it is in New England and the Middle States, so it is in the South, in the West, in California and Oregon and Washington; if we keep out the fires in the old choppings, the new growth will be ready before the old is gone—and the waste of today kept always damp by the young growth, brush, ferns, and vines, will rival in value the portion of the tree we are now able to market.

Again consumption in all things is in proportion to price. Advance the price of lumber, and you reduce the consumption. Stone, brick, concrete, and steel are ready substitutes, as the price of lumber advances. In Europe, lumber is no longer a necessity, only a luxury, and not one much cared for at that; this has been forced home to me in countries I have visited during the past six months. Six days from New York we touch the Azores, a land where no lumber is used except for floor-joists and rafters in the cheaper buildings; next we touched Madeira, and found a city of stone. So with Gibraltar, southern Spain, Egypt, Syria, Greece, Italy, France; not a lumber yard in all these countries that we could find. A cargo a year would supply the demand for all purposes. The wonder was not how these people get along without lumber, but how they use the 60 feet per annum they are reported to require. I do not think there is one shingle roof in all those

countries, and I expect a very good knowledge of Arabic would be needed to explain to those people what a shingle is and its use. In Constantinople we found a few miserable board shacks. Lumber comes to that market at a low price from the Black Sea, and it appears to be a detriment rather than a good. In Switzerland and southern Germany, some houses are built of wood above the lower story; but I think there are no shingle roofs. These countries are well timbered, with trees in rows showing they are planted. The price of common lumber is only a little higher than with us, but labor is cheap, and growing timber exempt from taxes. Trees there can find a profitable market, trunks, limbs, stump, and roots. It is then, little wonder mountain sides, impossible for agriculture, should be planted to timber. Those timber areas do not use much of their lumber. In Switzerland and Germany we found saw mills, some of them of fair capacity, and shipping by rail, but their towns were built of stone. The mills select the largest trees, and replace with new plants. In time we shall reach some of these same conditions, and plant our timber instead of allowing it to grow at will. All this will come about when proper laws are enacted.

American people will some time awaken to the fact, long since known in Europe, that timber is no necessity; only a makeshift. Bridges of rock, houses of brick and stone and steel, with roofs of tile, are for the centuries; buildings of wood are only for the years and the flames. Lumber is cheap in the new countries, and convenient for quick shelter; and it is there forests are found. Big timbers may become scarce, but their demand is also decreasing. Already our cities have fire limits. Bridges and spars are of steel; and if our farmers could obtain money at city rates, it is doubtful if it would not be cheaper for them to build fire-proof houses than to pay higher insurance on wooden buildings. Already roofs of shingles are in balance with roofs of other and safer material, and the price of shingles is fixed by this competition. As it is with shingles it will be with lumber, and is for many purposes; in many countries for nearly all purposes.

Do not think I underestimate the value of our timber, or fail to advocate its protection and reproduction; but he who says we are approaching the time when timber values are to be much greater than now, and he who predicts a timber famine, have both overlooked facts that will come to the front with the years. The cry of "Fire!" never stopped a conflagration. The cry of Conservation will never stop the waste of valueless commodities. Action is

needed in both instances, if results are to be attained. To conserve our timber we must give it value. Let the Government refuse to sell from its reserves except for cost of reproduction; also protect us from foreign competition. Educate our loggers to the enormity of the crime of burning choppings fit only for the timber crop. Let States impose rigid fire laws and make liberal appropriation for forest protection. Let our legislators see the folly and injustice of taxing the same crop year after year; a crop that can contribute nothing toward paying those taxes until marketed, a crop that is of far less value per acre than the yield of fruit gathered each year. Do not be afraid the few remaining timber owners are going to be benefited at the expense of the many; rather the benefit will be for our children and our children's children. Above all, remember the timber owner is not to blame, only fortunate that he bought timber that our Government was willing to part with for a song; and hold our laws and their makers responsible for results for which they, only, are to blame.

The forest fires of the West today are more often set by the railroads than by all others. Their locomotives are torches of demons, tearing through our forests, streaming fire from their stacks and leaving all behind in flames. From the rear platforms of trains I have seen hundreds of little fires spring up as we passed—this, when the woods were dry and conditions right. The timber they burn is their resources for freight. The destruction they create is a loss of millions to their own business. It would seem prosecution for damage done should follow their wanton torches, and that laws should be made for the protection of their own interests they so recklessly ignore. It is no longer the logger or the settler that causes our forest fires. Our laws and public opinion, and vast sums expended by timber owners prevent the setting of careless fires; but the railroad locomotives still scatter fire along their pathway through the woods. Let the railroads learn a lesson from the recent Montana fires that stretched along their lines on either side and crossed the rivers where they cross—fires that have destroyed millions of young pines that a few years hence would have yielded a freight of from \$10 to \$30 each tree for their transportation to market.

Let the loggers awaken to the fatal folly of allowing the first fire in their cuttings, and our legislators to the necessity of forest protection. Stop the first fire where land is only adopted to the timber crop.

Out in the West where our mountains are the highest; where our streams spring from the eternal glaciers and are fullest when the weather is warmest; where water falls the farthest; where our soils are most productive when moistened; where our fruit is the finest; where trees grow the largest; where our hills contain coal, iron, silver, copper, and gold; where our ocean is the greatest and our fisheries are most prolific, our people are all Conservationists. They are for Conservation that is practical and adapted to their peculiar conditions; Conservation that shall develop and utilize their resources, and that shall yield the greatest good to the greatest number, and to the future as well as the present.

Where all things are on so grand a scale, the people cannot be small and narrow. They are as are their woods, their mountains, and their torrents, grand and active; and they are to be trusted. They will solve the problem of conserving their timber. They will keep out fires. They will enact just tax laws. They will guard their holdings. They will encourage new growth. They will be first to awaken to the best methods of forest Conservation adapted to their needs. They will solve the problem of conserving our western forests.

FORESTS AND STREAM-FLOW

WILLIAM S. HARVEY
Philadelphia

Professor Willis L. Moore, Chief of the United States Weather Bureau, in his address before the Atlantic Deeper Waterways Convention in Providence, September 1, 1910, made the statement that the waterways were in no way affected by the forests; that he had records made for many years that clearly prove that the waterways have in no way been affected by the acts of man; that he was aware that he would destroy a popular impression by making this statement, and that he based his statement upon the facts as he knew them. The following eminent men in articles published in *American Forestry Magazine* for April, 1910, take exception to and refute the statements and claims made by Professor Moore (and which he had previously expressed): Professor Filibert Roth, University of Michigan, Forester; Professor L. C. Glenn, Vanderbilt University, Geologist; and Professor George F. Swain, Harvard University, Engineer. These gentlemen

represent geology, forestry, and engineering, and their training, knowledge, and experience qualify them to speak intelligently and with authority on this question of the influence and effect forests have upon streams.

Mill owners and operators on various rivers in New England have practical demonstration that denuding or partial denuding of the forests on the headwaters of the stream on which they are dependent for power has seriously impaired the uniformity of flow and lessened the amount of power which they are able to secure for the same number of days in a year: that denuding also allows the rainfall to run off rapidly, causing erosion, which erosion is filling and choking the streams and rivers and in seasons of flood depositing silt in valleys which have heretofore been of agricultural value, thus largely impairing or destroying their fertility. This condition equally applies to various streams and rivers in other sections of the United States. It is moreover denied and refuted by the greatest financial and manufacturing interests, who have spent and are spending hundreds of millions of dollars in the development of electric power on the waterways of the United States. They have in their employ the most competent engineers known, who have investigated the entire situation, studied the maximum and minimum rainfall for a long period of years, and conditions influencing the territory embraced on the streams and rivers upon which they propose to make and are making and have made developments. These great interests, vital to commerce and trade, emphatically state that the flow of streams is affected by the forest cover, and that they are most anxious and are earnest in efforts to have the forest cover protected in all territory in which they operate, claiming that if the hillsides or mountains on the headwaters of water-sheds are denuded the volume of power will be so diminished, impaired, or destroyed that the value of the bonds issued for the development of these powers, and heretofore considered one of the safest and most desirable investments, will be seriously imperilled.

In addition to the authorities above named, and to whose articles I have referred, there are others who have refuted and contradicted Professor Moore from his own premises and data. His Excellency M. J. J. Jusserand, Ambassador from France, publicly stated the absolute principle: "No forests, no waterways." Without forests regulating the distribution of water, rainfalls are at once carried to the sea, hurried sometimes, alas! across the country. After having devastated the neighboring fields, the rivers find

themselves again with little water and much sand; and with such rivers, how will you fill your canals?

The question is as clear as can be; do you want to have navigable rivers, or do you prefer to have torrents that will destroy your crops and never bear a boat? If you prefer the first, then mind your forests. If the Mississippi is the "Father of Waters," the forest is the father of the Mississippi. The French Ambassador, you will note, says, "We can tell you, for we know. France is now spending many millions of dollars to reforest the mountain-sides denuded many years ago, which have seriously affected her waterways."

Some of us feel it is unwise to take too seriously all the deductions and predictions that are made by academic, scientific, idealistic theorists, especially if the department of science with which they are most intimately identified relates almost exclusively to atmospheric conditions, which are still so imperfectly understood that they not infrequently elude prediction; though where the results of scientific deductions are proven correct and add to the fund of knowledge, they are deserving of our greatest respect and regard. We have much confidence, for example, in the conclusion of Gifford Pinchot and his staff of assistants, who have made a practical as well as scientific study of the effect of forest cover on the flow and supply of water in streams, which conclusions unqualifiedly refute the statements made by Professor Moore.

THE CONSERVATION OF MINERALS AND SUBTERRANEAN WATERS

GEORGE FREDERICK KUNZ, PH.D.
New York

The necessity for conserving the forests has been fully recognized, and it may be said that as to what is in the ground a clear and satisfactory distinction has been established between what must be conserved for the good of the people as a whole, and what can safely be left to the exclusive control, management, and ownership of individuals or corporations. In regard, however, to the material wealth that lies beneath the ground, whether diamonds, gold, silver, copper, oil, or clay, or, indeed, anything that has a material value and can be included as such in the domain of mining

statistics, there has been and still is a considerable difference of opinion touching what should be done.

The existence of these materials beneath the ground is not usually evident, and the judgment of the best experts is frequently required to determine whether they exist in a given tract or not; on the other hand they may sometimes be casually found where their presence was not suspected. The Government of the United States still owns great tracts of land, and it is most important that the whole people of the United States should receive the full benefit of all the mineral wealth that is below the ground—the invisible wealth of the Nation, as it may be termed.

In order to avoid any collusion on the part of officials engaged by the Government to make investigations, or of those who, though no longer in the Government service, might learn the results of these investigations and might in some manner try to obtain control of these lands before the Government knew they had a distinct value, it would seem that a Conservation Act should be passed making it imperative that all minerals contained in any land beneath the surface should forever remain the property of the Government. With lands containing minerals, there should further be an assurance that the deposits will be effectively worked, thus preventing an entire mineral supply from being locked up for many years, so as to maintain an artificial value for the material. Again, little-understood minerals, or those that have been very little worked and yet may have a value in the future, such as bauxite, which is valuable in the manufacture of aluminum; monazite sand, which is used in the making of the Welsbach incandescent light; and carnotite, whose value as a radium ore has been discovered within the past ten years—should all be made to yield royalties to the Government.

It is very evident that many minerals not considered to have any commercial value today may prove to be of the greatest industrial value in the future. Furthermore, as we are likely to discover new elements, and new uses for old minerals, the Conservation Act might be made to provide for a payment of 20, 30, or even 60 percent of the total value of the mineral as taken from the ground in royalty to the Government of the United States, exactly as the South African Government exacts as a royalty 60 percent of the product of all the diamond mines within its territory. This would be a

more generous treatment of private owners than was accorded them in some instances in the past. The French crown-deeds read in the Seventeenth Century that gold, silver, lapis-lazuli, etc, should belong exclusively to the crown. In reality, the Government should only sell to private owners what is in sight on the land and the right to what could be grown on it, not what is below the ground. The franchises of subways and tunnels and all mineral rights should be retained, as well as the right to condemn at a fair valuation any property needed for the development of a mine or a water-power.

The term "mineral" should apply to every substance found in the ground that is either a mineral or an associate of minerals, that is, rock, sand, clay, or even a swamp, that may have a value in the arts, sciences, agriculture, or any other monetary value. The word should be used in its broad sense and not in the more restricted scientific meaning of the word used by mineralogists, which is that a mineral must be a definite mineral compound.

The subterranean waters of the United States are a great and valuable asset of the Nation. Nearly all of our water companies sell water either for power or for consumption. As each owner of a piece of property ought to be entitled to an interest in the water under it, some provision in Conservation should be made for the actual ownership of the waters; not that they can be drained from under the property, for a series of springs could be threatened with ruin if this were done, just as were the famous springs in Saratoga. In other words Government lands should not be robbed of their subterranean waters to be in turn sold to those who have a joint right in them.

THE QUESTION OF LAND TITLES

FRANKLIN MCCRAY

Indianapolis

All the territory west of Mississippi river was acquired by the Government by three means, purchase, conquest, and treaty. This territory, having been obtained by the diplomacy and blood and treasure of our common country, belonged to the people of the whole country, and was held in trust by the Federal Government for them. It was subject only to their call for settlement.

The charge is made that practically all the looting of the public domain is in the Louisiana purchase, the territory wrested from Mexico, that acquired from Great Britain by the Ashburton-Webster treaty in the settlement of our northern boundary line, and that purchased from Russia. This land, being held in trust by the Federal Government for the people and being subject only to their call for actual settlement, it is charged, has been plundered through fraud and corruption of the trustee, the Government of the United States, in collusion with the grantees, who have obtained vast tracts and withdrawn the same from settlement by floating them into a different channel than that for which the Government held them in trust. By this corrupt and fraudulent method, it is charged that these vast estates have been monopolized by corporate greed and accumulated wealth, and that no less than 6,000,000 acres are now being held by two individuals alone within the State of California.

If this be true, then, under a well-settled principle of law, the Government has conferred no title upon such grantees, because fraud vitiates all contracts, and courts of equity have complete power under proper proceedings to follow this property, thus fraudulently obtained, in its labyrinthian processes and seize it by judicial decree, lay its stern hand upon it and restore it to its rightful owners, the people of the United States, and float it anew into the channel of settlement where it rested prior to its spoliation. I suggest that this Congress petition the United States Congress to investigate the titles of these grantees and, if found to be fraudulent, the Department of Justice should be instructed to institute proceedings calculated to restore the land to its rightful owners.

FOOTNOTES:

[1] The full report by Mrs Wilkinson appears on later pages.

[2] The members of the Resolution Committee were chosen by their respective State Delegations, and met at the call of a Temporary Chairman (and member-at-large), Ex-Governor George C. Pardee, of California, who was afterward chosen by the committee as Permanent Chairman. The full personnel of the committee follows:

ALABAMA, John L. Kaul, Birmingham
ARIZONA, B. A. Fowler, Phenix
ARKANSAS, John A. Fox, Blytheville
CALIFORNIA, Frank H. Short, Fresno
COLORADO, P. T. Coolidge, Colorado Spgs.
COLUMBIA (District of), W J McGee, Washington
CONNECTICUT, A. Fletcher Marsh, New Haven
FLORIDA, Cromwell Gibbons, Jacksonville
GEORGIA, C. L. Worsham, Atlanta
IDAHO, Jerome J. Day, Moscow
ILLINOIS, Alfred L. Baker, Chicago
INDIANA, William Holton Dye, Indianapolis
IOWA, Robert Hunter, Sioux City
KANSAS, Governor William R. Stubbs, Lawrence
KENTUCKY, C. C. Grassham, Paducah
LOUISIANA, Robert Roberts, Jr., Minden
MAINE, Cyrus C. Babb, Augusta
MARYLAND, Lynn R. Meekins, Baltimore
MASSACHUSETTS, C. A. Start, Boston
MICHIGAN, Francis King, Alma
MINNESOTA, E. W. Robinson, Minneapolis
MISSISSIPPI, H. L. Whitfield, Columbus
MISSOURI, George B. Logan, Saint Louis
MONTANA, F. L. Newman, Havre
 Rudolph Van Tolbel, Lewistown
NEBRASKA, Woodruff Ball, Valentine
NEW HAMPSHIRE, Geo. B. Leighton
NEW JERSEY, Frederick W. Kelsey, Orange
NEW MEXICO, W. A. Fleming Jones, Las Cruces
NEW YORK, J. S. Whipple, Salamanca
NORTH DAKOTA, James E. Boyle, Grand Forks
OHIO, Charles Lathrop Pack, Cleveland
OKLAHOMA, Benjamin Martin, Jr., Muskogee
OREGON, M. A. Moody, The Dalles
PENNSYLVANIA, M. I. McCreight, Dubois
RHODE ISLAND, H. A. Barker, Providence
SOUTH CAROLINA, E. W. Durant, Jr., Charleston

SOUTH DAKOTA, P. H. O'Neill, Faulkton
TEXAS, S. H. Cowan, Fort Worth
UTAH, Harden Bennion, Salt Lake City
VERMONT, George Aitkin, Woodstock
WASHINGTON, George H. Emerson, Hoquiam
WEST VIRGINIA, I. C. White, Morgantown
WISCONSIN, William Irvine, Chippewa Falls
WYOMING, E. H. Fourt, Lander

[3] The corrected list appears elsewhere (page [iv](#)).

[4] The reports submitted by States as mentioned in the responses to the Call of States are printed in the Supplementary Proceedings, beginning on page [327](#), and are entered in the Contents and Index of the volume.

[5] The formal report for Ohio appears in the Supplementary Proceedings.

[6] In the opinion he held that the plaintiff was entitled to judgment, declaring the Brooklyn Cooperage Company and Cornell University contract void, and directed the University to convey to the State of New York the 30,000 acres constituting the College Forest.

[7] The relative prices of forest lands sixty years ago and now may be judged from the fact that in 1850 a Law (Chapter 250) was passed providing that the State should not sell public land on Raquette river for less than 15 cents an acre. The State is now paying over \$7.00 an acre for the same kind of land.

[8] Declaration of Colonel David McClure in the Constitutional Convention of 1894.

INDEX

- Agriculture in schools, [265](#)
 Work of the Department of, [194](#)
- AITKIN, GEORGE, Report by, [373](#)
- Alaska coal lands, [21](#), [310](#)
- ALLEN, E. T., Report by, [367](#), [424](#)
- American Academy of Political and Social Science, Report of, [379](#)
- American Automobile Association, Report of, [380](#)
- American Civic Association, Report of, [383](#)
- American Forestry Association, Report of, [384](#)
- American Humane Association, Report of, [385](#)
- American Institute of Architects, Report of, [386](#)
- American Medical Association, Report of, [389](#)
 Work of, [255](#)
- American Paper and Pulp Association, Report of, [388](#)
- American Railway Engineering and Maintenance of Way Association,
 Report of, [392](#)
- American Railway Master Mechanics' Association, Report of, [393](#)
- American Scenic and Historic Preservation Society, Report of, [394](#)
- AMES, CHARLES W., Report by, [380](#)
- ANDERSON, A. T., Report by, [420](#)
- Arizona, Report from, [324](#)
- Arkansas, Report from, [314](#), [333](#)
- Association for the Protection of the Adirondacks, Report of, [397](#)
- Atlantic Deeper Waterways Association, Report of, [383](#)
-
- BABB, CYRUS C., Report by, [341](#)
- BABB, GEORGE W., Report by, [418](#)
- BABCOCK, GEORGE H., Report by, [411](#)
- BACKUS, E. W., Report by, [389](#)
- BAILEY, E. W. M., Report by, [411](#)
- BAILEY, LIBERTY HYDE, Address by, [203](#)
 Report by, [397](#)

BAKER, A. L., Address by, [222](#)
BAKER, BERNARD N., Introductory remarks by, [7](#)
 Report by, [314](#)
 Resolution of thanks to, [308](#)
BALDWIN, A. S., Report by, [393](#)
BALLINGER, R. A., Reference to work of, [76](#)
BANNISTER, O. B., Report by, [314](#)
BARKER, HENRY A., Report by, [324](#), [368](#), [384](#)
BARRETT, JOHN, Address by, [80](#), [237](#)
BARRETT, C. S., Report by, [411](#)
BENSON, R. DALE, Report by, [418](#)
BENTLEY, H. T., Report by, [394](#)
BEVERIDGE, Senator ALBERT J., Address by, [146](#)
BISSELL, R. M., Report by, [418](#)
BLANCHARD, Ex-Governor NEWTON C., Address by, [121](#)
BOARDMAN, Miss MABEL, Address by, [94](#)
BROOKS, Governor BRYANT B., Address by, [72](#)
BROWN, ELMER ELLSWORTH, Address by, [264](#)
 Letter from, [165](#)
BROWN, J. ADAMS, Report by, [416](#)
BRYCE, JAMES, Tribute to, [243](#)
BURNHAM, AUSTIN A., Report by, [420](#)
Business interest in Conservation, [222](#), [257](#)

California, Resources of, [116](#)
Call of States, Announcement of, [134](#), [171](#), [226](#), [298](#)
 Resumption of, [213](#), [299](#), [313](#)
Capital, Relation of, to resources, [226](#)
Carriage Builders' National Association, Report of, [410](#)
CASE, CHARLES R., Report by, [423](#)
CHAMPNEY, W. P., Report by, [411](#)
Cities, Wastes of, [309](#)
CLAPP, Senator MOSES E., Presiding Officer, [168](#)
CLEVELAND, GROVER, quoted on monopoly, [224](#)
Colorado, Report from, [334](#)
Columbia (District of), Report from, [324](#)
CONDRA, GEORGE E., Announcement by, [71](#)

Explanation by, [298](#)
Report by, [317](#)
Conference of Governors, Reference to work of, [87](#), [225](#), [245](#), [247](#),
[293](#), [379](#), [415](#)
CONNOLLY, MAURICE, Report by, [411](#)
Conservation
 of Minerals and Subterranean Waters, [429](#)
 of the Nation's Resources, [328](#)
 Practical Aspects of, [331](#)
 Problems, Address on, [106](#)
 Program, The, [292](#)
Constitution of the Congress, [ix](#)
 Adoption of, [79](#)
Country Life as related to Conservation, [203](#)
CRAIGHEAD, EDWIN BOONE, Address by, [168](#)
Credentials Committee, Appointment of, [79](#)
 Modification of, [106](#)
 Report of, [145](#)

DAMON, A. W., Report by, [418](#)
Daughters of the American Revolution, Work of, [270](#)
DAVIDSON, JAMES H., Address by, [132](#)
DAY, JEROME J., Report by, [336](#)
DEAN, WILLIAM B., Report by [380](#)
Delaware State Federation of Women's Clubs, Report of, [411](#)
DENEEN, Governor CHARLES E., Address by, [59](#)
Denmark, Successful agriculture of, [200](#)
DIXON, SAMUEL G., Reference to Work of, [301](#)
DORT, J. D., Report by, [411](#)
DRAPER, Mrs BELLE MERRILL, Reference to work of, [167](#)

EBERHART, Governor A. O., Address by [3](#), [298](#)
 Presiding Officer, [246](#)
EDSALL, Right Reverend SAMUEL COOK, Invocation by, [134](#)
Education and Conservation, [264](#)
EMERSON, GEORGE H., Report by, [320](#)
 "Timber Conservation," [424](#)

EVANS, POWELL, Report by, [382](#)
EVANS, R. D., Report by, [416](#)
Executive Committee, Report from, [317](#)
Executive powers, Definition of, [108](#)
Exports, Diminution of, [183](#)

Farm, Efficiency of the, [211](#)
 production, [136](#), [182](#), [190](#), [197](#)
 values, Losses in, [5](#)
Farmers' Union of America, Report of, [411](#)
FARQUHAR, A. B., "Practical Aspects of Conservation," [331](#)
Fertility, Conditions of, [210](#)
Forests
 and Stream Flow, [57](#), [216](#), [364](#), [400](#), [428](#)
 and the Nation, Address on, [214](#)
 Condition of, in Washington, [68](#)
 Conservation of, [199](#), [294](#), [398](#), [426](#)
 Destruction of, [11](#)
 National, Utilization of, [86](#), [261](#)
Forest Service, Reference to work of, [180](#)
FOSTER, J. ELLEN, Resolution in memory of, [276](#)
 Tribute to, [274](#)
FOURT, E. H., Report by, [324](#)
FOWLER, B. A., Report by, [324](#)
FINLEY, W. W., Address by, [135](#)
Fire Losses, Magnitude of, [418](#)
FISHER, IRVING, Reference to Work of, [249](#)
FISHER, WALTER L., Address by, [129](#)
Florida, Report from, [335](#)

GARFIELD, JAMES R., Address by, [106](#)
General Federation of Women's Clubs, Report of, [412](#)
Georgia, Report from, [325](#)
GIBBONS, CARDINAL, Greetings from, [3](#)
GIBBONS, CROMWELL, Report by, [335](#)
GILBERT, CASS, Report by, [387](#)
GIPE, JAMES C., Election of, as Recording Secretary, [307](#)

GLENN, L. C., Reference to work of, [428](#)
GOUDY, FRANK C., Report by, [334](#)
Government, Cost of, [185](#)
Governors, Western, Resolutions by, [72](#)
GRAVES, HENRY S., Address by, [214](#)
GREEN, F. R., Report by, [392](#)
GREEN, SAMUEL B., Resolutions in memory of, [313](#)
GREGORY, LUCIUS, Report by, [411](#)
GRIFFITH, E. M., Report by, [377](#)
GRIGGS, E. G., Report by, [375](#)

HALL, EDWARD HAGAMAN, Report by, [409](#)
HAMILTON, ALEXANDER, quoted on executive powers, [112](#)
HARDTNER, HENRY E., Report by, [339](#)
HARRIMAN, E. H., Reference to work of, [281](#)
HARVEY, WILLIAM S., "Forests and Stream Flow," [428](#)
 Report by, [299](#), [420](#)
HAY, Governor M. E., Address by, [64](#)
Health and Life as National assets, [247](#), [263](#)
HENEY, FRANCIS J., Address by, [278](#)
HIGLEY, WARREN, Report by, [409](#)
HILL, JAMES J., Address by, [177](#)
 Control of water-power by, [321](#)
 Explanation by, [194](#)
 Reference to Work of, [281](#)
 Tribute to, [13](#)
HOLLIDAY, CORNELIA R., Report by, [411](#)
HOLMES, J. A., Tribute to, [296](#), [325](#)
HORR, CHRISTOPHER G., Remarks by, [120](#), [313](#)
HOWARD, Mrs JAY COOKE, Address by, [167](#)
 Presentation by, [276](#)
HOWE, FREDERIC C., quoted on export trade, [184](#)
HUNT, G. M., Report by, [324](#)
 Resolution of thanks by, [307](#)

Idaho, Report from, [336](#)
Illinois, Resources of, [59](#)

Indiana, Report from, [314](#), [336](#)
Invocation by Archbishop Ireland, [1](#)
 Bishop Edsall, [134](#)
 Reverend J. S. Montgomery, [81](#)

Iowa, Report from, [337](#)
IRELAND, Archbishop, Invocation by, [1](#)
Iron ore of Minnesota, [12](#)
Irrigation, Extension of, [55](#)

JOHNS, WILLIAM DOUGLAS, Report by, [320](#)
JOHNSON, EMORY R., Report by, [380](#)
JONES, W. A. FLEMING, Report by, [347](#), [380](#)
 Resolution of thanks by, [307](#)
JUSSERAND, J. J., Quoted on forests and waterways, [428](#)

KAVANAUGH, W. K., Report by, [415](#)
KELLER, Mayor HERBERT E., Address of welcome by, [13](#)
KREMER, W. N., Report by, [418](#)
KRUEGER, A. W., Report by, [319](#)
KUNZ, GEORGE FREDERICK,
 "The Conservation of Minerals and Subterranean Waters," [429](#)

LACEY, JOHN F., Report by, [416](#)
Lakes-to-Gulf Deep Waterways Association, Report of, [413](#)
Land

 Laws, [17](#), [35](#)
 Laws, Needed reformation in, [288](#), [296](#), [425](#)
 taxation, [152](#)
 Titles, The Question of, [430](#)
Lands, Conservation and utilization of, [295](#)
Lands, Public, [16](#)
 Disposition of, [181](#)
LATCHAW, D. AUSTIN, Address by, [171](#)
 Election of, as Treasurer, [307](#)
LATTIMORE, GEORGE W., Report by, [319](#)
Laws That Should be Passed, [327](#)
LAZENBY, WILLIAM R., Report by, [364](#)

League of American Sportsmen, Report of, [415](#)
Legislative functions, Failure in performing, [280](#)
LEWIS, T. L., Report by, [424](#)
Life and Health, Address on, [247](#)
Louisiana, Report from, [339](#)
 Resources of, [124](#)
LOWE, E. N., Report by, [315](#)

MACKENZIE, ALEXANDER, Reference to work of, [414](#)
Maine, Report from, [341](#)
MARSHALL, Chief Justice JOHN, Opinions by, [109](#)
MARTIN, BEN J., JR., Report by, [319](#), [365](#)
 Resolution of thanks by, [326](#)
Massachusetts, Report from, [343](#)
MAXWELL, HU, Report by [376](#)
MCCRAY, FRANKLIN, "The Question of Land Titles," [430](#)
McDERMOTT, J. H., Report by, [416](#)
McENIRY, M. J., Report by, [422](#)
McGEE, W J, Reference to work of, [414](#)
McNEES, G. W., Report by, [301](#)
McVEY, FRANK L., Address by, [152](#)
MERRITT, E. T., Report by, [372](#)
MERSHON, WILLIAM B., Report by, [416](#)
METZGER, A. E., Report by, [336](#)
MILLER, A. C., Report by, [314](#), [337](#)
Minerals, Conservation of, [296](#), [429](#)
Minnesota, Deforestation of, [11](#)
 Farm Value in, [6](#)
 Iron Ore in, [12](#)
 Report from, [315](#)
Mississippi, Report from, [315](#)
 Resources of, [48](#)
Missouri, Report from, [315](#), [344](#)
 Valley River Improvement Association, Report of, [420](#)
 Waterways Commission, work of, [414](#)
MITCHELL, A. R., Report by, [392](#)
Monopoly, Growth of, [224](#)

of Power, [101](#)
of resources in California, [116](#)
of transportation, [191](#)
MOORE, WILLIS L., Reference to work of, [428](#)
Montana, Report from, [345](#)
Resources of, [52](#)
MONTGOMERY, Reverend J. S., Invocation by, [81](#)
MUCKLE, M. RICHARD, Report by, [386](#)

National Board of Fire Underwriters, Report of, [416](#)
Board of Trade, Report of, [419](#)
Business League of America, Report of, [420](#)
Conservation Association, Endorsement of, [371](#)
Conservation Commission, Reference to Report of, [247](#)
Efficiency, Promotion of, [296](#)
Nebraska, Report from, [317](#)
NELSON, Senator KNUTE, Address by, [35](#)
NESTOS, R. A., Report by, [318](#)
NEWELL, F. H., Reference to work of, [414](#)
NEWLANDS, Senator FRANCIS G., "Laws that Should be Passed," [327](#)
New Mexico, Report from, [347](#)
New York, Report from, [318](#), [347](#), [352](#)
NOBLE, JOHN W., Report and resolution by, [315](#)
NOEL, Governor EDMOND N., Address by, [48](#)
Tribute to, [315](#)
Nominating Committee, Creation of, [246](#)
Report of, [303](#)
NORRIS, Governor EDWIN L., Address by, [52](#)
North American Conservation Congress, Work of, [247](#)
North Dakota, Report from, [318](#), [362](#)

Ohio, Report from, [364](#)
Oklahoma, Report from, [365](#)
Oregon, Report from, [367](#)
OVEY, ESMOND, Address by, [243](#)

PAGE, THOMAS NELSON, Reference to work of, [271](#)

Pan-American Union, Work of, [237](#)
PARDEE, EX-GOVERNOR GEORGE C., Address by, [115](#)
 Report of Resolutions Committee by, [308](#)
PARRY, D. M., Report by, [411](#)
PARTRIDGE, JOHN, Report by, [386](#)
Pennsylvania, Report from, [299](#)
PERSONS, HENRY H., Report by, [352](#)
Phosphate lands, Withdrawal of, [29](#)
PINCHOT, GIFFORD, Address by, [292](#)
 Memorial resolution by, [313](#)
 Nominating address by, [304](#)
 Presentation of certificate to, [276](#)
 Reference to work of, [54](#), [166](#), [373](#)
 Response by, [152](#), [277](#)
 Tribute to, [48](#), [76](#), [117](#), [151](#), [159](#), [221](#), [242](#), [325](#)
PLUNKETT, Sir HORACE, Reference to work of, [87](#)
Power, Address on, [101](#)
Press (The), Address on, [171](#)
PRIESTLY, J. T., Report by, [392](#)
Property, Safeguarding public, [278](#)
PURDUE, A. H., Report by, [314](#)

Railway control of waterways, [64](#)
 subsidies, Magnitude of, [283](#)
Railways, Control of, to prevent forest fires, [351](#), [401](#)
 of the South and Conservation, [135](#)
RANE, FRANK WILLIAM, Report by, [343](#)
RATTERMANN, H., Report by, [411](#)
Reclamation Service, Reference to work of, [178](#)
Red Cross, Address on work of, [94](#)
REDDING, SID B., Report by, [333](#)
Republics, Address on American, [237](#)
Resolutions adopted by the Congress, [308](#)
Resolutions Committee, Announcement of, [80](#), [145](#)
 Composition of, [311](#)
 Report by, [308](#)
Resources as the basis for business, Address on, [222](#), [257](#)

Rhode Island, Report from, [324](#), [368](#)
RICHARDSON, GUY, Report by, [386](#)
RICHTER, JOSEPH C., Report by, [301](#)
Rivers, Federal control of, [125](#)
Roads, Value of good, [11](#)
ROBINSON, DOANE, Report by, [369](#)
ROOSEVELT, THEODORE, Address by, [82](#)
 cited on civilization, [351](#)
 Reference to work of, [242](#)
 Tribute to, [13](#), [38](#), [59](#), [76](#), [116](#), [151](#)
ROSS, E. W., Interlude by, [302](#), [312](#)
 Privileged statement by, [322](#)
ROTH, FILIBERT, Reference to work of, [428](#)
ROTHROCK, J. T., Reference to work of, [301](#)

SALISBURY, O. J., Report by, [372](#)
SARGENT, WILL L., Report by, [370](#)
SCOTT, Mrs MATTHEW T., Address by, [270](#)
Senate, Policies of the, [279](#)
SHELDON, CHARLES S., Report by, [392](#)
SHIPP, THOMAS R., Election of, as Secretary, [306](#)
SHORT, FRANK H., Address by, [226](#)
SHORTALL, JOHN L., Report by, [386](#)
SIMMONS, WALLACE D., Address by, [257](#)
SMITH, C. G., Report by, [418](#)
SMITH, HERBERT KNOX, Address by, [101](#)
SMYTHE, WILLIAM E., Address by, [127](#)
SNEATH, Mrs G. B., Address by, [166](#)
 Report through, [414](#)
SNYDER, WILLIAM A., Report by, [411](#)
Soil as a Resource, [6](#), [205](#), [259](#)
 Impoverishment of, [198](#)
South Carolina, Report from, [369](#)
South Dakota, Report from, [319](#), [369](#)
 Resources of, [77](#)
SPRAGUE, HENRY H., Report by, [343](#)
START, EDWIN A., Report by, [385](#)

STEVENS, Representative F. C., Address by, [201](#)

STEVENS, FREDERICK C., Report by, [380](#)

STILLMAN, WILLIAM O., Report by, [386](#)

STUBBS, Governor W. R., Address by, [75](#), [81](#)

Presiding Officer, [35](#)

SWAIN, GEORGE F., Reference to work of, [428](#)

TAFT, President, Address by, [14](#)

Cited on National functions, [116](#)

Tribute to, [13](#)

Taxation

of forests, [295](#)

of National resources, Address on, [152](#)

Texas, Report from, [370](#)

THORP, FREEMAN, Report by, [315](#)

Timber Conservation, [424](#)

TOMKIES, Mrs HOYLE, Address by, [163](#)

Trans-Mississippi Congress, Resolution concerning, [315](#)

Tree Planters of America, Proposed organization of, [420](#)

TWICHELL, JEROME, Report by, [421](#)

United Mine Workers, Report of, [424](#)

Upper Mississippi River Improvement Association, Report of, [421](#)

Utah, Report from, [372](#)

Vermont, Report from, [373](#)

VESSEY, Governor, Address by, [77](#)

Vice-Presidents, Ratification of the, [313](#)

VON TOBEL, RUDOLPH, Report by, [317](#), [345](#)

VON SCHRENK, HERMANN, Report by, [344](#)

WAGSTAFF, ALFRED, Report by, [386](#)

WALDRON, C. B., Report by, [362](#)

WALLACE, HENRY, Address by, [188](#), [305](#)

Nomination and election of as President, [304](#)

WARFIELD, R. EMORY, Report by, [418](#)

Washington, Report from, [320](#), [375](#)

Resources of, [64](#)

Sentiment of, [120](#)
Washington State Federation of Labor, Report of, [422](#)
Wastes of civilization, [309](#)
Water conservation, [404](#), [429](#)
Water-power, Address on, [101](#)
 Control of, [222](#), [365](#), [405](#)
 Disposition of, [30](#), [54](#), [59](#), [105](#), [129](#), [156](#), [294](#), [321](#), [338](#), [349](#), [356](#)
 Value of, [287](#)
Water resources of New York, [352](#)
Water supply, Conditions of, [343](#), [353](#)
Waterways, Development of, [163](#), [412](#)
 Utilization of, [83](#), [294](#), [364](#)
WATSON, E. J., Report by, [369](#)
WEIS, SAMUEL, Report by, [386](#)
WELCH, Mrs GEORGE O., Address by, [160](#)
WESBROOK, F. F., Address by, [247](#)
 Report by, [392](#)
WEST, WILLIAM L., Report by, [380](#)
Western Forestry and Conservation Association, Report of, [423](#)
West Virginia, Report from, [324](#), [376](#)
WHIPPLE, J. S., Report by, [347](#)
WHITE, L. C., Report by, [324](#)
WHITE, J. B., Address by, [158](#)
 "Conservation of the Nation's Resources," [328](#)
 Nominating Address by, [303](#)
 Report by, [316](#)
WHITNEY, MILTON, Reference to work of, [136](#)
Women's Clubs, Work of, [161](#)
WORSHAM, E. L., Report by, [325](#)
WILKINSON, Mrs J. D., Report by, [415](#)
WILLIAMS, Mrs LYDIA PHILLIPS, Reference to work of, [161](#)
WILSON, DANIEL T., Report by, [411](#)
WILSON, Secretary JAMES, Address by, [194](#)
 Tribute to, [183](#), [191](#)
Wisconsin, Report from, [377](#)
Wyoming, Report from, [324](#)
 Resources of, [72](#)

YOUNG, P. C., Report by, [314](#)

J. H. BERKSHIRE, President

O. W. FISHER, Vice-Pres.

J. B. WHITE, Sec'y-Treas. and Gen'l Mgr.

Missouri Lumber & Land Exchange Company

MISSOURI SOFT PINE
LONG LEAF YELLOW PINE
SHORT LEAF YELLOW PINE
OAK FLOORING

LOUISIANA MILLS:

LOUISIANA LONG LEAF LUMBER CO., FISHER, LA.
1 Pine and 1 Oak Flooring Mill

LOUISIANA LONG LEAF LUMBER CO., VICTORIA, LA.
1 Pine Mill

LOUISIANA CENTRAL LUMBER CO., CLARKS, LA.
2 Pine Mills

LOUISIANA CENTRAL LUMBER CO., STANDARD, LA.
1 Pine Mill

MISSOURI MILLS:

MISSOURI LUMBER & MINING CO., WEST EMINENCE, MO.
1 Pine Mill

OZARK LAND & LUMBER CO., WINONA, MO.
1 Pine and 1 Oak Mill

Annual Capacity 250,000,000 Feet

High Grades Quick Shipments

MAIN OFFICE 1111 LONG BUILDING
KANSAS CITY, MISSOURI

Missouri Lumber & Land Exchange Company

SHIPMENTS

We ship stock of our own manufacture exclusively. We carry large, well assorted stocks and quote only when we are sure of being able to comply with grade and time requirements. As our mills are located on four (4) different railroads we can always ship orders without delay.

FINISHING LUMBER

We specialize on FINISHING lumber and every care is exercised to make it THE BEST. By separating all widths and lengths, we are in a position to quote on any particular width or length desired. We carry in stock Finish as wide as 26 inches, in 4-4, 5-4, 6-4 and 8-4 thicknesses. Our MISSOURI FINISH, owing to its flexibility, lightness, softness and figure, is peculiarly adapted to all finishing work. We also carry all grades of Finish in 4, 6 and 8 feet lengths.

**If You want GOOD GRADES and PROMPT SERVICE, write the
"EXCHANGE"**

Missouri Lumber & Land Exchange Company

PINE YARD AND SHED STOCK

We carry a large stock of all yard and shed material, which is piled scientifically to avoid sap-staining and bluing and to insure dryness

Our upper grades are all
STEAM KILN DRIED
and a large percentage of our lower grades are either
STEAM KILN DRIED

or
SODA DIPPED

Either process precludes the possibility of sap-staining

**If You want GOOD GRADES and PROMPT SERVICE, write the
"EXCHANGE"**

Missouri Lumber & Land Exchange Company

OAK FLOORING

In the manufacture of our "DIAMOND BRAND" Oak Flooring the greatest possible care is taken in every department. Our Oak Flooring, owing to our method of drying and working, gives COMPLETE SATISFACTION in all climates. Due to the fact that our mill is located at the same point where we operate a pine plant we can ship Oak Flooring and Pine in the same car

Oak Timbers, Factory Stock, Etc.

We manufacture Oak Timbers, Ties, Car Framing, Grain Doors, Factory Stock, Etc. We can ship Oak, rough or dressed, in mixed cars with Yellow Pine

**If You want GOOD GRADES and PROMPT SERVICE, write the
"EXCHANGE"**

Missouri Lumber & Land Exchange Company

RAILROAD MATERIAL

We manufacture large quantities of Bridge Timbers, Cross Ties (Both Oak and Pine), Braces, Etc., with or without heart specifications, shipping same rough or surfaced as may be desired.

CAR MATERIAL

By specializing on car material we are in a position to handle satisfactorily orders for Sills, Framing (both Oak and Pine), Lining, Roofing, Siding, Insulation, Grain Doors, Etc.

**If You want GOOD GRADES and PROMPT SERVICE, write the
"EXCHANGE"**

Missouri Lumber & Land Exchange Company

SHORTS

We carry in stock 4, 6 and 8 foot lengths, in all grades from No. 3 Common to "A," in both 4-4 and 8-4 stock. All widths and lengths are *stacked separately*. We will work these "shorts" to any pattern desired or ship rough, trimming to any length specified

SPECIAL MATERIAL

We manufacture any special items desired, in Oak and Pine, such as Spreader Slats, Step-Ladder Material, Pulley Stiles, Crating Stock, Etc. If you use anything in "shorts," regardless of grade, length, width or pattern, send us your inquiry.

**If You want GOOD GRADES and PROMPT SERVICE, write the
"EXCHANGE"**

Transcriber's Notes

Spelling irregularities where there was no obviously preferred version were left as is. Variants include: derivatives of "Allegheny" and "Alleghany;" "antitoxin" and "anti-toxin;" "benefited" and "benefitted;" "Cæsar" and "Caesar;" "caretakers" and "care-takers;" "Chile" and "Chili;" derivatives of "criticise" and "criticize;" "drought" and "drouth;" "employes" and "employees;" "entry-man" and "entryman;" "endorse" and "indorse;" "fellowmen" and "fellow-men;" derivatives of "fiber" and "fibre;" "fulfil" and "fulfill;" "headwaters" and "head-waters;" derivatives of "homeseeker" and "home-seeker;" "horsepower" and "horse-power;" "inappreciable" and "unappreciable;" "lawmakers" and "law-makers;" "lifetime" and "life-time;" "livestock" and "live-stock;" "patrolmen" and "patrol-men;" derivatives of "payroll" and "pay-roll;" "Phoenix" and "Phenix;" "playgrounds" and "play-grounds;" derivatives of "postoffice" and "post-office;" "preeminently" and "pre-eminently;" derivatives of "preempt" and "pre-empt;" "prerequisite" and "pre-requisite;" "rainfall" and "rain-fall;" "Savior" and "Saviour;" derivatives of "short-sighted" and "shortsighted;" derivatives of "stockman" and "stock-man;" derivatives of "theater" and "theatre;" "Rudolph von Tobel," "Rudolf von Tobel" and "Rudolph van Tolbel;" "traveled" and "travelled;" "upbuilding" and "up-building;" "waterfalls" and "water-falls;" "watershed" and "water-shed," and their plurals; "workaday" and "work-a-day;" "workshops" and "work-shops."

Some of the index entries are not in strict alphabetical order, but they were left as is.

Changed "Mrs." to "Mrs" for consistency on page iii: "Mrs J. Ellen Foster."

Changed "Governer" to "Governor" on page v: "Ex-Governor Blanchard."

Changed "Thorpe" to "Thorp" on page vii.

Inserted missing "8." into caption of figure following page xii.

Changed "by" to "be" on page 2: "hallowed be Thy name."

Inserted missing comma after "fruit-raising" on page 10.

Changed "wth" to "with" on page 10: "with so many attractions."

Changed "non-irrigible" to "non-irrigable" on page 17: "applying to non-irrigable lands."

Changed "mattter" to "matter" on page 21: "set in this matter."

Changed "Phosphorous" to "Phosphorus" and "phosphorous" to "phosphorus" on page 29.

Changed "nothwithstanding" to "notwithstanding" on page 48: "notwithstanding the land."

Changed "believeing" to "believing" on page 49: "believing that our rights."

Changed "640 acre-tracts" to "640-acre tracts" on page 53.

Changed "reregarding" to "regarding" on page 59: "regarding this program."

Changed "Lakes-to-the-Gulf" to "Lakes-to-Gulf" on page 60: "the Lakes-to-Gulf deep waterway."

Changed "coöperate" to "cooperate" on page 63: "cooperate and work together."

Changed "attenton" to "attention" on page 68: "call your attention."

Changed "fadists" to "faddists" on page 71: "faddists, dreamers, and enthusiasts."

Changed "o'clock" to "oclock" on page 78, for consistency.

Changed "Tumultous" to "Tumultuous" on page 93: "Tumultuous applause."

Changed "Brifly" to "Briefly" on page 96: "Briefly, then."

Changed "subcommittee" to "sub-committee" on page 96: "A sub-committee, six of whom."

Changed "devasting" to "devastating" on page 98: "devastating hand."

Changed "prevent" to "prevents" on page 111: "that prevents all progress."

Changed "phase" to "phrase" to correct the quotation from Alexander Hamilton on page 112: "another phrase for a bad execution."

Changed "essenial" to "essential" on page 112: "not less essential."

Changed "differene" to "difference" on page 112: "a wide difference."

Changed "Mr." to "Mr" twice on page 132 for consistency: "Mr Bernard N. Baker" and "Mr Chairman."

Changed "re-forested" to "reforested" on page 142: "mountain sides are to be reforested."

Changed "multi-millionaries" to "multi-millionaires" on page 150.

Changed "queston" to "question" on page 158: "the question of taxation."

Changed "recources" to "resources" on page 160: "natural resources."

Changed "foresty" to "forestry" on page 161: "the line of forestry."

Changed "Mrs." to "Mrs" on page 165: "Mrs Hoyle Tomkies."

Changed "extravagancies" to "extravagances" on page 194: "all their extravagances."

Changed "offences" to "offenses" on page 227: "failures and offenses."

Changed "fertilty" to "fertility" on page 250: "the fertility of the soil."

Changed "gaities" to "gaieties" on page 242: "the gaieties of fashionable resorts."

Removed mismatched double quotation character on page 245, before "Gentlemen, please hold steady."

Changed "re-assembled" to "reassembled" on page 246: "The Congress reassembled."

Changed "his" to "this" on page 261: "this kind of revenue-making regulation."

Changed "responsibilties" to "responsibilities" on page 267: "their powers and responsibilities."

Changed "innoculate" to "inoculate" on page 271: "to inoculate them."

Changed "devasted" to "devastated" on page 275: "devastated by earthquake."

Changed "Lascruses" to "Las Cruces," "Lewiston" to "Lewistown" and "Aitken" to "Aitkin" in the footnote 2.

Changed "Mississippipi" to "Mississippi" on page 315: "We from Mississippi."

Inserted "as" on page 315: "prosecuted as rapidly as possible."

Changed "Washingon" to "Washington" on page 321: "the State of Washington and its officials."

Changed "sovereignity" to "soverignty" on page 327: "National soverignty."

Changed "dividuals" to "individuals" on page 330: "than can individuals."

Changed "extravagence" to "extravagance" on page 331: "extravagance and waste."

Changed "non-preventible" to "non-preventable" on page 332: "non-preventable accidents."

Changed "Metzer" to "Metzger" on page 336.

Changed "necesssary" to "necessary" on page 342: "the necessary steps."

Changed "perserved" to "preserved" on page 351: "preserved for all time."

Changed "runoff" to "run-off" on page 356: "run-off of streams."

Changed "Henry A. Baker" to "Henry A. Barker" on page 368.

Changed "Greely" to "Greeley" on page 376.

Changed "Agronomy" to "Agronomy" on page 380: "Chief in Agronomy and Chemistry."

Changed "viligance" to "vigilance" on page 396: "the vigilance of the Society."

Changed "his" to "this" on page 405: "The provisions of this section."

Changed "Mrs." to "Mrs" and "S. B." to "G. B." on page 413: "Mrs G. B. Sneath."

Changed "centre" to "center" on page 413: "center of the most fertile valley."

Changed "acretions" to "accretions" on page 414: "the rich accretions."

Changed "sportsmens" to "sportsmen's" on page 416: "sportsmen's travel."

Changed "kilowat" to "kilowatt" twice on page 421.

The index entry for "Blanchard" incorrectly references page 94, rather than page 121. Page 94 should be the reference for Miss Boardman, which is missing. These errors were corrected.

Changed "Emeperson" to "Emerson" in the index entry for "Emerson, George H."

Changed "Commitee" to "Committee" in the index entry for "Executive Committee, Report from."

Changed "Mathew" to "Matthew" in the index entry for "Scott, Mrs Matthew T."

Changed "Philips" to "Phillips" in the index entry for "Williams, Mrs Lydia Phillips."

*** END OF THE PROJECT GUTENBERG EBOOK PROCEEDINGS
OF THE SECOND NATIONAL CONSERVATION CONGRESS AT
SAINT PAUL, SEPTEMBER 5-8, 1910 ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.